

3D SPI Programming Manual

saki

[SJ00DCM06-04E]

First of all, thank you for purchasing the product of Saki Corporation.

Before using the product, please read this manual carefully together with the supplied manuals to ensure the prolonged life with the proper handling and the operation of this system.

Notations used in this manual

Notation	Meaning
DANGER	“DANGER” indicates an imminently hazardous situation which could be threat to life or cause serious injury.
WARNING	“WARNING” indicates a potentially hazardous situation which could be threat to life or cause serious injury.
CAUTION	“CAUTION” indicates a potentially hazardous situation which could result in injury or equipment (including peripheral) damage.
NOTE	“NOTE” indicates an advice on operation.

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Saki Corporation 3D SPI Programming Manual (Original Instructions)

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Basic Operation

1 Basic Operation of the Inspection Software

1.1 Starting/Quitting the Inspection Software

This section describes how to start/quit the inspection software.

1.1.1 Starting the Inspection Software

Step1: Activate the system according to **Instruction Manual**.

NOTE The PC in the machine is powered on in conjunction with the main breaker.

NOTE When using BF2-Editor, manually turn on the PC.

Step2: When the Operating System (hereinafter OS) is activated and the login screen appears on the LCD monitor, enter the following login name and password.

Login Name : sakicorp

Password : sakicorp

NOTE **Login Name** and **Password** are case-sensitive.



Figure 1-1 Login Screen

Step3: After succeeded login, the inspection software launches automatically.

NOTE If the software fails to start automatically, double-click the inspection machine's shortcut icon on the desktop to start the inspection software.
When using BF2-Editor, double-click the **BF2-Editor** shortcut icon on the desktop to start the inspection software.

Step4: The dialog box shown below appears.
Select Group and Board and click **Open**.

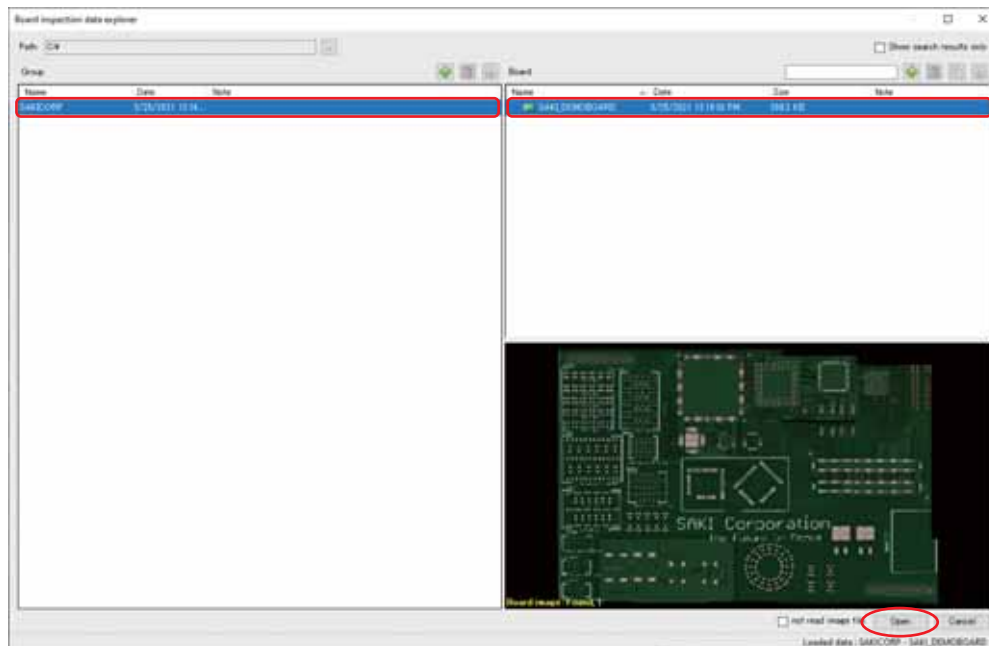


Figure 1-2 Loading an Inspection Data

Step5: If library has been changed, the dialog box shown below appears.
With a click of **All**, all shapes and recipes of inspection data to be opened can be updated with the library contents.
With a click of **Except local modified**, shapes and recipes which have not used **Save as local library** or **Save as local component** in the inspection data to be opened are updated with the library contents. The **Except local modified** button is available only for the inspection data which is using **Save as local library** or **Save as local component**.
With a click of **No**, inspection data is opened without updating from library.

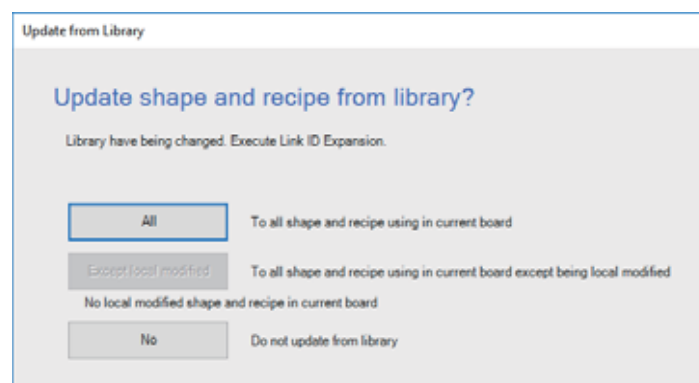


Figure 1-3 Library Update

NOTE Library exists per group and is shared to use one library in the same group. By using library, the modified contents of recipes and shapes in the inspection data can be developed easily to other inspection data in the same group.

NOTE For local modifications of library, refer to **Part II 2.7 Exit the Shape Editing Screen** and **Part II 2.10 Exit the Recipe Editing Screen**.

Step6: The inspection machine is in the emergency stop mode when it is activated.
Cancel the emergency stop mode according to **Instruction Manual**.

NOTE When BF2-Editor is used, this step is not necessary.

1.1.2 Quit the Inspection Software

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **Exit BF2**.

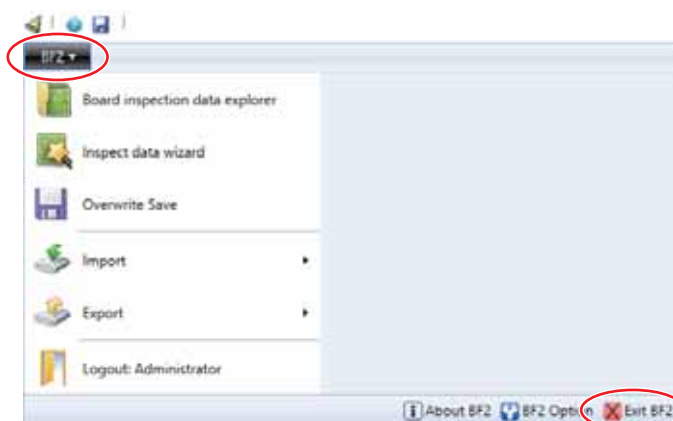


Figure 1-4 Quit the Inspection Software 1

Step2: The dialog box shown below appears.
Yes Goes to Step3.
No Returns to the original screen.

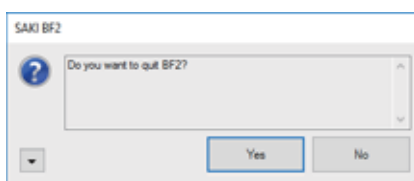


Figure 1-5 Quit the Inspection Software 2

Step3: The dialog box shown below appears.
Yes Overwrites the inspection data and quits the inspection software.
No Quits the inspection software without saving the inspection data.
Cancel Returns to the original screen without saving the inspection data.

NOTE Saving the inspection data is recommended when the data is modified.

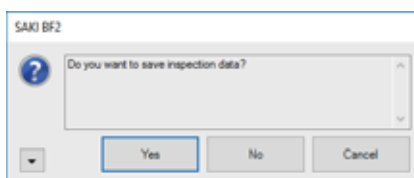


Figure 1-6 Quit the Inspection Software 3

Step4: From the desktop window, select  (Start button) > **Power** > **Shut down** and wait until OS of PC shutdowns.

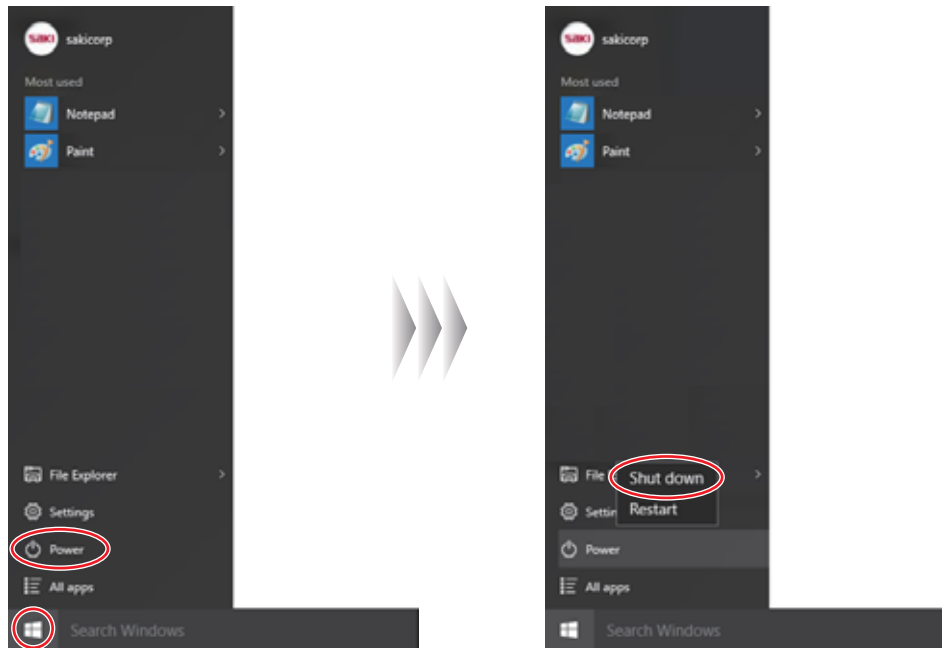


Figure 1-7 Shut down

Step5: Power off the inspection machine according to **Instruction Manual**.

NOTE In the case of using BF2-Editor, this step is not necessary.

1.2 Window Details

This section describes the inspection software window details.

NOTE The below illustration shows the default window layout.
The window layout can be changed freely.

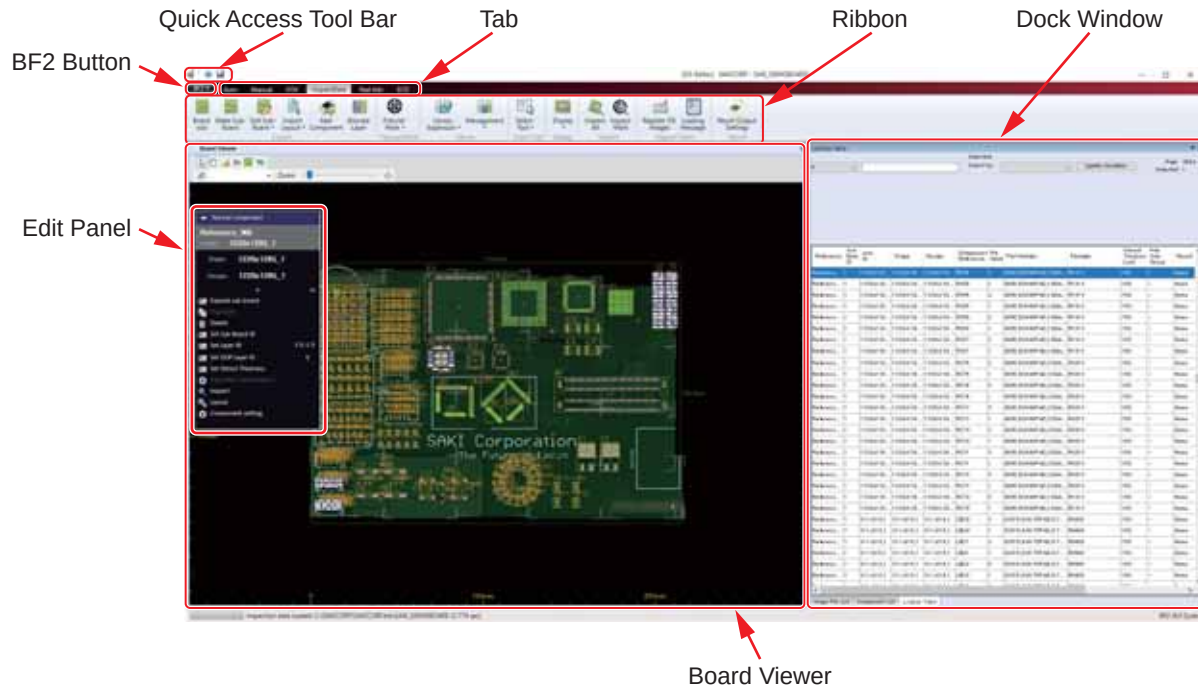


Figure 1-8 Window Details

Item	Description
BF2 Button	Used for saving and selecting an inspection data.
Quick Access Tool Bar	Used for Overwrite Save and displaying Help.
Tab	Switches the ribbon.
Ribbon	Allows the functions for inspections to be used.
Dock Window	Displays inspection data settings and inspection results.
Edit Panel	Used for editing and adding component data.
Board Viewer	Displays a PCB image and inspection data.

Table 1-1 Descriptions

1.3 Selecting an Inspection Data File

Step1: Make sure that there is no PCB on the scan table of the machine.

NOTE This step is not necessary when BF2-Editor is used.

Step2: Click the **BF2** button on the upper left side of the window to open the menu and click **Board inspection data explorer**.

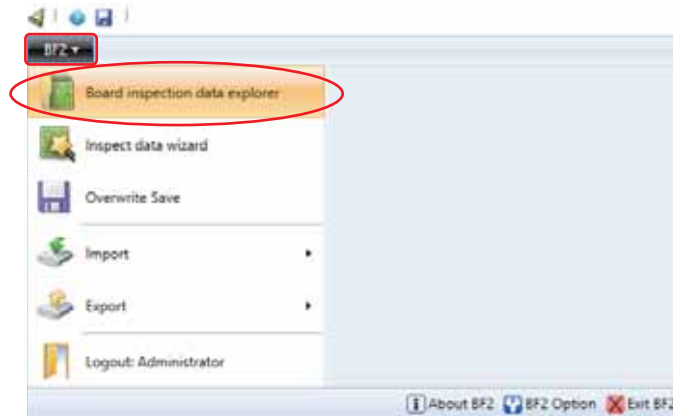


Figure 1-9 Selecting an Inspection Data File 1

Step3: Select **Group** and **Board** and click **Open**.

NOTE The file extension for an inspection group file is “.bfs”.

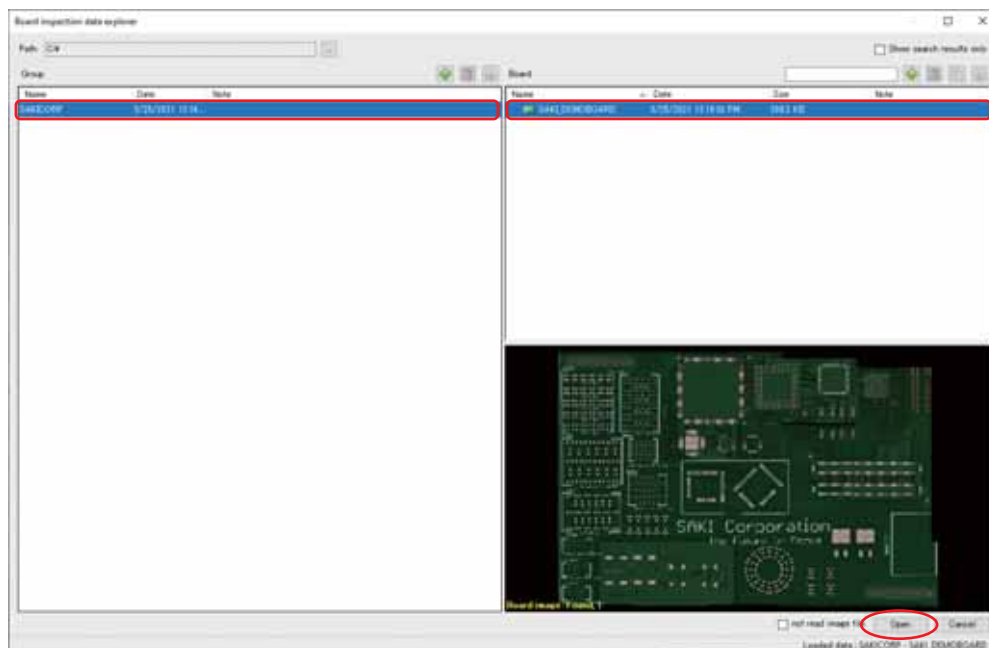


Figure 1-10 Selecting an Inspection Data File 2

Step4: If the library has been modified, the dialog box shown below appears.
For details, refer to Step5 of **1.1.1 Starting the Inspection Software**.

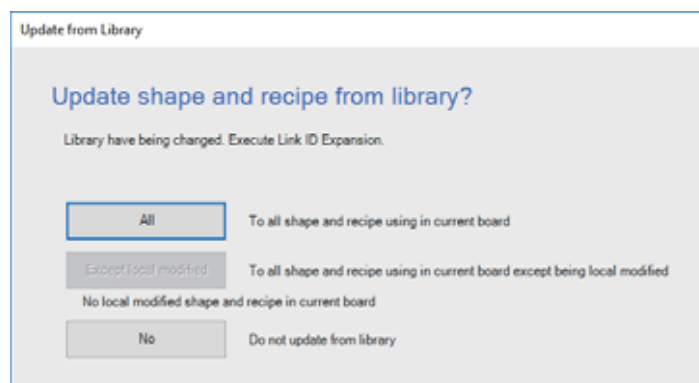


Figure 1-11 Library Update

Step5: Set the conveyor rail width of the machine according to **Instruction Manual**.

NOTE When BF2-Editor is used, this step is not necessary.

1.4 Saving the Inspection Data File

Click the **BF2** button on the upper left side of the window and select **Overwrite Save**.
Alternatively, click the  button on the quick access tool bar.

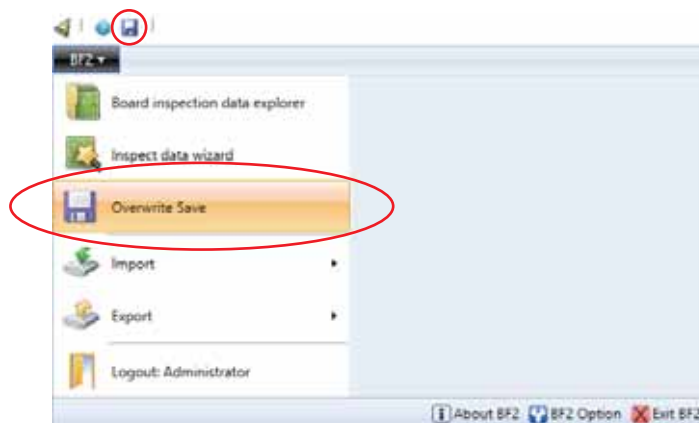


Figure 1-12 Overwriting an Inspection Data File

1.5 Basic Operation in Board Inspection Data Explorer

1.5.1 Window Detail

Here describes Board inspection data explorer details.

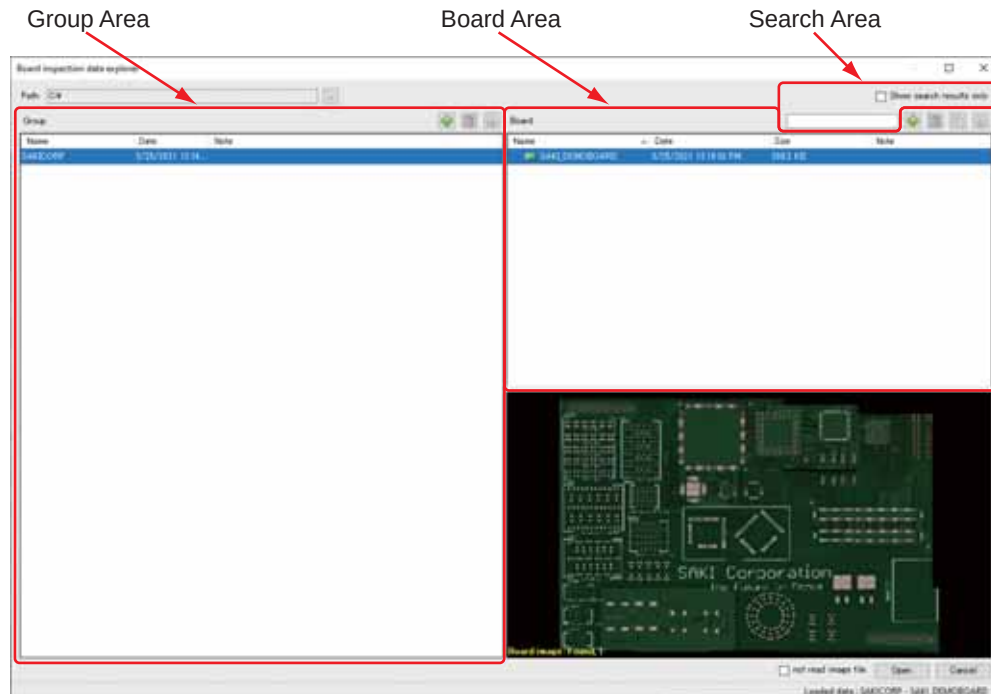


Figure 1-13 Window Details

Item		Description
Path		Displays the path of the data folders that is being expanded in Board inspection data explorer.
Group Area	Group	Displays group list.
		Create the new group and board data. For details, refer to 1.5.4 Creating a New Group .
		Delete the all data of selected group. For details, refer to 1.5.5 Deleting a Group .
	 Rename Edit note	Changes the name of the selected group. For details, refer to 1.5.6 Changing a Group Name. Edits the note of the selected group. For details, refer to 1.5.7 Editing a Group Note.
Search Area	Search	Search by board name.
	Show search results only	Check to show only search results in Board inspection data explorer.

Item		Description	
Board Area	Board	Displays the board data and folders that are currently selected in a list.	
		Creates new board data in the currently selected group. For details, refer to 1.5.15 Creating a Board Data .	
		Deletes selected board data or folder. For details, refer to 1.5.10 Deleting a Board Data and 1.5.11 Deleting a Folder .	
		Copies the currently selected board data and saves it with a different name. For details, refer to 1.5.16 Copying a Board Data .	
		Create folder	Creates a new folder in the currently selected group. For details, refer to 1.5.9 Creating a New Folder .
		Move	Moves the selected board data or folder to an arbitrary folder in the group. For details, refer to 1.5.17 Moving a Board Data and Folder .
		Rename	Changes the name of selected board data or folder. For details, refer to 1.5.12 Changing a Board Name and 1.5.13 Changing a Folder Name .
		Edit Note	Changes the name of selected board data or folder. For details, refer to 1.5.14 Editing a Board Data Note .
		Edit message	There is a function that displays the message passed to the next operator when the board data has finished being load. Edit the above message. For details, refer to 1.5.8 Setting a Displaying Message .
		Show board image	Check to show board images on Board inspection data explorer.
not read image file		Check to load the board data without image files set to Set as default . For details, refer to Part II 5.4.3 Image File Auto Loading Setting .	
Open		Loads the selected board data and displays the inspection software screen. For details, refer to Step4 to Step6 of 1.1.1 Starting the Inspection Software .	
Cancel		Closes Board inspection data explorer and moves to inspection software screen.	

Table 1-2 Descriptions

1.5.2 Displaying Board Inspection Data Explorer

Here describes how to display Board inspection data explorer.

Click **BF2** button on the upper left side of the window to open the menu and click **Board inspection data explorer**. Board inspection data explorer will appear.

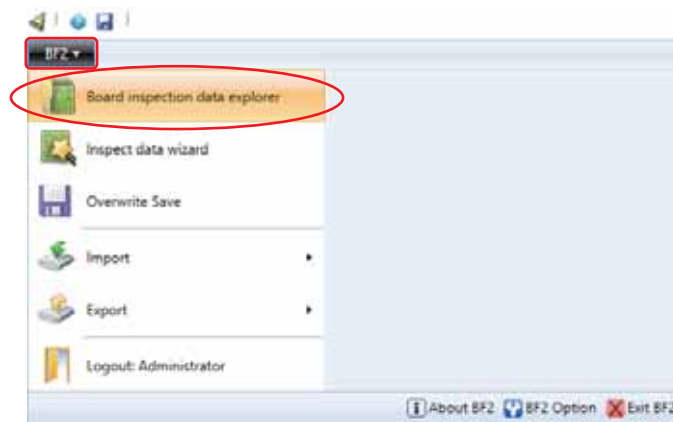


Figure 1-14 Displaying Board Inspect Data Explorer

1.5.3 Loading Board Data

For the procedures for loading an arbitrary board data, refer to **1.3 Selecting an Inspection Data File**.

1.5.4 Creating a New Group

Here describes how to create group and board data.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Click  on the Group area.

Step3: Enter a New Group name and a New board name and click **OK**.

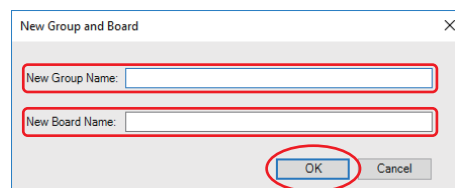


Figure 1-15 Create a New Group

1.5.5 Deleting a Group

Here describes how to delete the group.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a group and click  on the Group area.

NOTE

When a board of a group is being load,  cannot be clicked.

In this case, load the board of another group before Step 2.

For the procedures for loading board data, refer to **1.3 Selecting an Inspection Data File**.

Step3: The dialog box shown below appears. Click **OK**.

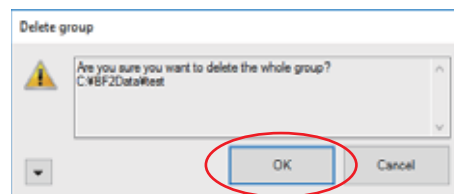


Figure 1-16 Delete a Group

1.5.6 Changing a Group Name

Here describes how to change the group name.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a group and select  > **Rename** on the Group area.

NOTE

When a board of a group is being load, **Rename** cannot be selected.

In this case, load the board of another group before Step 2.

For the procedures for loading board data, refer to **1.3 Selecting an Inspection Data File**.

Step3: The dialog box shown below appears. Enter a group name and click **OK**.

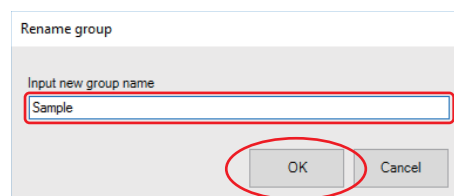


Figure 1-17 Changing a Group Name

1.5.7 Editing a Group Note

Here describes how to edit group note.

Placing an arbitrary comment allows for confirmation to be performed in Board inspection data explorer.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a group and select  > **Edit note** on the Group area.

Step3: Edit note and click **OK**.

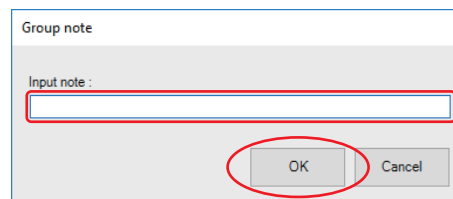


Figure 1-18 Editing Group Note

1.5.8 Setting a Displaying Message

There is a function that displays the message passed to the next operator when the board data has finished being load.

Here describes how to set the message and the procedures for displaying the message.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a board data and select  > **Edit message** on the Board area.

NOTE

Messages can also be edited from the **InspectData** tab.
Select **Edit message** from the **InspectData** tab.

Step3: The dialog box shown below appears. Set a message according to Table 1-3 and click **Save**.

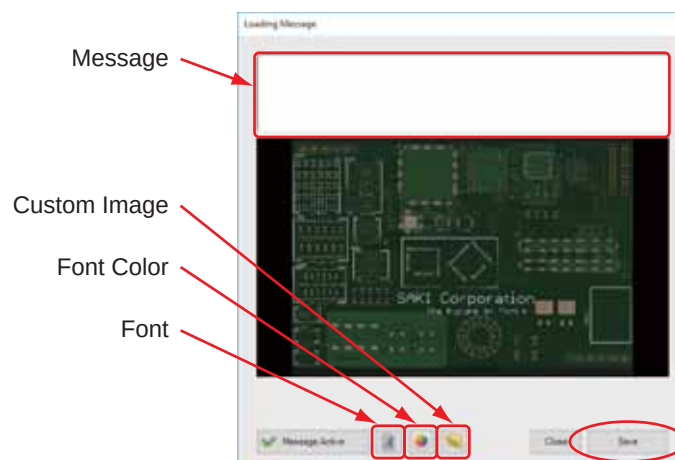


Figure 1-19 Setting and Displaying Message 1

Item	Description
Message	Enter message
Custom Image	Sets board image
Font Color	Sets font color of message
Font	Sets font of message
Message Active	Enables to display a message when a board data is loaded
Close	Finish editing message
Save	Saves message

Table 1-3 Descriptions

Step4: The set message is displayed when the inspection data is loaded while **Message Active** is checked.

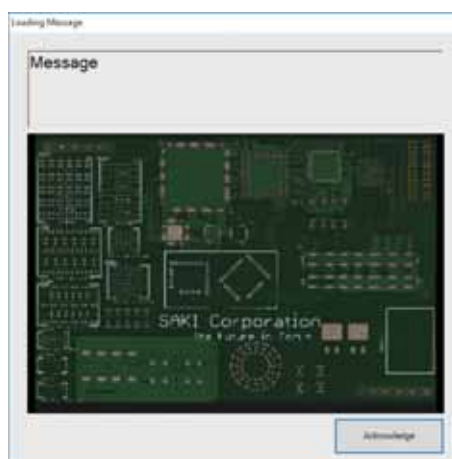


Figure 1-20 Setting and Displaying Message 2

1.5.9 Creating a New Folder

Here describes how to create a new folder.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a group and select  > **Create folder** on the Board area.

Step3: Enter a folder name and click **OK**.

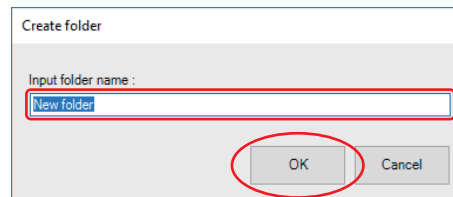


Figure 1-21 Create a New Folder

1.5.10 Deleting a Board Data

Here describes how to delete a board data.

Step1: Refer to **1.3 Selecting an Inspection Data File** to load board data of the same group of the data to be deleted.

Step2: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step3: Select a board data to delete and click  on the Board area.

1.5.11 Deleting a Folder

Here describes how to delete a folder.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a folder to delete and click  on the Board area.

NOTE

When deleting a folder, the board data moves to the upper directory without being deleted.

1.5.12 Changing a Board Name

Here describes how to change a board name.

- Step1: Load a board data to change the name according to **1.3 Selecting an Inspection Data File**.
- Step2: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.
- Step3: Select loaded board data and select  > **Rename** on the Board area.
- Step4: Enter a board name and click **OK**.

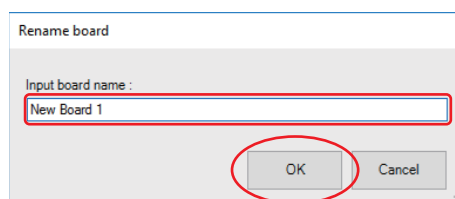


Figure 1-22 Changing Board Name

1.5.13 Changing a Folder Name

Here describes how to change a folder name.

- Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.
- Step2: Select a folder and select  > **Rename** on the Board area.
- Step3: Enter a folder name and click **OK**.

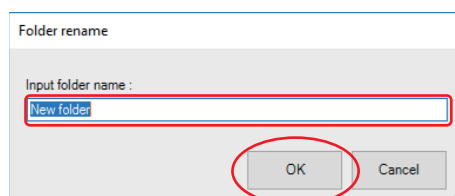


Figure 1-23 Changing a Folder Name

1.5.14 Editing a Board Data Note

Here describes how to edit a board data note.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a board data and select  > **Edit Note** on the Board area.

Step3: Edit note and click **OK**.

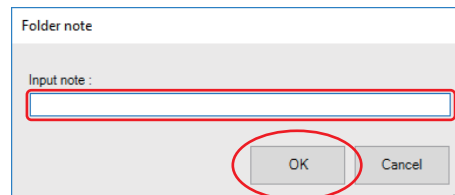


Figure 1-24 Editing a Board Data Note

1.5.15 Creating a Board Data

Here describes how to create board data.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Click  on the Board area.

NOTE

To create board data in an another group, load the board data of desired group before open Board inspection data explorer as described in the previous step.
For the procedures for loading board data, refer to **1.3 Selecting an Inspection Data File**.

Step3: The dialog box shown below appears.

Yes Overwrites the board data and proceeds to Step4.

No Proceeds to Step4 without saving the board data.

Cancel Returns to the original screen without saving the board data.

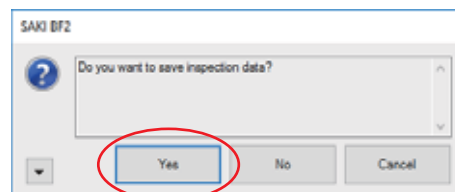


Figure 1-25 Create a Board Data 1

Step4: Enter a board name and click **OK**.

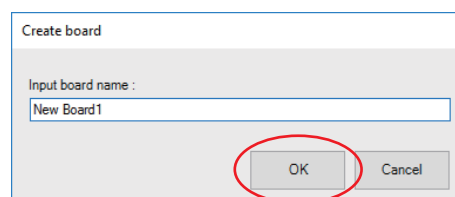


Figure 1-26 Create a Board Data 2

1.5.16 Copying a Board Data

Here describes how to copy a board data and save it in a different name.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a board data and click  on the Board area.

Step3: The dialog box shown below appears.

Yes Overwrites the board data and proceeds to Step4.

No Proceeds to Step4 without saving the board data.

Cancel Returns to the original screen without saving the board data.

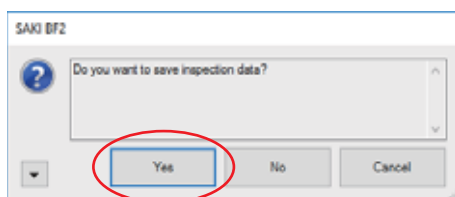


Figure 1-27 Copy a Board Data 1

Step4: Enter a new board name and click **OK**.

NOTE If not copying board image, check **Don't copy board image..**

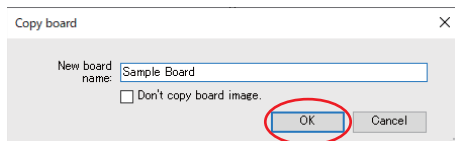


Figure 1-28 Copy a Board Data 2

1.5.17 Moving a Board Data and Folder

Here describes the procedures for moving board data and folders within the group.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Select a board data or folder and select  > **Move** on the Board area.

Step3: Specifies new path and click **OK**.

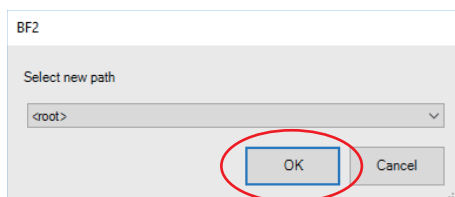


Figure 1-29 Moving a Board Data and Folder

1.5.18 Changing a Data Folder

Here describes how to change the data folder on Board inspection data explorer.

Step1: Open Board inspection data explorer according to **1.5.2 Displaying Board Inspection Data Explorer**.

Step2: Click  on the upper center of the dialog box.

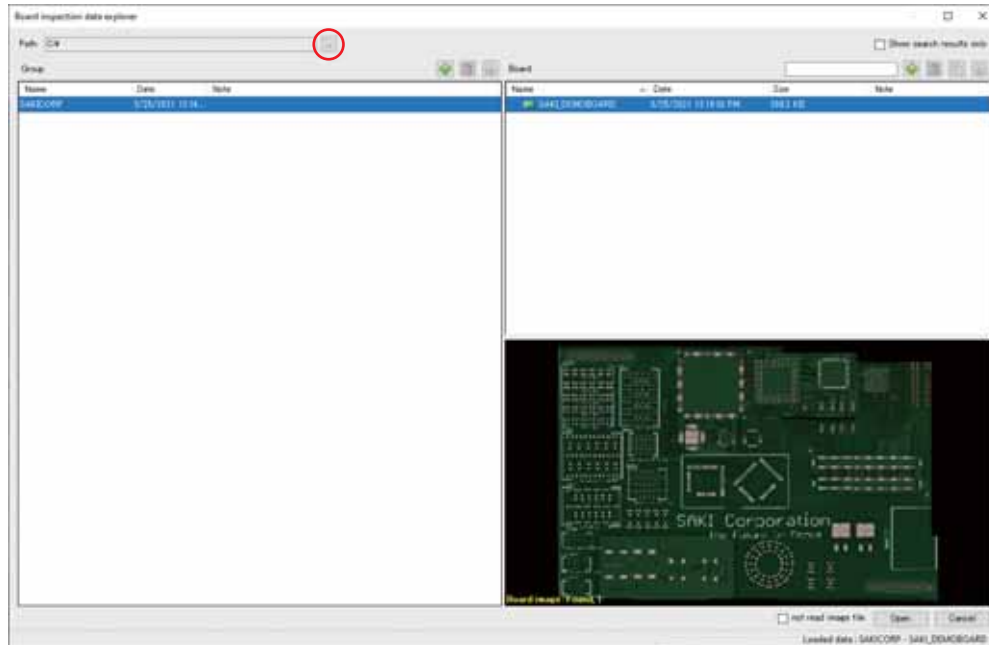


Figure 1-30 Changing a Data Folder 1

Step3: The dialog box shown below appears. Select a folder and click **OK**.

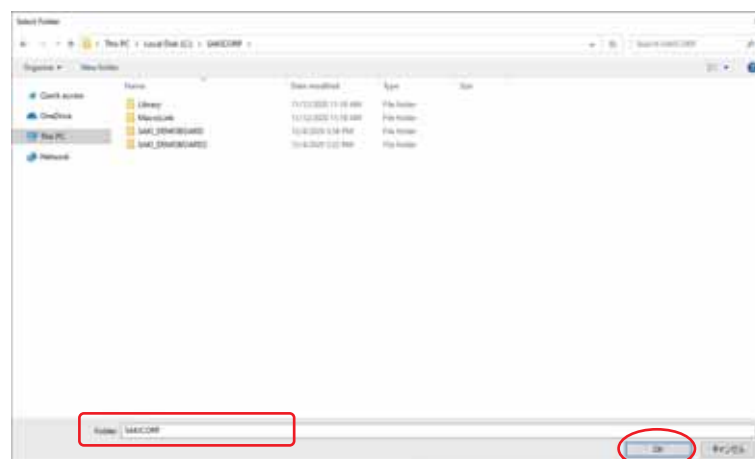


Figure 1-31 Changing a Data Folder 2

2 Operations in Board Viewer

2.1 Basic Operations in Board Viewer

This section describes the basic operations in **Board Viewer**.

2.1.1 Moving a Display Area

Step1: Click  (Hand tool) located on the top of **Board Viewer**.



Figure 2-1 Moving a Display Area

Step2: Drag the mouse on **Board Viewer**. The display area moves.

NOTE The display area can also be moved by dragging the mouse with pressing the mouse wheel or with pressing the space key.

NOTE In **Board Viewer** of the **InspectData** tab, the display area moves to a corner of the image by clicking  button.

The buttons appear when placing the mouse pointer on a corner of an image.

2.1.2 Zooming in/out an Image

Use the **Zoom** slide bar to zoom in/out an image. The slide bar is located on the top of **Board Viewer**.

NOTE An image is also zoomed in/out by scrolling the mouse wheel.

NOTE An image fits to the window size by clicking  (Fit to window) located on the right side of the Zoom slide bar.

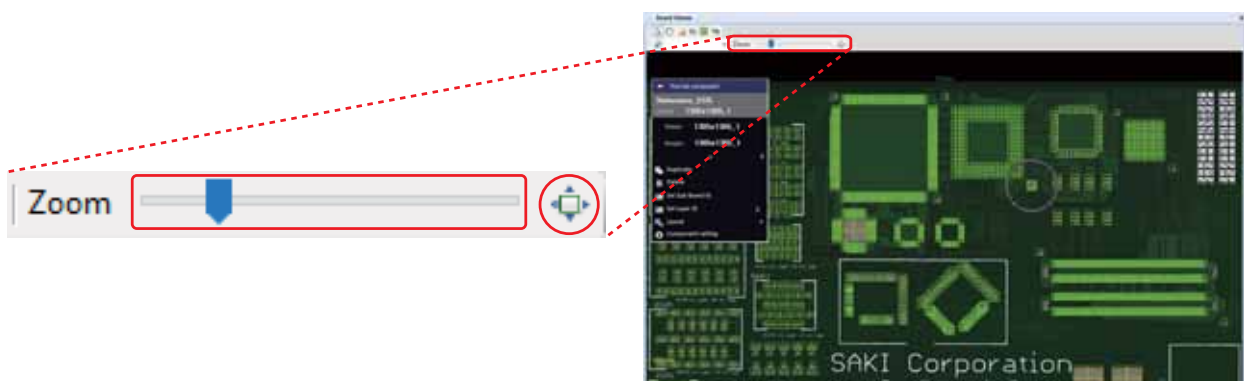


Figure 2-2 Zooming in/out an Image

2.1.3 Measuring Dimensions of the Selected Area

Step1: Click  (Ruler) located on the top of **Board Viewer**.

The dialog box shown in Figure 2-4 appears on the upper left side of **Board Viewer**.

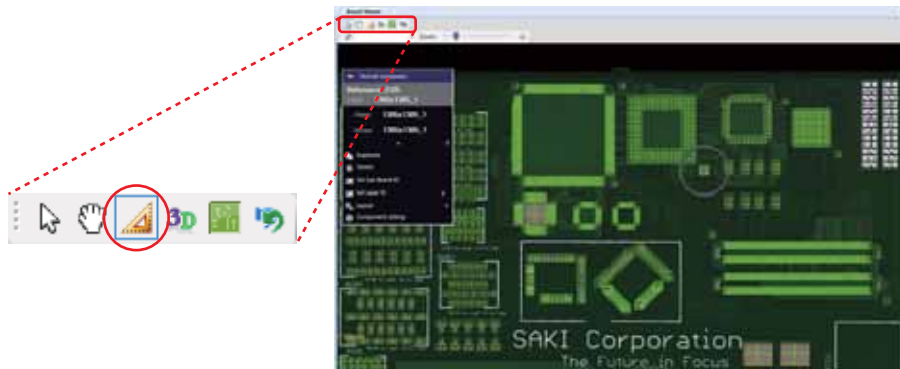


Figure 2-3 Measuring Dimensions 1

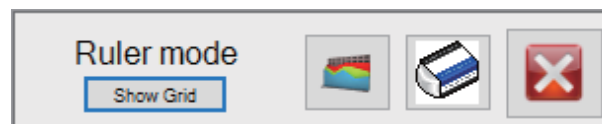


Figure 2-4 Measuring Dimensions 2

Step2: Select an area by mouse-dragging. The dimensions of the selected area are displayed on **Board Viewer**. To quit the measuring mode, click  (Close).

NOTE Clicking **Show Grid** displays a full-sized grid on screen.

NOTE If measuring multiple areas, select another area by mouse-dragging without quitting the measuring mode. To delete the dimension values, click .

NOTE To display the RGB color information of the selected area, click . (Refer to Figure 2-6)

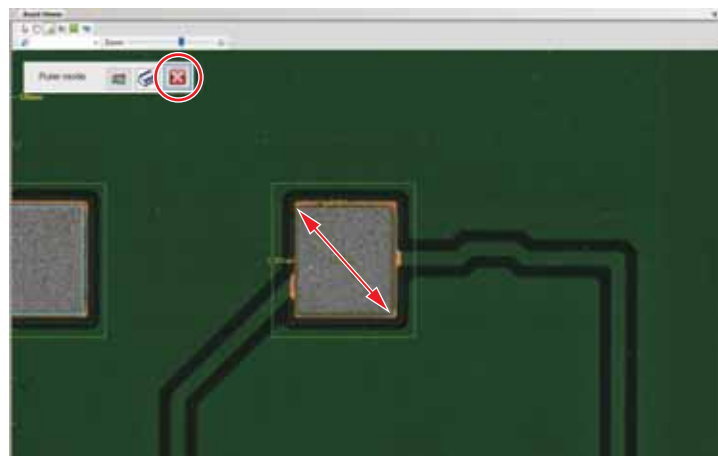


Figure 2-5 Measuring Dimensions 3

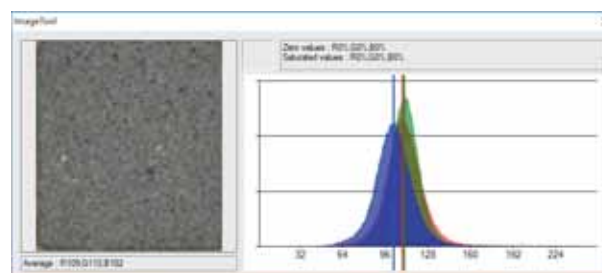


Figure 2-6 RGB Color Information

2.1.4 Confirming the Position of the Display Area on the PCB

Click  (Board Navigator) located on the top of **Board Viewer**. The dialog box shown in Figure 2-8 appears on the upper right side of **Board Viewer**. The yellow square on the board navigator indicates the position of the display area on the PCB.

NOTE

If the yellow square is moved by mouse-dragging on the board navigator, the display area also moves on **Board Viewer**.



Figure 2-7 Board Navigator 1



Figure 2-8 Board Navigator 2

2.1.5 Displaying the Command History

Click  (Command History) located on the top of **Board Viewer**. The dialog box shown in Figure 2-10 appears on the lower right side of Board Viewer. The dialog box shows the history of commands. The current state is marked with the ► icon and previous commands are shown above the current state.

To go back to a previous state, select a previous command.

NOTE

Press **Ctrl** + **Z** to undo the action, press **Ctrl** + **Y** to redo the action.

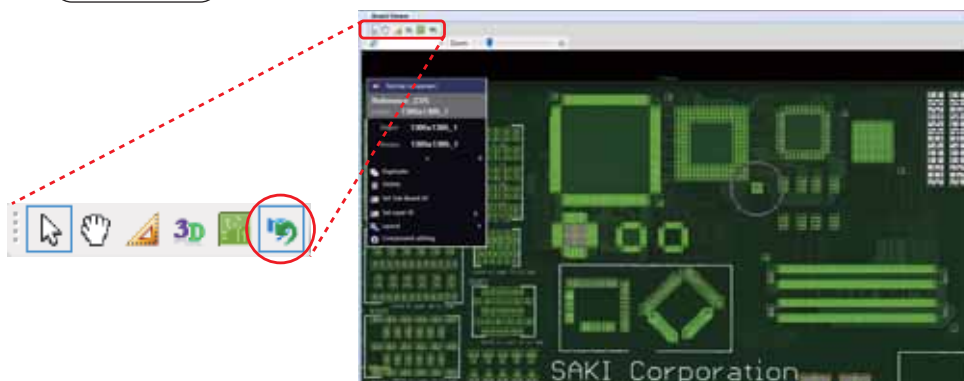


Figure 2-9 Command History 1

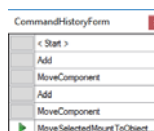


Figure 2-10 Command History 2

2.2 Switching the Image Display

2.2.1 Switching the Lighting

To switch the lighting of the image, select another lighting from the drop-down list located on the top of **Board Viewer**.



Figure 2-11 Switching the Lighting

2.2.2 3D Displaying of the Selected Area

Step1: Click **3D** (3D Viewer) located on the top of **Board Viewer**.

The dialog box shown in Figure 2-13 appears on the upper left side of **Board Viewer**.



Figure 2-12 3D Viewer 1

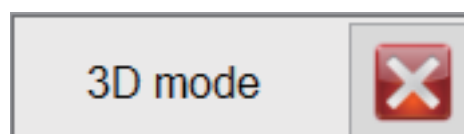


Figure 2-13 3D Viewer 2

Step2: Select an area by mouse-dragging.

Step3: Drag the selected area by the mouse. The image on the selected area rotates.

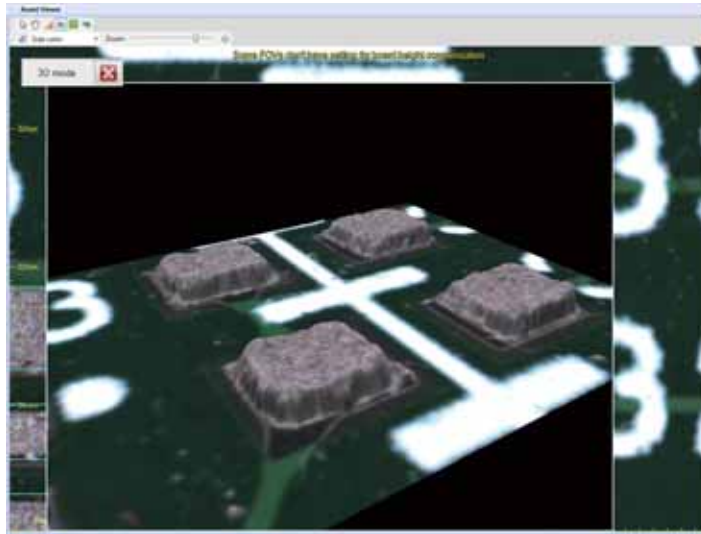


Figure 2-14 3D Viewer 3

Step4: To quit the 3D mode, click  (close).



Figure 2-15 Close

Inspection Data

1 Making an Inspection Data

1.1 Flow Chart for Data Making

Following data are included in an inspection data of the inspection software.

Figure 1-1 shows the flow chart of inspection data making.

Board Data

Board Data is a data used for inspections. Information of a board, pad, and FOV is included.

Library

Library is a series of data used for inspections, such as **Shape** and **Recipe**.

Group

Group is a series of **Board Data** which shares the same **Library**.

Shape

Shape is a shape data of a pad. Size data of a pad are included.

Shape and **Recipe** can be shared among multiple **Board Data** in the same **Group**.

Recipe

Recipe specifies an inspection area, algorithm, and its threshold of OK/NG/Warning.

Link ID

Link ID refers to a specific combination of a **Shape** and a **Recipe**.

Data with the same Link ID have the same Shape and Recipe, so a **Shape** and a **Recipe** created with one data can be easily expanded to data with the same Link ID.

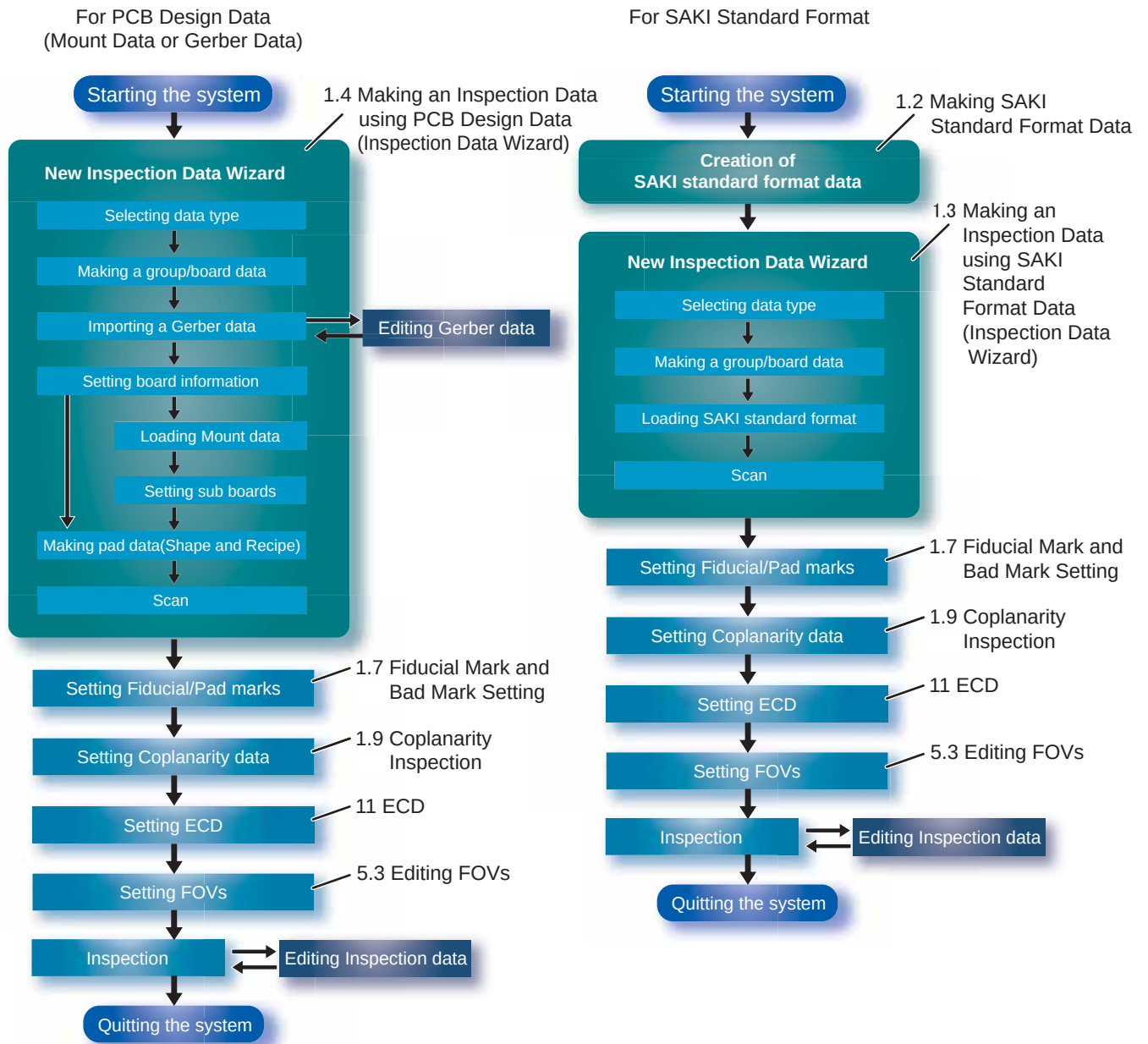


Figure 1-1 Flow Chart for Data Making

1.2 Making SAKI Standard Format Data

Using SAKI standard format is recommended to make inspection data easily.

The application ePM-AOI is used to make SAKI standard format data.

ePM-AOI is the application which reads PCB design data (e.g., Gerber data, Mount data, and ODB++) and convert it to SAKI standard format data (extension is “.json”).

The making flow of SAKI standard format data is as follows.

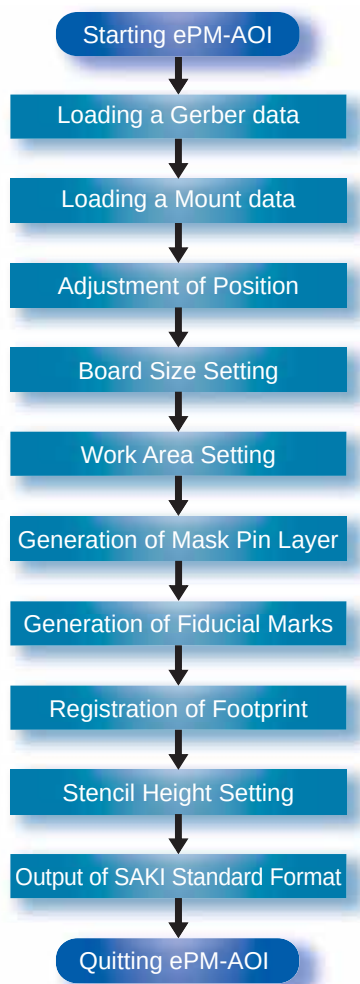


Figure 1-2 Flow Chart of Making SAKI Standard Format

1.2.1 ePM-AOI Window Details

Here describes the details of ePM-AOI window.

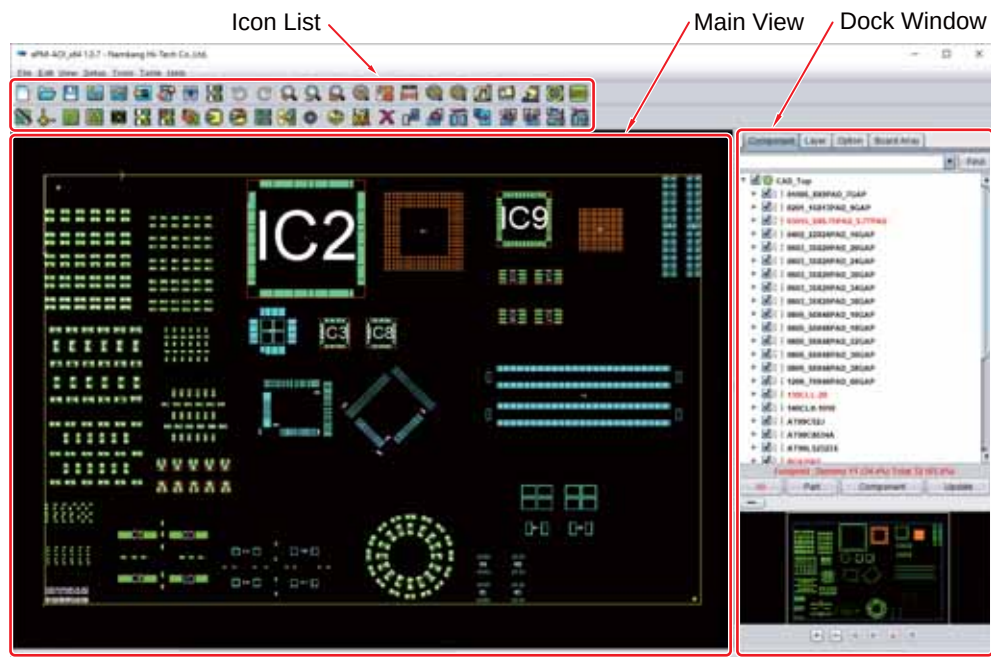


Figure 1-3 ePM-AOI Window

Item	Description
Icon List	Allows the functions for making SAKI standard format data to be used.
Main View	Displays PCB design data (Gerber data or mount data) which has been read.
Dock Window	Makes the setting for making SAKI standard format data.

Table 1-1 Descriptions

1.2.2 SAKI Standard Format Data Making Procedure

Step1: Double-click the ePM-AOI shortcut icon on the desktop.

NOTE

License key is necessary to use **ePM-AOI**.

Check the license key is inserted to USB port on PC.



Figure 1-4 ePM-AOI Shortcut Icon



Figure 1-5 ePM-AOI Startup Screen

Step2: Load the gerber data.

Click  (Gerber) on the icon list.

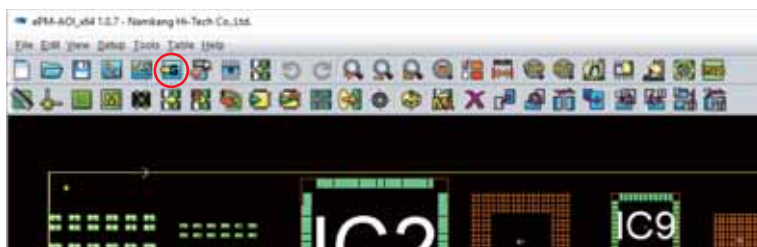


Figure 1-6 SAKI Standard Format Data Making Procedure 1

- Step3: The dialog box shown below appears.
Select the gerber data to load and click **Import**.

NOTE Gerber data can be loaded by drag and drop the file to main view.

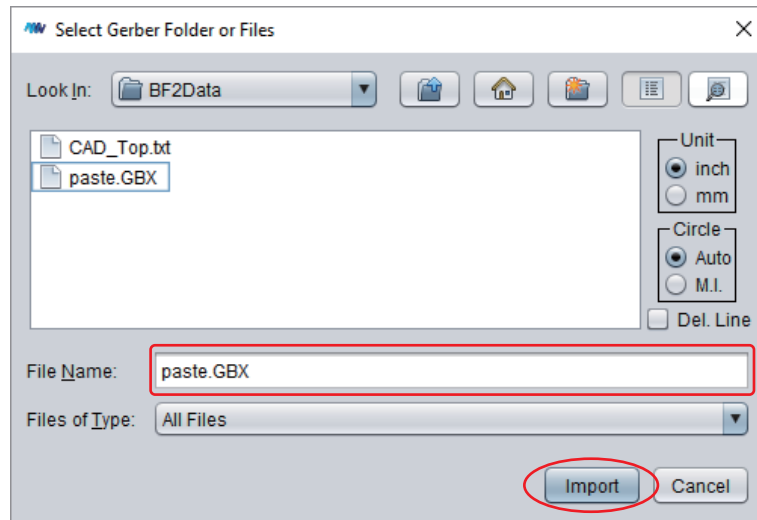


Figure 1-7 SAKI Standard Format Data Making Procedure 2

- Step4: Load the mount data.
Click  (CPL Wizard) on the icon list.



Figure 1-8 SAKI Standard Format Data Making Procedure 3

- Step5: The dialog box shown below appears.
Select the mount data to load and click **Open**.

NOTE Mount data can be loaded by drag and drop the file to main view.

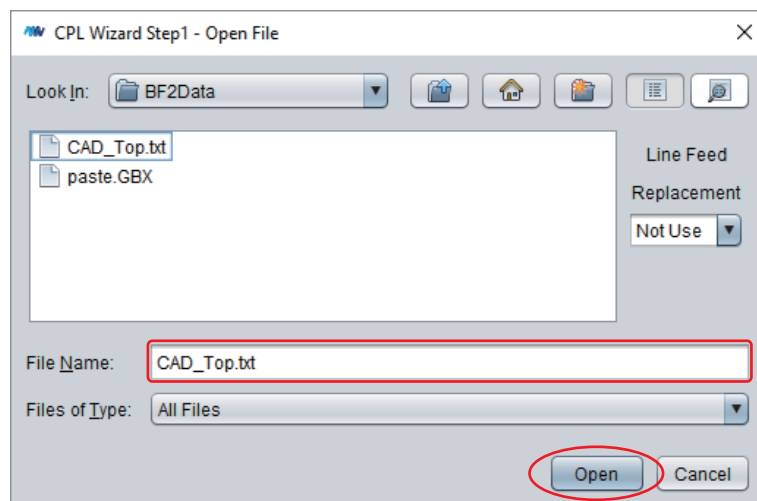


Figure 1-9 SAKI Standard Format Data Making Procedure 4

Step6: The dialog box shown below appears and loaded mount data is displayed on the upper side of the dialog box.

Check which row of the mount data are corresponding to the reference name and etc, and select item from the drop down list at the bottom of the dialog box.

After all settings are completed, click **Appending Import**.

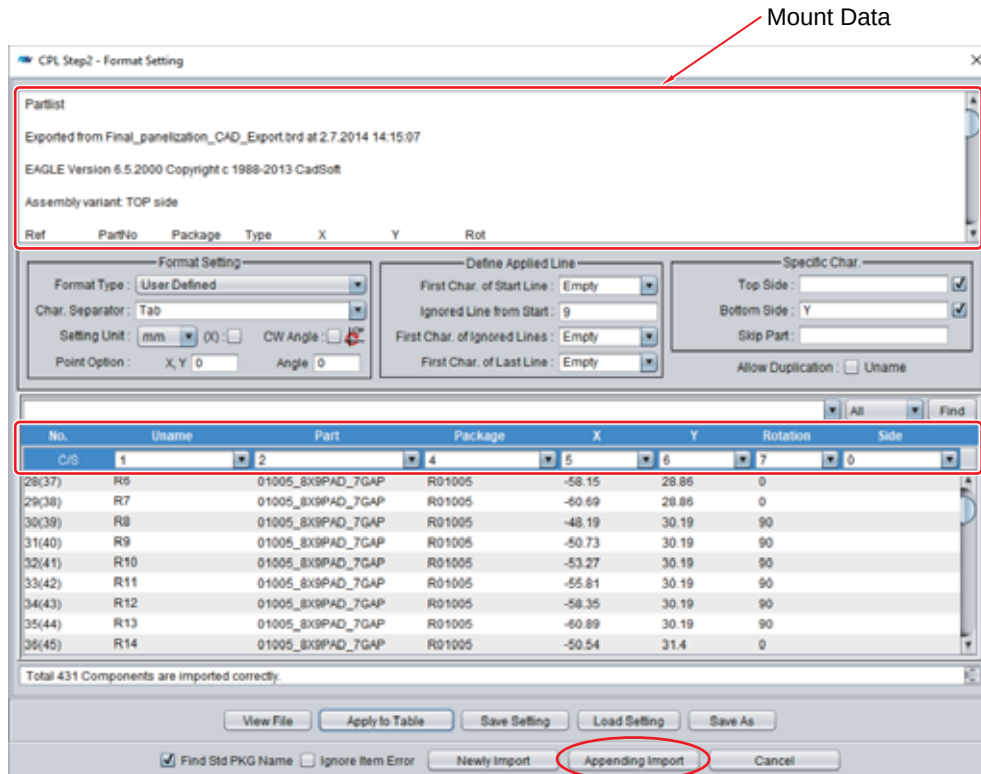


Figure 1-10 SAKI Standard Format Data Making Procedure 5

Item	Description
Uname	Select the row of reference name.
Part	Select the row of Link ID.
Package	Select the row of general component name.
X	Select the row of X coordinates.
Y	Select the row of Y coordinates.
Rotation	Select the row of component angle.
Side	Select the row of mount side.

Table 1-2 Descriptions

Step7: Align gerber data and mount data at the same coordinate.

Click  (Fit CAD Centroid & Gerber) on the icon list.

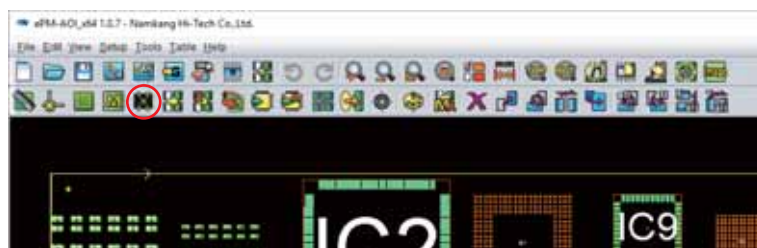


Figure 1-11 SAKI Standard Format Data Making Procedure 6

Step8: Click the component which to be basis of positioning on main view.

On the **Dock window > Option tab > Component Name**, the selected component name is displayed.

NOTE

Display area is moved by right-drag on main view.

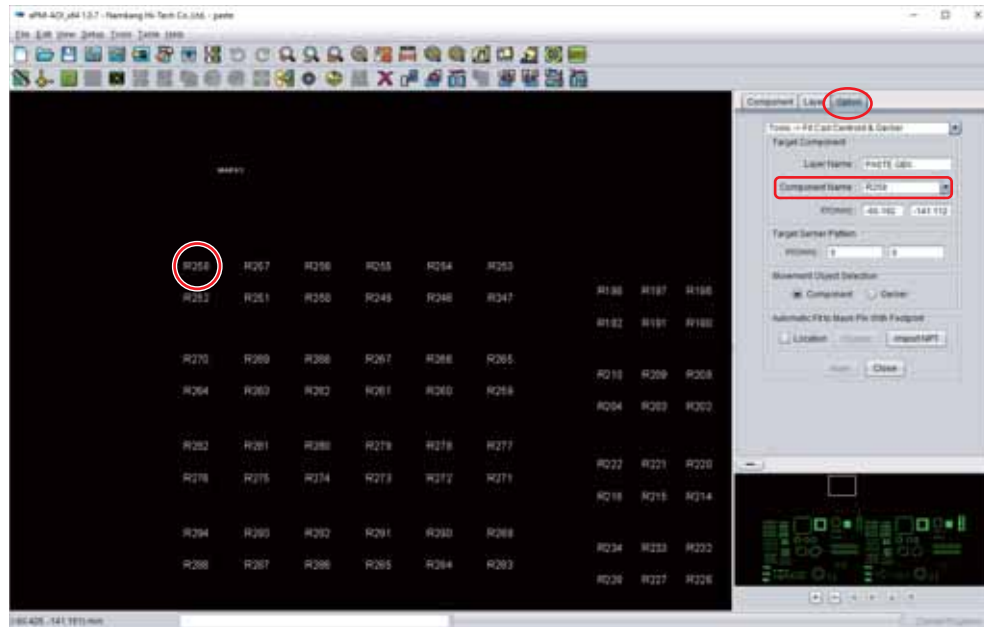


Figure 1-12 SAKI Standard Format Data Making Procedure 7

Step9: Select the footprint (*) which is corresponding to the component selected in Step8 by dragging on mainview. The selected footprint is colored red.

On the **Dock window > Option tab > Target Gerber Pattern**, the coordinate is displayed.

NOTE

(*) Here, it indicates the pad read from gerber data. Select the pad corresponding to the component.

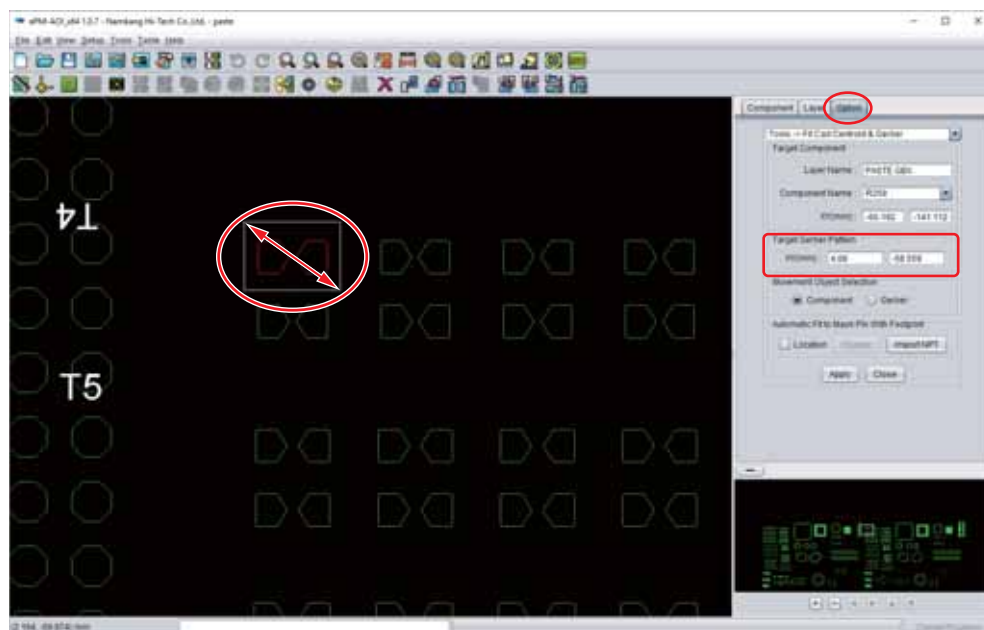


Figure 1-13 SAKI Standard Format Data Making Procedure 8

- Step10: On the **Dock window > Option tab > Movement Object Selection**, select the data to be moved from **Component** (Mount data) or **Gerber** (Gerber data).
After the setting is completed, click Apply.



Figure 1-14 SAKI Standard Format Data Making Procedure 9

- Step11: Specify board size.
Click  (Board Size) on the icon list.

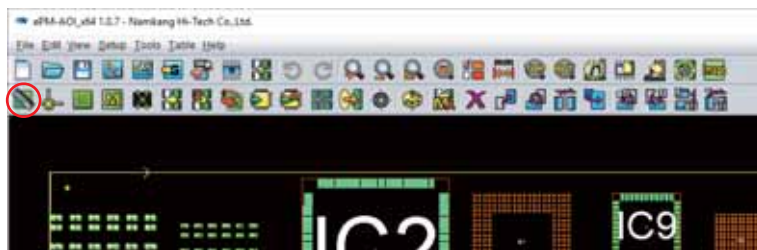


Figure 1-15 SAKI Standard Format Data Making Procedure 10

- Step12: On the **Dock window > Option tab > Size**, enter the board size and click **Apply**.

NOTE

The board size can be set by dragging on main view.

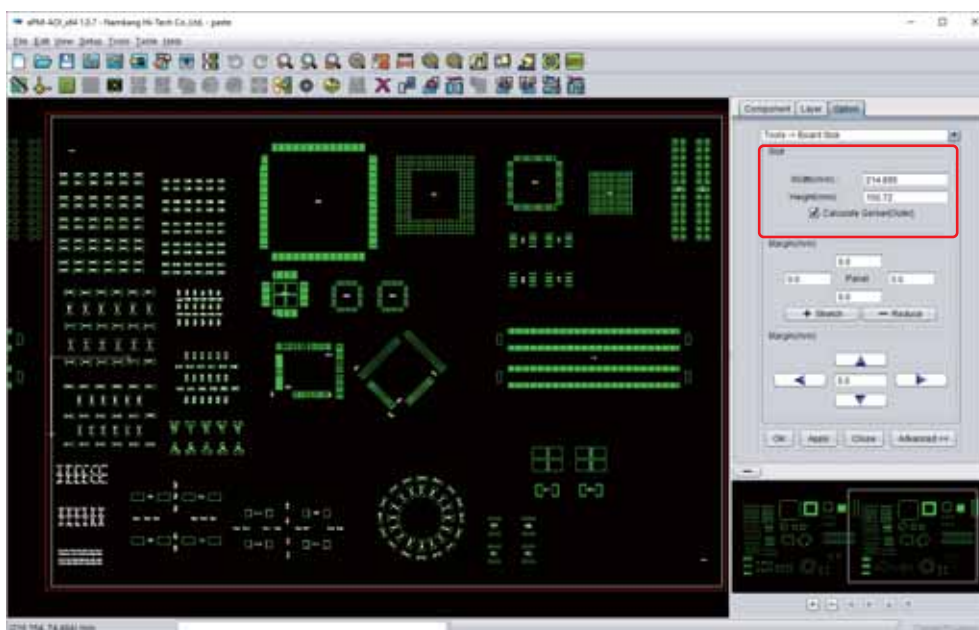


Figure 1-16 SAKI Standard Format Data Making Procedure 11

Step13: Specify work area.

Click  (Work Area) on the icon list.

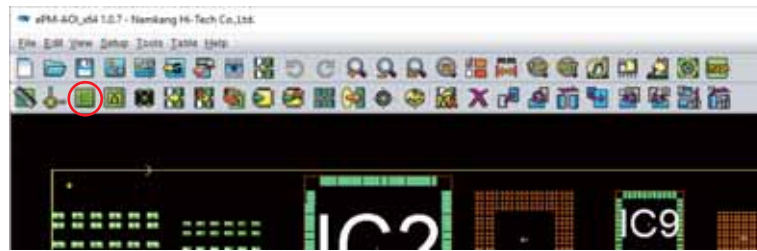


Figure 1-17 SAKI Standard Format Data Making Procedure 12

Step14: Set one sub board area on main view by dragging.

After the setting is completed, click **OK** on **Option** tab of **Dock window**.

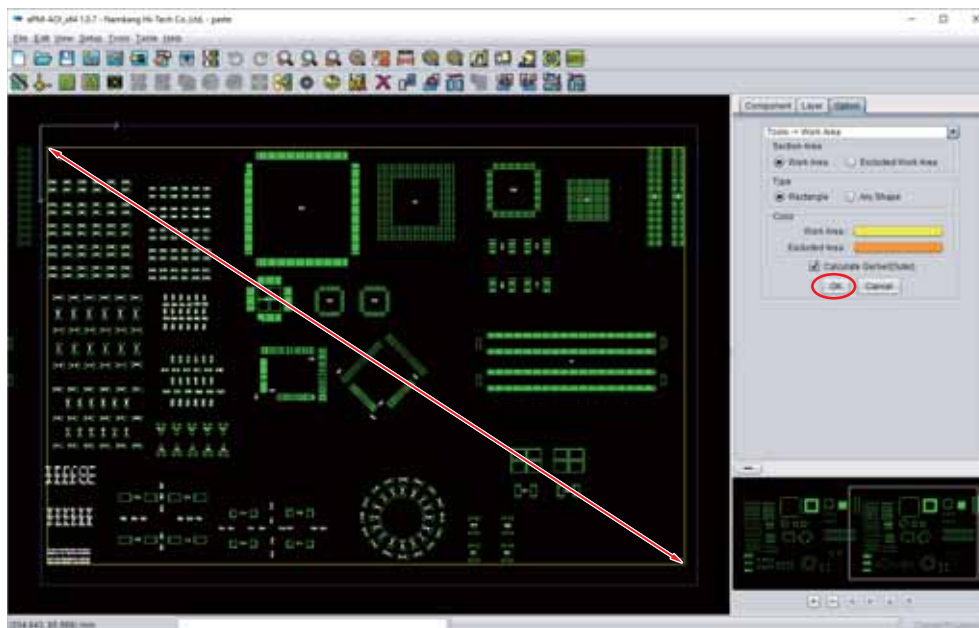


Figure 1-18 SAKI Standard Format Data Making Procedure 13

Step15: Generate mask pin layer.

Click  (Generate mask pin) on the icon list.

The mask pin is generated automatically and progress bar is displayed on the bottom of dialog box during generating mask pin.

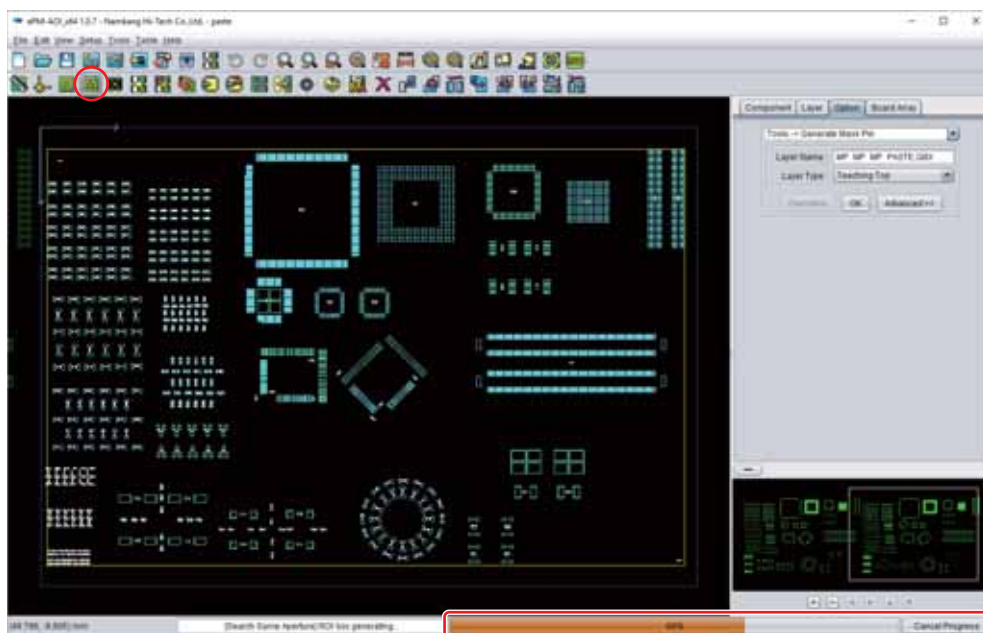


Figure 1-19 SAKI Standard Format Data Making Procedure 14

Step16: Create fiducial mark.

Click  (Fiducial mark) on the icon list.

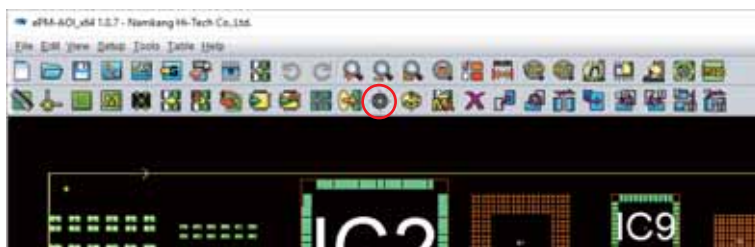


Figure 1-20 SAKI Standard Format Data Making Procedure 15

Step17: Select the fiducial mark on main view by dragging.

After the setting is completed, set the shape and size of fiducial mark on the **Dock window** > **Option** tab > **Feature(mm)**. For details, refer to Table 1-3.

After the setting is completed, click **Generation**.

Set the shape and size for all fiducial marks.

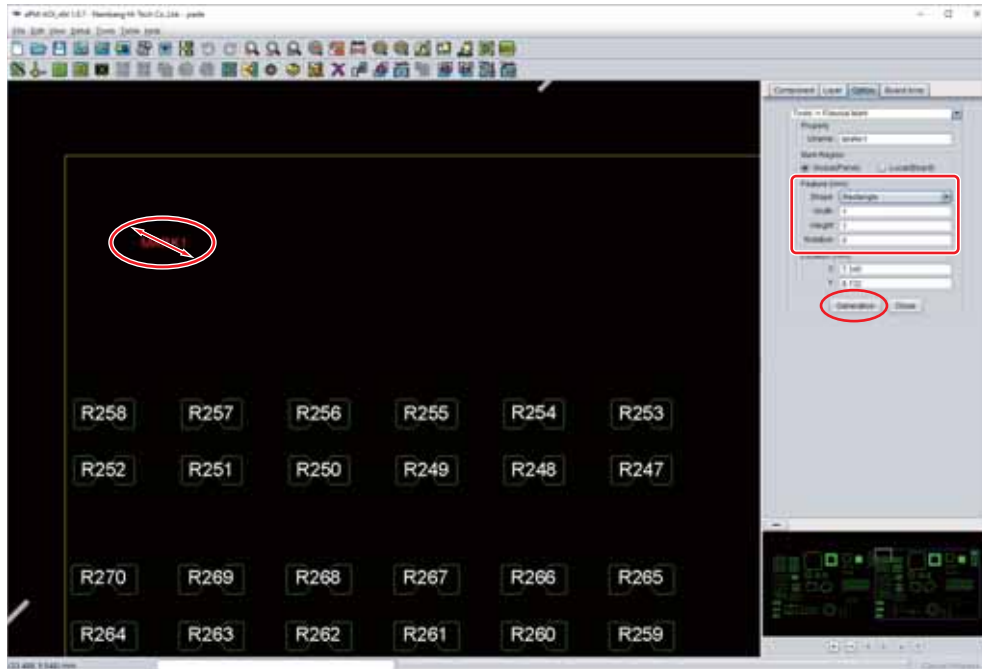


Figure 1-21 SAKI Standard Format Data Making Procedure 16

Item	Description
Shape	Select the shape of fiducial mark from Polygon , Oblong , Circle and Rectangle .
Width	Enter the size of fiducial mark. Unit is [mm].
Height	
Rotation	Enter 0 .

Table 1-3 Descriptions

Step18: Register footprint.

Click  (Automatic add footprint) on the icon list.

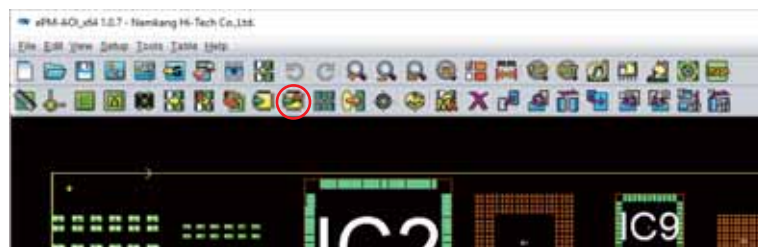


Figure 1-22 SAKI Standard Format Data Making Procedure 17

Step19: Click **Execution** on **Option** tab of **Dock window**.

The package information of component which is recognized from footprint is placed automatically.

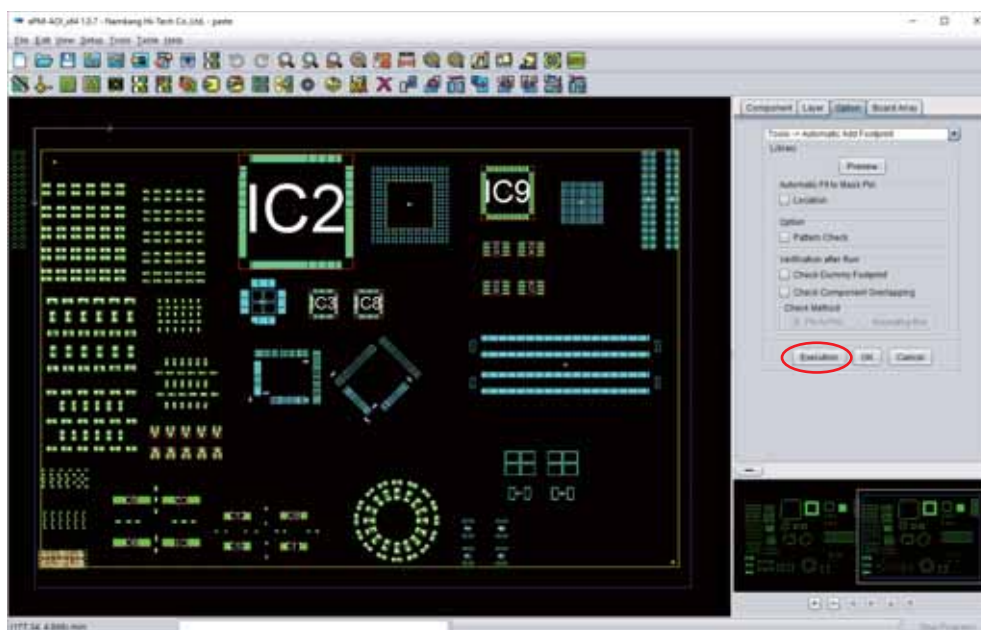


Figure 1-23 SAKI Standard Format Data Making Procedure 18

Step20: Assign the package information of component which information is not assigned automatically.

Click  (Add footprint) on the icon list.

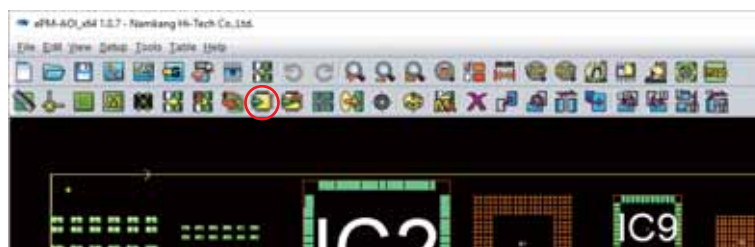


Figure 1-24 SAKI Standard Format Data Making Procedure 19

Step21: Select the pad set as package on main view by dragging.

After the setting is completed, set the package type on the **Dock window > Option tab > Package type**. And click **Add Footprint**.

Repeat this step until the package information is assigned to all component.

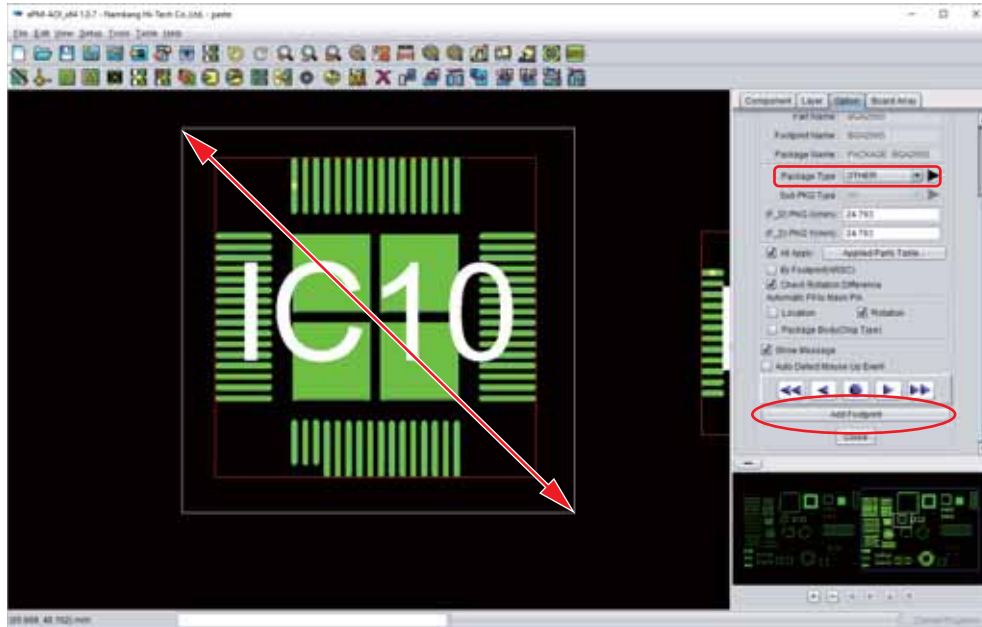


Figure 1-25 SAKI Standard Format Data Making Procedure 20

Step22: Set the stencil height.

From the menu, select **Setup > Stencil Height**.

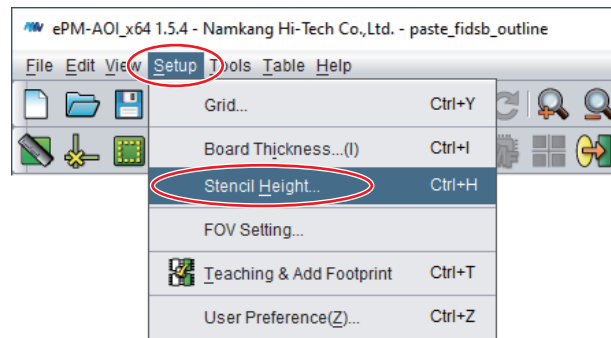


Figure 1-26 SAKI Standard Format Data Making Procedure 21

Step23: The dialog box shown below appears.

Enter the stencil height value in **Default Stencil Height(mm)**.

Click **Apply** and then **OK**.

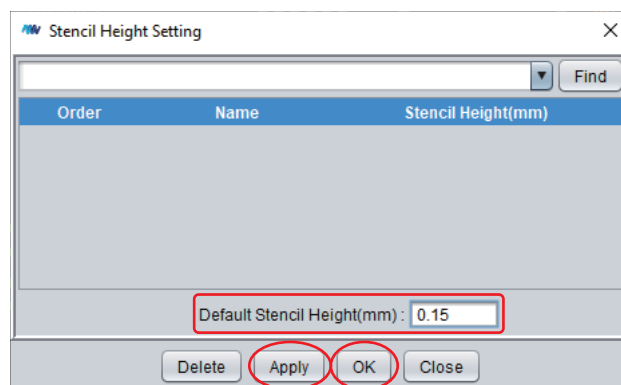


Figure 1-27 SAKI Standard Format Data Making Procedure 22

Step24: Output the SAKI Standard Format Data.

Click  (Export) on the icon list.

SAKI Standard Format Data is output.

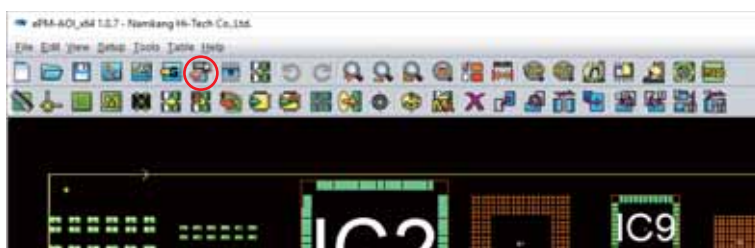


Figure 1-28 SAKI Standard Format Data Making Procedure 23

1.3 Making an Inspection Data using SAKI Standard Format Data (Inspection Data Wizard)

This section describes how to make an inspection data with SAKI Standard Format Data.

Step1: Click **BF2** on the upper left of the window and select **Inspect data wizard**.

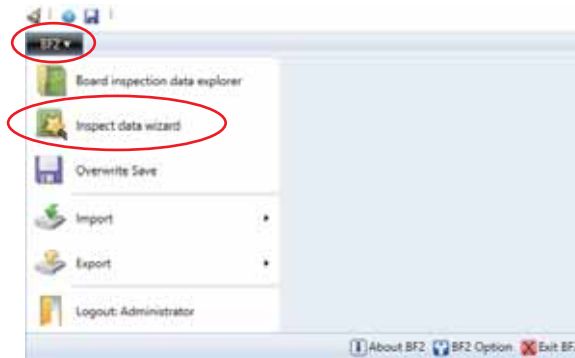


Figure 1-29 Making Inspection Data Wizard 1

Step2: If an inspection data is opened, the dialog box shown below appears.

- | | |
|---------------|--|
| Yes | Overwrites the inspection data and start Inspection Data Wizard. |
| No | Start Inspection Data Wizard without saving the inspection data. |
| Cancel | Returns to the original screen without saving the inspection data. |

NOTE Saving the inspection data is recommended when the data is modified.

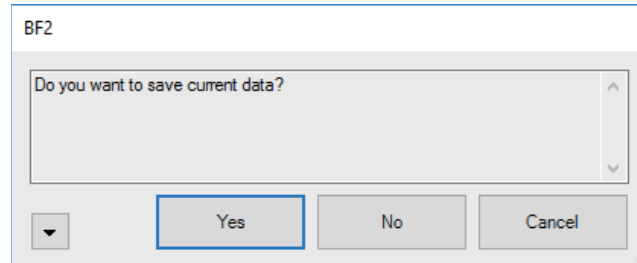


Figure 1-30 Making Inspection Data Wizard 2

Step3: The dialog box shown below appears.
Select **SAKI standard format data** and click **Next**.

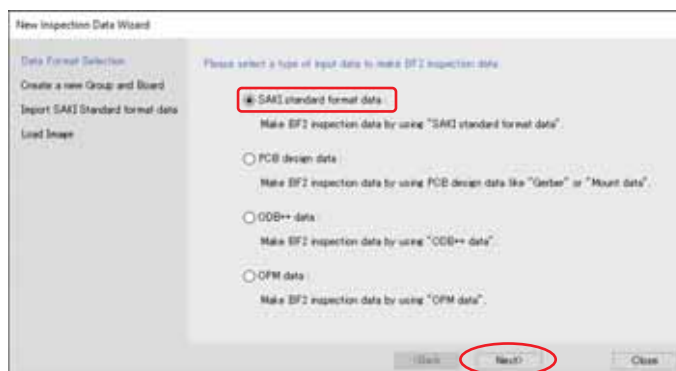


Figure 1-31 Data Format Selection

Step4: The dialog box shown below appears.

When Making a New Group

Specify a destination to save the group data on **Destination** by entering an address.

Alternately click  on the right side of the text-box and select a folder.

Fill out **Group** and **Board**. Click **Next**.

When Adding a Board Data to an Existing Group

Select an inspection data folder on **Destination** and select a group folder on **Group**.

To specify a destination, enter an address or click  on the right side of the text-box and select a folder. Fill out **Board** and click **Next**.

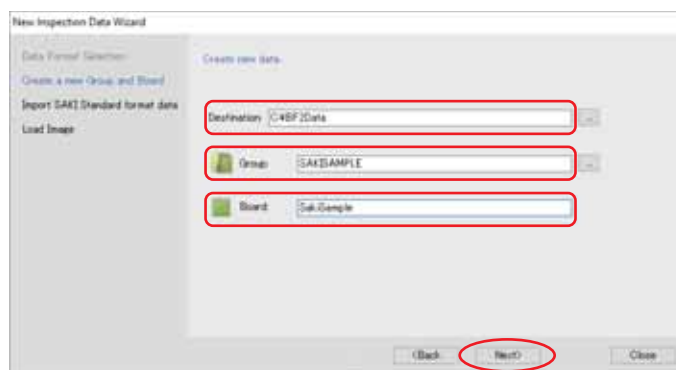


Figure 1-32 Create Group and Board Data

Step5: Specify SAKI standard format data (.json) to import on **SAKI format**, and click **Import**.

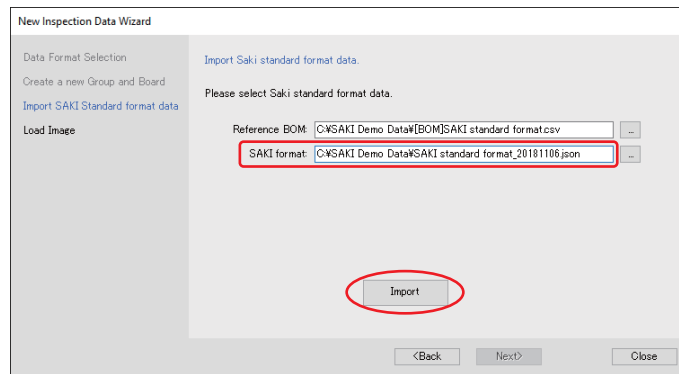


Figure 1-33 Import SAKI Standard Format Data 1

NOTE

The BOM file can be used to delete unnecessary components or change Link IDs and common component names. Click ☐ on the right side of **Reference BOM** and select the BOM file to load. The dialog box shown below appears. After setting it up according to **2.12.3 Import BOM Data**, click **OK**.

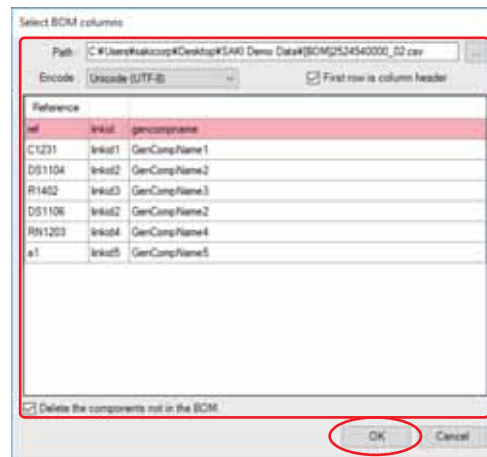


Figure 1-34 Set BOM Data

NOTE

Progress bar is displayed while importing SAKI standard format data.

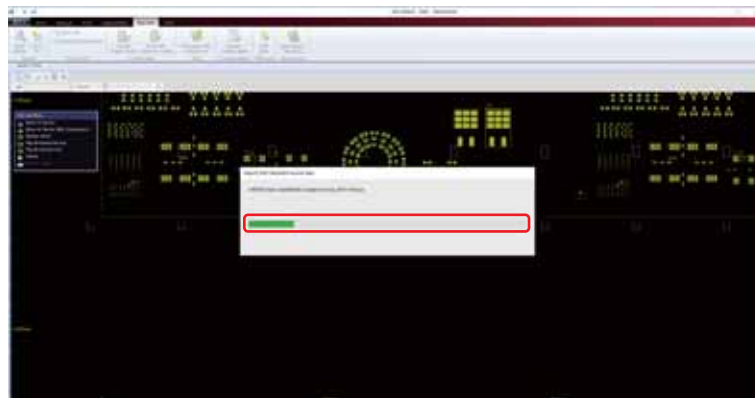


Figure 1-35 Import SAKI Standard Format Data 2

Step6: After importing SAKI standard format data is completed, click **Next**.

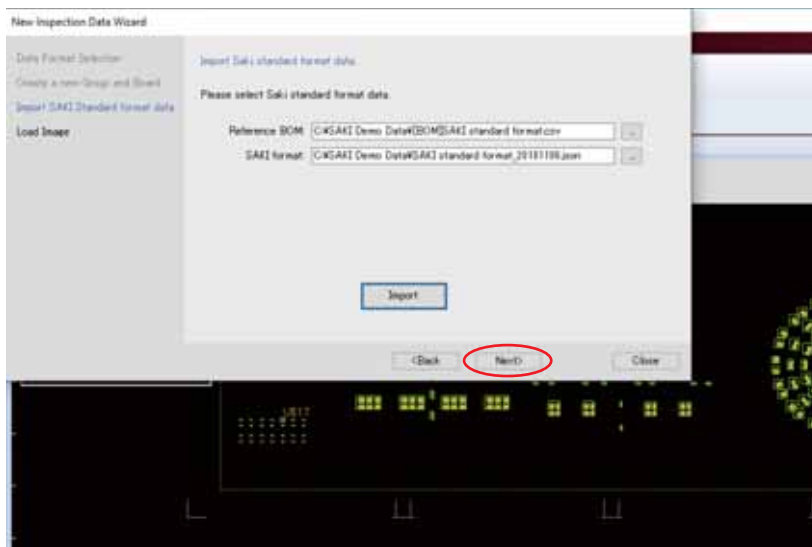


Figure 1-36 Import Complete

Step7: The dialog box shown below appears.
The board rail width is adjusted automatically. Click **OK**.

NOTE This dialog box is not displayed in BF2-Editor.

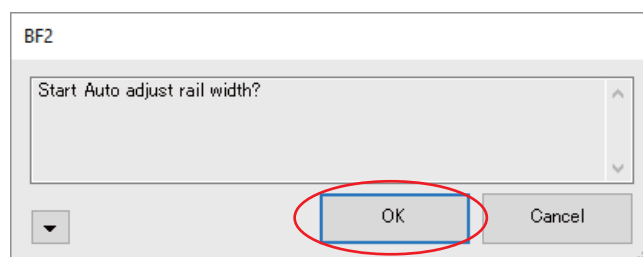


Figure 1-37 Auto Adjustment for the Rail Width

Step8: The dialog box shown below appears.

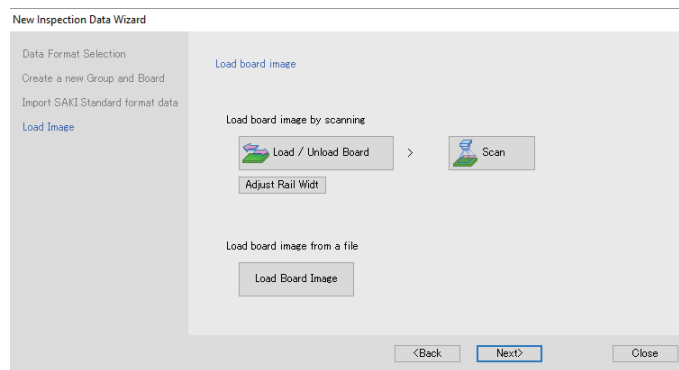


Figure 1-38 Scan Board Image

When Scanning a PCB

Click **Adjust Rail Width** on the dialog box shown in Figure 1-38.

The dialog box shown below appears.

Click **Board search**. Enter a board width into **Absolute** and click **Set**.

The conveyor rail width is adjusted.

When the adjustment is completed, click **Close** to close the dialog box.

CAUTION In BF2-Editor, PCB images cannot be acquired.

NOTE Images acquired from this step are only for making an inspection data. For inspections, FOV needs to be set separately. For details of FOV settings, refer to **5 Acquiring a PCB image**.

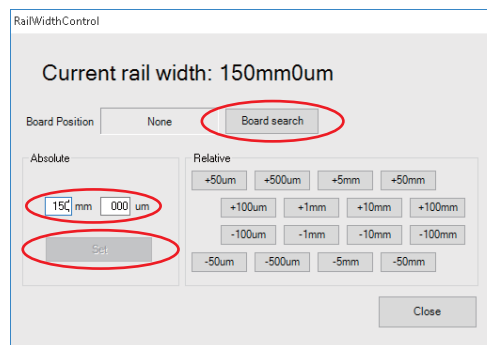


Figure 1-39 Adjusting the Rail Width

Place a PCB on the conveyor rail from the PCB inlet of the machine and click **Load/Unload Board** on the dialog box shown in Figure 1-38.

Click **Scan** and the PCB on the conveyor rail is scanned. Click **Next**.

When Loading an Existing Image File

Click **Load Board Image** on the dialog box shown in Figure 1-38 and select an image file. Click **Next**.

NOTE The extension for an image file is “.bdimg”.

After all steps are finished, the dialog box shown below appears. Click **Finish** and the wizard closes.

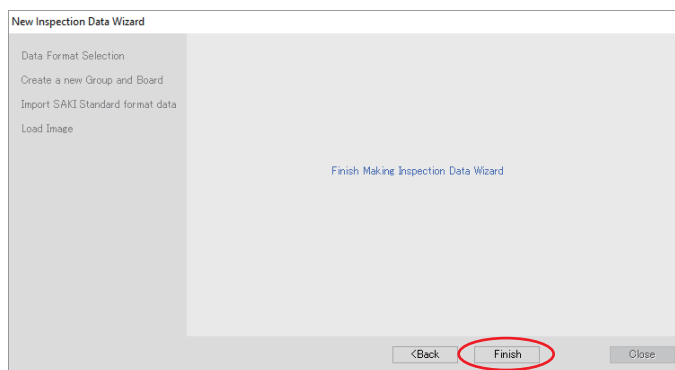


Figure 1-40 Finishing the New Inspection Data Wizard

NOTE After the wizard is closed, the dialog box and inspection data shown below appear.

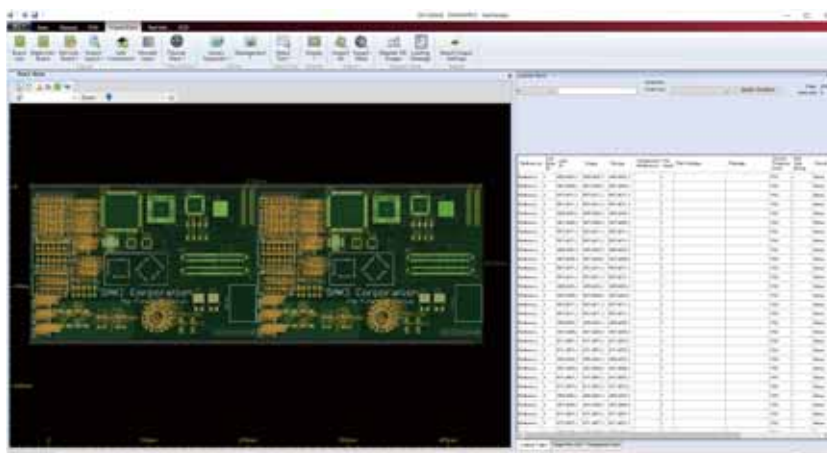


Figure 1-41 Inspection Data Dialog Box

1.4 Making an Inspection Data using PCB Design Data (Inspection Data Wizard)

This section describes how to make an inspection data by using PCB design data (gerber data or mount data).

NOTE

When using SAKI standard format data, refer to **1.2 Making SAKI Standard Format Data** and **1.3 Making an Inspection Data using SAKI Standard Format Data (Inspection Data Wizard)**.

NOTE

When editing the standard Gerber format (RS-274D) or Gerber data, use SAKI Standard Format Data .

1.4.1 Wizard Overview

When using PCB design data (gerber data or mount data), make an inspection data with Inspection Data Wizard.

In the Wizard, the dialog box shown below appears. Steps are shown in left side of dialog box and current step is highlighted in blue. Click **Next** to proceed to next step.



Figure 1-42 Making Inspection Data Wizard 1

1.4.2 Starting New Inspection Data Wizard

Step1: Click **BF2** button on the upper left of the window and select **Inspect data wizard**.

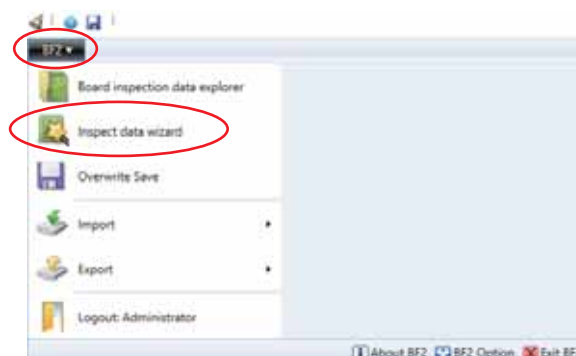


Figure 1-43 Making Inspection Data Wizard 2

Step2: If an inspection data is opened, the dialog box shown below appears.

Yes Overwrites the inspection data and start Inspection Data Wizard.

No Start Inspection Data Wizard without saving the inspection data.

Cancel Returns to the original screen without saving the inspection data.

NOTE Saving the inspection data is recommended when the data is modified.

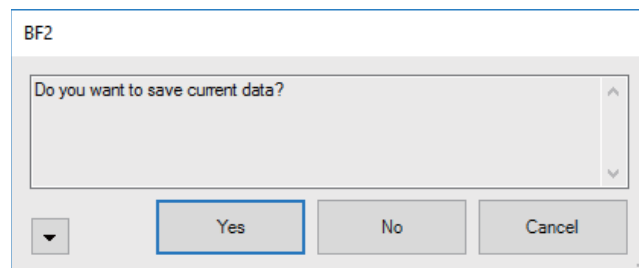


Figure 1-44 Making Inspection Data Wizard 3

Step3: The dialog box shown below appears.

Select **PCB design data** and click **Next**.

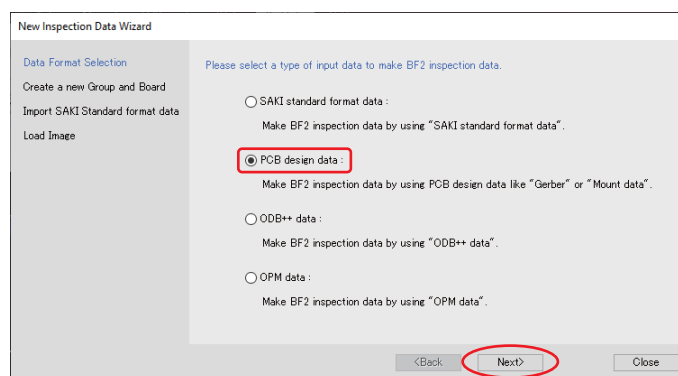


Figure 1-45 Data Format Selection

1.4.3 Create Group and Board Data

Here describes how to set a group and a board data.

When Making a New Group

Specify a destination to save the group data on **Destination** by entering an address.

Alternately click on the right side of the text-box and select a folder.

Fill out **Group** and **Board**. Click **Next**.

When Adding a Board Data to an Existing Group

Select an inspection data folder on **Destination** and select a group folder on **Group**.

To specify a destination, enter an address or click on the right side of the text-box and select a folder. Fill out **Board** and click **Next**.

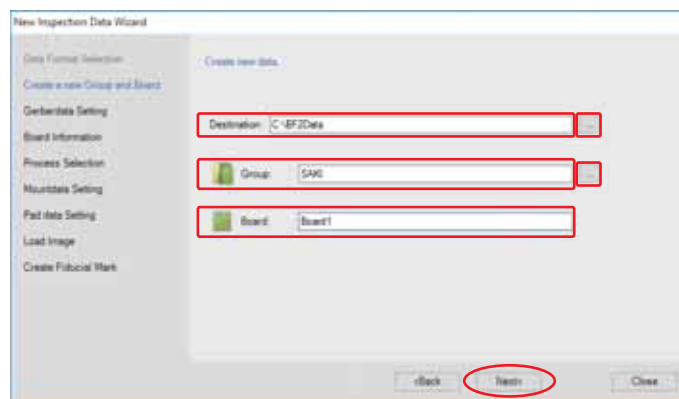


Figure 1-46 Create Group and Board Data

1.4.4 Importing Gerber Data

Here describes how to set a Gerber data.

Step1: Click  on the right side of the text-box and import Gerber data.
After selecting a Gerber data, click **Apply**.

NOTE Multiple data can be selected by selecting Gerber data with pressing **Ctrl**.

NOTE It is recommended to create the inspection data by using Extended Gerber format (RS-274X) or Gerber data of stencil mask.

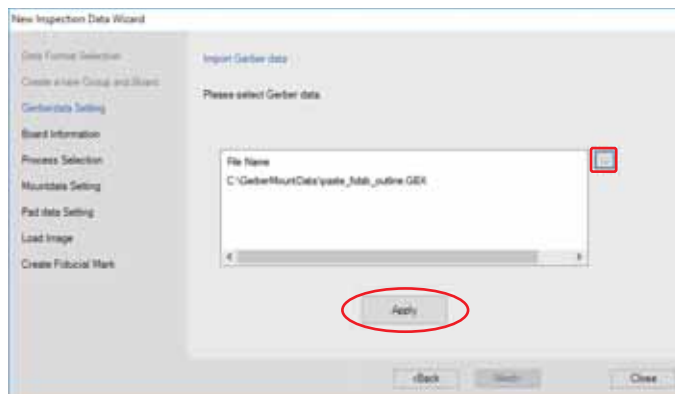


Figure 1-47 Import Gerber Data 1

Step2: The dialog box shown below appears. Confirm that the gerber data is displayed on the board image and adjust the position according to Table 1-4. Click **OK** to apply the Gerber data to **Board Viewer**.

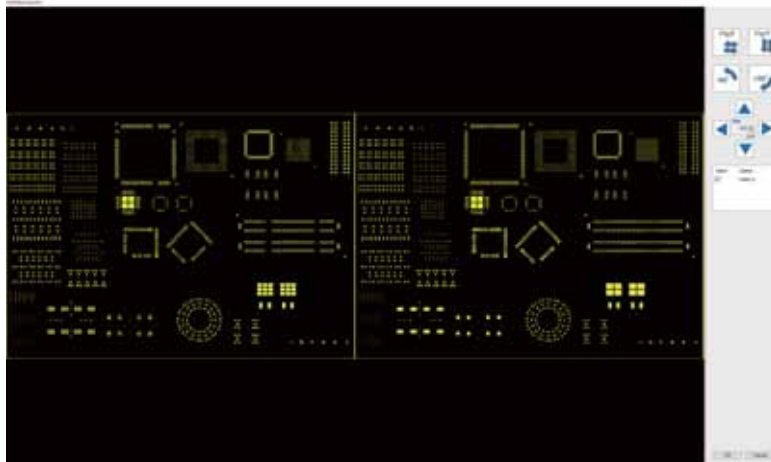


Figure 1-48 Import Gerber Data 2

Item	Description
FlipX	Reflect the Gerber data horizontally which is selected in text-box.
FlipY	Reflect the Gerber data vertically which is selected in text-box.
-90°	Rotate the Gerber data -90 degrees which is selected in text-box.
+90°	Rotate the Gerber data 90 degrees which is selected in text-box.
	Use when load the multiple Gerber data. Move the Gerber data which selected in text-box specified distance in Step .
Step[um]	Specify the distance to move above button.

Table 1-4 Descriptions

Step3: The dialog box shown in Figure 1-47 appears. Click **Next** to proceed next step.

1.4.5 Board Information Settings

Here describes how to set the board size and editing the Gerber data to fit the board size.

Step1: Set **Size X** and **Size Y** to the board size, and **Thickness** to the stencil mask. And click **Apply**.

NOTE Specified Board Size is displayed on **Board Viewer**.

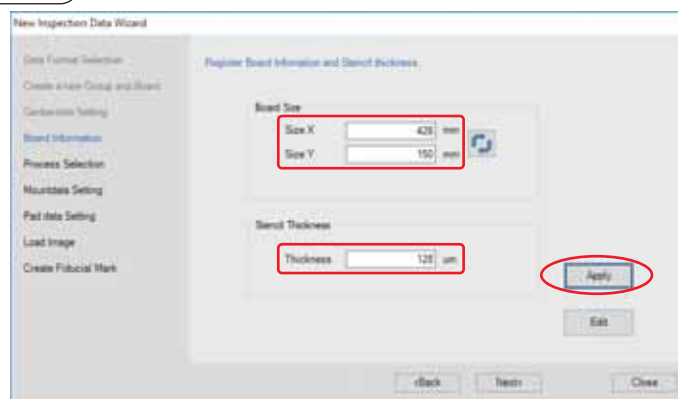


Figure 1-49 Board Information Settings 1

Step2: Click **Edit** to edit Gerber data. The dialog box shown in Figure 1-51 appears. Editing the Gerber data on **Board Viewer** according to Table 1-5. After editing is completed, click **Go back to Wizard**.

NOTE Delete the Gerber data which are not use in pad inspection.



Figure 1-50 Board Information Settings 2

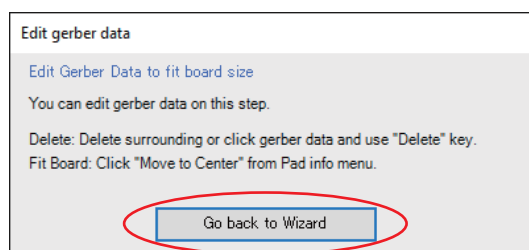


Figure 1-51 Board Information Settings 3

Action	Description
Delete Unnecessary Gerber Data	Select Gerber data by clicking or mouse dragging then press Del .
Move Gerber Data to the center of the Board	Select Move to Center from the Pad info menu.

Table 1-5 Action of Editing Gerber Data

Step3: The dialog box shown in Figure 1-50 appears. Click **Next** to proceed next step.

1.4.6 Selecting Data Create Process

Here describes selecting data create process.

When Using Only Gerber Data

Automatically create an inspection data of pad which displayed on **Board Viewer**.

NOTE

Adding component information is unable during wizard operation. To add component information, refer to **1.4.13 Adding Component Information after completion of wizard**.

Step1: Check **Gerber data** and click **Next**.

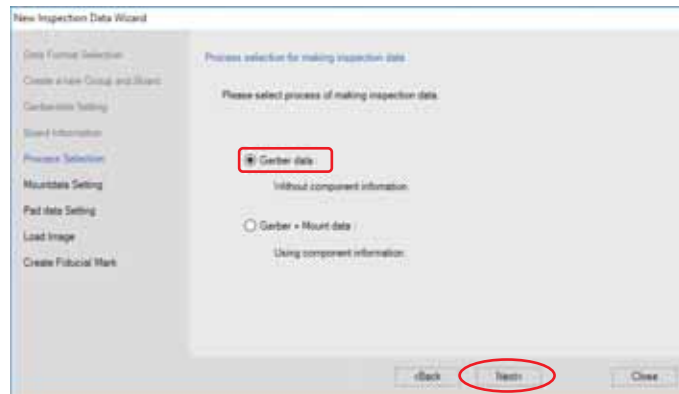


Figure 1-52 Selecting Data Create Process 1

Step2: The dialog box shown below appears. If creating the inspection data, click **OK**.

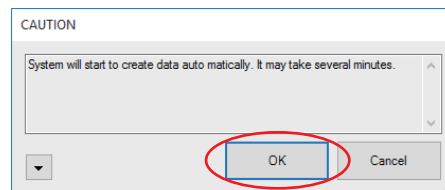


Figure 1-53 Data create from Gerber Data

Step3: After creating inspection data is completed, proceed to Import image procedure according to **1.4.10 Capturing Image**.

When Using Gerber Data and Mount Data

Can be manage the pad data of a components by importing Mount data.

Check **Gerber + Mount data** and click **Next**.

Proceed to mount data setting procedure according to **1.4.7 Importing Mount Data**.

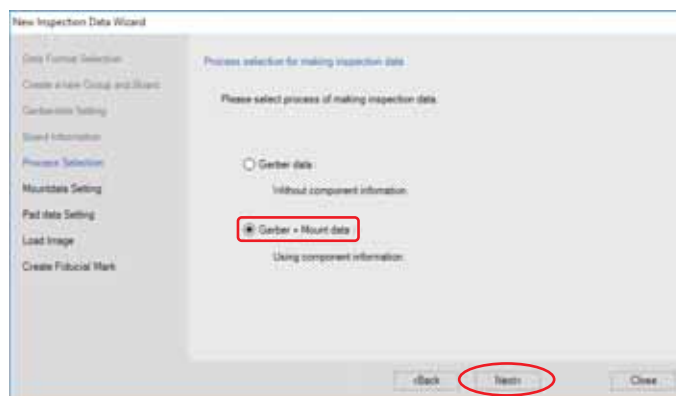


Figure 1-54 Selecting Data Create Process 2

1.4.7 Importing Mount Data

Here describes how to set a mount data.

Step1: The dialog box shown below appears.

Select the appropriate encoding setting from the **Encoding** drop-down list.

Click **Select** and select a mount data.

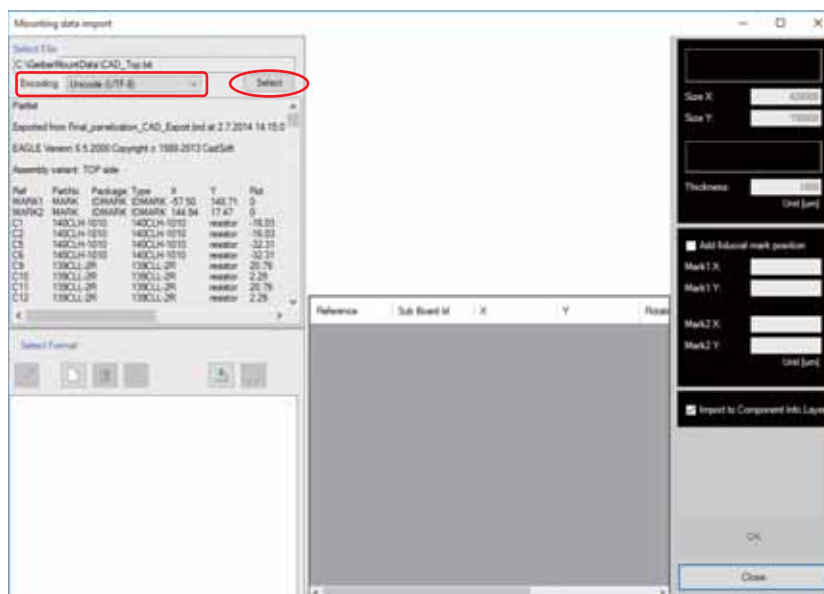


Figure 1-55 Import Mount Data 1

Step2: The preview of the selected data is displayed on the center of the dialog box.
Select a mount data format.

When Using an Existing File Format

Select a file format from the list-box on the lower left side of the dialog box and click **OK**.
The mount data opens. Proceed to Step13.

CAUTION The procedures in Step3 to Step12 are not necessary.

When Making a New File Format

Click  (New).

When Making a New File Format based on an Existing Format

Select a file format which can be the base for the new format and click  (Copy).

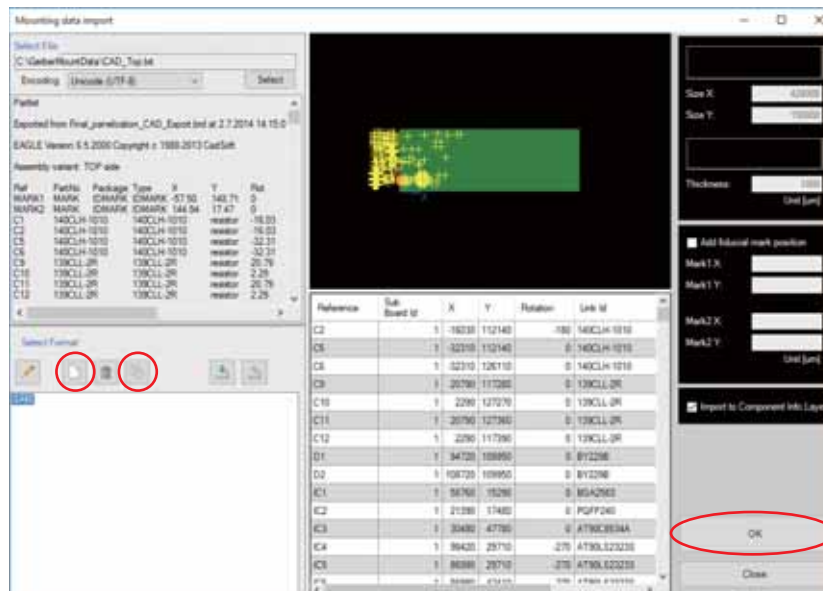


Figure 1-56 Importing a Mount Data 2

Step3: Enter a format name in the text-box and click **OK**.

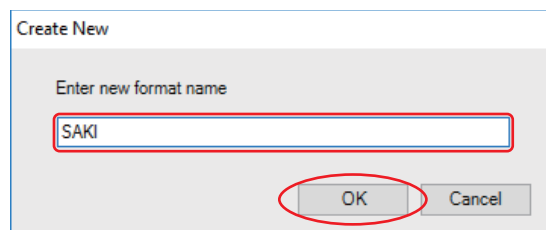


Figure 1-57 Creating a Mount Data Format

Step4: Select the newly created file format from the list-box on the lower left side of the dialog box and click  (Edit).

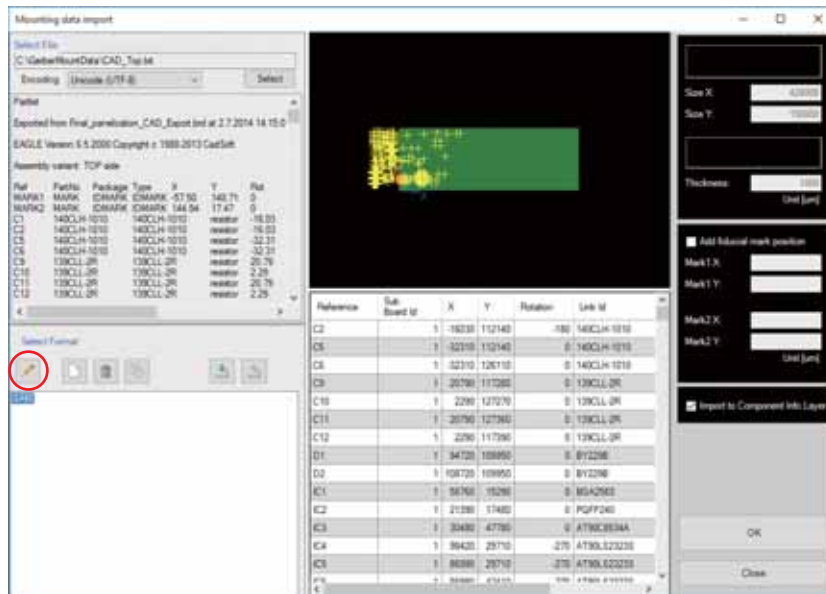


Figure 1-58 Edit

Step5: Contents of the imported mount data file are displayed in the **Line** list-box. Select the lines indicating component information. Information such as component names and positions are extracted from the mount data.

CAUTION Text-box is case-sensitive.

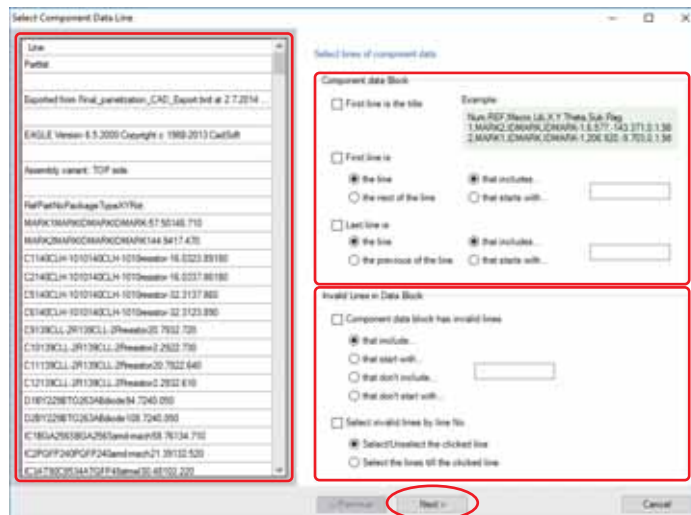


Figure 1-59 Selecting Components Data Line

Item		Description
Component Data Block	First line is the title	Check this item if the first line of the component data indicates contents of the data.
	First line is	Selects a line in accordance with the radio button settings and specifies the selected line as the top line of the component data.
	Last line is	Selects a line in accordance with the radio button settings and specifies the selected line as the bottom line of the component data.
	the line & that includes...	Searches for a line which includes a word entered in the text-box and selects the found line.
	the line & that starts with...	Searches for a line which starts with a word entered in the text-box and selects the found line.
	the next of the line & that includes...	Searches for a line which includes a word entered in the text-box and selects the next line of the found line.
	the next of the line & that starts with...	Searches for a line which starts with a word entered in the text-box and selects the next line of the found line.
Invalid Lines in Data Block	Component data block has invalid lines	Check this item if the component data has the line that does not include component information . Lines are grayed out in accordance with the radio button settings.
	that includes...	Selects lines which include a word entered in the text-box.
	that start with...	Selects lines which start with a word entered in the text-box.
	that don't include...	Selects lines which do not include a word entered in the text-box.
	that don't start with...	Selects lines which do not start with a word entered in the text-box.
	Select invalid lines by line No.	Check this item if specifying invalid lines by left-clicking. Lines are grayed out in accordance with the radio button settings. Since the lines are specified by the line numbers, the format may become inapplicable to another mount data.
	Select/Unselect the clicked line	Selects the clicked line. To cancel, click again.
	Select the lines till the clicked line	Selects the all lines from the top to clicked line.

Table 1-6 Descriptions

Mount Data	Board+ BoardSize_X,220.000 BoardSize_Y,150.000 Component+ (The first line of the component data is always the next line of Component+) Num,REF,Macro,Lib,X,Y,Theta,Sub,Flg (The title line is included in the component data) 1,MARK2,IDMARK,IDMARK-1,6.577,-143.371,0,1,98 2,MARK1,IDMARK,IDMARK-1,206.920,-9.703,0,1,98 ...
Setting	Check First line is . Enter Component+ into the text-box. Check the next of the line and that includes.... Check First line is the title .

Table 1-7 Example 1

Mount Data	BSIZEX150000Y300000 ORX5000Y5000 C"R101"X101223Y050234T090 (The component data always starts with C) C"C202"X059526Y189256T270 ... C"IC5" X033564Y253654T000 END (The last line is not the component data)
Setting	Check First line is . Enter C into the text-box. Check the line and that starts with.... Check Last line is . Enter C into the text-box. Check the line and that starts with....

Table 1-8 Example 2

Mount Data	... 374,U35,DS90LV031ATM,SOP8PIN,152.467,-27.459,90,1,2 (The component data always include ",") 375,U35,DS90LV031ATM,SOP8PIN,152.467,-27.459,90,1,2 <PAGE4> (The lines which do not include "," are not component data.) 376,U4,TL7705ACPS,SOP4PINS,36.094,-99.379,180,1,1 377,U5,TL7705ACPS,SOP4PINS,172.382,-101.121,0,1,2 ...
Setting	Check Component data block has invalid lines . Enter "," into the text-box. Check that don't include .

Table 1-9 Example 3

Step6: Make sure that the component data lines are highlighted in white and other lines are grayed out in the **Line**. Click **Next**.

NOTE If the title line is set, the line is highlighted in blue.

Step7: The component data lines are displayed in the **Line** list-box. Select the data format and define the parameters.

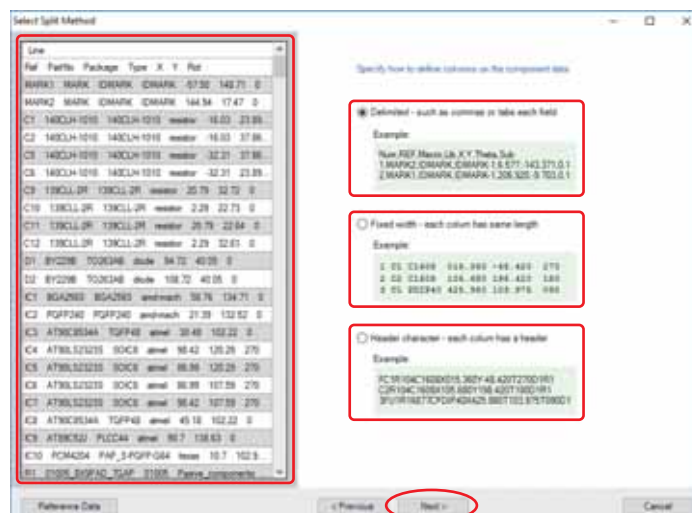


Figure 1-60 Selecting a Data Format

When the Data is Separated by Separator such as Commas, Tabs, or Spaces

Check **Delimited** in the dialog box shown in Figure 1-60 and click **Next**.

The dialog box shown below appears. Select a delimiter in **Separators**.

The data is automatically separated by parameters. After all settings are completed, click **Next**.

NOTE If double quotation marks are included in the mount data, check **Treat as one string when surrounded by double quotations**.

NOTE If two (or more) characters are used as one separator, check **Treat successive separators as one separator**.

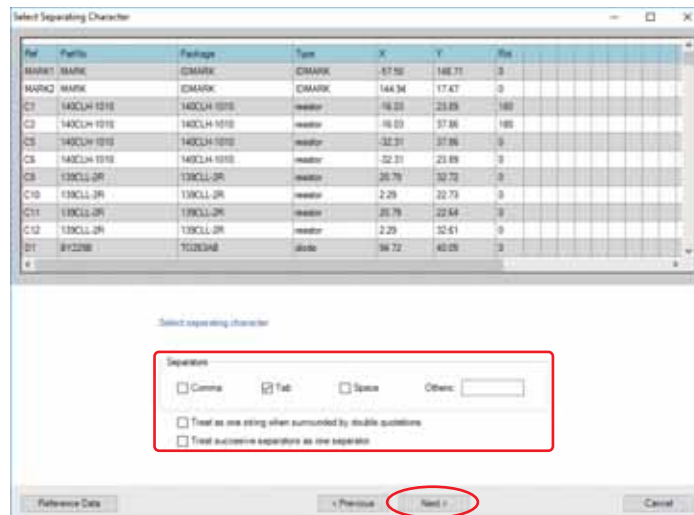


Figure 1-61 Selecting a Separator

When the Data is Separated by the Certain Width

Check **Fixed width** in Figure 1-60 and click **Next**. The dialog box shown below appears.

Adjust the column widths by dragging the boundary of the header to separate the data.

After all settings are completed, click **Next**.

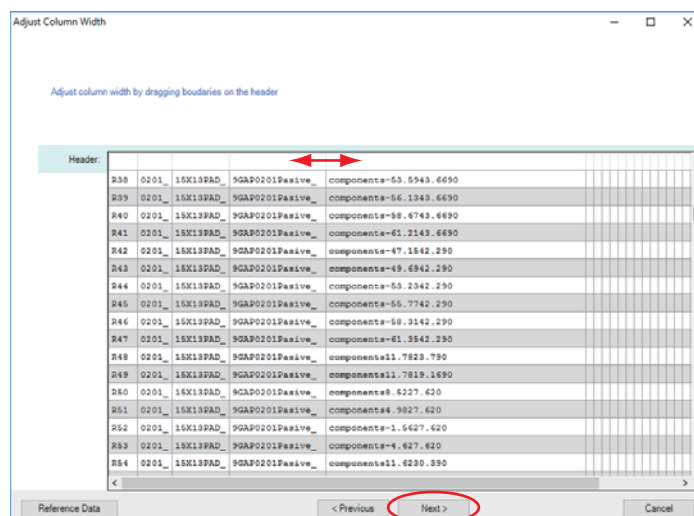


Figure 1-62 Adjusting the Column Widths

When the Data is Separated by the First Character

Check **Header character** in Figure 1-60 and click **Next**. The dialog box shown below appears. Enter the first characters in the **Head character** text-boxes located on the upper side of the dialog box.

The data is automatically separated by parameters.

After all settings are completed, click **Next**.

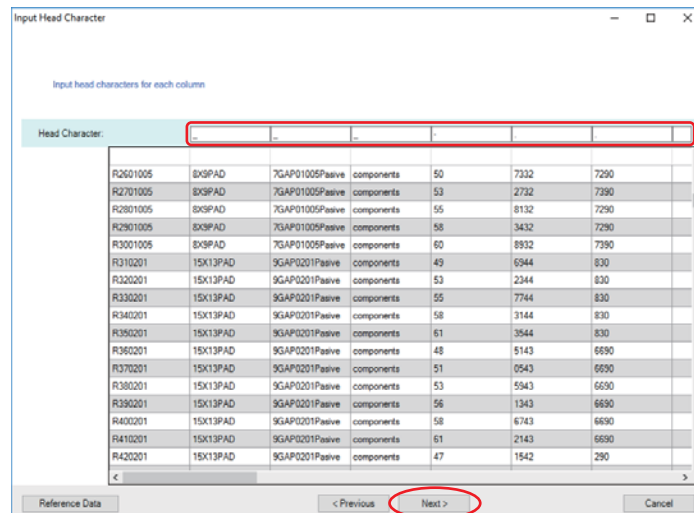


Figure 1-63 Entering the First Characters

Step8: Click a header of a column. Parameters are displayed for identifying the data. Assign a parameter to each column. After all settings are completed, click **Next**.

NOTE

Link IDs are using when creating the pad data (Shape and Recipe). Make sure to specify the Link IDs.

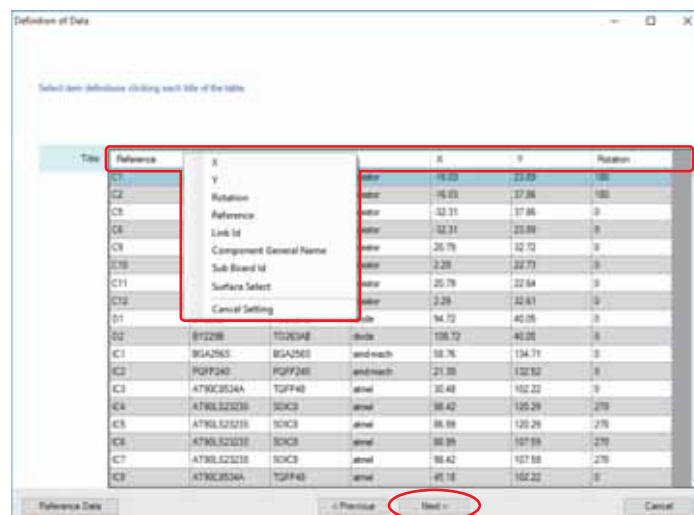


Figure 1-64 Defining the Data

Item	Description
X	Select if the column indicates X-axis positions of components.
Y	Select if the column indicates Y-axis positions of components.
Rotation	Select if the column indicates mounting angles of components.
Reference	Select if the column indicates references of components.
Link Id	Select if the column indicates LinkID of components. (e.g., Macro name)
Component General Name	Not available.
Sub Board Id	Select if the column indicates sub board numbers of components.
Surface Select	Select if the column indicates mounting surfaces of components.
Cancel setting	Select if canceling the selection.

Table 1-10 Descriptions

Step9: Select a conversion method for component coordinates and mounting angles from the drop-down lists of **XY Coordinate** and **Rotation**. After all settings are completed, click **Next**.

NOTE In **XY Coordinate** and **Rotation**, **Other** can be selected for entering a multiplier directly.

NOTE If a number is entered into **Offset angle**, the specified amount of the offset is added to all component installation angles.

NOTE If **Surface Select** (the parameter to specify the mounting surface for a double-side PCB) is set, enter the character which indicates the PCB side to be inspected into the text-box of **Surface**.

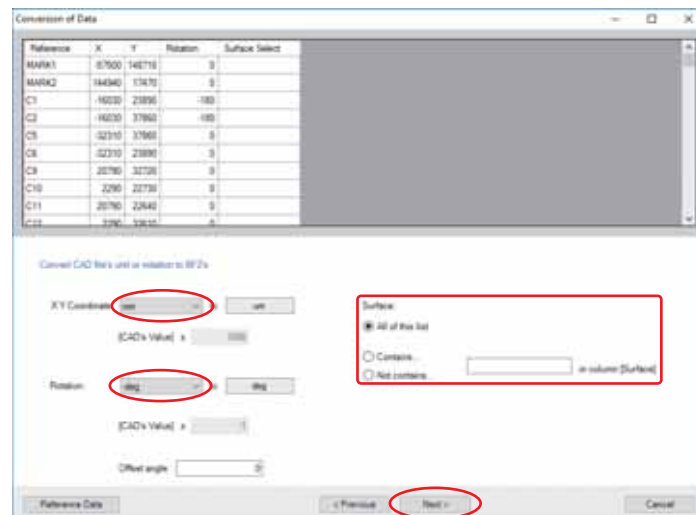


Figure 1-65 Data Conversion

Step10: The dialog box shown below appears.

If the PCB size is entered in **1.4.5 Board Information Settings**, no need to specify in this step. Click **Next**.

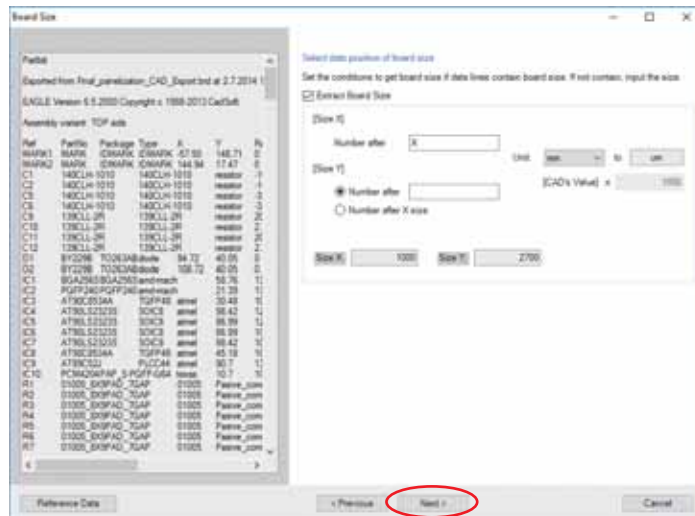


Figure 1-66 Acquiring the PCB Size and Thickness

Item			Description
Extract Board Size	Extract Board Size		Check this item to acquire the lengths in the X direction and Y direction of the PCB.
	[Size X]	Number after	Enter the preceding character of the X-axis length of the PCB. The extracted value is displayed in Size X .
	[Size Y]	Number after	Enter the preceding character of the Y-axis length of the PCB. The extracted value is displayed in Size Y .
		Number after X size	Check this item if the Y-axis length of the PCB is written directly after the X-axis length. The extracted value is displayed in Size Y .
	Unit		Select a unit used in the mount data from the drop-down list. To enter a multiplier, select Other from the drop-down list.

Table 1-11 Descriptions

Mount Data	Board+ BoardSize_X,220.000 BoardSize_Y,150.000	(The next value of BoardSize_X is always the X-axis length of the PCB) (The next value of BoardSize_Y is always the Y-axis length of the PCB)
Setting	Enter BoardSize_X into Number after of [Size X] . Enter BoardSize_Y into Number after of [Size Y] . (In this case, the number is also located on directly after the X-axis length. The same operation can be performed by checking Number after X size). Select mm from the drop-down list of Unit .	

Table 1-12 Example

Step11: Set the origin point and coordinates of the mount data so it matches to those of the inspection machine. After all adjustment is completed, click **Next**.



Figure 1-67 Coordinate Setting

Item		Description
Origin (CAD data)	Left Top	Check when the mount data origin is the upper left corner.
	Right Top	Check when the mount data origin is the upper right corner.
	Left Bottom	Check when the mount data origin is the lower left corner.
	Right Bottom	Check when the mount data origin is the lower right corner.
X Axial Direction (CAD data)	Top	Check when the X-axis value of the mount data increases upward.
	Bottom	Check when the X-axis value of the mount data increases downward.
	Left	Check when the X-axis value of the mount data increases leftward.
	Right	Check when the X-axis value of the mount data increases rightward.
	Flip Y Direction	Inverts the Y-axis of the mount data.
Offset X		Moves the mount data in the X-axis. The offset value can be set in the text-box on the right.
Offset Y		Moves the mount data in the Y-axis. The offset value can be set in the text-box on the right.

Table 1-13 Descriptions

Step12: Confirm that component data is displayed on the board image.

If all settings are completed, click **OK**.

NOTE

If fiducial mark data are not included in the component data, check **Add fiducial mark position** and enter coordinates. Fiducial marks are created on the specified positions.

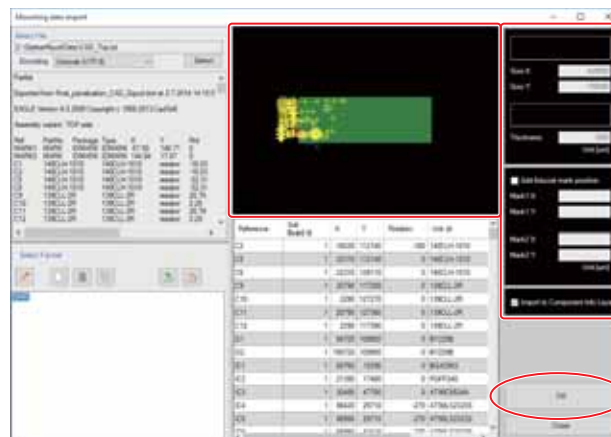


Figure 1-68 Opening the Mount Data

Step13: If invalid characters (e.g., "\", "/", ".", "*", "?", "<", ">", "|", or ".") are included in a reference name or Link ID, the dialog box shown in below appears and the reference names and Link IDs are corrected automatically.

Check the corrected names and edit them if necessary. Click **OK**.

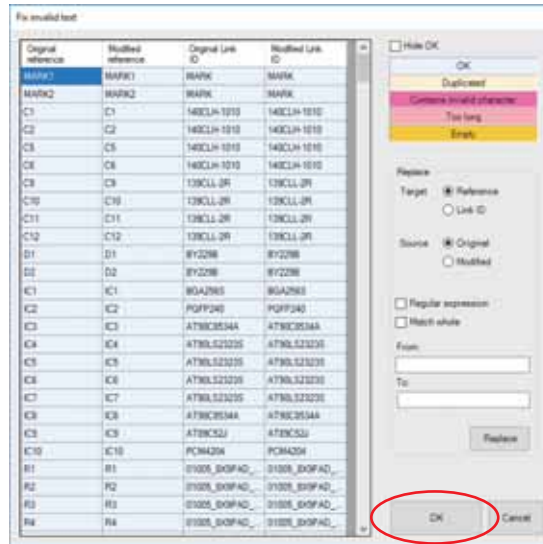


Figure 1-69 Correcting Reference Name and Link ID.

Item			Description
Original reference			Indicates the original reference names.
Modified reference			Indicates reference names modified by the auto-correction.
Original Link ID			Indicates the original Link IDs.
Modified Link ID			Indicates Link IDs modified by the auto-correction.
Hide OK			Hides reference names and Link IDs which do not need correction.
Replace	Target	Reference	Sets reference names as the search target.
		Link ID	Sets Link IDs as the search target.
	Source	Original	Sets the current reference names or Link IDs as the search target.
		Modified	Sets the modified reference names or Link IDs as the search target.
	Regular expression		Allows regular expressions to be used for searching.
	Match whole		Check this item when searching references or Link IDs which completely match to the word entered in From .
	From		Specifies the word to search.
	To		Specifies the word to replace.
	Replace		Searches the word entered in From among reference names or Link IDs, and replaces the found word by the word entered in To .

Table 1-14 Descriptions

Step14: Mount data is extract to **Board Viewer**.

If the library is existing in Inspection data group, the same Link ID data is loading automatically.

NOTE This product loading library by automatic.

1.4.8 Adjusting Mount Data

If importing mount data is completed, the dialog box shown below appears. After all settings are completed, click **Next**.

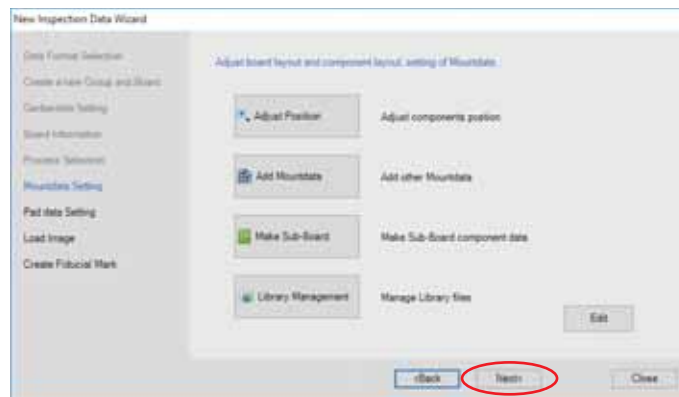


Figure 1-70 Adjusting Mount Data

When Adjusting the Component Data Positions

Adjust the Mount data matching it to the Gerber data.

Step1: Click **Adjust Position** on Figure 1-70 and dialog box shown below appears. Adjust the Mount data matching it to the Gerber data according to Table 1-15 of **Adjust Position**.

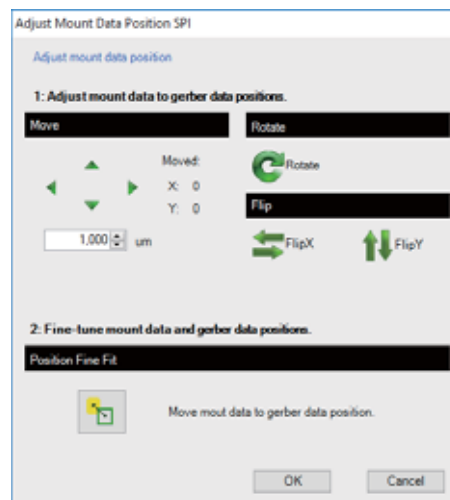


Figure 1-71 Adjusting the Position of Mount Data 1

Item		Description
Adjust position	Move	Moves the component data to the specified direction. The moving value can be specified in the text-box.
	Rotate	Rotates the component data 90 degrees to clockwise.
	Flip	FlipX Inverts the X-axis of the component data.
		Flip Y Inverts the Y-axis of the component data.
Fine-tune		Moves the component data to the center of the pad data on Gerber data to fine-tunes its position.

Table 1-15 Descriptions

Step2: After Adjusting the Mount data position and Gerber data position (for adjusting roughly), Click  to fine-tune the position.

NOTE Before the fine tuning, make sure to align the position in Step1 as much as possible.

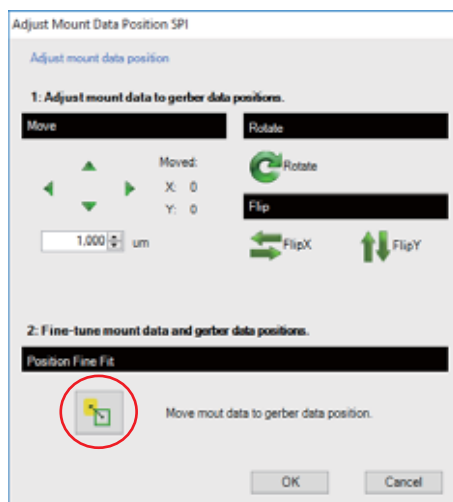


Figure 1-72 Adjusting the Position of Mount Data 2

Step3: Fine tuning of Step2 is completed, the dialog box shown below appears. Click **OK**.

NOTE Fine turning is completed, and Mount data is moved to the center of Gerber data shown in Figure 1-74.

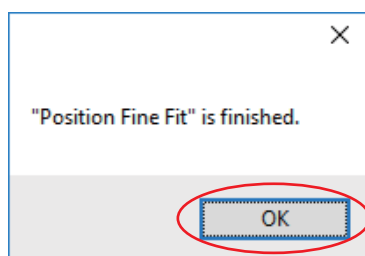


Figure 1-73 Finishing Adjusting the Position of Mount Data

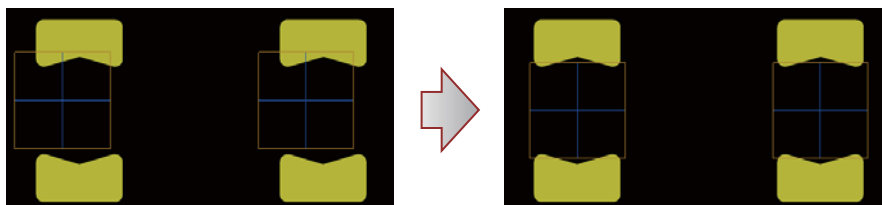


Figure 1-74 Adjusting the Position of Mount Data

Step4: After adjusting mounting position is completed, click **OK**.

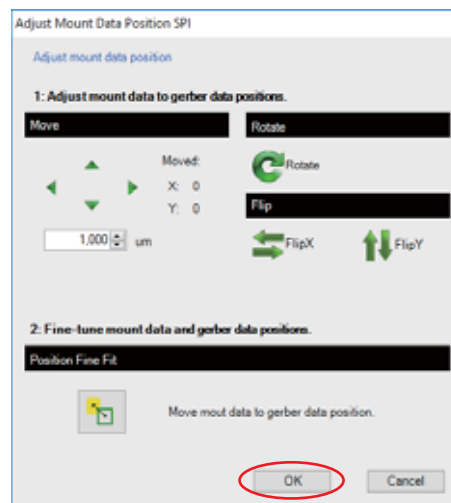


Figure 1-75 Adjusting the Position of Mount Data 3

When Adding a Mount Data

Click **Add Mountdata**. Mount data is added.

For detail, refer to **1.4.7 Importing Mount Data**.

When Making a Sub Board Data

Click **Make Sub-Board**. An interval between sub boards is acquired from Gerber data, and the sub boards are applied to inspection data.

For detail, refer to **1.8 Sub Board Data Setting**.

When Managing Library

Click **Library Management**. Libraries of created data can be managed.

For detail, refer to **2.11 Changing Library, Reference Name, Link ID, Shape, and Recipe**.

1.4.9 Settings of Pad Data

Here describes how to create a link between the Mount data (Component) and Gerber data (Pad), and how to make the pad data (Shape and Recipe).

The dialog box shown below appears. After creating pad data is completed, click **Next**.

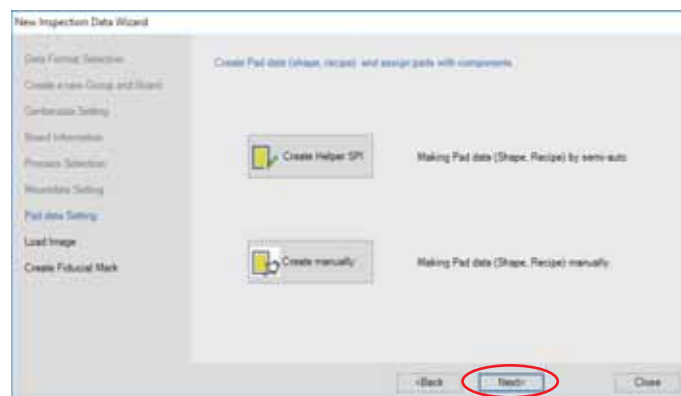


Figure 1-76 Settings of Pad Data

Create Helper SPI

Create Helper SPI displays Mount data in order of Link IDs, predicts pads belonging to the Mount data, and displays the pads. It also creates a link between Mount data (component) and gerber data (pad) for every Link ID, and makes pad data (Shape and Recipe).

Step1: Click **Create Helper SPI** to display the dialog box on upper left of **Board Viewer** shown below. Selecting Link ID's Mount data will display in center of window and cross-hair cursors appear. Then, the Gerber data predicted to belong to Mount data is automatically selected, and it blinks in red.

NOTE

The prediction of Gerber data occasionally fails. In this case, Gerber data does not blink. Select Gerber data by mouse dragging or mouse clicking with **Ctrl** key. Selected Gerber data will blink in red.

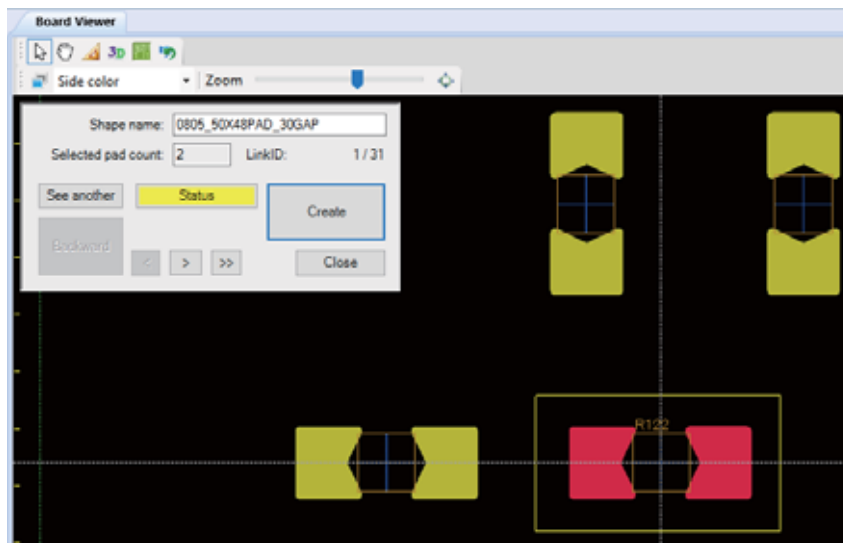


Figure 1-77 Create Helper SPI 1

Step2: Check the selected mount data and Gerber data. Check the Link ID name is in **Shape name** and the number of Gerber data is in **Selected pad count** according to Table1-16.

NOTE

Click **See Another** to confirm other components that use same Link ID as selected component data.

Item	Description
Shape name	Displays the selected Link ID that is selected in loading Mount data
Selected pad count	Displays the number of selected Gerber pad data.
Link ID	Displays the number of Link ID that contain loaded Mount data and Selecting Link ID number.
See another	Displays the same Link ID but different Mount data.
Status	The button color indicates pad data creation status. For details, click Status .
Create	Creates pad data (Shape, Recipe) to selecting Gerber pad data.
Backward	Displays the latest 10 mount data of Link ID which are created by Create .
< / >	Changes displaying Link ID.
>>	Displays the Link ID that is not create the pad data.
Close	Exits the Create Helper SPI and return to Wizard dialog box.

Table 1-16 Descriptions

Step3: Click **Create** to create pad data (Shape and Recipe).

NOTE

The same operation can also be performed by pressing **(Enter)** key.

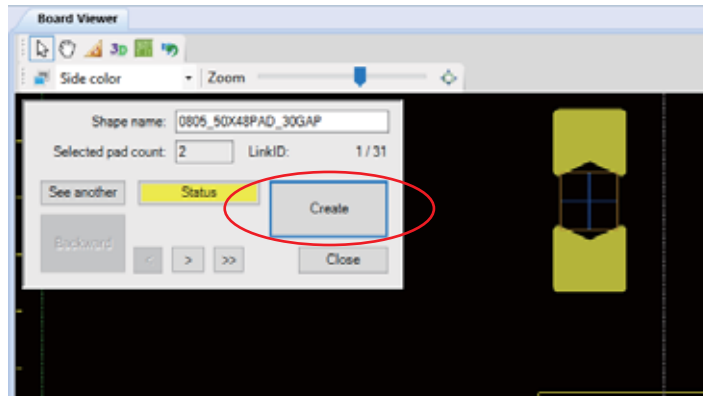


Figure 1-78 Create Helper SPI 2

Step4: When the Shape created in Step3 can be used with a different Link ID, the dialog box shown below appears.

Yes

Apply the pad data (Shape and Recipe) to other Link ID.

No

Does not apply the pad data (Shape and Recipe) to other Link ID.

NOTE

If the Shape is available to use and its shape is the same as the pad size, the same Recipe can be used.

NOTE

The Pad data is not applied to Link IDs that pad data (Shape and Recipe) is already created.

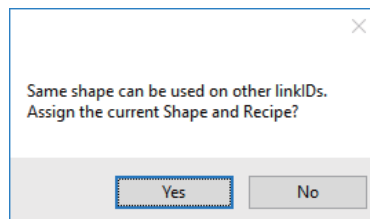


Figure 1-79 Create Helper SPI 3

Step5: The Mount data and Gerber data of Link IDs on which pad data (Shape and Recipe) is not created are displayed in the center.

NOTE To confirm the Mount data of Link ID on which pad data is already created, click  or  to display the Mount data in order of Link IDs.

NOTE To confirm the pad data of Link ID just created, click **Backward**.
When pad data is successfully created, the data is shown as below with Figure 1-80.

NOTE To recreate the data, select the data once again.

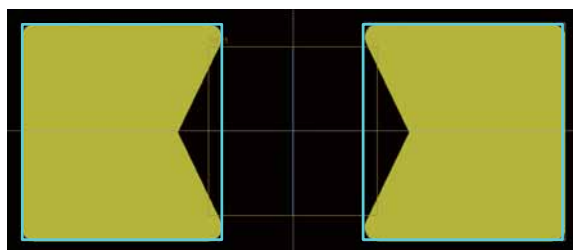


Figure 1-80 Example of created pad data

Step6: Create all pad data by repeating Step1 to Step5.

Step7: Click **Close** to exit the **Create Helper SPI**.

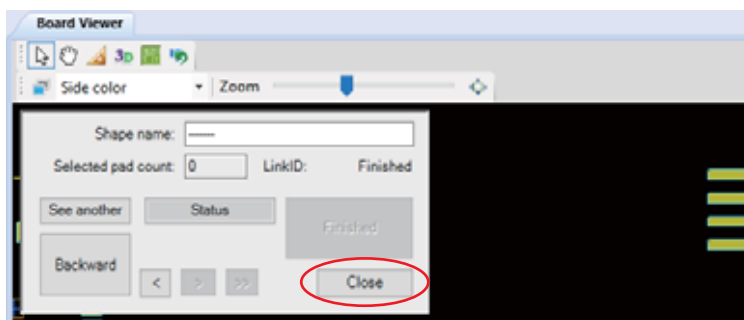


Figure 1-81 Exit Create Helper SPI

Step8: The dialog box shown below appears.

Yes	Create only for the pads linked with a component.
No	Create for all pads.
Cancel	Does not create.

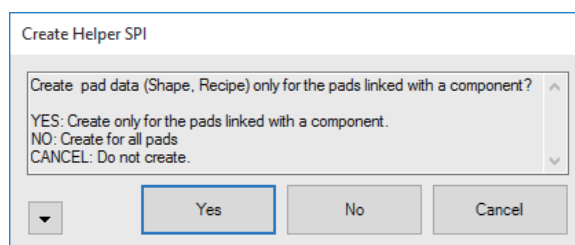


Figure 1-82 Create Helper SPI 4

Create Manually

Create Manually creates pad data by selecting arbitrary data. This function can be used when adjusting data created with **Create Helper SPI**, or when not creating data in order of Link IDs.

Step1: Click **Create Manually**. The dialog box shown below appears.

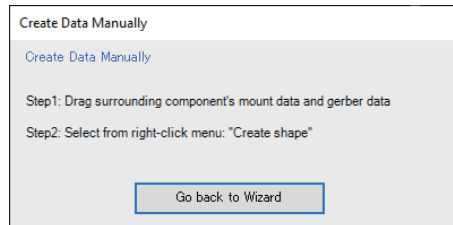


Figure 1-83 Create Manually 1

Step2: On **Board Viewer**, select the Mount data and Gerber data to create pad data by mouse dragging. Right click and select **Create Shape** shown in Figure1-84. Create the pad data (Shape and Recipe) in Link ID that has mount data.

NOTE Selected Gerber data is highlighted in light purple.

NOTE Multiple blocks of Gerber data can be selected by pressing **Ctrl**.

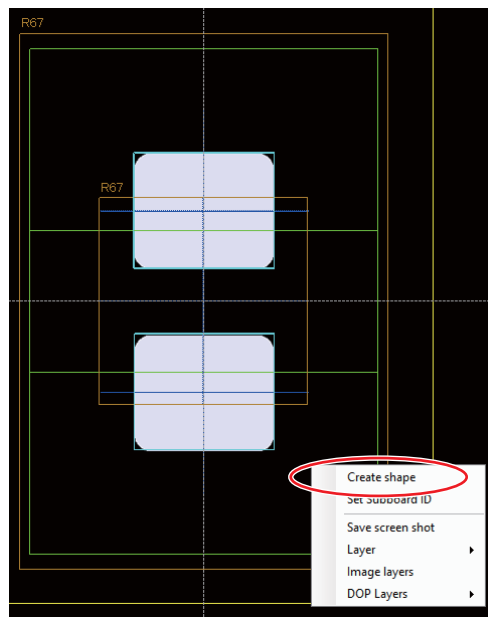


Figure 1-84 Create Manually 2

Step3: Refer to Step4 of **Create Helper SPI** to follow the same step.

Step4: Create pad data by repeating Step1 to Step3.

Step5: Creating pad data is completed, click **Go back to Wizard** to return to wizard.

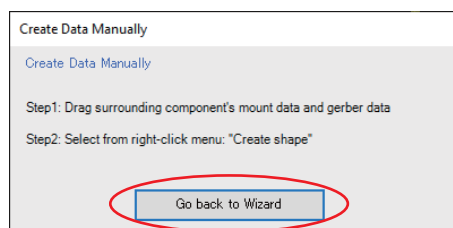


Figure 1-85 Create Manually 3

1.4.10 Capturing Image

Here describes capturing image procedure.

Step1: If creating pad data is completed, the dialog box shown below appears. Click **OK**.

NOTE This dialog box is not displayed in BF2-Editor.

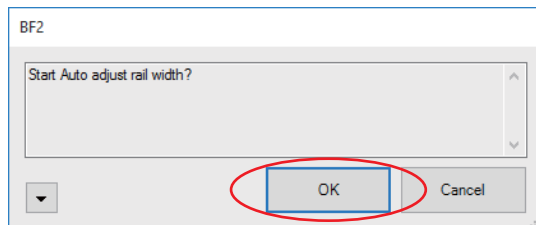


Figure 1-86 Auto Adjustment for the Rail Width 1

Step2: Click **Yes**.

NOTE This dialog box is not displayed in BF2-Editor.

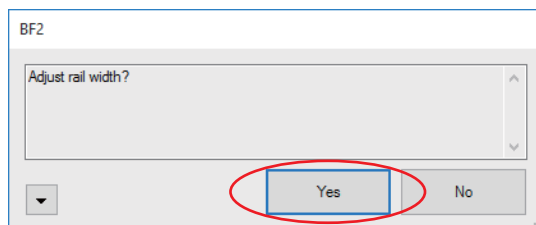


Figure 1-87 Auto Adjustment for the Rail Width 2

Step3: The dialog box shown below appears.



Figure 1-88 Scanning a Board

When Scanning a PCB

Click **Adjust Rail Width** on the dialog box shown in Figure 1-88.

The dialog box shown below appears.

Click **Board search**. Enter a board width into **Absolute** and click **Set**.

The conveyor rail width is adjusted.

When the adjustment is completed, click **Close** to close the dialog box.

CAUTION In BF2-Editor, PCB images cannot be acquired.

NOTE Images acquired from this step are only for making an inspection data. For inspections, FOV needs to be set separately. For details of FOV settings, refer to **5 Acquiring a PCB image**.

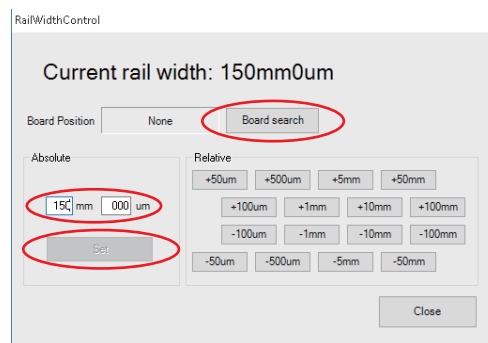


Figure 1-89 Adjusting the Rail Width

Place a PCB on the conveyor rail from the PCB inlet of the machine and click **Load/Unload Board** on the dialog box shown in Figure 1-88.

Click **Scan** and the PCB on the conveyor rail is scanned. Click **Next**.

When Loading an Existing Image File

Click **Load Board Image** on the dialog box shown in Figure 1-88 and select an image file.

Click **Next**.

NOTE The extension for an image file is ".bdimg".

1.4.11 Setting the Fiducial Marks

Can be select the fiducial mark form the Mount data.

Select the data to fiducial mark from **Component** list-box, click .

The data moves to **Mark** list-box.

After all settings are completed, click **Next**.

NOTE

If the no data to use as fiducial mark in Mount data, can be add a data by manual after finishing the wizard. For details, refer to **1.7 Fiducial Mark and Bad Mark Setting**.

NOTE

Multiple components can be selected by selecting component data with pressing **Ctrl**.

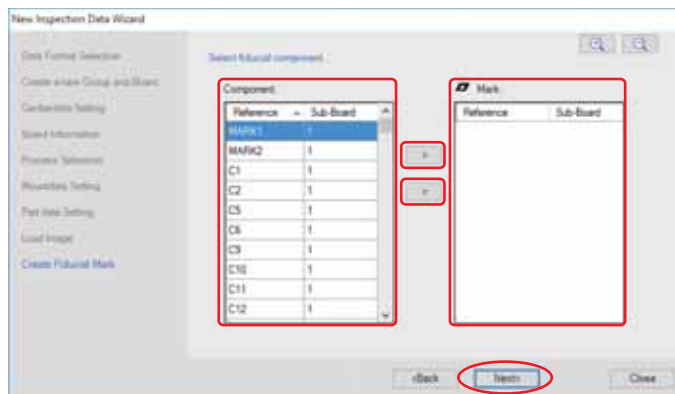


Figure 1-90 Setting the Fiducial Marks

1.4.12 Finishing the Wizard

After all steps are finished, the dialog box shown in below appears. Click **Finish** and the wizard closes.

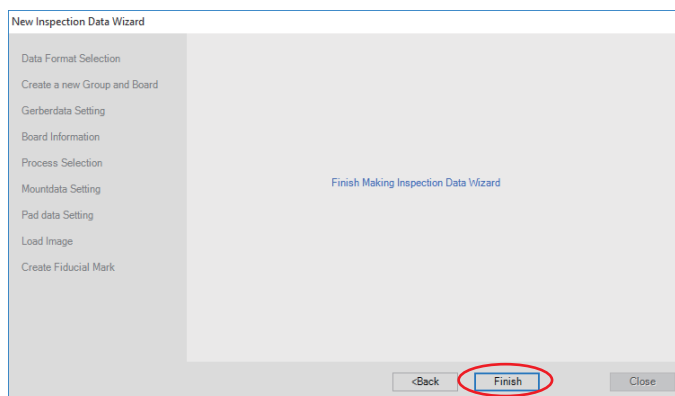


Figure 1-91 Finishing the Wizard

1.4.13 Adding Component Information after completion of wizard

Even if component information is not added during wizard operation (in the case of **When Using Only Gerber Data** in 1.4.6 Selecting Data Create Process) , that information can be added after completion of wizard operation.

Step1: From **Pad Info** tab, select **LookUp table** group > **Create LookUp table**.



Figure 1-92 Adding Component Information after completion of wizard 1

Step2: The dialog box shown below appears. Click **Component Info Wizard**.
For other buttons, refer to Table 1-17.

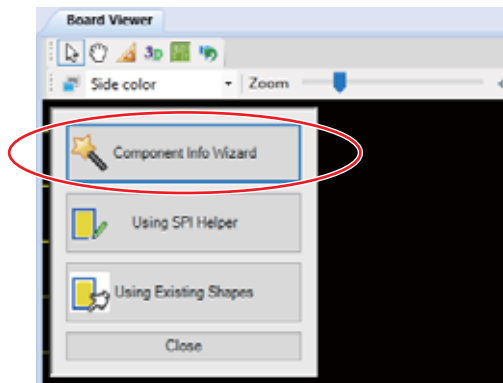


Figure 1-93 Adding Component Information after completion of wizard 2

Item	Description
Component Info Wizard	Imports component information into the existing data by using the wizard.
Using SPI Helper	When the mount data is expanded, assignment of pads to component data, creation of shapes, recipes and LookUp table are available. Please use this button in such case that Component info Wizard is canceled after mount data loading. When the mount data is not expanded, click Component Info Wizard .
Using Existing Shapes	Create or re-create the LookUp table. If the expanded mount data already has Shape information, re-creation of LookUp table is available without using Using SPI Helper .
Close	Closes the dialog box.

Table 1-17 Descriptions

Step3: The dialog box shown below appears.

Import the Mount Data according to **1.4.7 Importing Mount Data**.

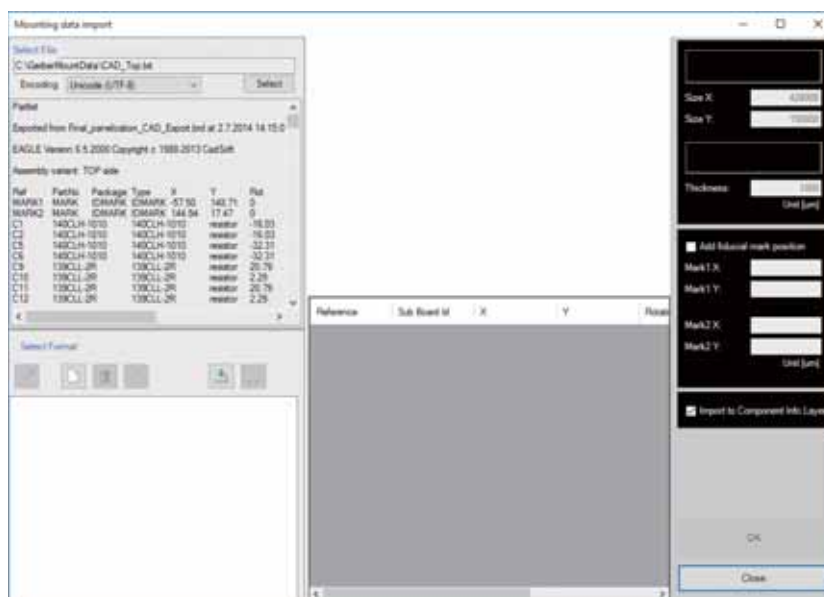


Figure 1-94 Adding Component Information after completion of wizard 3

Step4: The dialog box shown below appears.

Adjust the position of the mount data according to **1.4.8 Adjusting Mount Data**.

Also, if necessary, expand sub board and set library.

After all the settings are completed, click **Next**.

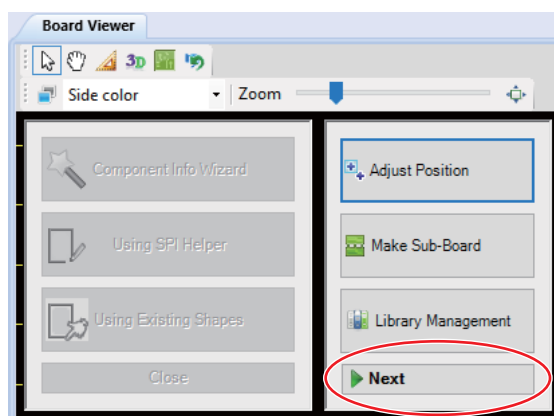


Figure 1-95 Adding Component Information after completion of wizard 4

Step5: The dialog box shown below appears.

Associate Mount data (component) with gerber data (pad) for each Link ID, and make pad data (Shape and Recipe) according to **Create Helper SPI** of 1.4.9 Settings of Pad Data.

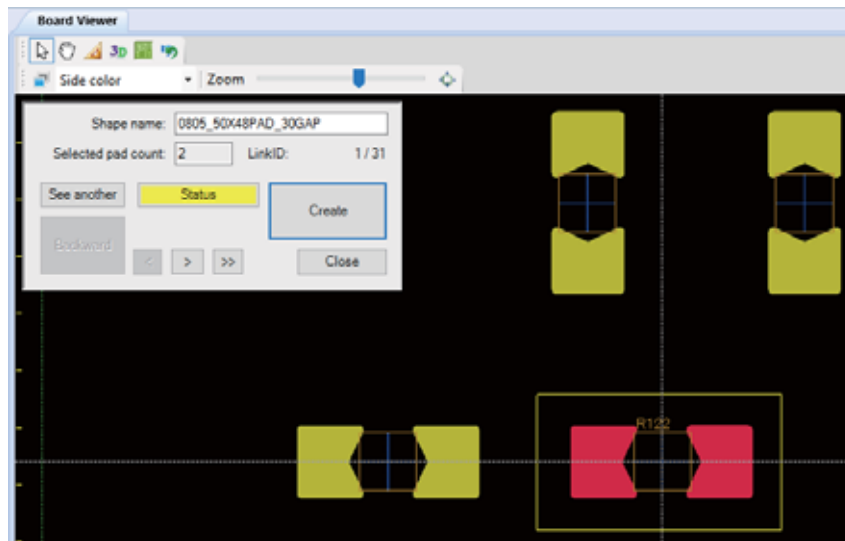


Figure 1-96 Adding Component Information after completion of wizard 5

Step6: The added component information is displayed on the **LookUp Table** window of the **InspectData** tab.

LookUp Table									
Component Inspection									
Reference	Sub Board ID	Link ID	Shape	Recipe	Component Reference	Pin Number	Part Number	Package	Result
Reference_255	1	1294x1188_1	1294x1188_1	1294x1188_1	R345	2	0805_50X48PAD_30GAP	None	
Reference_257	1	1294x1188_1	1294x1188_1	1294x1188_1	R295	2	0805_50X48PAD_30GAP	None	
Reference_258	1	1294x1188_1	1294x1188_1	1294x1188_1	R272	2	0805_50X48PAD_30GAP	None	
Reference_259	1	1294x1188_1	1294x1188_1	1294x1188_1	R275	2	0805_50X48PAD_30GAP	None	
Reference_260	1	1294x1188_1	1294x1188_1	1294x1188_1	R291	2	0805_50X48PAD_30GAP	None	
Reference_261	1	1294x1188_1	1294x1188_1	1294x1188_1	R305	2	0805_50X48PAD_30GAP	None	

Figure 1-97 Adding Component Information after completion of wizard 6

1.5 Making an Inspection Data using ODB++ (Inspection Data Wizard)

This section describes how to make an inspection data with **ODB++**.

NOTE

The ePM-AOI license key is necessary to use **ODB++**.
Make sure license key is inserted to a USB port on the PC.

Step1: Click **BF2** on the upper left of the window and select **Inspect data wizard**.

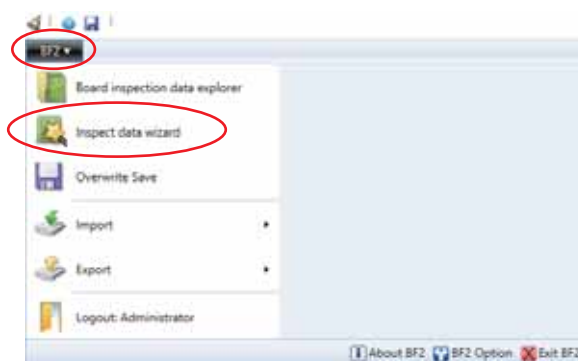


Figure 1-98 Making Inspection Data Wizard 1

Step2: If an inspection data is opened, the dialog box shown below appears.

Yes Overwrites the inspection data and start Inspection Data Wizard.

No Start Inspection Data Wizard without saving the inspection data.

Cancel Returns to the original screen without saving the inspection data.

NOTE

Saving the inspection data is recommended when the data is modified.

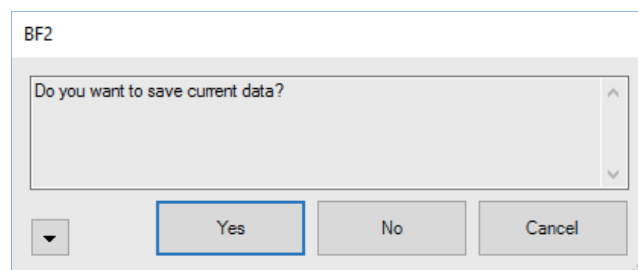


Figure 1-99 Making Inspection Data Wizard 2

Step3: The dialog box shown below appears.
Select **ODB++ data** and click **Next**.

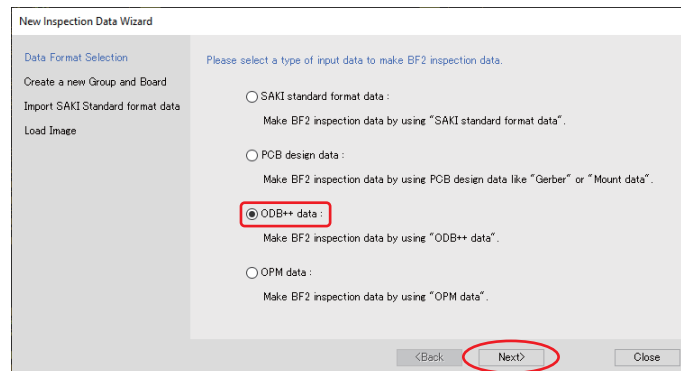


Figure 1-100 Data Selection

Step4: The dialog box shown below appears.

When Making a New Group

Specify a destination to save the group data on **Destination** by entering an address.
Alternately click  on the right side of the text-box and select a folder.
Fill out **Group** and **Board**. Click **Next**.

When Adding a Board Data to an Existing Group

Select an inspection data folder on **Destination** and select a group folder on **Group**.
To specify a destination, enter an address or click  on the right side of the text-box and select a folder. Fill out **Board** and click **Next**.

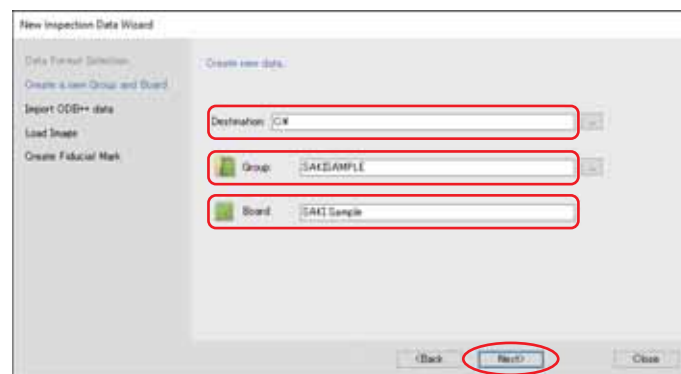


Figure 1-101 Create Group and Board Data

Step5: Specify ODB++ data to import on **ODB++ Path** and select **Board side**.
Click **Import**.

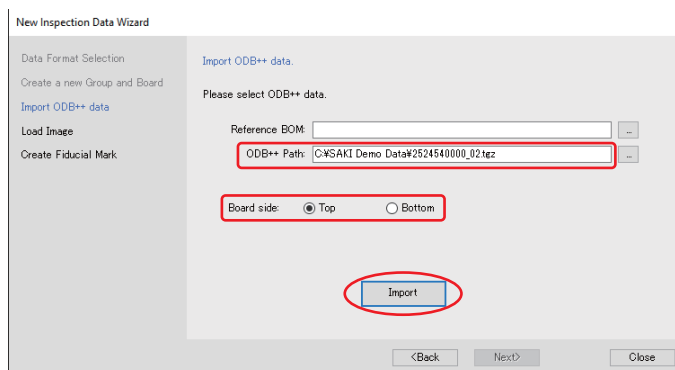


Figure 1-102 Import ODB++ 1

NOTE

The BOM file can be used to delete unnecessary components or change Link IDs and common component names. Click  on the right side of **Reference BOM** and select the BOM file to load. The dialog box shown below appears. After setting it up according to **2.12.3 Import BOM Data**, click **OK**.

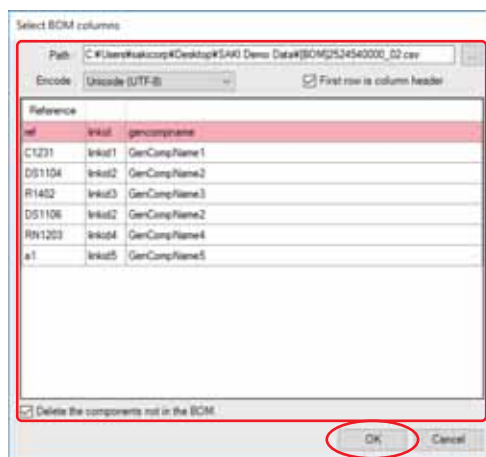


Figure 1-103 Set BOM Data

NOTE

Progress bar is shown while importing ODB++ data.

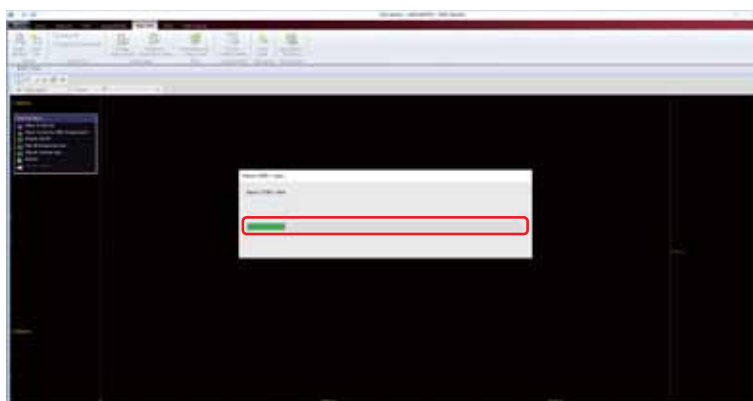


Figure 1-104 Import ODB++ Data 2

Step6: After importing ODB++ data, click **Next**.

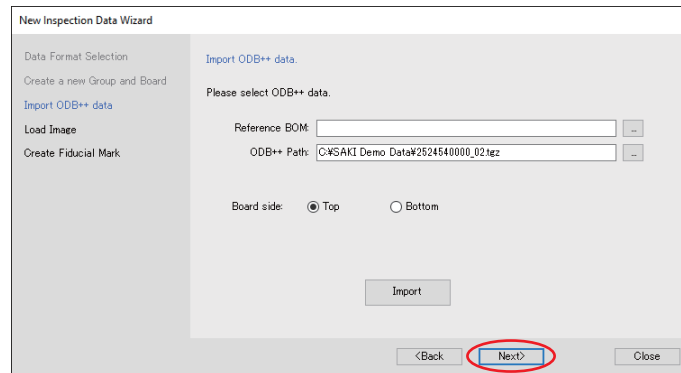


Figure 1-105 Import Complete

Step7: Complete Inspect Data Wizard according to Step7 and later in **1.3 Making an Inspection Data using SAKI Standard Format Data (Inspection Data Wizard)** and **1.4.11 Setting the Fiducial Marks**.

1.6 Making an Inspection Data using OPM (Inspection Data Wizard)

This section describes how to make an inspection data with **OPM**.

NOTE

The ePM-AOI license key is necessary to use **OPM**.
Make sure license key is inserted to a USB port on the PC.

Step1: Click **BF2** on the upper left of the window and select **Inspect data wizard**.

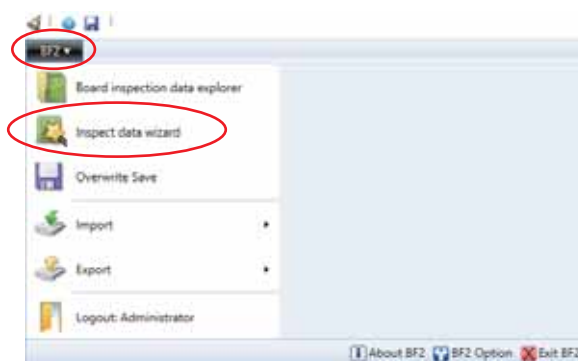


Figure 1-106 Making Inspection Data Wizard 1

Step2: If an inspection data is opened, the dialog box shown below appears.

- Yes** Overwrites the inspection data and start Inspection Data Wizard.
- No** Start Inspection Data Wizard without saving the inspection data.
- Cancel** Returns to the original screen without saving the inspection data.

NOTE

Saving the inspection data is recommended when the data is modified.

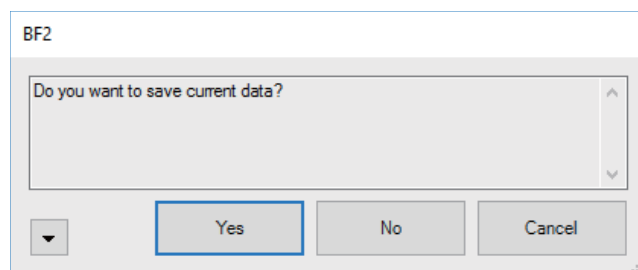


Figure 1-107 Making Inspection Data Wizard 2

Step3: The dialog box shown below appears.
Select **OPM data** and click **Next**.

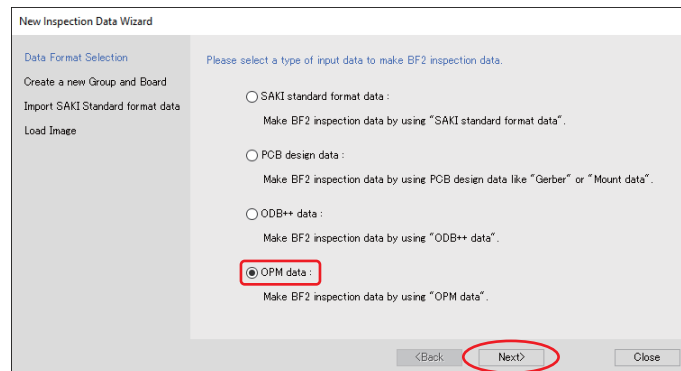


Figure 1-108 Data Selection

Step4: The dialog box shown below appears.

When Making a New Group

Specify a destination to save the group data on **Destination** by entering an address.
Alternately click  on the right side of the text-box and select a folder.
Fill out **Group** and **Board**. Click **Next**.

When Adding a Board Data to an Existing Group

Select an inspection data folder on **Destination** and select a group folder on **Group**.
To specify a destination, enter an address or click  on the right side of the text-box and select a folder. Fill out **Board** and click **Next**.

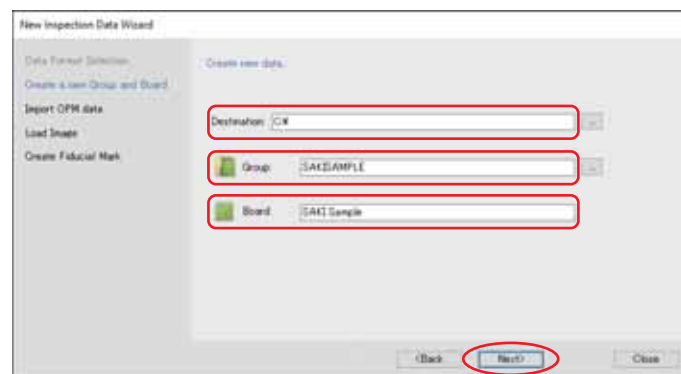


Figure 1-109 Create Group and Board Data

Step5: Specify OPM data to import on **OPM Path**, and select **Board side**.
Click **Import**.

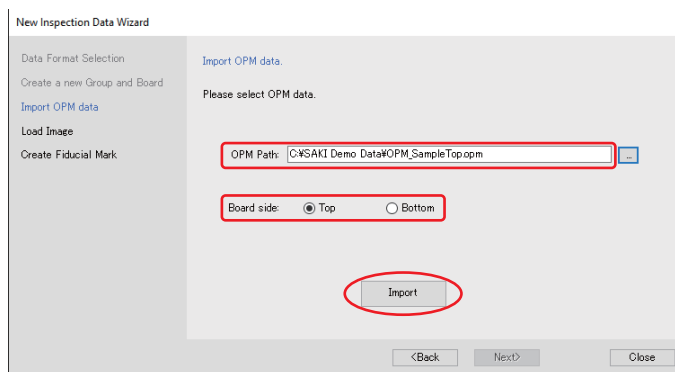


Figure 1-110 Import OPM 1

Step6: The dialog box shown below appears.
Select a layout to expand at **Board Name** and click **OK**.

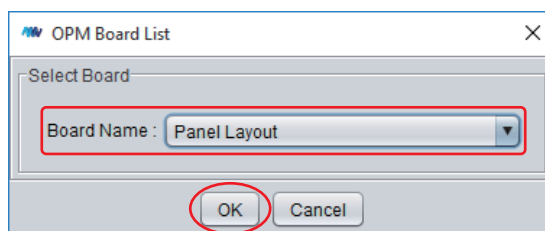


Figure 1-111 Import OPM 2

NOTE Progress bar is shown while importing OPM.

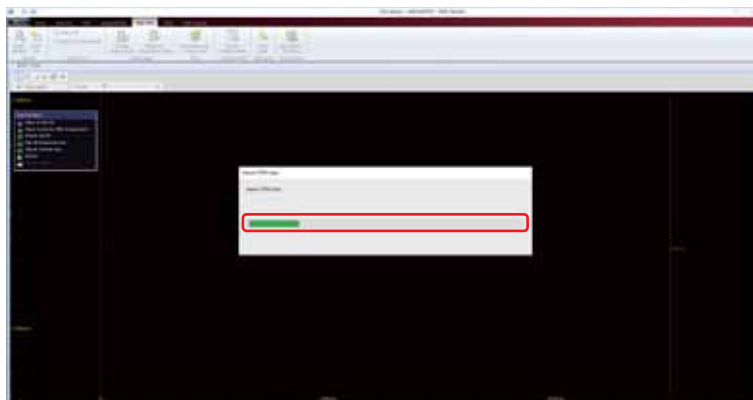


Figure 1-112 Import OPM 3

Step7: After importing OPM data, click **Next**.

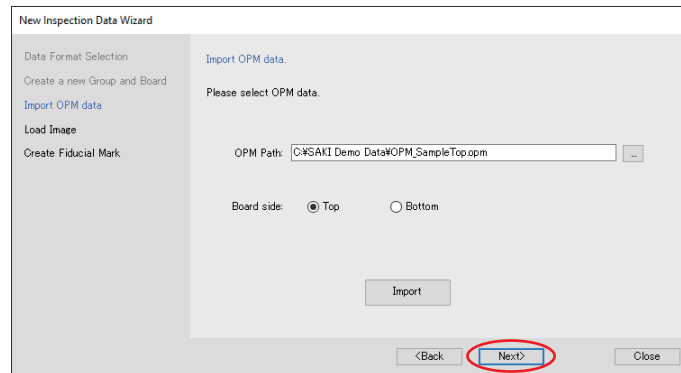


Figure 1-113 Import Complete

Step8: Complete Inspect Data Wizard according to Step7 and later in **1.3 Making an Inspection Data using SAKI Standard Format Data (Inspection Data Wizard)** and **1.4.11 Setting the Fiducial Marks**.

1.7 Fiducial Mark and Bad Mark Setting

1.7.1 Create Marks

Here describes how to set the fiducial marks and bad marks.

Step1: Select the **InspectData** tab.



Figure 1-114 InspectData

Step2: Select a data to be set as a fiducial mark or bad mark from **Board Viewer**.

NOTE

If there is no data which can be set as a fiducial mark or bad mark, create a new data for setting marks. For details, refer to **2.4.2 Adding a Data**.

Step3: The edit panel appears on **Board Viewer**.

Select **Normal Component > Fiducial mark** or **Bad mark**.

The dialog box shown in Figure 1-116 appears. Click **Yes**.

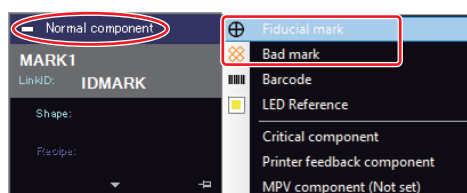


Figure 1-115 Edit Panel 1

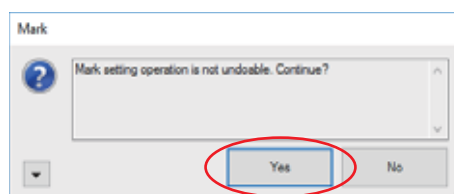


Figure 1-116 Dialog Box of Confirmation

Step4: Confirm that the selected data is set as the fiducial mark or bad mark and click **Recipe**.



Figure 1-117 Edit Panel 2

Step5: From the dock window, select **Window Tree > Library > Library**.
Click  on the upper side of window tree.

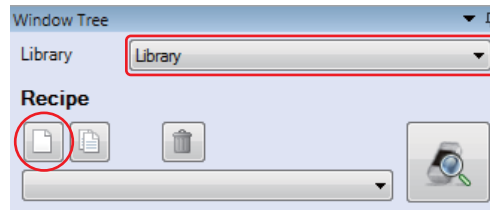


Figure 1-118 New

Step6: Enter a recipe name and click **OK**.

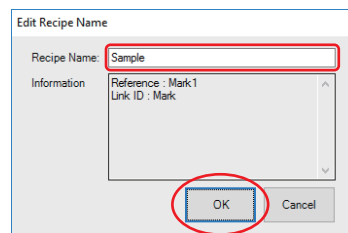


Figure 1-119 Editing a Recipe Name

Step7: Click **Assign** on **Recipe selection preview**.

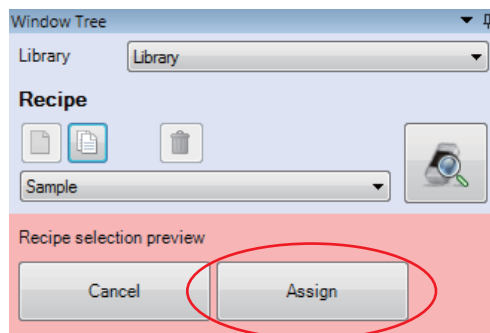


Figure 1-120 Assign

Step8: Click **Add** on the edit panel. A new inspection window is added.

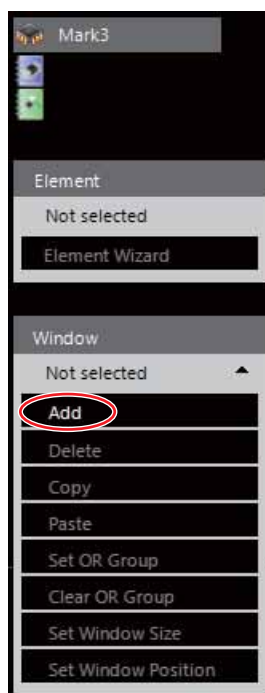


Figure 1-121 Edit Panel 3

Step9: From the dock window, select **Parameter Editor > Algorithm** to open the drop-down list. Select an algorithm which can detect a **Shift value** (e.g., **Circular Fiducial**, **Square Fiducial**).

NOTE For details of algorithms, refer to **Part III Inspection Algorithm**.

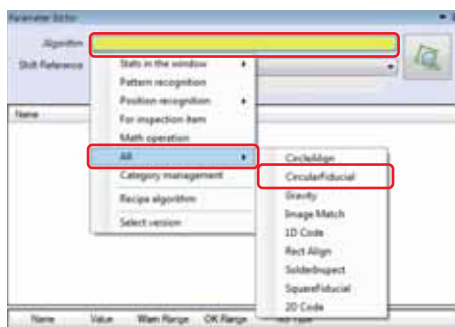


Figure 1-122 Selecting an Algorithm

Step10: Adjust the inspection window size so the window surrounds the fiducial mark/bad mark.

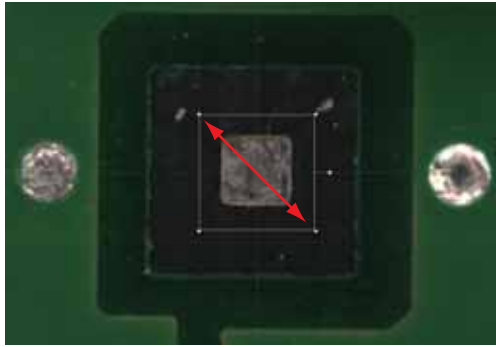


Figure 1-123 Adjusting the Inspection Window

Step11: Select a lighting from **Image filter** and click .

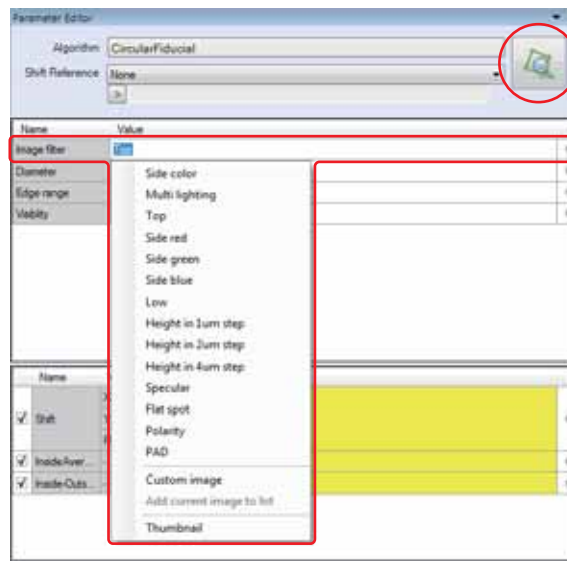


Figure 1-124 Selecting a Lighting

Step12: Confirm that the fiducial mark/bad mark is detected.

NOTE

If the fiducial mark/bad mark cannot be detected, change the lighting or adjust the parameters of the algorithm.

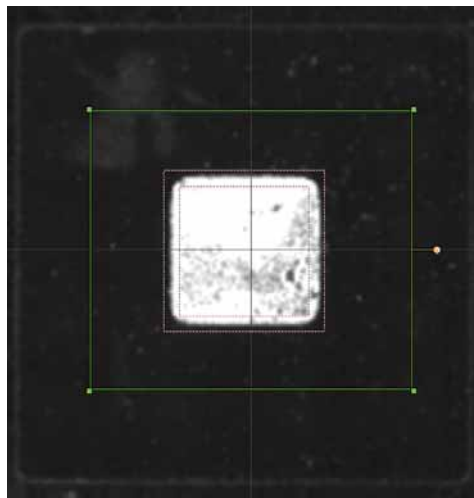


Figure 1-125 Detecting the Mark

Step13: From the ribbon, select the **Exit** group > **Exit**.

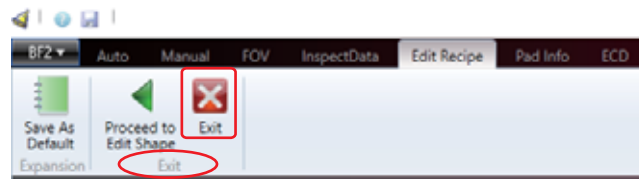


Figure 1-126 Finish Editing 1

Step14: Click **Yes**.

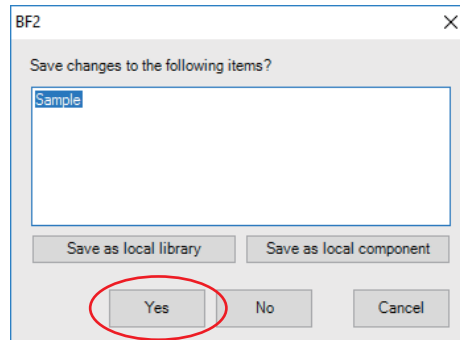


Figure 1-127 Finish Editing 2

Step15: If necessary, make more marks according to Step2 to Step14.

NOTE

If two fiducial marks are set on the PCB image, data positions and angles are adjusted when starting an inspection. If three marks are set, warpage of the PCB is also adjusted.

1.7.2 Manual Assignment of Mark

Here describes how to assign the created fiducial mark and bad mark by using the board example shown below.

The fiducial mark in orange is created as sub-board fiducial mark.

The Panel fiducial mark (The red mark in the below figure) revises each fiducial mark position.

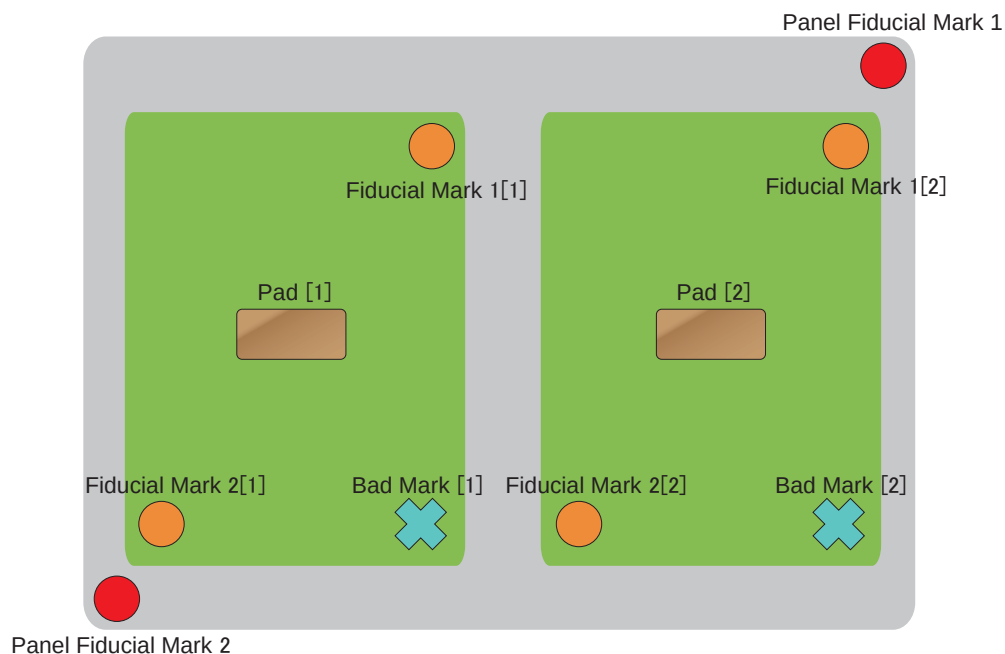


Figure 1-128 Assigning a Fiducial Mark

Step1: From **InspectData** tab, select the **Fiducial Mark** group > **Fiducial Mark**.
The dialog box shown in Figure 1-130 appears.



Figure 1-129 Manual assignment of mark 1

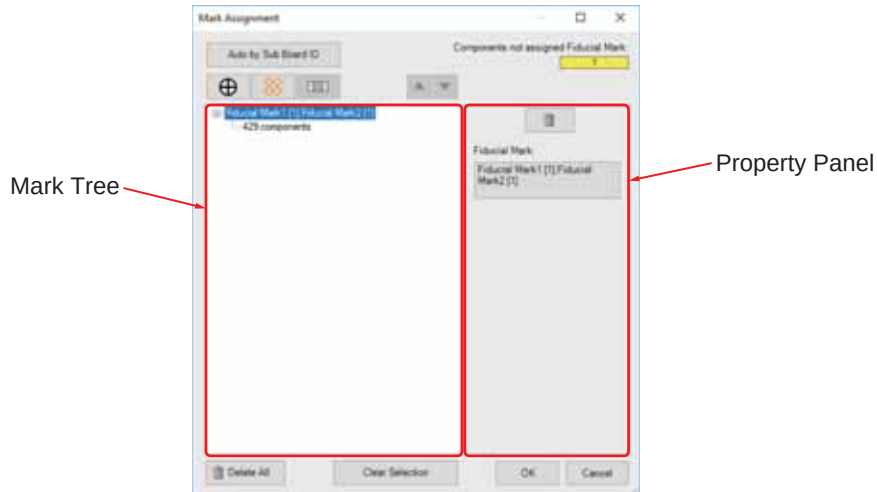


Figure 1-130 Manual assignment of mark 2

Item	Description
Auto by Sub Board ID	Set fiducial mark and bad mark to data.
Mark Tree	Displays relation between fiducial mark, bad mark, and data.
Property Panel	Edits selected item in mark tree.
	Adds a fiducial mark in specified place in mark tree.
	Adds a bad mark in specified place in mark tree.
	Adds a data in specified place in mark tree.
	Moves the position in mark tree.
Components not assigned Fiducial Mark	Displays the number of data not assigned fiducial mark.
	Deletes selected item in mark tree.
Delete All	Deletes all items in mark tree.
Clear Selection	Cancels item select. When adding new item to top of mark tree, deselect the item is necessary.

Table 1-18 Descriptions

Step2: Click  (Add fiducial mark) and check panel fiducial mark and click **OK**.
Although Bad mark select dialog box appears automatically, click **Cancel**.

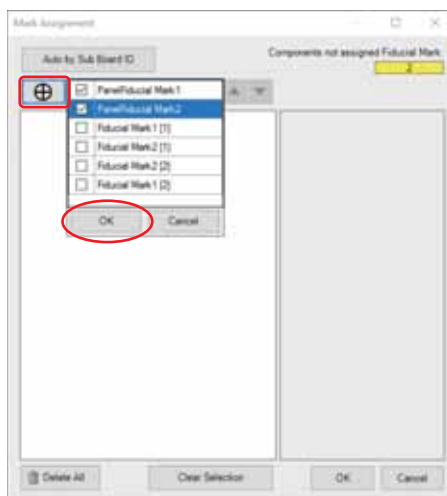


Figure 1-131 Manual Assignment of Mark 3

Step3: Assigning fiducial mark for sub board 1 by manual.
Click  (Add fiducial mark) and check fiducial marks for sub board 1.
Click **OK**.

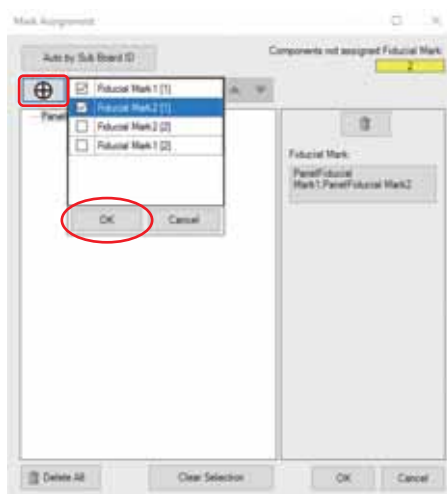


Figure 1-132 Manual Assignment of Mark 4

- Step4: Bad mark select dialog box appears.
Check bad mark for sub board 1, and click **OK**.

NOTE

When put bad mark manually, click  (Add bad mark).

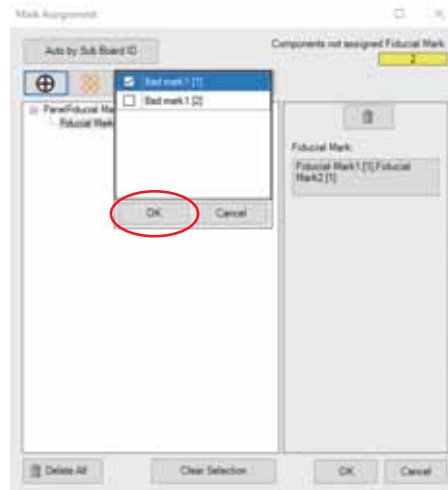


Figure 1-133 Manual Assignment of Mark 5

- Step5: The dialog box shown below appears.
Check **Sub board** and set to **1**.
After all the setting is completed, click **OK**.
Assigning marks sub board 1 is complete.

NOTE

When set components to sub board manually, click  (Add component).

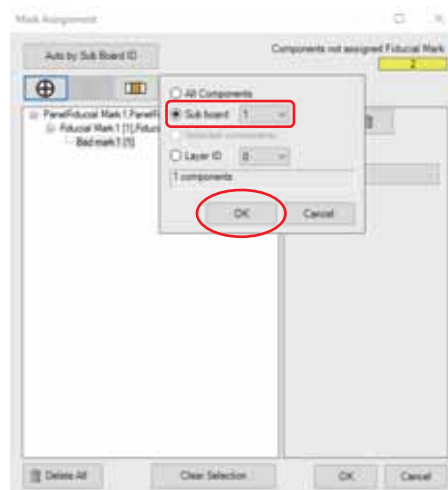


Figure 1-134 Manual Assignment of Mark 6

Step6: Assigning marks for sub board 2 by manual.

Select **PanelFiducial Mark1**, **PanelFiducial Mark2** on the top of mark tree.

Assign the marks of sub board 2 according to Step2 to Step 5.

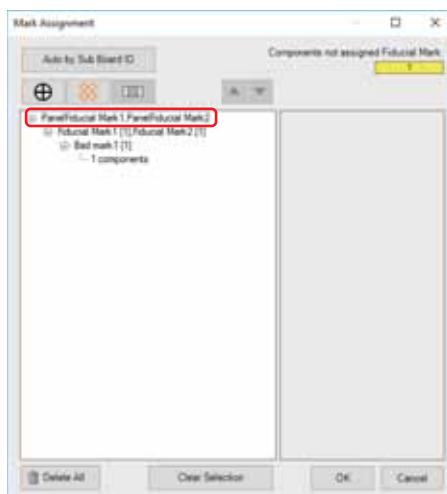


Figure 1-135 Manual Assignment of Mark 7

Step7: Assigning marks for sub board 2 is completed, click **OK** to close Mark Assignment dialog box.



Figure 1-136 Manual Assignment of Mark 8

1.7.3 Automatic Assignment of Mark

Assign proper mark to each data automatically based on sub board ID of data.

Step1: From **InspectData** tab, select the **Fiducial Mark** group > **Fiducial Mark**.



Figure 1-137 Automatic Assignment of Mark 1

Step2: The dialog box shown below appears.

Click **Auto by Sub Board ID**.

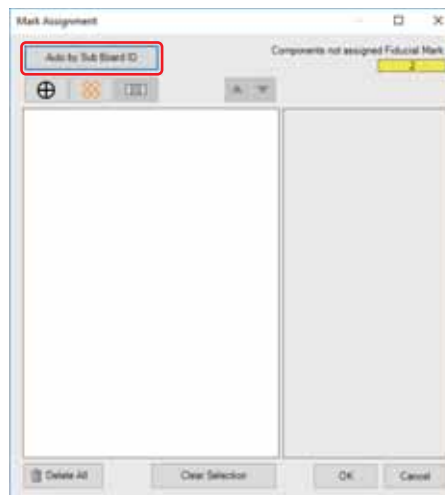


Figure 1-138 Automatic Assignment of Mark 2

Step3: The dialog box shown below appears and marks are assigned to all data automatically.

Click **OK** to close dialog box.

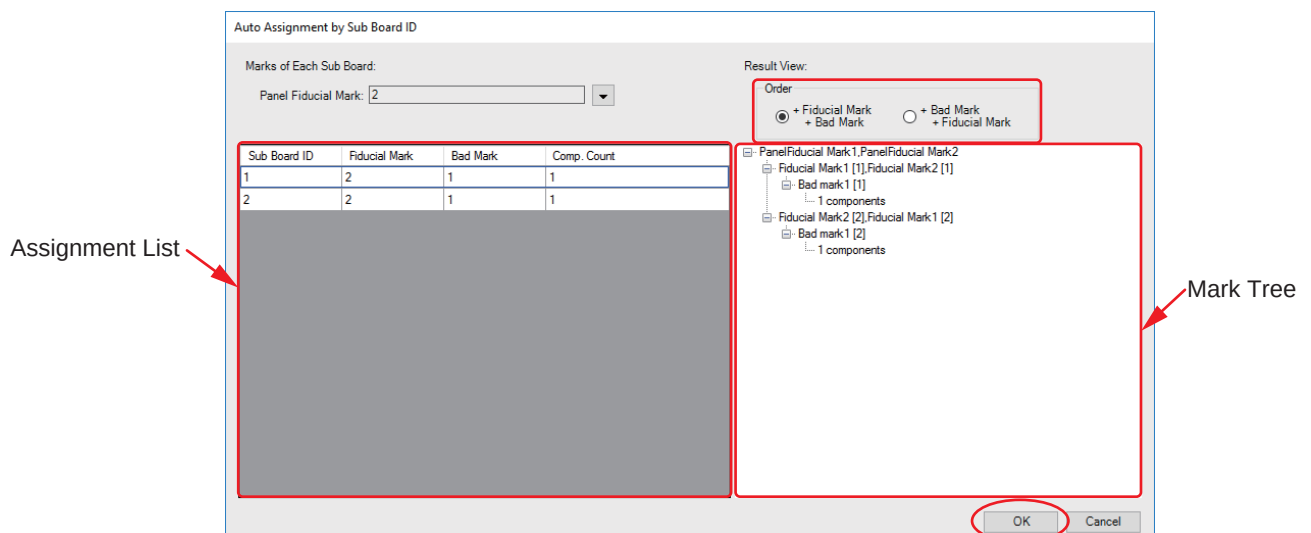


Figure 1-139 Automatic Assignment of Mark 3

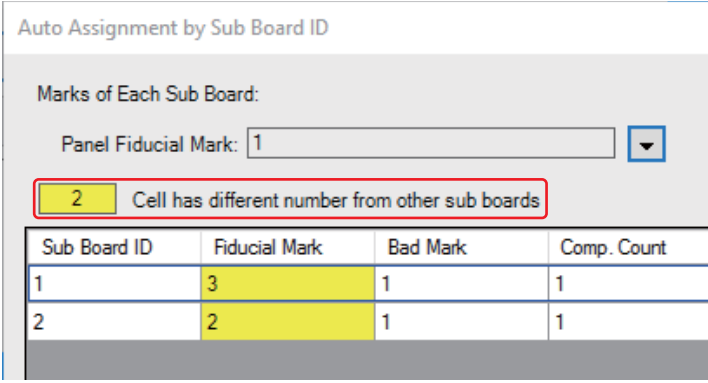
Item	Description
Cell has different number from other sub boards	Appears when mark assignment is not completed normally. CAUTION Mark assignment is not completed normally if any mark is not existing in sub board, or mark setting has any mismatch. For details, refer to 1.7.4 Irregular Case.  Figure 1-140 Automatic Assignment of Mark 4
Panel Fiducial Mark	The number of panel fiducial mark is displayed.
Order	Specifies inspection order for fiducial mark and bad mark.
Assignment List	The numbers of fiducial mark, bad mark, and data for each sub board are displayed. When the number of mark on a sub board and on other sub board are different, corresponding cell is highlighted in yellow.
Mark Tree	Automatic mark assignment result is displayed.

Table 1-19 Descriptions

1.7.4 Irregular Case

When Mark is not Existing on Sub Board

When mark is not existing on sub board, or mark setting is not correct, automatic assignment of mark is not completed normally.

When the number of mark on a sub board is different from the one on other sub board, the corresponding cell is highlighted in yellow. And the number of yellow cell is shown in red frame as below.

Check if mark is not deleted accidentally, or mark is set to sub board ID properly.

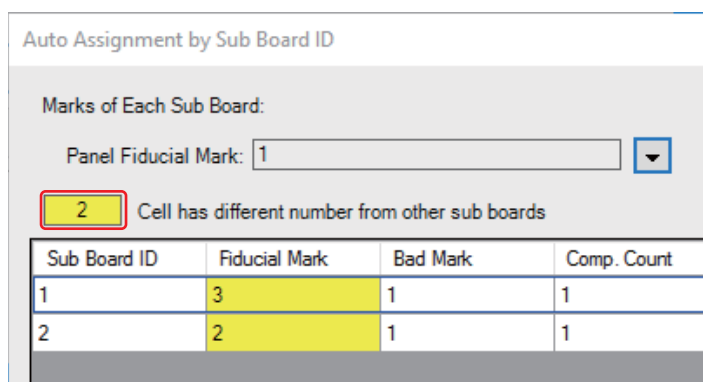


Figure 1-141 Irregular Case 1

When the Sub Board ID for Panel Fiducial Mark and the Sub Board ID for Normal Fiducial Mark are Duplicated

When the sub board ID for panel fiducial mark and the sub board ID for normal fiducial mark are duplicated, automatic assignment of mark is not completed normally.

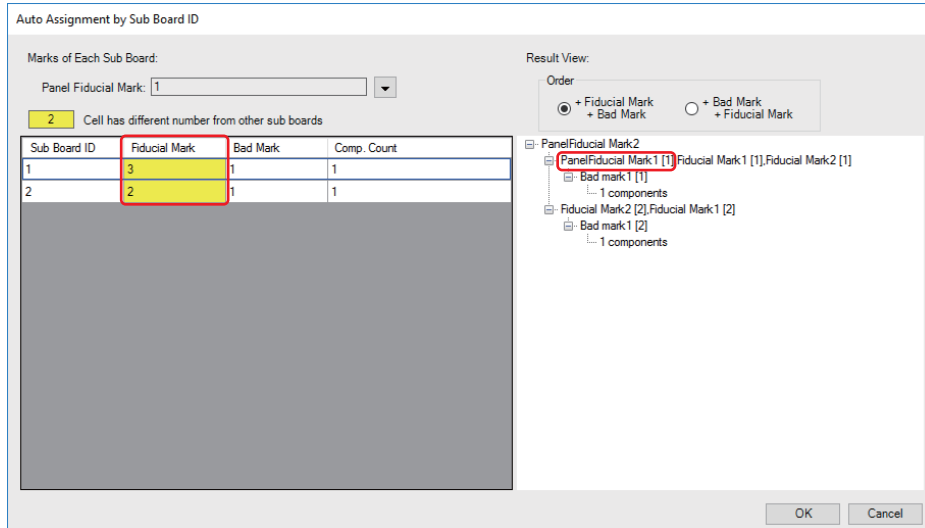


Figure 1-142 Irregular Case 2

The sub board ID of panel fiducial mark should be different from the sub board ID of other fiducial mark.

But panel fiducial mark setting is changed without changing sub board ID by clicking  in right side of panel fiducial mark.

The list of fiducial mark is displayed. Check panel fiducial mark and click **OK**.

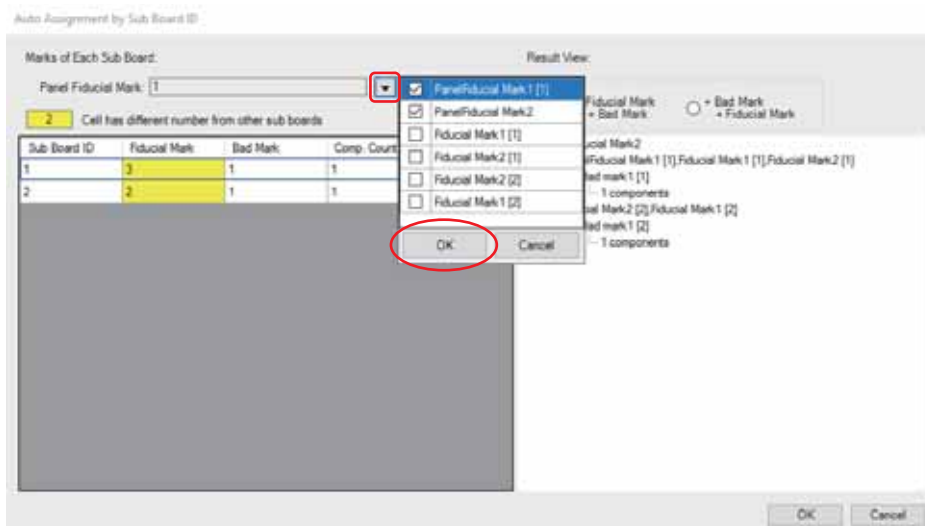


Figure 1-143 Irregular Case 3

The panel fiducial mark is assigned correctly.

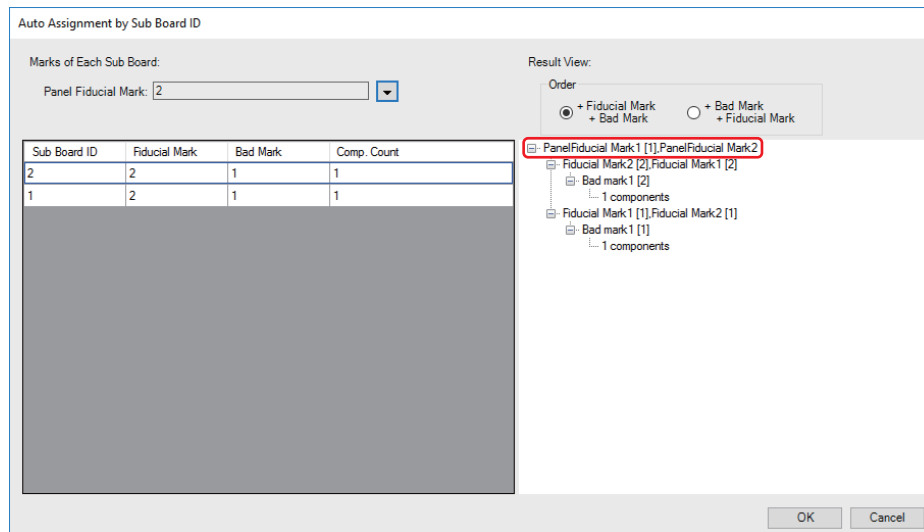


Figure 1-144 Irregular Case 4

1.7.5 Makes Fiducial Mark Adjustment

If data which can be set as a fiducial mark is not present, data needs to be manually added as a fiducial mark based on the captured image of the actual Board.

A gap with the original fiducial mark location occurs when data is manually added as a fiducial mark. Use this function to correct the gap described above.

Step1: Create a new fiducial mark at the position of the fiducial mark on the image according to **1.7.1 Create Marks**.

Step2: From **InspectData** tab, select the **Fiducial Mark** group > **Fiducial Mark** > **Adjust Mark Position**.



Figure 1-145 Adjust Mark Position

Step3: The dialog box shown below appears.
 Drag the green-colored window to match the center position of the fiducial mark.
 After all the setting is completed, click **Next**.

NOTE Scroll the mouse to zoom in/out the screen.



Figure 1-146 Adjusting Mark Position 1

Step4: The dialog box shown below appears.
 The data closest to the fiducial mark set in Step3 is displayed on the screen.
 Adjust the element size by mouse-dragging its corner so that it matches to the data image.
 Click **Other Candidate Component** if the shape of the data appears distorted and does not match the center position.
 The data which is the second closest to the fiducial mark set in Step3 is displayed.
 Clicking **Other Candidate Component** displays a maximum of ten candidates before returning to the first candidate. Click **Back** to return to Step3.
 After all the setting is completed, click **Next**.

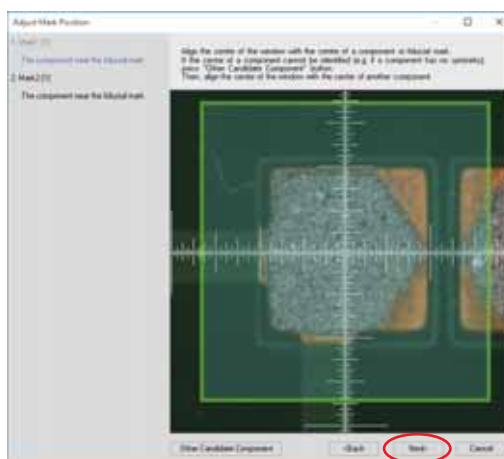


Figure 1-147 Adjusting Mark Position 2

Step5: Repeat Step3 and Step4 as per the number of fiducial marks.

The dialog box shown below appears after completing the setting. Click **OK**.

Fiducial mark moves to appropriate coordinate. After all the setting is completed, set the fiducial mark according to **1.7.2 Manual Assignment of Mark** and **1.7.3 Automatic Assignment of Mark**.

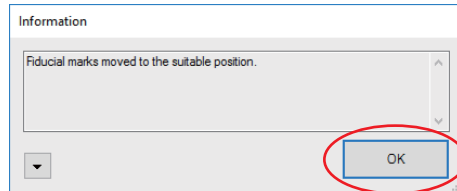


Figure 1-148 Adjusting Mark Position 3

1.8 Sub Board Data Setting

This section describes how to make a sub board data.

NOTE For inspection data creation, applying Gerber data of stencil mask is recommended.

Step1: From **InspectData** tab, Select the **Layout** group > **Make Sub-Board**.



Figure 1-149 Make Sub-Board

Step2: The dialog box for setting a sub board appears.

Click **Pad selection mode**.

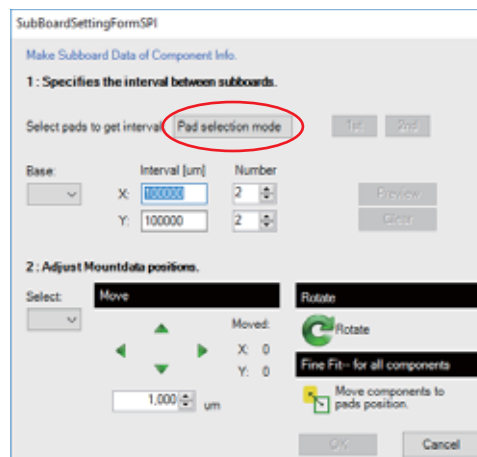


Figure 1-150 Sub Board Setting

Step3: Select a pad in reference sub board by dragging diagonally across the pad as below.
After selecting a pad, click **1st**.

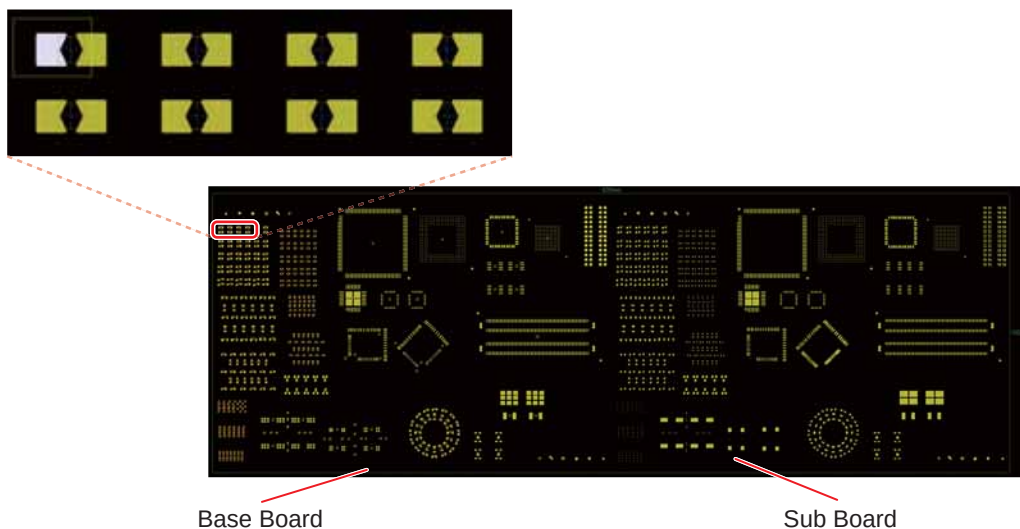


Figure 1-151 Pad Select 1

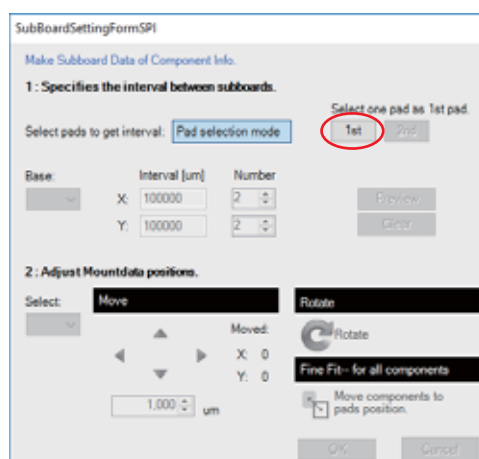


Figure 1-152 Registration of 1st Pad

- Step4: Select a pad which is corresponding to selected pad in Step3 on sub board to expand data.
 After selecting a pad, click **2nd**.
 The number of **Interval** is input automatically.

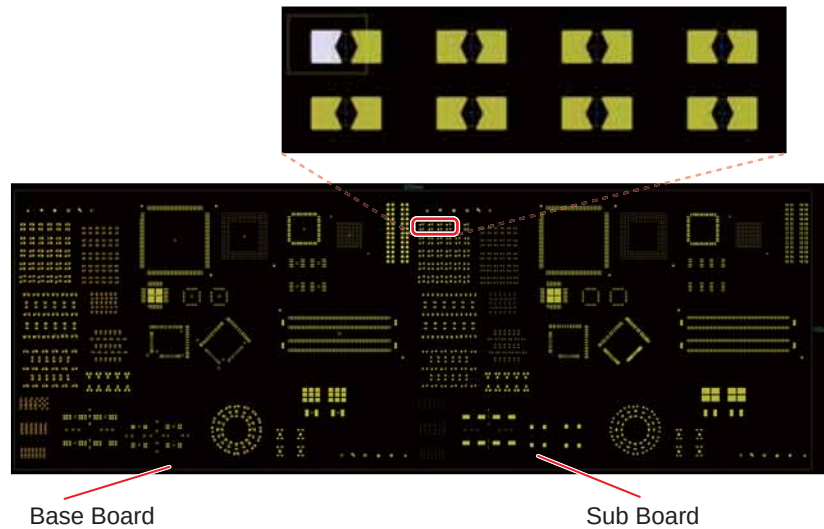


Figure 1-153 Pad Select 2

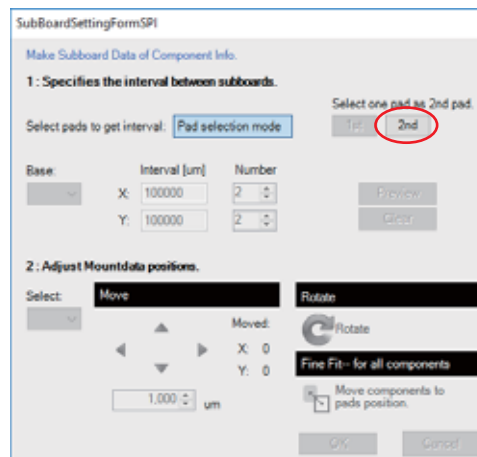


Figure 1-154 Registration of 2nd Pad

Step5: Select sub board ID from **Base** drop-down list, and click **Preview**.
Sub board data is displayed on **Board Viewer** as below.

NOTE Displayed sub board data can be cleared by clicking **Clear**.

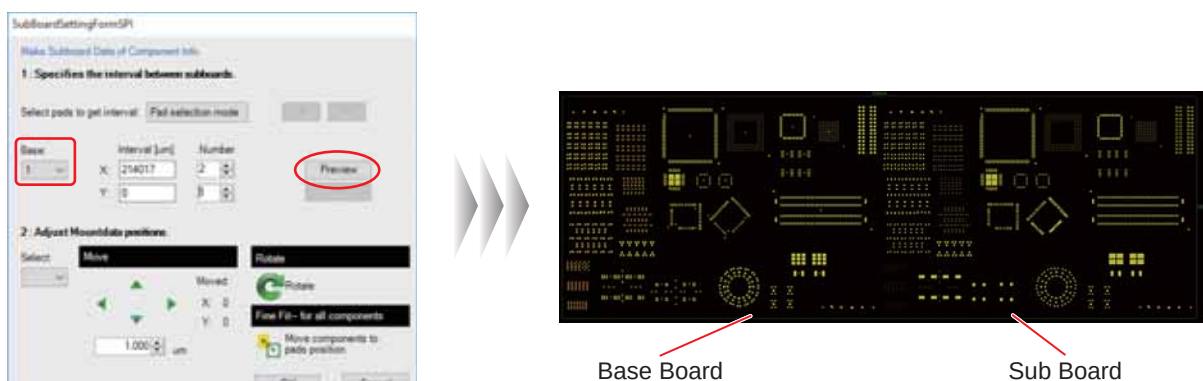


Figure 1-155 Displaying sub board mount data

Step6: Select a sub board ID to adjust position from **Select** drop-down list.
Adjust position according to Table1-20.

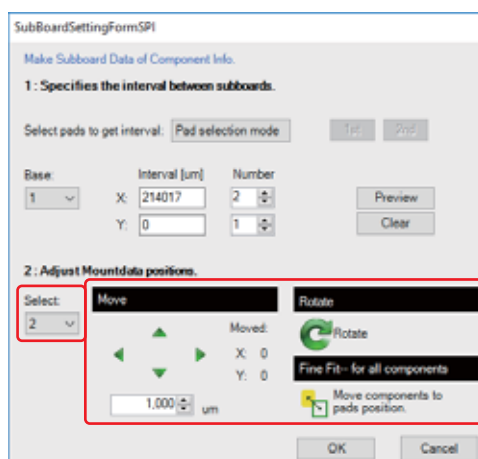


Figure 1-156 Adjustment of sub board mount data

Item		Description
Specifies the interval between subboards.	Pad selection mode	When pad of sub board is existing the gap between sub boards is acquired from pad.
	Base	Specifies a sub board number to be copied.
	Interval	X Specifies the X-axis interval between sub boards.
		Y Specifies the Y-axis interval between sub boards.
	Number	X Specifies a number of sub boards in the X-axis.
		Y Specifies a number of sub boards in the Y-axis.
	Preview	Displays the preview of a setting.
	Clear	Reset the preview screen.
Adjust Mountdata positions.	Select	Displays sub board ID to be expanded. Select sub board ID to be adjusted position.
		Moves the selected sub board to the specific direction.(*)
		Rotates the selected sub board 90 degrees.(*)
	Move components to pads position.	Adjusts data to the center of gerber data.

Table 1-20 Descriptions

CAUTION

(*) Before move or rotate mount data, click **Preview** to display sub board setting result. And select target sub board.

1.9 Coplanarity Inspection

1.9.1 Overview of Coplanarity Inspection

The coplanarity inspection defines multiple solder paste as a group.

And the differences of sample value between the maximum value and the minimum value for height, volume, and area in the group is inspected.

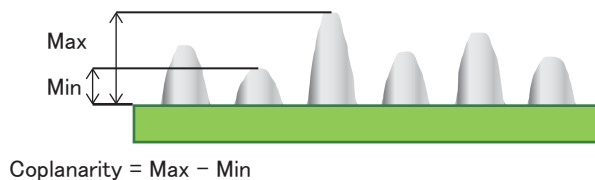


Figure 1-157 Coplanarity Inspection 1

1.9.2 Setting Coplanarity Inspection

Here describes how to make the setting for coplanarity inspection.

Adding Coplanarity Inspection Data

Step1: Select solder paste data to perform coplanarity inspection from **Board Viewer** or **LookUp Table Window**.

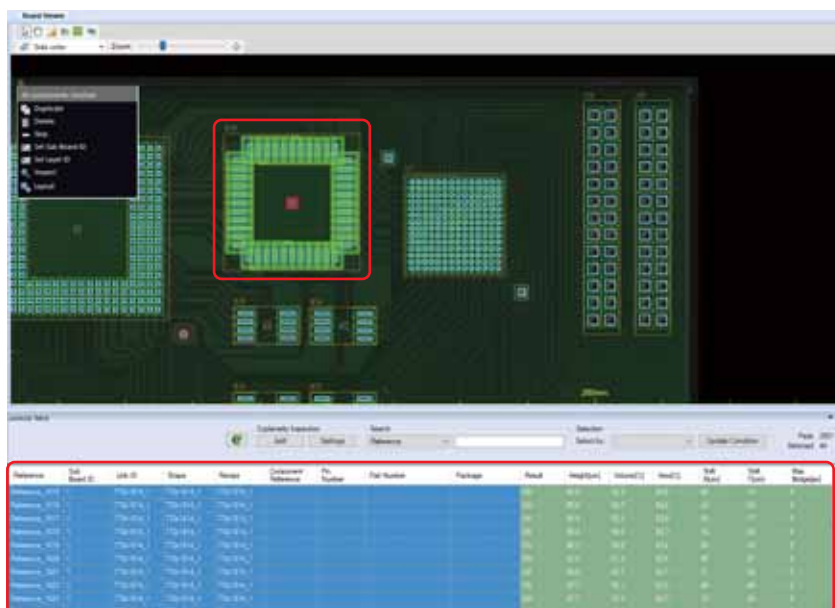


Figure 1-158 Coplanarity Inspection 2

Step2: From **LookUp Table** window, Click **Coplanarity Inspection > Add**.



Figure 1-159 Coplanarity Inspection 3

Step3: The dialog box shown below appears. Make the settings according to Table 1-21.
After all settings are completed, click **Add Coplanarity**.

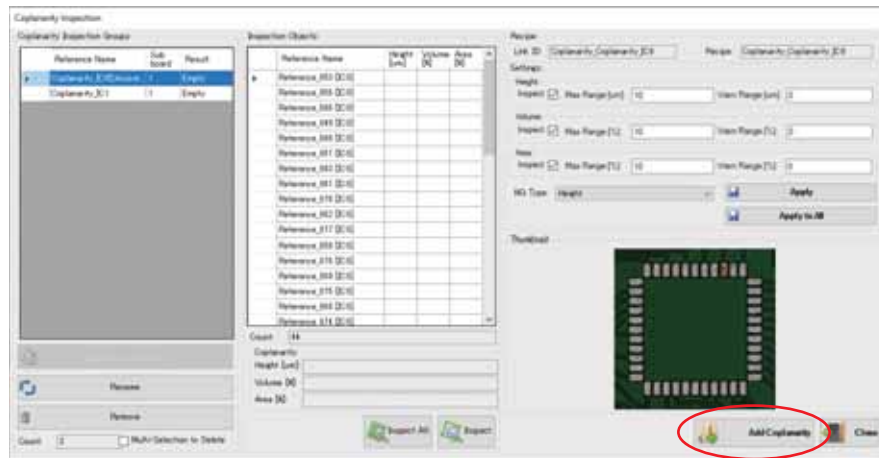


Figure 1-160 Coplanarity Inspection 4

Item	Description
Coplanarity Inspection Groups	Displays coplanarity inspection groups of the current inspection data. The below items are displays in the table. - Reference Name: name of coplanarity inspection - Sub board: number of sub board - Result: inspection result
Copy To Other Blocks	If there is information of component, copy the coplanarity inspection settings to corresponding data of another sub board.
Rename	The rename dialog box appears. "Coplanarity_" is automatically add before the entered name.
Remove	Delete the selected coplanarity inspection group.
Multi-Selection to Delete	Check this item if delete multiple coplanarity inspection groups. Multiple items can be selected while pressing (Ctrl).
Inspection Objects	Displays the contents of the selected coplanarity inspection group. The contents are displayed in height [um], volume [%], area [%].
Inspect All	Performs all the coplanarity inspection. Inspection result is displayed in Inspection Objects .
Inspect	Performs the selected coplanarity inspection. Inspection result is displayed in Inspection Objects .
Height	Sets the maximum range and warning range of the coplanarity inspection at the height. If unchecked Inspect , the coplanarity inspection at the height will be skipped.
Volume	Sets the maximum range and warning range of the coplanarity inspection at the volume. If unchecked Inspect , the coplanarity inspection at the volume will be skipped.
Area	Sets the maximum range and warning range of the coplanarity inspection at the area. If unchecked Inspect , the coplanarity inspection at the area will be skipped.
NG Type	Sets the NG type.
Apply	Saves edit contents of the selected coplanarity inspection.
Apply to All	Saves edit contents of the all coplanarity inspection.
Add Coplanarity	Adds the new coplanarity inspection data.
Close	Closes the dialog box. The edit contents will not be saved before Apply or Apply to All are not clicked.

Table 1-21 Descriptions

Editing Coplanarity Inspection Data

Step1: From **LookUp Table** window, click **Coplanarity Inspection > Settings**.



Figure 1-161 Coplanarity Inspection 5

Step2: The dialog box shown below appears. Edit coplanarity inspection data according to Table 1-21. After all editings are completed, click **Close** to close the dialog box.

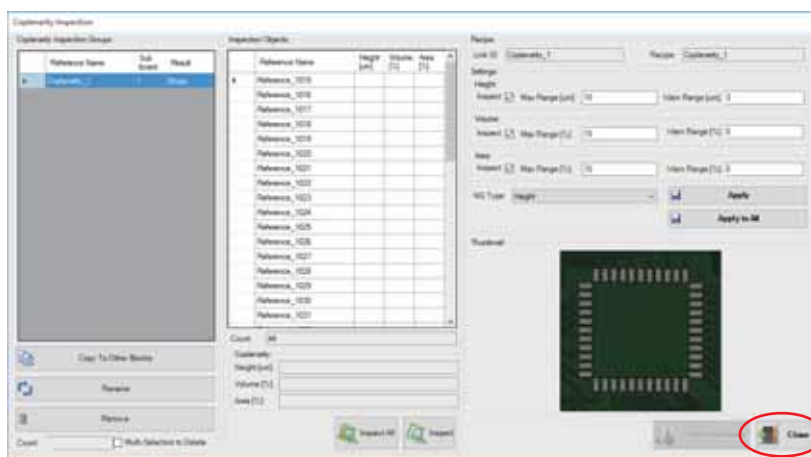


Figure 1-162 Coplanarity Inspection 6

1.9.3 Inspection Result of Coplanarity Inspection

Coplanarity inspection is performed as same as normal solder paste inspection.

In Manual Inspection

The result of the coplanarity inspection can be confirmed on **Coplanarity Inspection** dialog box (Figure 1-163) and **Component List** of the **InspectData** tab (Figure 1-164).

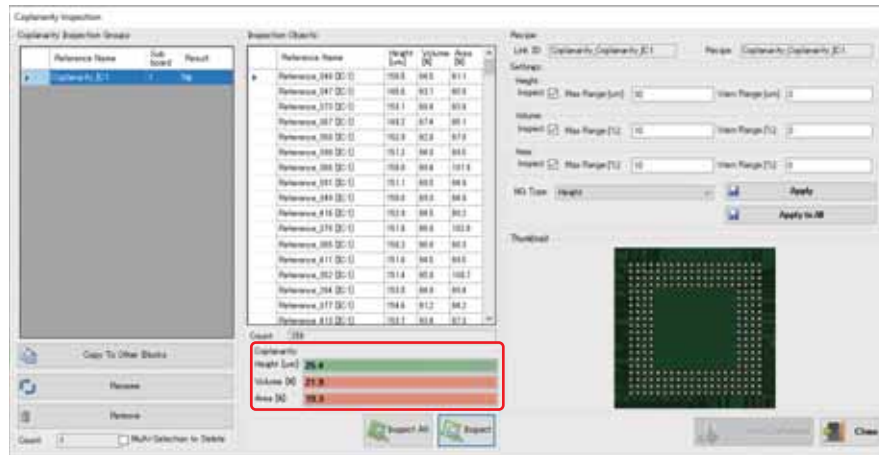


Figure 1-163 Coplanarity Inspection 7



Figure 1-164 Coplanarity Inspection 8

In Auto Mode

The result of the coplanarity inspection is displayed on **Sequence State Viewer** (Figure 1-165). Also, it can be judged on the judgement window of BF2-Monitor (Figure 1-166).

NOTE

The coplanarity inspection data is not displayed on the Edit Recipe Screen. So that coplanarity inspection data can not be used in the Debug mode.



Figure 1-165 Coplanarity Inspection 9



Figure 1-166 Coplanarity Inspection 10

2 Edit Data

2.1 Data Structure

An inspection data of this product is mainly classified into three data.

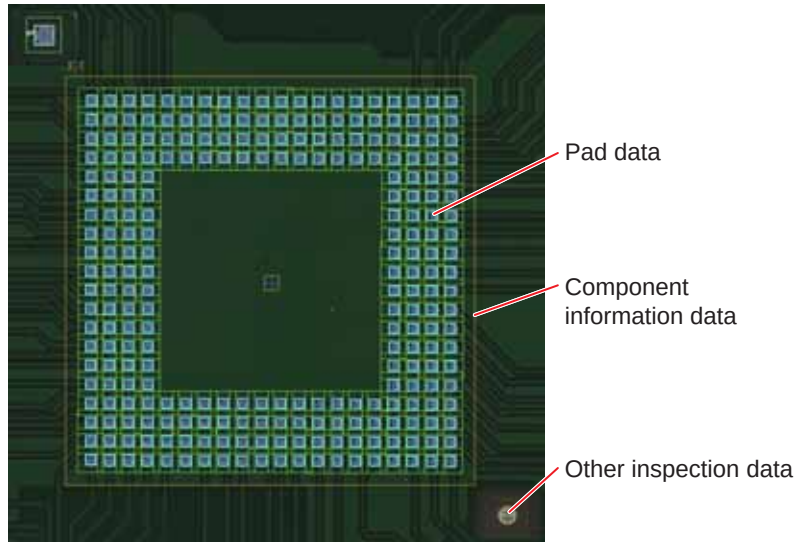


Figure 2-1 Inspection Data Structure

Item	Description
Pad data	Pad data is used in normal inspection such as solder paste inspection. Pad data includes Shape and Recipe. For details of Shape and Recipe, refer to 2.5.1 Overview of Shape and 2.8.1 Overview of Recipe .
Component information data	Displays the component information based on mounter data. Not available in inspection.
Other inspection data	Other inspection data is used for fiducial mark adjustment or barcode inspection.

Table 2-1 Descriptions

2.2 List Display of Data (Pad Data)

To display the **LookUp Table window**, select the **InspectData** tab > **dock window** > **LookUp Table**.

Pad data, inspection results and sample values are displayed on **LookUp Table**.

On **LookUp Table**, pad data searching, editing parameter and setting coplanarity inspection are available.

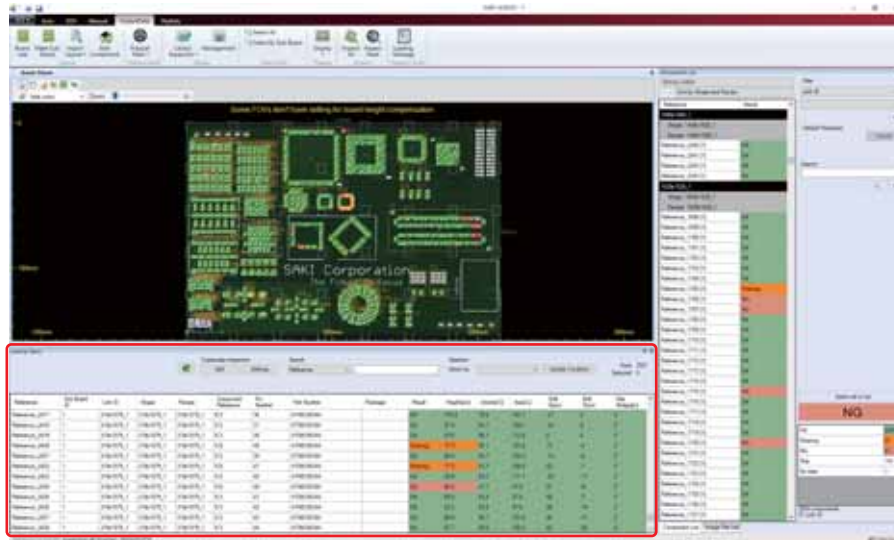


Figure 2-2 LookUp Table Window

2.2.1 Overview of LookUp Table

Here describes the LookUp Table.

NOTE

The color of the inspection result and sample value column in the LookUp Table will be changed depending on the latest inspection result (OK : green, Warning : orange, NG : red).

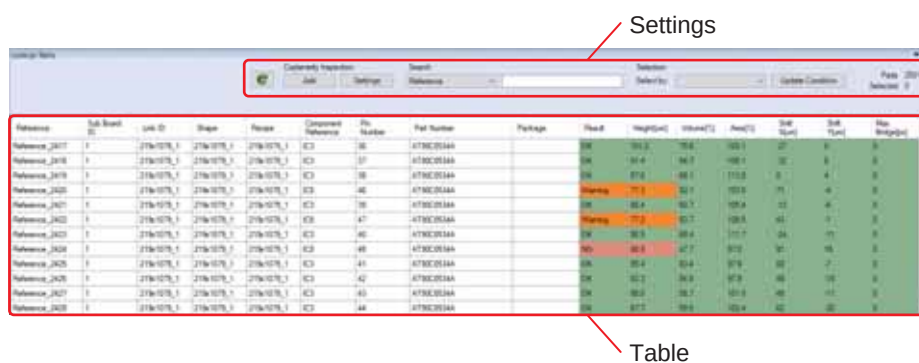


Figure 2-3 LookUp Table

Item	Description
Settings	Settings for searching pad data, applying parameter and setting coplanarity inspection. For applying parameter, refer to 4. Parameter Editor . For coplanarity inspection, refer to 1.9 Coplanarity Inspection .
Table	Displays pad data. For details, refer to Table 2-3.

Table 2-2 Descriptions

Item	Description
Reference	Displays the reference name of pad data.
Sub Board ID	Displays the sub board ID of pad data.
Link ID	Displays the Link ID of pad data.
Shape	Displays the shape name of pad data.
Recipe	Displays the recipe name of pad data.
Component Reference	Displays the reference name of component information (mount data) corresponding to the pad data.
Pin Number	Displays the pin number of pad data.
Part Number	Displays the component ID (link ID) of component information (mount data).
Package	Displays the package type (Component Common Name) of component information (mount data).
Result	Displays the latest inspection result.
Height[μm]	Displays the height value (sample value of Solder Paste Inspection algorithm) in the latest inspection.
Volume[%]	Displays the volume value (sample value of Solder Paste Inspection algorithm) in the latest inspection.
Area[%]	Displays the area value (sample value of Solder Paste Inspection algorithm) in the latest inspection.
Shift X[μm]	Displays the shift value of X-axis (sample value of Solder Paste Inspection algorithm) in the latest inspection.
Shift Y[μm]	Displays the shift value of Y-axis (sample value of Solder Paste Inspection algorithm) in the latest inspection.
Max Bridge[px]	Displays the Maximum value of Left , Right , Top , and Bottom (sample value of Solder Paste Inspection algorithm) in the latest inspection.

Table 2-3 Displays item on the LookUp Table

2.2.2 Selecting a Data (LookUp Table)

Click a specific data row on **LookUp Table** to select the corresponding pad data.

By selecting item from the **Selection > Select by**, multiple pad data with the same information is selected. For details, refer to Table 2-4.

NOTE Double click on the data row to display the Recipe editing screen.

NOTE Algorithm parameters can be updated for data selected in the LookUp Table. For details, refer to **4. Parameter Editor**.

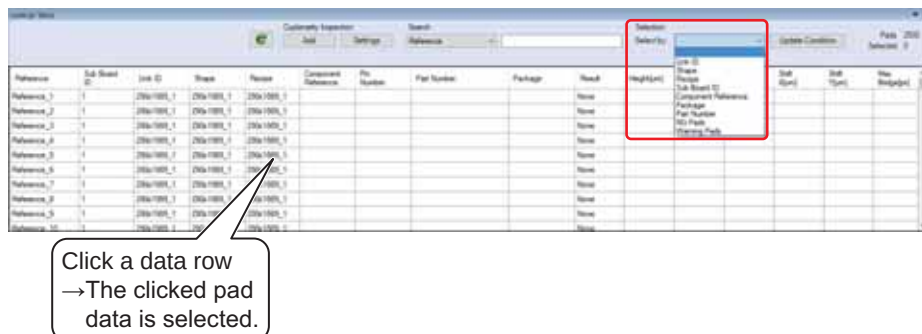


Figure 2-4 Selecting a Data (LookUp Table)

Item	Description
Link ID	Data with the same link ID as the selected pad data is selected.
Shape	Data with the same shape name as the selected pad data is selected.
Recipe	Data with the same recipe name as the selected pad data is selected.
Sub Board ID	Data with the same sub board ID as the selected pad data is selected.
Component Reference	Data with the same component reference as the selected pad data is selected.
Package	Data with the same package name (component common name of Mount data.) as the selected pad data is selected.
Part Number	Data with the same component ID (link ID of mount data) as the selected pad data is selected.
NG Pads	All NG judged pad data is selected.
Warning Pads	All Warning judged pad data is selected.

Table 2-4 Descriptions

2.2.3 Searching a Data (LookUp Table)

Pad data can be searched in the **LookUp Table**.

Select item from **Search** and enter a word to search in the text box. For details, refer to Table 2-5.

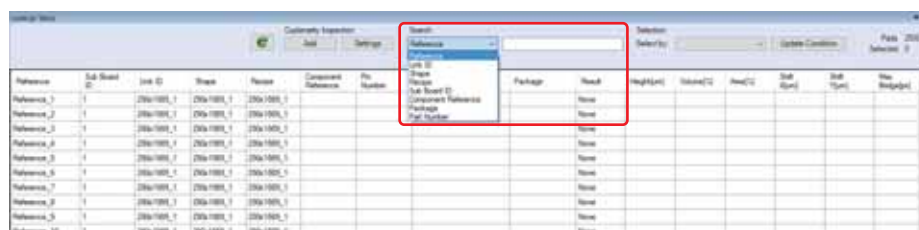


Figure 2-5 Searching a Data (LookUp Table)

Item	Description
Reference	Searches reference name of pad data.
Link ID	Searches link ID of pad data.
Shape	Searches shape name of pad data.
Recipe	Searches recipe name of pad data.
Sub Board ID	Searches sub board ID of pad data.
Component Reference	Searches reference name of Mount data (Component information) to which the pad data belongs.
Package	Searches package name (component common name of Mount data.) of Mount data (Component information) to which the pad data belongs.
Part Number	Searches part number (link ID of mount data) of Mount data (Component information) to which the pad data belongs.

Table 2-5 Descriptions

2.3 List Display of Data (All Data)

To display the **Component List** window, select the **InspectData** tab > **dock window** > **Component List**. All data of inspection target (pad data and other inspection data) and inspection results are displayed on **Component List**. A specific data can also be searched from this window.



Figure 2-6 Component List

2.3.1 Selecting a Data (Component List)

When a specific data row is clicked on **Component List**, the corresponding data is selected. When another row of **Component List** is clicked, all data which share the clicked Link ID are selected. When an inspection result is clicked, a data corresponding to clicked inspection result (OK/NG/Warning/Skip/None) is selected in turn.

NOTE

Double click on the data row to display the Edit recipe screen.

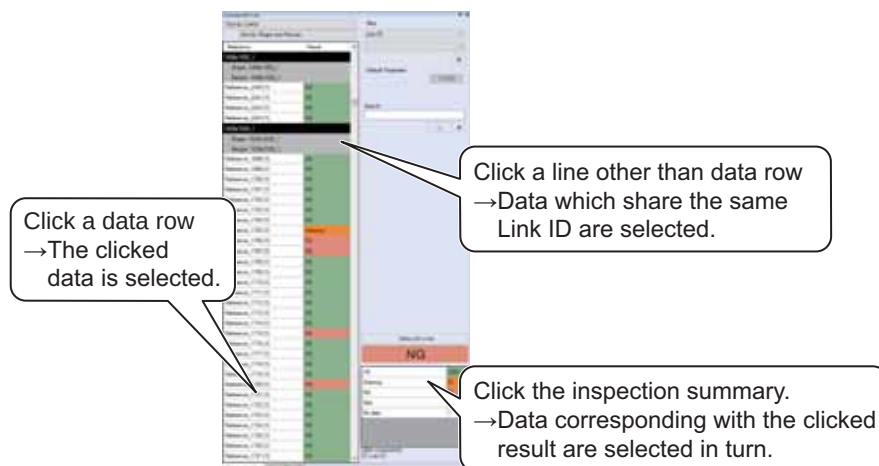


Figure 2-7 Selecting a Data (Component List)

2.3.2 Sorting Data

Sorts data in accordance with the specified conditions.



Figure 2-8 Sorting Data

Item	Description
Sort by Link ID	Sorts data in ascending order based on Link ID.
Sort by Shape and Recipe	Sorts data in ascending order based on Shape names. If Shape names are the same, sorts based on Recipe names. If Shape names are not set, data are not sorted and displayed on the top of the list.

Table 2-6 Descriptions

2.3.3 Filtering Data

The filtering function displays only the data match to the specified conditions.

Step1: Select an item from **Filter** drop-down list on the upper right of **Component List**.

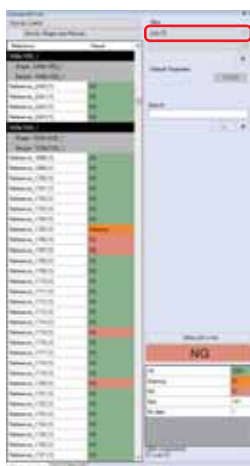


Figure 2-9 Filtering Data 1

Step2: Select an item from the lower drop-down list of **Filter**.
Only the data matched to the specified condition are displayed on **Component List**.

NOTE

Click  button to reset the filtering conditions.

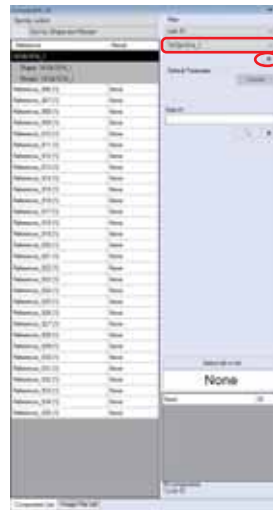


Figure 2-10 Filtering Data 2

2.3.4 Searching a Data

Enter a word to the text-box under **Search** and click . Data which include the word in their names are searched and highlighted in yellow.

CAUTION

Text-box is case-sensitive.



Figure 2-11 Searching a Data

2.4 Editing a Data

2.4.1 Selecting a Data (Board Viewer)

Here describes how to select a data from **Board Viewer**.

Step1: Select the **InspectData** tab.



Figure 2-12 Inspection Data 1

Step2: **To Select a Specific Data**

On **Board Viewer**, click a data or drag the mouse pointer to surround a data.
The selected data is displayed with thick green lines.

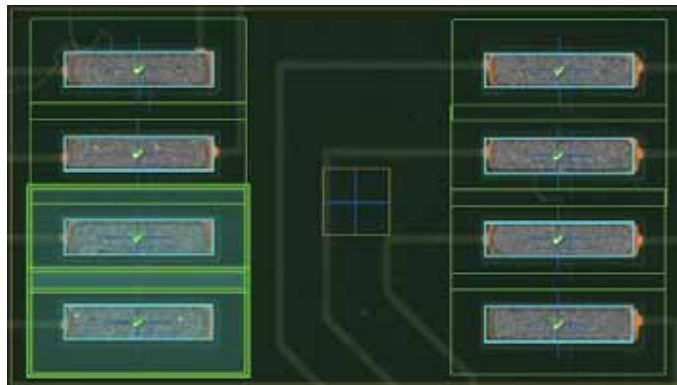


Figure 2-13 Selecting a Specific Data

To Select All Data

From the ribbon, select the **Select Tool** group > **Select Tool** > **Select all**.

NOTE The same operation can also be performed by pressing **Ctrl** + **A**.

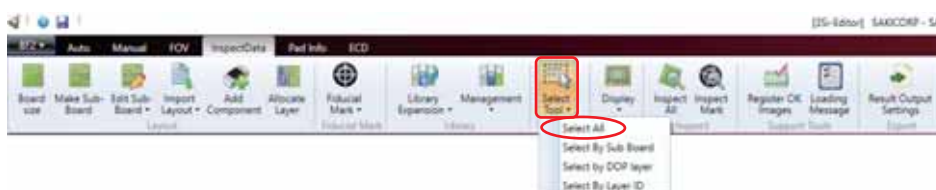


Figure 2-14 Inspection Data 2

To Select All Data in the Selected Sub Board

From the ribbon, select the **Select Tool** group > **Select Tool** > **Select By Sub Board**.
Enter a number into **Sub Board ID** and click **OK**.

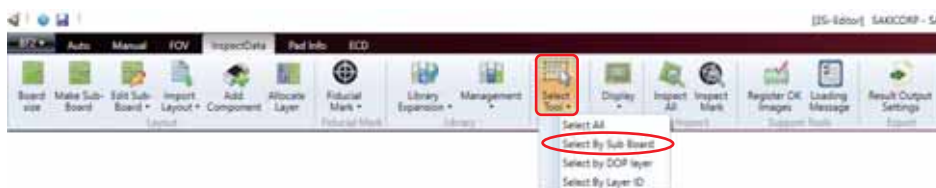


Figure 2-15 Selecting All Data in the Selected Sub Board 1

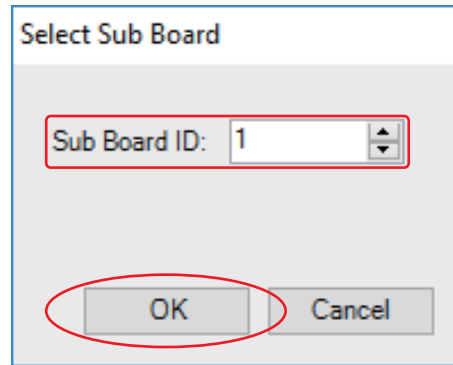


Figure 2-16 Selecting All Data in the Selected Sub Board 2

2.4.2 Adding a Data

Here describes how to add data manually.

Step1: Select the **InspectData** tab.



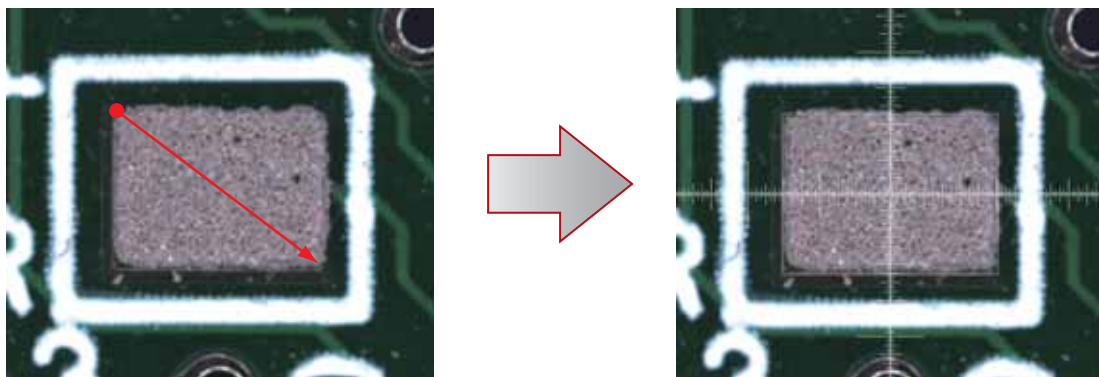
Figure 2-17 InspectData

Step2: From the ribbon, select the **Layout** group > **Add Component**.



Figure 2-18 Adding a Data 1

Step3: Drag the mouse on where a data to be added.
A data is added to the center of the white cross line.



Drag the mouse from a corner
to the opposite corner of the pattern.

A data is added to
the center of the pad.

Figure 2-19 Adding a Data 2

2.4.3 Skip Setting for Data

Here describes how to make the skip setting for the specified data.

Step1: Select a data according to **2.3.1 Selecting a Data (Component List)** or **2.4.1 Selecting a Data (Board Viewer)**.

Step2: **For a Single Data**

Click  on the upper left side of the edit panel.

For Multiple Data

Select **Skip** on the edit panel.

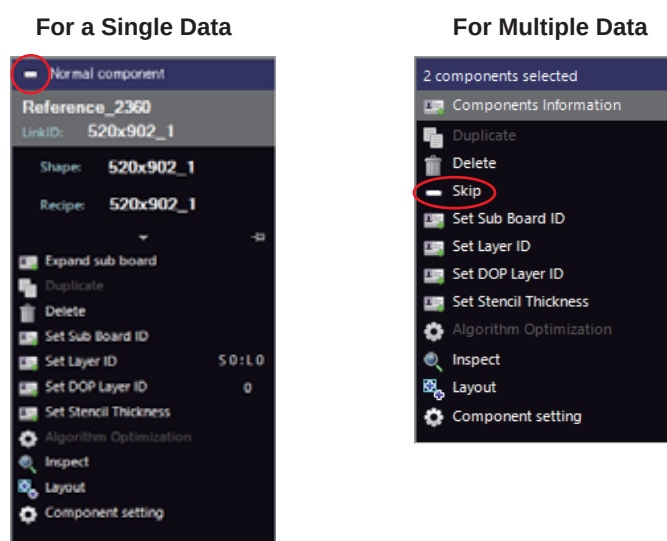


Figure 2-20 Skip Setting for Data

2.4.4 Deleting a Data

Here describes how to delete a data.

Step1: Select a data according to **2.3.1 Selecting a Data (Component List)** or **2.4.1 Selecting a Data (Board Viewer)**.

Step2: Select **Delete** on the edit panel.

CAUTION The panel display may differ according to if single data is selected or multiple data are selected.

NOTE The same operation can also be performed by pressing **[Del]**.

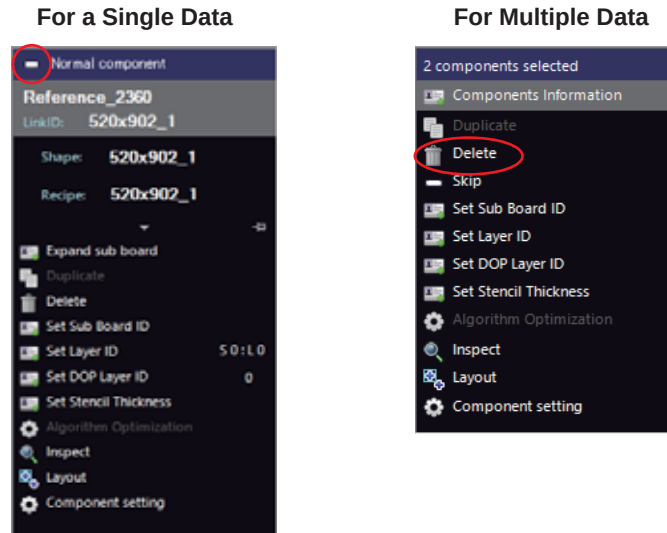


Figure 2-21 Deleting a Data 1

Step3: Click **Yes**.

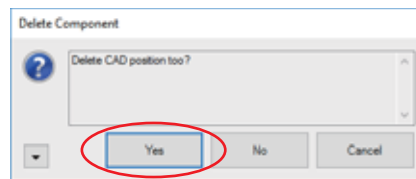


Figure 2-22 Deleting a Data 2

2.4.5 Moving a Data

Step1: Select a data to be moved according to **2.3.1 Selecting a Data (Component List)** or **2.4.1 Selecting a Data (Board Viewer)**.

Step2: **To Move a Data Manually**

Drag the mouse and move the data to the desired point.

The dialog shown below appears. Click .

NOTE To return the data to the original position, click **Cancel**.

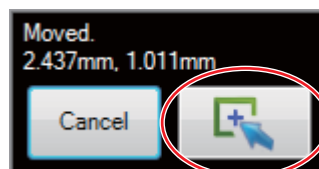


Figure 2-23 Moving a Data Manually

To Move a Data to the Specific Coordinates

Select **Layout** on the edit panel and enter numbers into the text-boxes under **Coordinate**.

CAUTION This function is not available when selecting multiple data.

NOTE To move gerber data at the same time, click **Move gerber with component**.

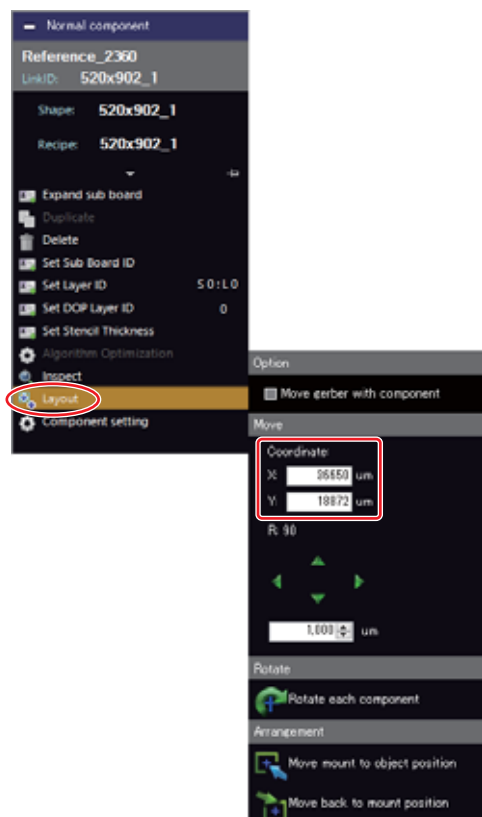
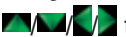


Figure 2-24 Moving a Data to the Specific Coordinates

To Move a Data to the Specific Direction by the Specific Value

Select **Layout** on the edit panel and enter numbers to the text-boxes under .

Click  to move the data.

CAUTION The panel display may differ according to if single data is selected or multiple data are selected.



Figure 2-25 Moving a Data to the Specific Direction by the Specific Value

To Move a Data to the Mount Data Position

Select **Layout** on the edit panel and click **Move mount to object position**.

CAUTION

The panel display may differ according to if single data is selected or multiple data are selected.

NOTE

To move the mount data to the data position, click **Move back to mount position**.

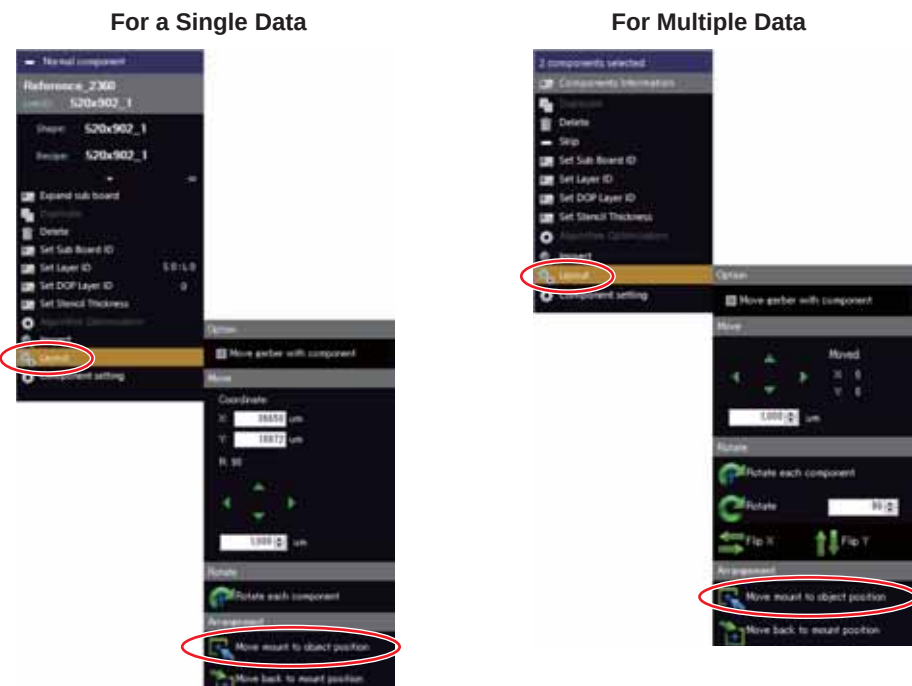


Figure 2-26 To Move a Data to the Mount Data Position

2.4.6 Rotating Data

Step1: Select a data to be rotated according to **2.3.1 Selecting a Data (Component List)** or **2.4.1 Selecting a Data (Board Viewer)**.

Step2: **To Rotate a Single Data**

Select **Layout** > **Rotate each component** on the edit panel and proceed to Step3.

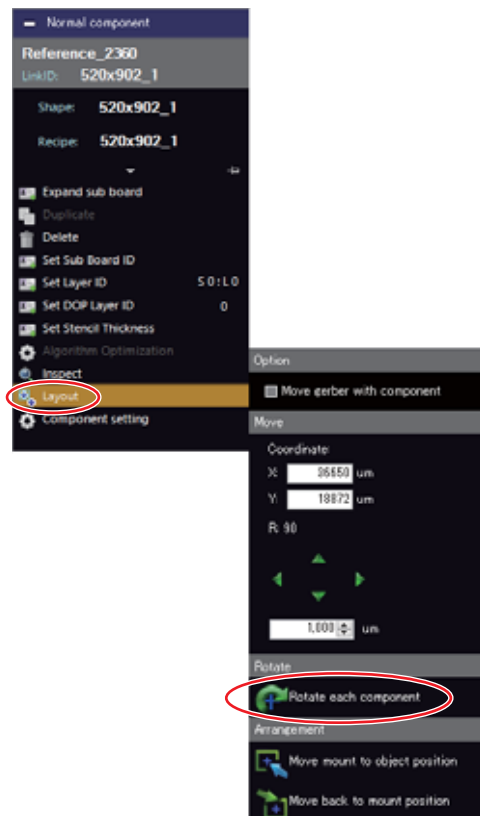


Figure 2-27 Rotating a Single Data

To Rotate Multiple Data 90 Degrees

Select **Layout** > **Rotate**. Data rotate each time the button is clicked.

CAUTION

This function is not available when selecting a single data.

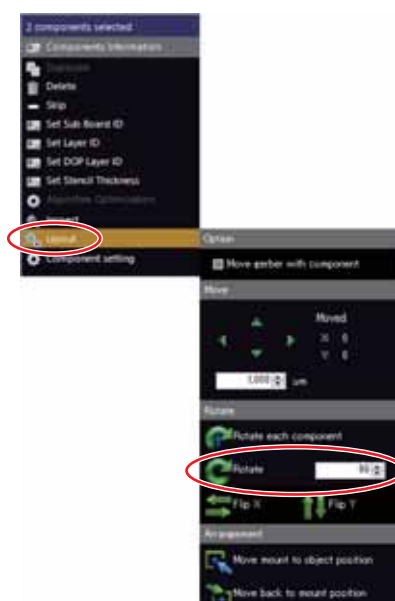


Figure 2-28 Rotating Multiple Data 1

To Rotate Multiple Data by the Specific Degree

Select **Layout > Rotate each component** on the edit panel and proceed to Step3.

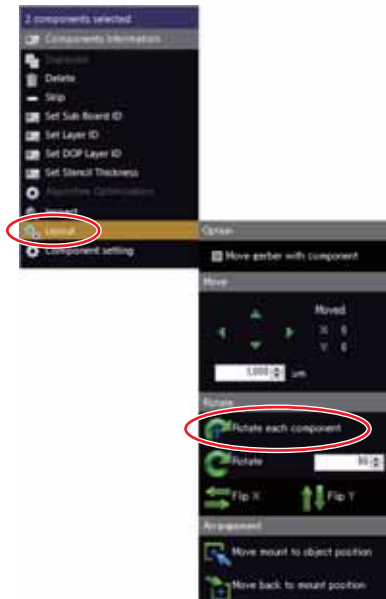


Figure 2-29 Rotating Multiple Data 2

Step3: Make the settings according to Table 2-7 and click **Execute**.

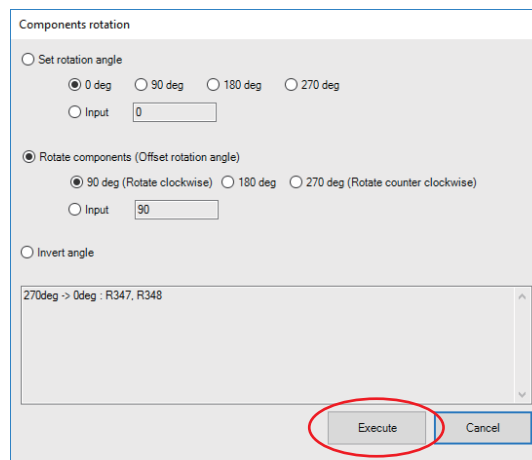


Figure 2-30 Rotation Angle Setting

Item	Description
Set rotation angle	Changes the data angle to the specified value.
Rotate components (Offset rotation angle)	Rotates the data by the specified angle.
Invert angle	Reverses the rotation direction.

Table 2-7 Descriptions

2.4.7 Inverting Data Positions

CAUTION This function is not available when selecting a single data.

Step1: Select two or more data according to **2.3.1 Selecting a Data (Component List)** or **2.4.1 Selecting a Data (Board Viewer)**.

Step2: **To Invert the X-axis of the Data Position**
From the edit panel, select **Layout > Flip X**.

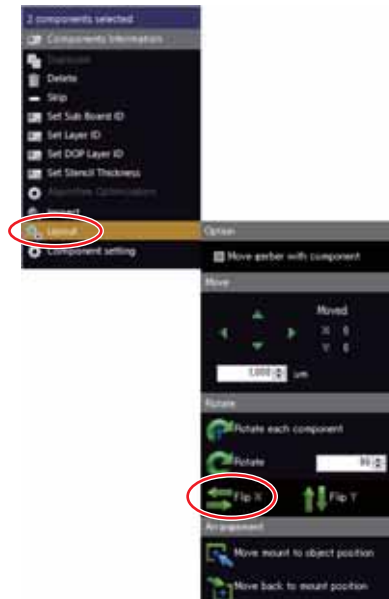


Figure 2-31 Inverting the X-axis of the Data Position

To Invert the Y-axis of the Data Position

From the edit panel, select **Layout > Flip Y**.

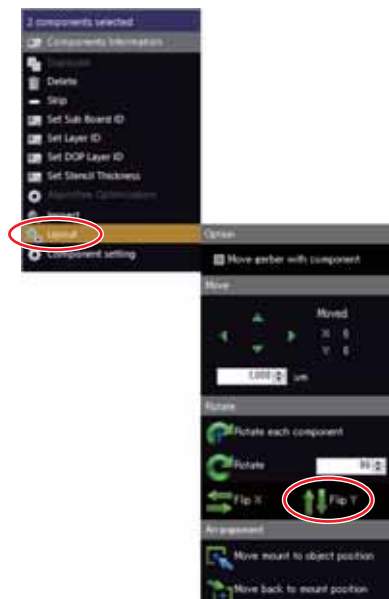


Figure 2-32 Inverting the Y-axis of the Data Position

2.5 Assigning a Shape

This section describes how to assign the shape.

NOTE

This software creates inspection data basically from Gerber data by using inspection data wizard. Also shape is created by automatic. This section describes how to adjust the shape data if necessary.

2.5.1 Overview of Shape

Shape is a shape data of a pad mounted on a board.

A Shape contains data of pad, and each data is called **Element**.

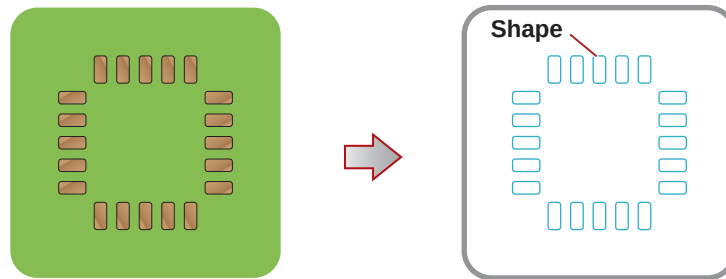


Figure 2-33 Overview of Shape

Each **Element** has an **Element ID**. When one of **Elements** is edited, the same modification also applies to **Elements** which share the same **Element ID**.

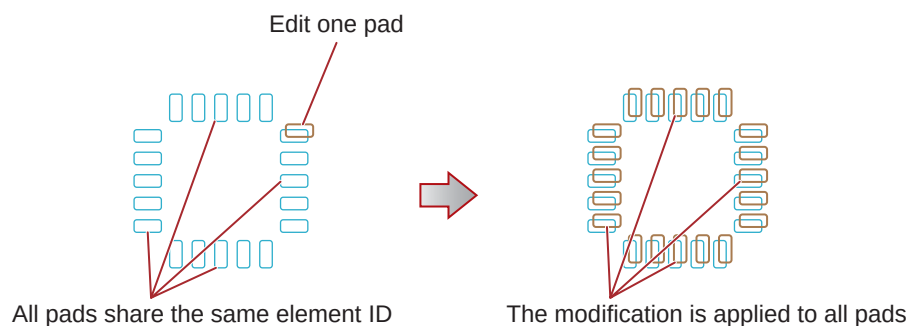


Figure 2-34 Editing a Shape

The following chart shows the operation procedure for assigning a shape by manual.

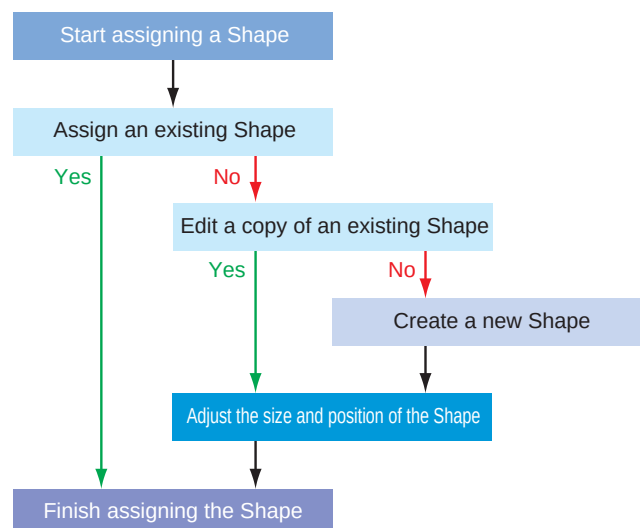


Figure 2-35 Assigning a Shape

2.5.2 Window Details

Here describes and illustrates the shape editing screen.

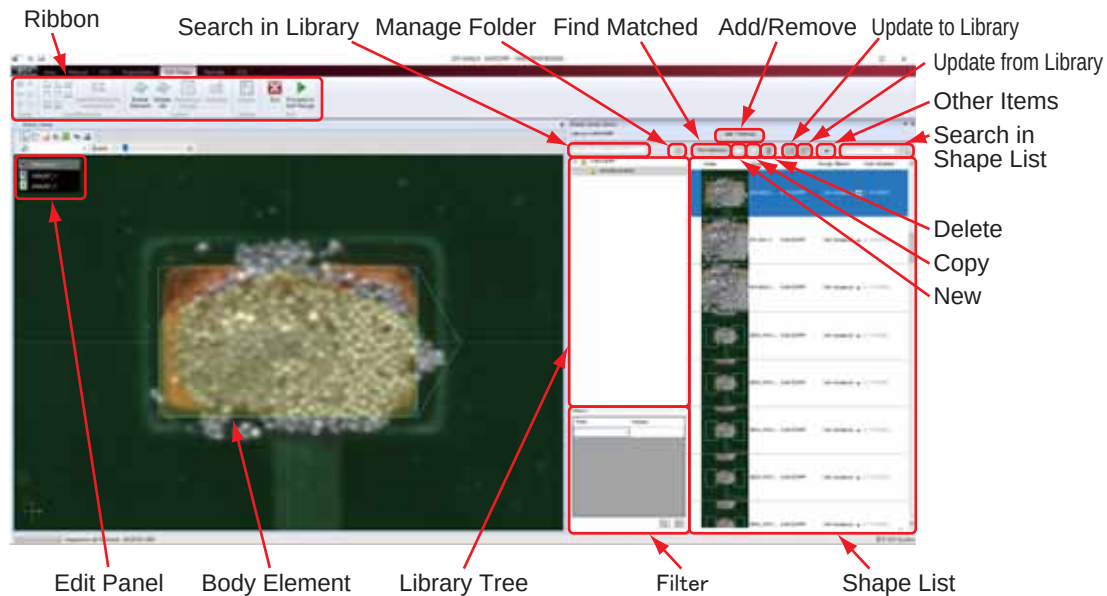


Figure 2-36 Window Details

Item			Description
Body Element			Indicates an element of a component body. The element is also used for adjusting a Shape. For details, refer to 2.6 Adjusting a Shape .
Ribbon	Body		Not available
	Lead/Electrode		
	Layout		
	Option		
	Exit	Exit	Finishes editing a Shape and return to the InspectData tab. For details, refer to 2.7 Exit the Shape Editing Screen .
Proceed to Edit Recipe		Finishes editing a Shape and proceed to the Recipe editing screen. For details, refer to 2.7 Exit the Shape Editing Screen .	
Edit Panel			Used for adjusting a Shape. For details, refer to 2.6 Adjusting a Shape .
Add/Remove			Adds or deletes reference libraries.
Search in Library			Searches a Shape whose name includes the text entered from the selected library.
Find Matched			Not available
New	Blank		Adds a square element as a Shape.
	Shape wizard		Not available
Copy	Copy shape only		Copies the selected Shape.
	Copy recipes together		Copies the selected Shape together with a Recipe.
Delete			Copies the selected Shape.

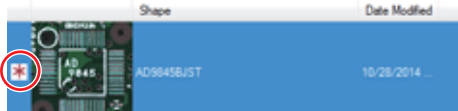
Item			Description
Update to Library			When exit the editing and click Local Save, the changes are saved in only for opening inspection data not a library. In the case of the changes that is not registered with a library,  icon is displays on left side of the shape list.
Update from Library			 Figure 2-37 Local save Only changes that is not registered with a library existing, Update to Library and Update from Library become enable. Click Update to Library to update the library. Click Update from Library to update the only for opening inspection data with library.
Other Items	Rename		Changes the name of the selected Shape.
	Move		Moves the selected Shape to the selected folder.
	View	List	Displays Shape names and updated dates.
		Image List	Displays Shape names, updated dates, and images of Shapes.
Search in Shape List			Searches for Shapes that contain the entered texts in the names from the library.
Library Tree			Displays folders saved in the selected library.
Shape List			Displays Shapes saved in the selected folder.
Manage Folder	New Folder		Adds a new folder to the library tree.
	Delete Folder		Deletes the selected folder from the library tree.

Table 2-8 Descriptions

2.5.3 Opening the Shape Editing Screen

Step1: Select the **InspectData** tab.



Figure 2-38 InspectData

Step2: Select a component data which a Shape to be edited from **Board Viewer**.
The edit panel appears on **Board Viewer**. Select **Shape**.

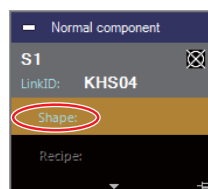


Figure 2-39 Edit Panel

2.5.4 Assigning an Existing Shape

Here describes how to assign a Shape saved in a library to a component data.

CAUTION Changing an existing Shape setting may affect to other components inspection result. If a component data is similar but not identical to an existing Shape, copy the existing Shape according to **2.5.5 Copying an Existing Shape** and edit the copied Shape.

Step1: Open the Shape editing screen according to **2.5.3 Opening the Shape Editing Screen**.

Step2: Select a Shape from the Shape list.

NOTE To search a Shape by its name, enter the name in the Search in Library text-box on the upper left side of **Shape Library Editor**.

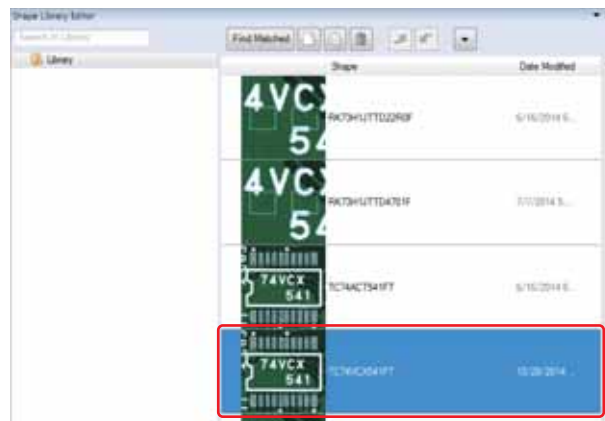


Figure 2-40 Assigning an Existing Shape 1

Step3: Click **Assign** in **Shape selection preview**.

The Shape is assigned to all component data sharing the same Link ID.

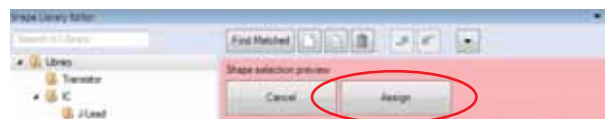


Figure 2-41 Assigning an Existing Shape 2

Step4: Close the Shape editing screen according to **2.7 Exit the Shape Editing Screen**.

2.5.5 Copying an Existing Shape

Here describes how to copy a Shape saved in a library and assign to selected component data. If a component data is similar but not identical to an existing Shape, copy the existing Shape according to the following procedure and edit the copied Shape.

Step1: Select a Shape to be copied according to Step1 to Step3 of **2.5.4 Assigning an Existing Shape**.

Step2: To copy only Shape, select  (Copy) > **Copy shape only** in **Shape Library Editor**.
To copy Recipes together, select  (Copy) > **Copy recipes together** in **Shape Library Editor**.



Figure 2-42 Copying an Existing Shape 1

Step3: To copy Recipes together, check Recipes to copy and click **OK**.

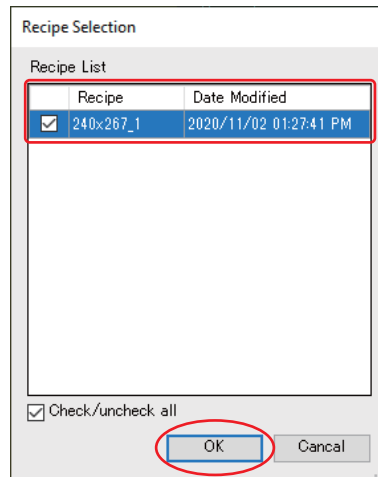


Figure 2-43 Copying an Existing Shape 2

Step4: Enter a Shape name and click **OK**.

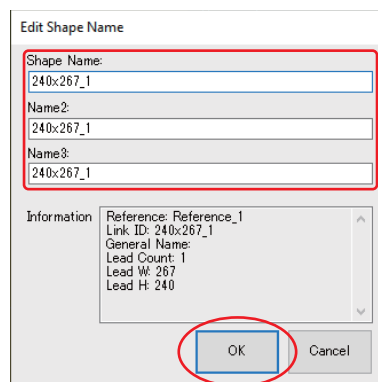


Figure 2-44 Copying an Existing Shape 3

Step5: Click **Assign** in **Shape selection preview**.

The Shape is assigned to all component data sharing the same Link ID.

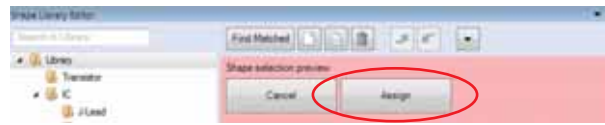


Figure 2-45 Copying an Existing Shape 4

Step6: Adjust the Shape according to **2.6 Adjusting a Shape**.

Step7: Close the Shape editing screen according to **2.7 Exit the Shape Editing Screen**.

2.5.6 Making a Shape Manually

Here describes how to make a Shape manually.

CAUTION In this function, only the square Body element is made. Adjust the Shape according to **2.6 Adjusting a Shape** after this procedure.

NOTE This software make inspection data basically from Gerber data.
Use this function only when you must make inspection data by manual exceptionally.

Step1: Open the Shape editing screen according to **2.5.3 Opening the Shape Editing Screen**.

Step2: Select **New > Blank**.



Figure 2-46 Making a Shape Manually 1

Step3: Enter a shape name and click **OK**.

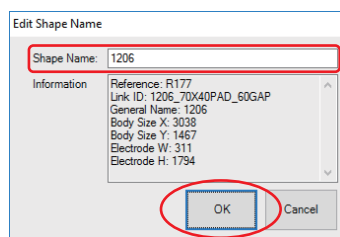


Figure 2-47 Making a Shape Manually 2

Step4: Click **Assign** in **Shape selection preview** of **Window Tree**.

The Shape is assigned to all component data sharing the same Link ID.

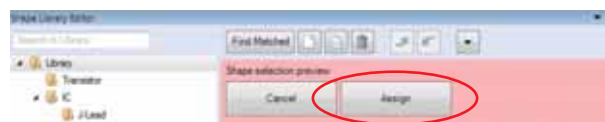


Figure 2-48 Making a Shape Manually 3

Step5: Adjust the Shape according to **2.6 Adjusting a Shape**.

Step6: Close the Shape editing screen according to **2.7 Exit the Shape Editing Screen**.

2.6 Adjusting a Shape

This section describes how to edit and adjust a Shape.

CAUTION

This software creates inspection data basically from Gerber data by using inspection data wizard. Also shape is created by automatic. This section describes how to adjust the shape data if necessary.

2.6.1 Window Details

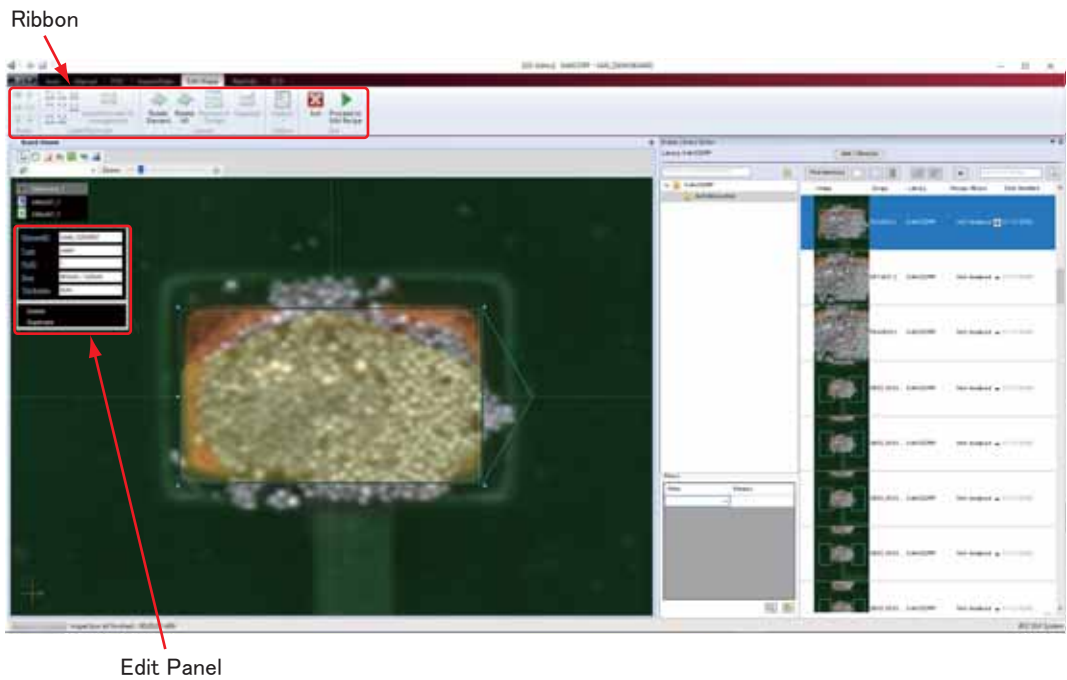


Figure 2-49 Window Details

Item			Description
Ribbon	Body		Not available
			
			
			
			
			
	Lead/ Electrode		
			
			
			
			
			
			
			
	Lead/Electrode ID management		

Item				Description	
Ribbon	Layout	Rotate Element		Rotates the selected element 90 degrees clockwise with the center of each element as the origin.	
		Rotate All		Rotates all Shapes 90 degrees clockwise with the center of the component as the origin.	
		Numerical Design		Not available	
		Organize		Not available	
	Option	Stick to body		Not available	
Edit Panel	Element ID	Select or input		Used for changing an element ID.	
		Apply the same value to...	Copy selected element to same type of element (all)	Apply the same modification to all elements on the same side and sharing the same Type.	
			Copy selected element to same type of element (on this side)	Apply the same modification to all elements with the same Type.	
	Type			Indicates an element type. Type can be changed by clicking the text-box.	
	Pin ID	Set this as #1 and count up clock-wise		Sets the Pin ID of the selected element to 1 and give numbers to other pins clockwise.	
		Set this as #1 and count up counter-clock-wise		Sets the Pin ID of the selected element to 1 and give numbers to other pins counter-clockwise.	
		Input		Assigns the Pin ID entered in the text-box to the selected element.	
	Size	Input		Changes the selected element size to the entered value.	
		Apply the same value to...	all elements on this side with the same type	Apply the same modification to all elements on the same side and sharing the same Type.	
			all elements with the same type	Apply the same modification to all elements with the same Type.	
	Thickness	Input		Changes the selected element thickness to the entered value.	
		Apply the same value to...	all elements on this side with the same type	Applies the same modification to all elements on the same side and sharing the same Type .	
			all elements with the same type	Applies the same modification to all elements with the same Type .	
	Delete				Deletes the selected element.
	Duplicate				Copies the selected element.

Table 2-9 Descriptions

2.7 Exit the Shape Editing Screen

This section describes how to close the Shape editing screen.

After closing the Shape editing screen, it can proceed to the Recipe editing screen or return to the main screen.

NOTE

It is possible to open the Recipe editing screen from the main screen.

Step1: To proceed to the **Recipe Editing Screen**, from the ribbon, select the **Exit** group > **Proceed to Edit Recipe**.

To return to the **InspecData** tab, from the ribbon, select the **Exit** group > **Exit**.



Figure 2-50 Exit Adjusting a Shape1

Step2: The dialog shown below appears. Select Item.

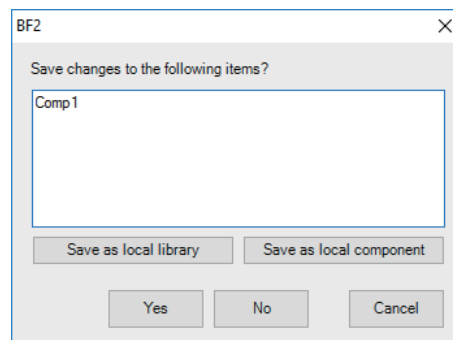


Figure 2-51 Exit Adjusting a Shape 2

Save as local library

Save the Changes only to the opened inspection data and not to the library. In this case, changes are not applied to other inspection data within the same group.

Save as local library targets inspection data currently opened, and the changes are applied to other inspection windows using the same shapes and recipes.

Save as local component

Save the Changes only to the opened inspection data and not to the library. In this case, changes are not applied to other inspection data within the same group. **Save as local component** applies changes only to the inspection window currently being saved.

Yes

Enters the changes into the library. The changes are applied to other inspection data in the same group when the changes are entered to the library.

No

Discards all the changes.

Cancel

Returns to shape editing.

2.8 Assigning a Recipe

2.8.1 Overview of Recipe

Recipe is inspection settings for each **Shape**.

In Recipe, an inspection window is set to each element, and an algorithm is assigned to the inspection window. The window size can also be adjusted.

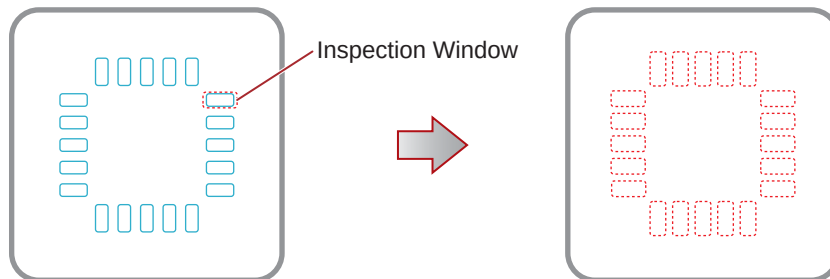


Figure 2-52 Overview of Recipe

2.8.2 Window Details

Here describes and illustrates the Recipe editing screen.

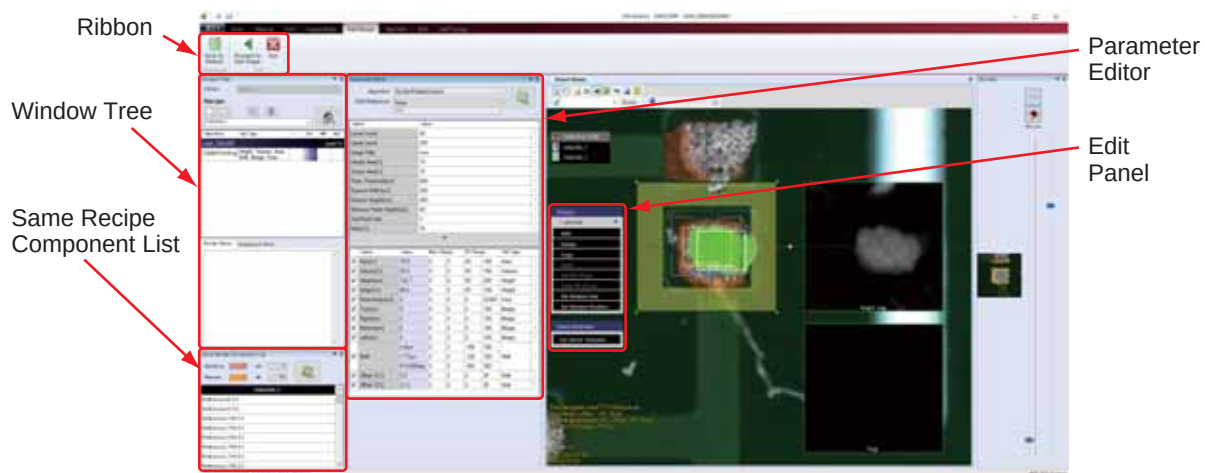


Figure 2-53 Window Details

Item			Description
Ribbon	Expansion	Save As Default	The set parameters are registered as a template. For details, refer to 4.4.3 Register the Recipe Parameter .
	Exit	Exit	Finishes editing a Shape and return to the InspectData tab. For details, refer to 2.10 Exit the Recipe Editing Screen .
Window Tree			Creates a new Recipe.
			Copies the Recipe currently assigned.
			Renames the Recipe currently assigned.
			Delete a Recipe. The Recipe currently assigned cannot be deleted.

Item		Description
Window Tree		Click Local Save to save the changes to the opened inspection data and not the library when exiting and  is displayed on upper part of window tree. If opened inspection data and library information are different,  is displayed on upper part of window tree. If meet the both conditions,  is displayed on upper part of window tree and click this ICON, the dialog shown below appears.
		
		
		
	Recipe	Click Update from Library to overwrite the changes in the currently opened inspection data with the contents of the library. Click Update to Library to enter the change in the library.
	Inspection Window List	Inspects all inspection windows included in the Recipe. Selects a Recipe from the drop-down list. This menu item is available only when multiple Recipes are assigned. Displays an algorithm, NG type, and inspection result of each inspection window. When an inspection is finished, each cell is highlighted in a color that corresponds to its inspection result. If an inspection result is OK, the cell is highlighted in blue. If a result is NG, it is highlighted in red. If a result is Warning, it is highlighted in orange. If an inspection has not been performed, it is highlighted in gray.
		Figure 2-54 Exit Editing
Same Recipe Component List	NG/Error	Click Update from Library to overwrite the changes in the currently opened inspection data with the contents of the library. Click Update to Library to enter the change in the library.
	Warning	Inspects all inspection windows included in the Recipe.
	OK	Selects a Recipe from the drop-down list. This menu item is available only when multiple Recipes are assigned.
	All	Displays an algorithm, NG type, and inspection result of each inspection window. When an inspection is finished, each cell is highlighted in a color that corresponds to its inspection result. If an inspection result is OK, the cell is highlighted in blue. If a result is NG, it is highlighted in red. If a result is Warning, it is highlighted in orange. If an inspection has not been performed, it is highlighted in gray.
	Inspection Result List	Inspects all data which the same Recipe is assigned. Inspection results are displayed in Inspection Result List .
		
Parameter Editor		Makes the setting for the selected inspection window. For details, refer to 2.9 Adjusting a Recipe .
Edit Panel	Window	Used for adjusting an inspection window. For details, refer to 2.9 Adjusting a Recipe .
	Stencil thickness	Set the stencil thickness of the selected pad.
3D View (*1) (*2)		Display the theoretical 3D image that calculate from shape area and stencil thickness.
		Display the height scale.

Table 2-10 Descriptions

NOTE(*1) Select **BF2 Option > Editor > Editor**, set the **Is3DWindow Enabled** to **True**(*2) Select **BF2 Option > Editor > Editor**, can be change the displaying size of 3D image by **SizeRaito of 3DWindow**.

2.8.3 Opening the Recipe Editing Screen

Here describes how to open the Recipe editing screen.

NOTE When moved directly from the Shape editing screen, this step is not necessary.

Step1: Select the **InspectData** tab.



Figure 2-55 InspectData

Step2: Select a data which a Shape has been assigned from **Board Viewer**.

Step3: The edit panel appears on **Board Viewer**.
Select **Recipe**. The Recipe editing screen appears.

NOTE The screen also appears when double-clicking a component data which a Shape has been assigned.

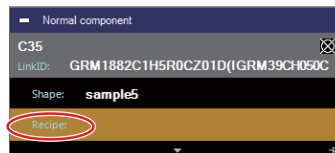


Figure 2-56 Edit Panel

2.8.4 Making a Recipe

Here describes how to make a new Recipe.

Step1: Open the Recipe editing screen according to **2.8.3 Opening the Recipe Editing Screen**.

Step2: Click  on the top of **Window Tree**.

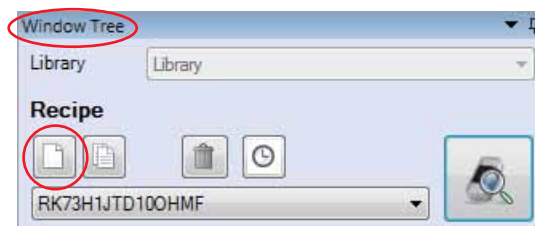


Figure 2-57 Making a Recipe 1

Step3: Enter a Recipe name and click **OK**.

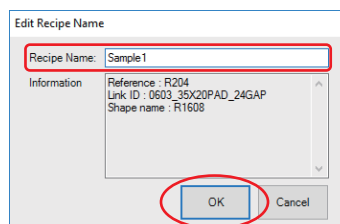


Figure 2-58 Making a Recipe 2

- Step4: Click **Assign** in **Recipe selection preview** of **Window Tree**.
The Recipe is assigned to all component data sharing the same **Link ID**.

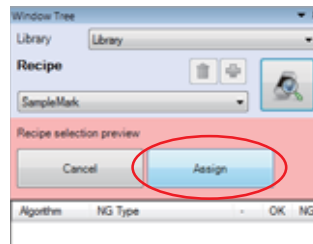


Figure 2-59 Making a Recipe 3

2.8.5 Setting Inspection Items Manually

Here describes how to set inspection items by manual.

- Step1: Assign the Recipe to a Shape according to **2.8.4 Making a Recipe**.

- Step2: Select an element and select **Add** from the edit panel.

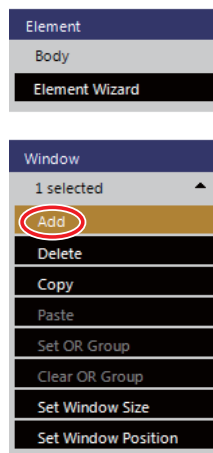


Figure 2-60 Adding an Inspection Window

- Step3: From the **Dock window > Parameter Editor > Algorithm**, select an algorithm used for the inspection.

NOTE For details of algorithms, refer to **Part III Inspection Algorithm**.

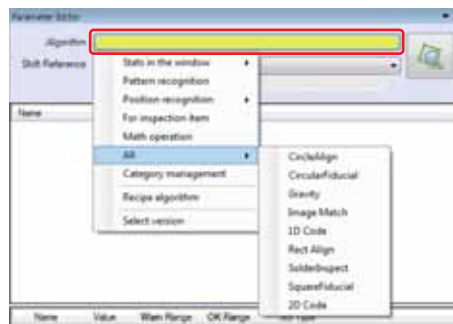


Figure 2-61 Selecting an Algorithm

- Step4: Click  of **Parameter Editor** and check if the inspection is completed properly. If the inspection is failed, adjust the parameters and inspection window sizes.

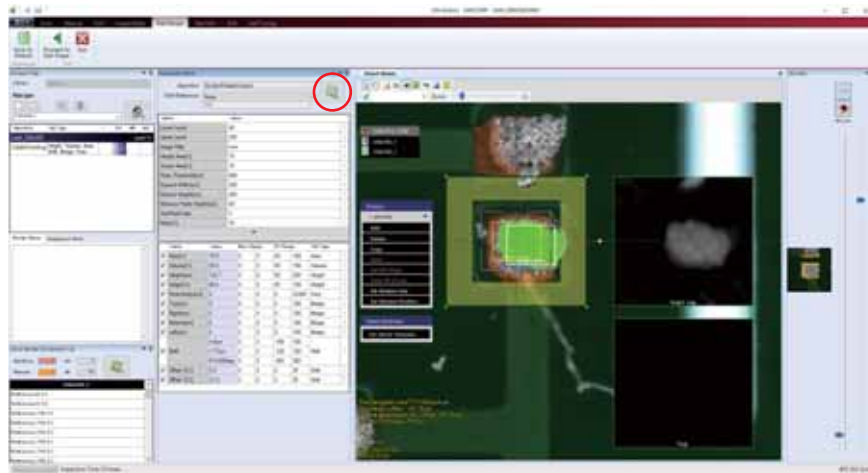


Figure 2-62 Checking the Inspection Window Settings

- Step5: Repeat Step2 to Step4 and set inspection items to all elements.

NOTE If an element shares the same element ID with another element, the same inspection items are set to both elements automatically.

- Step6: Click  of **Window Tree** and check if inspections are completed properly in all inspection windows.

If one of inspections is failed, adjust the parameters and sizes of the inspection window.

NOTE When an inspection is finished, each cell is highlighted in a color that corresponds to its inspection result. If an inspection result is OK, the cell is highlighted in blue. If a result is NG, it is highlighted in red. If a result is Warning, it is highlighted in orange. If an inspection has not been performed, it is highlighted in gray.

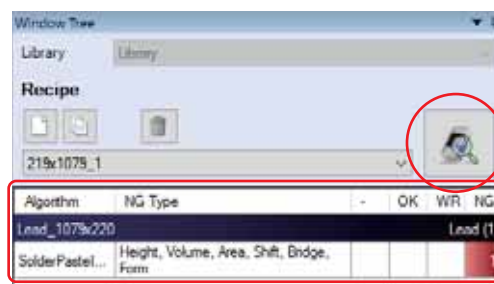


Figure 2-63 Checking the Data

Step7: Click  of **Same Recipe Component List** and check if inspections are completed properly in all component data. If one of inspections is failed, adjust the parameters and inspection window sizes of the corresponding element.

NOTE If an inspection result is NG, the cell is highlighted in red.

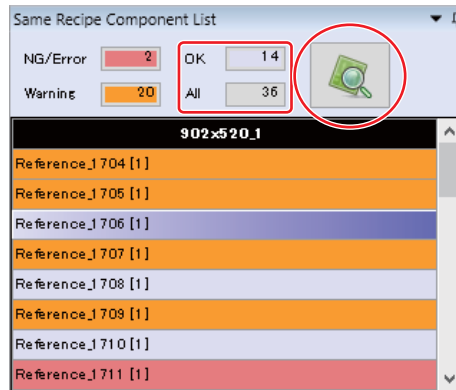


Figure 2-64 Checking the Same Recipe Components

Step8: Exit the Recipe editing screen according to **2.10 Exit the Recipe Editing Screen**.

2.9 Adjusting a Recipe

This section describes how to edit and adjust a Recipe.

For editing a Recipe, Window Tree, Edit panel, and Parameter Editor are mainly used.

2.9.1 Window Details

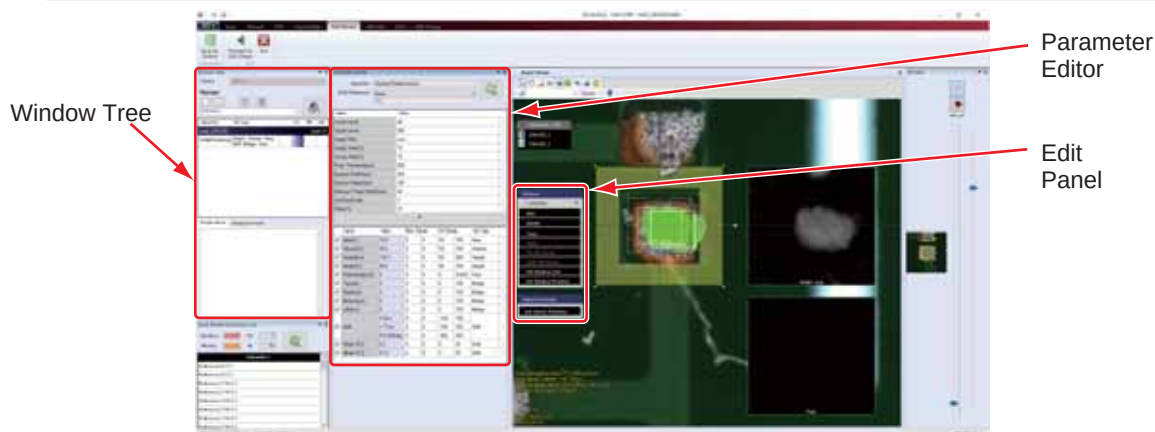


Figure 2-65 Window Details

Item			Description
Window Tree			Used for operations such as adding a Recipe and selecting an inspection window. For details, refer to 2.8 Assigning a Recipe .
Parameter Editor	Algorithm		The drop-down list opens when clicked. Select an algorithm.
	Shift Reference		Not available
			Inspects the selected inspection window.
	(Upper table)	Name	Displays parameters for an algorithm.
		Value	Used for setting thresholds and settings to each parameter. If a inspection target is not recognized or false call happens, adjust these values.
	(Lower table)	Name	Displays inspection items. When checked, the item is judged in an inspection. When unchecked, the item is not judged in an inspection.
		Value	Displays values detected by an inspection. If an inspection has not been performed, – is displayed.
		Warning Range	Specifies the range which is judged as Warning.
		OK Range	Specifies the range which is judged as OK (good).
NG Type		Specifies an NG type for when an inspection result is NG (defect, bad).	
Edit Panel	Window	Add	Add an inspection window to the selected element.
		Delete	Deletes the selected inspection window.
		Copy	Copies the selected inspection window.
		Paste	Pastes the copied inspection window.
		Set OR Group	Not available
		Clear OR Group	
		Set Window Size	Used for adjusting an inspection window size. For details, refer to 2.9.3 Adjusting an Inspection Window Size .
		Set Window Position	Not available
	Stencil thickness		Set the stencil thickness of the selected pad.

Table 2-11 Descriptions

2.9.2 Parameter Adjustment

If inspection accuracy is low or too many false calls occurred, adjust parameters of a Recipe. Here describes how to adjust parameters of a Recipe.

CAUTION

The procedure described here is the most commonly used one in practice. According to an algorithm, the procedure may differ slightly. For details of algorithms, refer to **Part III Inspection Algorithm**.

NOTE

The procedure for adding inspection items is the same as the manual setting of inspection items. For details, refer to **2.8.5 Setting Inspection Items Manually**.

Step1: Select an inspection data which Recipes are already assigned according to **Part I 1.3 Selecting an Inspection Data File**.

Step2: Scan a good PCB according to **5.2 Scanning a PCB**.

Step3: Select the **InspectData** tab.



Figure 2-66 InspectData

Step4: From the ribbon, Select the **Inspect** group > **Inspect All** to start the inspection.



Figure 2-67 Inspect All

Step5: Inspection results are displayed on **Board Viewer**, **Component List**, and **LookUp Table**.

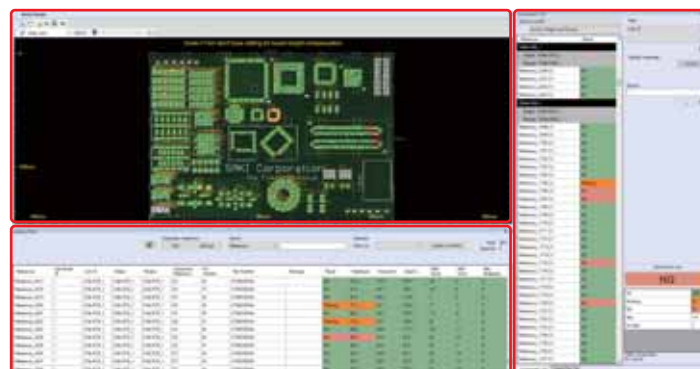


Figure 2-68 Inspection Results

Step6: Select an NG data from **Board Viewer** and select **Recipe** from the edit panel.

NOTE The same operation can also be performed by double-clicking a reference name on **Component List** and **LookUp Table**.

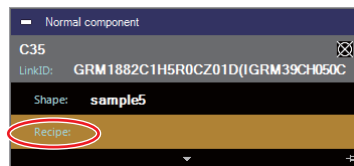


Figure 2-69 Edit Panel

Step7: Click the value of **NG** in **Window Tree**.

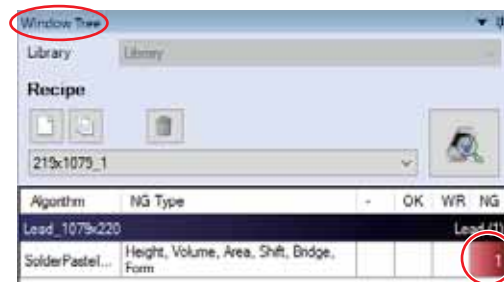


Figure 2-70 Window Tree

Step8: Adjust the OK Range settings of the algorithm set to the inspection window. On the lower table of **Parameter Editor**, click a cell which is highlighted in red.

NOTE Barcode inspection algorithm is different. For details, refer to **Part III Inspection Algorithm**.

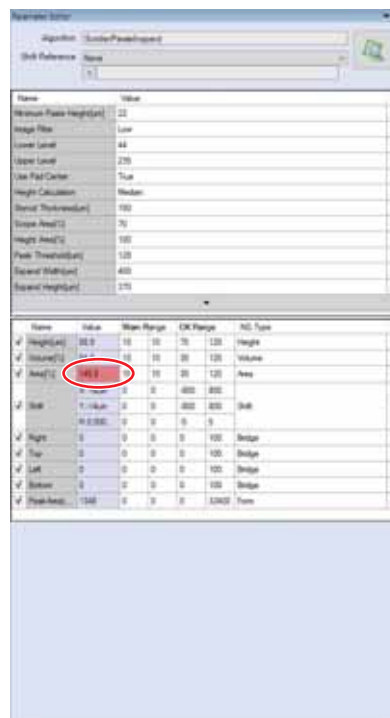


Figure 2-71 Parameter Editor

Step9: The dialog shown below appears.

CAUTION According to an algorithm, the dialog displayed may differ slightly.

When Adjusting the Parameter Manually

Slide the slider on the left side of the histogram or enter a value to the text-box to adjust the threshold. Adjustment is completed, click **OK**.

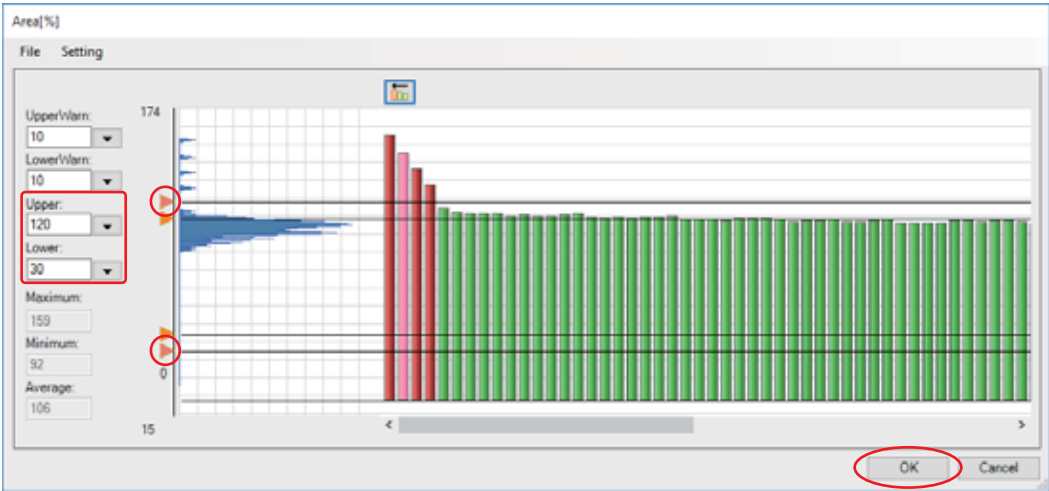


Figure 2-72 Adjusting the Sample Value 1

	OK	NG
Sample value of the selected inspection window		
Sample values of inspection windows sharing the same Recipe		

When Adjusting the Parameter with the Advanced Function

Click  on the right side of **Upper** or **Lower**.

The drop-down list appears for selecting a calculation method of a threshold.

Make the setting according to Table 2-12. After all settings are completed, click **OK**.

NOTE A sigma ratio and percent ratio can be changed from **Setting** located on the top of the dialog.

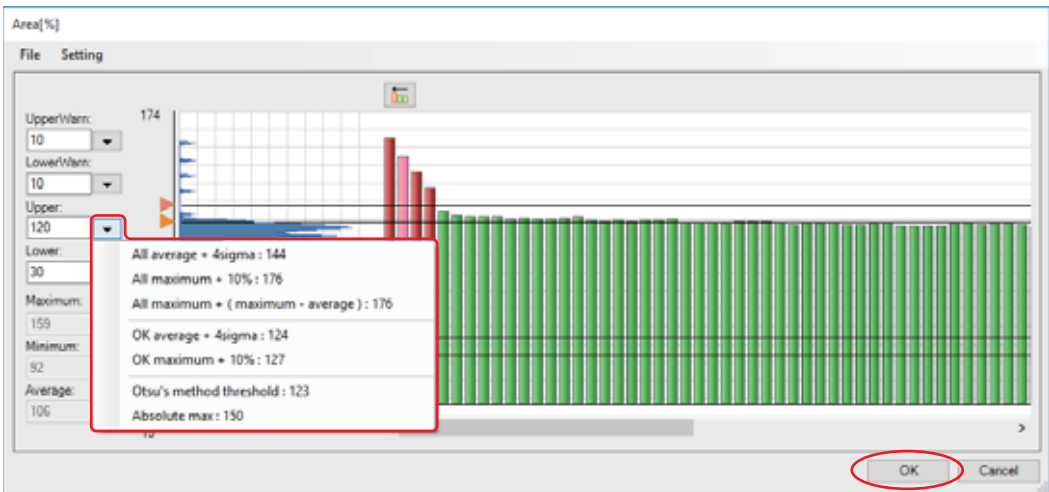


Figure 2-73 Adjusting the Sample Value 2

Item		Calculation Target	Description
Upper limit	All average + 4sigma	All sample value	Calculates the upper limit of the sample value based on a sigma ratio and the average of the current sample values.
	All maximum + 10%		Calculates 110% of the greatest value in the current sample values. The result is set as the upper limit of the sample value.
	All maximum + (maximum - average)		In this option, it calculates the difference between the greatest and average of the current sample values first. Then, it adds the calculated value to the greatest of the sample values. The result is set as the upper limit of the sample value.
	OK average + 4sigma	Sample values of the OK windows	Calculates the upper limit of the sample value based on a sigma ratio and the average of the sample values of the inspection windows whose inspection results are OK.
	OK maximum + 10%		Calculates 110% of the greatest value in the sample values of the inspection windows whose inspection results are OK. The result is set as the upper limit of the sample value.
	Otsu's method threshold	All sample value	Calculates the upper limit of the sample value based on Otsu's method (histogram shape-based image thresholding).
	Absolute max		The default value of the algorithm is set as the upper limit of the sample value.
Lower limit	All average - 4sigma	All sample value	Calculates the lower limit of the sample value based on a sigma ratio and the average of the current sample values.
	All minimum - 10%		Calculates 90% of the smallest value in the current sample values. The result is set as the lower limit of the sample value.
	All minimum - (average - minimum)		In this option, it calculates the difference between the average and smallest of the current sample values first. Then, it subtracts the calculated value from the smallest of the sample values. The result is set as the lower limit of the sample value.
	OK average - 4sigma	Sample values of the OK windows	Calculates the lower limit of the sample value based on a sigma ratio and the average of the sample values of the inspection windows whose inspection results are OK.
	OK minimum - 10%		Calculates 90% of the smallest value in the sample values of the inspection windows whose inspection results are OK. The result is set as the lower limit of the sample value.
	Otsu's method threshold	All sample value	Calculates the lower limit of the sample value based on Otsu's method (histogram shape-based image thresholding).
	Absolute min		The default value of the algorithm is set as the lower limit of the sample value.

Table 2-12 Descriptions

Step10: Click  of **Parameter Editor** and check if the inspection is completed properly.
If the inspection is failed, adjust the parameters again.

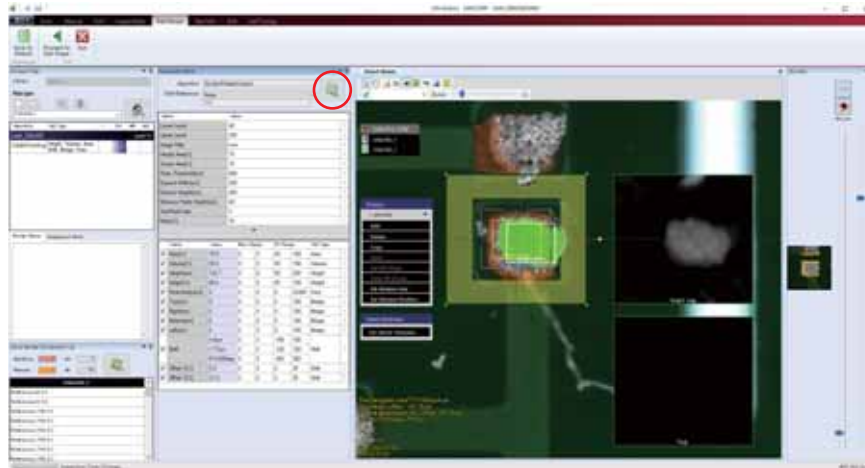


Figure 2-74 Checking the Inspection Window Settings

Step11: Click  of **Window Tree** and check if all inspection windows are judged as OK. If NG is detected, adjust the parameters again.

NOTE

If an inspection result is OK, the cell is highlighted in blue. If a result is NG, it is highlighted in red. If a result is Warning, it is highlighted in orange. If an inspection has not been performed, it is highlighted in gray.

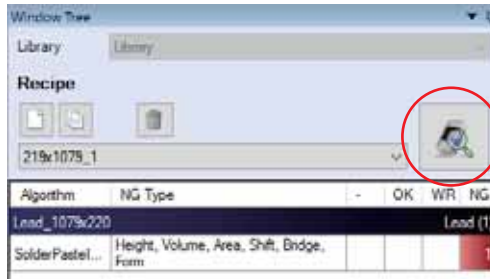


Figure 2-75 Checking the Data

Step12: Click  of **Same Recipe Component List** and check if all inspection targets are judged as OK. If NG or Warning is detected, adjust the parameters again.

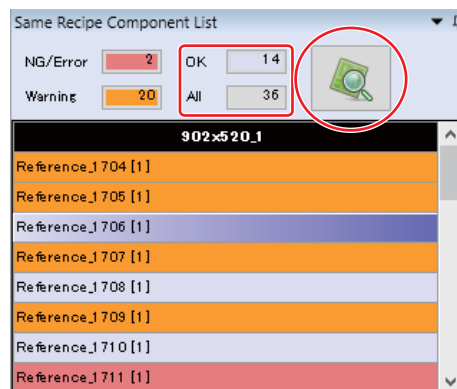


Figure 2-76 Checking the Same Recipe data

Step13: Adjust Recipes of other NG inspection targets by repeating Step5 to Step12.

2.9.3 Adjusting an Inspection Window Size

Here describes how to adjust an inspection window size.

NOTE

When use **SolderPasteInspect** algorithm, inspection window size can be adjusted other than the method described here. For details, refer to **Part III 1 Solder Paste Inspection**.

Step1: Select the **InspectData** tab.



Figure 2-77 InspectData

Step2: Select a data to adjust the window size from **Board Viewer**.

Step3: The edit panel appears on **Board Viewer**. Select **Recipe**.

NOTE

The same operation can also be performed by double-clicking a reference name on **Component List** and **LookUp Table**.



Figure 2-78 Edit Panel

Step4: Select an inspection window to adjust the window size.

Step5: From the edit panel, select **Window > Set Window Size**.

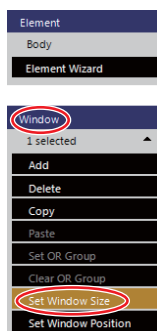


Figure 2-79 Adjusting an Inspection Window Size 1

Step6: Enter values to **X [um]** and **Y [um]**. Click **Element size** to match the inspection window size to the element size. After all settings are completed, click **OK**.

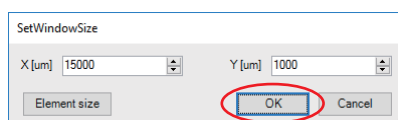


Figure 2-80 Adjusting an Inspection Window Size 2

2.10 Exit the Recipe Editing Screen

This section describes how to close the Recipe editing screen.

Return to the main screen after exiting the recipe adjustment.

Step1: From the ribbon, select the **Exit** group > **Exit**.

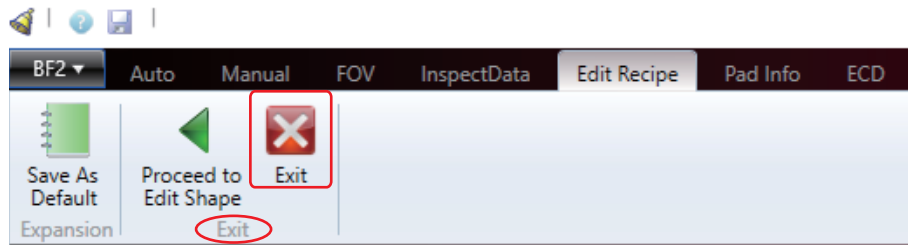


Figure 2-81 Exit Adjusting a Recipe 1

Step2: The dialog shown below appears. Select Item.

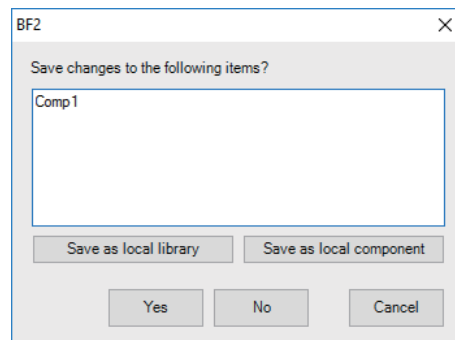


Figure 2-82 Exit Adjusting a Recipe 2

Save as local library	Save the Changes only to the opened inspection data and not to the library. In this case, changes are not applied to other inspection data within the same group. Save as local library targets inspection data currently opened, and the changes are applied to other inspection windows using the same shapes and recipes.
Save as local component	Save the Changes only to the opened inspection data and not to the library. In this case, changes are not applied to other inspection data within the same group. Save as local component applies changes only to the inspection window currently being saved.
Yes	Enters the changes into the library. The changes are applied to other inspection data in the same group when the changes are entered to the library.
No	Discards all the changes.
Cancel	Returns to recipe editing.

Step3: If multiple Recipes are assigned to one Shape, the dialog shown in below appears.
Make the setting according to Table 2-13.

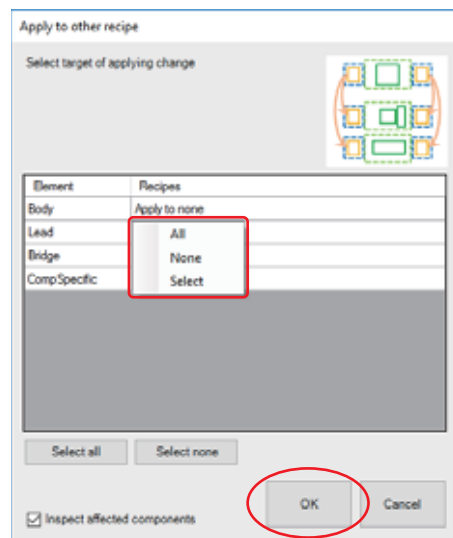


Figure 2-83 Applying the Changes to Other Recipes

Item		Description
Element		Display the element name.
Recipe	All	Applies the changes to all Recipes which assigned to the same Shape.
	None	Cancels the selection.
	Select	Applies the changes to the selected Recipes.
Select all		Sets Apply to all to all Recipes.
Select none		Sets Apply to none to all Recipes.
Inspect affected components		After click OK , start inspecting data which applied a changes.

Table 2-13 Descriptions

2.11 Changing Library, Reference Name, Link ID, Shape, and Recipe

2.11.1 Changing the Assignment on Board Viewer

Here describes how to change the assignments for reference name, Link ID, Shape, and Recipe by the edit panel of **Board Viewer**.

CAUTION Libraries cannot be changed in **Board Viewer**.

Step1: Select the **InspectData** tab.



Figure 2-84 InspectData

Step2: Select a data from **Board Viewer**.

Step3: The edit panel appears on **Board Viewer**.

Click the upper side of the panel in which the reference name and Link ID is displayed.

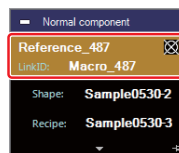


Figure 2-85 Edit Panel

Step4: Make the settings according to Table 2-14 and click **OK**.

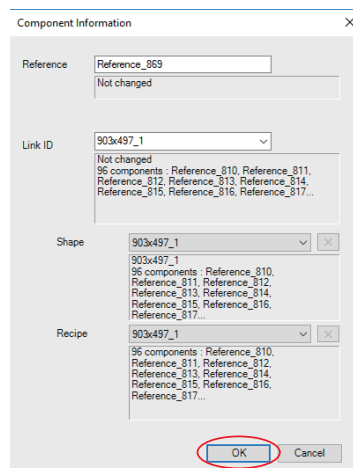


Figure 2-86 Component Information

Item	Description
Reference	Specifies the reference name of the selected component data.
Link ID	Specifies the Link ID of the selected component data.
Shape	Specifies the Shape of the selected component data.
Recipe	Specifies the Recipe of the selected component data.

Table 2-14 Descriptions

NOTE

Change the Link ID in order to change the shape and recipe of a specific data among those which have the same Link ID.

2.11.2 Displaying the Link ID Table

The Link ID table which includes the list of library, reference name, Link ID, Shape, and Recipe is displayed by selecting **Management** from the ribbon.

Step1: From the **InspectData** tab, select the **Library** group > **Management** > **Library management**.

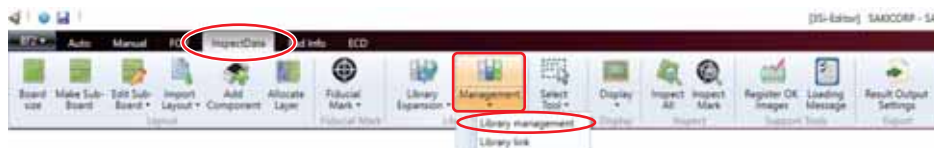


Figure 2-87 Library Management

Step2: Click **OK**.

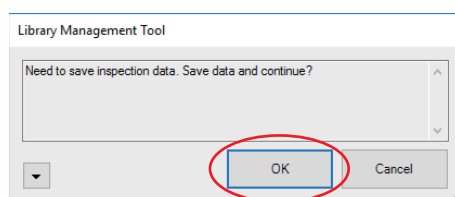


Figure 2-88 Library Management Tool 1

Step3: Click **Link ID Table**.

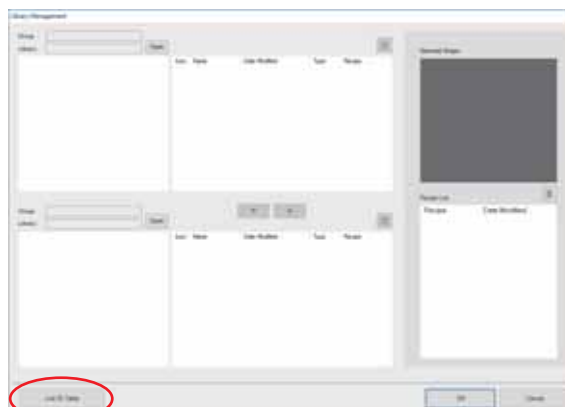


Figure 2-89 Library Management Tool 2

Step4: Click **OK**.

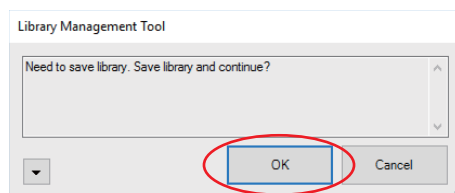


Figure 2-90 Library Management Tool 3

Step5: The Link ID table appears. The table displays the list of reference name, Link ID, Shape, and Recipe. To close the dialog, click **Close**.

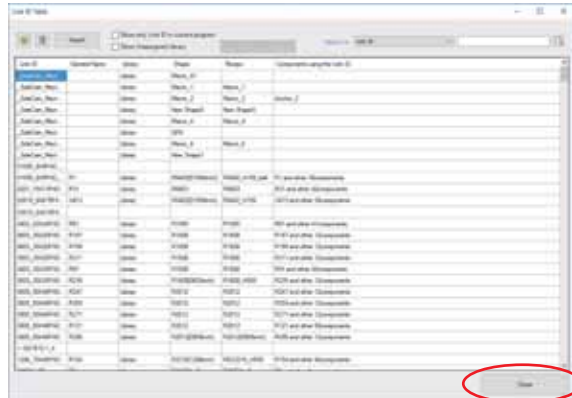


Figure 2-91 Link ID table

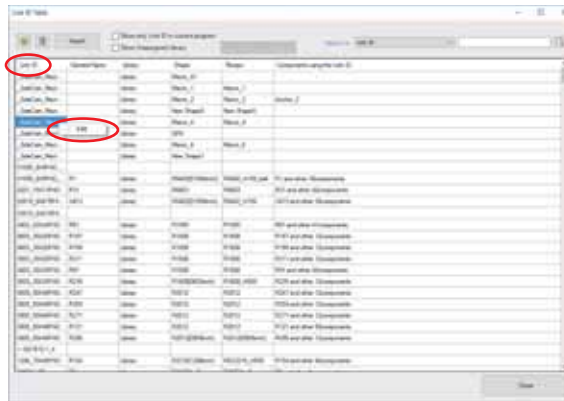
Item	Description
	Adds Link ID.
	Deletes Link ID. If the Link ID is currently assigned to a component, it cannot be deleted.
Import	Imports Link IDs from another inspection data. For details, refer to 2.11.8 Importing Link IDs .
Search in	Specifies the search target from drop-down list and searches a word entered in the right text-box
Show only Link ID in current program	Check this item to displays only the link IDs included in the current open inspection data in the list.
Show Unassigned Library	Check this item to displays unassigned libraries in the list.
Link ID	Displays all Link IDs included in the current inspection data in the list. To sort data in ascending/descending order, click a header cell. Link IDs can also be changed by clicking cells. For details, refer to 2.11.3 Changing a Link ID on the Link ID Table .
General Name	Displays the general name, library, Shapes, and Recipes assigned to the Link ID. To sort data in ascending/descending order, click a header cell. The setting can be changed by clicking a cell. For details, refer to 2.11.4 Changing Library, Shape, and Recipe on the Link ID Table .
Library	
Shape	
Recipe	
Components using this Link ID	Displays a component data which a Link ID is assigned by a reference name. To sort data in ascending/descending order, click a header cell. An assigned component can be changed by clicking a cell. For details, refer to 2.11.5 Changing Assignment of a Link ID on the Link ID Table .

Table 2-15 Descriptions

2.11.3 Changing a Link ID on the Link ID Table

Step1: Open the Link ID table according to **2.11.2 Displaying the Link ID Table**.

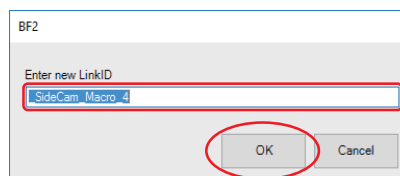
Step2: Click one of the **Link ID** cells and click **Edit**.



Link ID	Name	Shape	Recipe	Comments
1	1000000000	1000	1000	1000000000
2	1000000000	1000	1000	1000000000
3	1000000000	1000	1000	1000000000
4	1000000000	1000	1000	1000000000
5	1000000000	1000	1000	1000000000
6	1000000000	1000	1000	1000000000
7	1000000000	1000	1000	1000000000
8	1000000000	1000	1000	1000000000
9	1000000000	1000	1000	1000000000
10	1000000000	1000	1000	1000000000
11	1000000000	1000	1000	1000000000
12	1000000000	1000	1000	1000000000
13	1000000000	1000	1000	1000000000
14	1000000000	1000	1000	1000000000
15	1000000000	1000	1000	1000000000
16	1000000000	1000	1000	1000000000
17	1000000000	1000	1000	1000000000
18	1000000000	1000	1000	1000000000
19	1000000000	1000	1000	1000000000
20	1000000000	1000	1000	1000000000
21	1000000000	1000	1000	1000000000
22	1000000000	1000	1000	1000000000
23	1000000000	1000	1000	1000000000
24	1000000000	1000	1000	1000000000
25	1000000000	1000	1000	1000000000
26	1000000000	1000	1000	1000000000
27	1000000000	1000	1000	1000000000
28	1000000000	1000	1000	1000000000
29	1000000000	1000	1000	1000000000
30	1000000000	1000	1000	1000000000
31	1000000000	1000	1000	1000000000
32	1000000000	1000	1000	1000000000
33	1000000000	1000	1000	1000000000
34	1000000000	1000	1000	1000000000
35	1000000000	1000	1000	1000000000
36	1000000000	1000	1000	1000000000
37	1000000000	1000	1000	1000000000
38	1000000000	1000	1000	1000000000
39	1000000000	1000	1000	1000000000
40	1000000000	1000	1000	1000000000
41	1000000000	1000	1000	1000000000
42	1000000000	1000	1000	1000000000
43	1000000000	1000	1000	1000000000
44	1000000000	1000	1000	1000000000
45	1000000000	1000	1000	1000000000
46	1000000000	1000	1000	1000000000
47	1000000000	1000	1000	1000000000
48	1000000000	1000	1000	1000000000
49	1000000000	1000	1000	1000000000
50	1000000000	1000	1000	1000000000

Figure 2-92 Link ID table

Step3: Enter a Link ID into the text-box and click **OK**.



BF2

Enter new LinkID

SideCam_Macro

OK Cancel

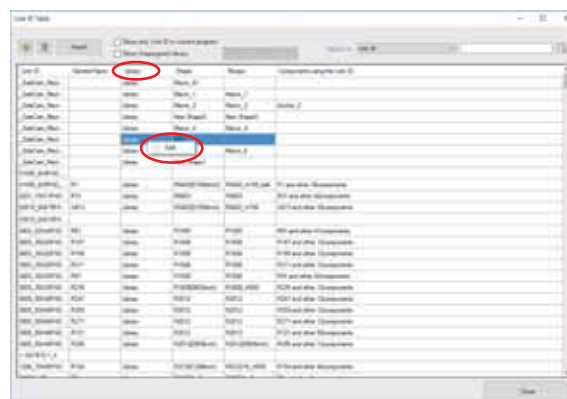
Figure 2-93 Changing a Link ID

2.11.4 Changing Library, Shape, and Recipe on the Link ID Table

Step1: Open the Link ID table according to **2.11.2 Displaying the Link ID Table**.

Step2: Click one of the **Library** cells and click **Edit**.

NOTE The same operation can also be performed by clicking one of the **Shape** or **Recipe** cells.



Link ID	Library	Shape	Recipe	Comments
1	1000000000	1000	1000	1000000000
2	1000000000	1000	1000	1000000000
3	1000000000	1000	1000	1000000000
4	1000000000	1000	1000	1000000000
5	1000000000	1000	1000	1000000000
6	1000000000	1000	1000	1000000000
7	1000000000	1000	1000	1000000000
8	1000000000	1000	1000	1000000000
9	1000000000	1000	1000	1000000000
10	1000000000	1000	1000	1000000000
11	1000000000	1000	1000	1000000000
12	1000000000	1000	1000	1000000000
13	1000000000	1000	1000	1000000000
14	1000000000	1000	1000	1000000000
15	1000000000	1000	1000	1000000000
16	1000000000	1000	1000	1000000000
17	1000000000	1000	1000	1000000000
18	1000000000	1000	1000	1000000000
19	1000000000	1000	1000	1000000000
20	1000000000	1000	1000	1000000000
21	1000000000	1000	1000	1000000000
22	1000000000	1000	1000	1000000000
23	1000000000	1000	1000	1000000000
24	1000000000	1000	1000	1000000000
25	1000000000	1000	1000	1000000000
26	1000000000	1000	1000	1000000000
27	1000000000	1000	1000	1000000000
28	1000000000	1000	1000	1000000000
29	1000000000	1000	1000	1000000000
30	1000000000	1000	1000	1000000000
31	1000000000	1000	1000	1000000000
32	1000000000	1000	1000	1000000000
33	1000000000	1000	1000	1000000000
34	1000000000	1000	1000	1000000000
35	1000000000	1000	1000	1000000000
36	1000000000	1000	1000	1000000000
37	1000000000	1000	1000	1000000000
38	1000000000	1000	1000	1000000000
39	1000000000	1000	1000	1000000000
40	1000000000	1000	1000	1000000000
41	1000000000	1000	1000	1000000000
42	1000000000	1000	1000	1000000000
43	1000000000	1000	1000	1000000000
44	1000000000	1000	1000	1000000000
45	1000000000	1000	1000	1000000000
46	1000000000	1000	1000	1000000000
47	1000000000	1000	1000	1000000000
48	1000000000	1000	1000	1000000000
49	1000000000	1000	1000	1000000000
50	1000000000	1000	1000	1000000000

Figure 2-94 Link ID table

Step3: Select a library in **Library**. Select a Shape in the left side and center list-box. Select a Recipe in the right side list-box. The selected library, Shape, and Recipe are displayed on the lower side of the dialog. After all the settings are completed, click **OK**.

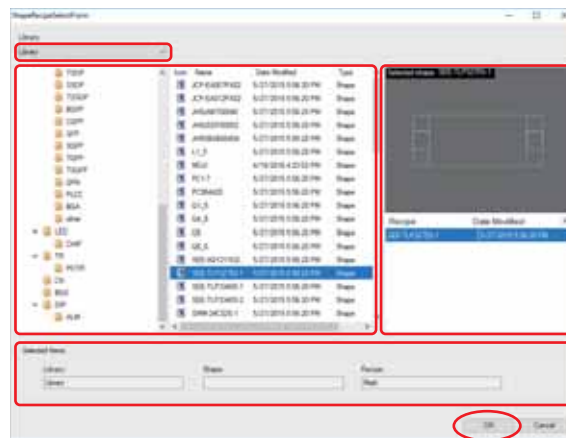


Figure 2-95 Changing Assignments (Library/Shape/Recipe)

2.11.5 Changing Assignment of a Link ID on the Link ID Table

Step1: Open the Link ID table according to **2.11.2 Displaying the Link ID Table**.

Step2: Click one of the **Components using this Link ID** cells and click **Edit**.

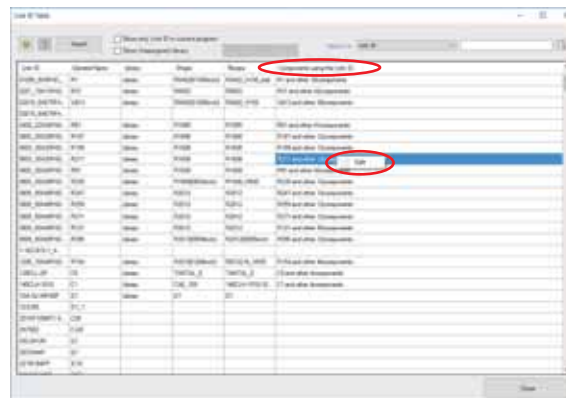


Figure 2-96 Link ID table

Step3: The left side list-box displays components which the selected Link ID is assigned. The right side list-box displays all component data included in the inspection data. Move Link IDs by clicking < / > and click **OK**.

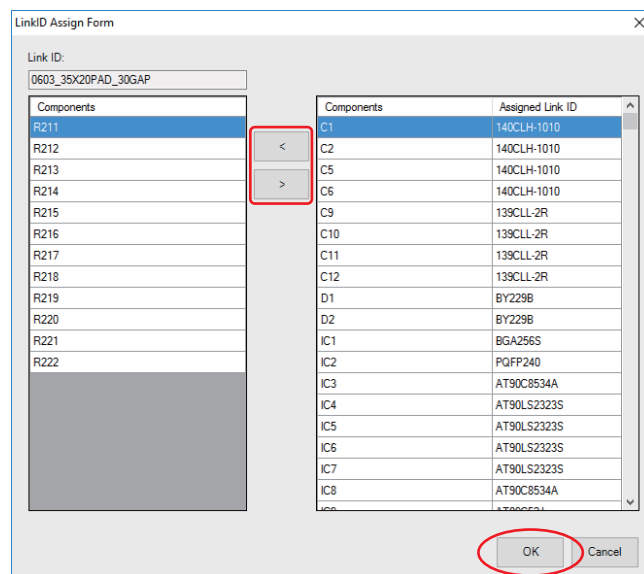


Figure 2-97 Changing Assignment (Link ID)

2.11.6 Importing/Exporting a Library

Step1: From the **InspectData** tab, select the **Library** group > **Management** > **Library management**.

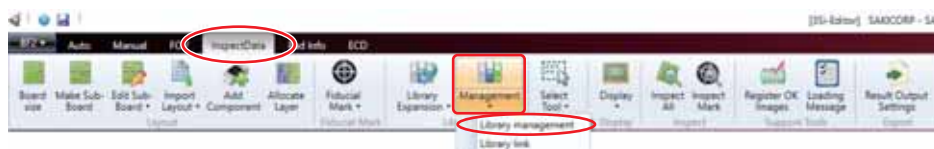


Figure 2-98 Library Management

Step2: Click **OK**.

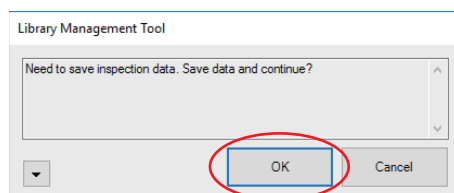


Figure 2-99 Library Management Tool 1

Step3: Click **Open** on the upper side of the dialog.

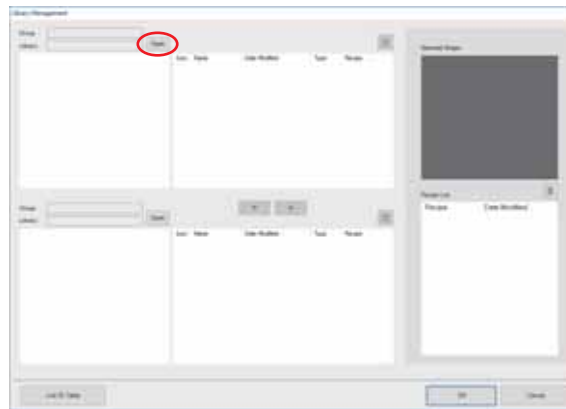


Figure 2-100 Library Management Tool 2

Step4: The current inspection group is displayed on **Opened Groups**.

To Import/Export the Library from/to the Current Inspection Data

Click **OK**.

For Other Cases

Click **Add** and select an inspection group file (bfs file) whose library to be imported/exported. Click **OK**.

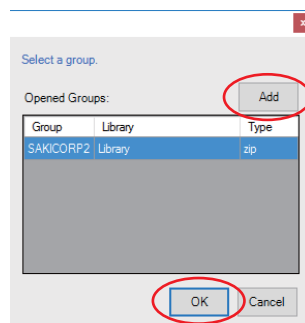


Figure 2-101 Library Management Tool 3

Step5: The library of the selected group file appears on the upper side of the dialog. Click **Open** on the lower side of the dialog.

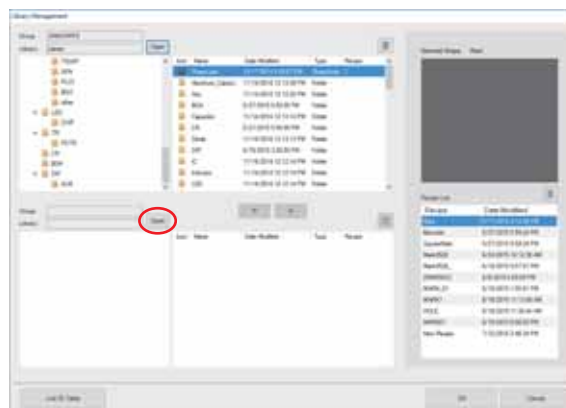


Figure 2-102 Library Management Tool 4

Step6: The current inspection group is displayed on **Opened Groups**. Click **Add** and select an inspection group file (bfs file) whose library to be imported/exported. Click **OK**.

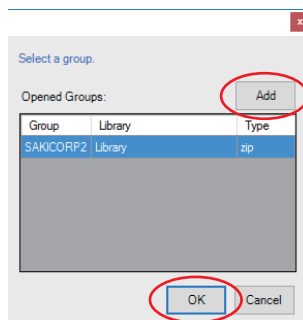


Figure 2-103 Library Management Tool 5

Step7: The library of the selected group file appears on the lower side of the dialog. Select a library and click ▼ or ▲. When ▼ is clicked, a library on the upper side table is copied to the lower side table. When ▲ is clicked, a library on the lower side table is copied to the upper side table. After the copies are completed, click **OK**.

CAUTION Changes are not applied until **OK** is clicked.

CAUTION Link IDs are not assigned to component data until the Link IDs are imported from the other inspection group. For details, refer to **2.11.8 Importing Link IDs**.

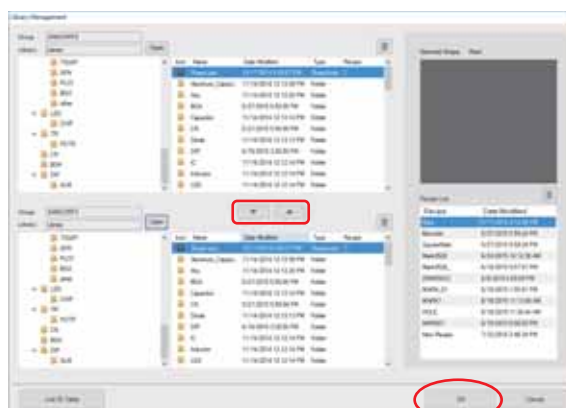


Figure 2-104 Library Management 6

2.11.7 Delete Library

Step1: From the **InspectData** tab, select the **Library** group > **Management** > **Library management**.

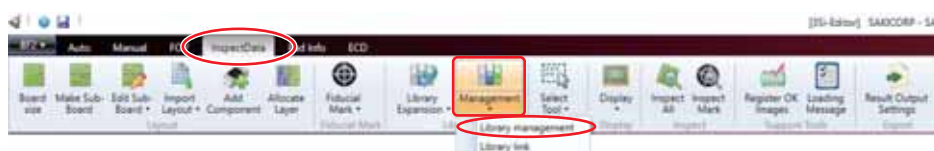


Figure 2-105 Delete Library 1

Step2: The dialog shown below appears.
Click **OK**.

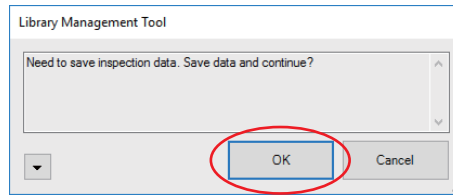


Figure 2-106 Delete Library 2

Step3: Click **Open** on the upper side of the dialog.

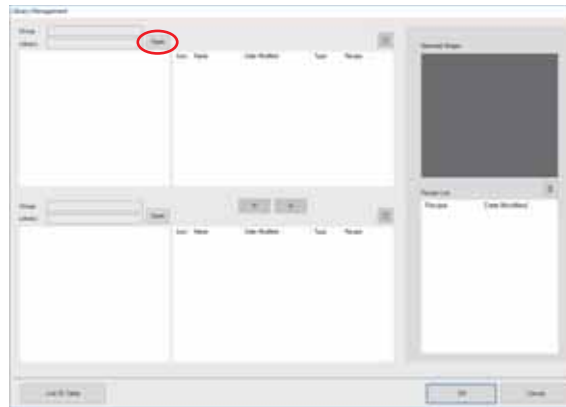


Figure 2-107 Delete Library 3

Step4: The current inspection group is displayed on Opened Groups.

To Delete Libraries from the Current Inspection Data

Click **OK**.

For Other Cases

Click **Add** and select an inspection group file (bfs file) whose library to be imported/exported.
Click **OK**.

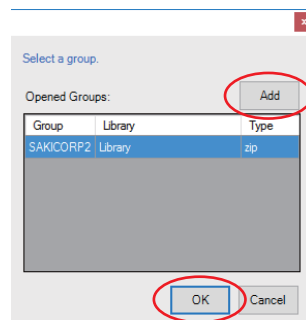


Figure 2-108 Delete Library 4

Step5: The shape whose recipe column is 0 is not used.
Select the library which is judged as unnecessary from the list.
Click  (Delete).

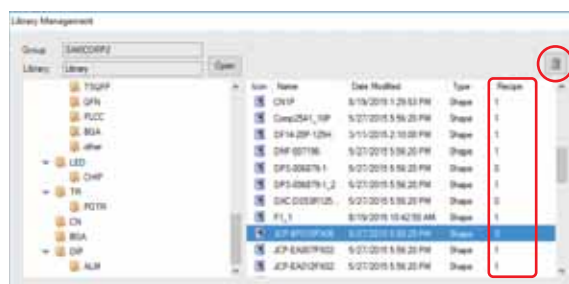


Figure 2-109 Delete Library 5

Step6: Click **Link ID Table**.

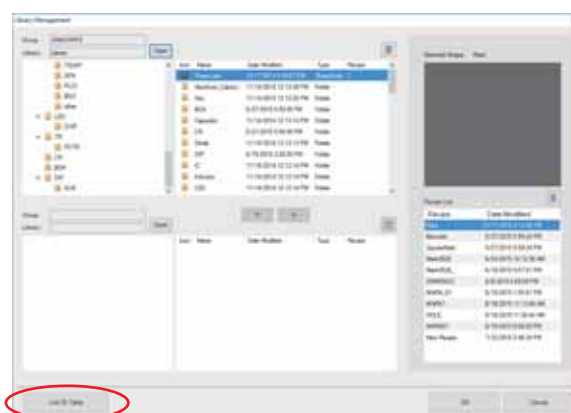


Figure 2-110 Delete Library 6

Step7: The dialog shown below appears.
Click **OK**.

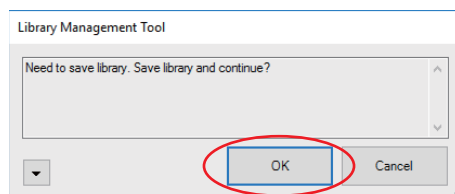


Figure 2-111 Delete Library 7

Step8: The dialog shown below appears.
Check **Show Unassigned Library**.
The unassigned libraries are displayed in list and **Delete Unassigned Library** button becomes active.



Figure 2-112 Delete Library 8

Step9: Click **Delete Unassigned Library**.

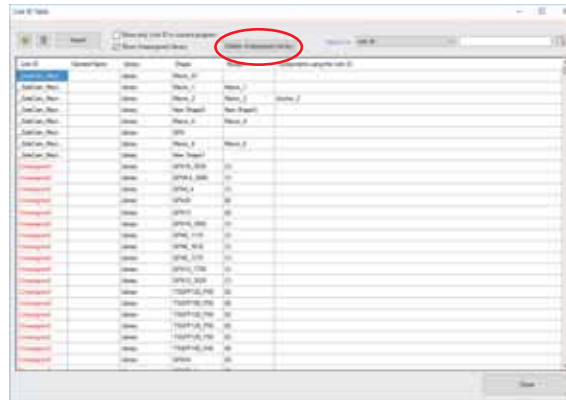


Figure 2-113 Delete Library 9

Step10: The dialog shown below appears.
Click **Yes** to delete all unassigned libraries.

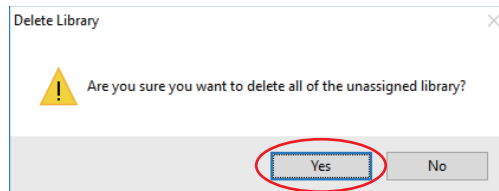


Figure 2-114 Delete Library 10

2.11.8 Importing Link IDs

Step1: Open the Library Management dialog according to **2.11.6 Importing/Exporting a Library**.

Step2: Click **Link ID Table**.

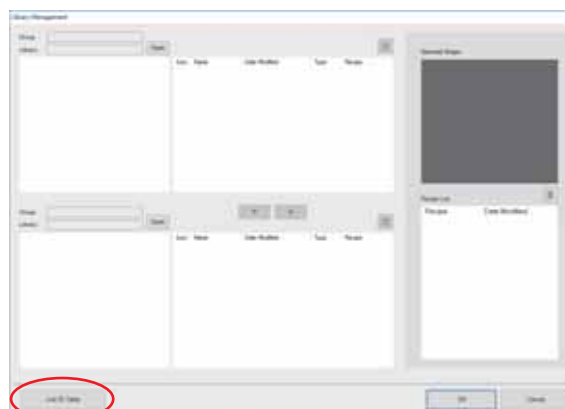


Figure 2-115 Importing Link IDs 1

Step3: The dialog shown below appears.
Click **OK**.

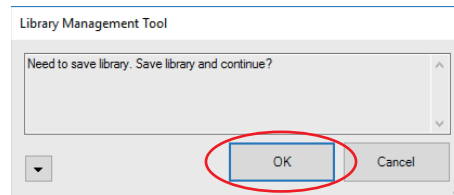


Figure 2-116 Importing Link IDs 2

Step4: Click **Import** and select an inspection group file (bfs file) to load Link IDs.
The data of the selected group file appears.
Click **Close** to close the dialog.

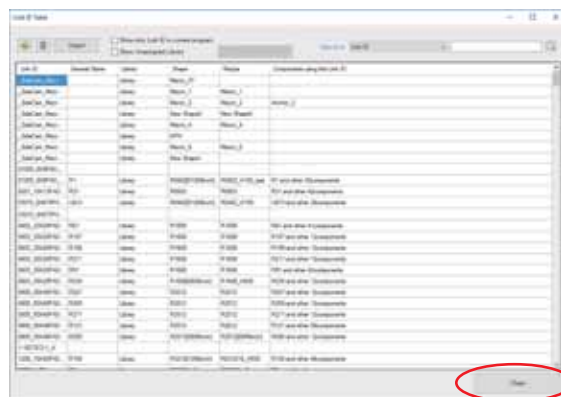


Figure 2-117 Importing Link IDs 3

2.11.9 Library Expansion

Changes to the recipes and shapes in the inspection data can be saved in the library. The library exists per group, and one library is shared and used within the same group. When using a library, changes in the recipes and shapes within the inspection data can be easily expanded to other inspection data in the same group.

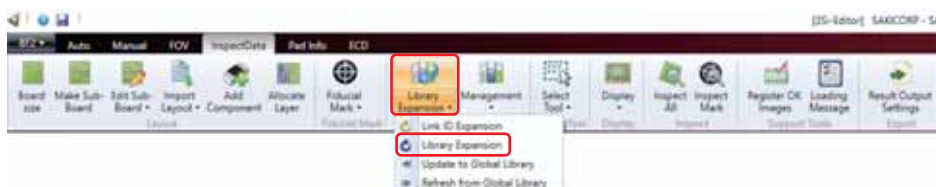


Figure 2-118 Library Expansion

Library Expansion

For components which use the same shapes and recipes within the group, use **Library Expansion** to apply the changes (e.g., algorithm sample value setting) entered in the library.

Link ID Expansion

For components which use the same Link ID within the group, use **Link ID Expansion** to import and assign a Link ID from the library. The shapes and recipes tied to the Link ID are also assigned when the Link ID is assigned.

2.12 Import Layout

This section describes how to import layout to the current inspection data.

2.12.1 Import Mount Data

Step1: From **InspectData** tab, select the **Layout** group > **Import Layout** > **Mount data**.

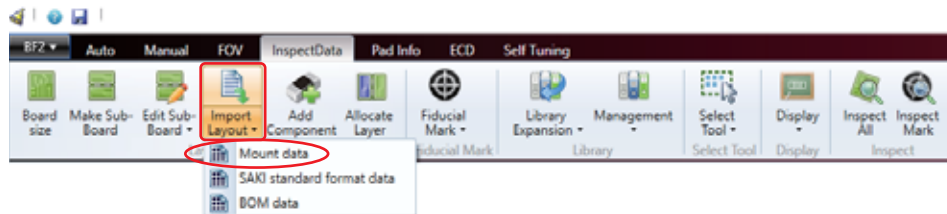


Figure 2-119 Import Mount Data 1

Step2: Select the appropriate encode setting from the **Encoding** drop-down list. Click **Select** and select a CAD data.

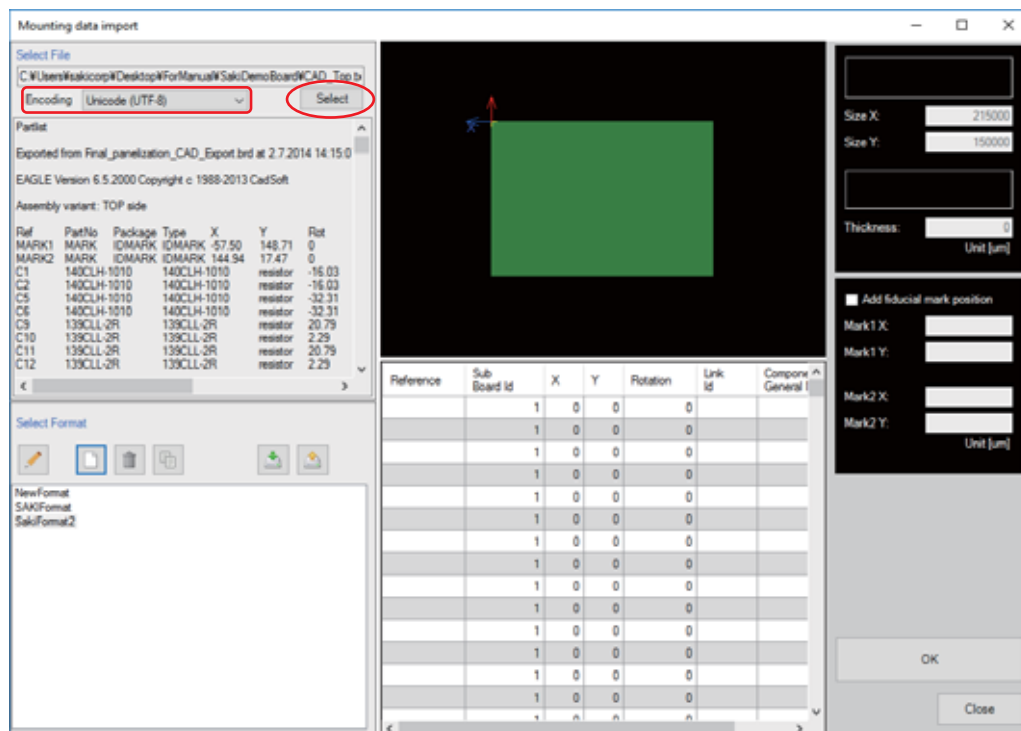


Figure 2-120 Import Mount Data 2

Step3: Make the setting according to **1.4.7 Importing Mount Data**.

2.12.2 Import SAKI Standard Format Data

Step1: From **InspectData** tab, select the **Layout** group > **Import Layout** > **SAKI standard format data**.

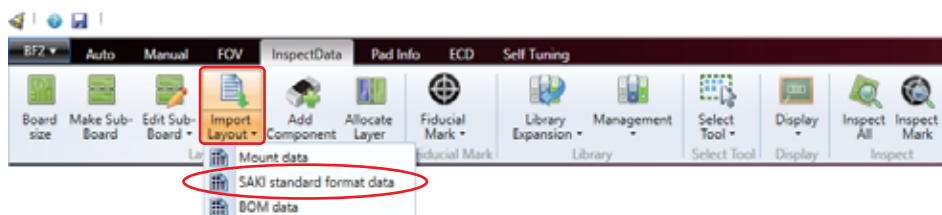


Figure 2-121 Import SAKI Standard Format Data 1

Step2: The dialog shown below appears.
Select SAKI standard format data and click **Open**.

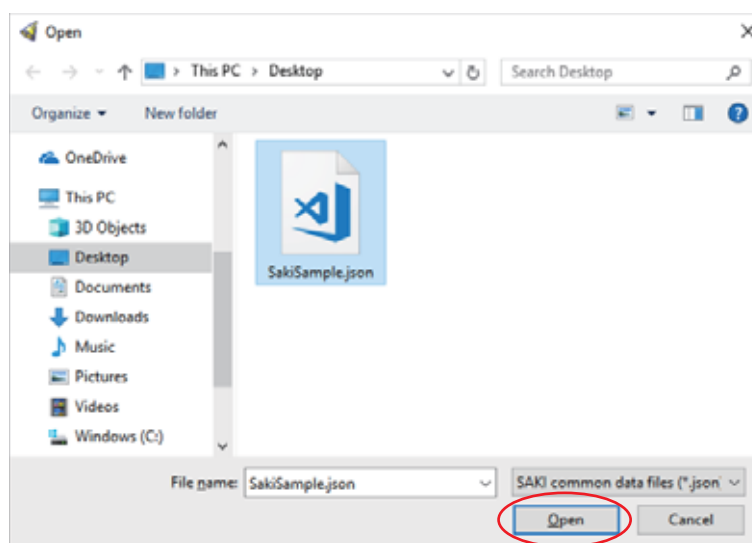


Figure 2-122 Import SAKI Standard Format Data 2

2.12.3 Import BOM Data

The BOM files can be used to delete unused component data from inspection data and change Link IDs.

Step1: From **InspectData** tab, select the **Layout** group > **Import Layout** > **BOM data**.

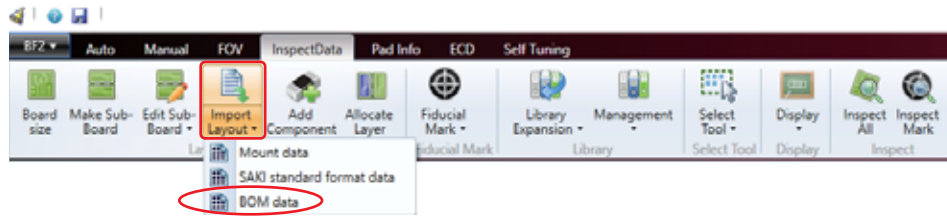


Figure 2-123 Import BOM Data 1

Step2: The dialog shown below appears.
Select a BOM file and click **OK**.

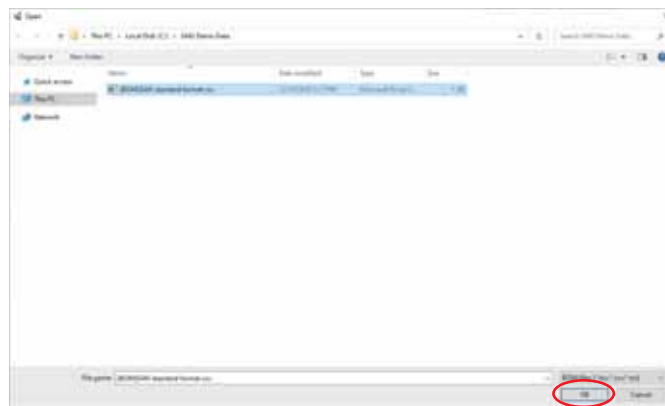


Figure 2-124 Import BOM Data 2

Step3: The dialog shown below appears.
Set each items according to Table 2-16. After all settings are complete, click **OK**.

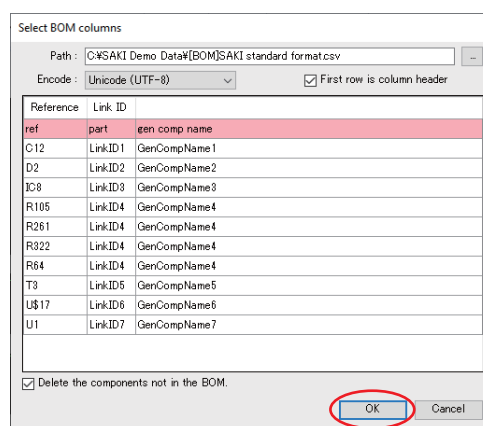


Figure 2-125 Select BOM Columns

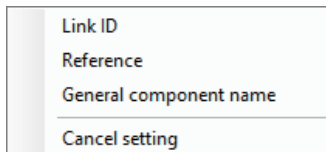
Item	Description
Path	Shows the path of the selected BOM file. Click  on the right side of the text-box to reselect the BOM file to load. Create a BOM file in comma-delimited or tab-delimited format.
Encode	Select a BOM file encoding.
First row is column header	Check to make the first line of the BOM file a header line so that it is not converted. Reference name and Link ID are also automatically set in the header part of the BOM table by the keyword of the header line. For details, refer to Table 2-17.
BOM table	Shows the contents of a BOM file. Click the column header part to select Link ID , Reference , General component name , or Cancel settings . Reference and Link ID are required to be selected. General component name is optional.  Figure 2-126 BOM Table
Delete the components not in the BOM.	Check to delete component data not included in the BOM file. If unchecked, component data not included in the BOM file remains unchanged without changing the Link ID and General component name.

Table 2-16 Descriptions

Keyword	Description
partnumber, partnumbers, part number, part numbers, part, parts, parts., component item, 部品コード, 部品コード'	Shows Link ID in the column header of the BOM table.
reference designators, reference name, reference names, references, reference, ref, ref., Location, Loc, Loc., 回路番号, 回路, 部品名, 部品, リファレンス, リファレンス	Shows Reference name in the column header of the BOM table.

Table 2-17 BOM File Header Part Keyword

3 Manual Inspection

3.1 Inspect Mark

This section describes the procedures for the mark inspection.

By making fiducial marks and perform **Inspect Mark**, component data positions are adjusted. By setting two marks on a PCB image, component data positions and angles are adjusted when starting an inspection. If three marks are set, warpage of a PCB is also adjusted.

NOTE

Inspect Mark is included in the process of **Inspect All**.

If **Inspect All** is going to be performed, it is not necessary to perform **Inspect Mark**.

3.1.1 Mark Inspection Procedure

Step1: Select the **InspectData** tab.



Figure 3-1 InspectData

Step2: From the ribbon, select the **Inspect** group > **Inspect Mark** to start the inspection.



Figure 3-2 Inspect Mark

3.1.2 Disable Mark Adjustment

Step1: Select the **InspectData** tab.



Figure 3-3 InspectData

Step2: From the ribbon, select the **Fiducial Mark** group > **Fiducial Mark** > **Disable Mark Adjustment**.

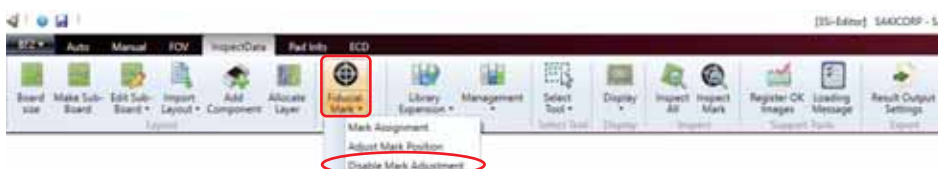


Figure 3-4 Disable Mark Adjustment

Step3: Disable the mark adjustment.

3.2 PCB Inspection

This section describes the procedure for manually starting an inspection and displaying the inspection result. A PCB inspection can be performed after making an inspection data.

3.2.1 Starting an Inspection

Step1: Select the **InspectData** tab.



Figure 3-5 InspectData

Step2: From the ribbon, select the **Inspect** group > **Inspect All** to start the inspection.



Figure 3-6 Inspect All

Step3: The inspection results are displayed on **Board Viewer**, **Component List**, and **LookUp Table**.

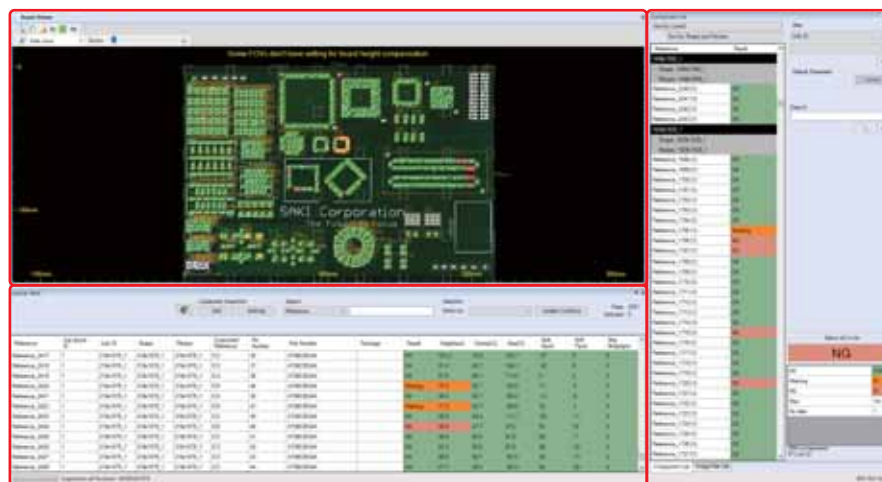


Figure 3-7 Inspection Results

3.2.2 Inspection Result Display

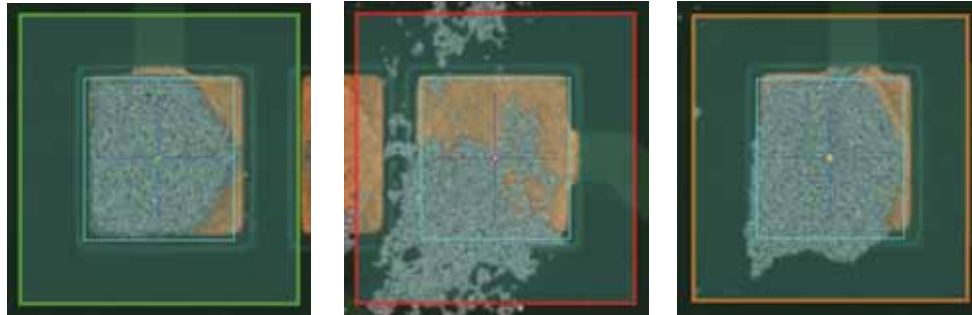
After an inspection, inspection results appear as shown below.

Board Viewer

OK components data are displayed in green rectangles (with  mark on the center of a component).

NG components data are displayed in red rectangles (with  mark on the center of a component).

Warning components are displayed in orange rectangles (with  mark on the center of a component).



Judged as OK

Judged as NG

Judged as Warning

Figure 3-8 Board Viewer

Component List

If an inspection result is OK (good), **OK** is displayed in green. If an inspection result is NG (detect, bad), **NG** is displayed in red. If an inspection result is Warning, **Warning** is displayed in orange.

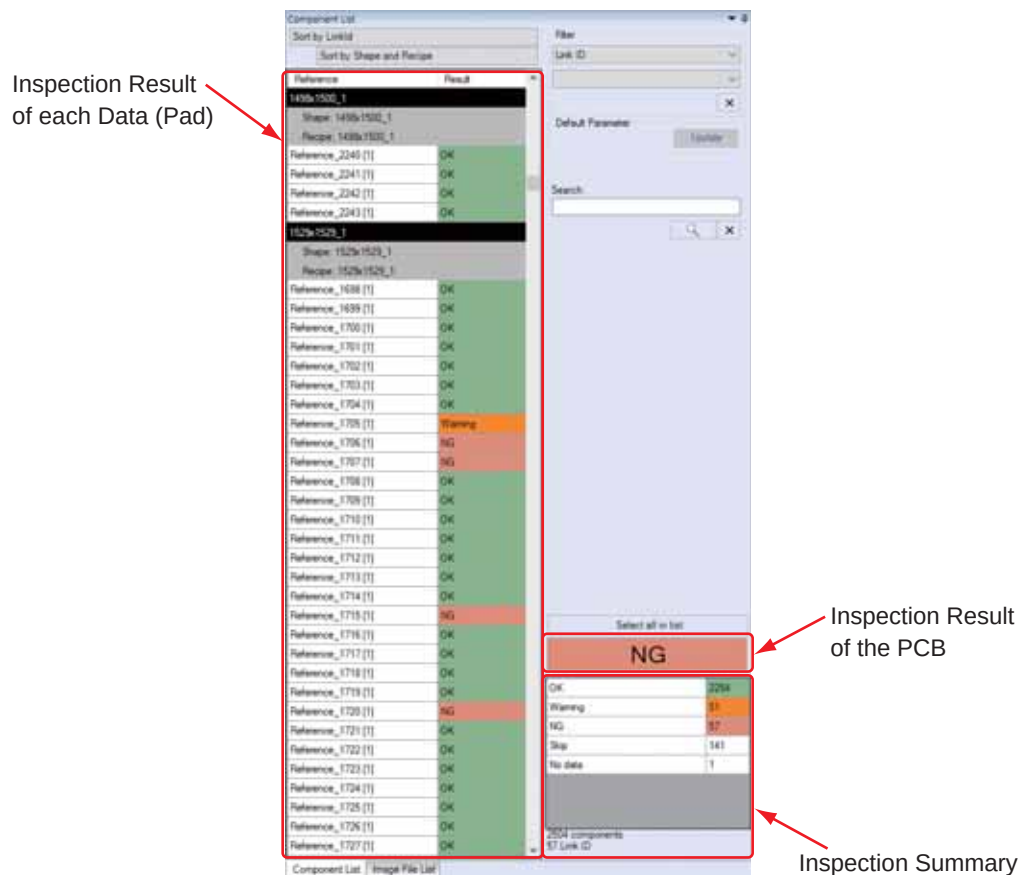


Figure 3-9 Component List

LookUp Table

The color of the cell changes according to the inspection result.

If an inspection result is OK (good), displayed green. If an inspection result is NG (detect, bad), displayed red. If an inspection result is Warning, displayed orange.

Reference	Sub Board ID	Link ID	Shape	Recipe	Component Reference	Pin Number	Part Number	Package	Result	Height[um]	Volume[mm³]	Area[mm²]	Drift X[um]	Drift Y[um]	Max Bridge[um]
Reference_2417	1	219b1079_1	219b1079_1	219b1079_1	IC3	36	A790C3E34A		OK	101.2	70.6	100.1	-27	0	0
Reference_2418	1	219b1079_1	219b1079_1	219b1079_1	IC3	37	A790C3E34A		OK	91.8	64.7	100.1	-52	0	0
Reference_2419	1	219b1079_1	219b1079_1	219b1079_1	IC3	38	A790C3E34A		OK	97.6	40.1	110.0	0	4	0
Reference_2420	1	219b1079_1	219b1079_1	219b1079_1	IC3	40	A790C3E34A		Warning	77.3	52.1	100.0	71	4	0
Reference_2421	1	219b1079_1	219b1079_1	219b1079_1	IC3	39	A790C3E34A		OK	95.4	62.7	100.4	-12	4	0
Reference_2422	1	219b1079_1	219b1079_1	219b1079_1	IC3	47	A790C3E34A		Warning	77.2	51.7	100.0	42	1	0
Reference_2423	1	219b1079_1	219b1079_1	219b1079_1	IC3	40	A790C3E34A		OK	99.5	68.6	111.7	-24	-10	0
Reference_2424	1	219b1079_1	219b1079_1	219b1079_1	IC3	40	A790C3E34A		NG	68.9	47.7	97.0	91	14	0
Reference_2426	1	219b1079_1	219b1079_1	219b1079_1	IC3	41	A790C3E34A		OK	96.4	63.4	97.9	64	1	0
Reference_2428	1	219b1079_1	219b1079_1	219b1079_1	IC3	42	A790C3E34A		OK	91.3	54.8	97.9	48	-19	0
Reference_2427	1	219b1079_1	219b1079_1	219b1079_1	IC3	43	A790C3E34A		OK	99.0	58.7	101.0	34	-11	0
Reference_2428	1	219b1079_1	219b1079_1	219b1079_1	IC3	44	A790C3E34A		OK	97.7	66.0	102.8	62	-20	0

Figure 3-10 LookUp Table

4 Parameter Editor

4.1 Parameter Setting Function

Parameters of inspection algorithm can be set and they can be applied to data at once.

The functions shown below are used for setting default values of parameters to be applied at data creation (*1) and parameter applying at tuning (*2).

NOTE

(*1) For details, refer to **4.2 Default Value Settings of Algorithm**.

(*2) For details, refer to **4.3 Applying the Parameter**.

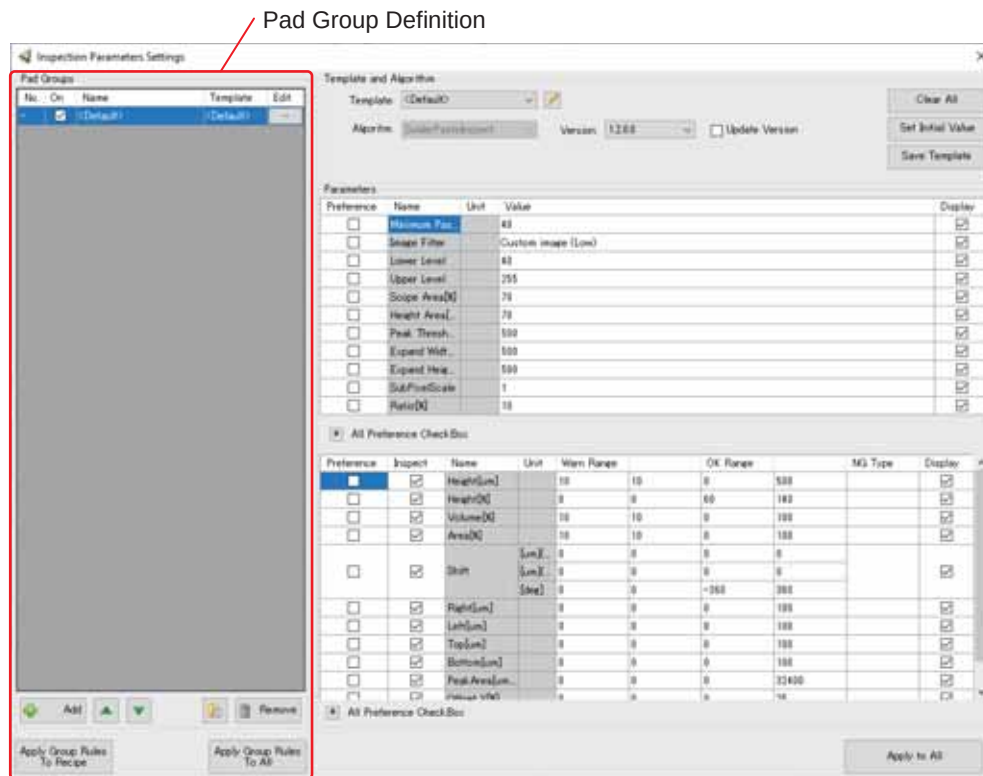


Figure 4-1 Parameter Setting Function

Item	Description
Pad Group Definition	Set pad groups. Refer to 4.5 Pad Group Settings .
Template	Select a template. Refer to 4.4 Storing the Multiple Parameter .
Algorithm	Select an algorithm to be edit parameter.
Version	Select the version of algorithm to be register.
Update Version	Turn on the check box to apply the selected algorithm version.
Clear All	Does not display the parameter.
Set Initial Value	Enter the default value of algorithm which are saved.
Save	Save the Parameter.

Item		Description
Parameters	Preference	Set the applying parameter or not. Checked item will assigning the parameter.
	Name	Display the name of setting items. For details, refer to Part III Inspection Algorithm .
	Unit	Display the unit of setting items.
	Value	Display the value of setting items.
	Display	Set the displaying parameter on Recipe edit screen. Does not display the unchecked items.
	Preference	Set the applying parameter or not. Checked item will assigning the parameter.
	Inspect	Specify the inspect or not. Unchecked item will not inspect.
	Name	Display the name of setting items. For details, refer to Part III Inspection Algorithm .
	Unit	Display the unit of setting items.
	Warn Range	Set the threshold value when sample value to be Warning judgment. For details, refer to Part VII Appendices and References .
	OK Range	Set the threshold value when sample value to be OK judgment.
	NG Type	Set the NG type.
	Display	Specify the displays on Recipe edit screen. Unchecked item will not displays.
	All Preference CheckBox	Click to turn all of Preference on or off.

Table 4-1 Descriptions

4.2 Default Value Settings of Algorithm

The default value of parameters in inspection algorithm are can be changed.

NOTE

When creating new data, the default value set here will be used.
However, if a library already exists in the group, the library takes precedence.

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

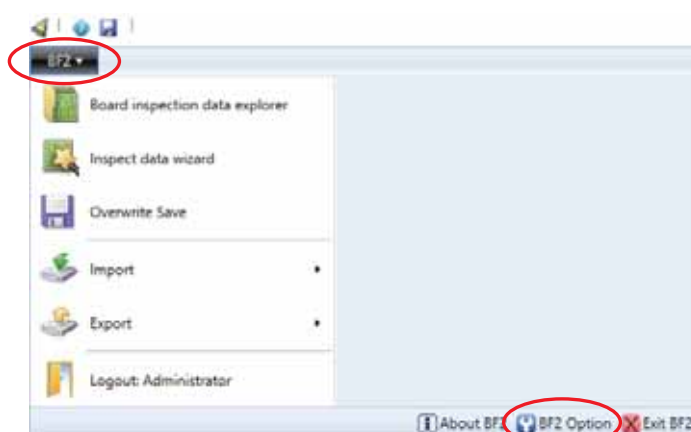


Figure 4-2 Default Value Settings of Algorithm 1

Step2: The dialog box shown below appears.
Click **Inspection Data**.

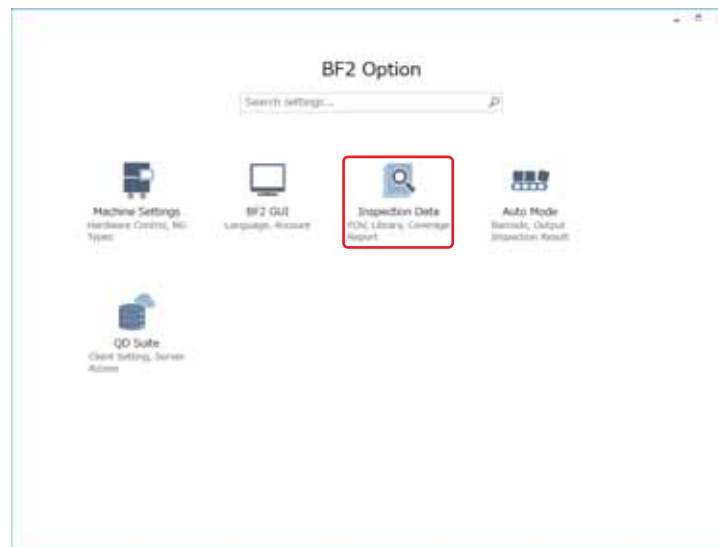


Figure 4-3 Default Value Settings of Algorithm 2

Step3: Select **Algorithm Default Parameters**.
The dialog box shown below appears.

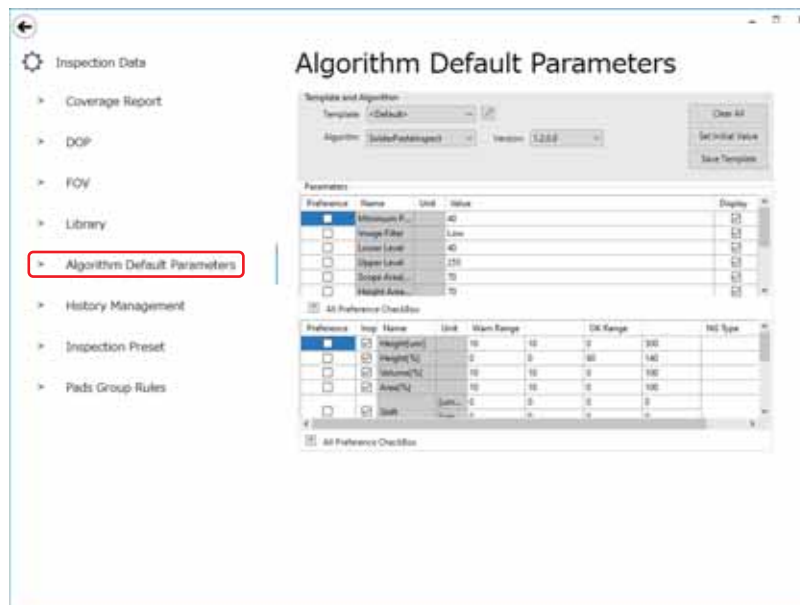


Figure 4-4 Default Value of Settings Algorithm 3

Step4: Set **<Default>** for **Template** and select the algorithm to set for **Algorithm**.
Set the Default value of algorithm according to Table 4-1 of **4.1 Parameter Setting Function**.

NOTE For details of algorithms, refer to **Part III Inspection Algorithm**.

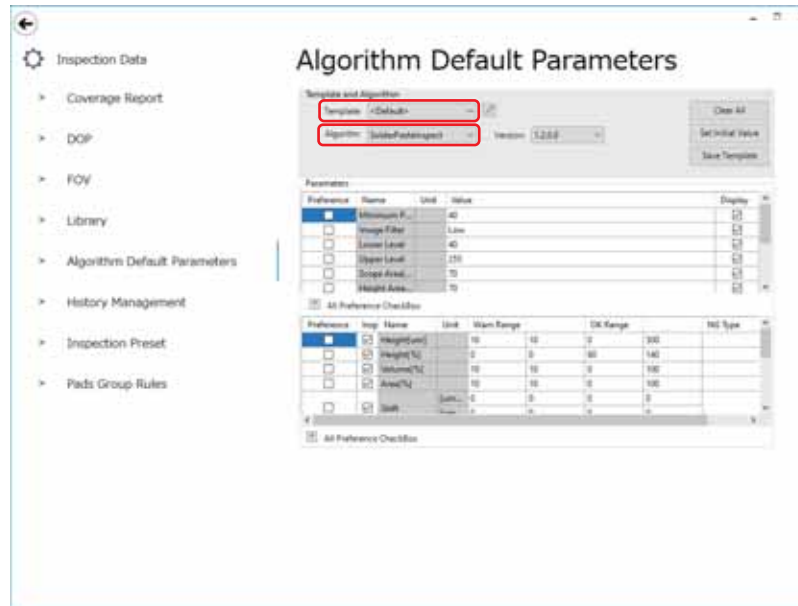


Figure 4-5 Default Value Settings of Algorithm 4

Step5: After all settings are completed, click **Save Template** to save as default settings.

NOTE Specified values are saved in **C:\BF2\SystemData\DefaultAlgParam.xml**.
If necessary, make a backup.

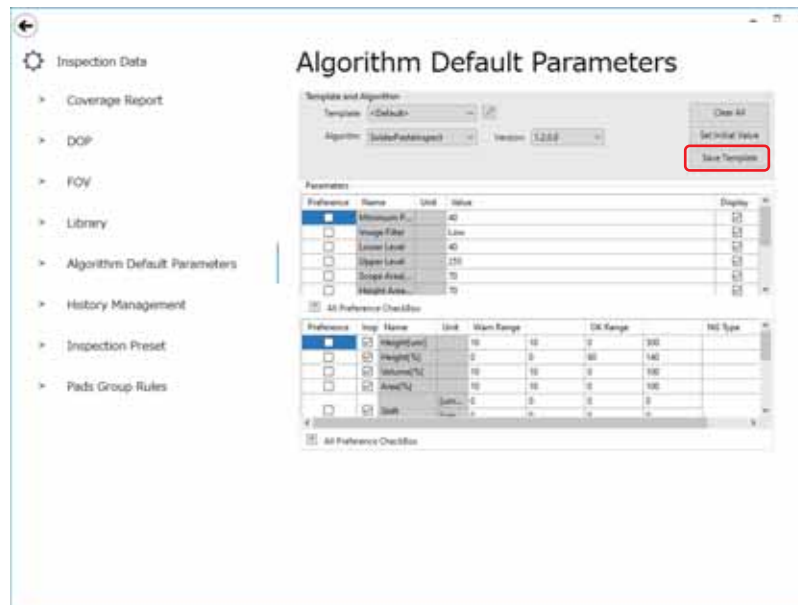


Figure 4-6 Default Value Settings of Algorithm 5

Step6: After all the settings are completed, click **X** on the upper right side of the dialog box and click **Yes** to close dialog box.

NOTE Specified default settings does not apply to created inspection data.
To apply the settings, refer to **4.3 Applying the Parameter**.

4.3 Applying the Parameter

Here describes how to apply the default parameter of inspection algorithm to created inspection data.

NOTE To specify the default parameter of inspection algorithm, refer to **4.2 Default Value Settings of Algorithm**.

4.3.1 Pad Data

Here describes how to apply parameters to pad data for solder paste inspection.

Applying to all data

Step1: Select the **InspectData** tab.

Confirm that no data is selected, click **Update Condition** on the **LookUp Table**.

NOTE To deselect the data, click the **Board Viewer** where no data is exist.

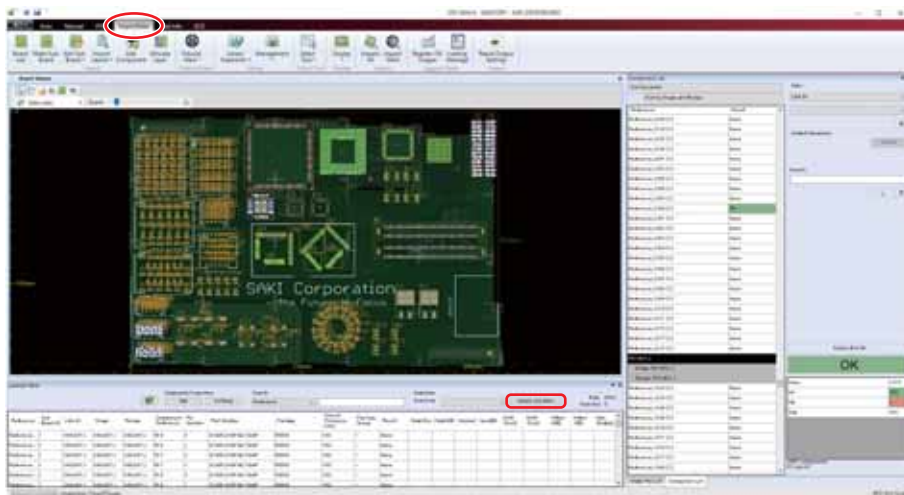


Figure 4-7 Applying Parameter - Pad Data 1

Step2: The dialog box shown below appears.

Set the parameter of algorithm according to Table 4-1 of **4.1 Parameter Setting Function**, and click **Apply to All**.

Parameters are applied to all data.

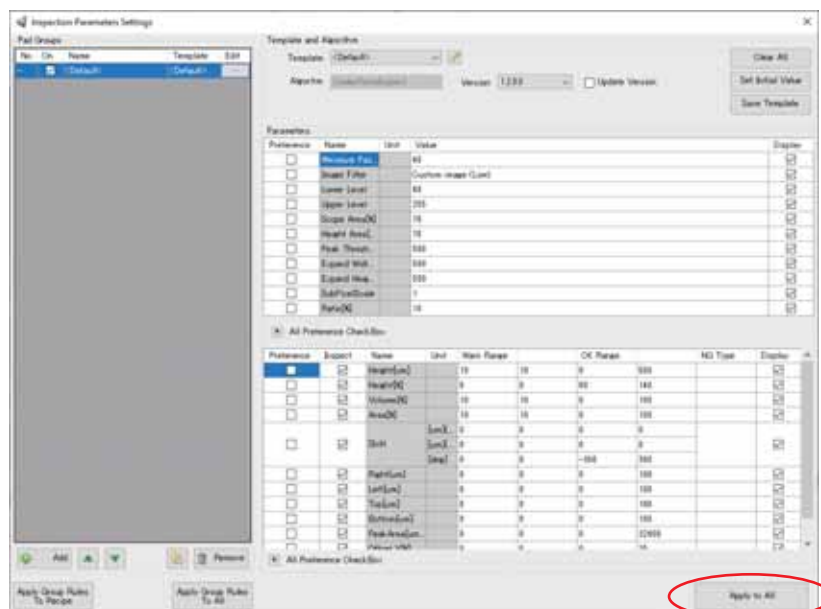


Figure 4-8 Applying Parameter - Pad Data 2

Applying to selected data

Step1: Select the **InspectData** tab.

Select the data to apply parameters on the **LookUp Table**.

NOTE For details of **LookUp Table**, refer to **2.2 List Display of Data (Pad Data)**.

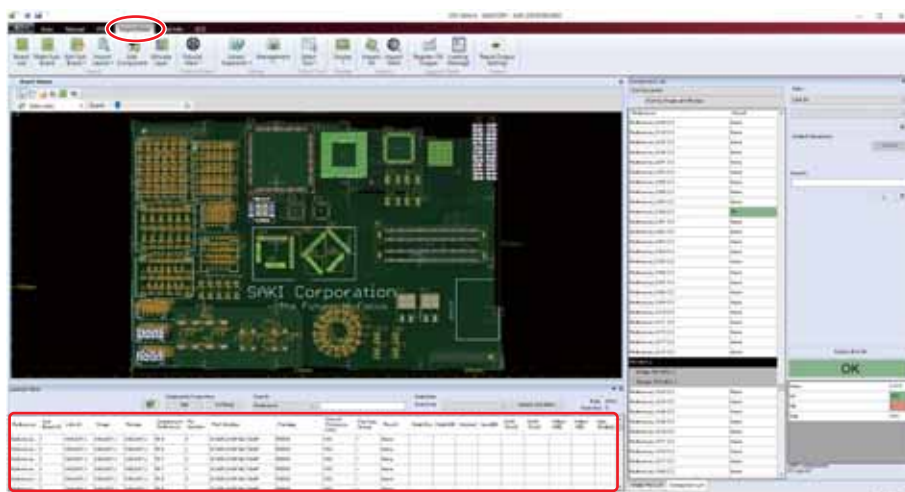


Figure 4-9 Applying Parameter - Pad Data 3

Step2: Click **Update Condition**.

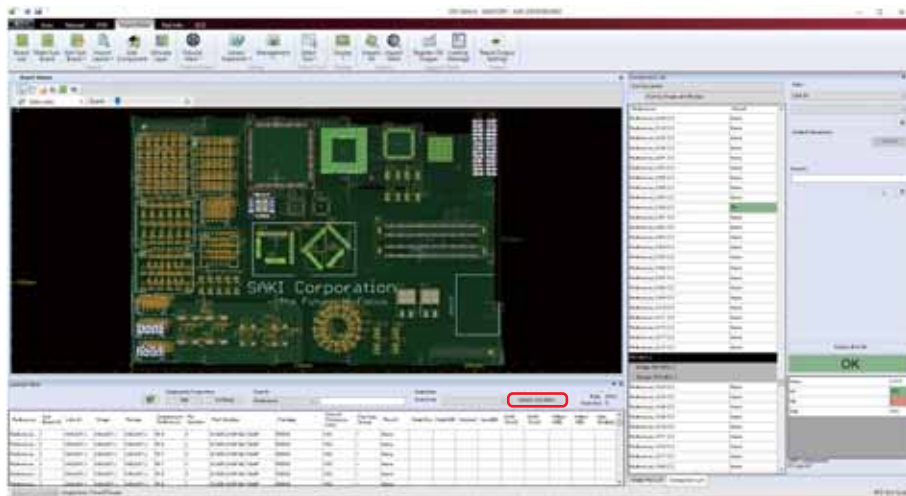


Figure 4-10 Applying Parameter - Pad Data 4

Step3: The dialog box shown below appears.

Set the parameter of algorithm according to Table 4-1 of **4.1 Parameter Setting Function**.

Select the applying method according to Table 4-2.

Parameters are applied to pad data with selected the applying method.

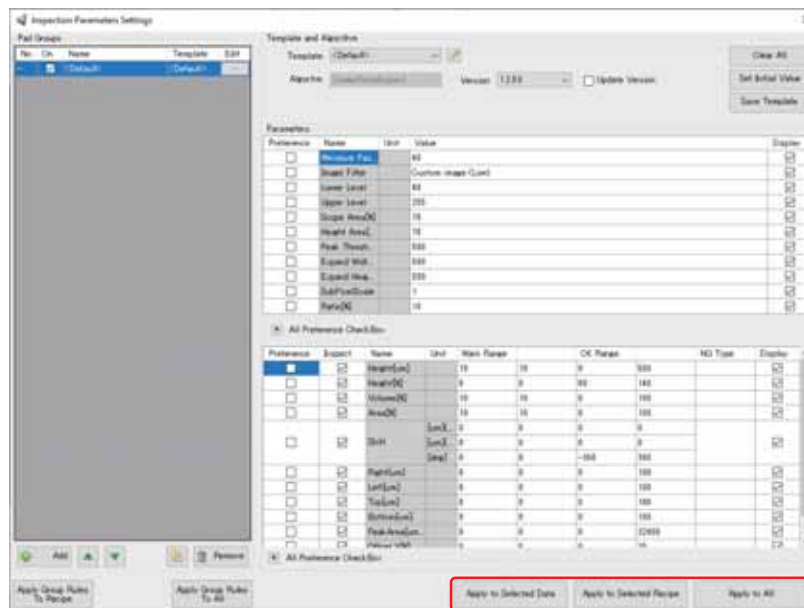


Figure 4-11 Applying Parameter - Pad Data 5

Item	Description
Apply to Selected Data	Applies the parameters only for the selected data. After applying, local save is performed.
Apply to Selected Recipe	Applies the parameters to the recipe of the selected data.
Apply to All	Applies the parameters to all the data.

Table 4-2 Descriptions

4.3.2 Other Data

Here describes how to apply parameters to data other than pad data.

NOTE Parameters can be applied to the pad data in the same procedure.

Applying to all data which algorithm is assigned

Step1: Select the **InspectData** tab. Select from **Component List** window, **Filter > Algorithm**.

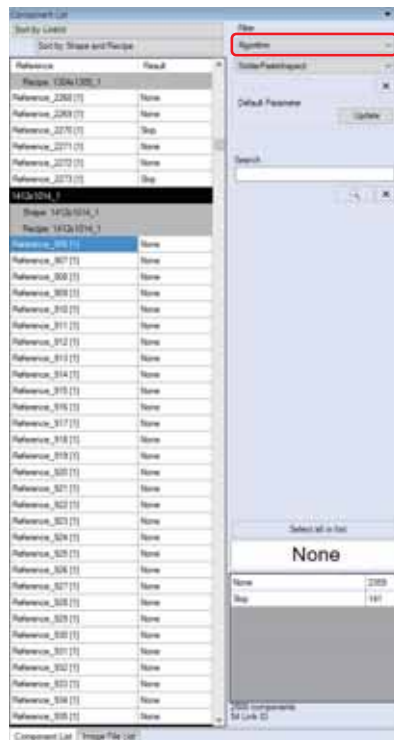


Figure 4-12 Applying Parameter 1

Step2: Select algorithm to applying parameter.

Confirm that no data is selected, click **Update** on **Default Parameter**.

NOTE To deselect the data, click the **Board Viewer** where no data is exist.

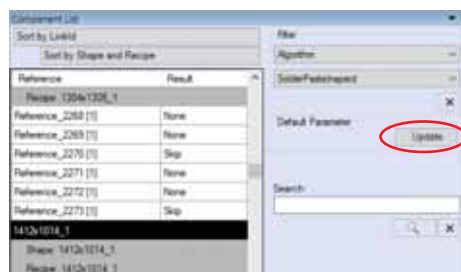


Figure 4-13 Applying Parameter 2

Step3: The dialog box shown below appears.

Set the parameter of algorithm according to Table 4-1 of **4.1 Parameter Setting Function**, and click **Apply to All**.

Apply the value to all data which algorithm is assigned.

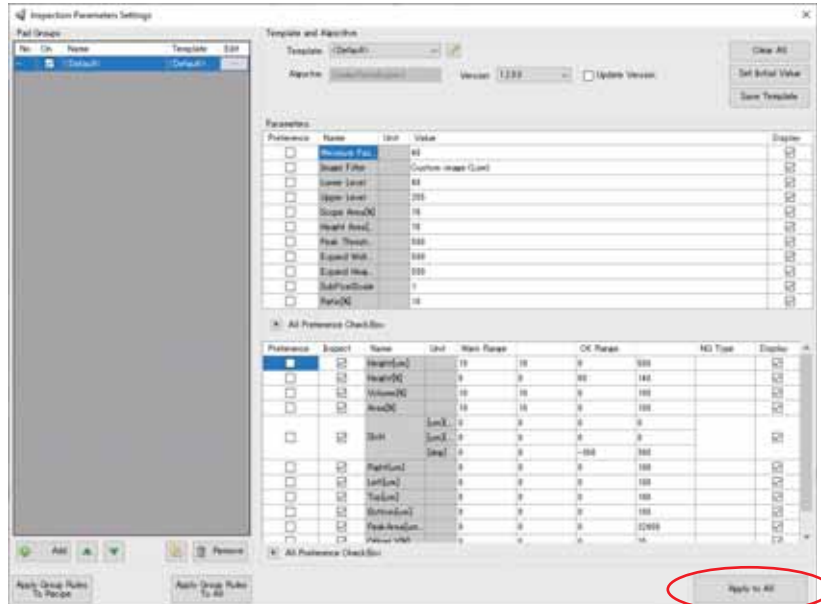


Figure 4-14 Applying Parameter 3

Applying to selected data

Step1: Select the **InspectData** tab. Select from **Component List** window, **Filter** > **Algorithm**.

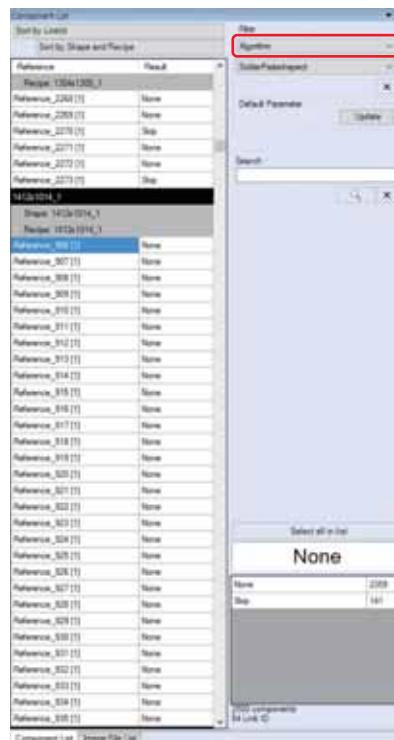


Figure 4-15 Applying Parameter 4

Step2: Select the Algorithm to applying the parameter.

Select the data to applying the parameter, and click **Update** on **Default Parameter**.

NOTE

Selecting the data is from clicking data on **Board Viewer** or from **Component list** window. For details, refer to **2.3 List Display of Data (All Data)**.

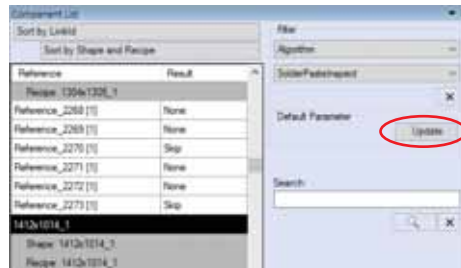


Figure 4-16 Applying Parameter 5

Step3: The dialog box shown below appears.

Set the parameter of algorithm according to Table 4-1 of **4.1 Parameter Setting Function**.

Select the applying method according to Table 4-3.

Parameters are applied to data with selected the applying method.

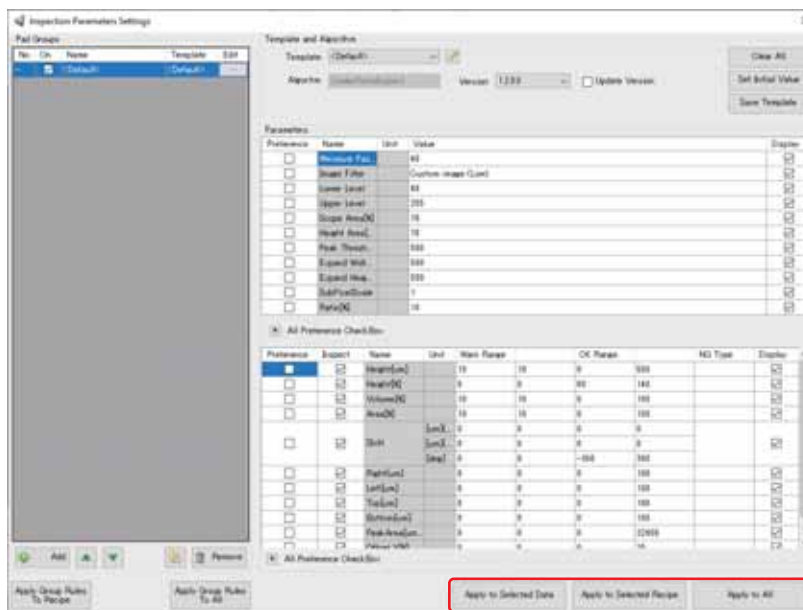


Figure 4-17 Applying Parameter 6

Item	Description
Apply to Selected Data	Applies the parameters only for the selected data. After applying, local save is performed.
Apply to Selected Recipe	Applies the parameters to the recipe of the selected data.
Apply to All	Applies the parameters to all the data.

Table 4-3 Descriptions

4.4 Storing the Multiple Parameter

Inspection algorithm can store the multiple parameters in addition to default value.

4.4.1 Register a Template

Template is parameters set which included in inspection algorithm.

Here describes how to register a template.

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

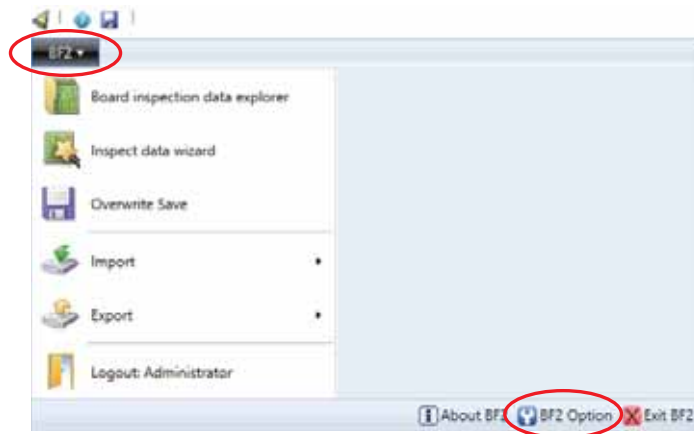


Figure 4-18 Register a Template 1

Step2: The dialog box shown below appears.
Click **Inspection Data**.

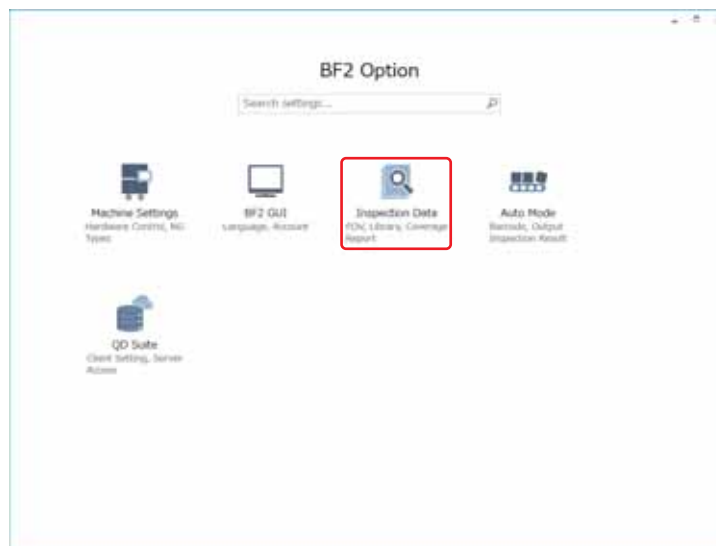


Figure 4-19 Register a Template 2

Step3: Select **Algorithm Default Parameters**.

The dialog box shown below appears. Click the  button.

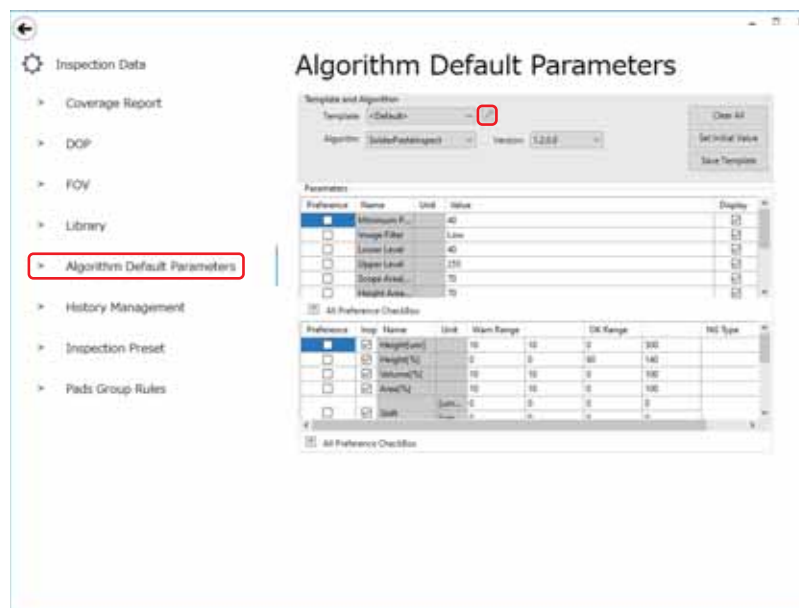


Figure 4-20 Register a Template 3

Step4: The dialog box shown below appears.

Enter a template name in the text-box and click **Add Template**.

NOTE The existing template name cannot be registered.

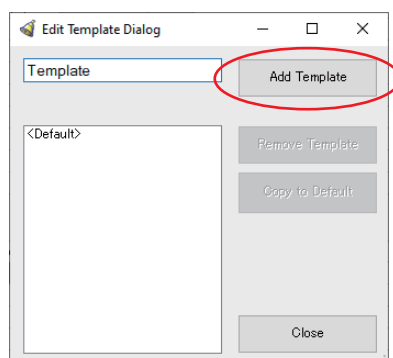


Figure 4-21 Register a Template 4

Step5: The template name entered in Step4 is shown in the list. Click **Close** to close the dialog box.

NOTE The existing template name cannot be registered.

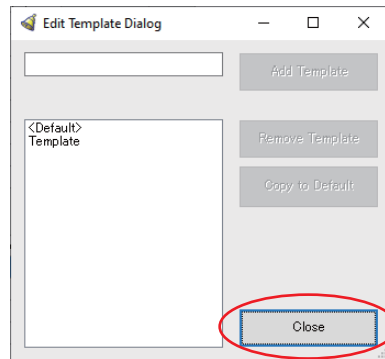


Figure 4-22 Register a Template 5

Step6: Select the template which entered in Step4 from **Template** and set the parameters. For details, refer to Table 4-1 of **4.1 Parameter Setting Function**.

NOTE Specified value is saved in **C:\BF2\SystemData\DefaultAlgParamType**
If necessary, make a backup.

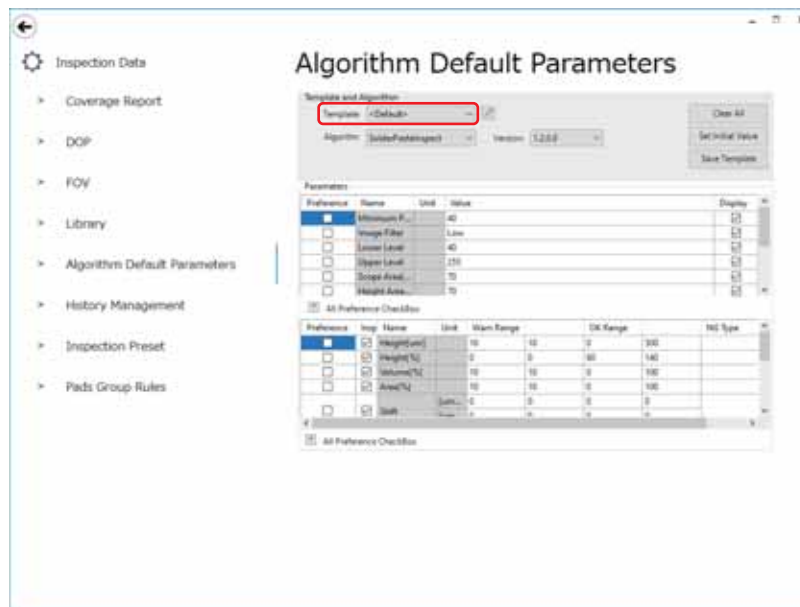


Figure 4-23 Register a Template 6

4.4.2 Editing Templates

Registered templates can be delete or copy to default parameter.

Delete a Template

Step1: Proceed Step1 to Step3 of **4.4.1 Register a Template**.

Step2: The dialog box shown below appears. Select the template to delete from the list and click **Remove Template**. The selected template is deleted.

NOTE

The template registered in the pad group cannot be deleted.

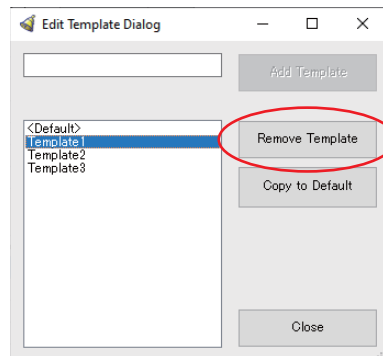


Figure 4-24 Editing Templates 1

Copy a Template to Default Parameter

Step1: Proceed Step1 to Step3 of **4.4.1 Register a Template**.

Step2: The dialog box shown below appears. Select the template to copy from the list and click **Copy To Default**. The selected template is copied to **<Default>**.

NOTE

Proceeding this procedure, default value is overwritten.

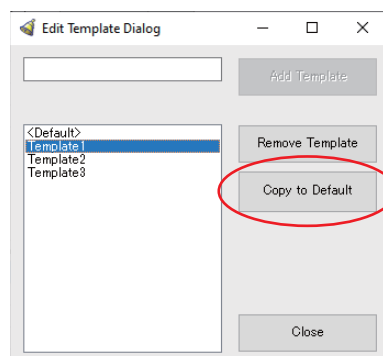


Figure 4-25 Editing Templates 2

4.4.3 Register the Recipe Parameter

Parameters set in Recipe editing screen is able to registered as a template.

Step1: Open the Recipe editing screen according to **2.8 Assigning a Recipe**.

Step2: From **Window Tree**, select the parameters of inspection window to register as a template, click **Save As Default**.



Figure 4-26 Register Recipe Parameters as a Template 1

Step3: The dialog box shown below appears. Enter a template name into the text-box and click **Add Template**. The template is registered in the list.

NOTE

The existing template name cannot be registered.

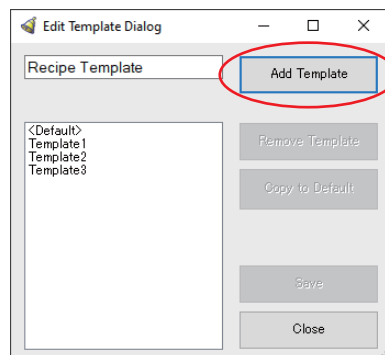


Figure 4-27 Register Recipe Parameters as a Template 2

Step4: Click **Save** to save the template.

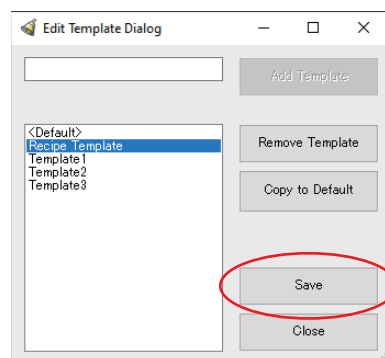


Figure 4-28 Register Recipe Parameters as a Template 3

4.5 Pad Group Settings

This section describes how to apply the default parameter of inspection algorithm to created inspection data using pad groups.

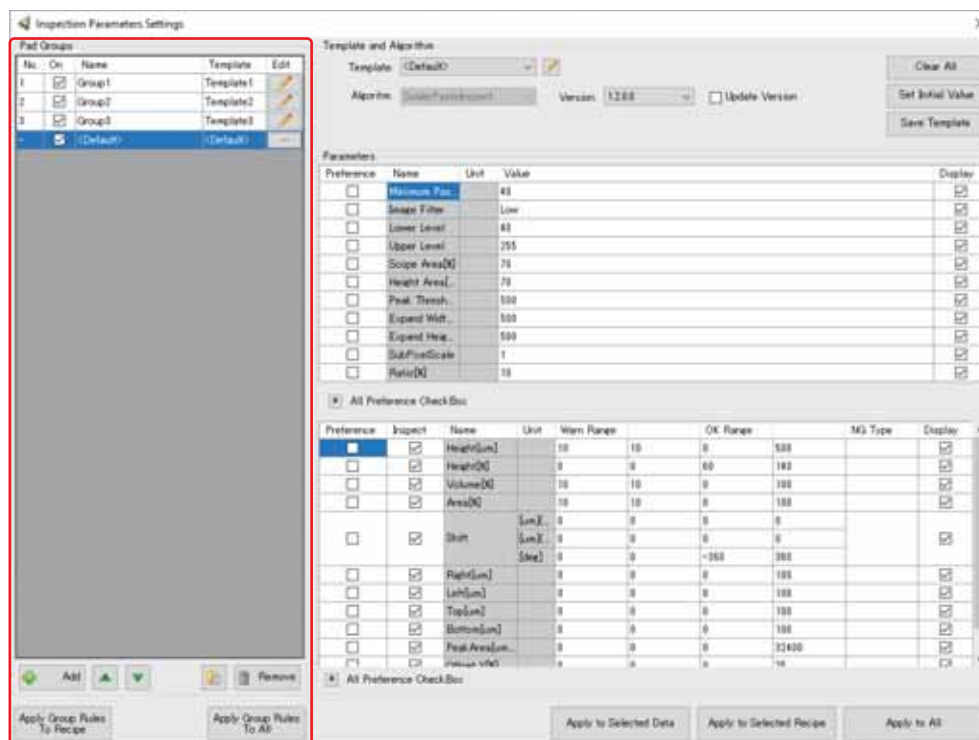


Figure 4-29 Pad Group Settings

Item		Description
Pad Groups	No.	Shows the priority of pad group.
	On	Check to enable pad group rules.
	Name	Shows pad group names
	Template	Shows templates applied to pad groups.
	Edit	Click  to show the Add/Modify Pad Group dialog box for the line clicked. <Default> cannot be edited.
		Click to show the Add/Modify Pad Group dialog box.
		Changes the priority of the selected pad group. <Default> is fixed to the lowest.
	Export	Exports pad group settings to the specified folder.
	Import	Imports pad group settings in the specified folder.
		Removes the selected pad group. <Default> cannot be removed.
Apply Group Rules To Recipe		Click to apply the rules for the created pad group to the recipe selected in LookUp Table.
Apply Group Rules To All		Click to apply rules for the created pad group to all recipes.

Table 4-4 Descriptions

4.5.1 Adding a Pad Group

Here describes how to add a pad group.

- Step1: Select the **InspectData** tab.
Click **Update Condition** on **LookUp Table**.

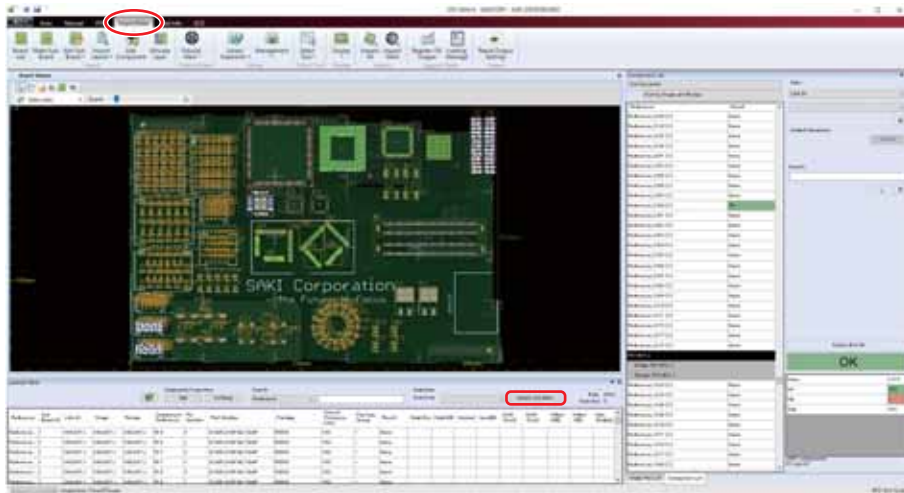


Figure 4-30 Adding a Pad Group 1

- Step2: The dialog box shown below appears.
Click **Add**.

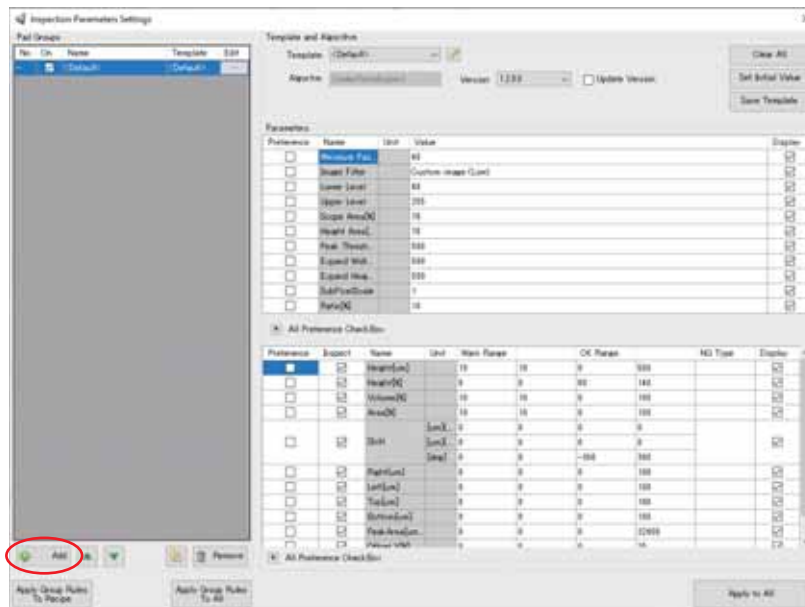


Figure 4-31 Adding a Pad Group 2

Step3: The dialog box shown below appears.
Set pad group rules according to Table 4-5.



Figure 4-32 Adding a Pad Group 3

Item	Description
Priority	Set the priority of pad group. Apply group rules of a pad group in order from the lowest priority numbers.
Group Name	Set the pad group name.
Parameters Template	Select the template to apply to a pad group
	Adds a template. Refer to 4.4 Storing the Multiple Parameter .
Active	Check to enable group rules of a pad group.
Pad Size	Click to add group rules for pad size. Select AreaSize , PadLength , or PadWidth to set Upper and Lower .
Pad Shape	Click to add group rules for pad shape. Select Circle , Rectangle , or Other .
Stencil	Click to add group rules for stencil thickness. Set Upper and Lower of StencilHeight ,
Save	Click to save pad group settings.

Table 4-5 Descriptions

NOTE Operator is fixed to **And**.

NOTE Click  to delete the added group rules.

Step4: After settings are complete, click **Save** to save the settings.



Figure 4-33 Adding a Pad Group 4

4.5.2 Applying Group Rules of the Pad Group

Here describes how to apply group rules of the created pad group.

Applying group rules to all data

Step1: Select the **InspectData** tab.

Make sure no data is selected and click **Update Condition** on **LookUp Table**.

NOTE

If any data is selected, click where no data exists on Board Viewer to cancel it.

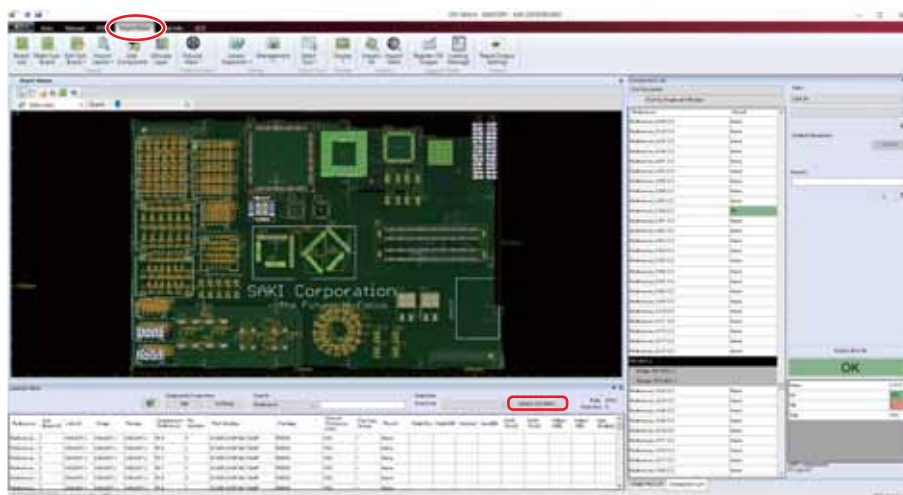


Figure 4-34 Applying Group Rules of the Pad Group 1

Step2: The dialog box shown below appears.

Set pad groups according to Table 4-5 of **4.5.1 Adding a Pad Group**, and click **Apply Group Rules To All**.

Group rules are applied to all data.

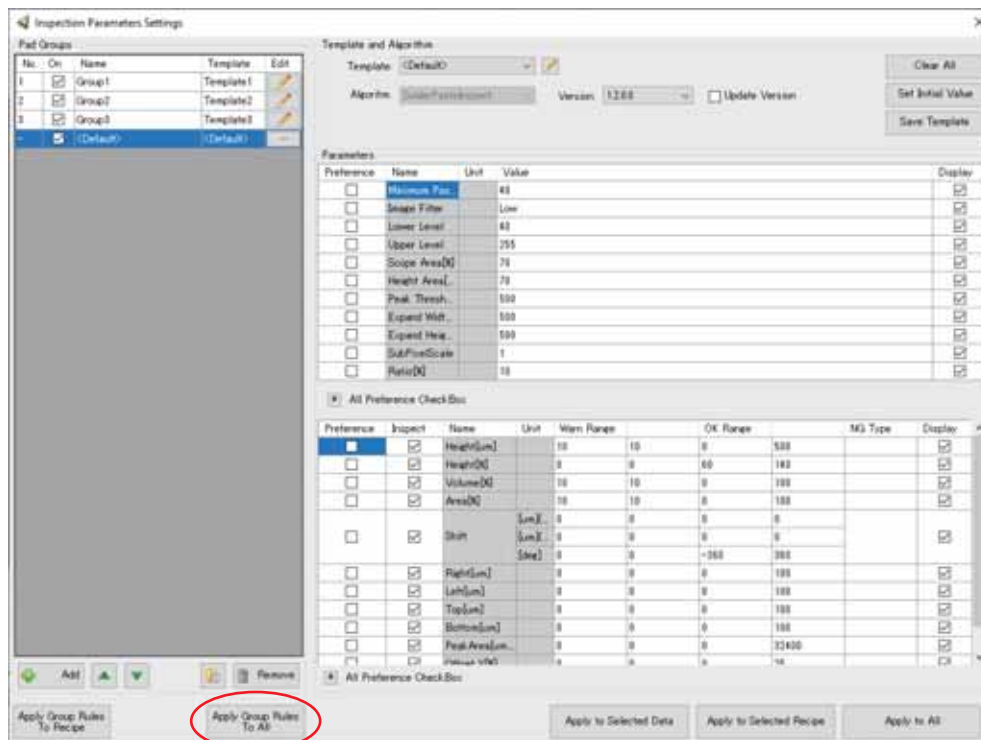


Figure 4-35 Applying Group Rules of the Pad Group 2

Applying group rules to the selected data

Step1: Select the **InspectData** tab.

Select the data to apply parameters on **LookUp Table**.

NOTE For details on **LookUp Table**, refer to **2.2 List Display of Data (Pad Data)**.

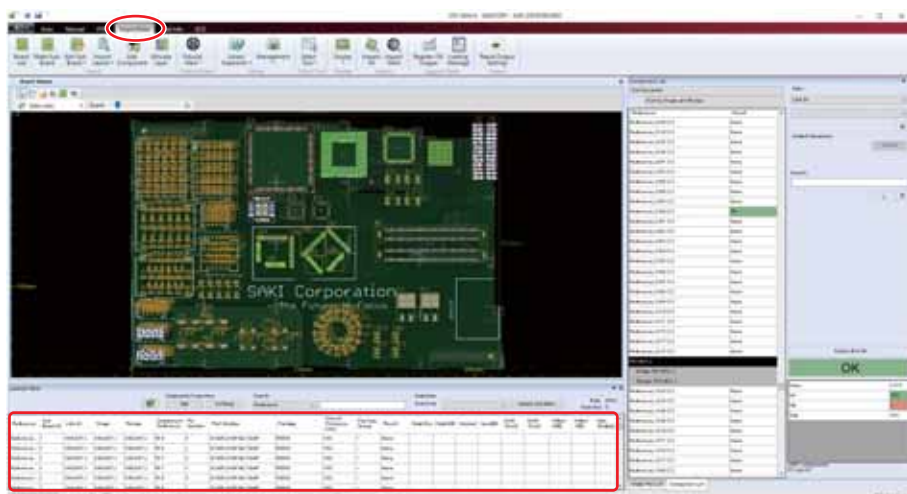


Figure 4-36 Applying Group Rules of the Pad Group 3

Step2: Click **Update Condition**.

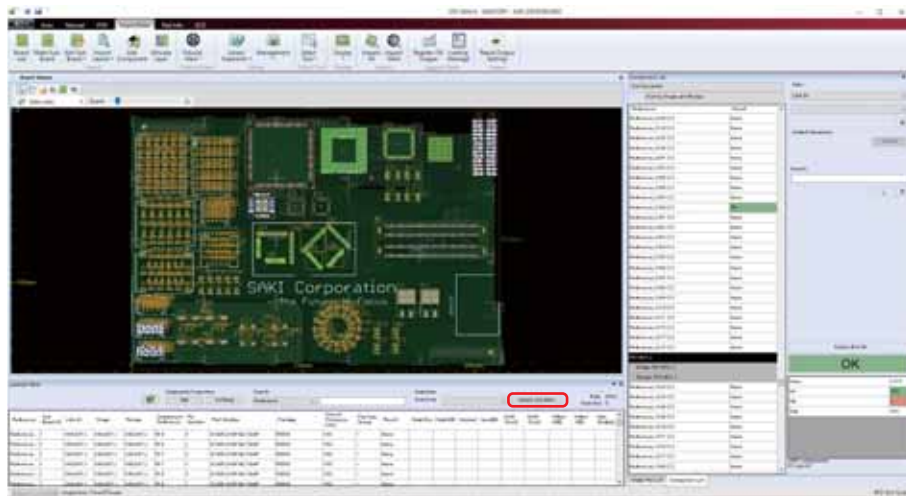


Figure 4-37 Applying Group Rules of the Pad Group 4

Step3: The dialog box shown below appears.

Set pad groups according to Table 4-5 of **4.5.1 Adding a Pad Group**, and click **Apply Group Rules To Recipe**.

Group rules are applied to the selected data.

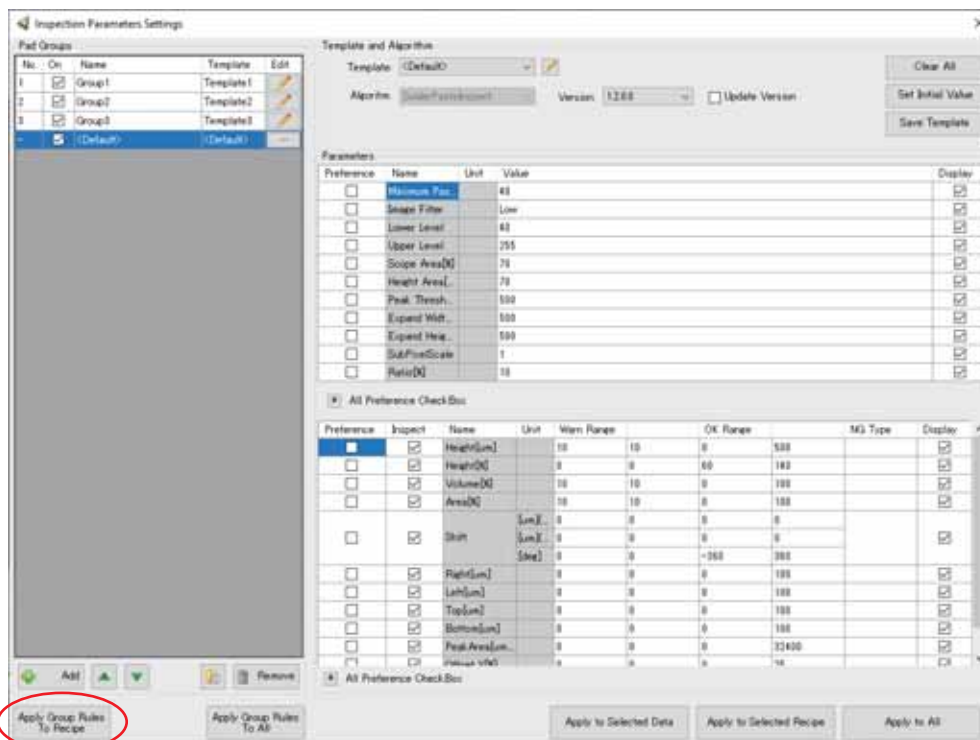


Figure 4-38 Applying Group Rules of the Pad Group 5

4.5.3 Automatically Processing Group Rules of the Pad Group

Here describes how to automatically process group rules of the created pad group when inspection data are loaded.

Step1: Click the **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

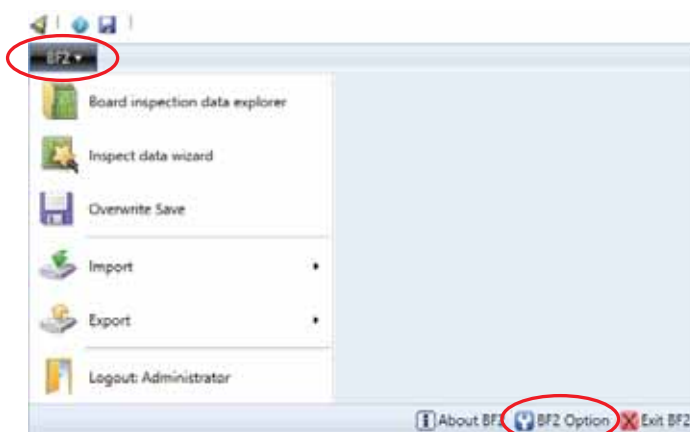


Figure 4-39 Automatically Processing Group Rules of the Pad Group 1

Step2: The dialog box shown below appears.
Click **Inspection Data**.

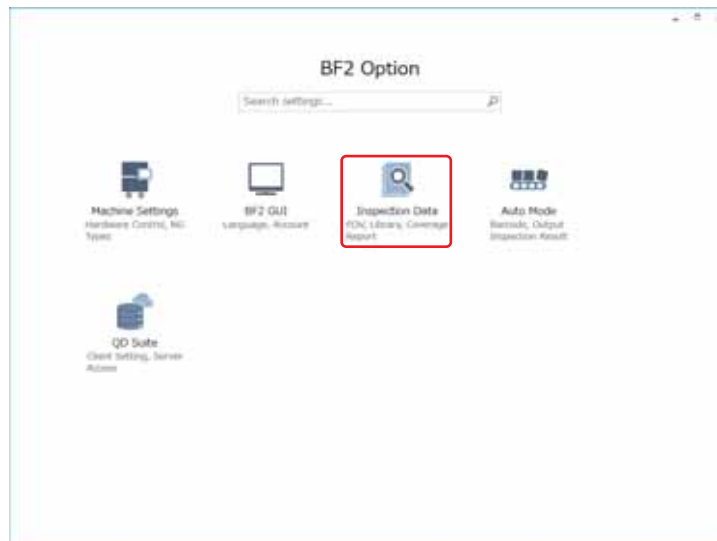


Figure 4-40 Automatically Processing Group Rules of the Pad Group 2

- Step3: Select **Pad Group Rules**.
Check **Process Pad Size Group Rules**.

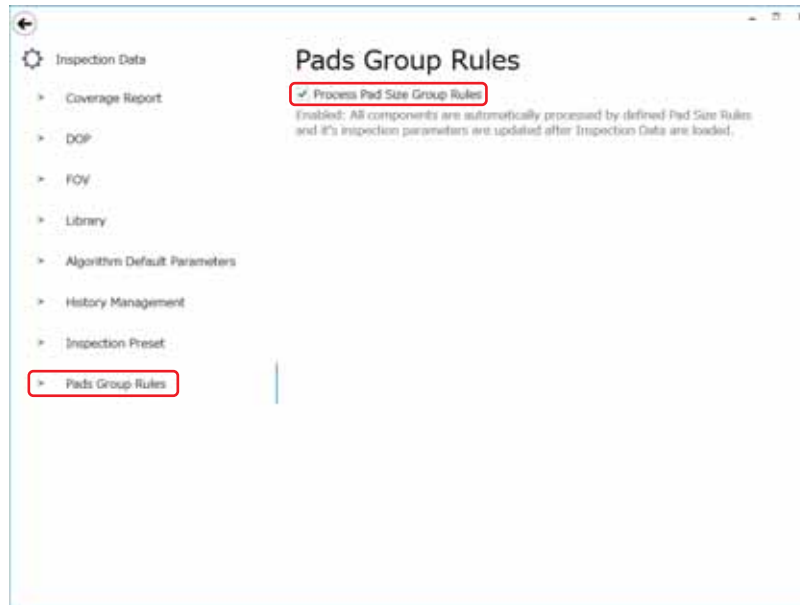


Figure 4-41 Automatically Processing Group Rules of the Pad Group 3

- Step4: After all settings are complete, click on the upper right side of the dialog box and click **Yes** to close the dialog box.

5 Acquiring a PCB image

5.1 Making an FOV File

FOV stands for **F**ield **O**f **V**iew and indicates an image area which a camera can capture at once.

In the inspection machine, the FOV size is fixed so the machine uses multiple FOVs to take an image.

Therefore, an FOV file, which defines the FOV allocation, is necessary for scanning.

CAUTION When an inspection data is made by New Inspection Data Wizard (refer to **1.3 Making an Inspection Data using SAKI Standard Format Data (Inspection Data Wizard)** and **1.4 Making an Inspection Data using PCB Design Data (Inspection Data Wizard)**, FOVs are automatically set and an FOV file is automatically made so the procedure in this section is not necessary except when making multiple FOV files.

5.1.1 Making an FOV File

Here describes how to make an FOV file.

Step1: Select the **FOV** tab.



Figure 5-1 FOV

Step2: Click **New** on **FOV File List**.

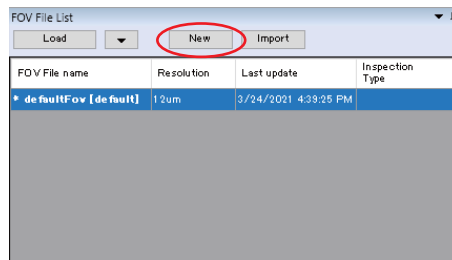


Figure 5-2 Making an FOV File 1

Step3: Enter a file name and click **OK**.

An FOV file is added to **FOV File List**.

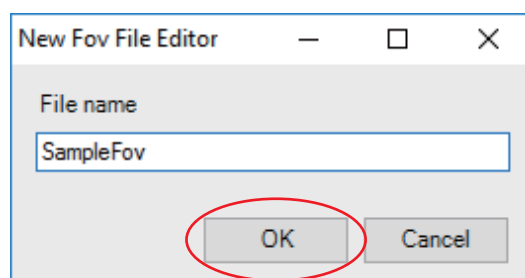


Figure 5-3 Making an FOV File 2

5.1.2 Easy FOV Allocation

Here describes how to use the easy FOV allocation function. When it needs to take a PCB image before setting Shapes and Recipes, allocate FOVs on the inspection data by the following procedure.

CAUTION

By the easy FOV allocation function, FOVs are allocated to the points where not to be inspected. To reduce the scanning time, make sure to optimize the FOV allocations by **FOV allocation** (refer to **5.3.3 Optimizing FOV Allocation (Auto Allocation)**) and **Path optimize** (refer to **5.3.7 FOV Path Optimize**) after setting Recipes.

NOTE

In the default setting, the projector setting is **NS** (North-South projection). To reduce tact time, change the projector setting by reference to **5.3.10 Projector Setting**.

Step1: Select the **FOV** tab.



Figure 5-4 Easy FOV Allocation 1

Step2: Drag the mouse and surround the area to set FOVs on **Board Viewer**.

When the mouse button is released, the icons shown in Figure 5-5 appears on the lower right side of the window.

Refer to Table 5-1 and select an icon.

FOVs are allocated in accordance with the selected icon.

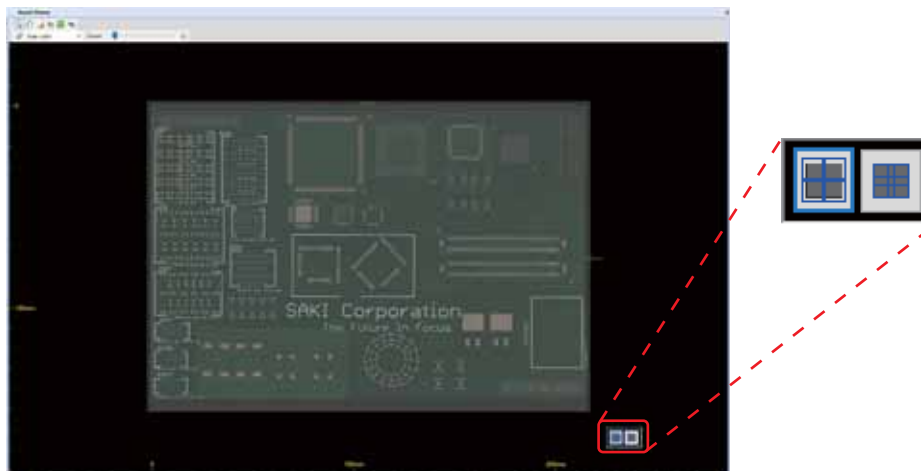


Figure 5-5 Easy FOV Allocation 2

Item	Description
	Allocates FOVs to include the selected area and not to overlap each other.
	Allocates FOVs to match with the selected area (FOVs may overlap each other).

Table 5-1 Descriptions

5.2 Scanning a PCB

This section describes how to scan a PCB manually.

NOTE For scanning by the Auto inspection, refer to **Instruction Manual**.

Step1: Select an inspection data by reference to **Part I 1.3 Selecting an Inspection Data File**.

Step2: From **Manual** tab, select the **Machine control** group > **Load/Unload** to load a PCB to the machine.
Check that the PCB is set to the right position and firmly fixed.

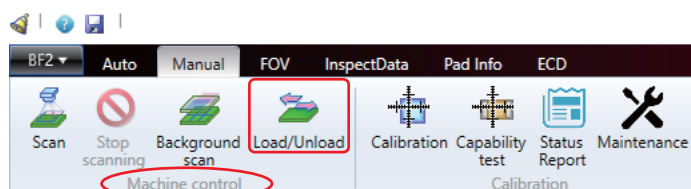


Figure 5-6 Load PCB

Step3: From the ribbon, select the **Machine control** group > **Scan**. The loaded PCB is scanned.

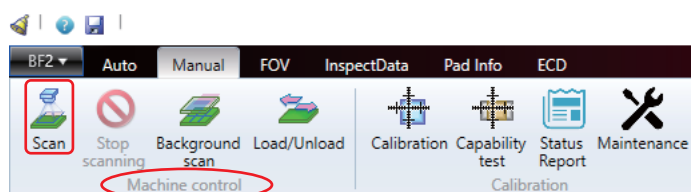


Figure 5-7 Scan

Step4: From the ribbon, select the **Machine control** group > **Load/Unload** to unload the PCB from the machine.

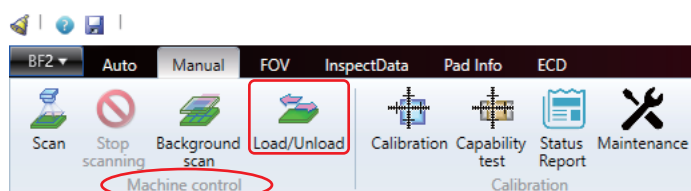


Figure 5-8 Unload PCB

5.3 Editing FOVs

5.3.1 Selecting an FOV

Here describes how to select an FOV from **Board Viewer**.

Step1: Select the **FOV** tab.



Figure 5-9 FOV

Step2: **To Select All FOV**

From the ribbon, select the **Select Tool** group > **Select all**.

NOTE The same operation can also be performed by pressing **Ctrl** + **A**.



Figure 5-10 Selecting All FOV

To Select All FOV with Fiducial Mark

From the ribbon, select the **Select Tool** group > **Select fiducial FOV**.



Figure 5-11 To Select All FOV with Fiducial Mark

To Disable the Selected FOV

From the ribbon, select the **Select Tool** group > **Clear selection**.



Figure 5-12 To Disable the Selected FOV

5.3.2 Locking the FOV

Here describes how to use the FOV lock function.

Lock the FOV to disable movement and deletion of the FOV.

Step1: Select the **FOV** tab.



Figure 5-13 FOV

Step2: Select an FOV to lock from **Board Viewer**.

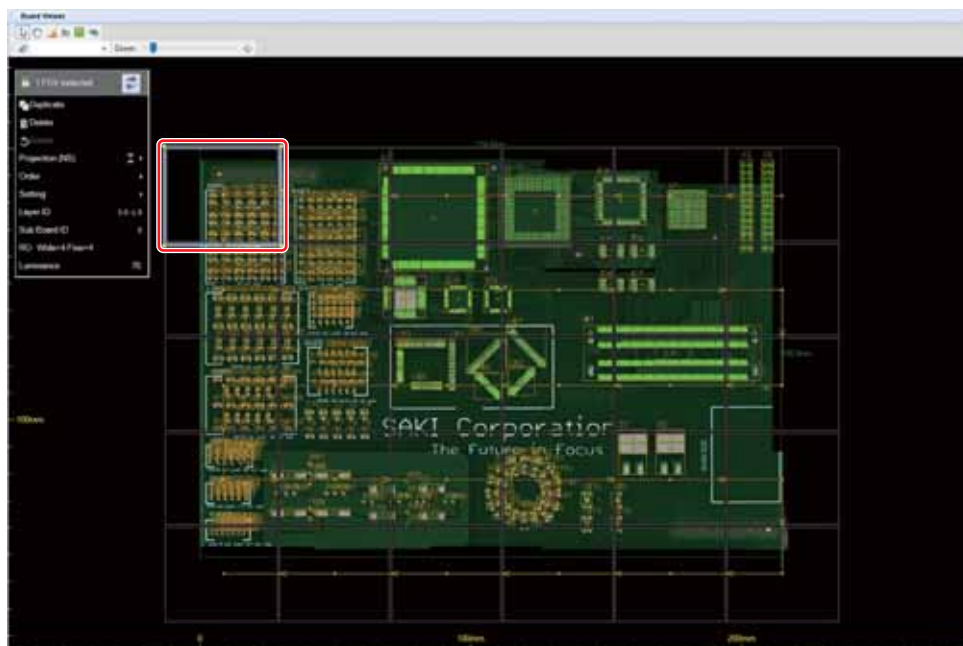


Figure 5-14 Select FOV

Step3: Click  (FOV lock icon) on the **Edit Panel**. FOV is locked.
The FOV lock icon is displayed at the center of the locked FOV.

NOTE

To release the FOV lock, click  (FOV lock icon) again.

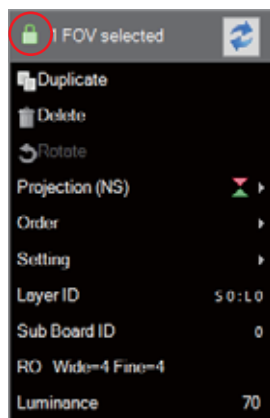


Figure 5-15 FOV Lock

5.3.3 Optimizing FOV Allocation (Auto Allocation)

Here describes how to optimize the FOV allocation by the auto allocation function.

By the auto allocation function, FOVs are automatically allocated to data on which a Recipe is set. As FOVs are not allocated to points where no Recipe is set (= no inspections are performed), the scanning time is reduced.

CAUTION FOV allocation does not work correctly if the Recipe setting is not completed. Make sure to complete the Recipe setting in advance.

CAUTION If there is a locked FOV, it is not affected by FOV allocation.

NOTE In the default setting, the projector setting is **NS** (North-South projection). To reduce tact time, change the projector setting by reference to **5.3.10 Projector Setting**.

Step1: Select an inspection data on which Recipes are already assigned by reference to **Part I 1.3 Selecting an Inspection Data File**.

Step2: From **FOV** tab, select the **Automatic** group > **FOV allocation**.
FOVs are allocated automatically.



Figure 5-16 Auto FOV Allocation

5.3.4 Add FOV

Step1: From **FOV** tab, select the **Edit FOV** group > **Add FOV**.



Figure 5-17 Add FOV

Step2: Specify an area to add an FOV from the **Board Viewer**.

5.3.5 Adding an FOV for a Specified Area

Step1: From **FOV** tab, select the **Edit FOV** group > **Addition area**.



Figure 5-18 Additional Area 1

Step2: Specify the area to add an FOV.

Click in the **Board Viewer** window to draw an orange-colored straight line from the clicking point.

Click once again to draw a 90-degree right angled line.

Move the mouse to draw an orange line. Then specify the area so that all the windows of Recipes are included.

Click the origin point to finish specifying the area and automatically add FOVs to the specified area.

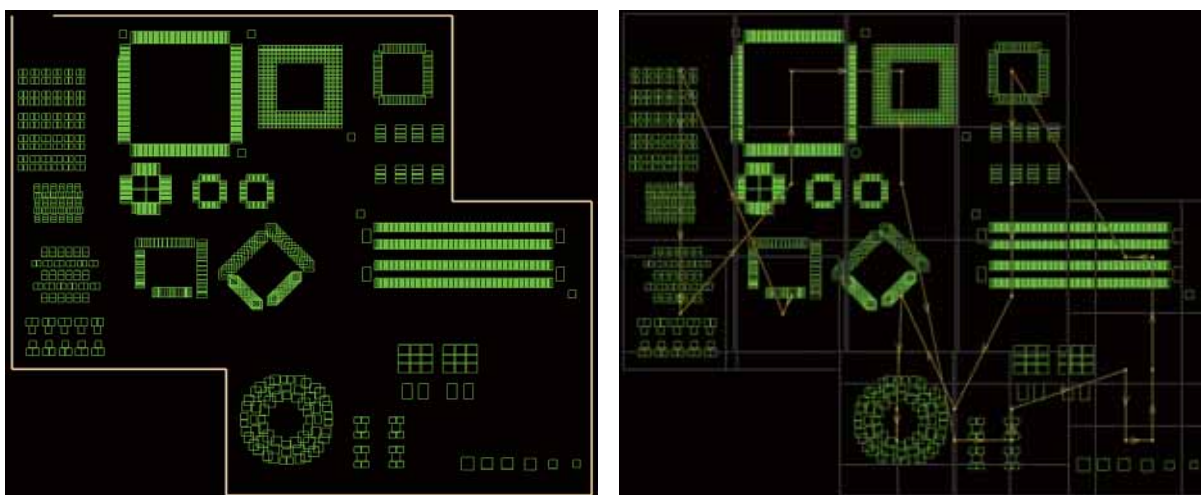


Figure 5-19 Additional Area 2

5.3.6 Checking the FOV Allocation

After changing the FOV allocation, make sure to check the FOV allocation.

Step1: From **FOV** tab, select the **Validate** group > **FOV allocation**.



Figure 5-20 Checking the FOV Allocation 1

Step2: **When all data are Included in FOVs**

The dialog shown below appears. Click **OK**. The FOV allocation is completed.

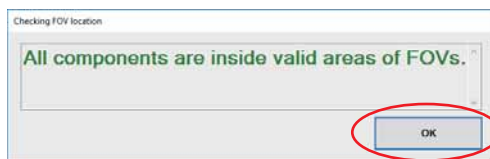


Figure 5-21 Checking the FOV Allocation 2

When there is a data which is not Included in FOVs

The dialog shown below appears. Click **OK** and proceed to Step3.

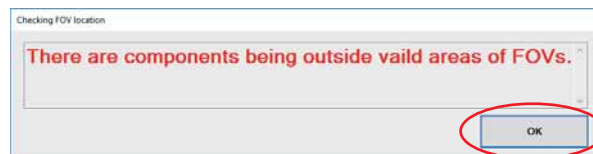


Figure 5-22 Checking the FOV Allocation 3

Step3: Data not included in FOVs or on the boundary are displayed in red on **Board Viewer**.

Adjust the FOV positions or add FOVs to eliminate the error (reddened data).



Figure 5-23 Checking the FOV Allocation 4

5.3.7 FOV Path Optimize

Path Optimize is the function for optimizing the camera moving path.
By optimizing the FOV path, scanning time may be shortened.

Step1: Select the **FOV** tab.



Figure 5-24 FOV

Step2: From the ribbon, select the **Automatic** group > **Path optimize**.
The FOV path is optimized in accordance with the FOV position.

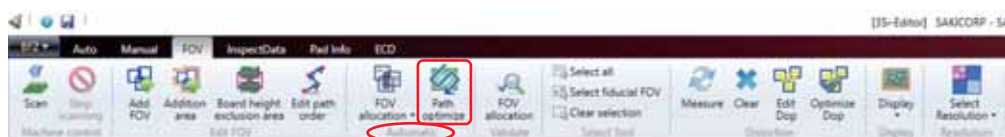


Figure 5-25 Path Optimize

5.3.8 Board Height Exclusion Area

When there is a slit or transparent area on a PCB, a PCB height may not be measured correctly. By setting the **Board Height Exclusion Area** on an FOV, the height can be measured more accurately.

Step1: From **FOV** tab, select the **Edit FOV** group > **Board height exclusion area**.



Figure 5-26 Board Height Exclusion Area

Step2: The edit panel shown below appears.

Refer to Table 5-2 to configure the settings.

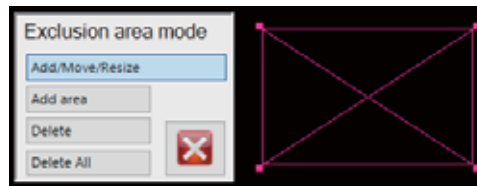


Figure 5-27 Adding the Board Height Exclusion Area

Item	Description
Add/Move/Resize	Click on any point on the board with Add/Move/Resize selected and drag to display a thick pink frame. Drag the created frame to a specified location. Drag the four corner points to change the size.
Add area	Click on any point on the board with Add area selected to draw a straight, white line from the origin point. Click once again to draw a 90-degree right angled line. Move the mouse to draw a white line and specify the exclusion area so that it is surrounded by the white line. Click the origin point once again to finish specifying the area and automatically add a Board Height Exclusion Area to the specified area.
Delete	Click the existing Board Height Exclusion Area with Delete selected to delete the selected Board Height Exclusion Area.
Delete All	Deletes all the existing Board Height Exclusion Areas.

Table 5-2 Descriptions

5.3.9 Individual FOV Setting

Select an FOV to appear the edit panel as below.

Use this edit panel to set the individual FOVs.

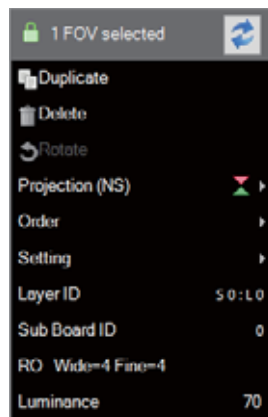
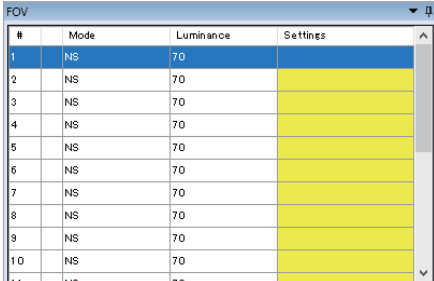


Figure 5-28 Edit Panel

Item		Description
n FOV selected		Displays the number of currently selected FOVs.
		Captures an image of the selected FOV only.
Duplicate		Duplicates selected FOVs.
Delete		Deletes selected FOVs.
Rotate		Rotates the selected FOV 90 degrees. This menu item is available only when multiple FOVs are assigned.
Projection		Sets the projector unit to use for image capture. For details, refer to 5.3.10 Projector Setting .
Order	To first	 Figure 5-29 FOV Path Order Selects the FOV to display it highlighted in blue in the FOV ordered list on the right side. Image capture of the FOVs is performed in order starting from the top. Click To first to place the selected FOV at the top of the list. Click To last to place the selected FOV at the bottom of the list. Click Reverse to reverse the order of the selected FOV. Click Optimize to optimize the order of the selected FOV. Reverse and Optimize are only valid when multiple FOVs are selected.
	To last	
	Reverse	
	Optimize	

Item		Description
Setting	Height offset(0)	The item is for Saki engineers only. Do not change the setting.
	Board area setting	
	Board color setting	Set the FOV board height compensation. For details, refer to 5.3.12 Board Height Compensation Setting .
	Board color extraction setting	Extracts the board color from the specified area during each scan. This function is used when it is necessary to inspect boards whose colors are different because of different board manufacturers in a same line. For details, refer to 7.4.2 Board Color Extraction Setting .
	Distortion Measurement Setting	Set the DOP compensation. For details, refer to 7 DOP Compensation .
Layer ID		Set the layer number. For details of layer, refer to 8 Making Inspection Data of a Large Board .
Sub Board ID		Normally not available.
RO		The item is for Saki engineers only. Do not change the setting.
Luminance		Select the luminance to scan. For details, refer to 5.3.11 Luminance Selecting .

Table 5-3 Descriptions

5.3.10 Projector Setting

For measuring height, 2 projectors are installed to this machine.

Can be set the 1 projection and 2 projections on each FOVs.

Step1: Select an FOV to change the setting. The Edit panel shown below appears.

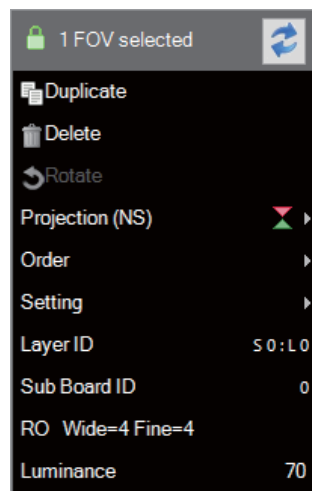


Figure 5-30 Selecting an FOV

Step2: Select the appropriate setting from **Projection** drop-down list.

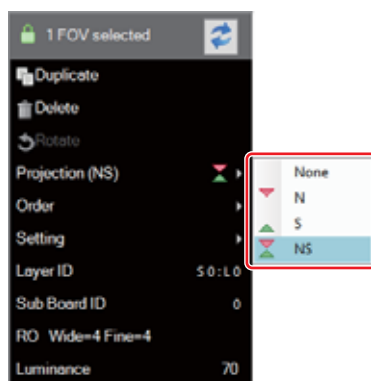


Figure 5-31 Projector Setting

Item	Description
None	Only uses 2D lightings.
N	Uses the north side (machine rear side) projector with 2D lightings.
S	Uses the south side (machine front side) projector with 2D lightings.
NS	Uses the north and south side projectors with 2D lightings.

Table 5-4 Descriptions

5.3.11 Luminance Selecting

Depending on the color of the board to be inspected, the luminance of projected striped image need to be adjusted.

By registering some **Luminance** in advance and changing them according to the board to be inspected, stable height measurement is possible.

Step1: Perform calibration refer to **Instruction Manual**.

NOTE Register the luminance want to use in **Illumination Balance** calibration.

Step2: Select an FOV to change the setting. The Edit panel shown below appears.

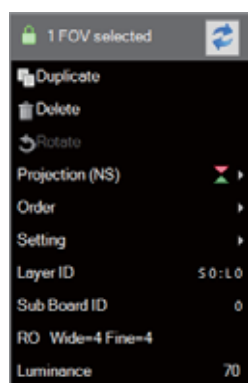


Figure 5-32 Luminance Selecting 1

Step3: Select the appropriate setting from **Luminance** drop-down list.

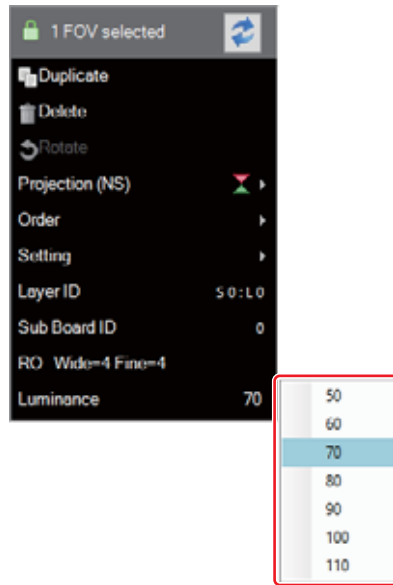


Figure 5-33 Luminance Selecting 2

Step4: The selected luminance is set for each FOV.
The set luminance can be confirmed on edit panel and **FOV** window.

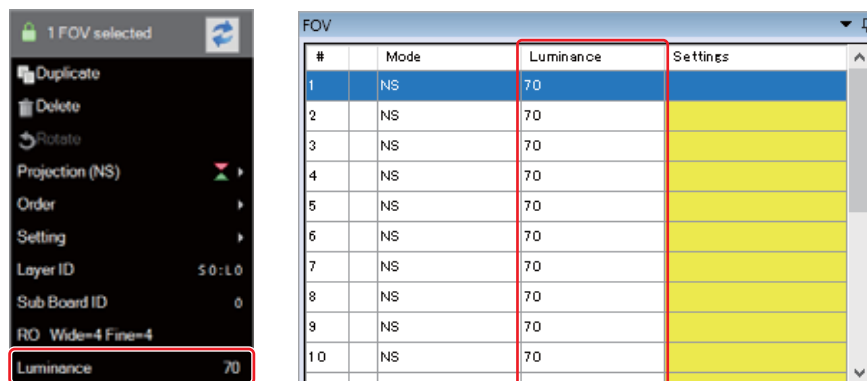


Figure 5-34 Luminance Selecting 3

5.3.12 Board Height Compensation Setting

Set the board surface (zero height point) to make accurate height measurements.

Here describes how to use the board color to set the zero height point and the procedure for performing board height compensation.

Step1: Select the FOV for board height compensation and display the **edit panel**.

Select **Setting > Board Color Setting**.

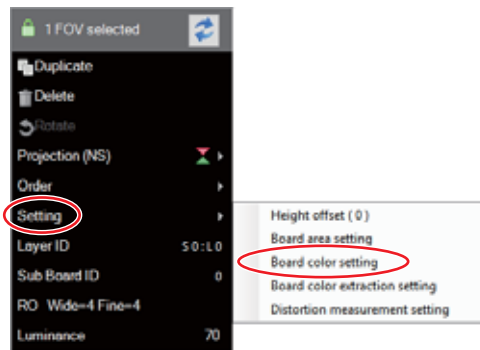


Figure 5-35 Board Height Compensation Settings 1

Step2: The dialog shown below appears.

Here, specify the color of the board and other colors.

Check **Use board color setting** and click the board color section.

Select **Apply average as board color**.

A list of board colors is added on the right side of **Board color** field.

Click an area which has a color that is different from the board color and select **Apply average as anti board color**.

A list of anti board colors is added on the right side of **Anti Board color** field.

NOTE

Select the wiring patterns for the board color.

Select the largest range possible around an area in which there are few nearby solder paste. For colors other than the board color, select solder paste, silk, holes, insulation material areas, etc.

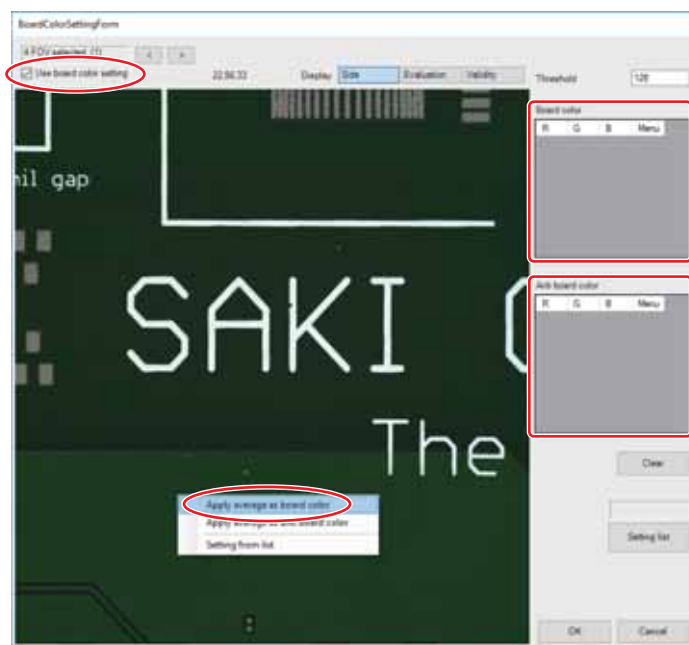


Figure 5-36 Board Height Compensation Settings 2

Step3: Fine tune the color specification set in Step2.

Select **Evaluation** on the right side of **Display**.

The area registered as the board color is shown in white depending on the **Board color** and **Anti Board color** settings.

Select **Validity** on the right side of **Display** to display areas with a brightness greater than the **Threshold** in white and all other areas in black on the **Evaluation** image.

Areas displayed in white when using **Validity** are recognized as the board surface.

Set the **Board color**, **Anti Board color**, and **Threshold** to display only the board color area in white.

For detail settings, refer to Table 5-5.



Figure 5-37 Board Height Compensation Settings 3

Item		Description
< / >		Switches an FOV.
Use board color setting		Check this item to perform board height compensation.
Display	Side	Displays the images of SideLight.
	Evaluation	Displays the board color area in white based on the Board color and Anti Board color settings.
	Validity	Areas with a brightness greater than the Threshold are displayed in white and all other areas in black on the Evaluation image.
Threshold		Sets the threshold value when converting the Evaluation image to the Validity image. White areas in the Validity image are recognized as the board surface.
Board color		A list of board colors set with Apply average as board color .
Anti board color		A list of board colors set with Apply average as anti board color .
Clear		Deletes all Board color and Anti Board color settings.
Setting list	Load	Loads all saved settings for Board color and Anti board color .
	Save	Saves all Board color and Anti Board color settings.
	Save as	Overwrites all Board color and Anti Board color settings.
	Delete	Deletes all saved settings for Board color and Anti board color .

Table 5-5 Descriptions

Step4: After all the settings are completed, click **OK**.

If the multiple FOVs are selected, the dialog shown below appears.

Select the check boxes of the FOV images for applying board height compensation and click **OK**.

NOTE

Click **Select all** to select all FOVs.

Click **Clear Selection** to deselect all FOVs.

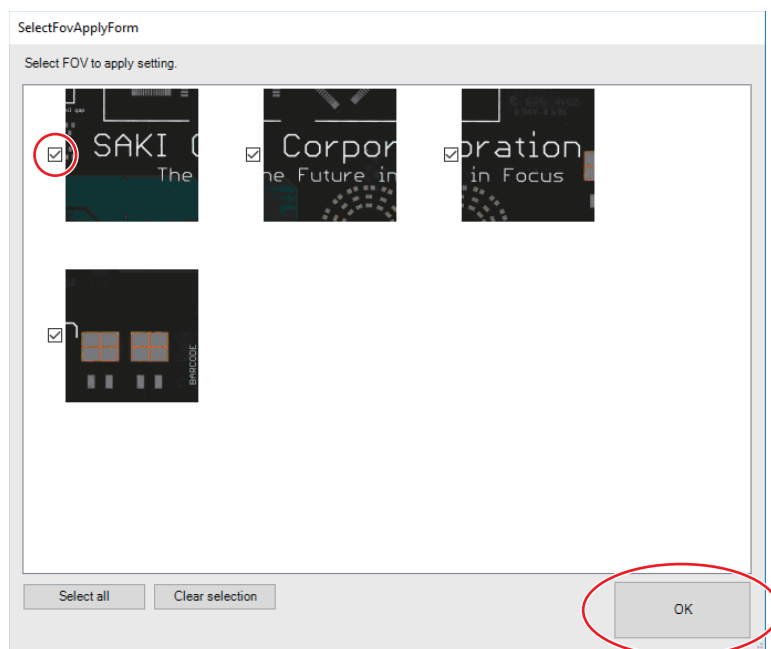


Figure 5-38 Board Height Compensation Settings 4

5.3.13 Saving an FOV File by Another Names

Here describes how to save an FOV file by another names.

By saving FOV files in different names, multiple FOV settings can be used in one inspection data.

NOTE

FOV files are overwritten when the inspection data is saved.

Step1: Select the **FOV** tab.



Figure 5-39 FOV

Step2: Click  on the right side of **Load** button.

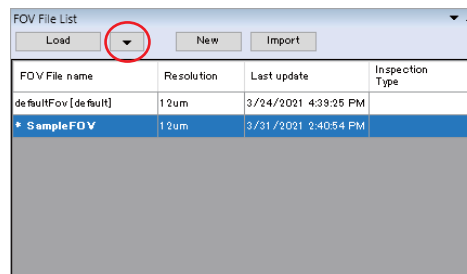


Figure 5-40 Saving an FOV File 1

Step3: The pop-up menu appears. Select **Save As**.

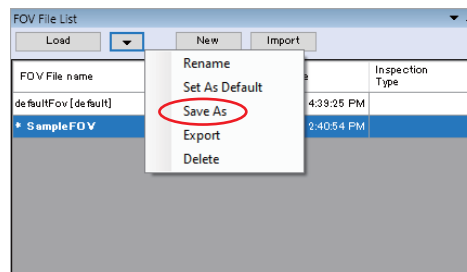


Figure 5-41 Saving an FOV File 2

Step4: Enter a file name and click **OK**.

An FOV file is added to **FOV File List**.

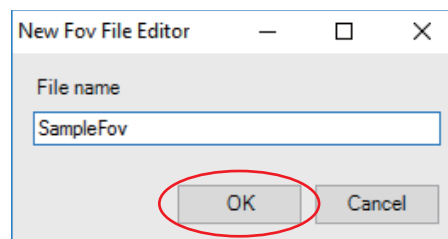


Figure 5-42 Saving an FOV File 3

5.3.14 Selecting an FOV File

Here describes how to select another FOV setting.

The selected setting is applied to the inspection data temporary.

CAUTION

The FOV file set as the default file is opened when the inspection data is restarted. To use another FOV setting in the future inspections, change the default setting by reference to **5.3.15 FOV File Auto Loading Setting**.

Step1: Select the **FOV** tab.



Figure 5-43 FOV

Step2: Select an FOV file from **FOV File List** and click **Load**.

The selected FOV setting is applied to the inspection data.

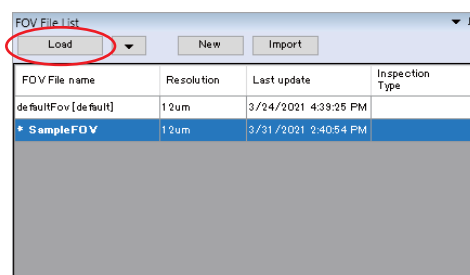


Figure 5-44 FOV File List

5.3.15 FOV File Auto Loading Setting

In this machine, an FOV file is automatically loaded when an inspection data is opened.

Here describes how to change the file to be loaded.

To use a new FOV file in the future inspections, change the setting by following the procedure below.

Step1: Select the **FOV** tab.



Figure 5-45 FOV

Step2: Select an FOV file from **FOV File List** and click  on the right side of **Load**.

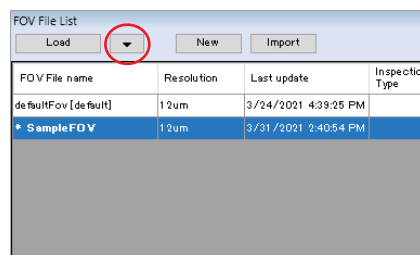


Figure 5-46 FOV File Auto Loading Setting 1

Step3: The pop-up menu appears. Select **Set As Default**.

The selected file is set as the file automatically loaded when opening the data.

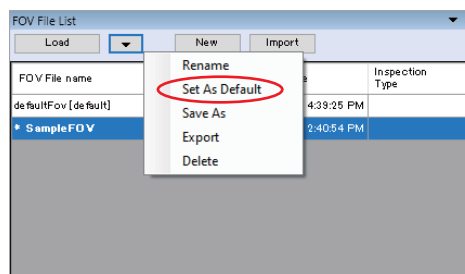


Figure 5-47 FOV File Auto Loading Setting 2

5.3.16 Changing an FOV File Name

Step1: Select the **FOV** tab.



Figure 5-48 FOV

Step2: Select an FOV file from **FOV File List** and click  on the right side of **Load**.

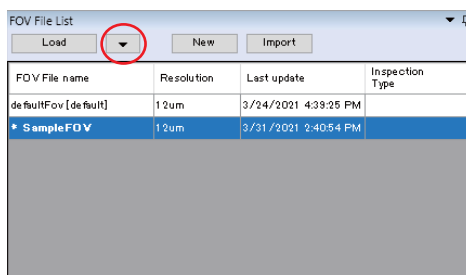


Figure 5-49 Changing an FOV File Name 1

Step3: The pop-up menu appears. Select **Rename**.

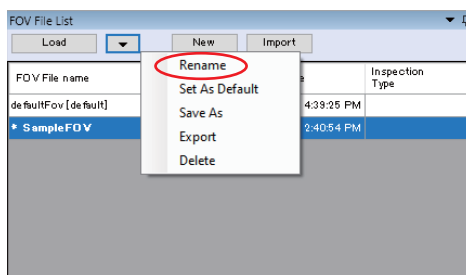


Figure 5-50 Changing an FOV File Name 2

Step4: Enter a file name and click **OK**. The selected FOV file name is changed.

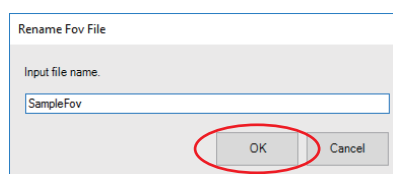


Figure 5-51 Changing an FOV File Name 3

5.3.17 Deleting an FOV File

Step1: Select the **FOV** tab.



Figure 5-52 FOV

Step2: Select an FOV file from **FOV File List** and click ▼ on the right side of **Load**.

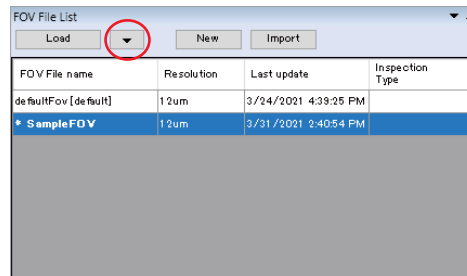


Figure 5-53 Deleting an FOV File 1

Step3: The pop-up menu appears. Select **Delete**.

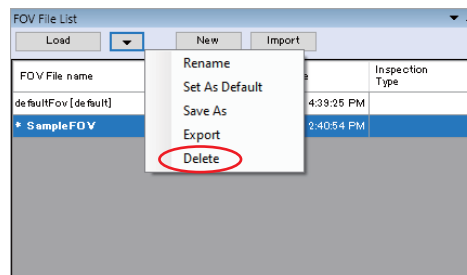


Figure 5-54 Deleting an FOV File 2

Step4: Click **OK**. The selected FOV file is deleted from **FOV File List**.

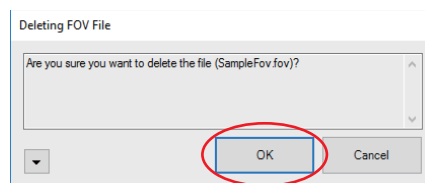


Figure 5-55 Deleting an FOV File 3

5.3.18 Importing an FOV File

Here describes how to import an FOV file from another inspection data.

To import an FOV file made on another inspection machine or BF2-Editor, follow the procedure below.

Step1: Select the **FOV** tab.



Figure 5-56 FOV

Step2: Click **Import** on **FOV File List**.

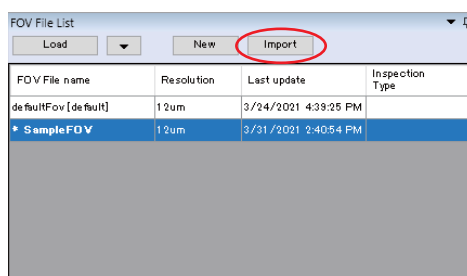


Figure 5-57 Importing an FOV File 1

Step3: Select an FOV file and click **OK**.

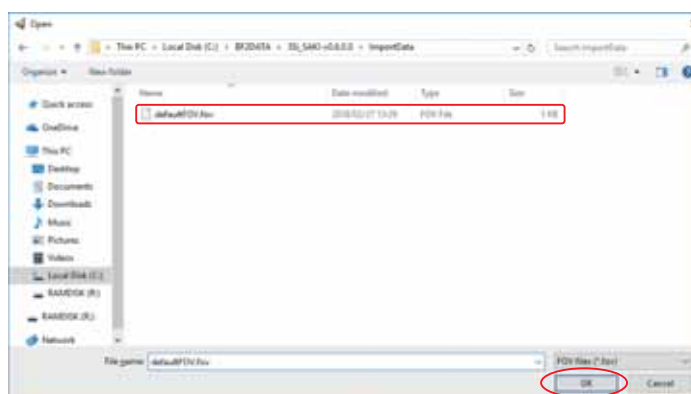


Figure 5-58 Importing an FOV File 2

5.3.19 Exporting an FOV File

Here describes how to export an FOV file to another inspection data. To export an FOV file to another inspection machine or BF2-Editor, follow the procedure below.

Step1: Select the **FOV** tab.



Figure 5-59 FOV

Step2: Click  on the right side of **Load** button.

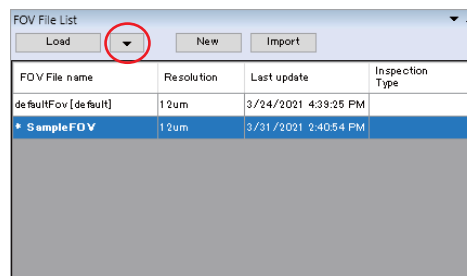


Figure 5-60 Exporting an FOV File 1

Step3: The pop-up menu appears. Select **Export**.

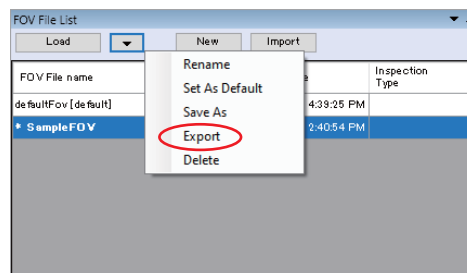


Figure 5-61 Exporting an FOV File 2

Step4: Select a directory to save the FOV file and click **OK**.

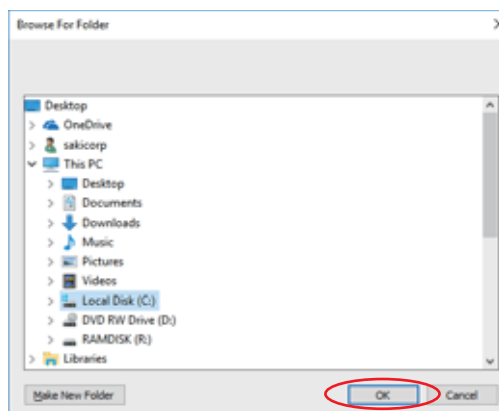


Figure 5-62 Exporting an FOV File 3

5.3.20 Changing the FOV File Resolution

Here describes how to change the resolution of FOV files.

From **Select Resolution** on the **FOV** tab, select the resolution to change and rearrange the existing FOV location according to the selected resolution.

NOTE

Select Resolution is shown only for BF2-Editor.



Figure 5-63 Changing the FOV File Resolution

5.4 Image File Management

5.4.1 Adding a PCB image

Multiple image files can be saved in an inspection data.

Here describes how to add a PCB image to an inspection data.

Step1: Scan a PCB by reference to **5.2 Scanning a PCB**.

Step2: Select the **InspectData** tab.



Figure 5-64 InspectData

Step3: Click **Add current image** on **Image File List**.

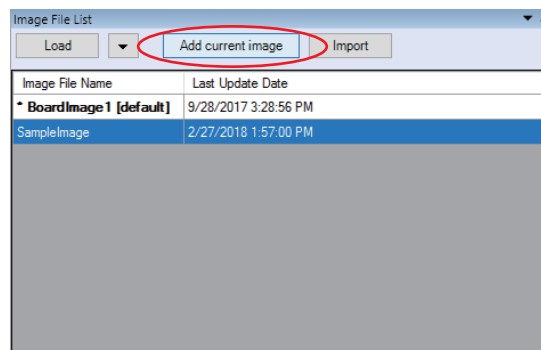


Figure 5-65 Add current image

Step4: Enter a file name and click **OK**.

The image file is added to **Image File List**.

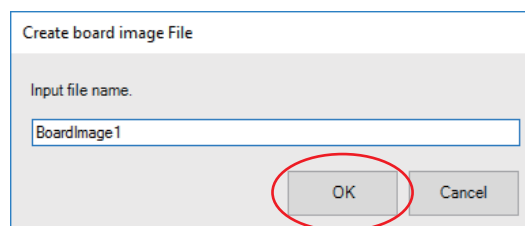


Figure 5-66 Entering the Image File Name

5.4.2 Changing the Displayed PCB Image

Here describes how to change an image file to be displayed.

CAUTION

The image file set as the automatically loaded file is displayed when the inspection data is restarted. To display the selected image in the future inspections, change the auto loading setting by reference to **5.4.3 Image File Auto Loading Setting**.

Step1: Select the **InspectData** tab.



Figure 5-67 InspectData

Step2: Select an image file on **Image File List** and click **Load**.
The selected image file is loaded and displayed on the screen.

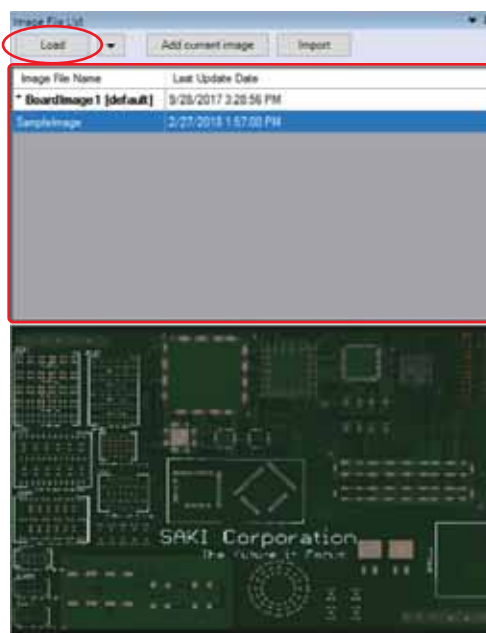


Figure 5-68 Image File Name List

5.4.3 Image File Auto Loading Setting

In this machine, the image file which is set as the automatically loaded file is displayed when an inspection data is opened.

To use a new image file in the future inspections, change the setting by following the procedure below.

Step1: Select the **InspectData** tab.



Figure 5-69 InspectData

Step2: Select an image file on **Image File List** and click  on the right side of **Load**.

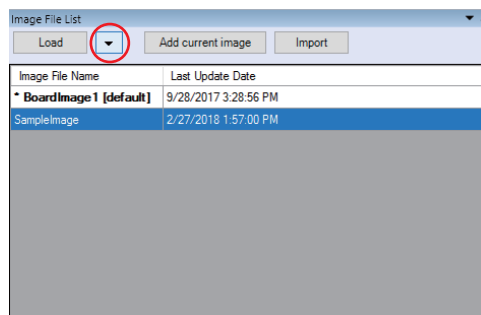


Figure 5-70 Image File Auto Loading Setting 1

Step3: Select **Set As Default > Set as default** from the pop-up menu.
The selected image file is set as the automatically loaded file.

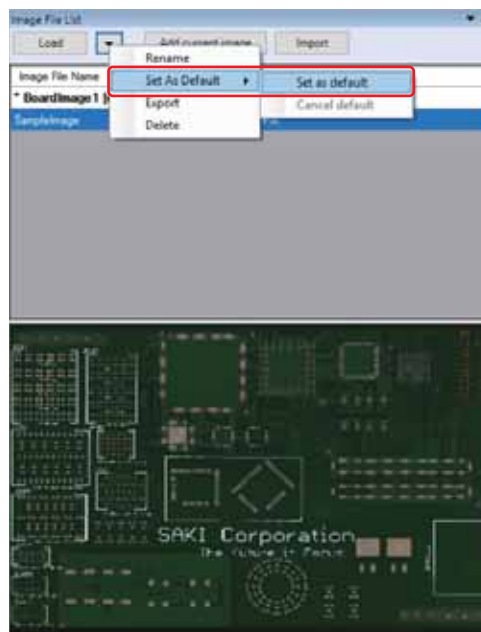


Figure 5-71 Image File Auto Loading Setting 2

5.4.4 Changing an Image File Name

Here describes how to change an image file name.

Step1: Select the **InspectData** tab.



Figure 5-72 InspectData

Step2: Select an image file on **Image File List** and click  on the right side of **Load**.

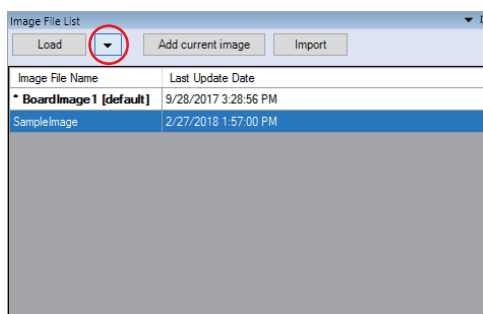


Figure 5-73 Changing an Image File Name 1

Step3: The pop-up menu appears. Select **Rename**.



Figure 5-74 Changing an Image File Name 2

Step4: Enter a file name and click **OK**.
The selected image file name is changed.

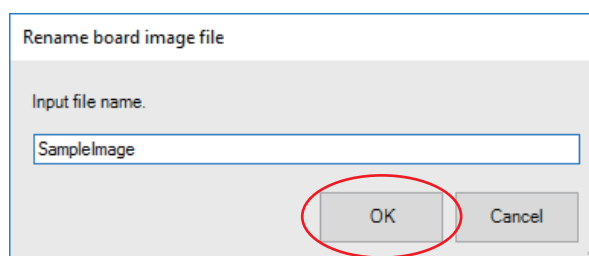


Figure 5-75 Changing an Image File Name 3

5.4.5 Deleting an Image File

Here describes how to delete an image file.

Step1: Select the **InspectData** tab.



Figure 5-76 InspectData

Step2: Select an image file on **Image File List** and click  on the right side of **Load**.

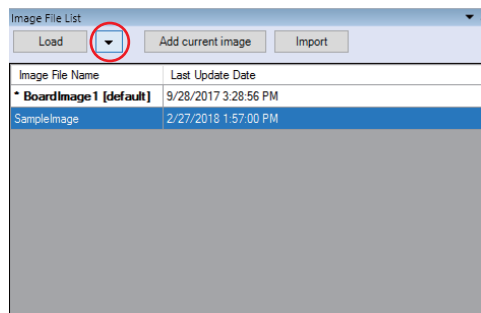


Figure 5-77 Deleting an Image File 1

Step3: The pop-up menu appears. Select **Delete**.



Figure 5-78 Deleting an Image File 2

Step4: Click **OK**. The selected image file is deleted from **Image File List**.

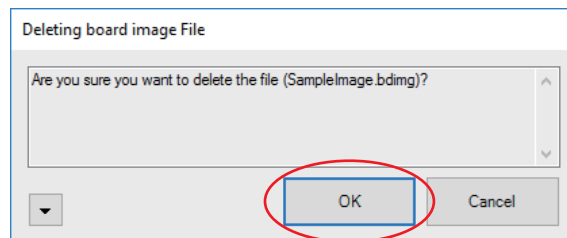


Figure 5-79 Deleting an Image File 3

5.4.6 Importing an Image File

Here describes how to import an image file from another inspection data.

To use an image file scanned by another inspection machine, follow the procedure below.

Step1: Select the **InspectData** tab.



Figure 5-80 InspectData

Step2: Click **Import** on **Image File List**.

NOTE

The same operation can be performed by selecting **Import > Import Board Image** from the **BF2** menu on the upper left side of the window

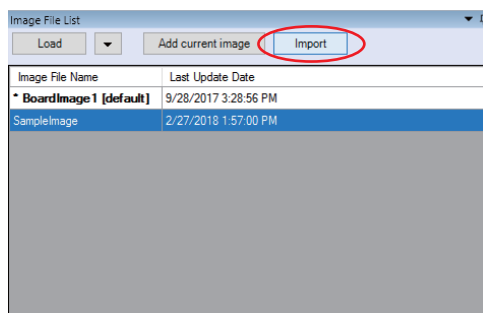


Figure 5-81 Importing an Image File 1

Step3: Select an image file and click **OK**.

NOTE

The extension for an image file is ".bdimg".

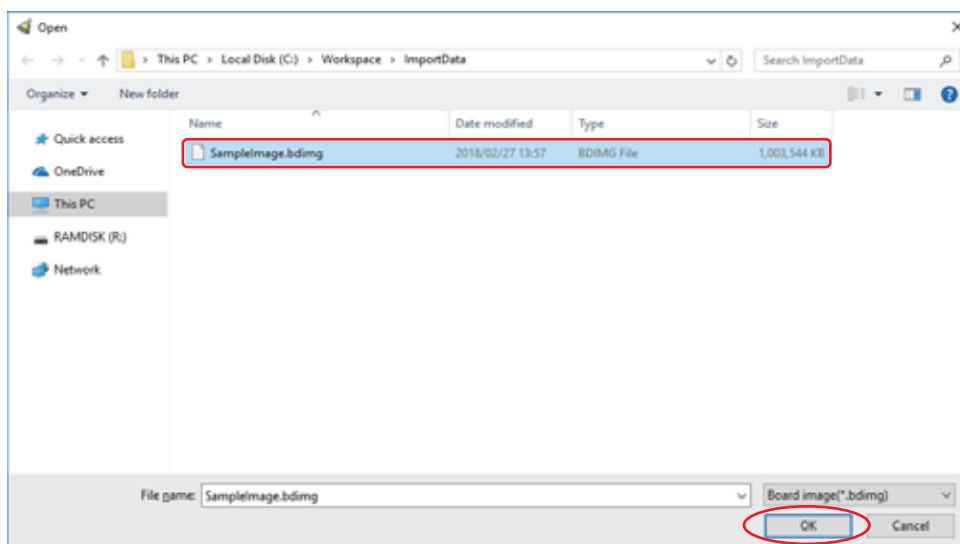


Figure 5-82 Importing an Image File 2

5.4.7 Exporting an Image File

Here describes how to export an image file to another inspection data.

To use an image file in another inspection machine or BF2-Editor, follow the procedure below.

Step1: Select the **InspectData** tab.



Figure 5-83 InspectData

Step2: On **Image File List**, click  on the right side of **Load**.
Select **Export** from the pop-up menu.

NOTE

The same operation can be performed by selecting **Export > Export Board Image** from the **BF2** menu on the upper left side of the window



Figure 5-84 Exporting an Image File 1

Step3: Select a directory to save the image file and click **Save**.

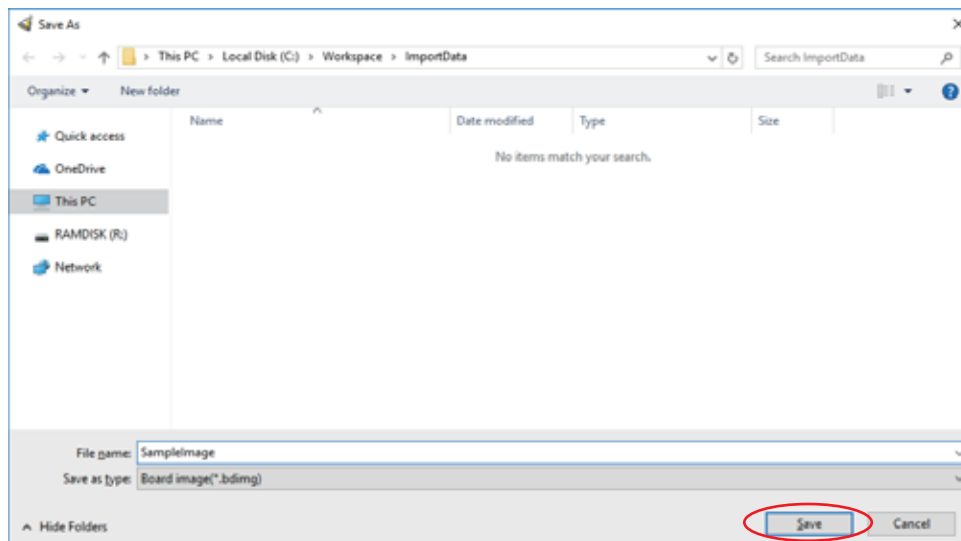


Figure 5-85 Exporting an Image File 2

6 Bare Board Teaching

6.1 Overview of Bare Board Teaching

Bare Board Teaching is the function which measures the height of board and pad with bare board and applies offset to the height of measured solder paste.

By considering the height of the pad, the actual height and volume of the solder paste can be measured more accurately.

NOTE Bare Board Teaching function is an effective function in the algorithm **Solder Paste Inspection**.

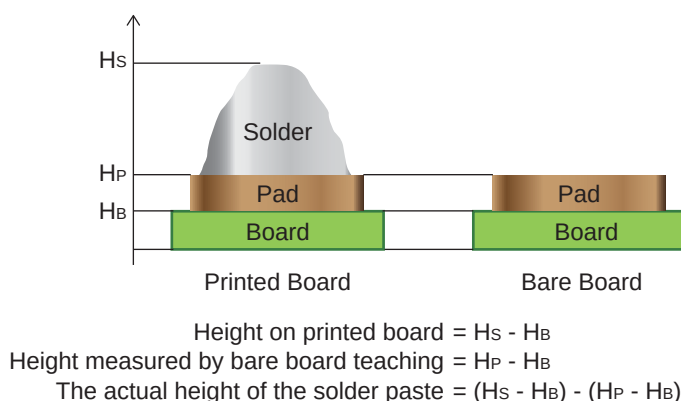


Figure 6-1 Conceptual Diagram of Bare Board Teaching

6.2 Setting Bare Board Teaching

Here describes setting procedure for using Bare Board Teaching function.

NOTE When using this function, be sure to use the bare board that is not placed solder paste.

Step1: Make the inspection data according to **1 Making an Inspection Data**.

NOTE This function can also be used for already created inspection data.

Step2: Select **PadInfo** tab.



Figure 6-2 Setting Bare Board Teaching 1

Step3: Select **Bare Board** group > **Bare Board Teaching**.



Figure 6-3 Setting Bare Board Teaching 2

Step4: The dialog box shown below appears.
Make the setting according to Table 6-1.

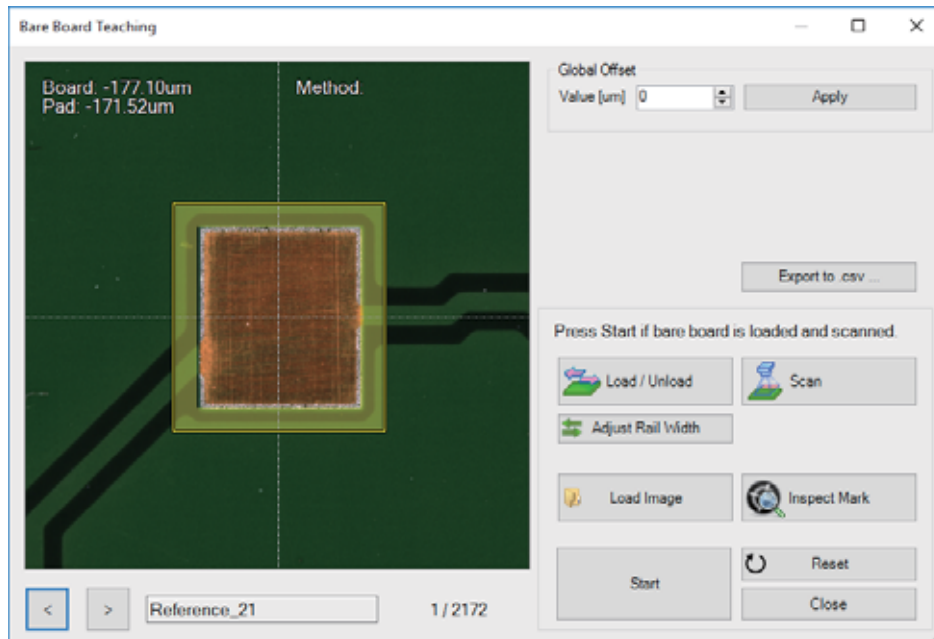


Figure 6-4 Setting Bare Board Teaching 3

Item	Description
Global Offset	Instead of using the bare board teaching, the set value can be applied on all pads. (*1)
Export to .csv	Exports bare board teaching data in CSV format.
Load / Unload	Loads / Unloads the board. (*2)
Scan	Scans the board. (*2)
Adjust Rail Width	Displays rail width adjustment dialog box. (*2)
Load Image	Loads the board image.
Inspect Mark	Performs mark inspection to adjust board position.
>	Displays next pad.
<	Displays previous pad.
Start	Start bare board teaching.
Reset	Resets all offset value.
Close	Closes the dialog box.

Table 6-1 Descriptions

NOTE

(*1) When using **Global Offset**, only click **Apply** and do not click **Start**. When using bare board teaching, the default value is **0**.

(*2) This button is not displayed in BF2-Editor.

7 DOP Compensation

DOP (Distortion Observation Point) compensation is a function to compensate the board height. Shifts in height, board tilt and warpage of the board because of board clamp conditions are corrected, in order to set the height of the board to $0\mu\text{m}$ and level the board. This avoids height differences between FOVs, and applies more accurate compensation to boards that is easily bent.

7.1 Conventional Board Height Compensation Method

Due to various factors such as warpage of the board or allocation of pads or cut-out, the board is captured with it warpage.

In order to accurately calculate the height of the solder paste, the height datum to the board needs to be treated as zero reference by compensating the warpage of the board. Conventionally, the board height compensation was performed by defining the reference plane per FOV and adjusting the height of the board to zero reference.

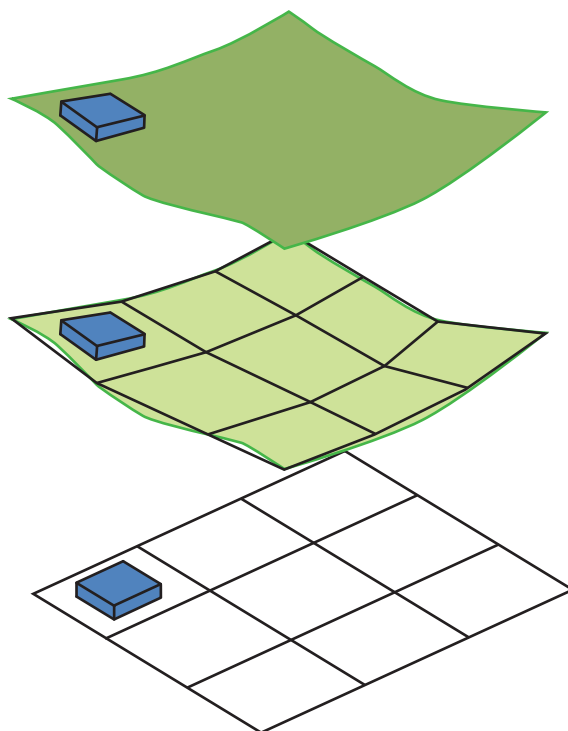


Figure 7-1 Conventional Board Height Compensation Method

However, defining the reference plane per FOV follows the warpage of the board and tilts each FOV, resulting in the occurrences of differences in contours between adjacent FOVs. When a bigger footprint pad spans multiple FOVs, it is impossible to accurately measure the height of the solder paste. The solution to this problem is by implementing DOP compensation methods.

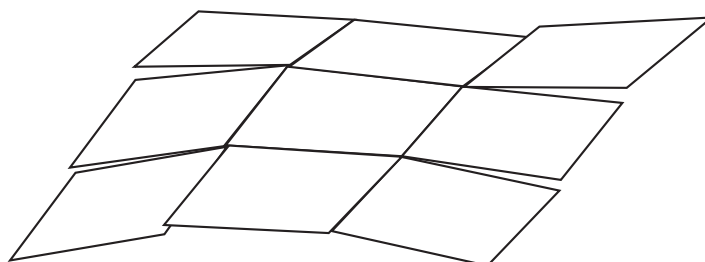


Figure 7-2 Steps Between FOVs

7.2 DOP Compensation Type

There are two methods of DOP compensation, **On FOV** and **On Edge**.

Default is **On FOV**.

7.2.1 On FOV

In **On FOV**, board surfaces are detected per FOV.

So far, this is the same as the convention board height compensation.

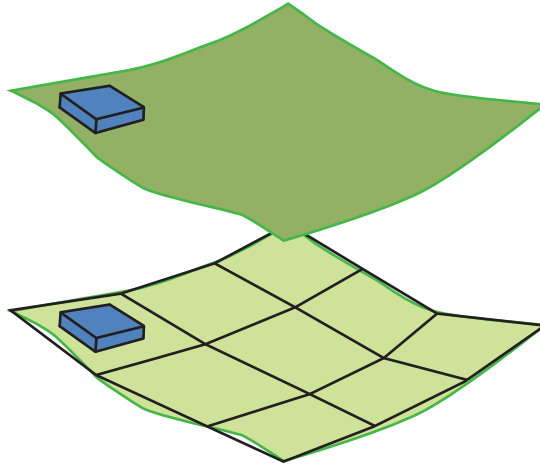


Figure 7-3 On FOV 1

In **On FOV**, when a board is scanned, regular triangles are automatically allocated.

These triangles are called DOPs (Distortion Observation Points), and each DOP contains board height information.

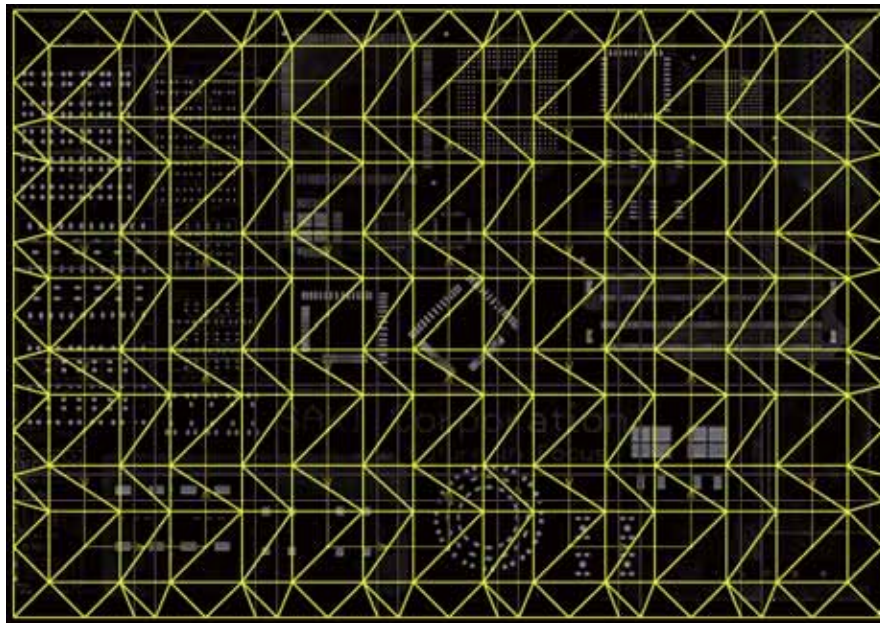


Figure 7-4 On FOV 2

Heights are measured at DOP, not for each FOV, and warpage of the board is compensated for each triangle surrounded by DOPs. Because compensation is not performed per FOV, no differences in the contours between FOVs are generated.

In **On FOV**, each DOP acquires height information of a boards which is compensated with the conventional method. DOPs for **On FOV** cannot be added or deleted, since they are automatically allocated.

7.2.2 On Edge

In **On Edge** method, DOPs are allocated at slightly outside the four corners of the pad.

Warpage of the board is compensated by treatment slightly outside the pads surrounding as the board height datum zero reference.

Because compensation is not performed per FOV, no differences in the contours between FOVs are generated.

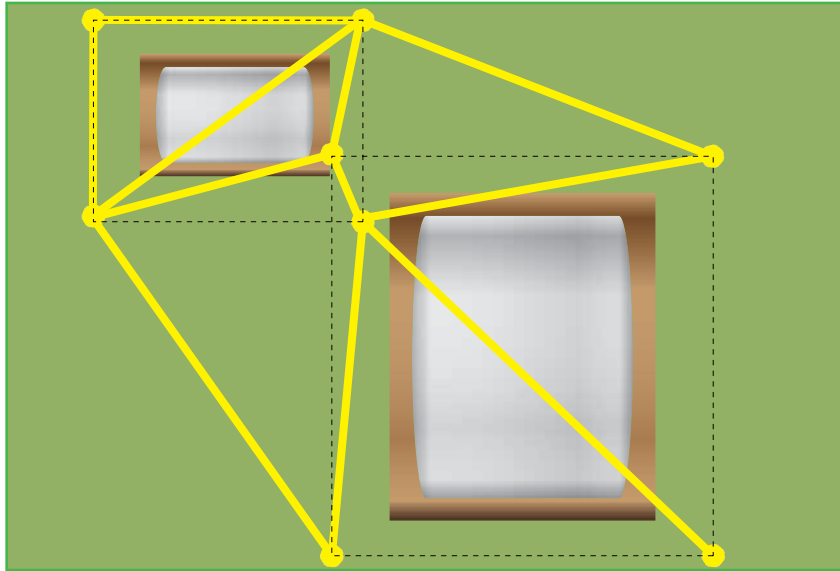


Figure 7-5 On Edge

DOPs for **On Edge** cannot be added or deleted, since they are automatically allocated.

And DOPs for **On Edge** are allocated based on pad allocation. So DOPs are not allocated when the pad does not have recipe.

7.2.3 Uses of DOP Compensation

Normally use **On FOV**.

When DOPs are not allocated by **On Edge**, DOPs are automatically allocated with **On FOV**. In the most cases, compensation is accurately performed with **On FOV**. However, **On Edge** is suitable the following circumstances.

Item	Description
Easily warped boards	Boards that will be greatly warped within the field of a single FOV cannot be accurately compensated with On FOV .
Some boards with sub boards	Each sub board may be inclined in different ways depending on the designs of the cut-out. If multiple sub boards are inclined in different ways and coexist in the same single FOV, it is not possible to accurately perform compensation with On FOV .
Boards upon which boards are installed	Because of the same reason as the above case of sub boards, compensation cannot be accurately performed with On FOV .
FOVS containing many semitransparent fields	Compensation cannot be accurately performed with On FOV in some cases.

Table 7-1 Descriptions

7.3 DOP Compensation Setting

7.3.1 Switching DOP Compensation

Step1: Click **BF2** button on the upper left side of the window to open the menu and Click **BF2 Option**.



Figure 7-6 Switching DOP Compensation 1

Step2: The dialog shown below appears.
Select **Inspection Data**.

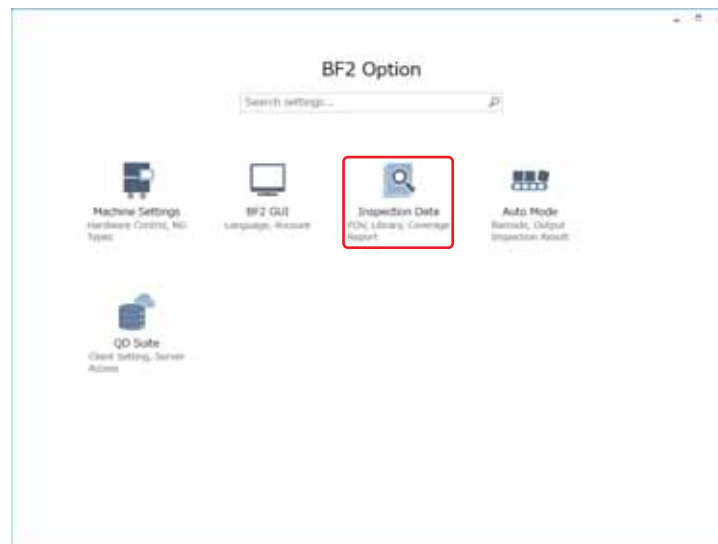


Figure 7-7 Switching DOP Compensation 2

Step3: Select **DOP** and check **Distortion Measurement**.

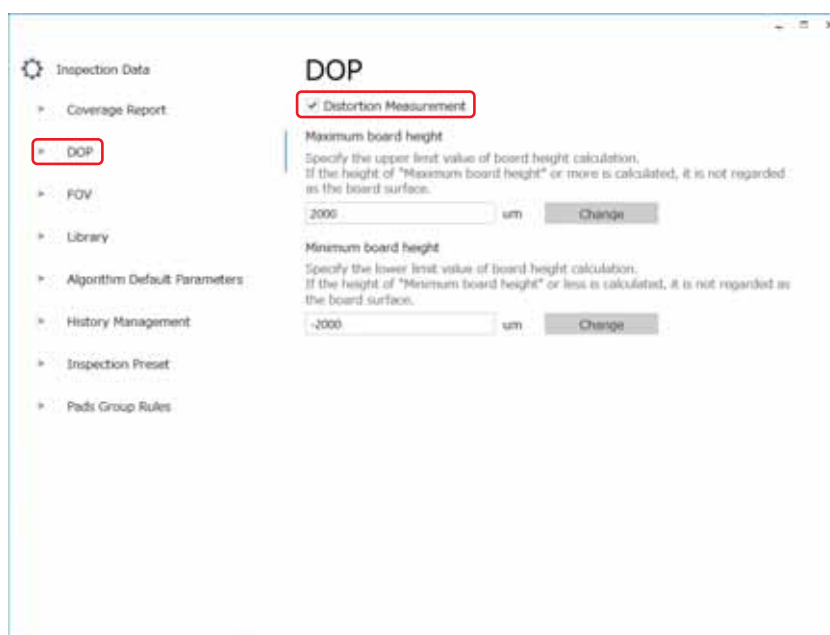


Figure 7-8 Switching DOP Compensation 3

7.3.2 Editing DOP of On Edge

Here describes how to edit the DOP of **On Edge**.

CAUTION DOPs of **On FOV** cannot be edited because they are allocated automatically.

CAUTION DOPs of **On Edge** are allocated based on pad allocation. So DOPs are not allocated when the pad does not have recipe.

Step1: Select **FOV** tab.



Figure 7-9 Editing DOP of On Edge 1

Step2: Edit panel appears.

Select **Setting** > **Distortion measurement setting**.

NOTE

When **Distortion measurement setting** is changed, the changes are applied to all other FOVs too.

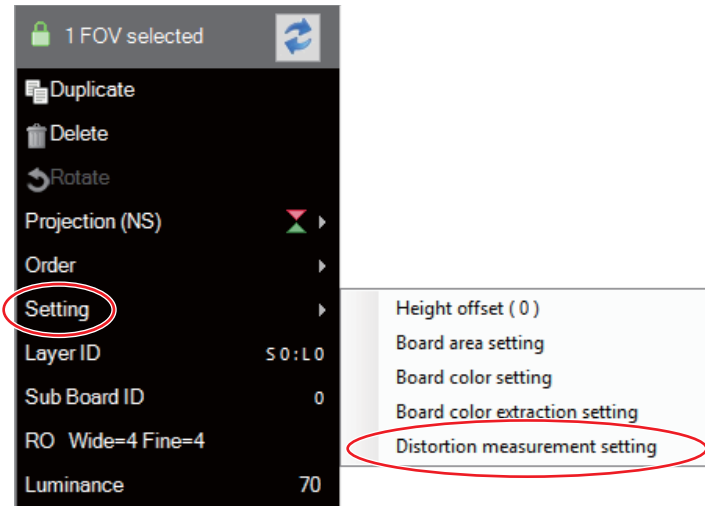


Figure 7-10 Editing DOP of On Edge 2

Step3: The dialog shown below appears. For details, refer to table 7-2.

Select **On Edges** from **Component DOP type** drop-down list, and click **OK**.

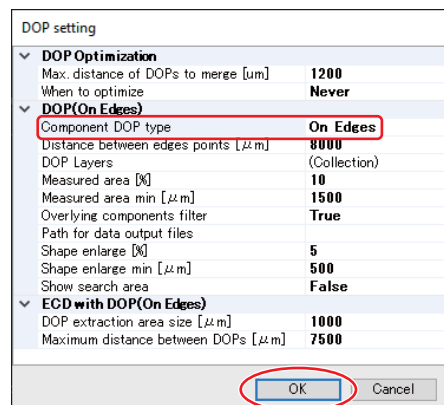


Figure 7-11 Editing DOP of On Edge 3

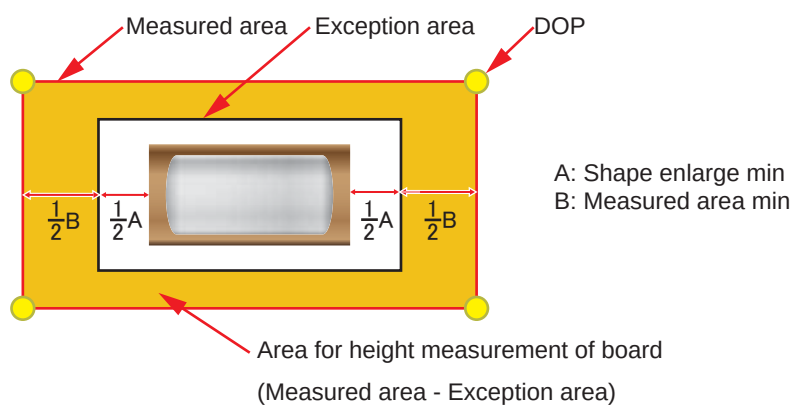
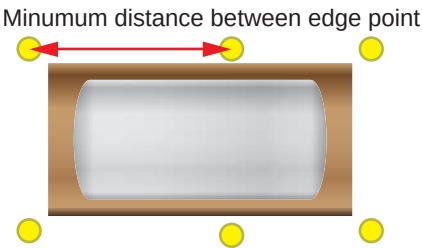


Figure 7-12 Editing DOP of On Edge 4

Item	Description
DOP Layers	Not available
Distance between edges points[μm]	When distance between DOPs is long, difficult to properly correction. DOP is added automatically when the original distance between DOPs is longer than Distance between edges points.  Figure 7-13 Editing DOP of On Edge 5
Component DOP type	Set On Edge to use On Edge correction. The default value is On FOV .
Path for data output files	Specifies output path for log file.
Shape enlarge min [μm]	Specifies the minimum size of the exception area of board height calculation. In case the pad size is small and only Shape enlarge[%] setting is used, the exception area may be too small to properly correct. When the area specified by Shape enlarge min[μm] is bigger than the area specified by Shape enlarge[%], the Shape enlarge min[μm] area is used as the exception area. In that case, the outside of pad shape by half of Shape enlarge min[μm] is used as the exception area. For details, refer to Figure 7-12.
Measured area min [μm]	Specifies the minimum size for board height calculation. In case the pad size is small and only Measured area[%] setting is used, the measurement area may be too small to properly correct. When the area specified by Measured area min[μm] is bigger than the area specified by Measured area[%], Measured area min[μm] area is used as the measurement area. In that case, the outside of the exception area by half of Measured area min[μm] is used as the measurement area. For details, refer to Figure 7-12.
Shape enlarge[%]	Specifies the area to exclude from calculation of board height. There may be any noises near pad. So it is necessary to exclude that area from calculation of board height. The area enlarged from the shape of pad with specified percentage is exclude. Recommended setting is 5%.
Measured area[%]	Specifies the area to calculate board height. The expanded pad shape size with specified percentage is the area to calculate board height. Recommended setting is 10%.

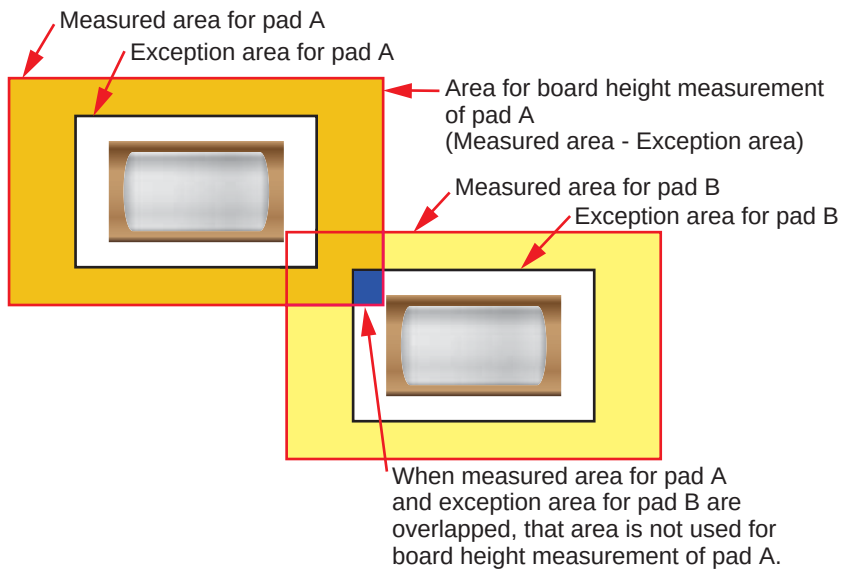
Item	Description
Overlying components filter	If an exclusion area of another part is included in the measurement area, exclude that area from the height calculation of the board surface. Default is True.  Figure 7-14 Editing DOP of On Edge 6
Show search area	Show measurement area as red frame.
Max. distance of DOPs to merge [μm]	The set value is used for the optimization of number of DOP. DOPs are merged when the distance between DOPs which can be merged is shorter than Max. distance of DOPs to merge [μm]. Default value is 1200[μm].
When to optimize	The set value is used for the DOP optimization. Default value is Never.
DOP extraction area size [μm]	Not available
Maximum distance between DOPs [μm]	Not available

Table 7-2 Descriptions

Step4: From the **FOV** tab, select **Distortion** group > **Optimize DOP**.

CAUTION DOPs of **On Edge** are not allocated when the pad does not have recipe.

NOTE **Optimize DOP** appears only when **Component DOP type** is set to **On Edges**.



Figure 7-15 Editing DOP of On Edge 7

Step5: From the **FOV** tab, select **Display** group > **Display** > **Distortion compensation**.

Allocated DOPs of **On Edge** is displayed.

Check DOPs are allocated.

CAUTION When DOPs are not allocated, the pad may not have any recipe.



Figure 7-16 Editing DOP of On Edge 8

Step6: After scanning the board, set image filter as **Board height**.

And check the board part is colored green or yellow.

When board part is colored black, blue, orange, or red, adjust parameters by reference to Table 7-2.

If the area which height is different from board height is included in **Area for height measurement of board** of Figure 7-12, compensation does not work well.

NOTE When the pad is small, adjust the parameters of **Shape enlarge min[μm]** and **Measured area min[μm]**.

Step7: From the **FOV** tab, select **Distortion** group > **Measure**.

If board part is colored black, blue, orange, or red, return to Step6 and adjust parameters again.



Figure 7-17 Editing DOP of On Edge 9

7.4 Settings When Using On FOV

When using **On FOV**, the following settings may be necessary.

NOTE These settings are not necessary when using **On Edge**.

7.4.1 Board Height Exclusion Area

When there is a slit or transparent part on a PCB, a PCB height may not be measured correctly. By setting the Board Height Exclusion Area on an FOV, the height measurement becomes more accurate.

For details of board height exclusion area settings, refer to **5.3.8 Board Height Exclusion Area**.

7.4.2 Board Color Extraction Setting

This is a function to extract the board color from the specified area during each scan.

This function is used when it is necessary to inspect boards having different board colors because of different board manufacturers on the same line.

Step1: Select the FOV for board height compensation and display the edit panel.

Select **Setting > Board color extraction setting**.

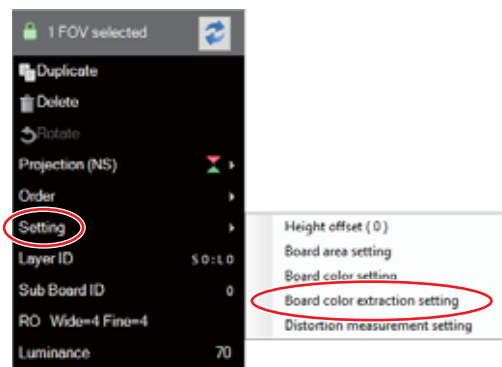


Figure 7-18 Board Color Extraction Setting 1

Step2: The dialog shown below appears.

Select an FOV to extract board color by clicking and .

After FOV is selected, check **Enable Board Color Rects**.

NOTE

One FOV only from which a board color can be extracted.

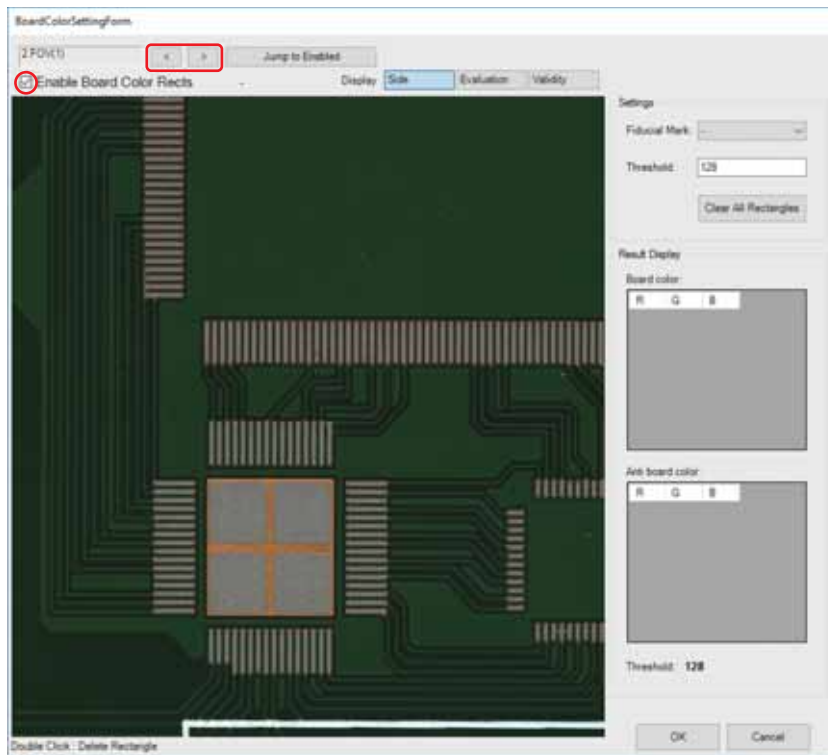


Figure 7-19 Board Color Extraction Setting 2

Step3: Specify the color of the board and other colors.

Check **Enable Board Color Rects** and click the board color section. Select **Apply average as board color** and a list of board colors is added to the **Board color** on the right side.

Click an area which has a color that is different from the board color and select **Apply average as anti board color**.

A list of anti board colors is added on the right side of **Anti Board color** field.

NOTE

Select the wiring patterns for the board color.

Select an area that there are few pad around. The area should be as large as possible.

As colors other than the board color, select colors of pads, silk, holes, insulation material areas, etc.

NOTE

The areas specified as board colors are shown in blue squares, and the areas specified as colors other than board colors are shown in white squares.

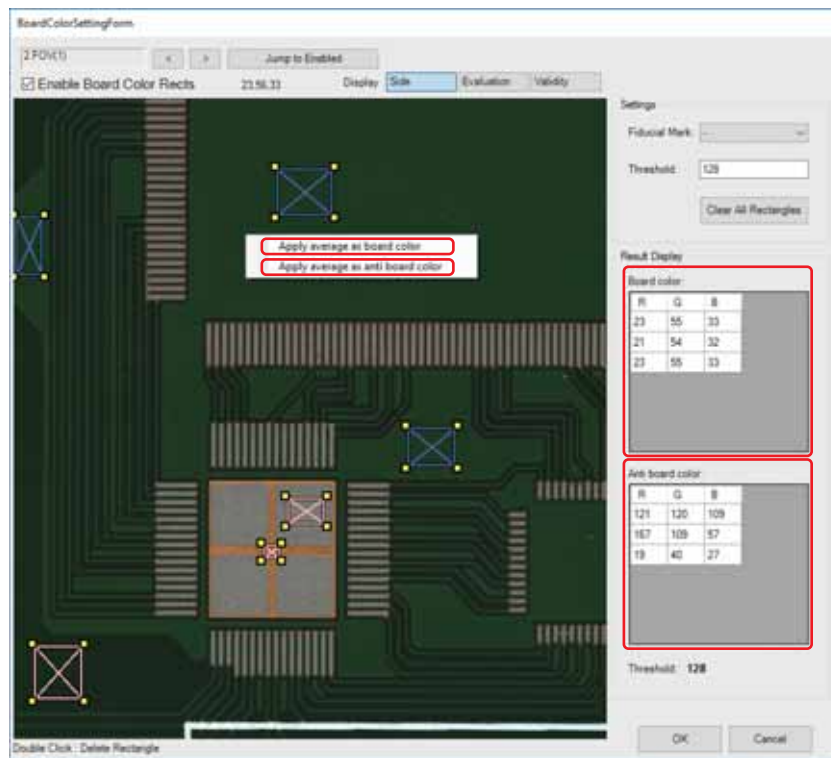


Figure 7-20 Board Color Extraction Setting 3

Step4: Fine tune the color specification set in Step3.

Select **Evaluation** on the right side of **Display**.

The area registered as the board color is shown in white depending on the Board color and Anti Board color settings.

Select **Validity** on the right side of **Display** to display areas with a brightness greater than the **Threshold** in white and all other areas in black on the **Evaluation** image.

Areas displayed in white when using **Validity** are recognized as the board surface.

Set the **Board color**, **Anti Board color**, and **Threshold** to display only the board color area in white.

For details settings, refer to Table 7-3.

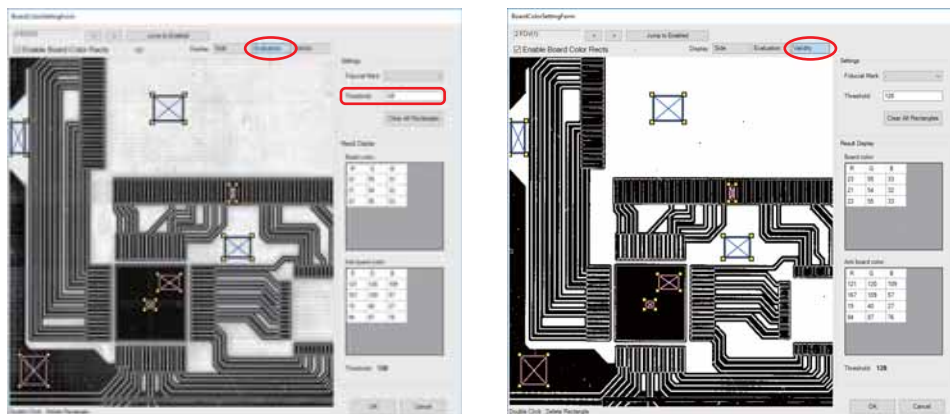


Figure 7-21 Board Color Extraction Setting 4

Item		Description
		Switches an FOV.
Jump to Enabled		Switches to an FOV for which Enable Board Color Rects is selected.
Enable Board Color Rects		Select after switching to an FOV from which the board color is to be extracted. A board color can be extracted from one FOV only.
Display	Side	Displays the images of SideLight.
	Evaluation	Displays the board color area in white based on the Board color and Anti Board color settings.
	Validity	Areas in the Evaluation image with a brightness greater than the Threshold are displayed in white and all other areas in black.
Fiducial Mark		Assigns the fiducial mark to use for misalignment adjustment. Fiducial marks not used for adjusting a misalignment of the pad position are not displayed.
Threshold		Sets the threshold value when converting the Evaluation image to the Validity image. White areas in the Validity image are recognized as the board surface.
Clear All Rectangles		Deletes all Board color and Anti Board color settings.
Board color		A list of board colors set with Apply average as board color.
Anti board color		A list of board colors set with Apply average as anti board color.

Table 7-3 Board Color Extraction Setting

Step5: After assigning a fiducial mark to use for misalignment adjustment, click **OK**.

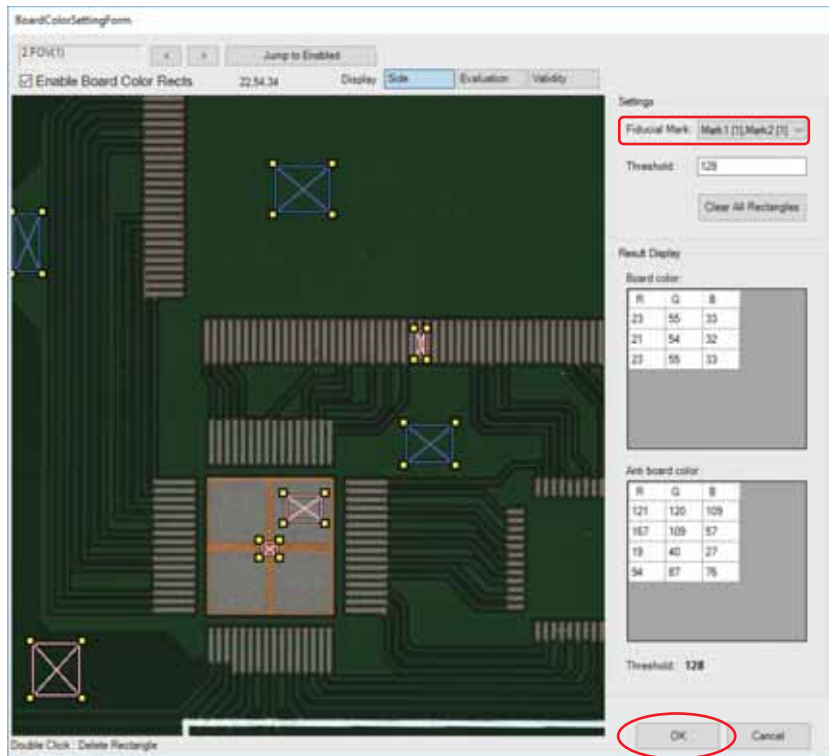


Figure 7-22 Board Color Extraction Setting 5

7.5 Board Height Compensation Tuning

Here describes tuning of board height compensation.

Step1: After allocation of FOVs has completed, capture the board before setting DOP compensation.

From **FOV** tab, select **Machine control** group > **Scan**.

At this point in time, DOP compensation is performed with **On FOV**.



Figure 7-23 Board Height Compensation Tuning 1

Step2: Change the Image Filter to **Board height** and confirm that the board surfaces are green or yellow. If the board surfaces are black, blue, orange, and/or red, proceed to Step 3.

NOTE

The color changes from blue to black as the height becomes lower than Board surface.

Also, the color changes from orange to red as the height becomes higher than Board surface.

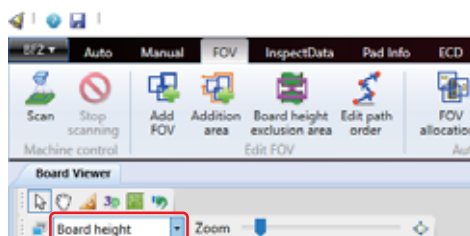


Figure 7-24 Board Height Compensation Tuning 2

Step3: Set the board color. For details, refer to **5.3.12 Board Height Compensation Setting**.

After all the setting of the board colors is completed, select **Distortion** group > **Measure** from the ribbon to perform the DOP compensation processing.

If the board surfaces are black, blue, orange, and/or red, make the settings by reference to Table 7-4.

NOTE

From the ribbon, select **Distortion** group > **Clear** to cancel the DOP compensation.



Figure 7-25 Board Height Compensation Tuning 3

Item	Description
The height of an FOV having many wiring patterns and the height of an FOV having few wiring patterns are different	Set the board color extraction setting. For details, refer to 7.4.1 Board Height Exclusion Area .
A lot of semitransparent parts are included in an FOV, and compensation cannot performed correctly.	Set the board color extraction setting. For details, refer to 7.4.1 Board Height Exclusion Area .
The warpage at the edge of the board is large and the height of the nearby pads is low or high	Sets the On Edge setting. For details, refer to 7.3.2 Editing DOP of On Edge .
The inclination between sub boards is different because of the slit design, and the heights of the nearby pads are low or high	Sets the On Edge setting. For details, refer to 7.3.2 Editing DOP of On Edge .

Table 7-4 Descriptions

8 Making Inspection Data of a Large Board

Inspection machine for large boards can inspect boards having a length of 510 mm or longer.

If a board has a length of 510 mm or longer, shift the board clamp position after scanning the board and scan the remaining area.



Figure 8-1 Scan Method of Large Board

8.1 Assigning the Layer of Data

This section describes how to assign the layer of data.

The FOV scanned with the first scan and the FOV scanned with the second scan are separated by layer numbers.

The FOV scanned with the first scan is Layer 1, and the FOV scanned with the second scan is Layer 2.

Data also have the concept of layers, so Layer 1 of a data is inspected with an image scanned at Layer 1 of the FOV and Layer 2 of the data is inspected with an image scanned at Layer 2 of the FOV.

NOTE The initial layer of an FOV exceeding 510 mm is Layer 2.

Step1: From **InspectData** tab, select **Layout** group > **Allocate Layer** to assign layers to each data automatically.

Layer1 data are highlighted in green and Layer2 data are highlighted in purple.



Figure 8-2 Assigning the Layer of Data 1

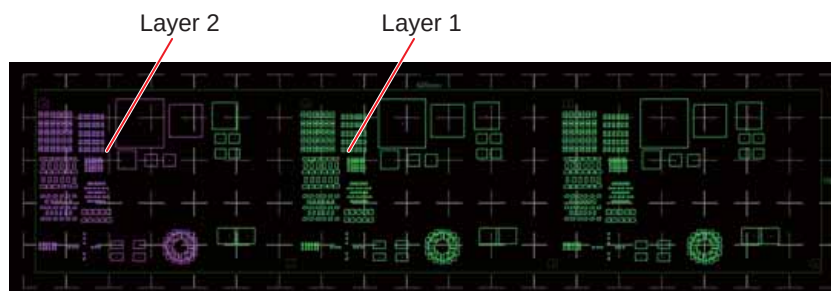


Figure 8-3 Assigning the Layer of Data 2

Step2: To change layer ID manually, click **Set Layer ID** from edit panel.
Layer ID setting dialog box appears. Enter layer ID to set and click **OK**.

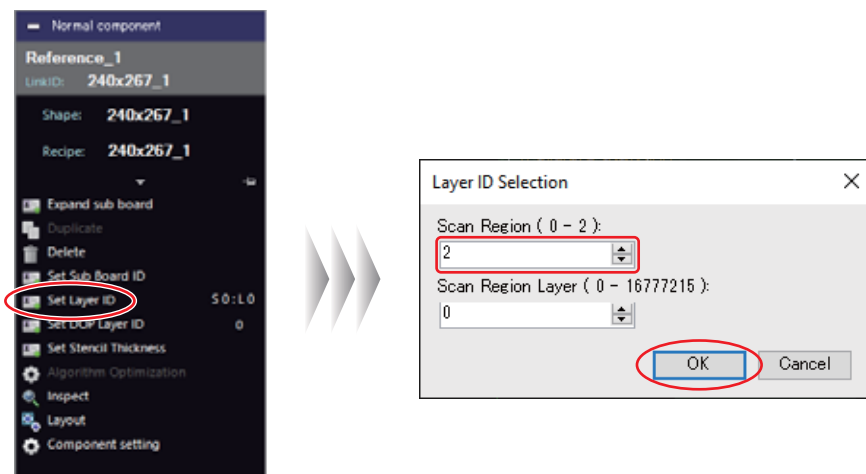


Figure 8-4 Assigning the Layer of Data 3

8.2 Assigning the Layer of FOV

Perform automatic allocation of the FOVs according to **5.3.3 Optimizing FOV Allocation (Auto Allocation)**.

The FOV settings in consideration of the layers are automatically set.

Layer1 FOVs are highlighted in green and Layer2 FOVs are highlighted in purple.

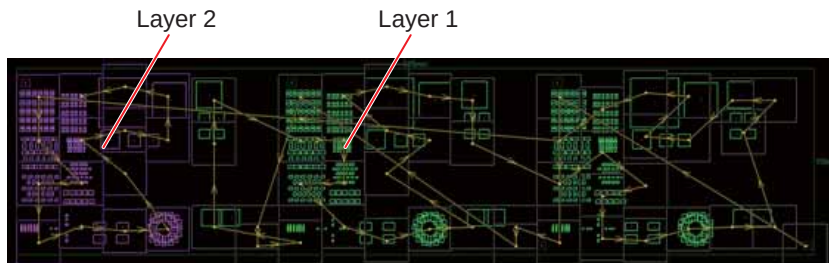


Figure 8-5 Assigning the Layer of FOV 1

To change layer ID of FOV, select any FOV on **Board Viewer**.

Click **Layer ID** and layer ID setting dialog box appears. Enter layer ID and click **OK**.

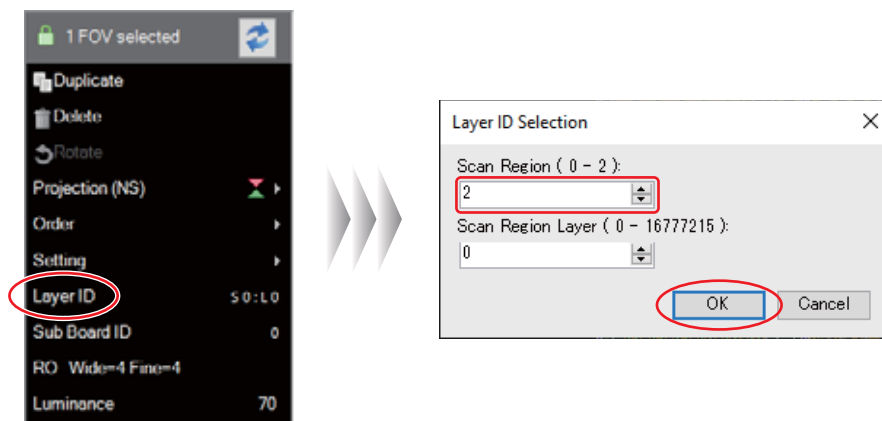


Figure 8-6 Assigning the Layer of FOV 2

8.3 Adding the Fiducial Mark

Regarding large board 510 mm or longer, two or more fiducial marks are necessary for each layer. Depends on the board, It is necessary to add fiducial mark to the boundary of layers by manual.

Step1: Add a fiducial mark to the boundary between Layer 1 and Layer 2 by manual.

Create new fiducial marks at the positions to be treated as the fiducial marks according to **1.7.1 Create Marks**.

NOTE

When it is difficult to create mark in Layer2, add a fiducial mark on the range of 410 mm to 510 mm from board edge.

Add a fiducial mark to the boundary between Layer 1 and Layer 2

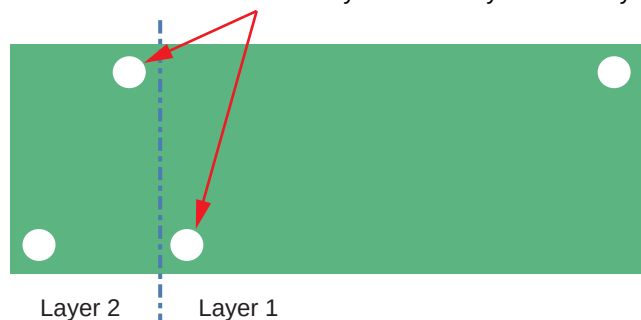


Figure 8-7 Adding the Fiducial mark 1

Step2: Select added fiducial mark and click **Set Layer ID** on edit panel.

The layer ID setting dialog box appears. Enter layer ID to set and click **OK**.

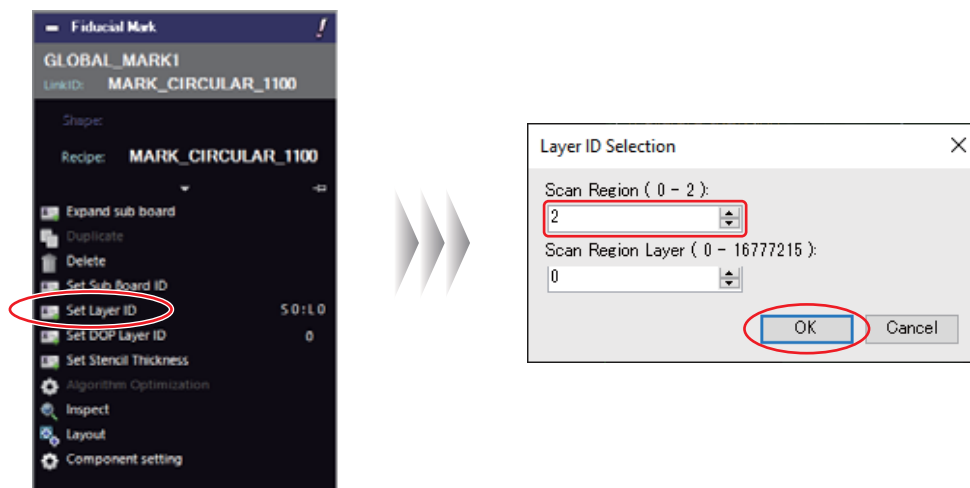


Figure 8-8 Adding the Fiducial mark 2

Step3: Confirm sub board ID and layer ID are correct, and assign fiducial mark according to **1.7.3 Automatic Assignment of Mark**.

Step4: Adjust the positions of the added fiducial marks.

Adjust the positions of fiducial marks according to **1.7.5 Makes Fiducial Mark Adjustment**.

8.4 Registering the Template Image for Detection of the Board Edge

When scanning a large board, the amount of shift of the board stop position is sought by scanning the edge of the board while the board is clamped before scanning and then matching this image with the template image registered in advance.

The scan position is compensated based on the results of the calculation.

If the template image has not been registered, the following template image registration screen is displayed when the board is clamped before scanning.

Move the cursor over the edge of the board to register the template image and click **Template Registration**.

NOTE

Right-click at the **Board Viewer** of the **FOV** tab and select **Clear template for detecting board edge** to clear the template image and it will be unregistered state.

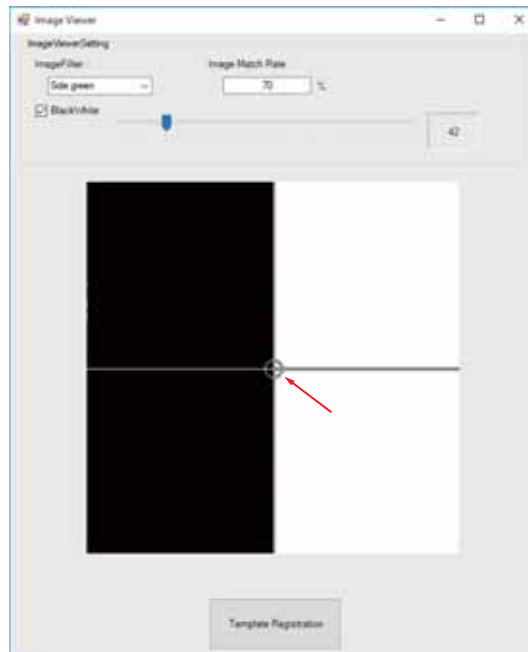


Figure 8-9 Adding Template Image

Item	Description
Image Filter	Select a lighting that displays the board clearly.
Image Match Rate	Sets an image match rate to use when detecting the edge of a board.
Scan Retry Count	If the image match rate is equal to or less than the specified rate, only during detection of the board edge of the rear end, the camera moves 10 mm to the right and the detection of the edge of the board is performed again. Retries while moving to the right for the set number of times.
BlackWhite	Check to display BlackWhite. Threshold is changed by using sliderbar.
Template Registration	Adds the template image.

Table 8-1 Adding Template Image

8.5 Display by layer

Scan data is displayed by each layer.

Step1: Select **InspectData** tab and right-click on **Board viewer**.

Select **Image layers** from the pop-up menu. The list of layer ID appears.

Check layer to display image.

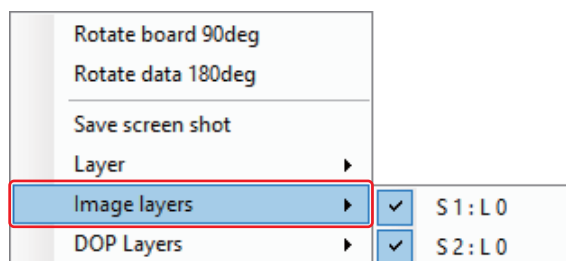


Figure 8-10 Display by layer 1

When Layer1 and Layer2 overlap, FOV image of Layer2 is displayed.

Because of the differences of board clamped state between Layer1 and Layer2, the joint of window may be shift.

But it is no problem since only the FOV image of the same layer as the data layer is used in inspection.

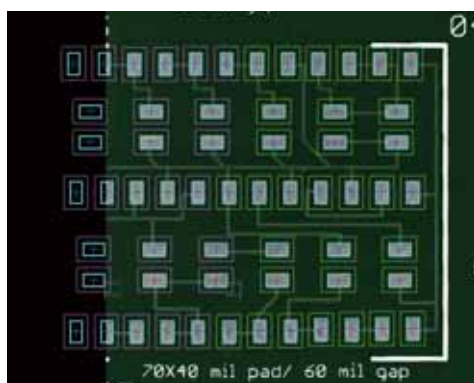


Image of layer 1

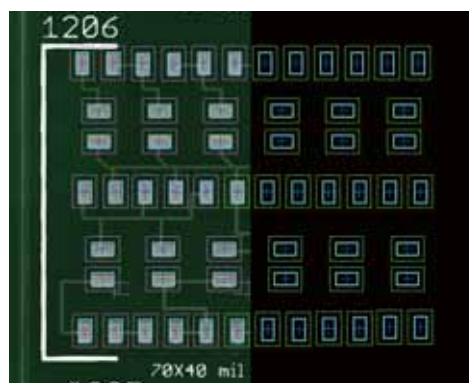


Image of layer 2

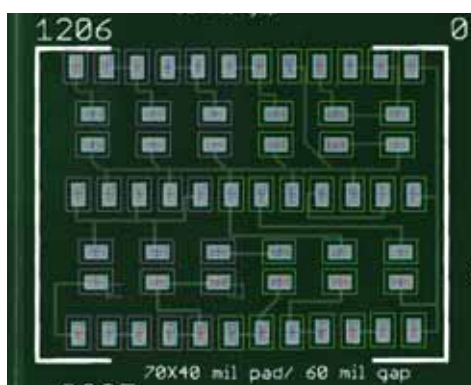


Image of all layers

Figure 8-11 Display by layer 2

9 Offline Debug

Offline Debug is a function to outputs pad images during automatic operation and uses those images to conduct parameter tuning of the inspection window.

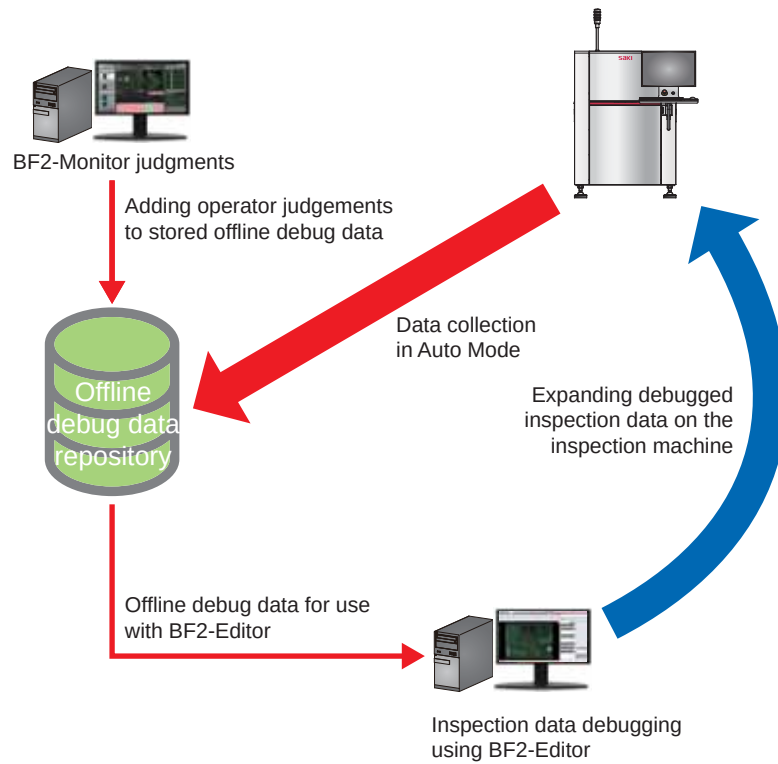


Figure 9-1 Offline Debug

9.1 Specifying the Output Destination

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

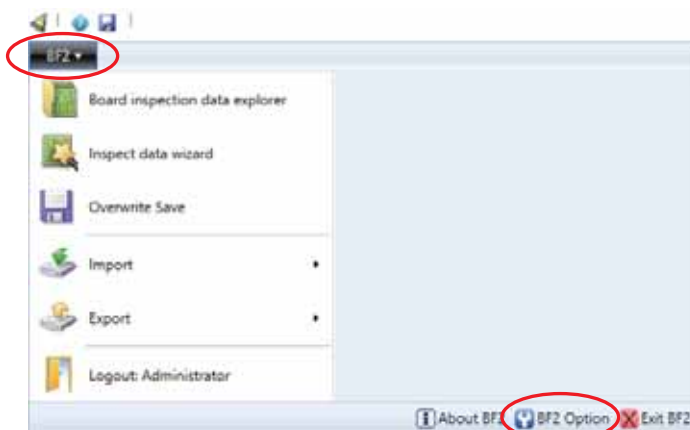


Figure 9-2 Specifying the Output Destination 1

Step2: The dialog shown below appears.
Click **Auto Mode**.

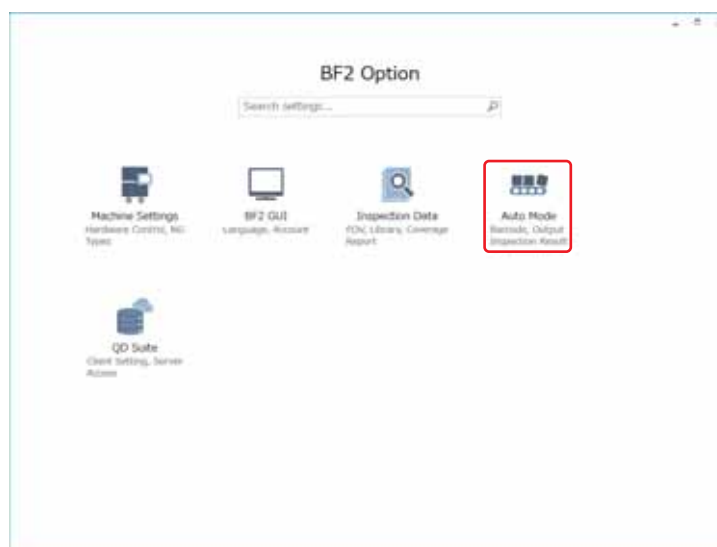


Figure 9-3 Specifying the Output Destination 2

Step3: Select **Output Inspection Result** and set toggle switch of **Offline Debug** to **ON**.

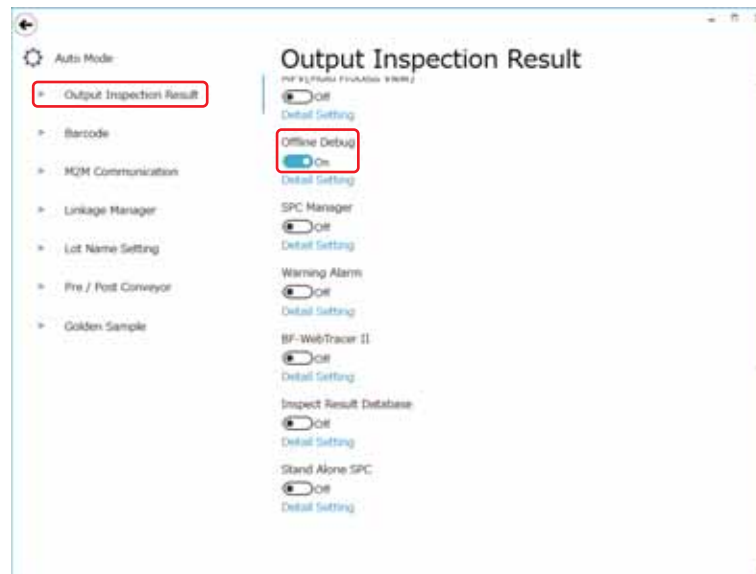


Figure 9-4 Specifying the Output Destination 3

Step4: Click **Detail Setting** under the toggle switch of **Offline Debug** to set detailed items. Specify **Offline debug data destination path**.

NOTE Default setting is C:\Bf2OfflineDebugData.

9.2 Pad Image Output Settings

Step1: Select the **Auto** tab.

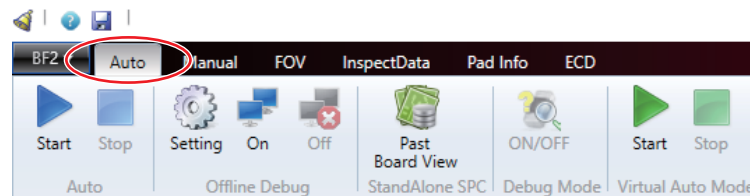


Figure 9-5 Pad Image Output Settings 1

Step2: From the ribbon, select the **Offline Debug** group > **Setting**.

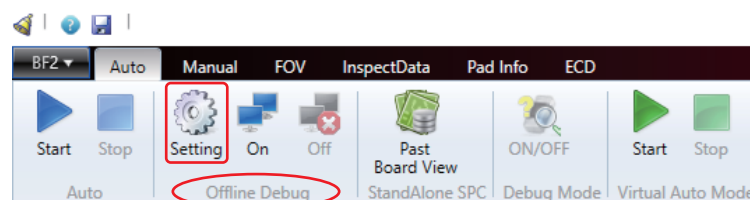


Figure 9-6 Pad Image Output Settings 2

Step3: The dialog shown below appears.

Select each folder in the Library directory to display the list of shapes included in the selected folders in the Shape list under the **Shape/recipe list**.

Click  and  arrows to move the **Shapes/recipes list** to the **Selected shape/recipe list** and edit the **Output option**.

For details, refer to Table 9-1.

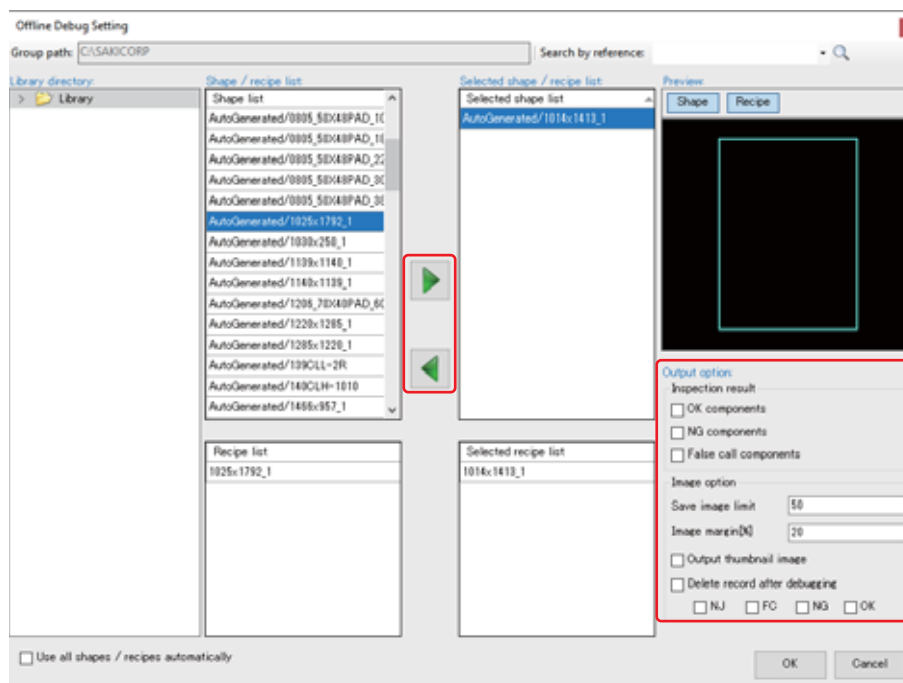


Figure 9-7 Pad Image Output Settings 3

Item	Description
Group path	The path for the inspection data which is currently open.
Search by reference	Click the ▼ arrow next to Search by reference to display the reference list for the currently opened inspection data. Select a reference from the list to search the shapes/recipes or check the settings in the specified reference.
Library directory	This is the same list as the Library Tree described in 2.5.2 Window Details . Select each folder to display the list of shapes included in the selected folders in the Shape list under the Shape/recipe list .
Shape / recipe list	The list of shapes included in the folder selected from the Library directory is displayed in the Shape list. Select a shape from the Shape list to display a list of the included recipes in the Recipe list.
 , 	Click  to move the shape/recipe selected from the Shape/recipe list to the Selected shape/recipe list . Click  to move the shape/recipe selected from the Selected shape/recipe list to the Shape/recipe list .
Selected shape / recipe list	The shapes and recipes in the Selected Shape/Recipe list are modified by the Output option settings. To exclude a particular shape and recipe from being output, select the shape and recipe and click  .

Item		Description
Preview	Shape	Displays the shape selected in the Selected shape/recipe list with a solid line.
	Recipe	Displays the shape of the recipe selected in the Selected shape/recipe list with a dotted line.
Output option	OK components	Check this option to save pad images for shapes and recipes in the Selected shape/recipe list when the inspection result is OK.
	NG components	Check this option to save pad images for shapes/recipes in the Selected shape/recipe list when the inspection result is NG.
	False call components	Check this option to save pad images for shapes/recipes in the Selected shape/recipe list when the inspection result is False call.
	Save image limit	Specifies the upper limit on the number of pad images output for one recipe. Older images are deleted and new images are output when the upper limit is reached.
	Image margin[%]	Specifies the image margin when outputting pad images.
	Output thumbnail image	Saves the thumbnail images to the folder specified in 9.1 Specifying the Output Destination .
	Delete record after debugging	In 9.4 Parameter Tuning Using the Offline Debug Function , the inspection is performed using images output with the Offline Debug function to tune parameters. Check to delete offline debug images with the judgement result OK when the recipe edit screen is closed. Select the offline debug images to delete from NJ, FC, NG, or OK . NJ: Deletes offline debug images of boards before judgement on BF2-Monitor. FC: Deletes offline debug images of boards that are judged as False Call on BF2-Monitor. NG: Deletes offline debug images of boards that are judged as NG on BF2-Monitor. OK: Deletes offline debug images of boards that are judged as OK on the inspection machine.

Table 9-1 Pad Image Output Settings

Step4: After all the settings are completed, click **OK**.

Check that **On** is selected in the **Offline Debug** group.

Starting automatic operation under these conditions saves the pad images based on the Output options.

NOTE The **On** and **Off** settings may be changed during automatic operation.

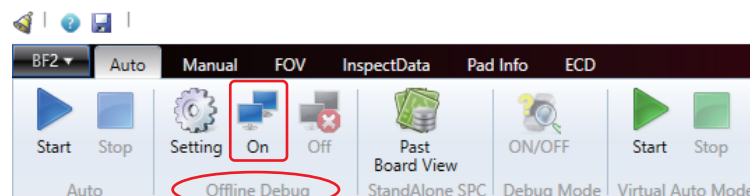


Figure 9-8 Pad Image Output Settings 4

9.3 Listing the Pad Image Output

Starting Auto inspection after make the settings of **9.2 Pad Image Output Settings**, saves the pad images based on the Output settings.

The output pad images can be confirmed after stopping automatic operation.

Step1: Select the **InspectData** tab.



Figure 9-9 InspectData

Step2: Select the **Offline Debug** tab on bottom part of dock window.

Click  to display a list of output pad images.

Select a reference from the list and double-click it to display the Recipe edit screen for that reference.

For details, refer to Table 9-2.



Figure 9-10 Listing the Pad Image Output

Item		Description
		Updates the list.
	Delete the selected reference image	Deletes the reference image selected in the list.
	Delete all of the board reference images	Deletes all the reference images in the currently opened board. References which are listed as Others belong to other boards within the same group and are excluded from deletion.
	Delete all of the group reference images	Deletes all of the reference images in the group. References listed as Others are also deleted.
Select all		Selects all of the list items.
View	Hide references with no output images	References which have shapes/recipes set in 9.2 Pad Image Output Settings are displayed in the list. Check this item to hide references with 0 output images.
	Display references with output images at full size	Displays references with output images with a dotted yellow line at the full image size.
Display condition		Filters by Board number, From last, judgement results, or the specified date and time. Check Board number and enter the number of boards in the text-box on the right to show offline debug images of the boards that were recently inspected for the specified number of the boards. Check From last and enter the date in the text-box on the right to show offline debug images of the board from the current time to the specified date and time. NJ, FC, NG, and OK indicate the judgement results. NJ: Shows offline debug images of the board before judgement on BF2-Monitor. FC: Shows offline debug images of the board that are judged as False Call on BF2-Monitor. NG: Shows offline debug images of the board that are judged as NG on BF2-Monitor. OK: Shows offline debug images of the board that are judged as OK on the inspection machines. Specify the date and time in From and To to show offline debug images of the board inspected during the specified period.
Search		Enter the shape/recipe/link ID/reference name into the text-box and click  to highlight the search items in the list.

Table 9-2 Listing the Pad Image Output

9.4 Parameter Tuning Using the Offline Debug Function

9.4.1 Image List

Double-click the desired reference from the list described in **9.3 Listing the Pad Image Output** to display the Recipe edit screen for that reference.

A list of pad images output via the Offline Debug function within identical recipes is displayed in the **Image List** on the right side of the Recipe edit screen.

Select the reference from the list to switch to full-size image display and show the selected reference image.

The **Offline Debug Mode** panel is also displayed to the upper right of the full-size image.

Click the  on the right side of the **Offline Debug Mode** panel to return to the original image.

For details, refer to Table 9-3.

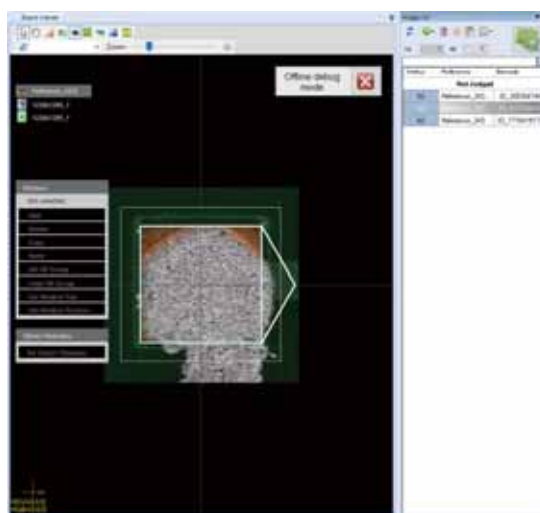


Figure 9-11 Image list

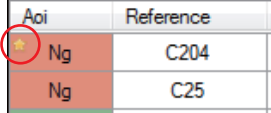
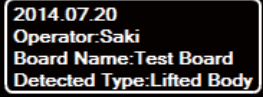
Item		Description
Offline debug mode		Select the reference from the list to display it. Click  on the right side to return to the original image.
		Updates the list.
	Add the current component image to the list	Adds the reference images in the currently opened Recipe edit screen to the list.
	Add the component image from the identical recipe component list to the list	Adds the images for each pad in the Identical Recipe Component List in the lower left of the Recipe edit screen to the list.
		Deletes the reference image selected in the list.
		Bookmarks the reference image selected in the list. Bookmarked images are displayed with a yellow, star-shaped icon. The upper limit of the number of images displayed in the list depends on the Save image limit set in 9.2 Pad Image Output Settings. Older images are deleted and new images are output when the upper limit is reached, except bookmarked images are excluded. Click it once again to clear the bookmark.  Figure 9-12 Bookmark
		Adds a comment to the reference image selected in the list. The added comment is displayed in the lower right of the full-screen display.  Figure 9-13 Comment
		Shows components grouped by judgement results. Ungrouped : Cancels grouping of components. Grouped : Shows grouped components.
		Inspects the images output by the Offline Debug function using the parameters of the currently opened recipe. For details. Refer to 9.4.4 Inspecting Images Output with the Offline Debug Function .
Search		Enter the reference name into the text-box and press  to highlight the search items in the list.
Image List		The Recognition Result is the inspection result when the image is output during automatic operation. The inspection result is displayed in green if it is OK or red if it is NG. Click  to inspect the images output by the Offline Debug function using the parameters of the currently opened recipe. The color of the Reference Name and Bar Code changes based on the inspection result. The inspection result is displayed in light-blue if it is OK or red if it is NG.
Status		Shows judgement results of components. For details, refer to 9.4.2 Judgement Setting .

Table 9-3 Image List

9.4.2 Judgement Setting

The component judgement in **Image List** can be changed.

Step1: Right-click a component name in **Image List**.

Step2: Select **Set Judgement** from the pop-up menu.

After the judgement list is shown, select the judgement to set.

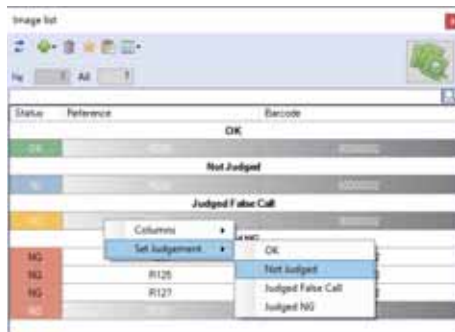


Figure 9-14 Judgement Setting

Item	Description
OK	Sets a pad as OK .
Not Judged	Sets a pad as Not Judged .
Judged False Call	Sets a pad as Judged False Call .
Judged NG	Sets a pad as Judged NG .

Table 9-4 Descriptions

9.4.3 Showing Algorithm Expansion and Inspection Preset Columns

The **Algorithm Expansion** and **Inspection Preset** columns can be shown in **Image List**.

Step1: Right-click a reference name in **Image List**.

Step2: Select **Columns** from the pop-up menu.

After the column list is shown, check **Algorithm Expansion** and **Inspection Preset**.

NOTE

To hide **Algorithm Expansion** and **Inspection Preset**, uncheck **Algorithm Expansion** and **Inspection Preset**.

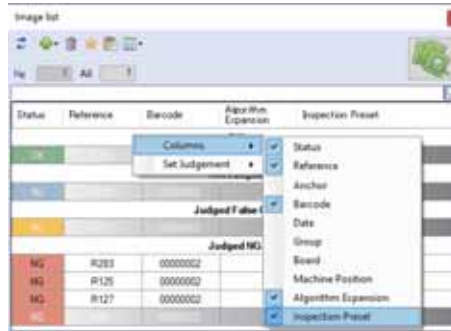


Figure 9-15 Showing Algorithm Expansion and Inspection Preset Columns

9.4.4 Inspecting Images Output with the Offline Debug Function

Click  to inspect the images output by the Offline Debug function using the parameters of the currently opened recipe.

The color of the **Reference Name** and **Bar Code** changes based on the inspection result.

The inspection result is displayed in light-blue if it is **OK** or red if it is **NG**.

Step1: Click  to start inspection.

Step2: Perform the parameter tuning following the directions in **2.9.2 Parameter Adjustment**.

The appropriate parameter threshold value can be set by storing pad images for NG inspection results.

9.5 Self Tuning Setting

To view Offline Debug Data on the **Self Tuning** tab, the **Offline Debug data** option needs to be selected.

Step1: Select the **Self Tuning** tab.

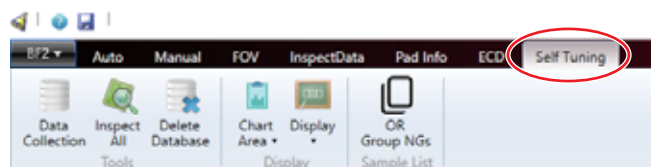


Figure 9-16 Self Tuning Setting 1

Step2: Select the **Offline Debug** tab.

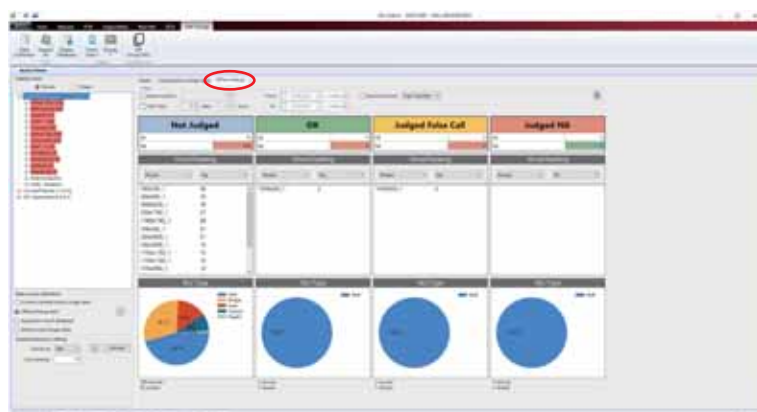


Figure 9-17 Self Tuning Setting 2

Step3: Check **Offline Debug data** of **Data source definition**.

Click .



Figure 9-18 Self Tuning Setting 3

Step4: The dialog box below appears.

After the component status for use is checked, click **Ok**.

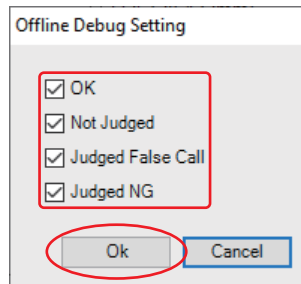


Figure 9-19 Self Tuning Setting 4

Item	Description
OK	Debug data judged as OK by the machine, or debug data set as OK on the recipe edit screen
Not Judged	Debug data set as Not Judged by the machine
Judged False Call	Debug data set as False Call , either by real-time judgement or on the recipe edit screen
Judged NG	Debug data set as NG , either by real time judgement or on the recipe edit screen

Table 9-5 Descriptions

Step5: Click **Inspect All**.



Figure 9-20 Self Tuning Setting 5

NOTE

The offline debug data can be filtered by the number of boards starting from the last board, the number of days and hours based on the current date and time, or a time interval. Check **Record Number** to select the number of records to load from **Top 5 records**, **Top 10 records** or **Top 15 records**.

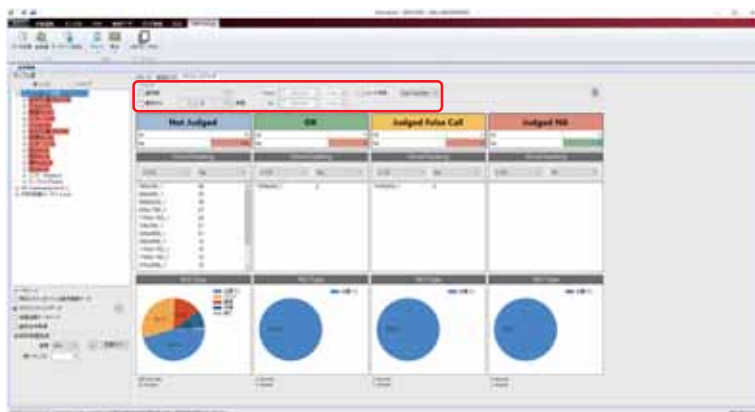


Figure 9-21 Self Tuning Setting 6

9.5.1 Offline Debug Data View

The background colors of the Ok and Ng cells will change according to the position and status.

Not Judged		OK		Judged False Call		Judged NG	
Ok	15	Ok	0	Ok	0	Ok	0
Ng	322	Ng	2	Ng	2	Ng	2

Figure 9-22 Offline Debug Data View

Item	Description
Not Judged	Shows the results of data that have not been judged by Offline Debug after Inspect All is performed. If the inspection result is NG, the Ng cell turns red.
OK	Shows the results of data judged as OK by Offline Debug after Inspect All is performed. If the inspection result is NG, the Ng cell turns red. If all inspection results are OK, the Ok cell turns green.
Judged False Call	Shows the results of data judged as False Call by Offline Debug after Inspect All is performed. If the inspection result is NG, the Ng cell turns red. If all inspection results are OK, the Ok cell turns green.
Judged NG	Shows the results of data judged as NG by Offline Debug after Inspect All is performed. If the inspection result is OK, the Ok cell turns red. If all inspection results are NG, the Ng cell turns green.

Table 9-6 Descriptions

9.5.2 Worst Ranking

Worst Ranking shows the recipes with most OK or NG results, according to the operator's judgement. Double-click a recipe from the recipe list to open the recipe editor screen for the selected recipe.

Worst Ranking	
Recipe	Ng
392x336_1	48
564x564_1	33
2600x250_1	30
250x1760_1	27
1742x1742_1	24
336x392_1	21
250x2600_1	21
392x2090_1	15
1755x1755_1	15
1725x1725_1	12
375x2088_1	12

Double-click to open the recipe editor screen

Figure 9-23 Worst Ranking

9.5.3 Pie Chart

The pie chart shows the percentage of NG types that caused the NG components. Click the target slice of the pie chart to open the self tuning chart for the selected NG type.

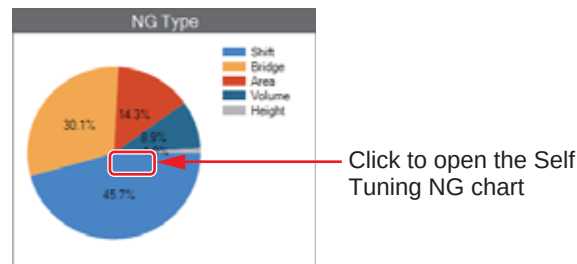


Figure 9-24 Pie Chart

9.6 Feedback to Offline Debug Data

BF2-Monitor can be set to automatically update offline debug data.

Step1: Activate BF2-Monitor.

Right-click the **BF2-Monitor** icon on the task bar and select **Options**.

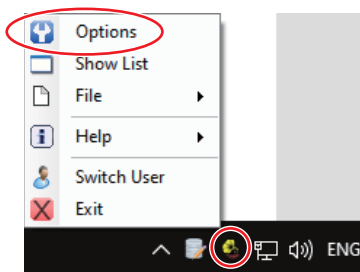


Figure 9-25 Feedback to Offline Debug Data 1

Step2: The dialog box shown below appears.

Click **Export Data**.

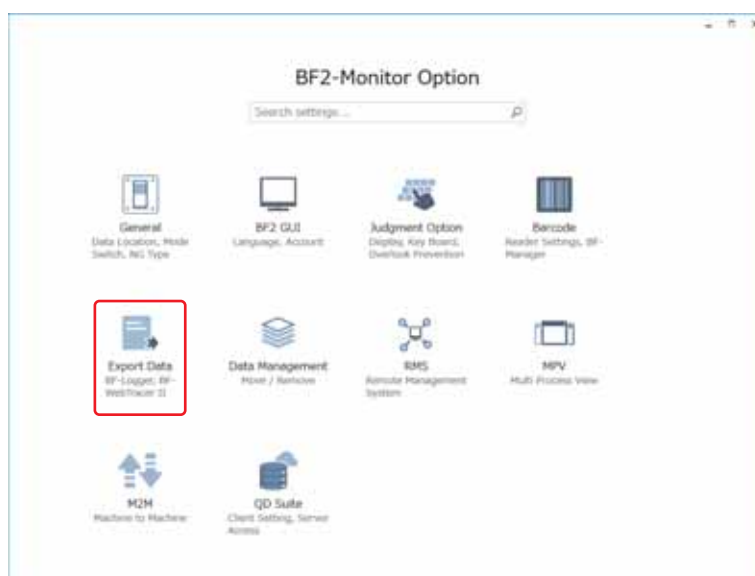


Figure 9-26 Feedback to Offline Debug Data 2

Step3: The dialog box shown below appears.
Click **Offline Debug**.



Figure 9-27 Feedback to Offline Debug Data 3

Item	Description
Allow Offline Debug feedback	Check to enable the feedback operation to offline debug data.
Show message when Offline Debug data path is not reachable	Check to show a message when BF2-Monitor cannot access offline debug data to update inspection results.

Table 9-7 Descriptions

NOTE

The offline debug data location is delivered to BF2-Monitor automatically with the inspection result data. In order to detect the feedback, the offline debug data in BF2 Option of a machine should be set via the shared network path. The destination must be also shared and accessible by BF2-Monitor.

10 Debug Mode

Debug Mode is a function to pause Auto Mode to perform tuning of inspection data by pad and manual judgement of inspection results when an NG result has been detected in Auto Mode.

10.1 Window Details

This section describes and illustrates the debug mode screen.

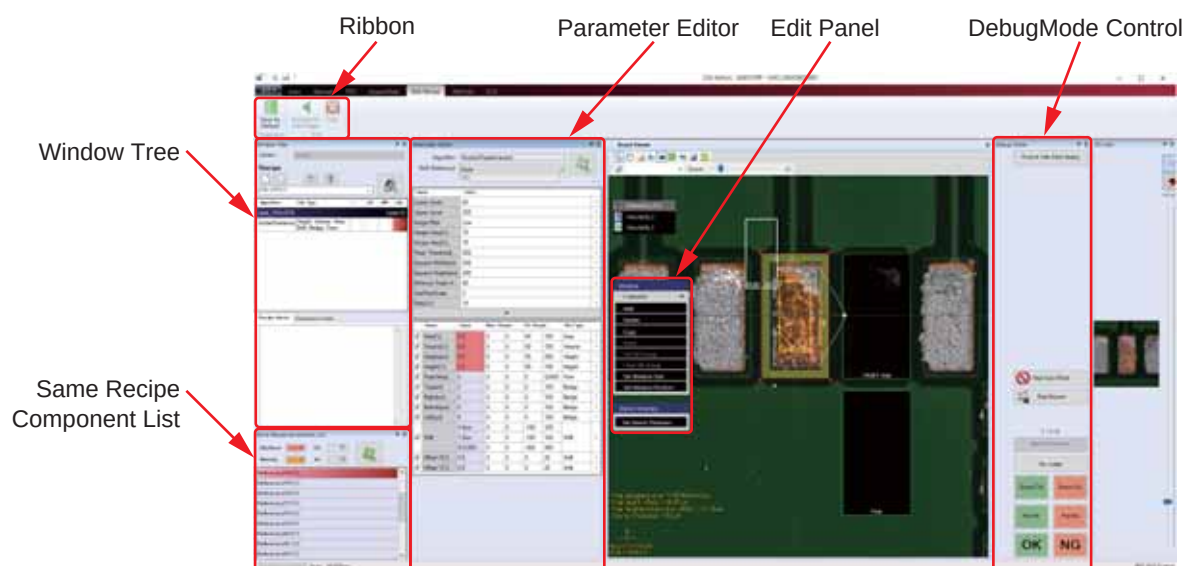


Figure 10-1 Window Details

Item		Description
Ribbon		Used when tuning inspection data. For details, refer to 2.9 Adjusting a Recipe .
Window Tree		
Parameter Editor		
Same Recipe Component List		
Edit Panel		
DebugMode Control	Fixed in Side Color display	Fix the display image in side color on Debug Mode. The comparison with actual board can be easily.
	Stop Auto Mode	Move to Auto Mode tab and stop Auto Mode.
	Stop Buzzer	Stop the buzzer.
	Back to Previous	Back to previous judgement window to retry judgement.
	No Judge	Use the manual judgment. For details, refer to 10.4 Manual Judgement of Inspection Result .
	Board OK	
	Board NG	
	Pad OK	
	Pad NG	
	OK	
	NG	

Table 10-1 Descriptions

10.2 Settings of Debug Mode

Step1: Click **BF2** button on the upper left side of the window to open the menu and Click **BF2 Option**.



Figure 10-2 Setting of Debug Mode 1

Step2: The dialog shown below appears.
Click **Auto Mode**.

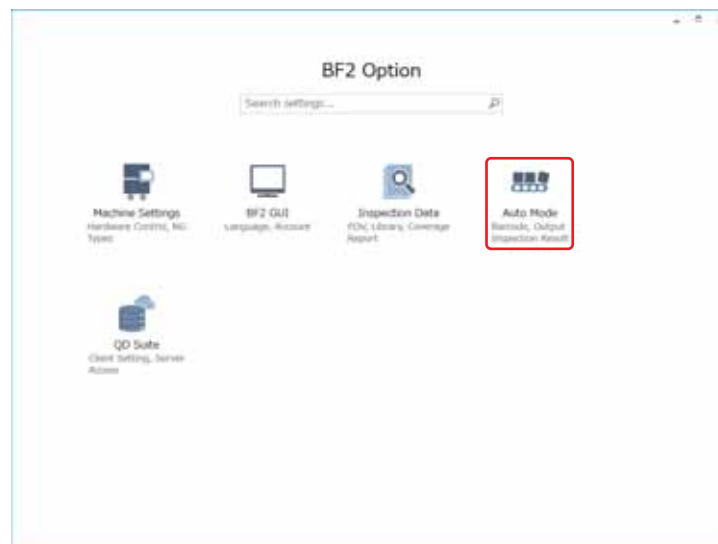


Figure 10-3 Setting of Debug Mode 2

Step3: Select **Output Inspection Result** and set toggle switch of **Debug Mode** to **ON**.

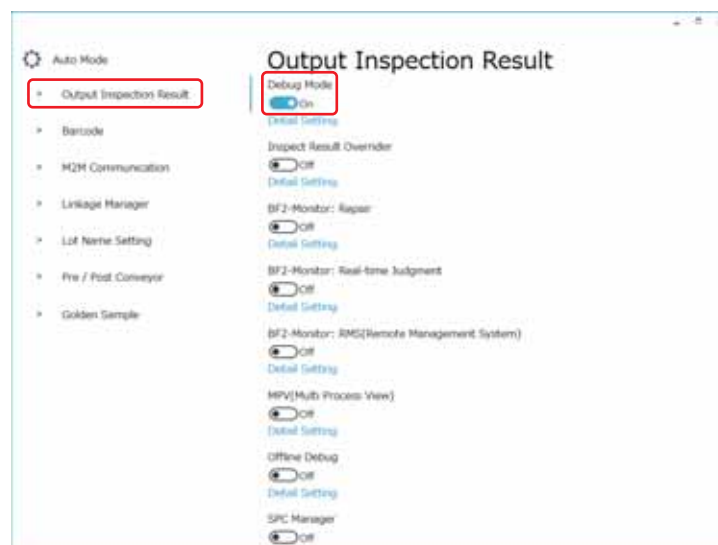


Figure 10-4 Setting of Debug Mode 3

10.3 Debugging Inspection Data

Inspection data can be debugged in the DebugMode screen.

Tune the inspection data if necessary by reference to **2.9 Adjusting a Recipe**.

NOTE

Coplanarity inspection data cannot be tuned in Debug Mode.

For editing coplanarity inspection data, refer to **1.9 Coplanarity Inspection**.

10.4 Manual Judgement of Inspection Result

Inspection result can be judged in the DebugMode screen.

In the Debug Mode operation screen, click an arbitrary judgement button.

For details, refer to Table 10-2.

Judge all the NG windows of a board or click **No Judge** to terminate the Debug Mode and resume Auto Mode.



Figure 10-5 Manual Judgement of Inspection Result 1

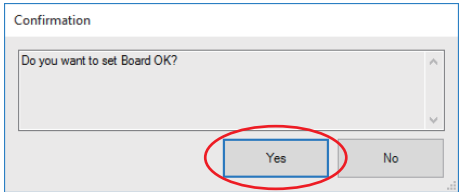
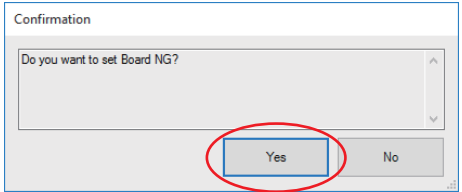
Item	Description
No Judge	Manual judgement is not necessary if you would only like to perform tuning of the inspection data. When manual judgement is not to be made, click No Judge.
Board OK	Judges all current boards as false calls at once. Click Board OK. The dialog shown below appears. Click Yes to exit debug mode and restart Auto Mode.  Figure 10-6 Manual Judgement of Inspection Result 2
Board NG	Judges all current boards as NG at once. Click Board NG. The dialog shown below appears. Click Yes to exit debug mode and restart Auto Mode.  Figure 10-7 Manual Judgement of Inspection Result 3
Pad OK	Judges current pads as false calls. If there is another NG pad, the screen switches to the Debug Mode screen of the next pad.
Pad NG	Judges current pads as NG. If there is another NG pad, the screen switches to the Debug Mode screen of the next pad.
OK	Click if the inspection result of the window is NG but the pad is indeed good. If there is another NG window, the window switches to the next NG window.
NG	Click if the inspection result is NG and the pad is indeed defective. If there is another NG window, the window switches to the next NG window.

Table 10-2 Descriptions

11 ECD

ECD (Extra Component Detection) is a function to detect drop-off foreign matters or other debris due to some reason from solder prints.

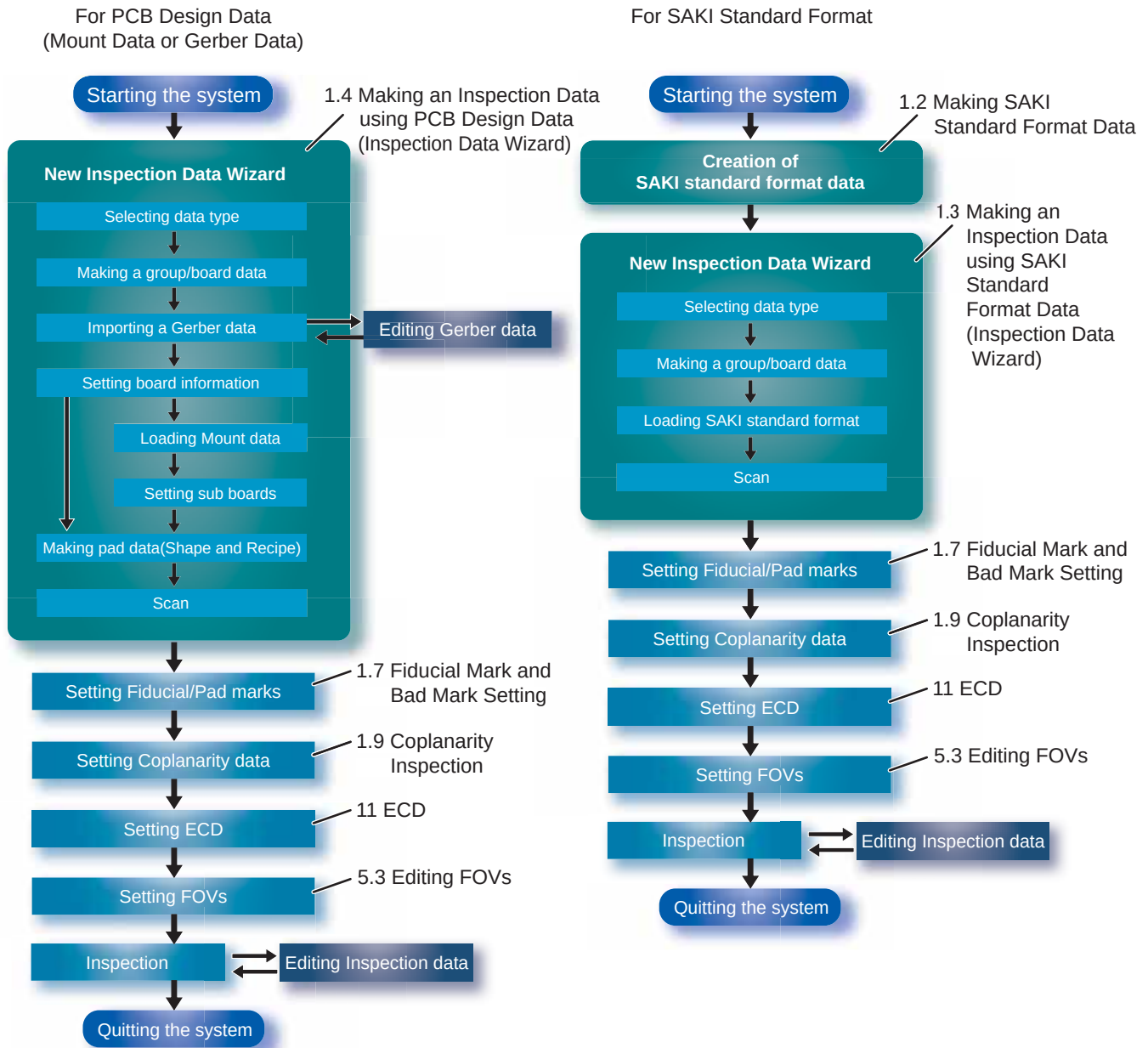


Figure 11-1 Flow chart of data creation

11.1 ECD Setting Procedure Overview

The figure shown below is the procedure of ECD creation.

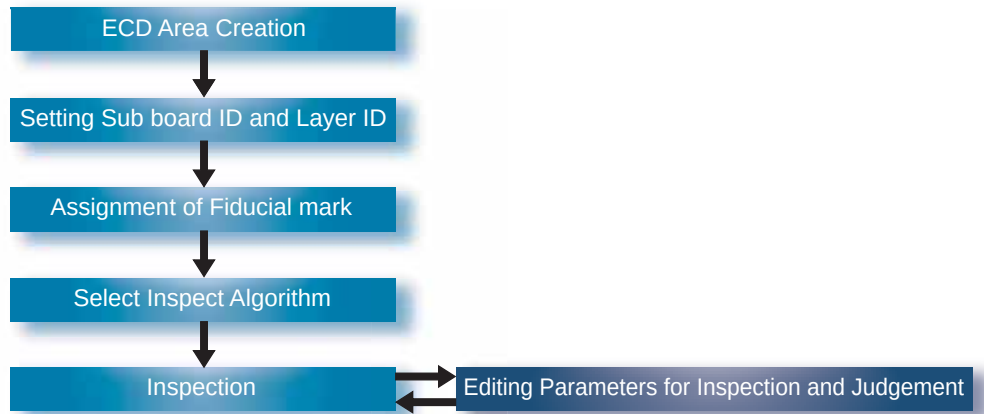


Figure 11-2 ECD Creation Procedure

11.2 ECD Inspection Overview

In ECD inspection, perform detection and judgement of foreign matter.

The ECD inspection procedure is shown below.



Figure 11-3 ECD Inspection Procedure

Item	Description
Detection	Detect the candidate area of foreign matter.
Judgement for foreign matter	Judge if the candidate area is foreign matter or not.

Table 11-1 Descriptions

11.3 ECD Setting Procedure

After completing the setting till **Setting Coplanarity data** in Figure 11-1, proceed to the ECD setting.

Step1: Select **ECD** tab.

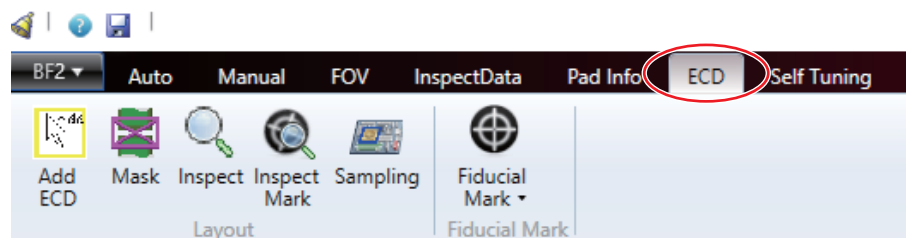


Figure 11-4 ECD Setting 1

Step2: From the ribbon, select **Layout** group > **Add ECD**.

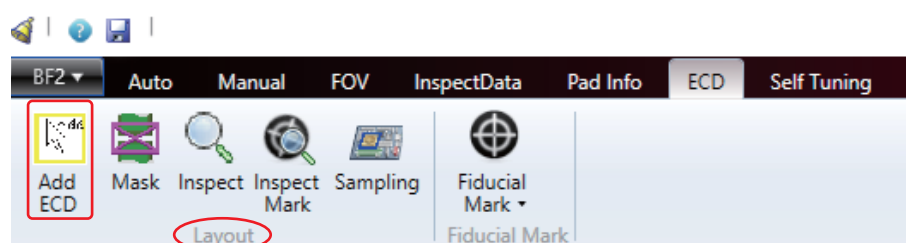


Figure 11-5 ECD Setting 2

Step3: **Add area mode** dialog appears on **Board Viewer**.
Select **Add/Move/Resize**.

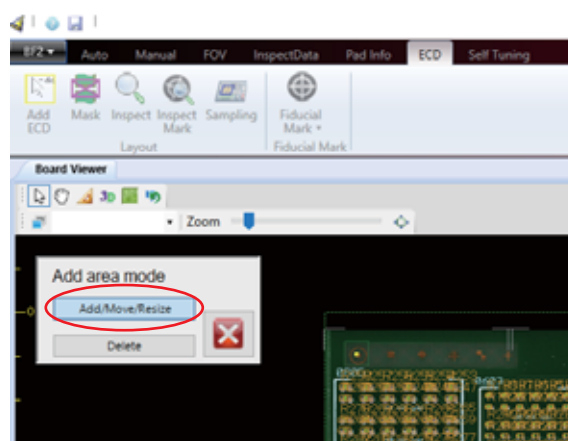


Figure 11-6 ECD Setting 3

Step4: Surround the area to inspect by mouse-dragging.

ECD area is set.

After all the setting is completed, click .

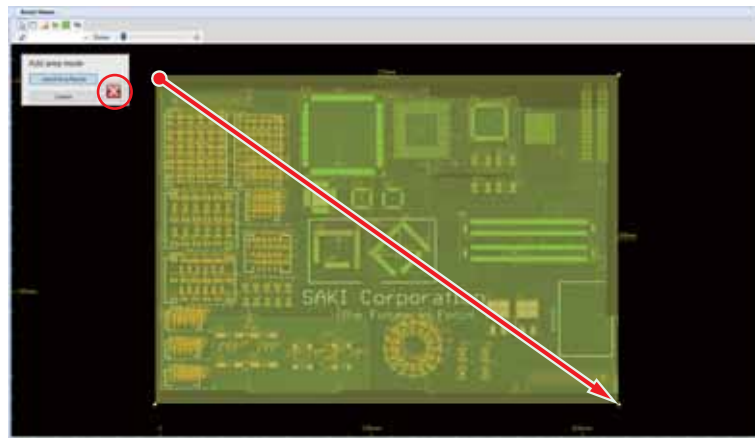


Figure 11-7 ECD Setting 4

Step5: Added ECD area is displayed on **ECD Area List**.

Select ECD area which is created in Step4.

ECD Area List				
ECD Area Name	Required Number	Sampling Number	Status	Result
ECD Area 1	10	0	Sampling	No Image
ECD Area 2	10	10	Inspection	NG
ECD Area 3	10	10	Inspection	OK
ECD Area 4	—	0	Not Ready	Not Inspection
ECD Area 5	10	0	Skip	Not Inspection

Figure 11-8 ECD Setting 5

Item	Description
ECD Area Name	Display name of created ECD area
Required Number	The item is for Saki engineers only. Do not change the setting.
Sampling Number	The item is for Saki engineers only. Do not change the setting.
Status	Displays status of ECD inspection. For details of status, refer to Table 11-3.
Result	Displays result of ECD inspection. For details of result, refer to Table 11-4.

Table 11-2 Descriptions

Item	Description
Sampling	Shows the status that sampling is required.
Inspection	Shows the status that inspect is available.
Not Ready	Shows the algorithm is not yet set for ECD inspection. ECD area is existing but inspection is not performed.
Skip	Shows that ECD area is set as skip.

Table 11-3 Descriptions of Status

Item	Description
OK	Shows there is no foreign matter.
NG	Shows foreign matter is found.
Not Inspection	Shows inspection is not performed. This is set when Status is Not Ready or Skip .
No Image	Shows there is no images in ECD area. Inspection is not performed.

Table 11-4 Descriptions of Result

Step6: When select ECD area, the edit panel appears.

Set **ECD area name**, **Sub Board ID**, **Layer ID**, and **DOP Layer ID** if necessary.

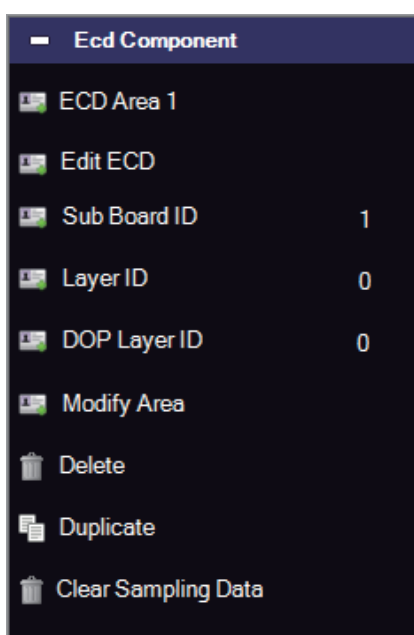


Figure 11-9 ECD Setting 6

Item	Description
	Click to toggle between Skip or Inspect.
ECD area name	ECD area name is displayed. This can be edited by selecting.
Edit ECD	By selecting this, enter to ECD edit display.
Sub Board ID	Displays sub board ID of ECD layer. Sub board ID is changed by clicking.
Layer ID	Displays layer ID of ECD area. Layer ID is changed by clicking.
DOP Layer ID	Displays DOP layer ID of ECD area. DOP layer ID can be changed by clicking. Normally not available.
Modify Area	ECD area size is changed.
Delete	ECD area is deleted.
Duplicate	ECD area is duplicated. Algorithm setting is not copied.
Clear Sampling Data	The item is for Saki engineers only. Do not change the setting.

Table 11-5 Descriptions

Step7: To correct ECD area position of each board, assign fiducial mark by reference to **1.7 Fiducial Mark and Bad Mark Setting**.

Step8: Set mask to avoid false call.
From **ECD** tab, select **Layout** group > **Mask**.

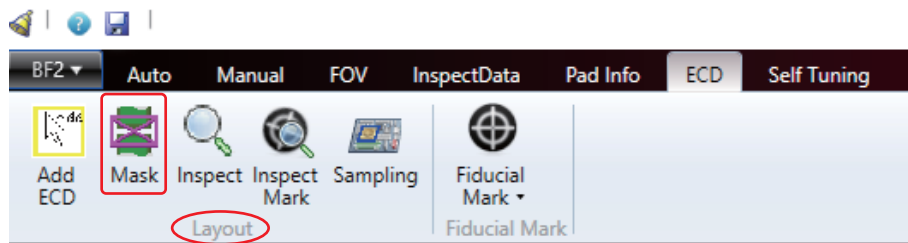


Figure 11-10 ECD Setting 7

Step9: The dialog shown below appears.
Click **Add Mask for All Component**.

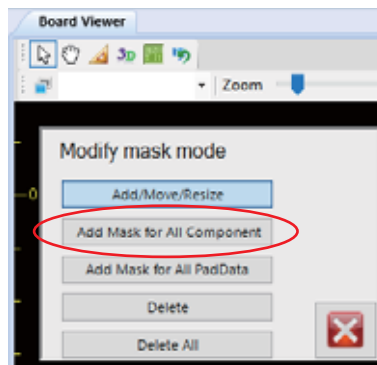


Figure 11-11 ECD Setting 8

Item	Description
Add/Move/Resize	By clicking this, the mask can be added ,moved, resized.
Add Mask for All Component	Make mask for all component based on all inspect data information. The shape of component size becomes base of mask making. This is suitable when correctly mounted components are judged as foreign matter. For details, refer to Step10.
Add Mask for All PadData	Make mask based on pad data. This is suitable when false call happened on pad.
Delete	Switch to mask delete mode. Delete mask by click.
Delete All	Delete all mask.

Table 11-6 Descriptions

Step10: The dialog shown below appears.

Set the mask area (Detection exception area) by reference to Table 11-7.

The created mask is shown as pink line as Figure 11-13.

The internal area indicated in pink is excluded from ECD inspection.

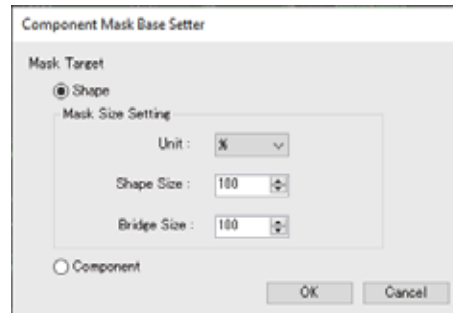


Figure 11-12 ECD Setting 9

Item	Description		
Shape	Make mask according to shape.		
	Mask Size Setting	Unit	Specifies of unit for Shape Size and Bridge Size . % or μm can be set.
		Shape Size	Set percentage of mask based on shape size except bridge.
		Bridge Size	Set percentage of mask for bridge.
Component	Make mask to fit inspection data outline.		

Table 11-7 Descriptions

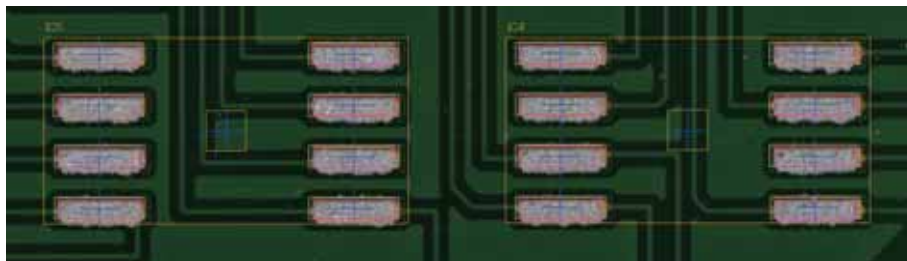


Figure 11-13 Component Mask

Step11: Select **ECD** tab, and select ECD area to edit from **ECD Area List**.
The edit panel appears, and click **Edit ECD** from the panel.

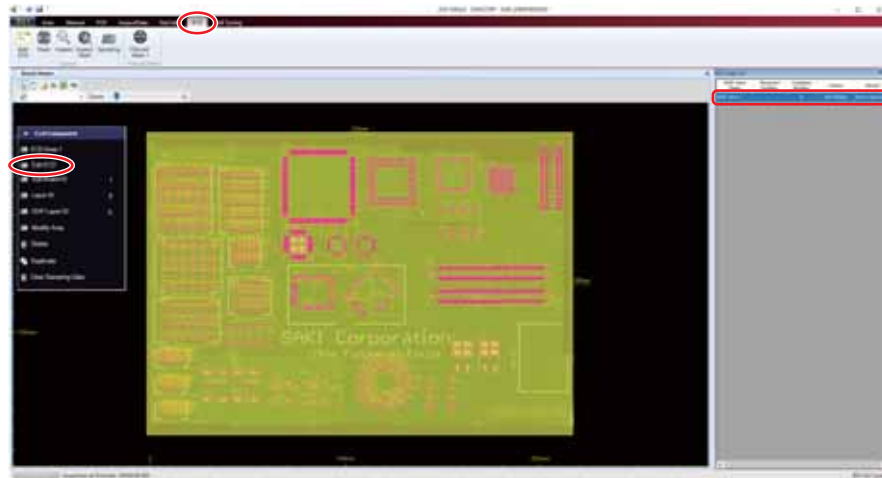


Figure 11-14 ECD Setting 10

Step12: The ECD editing screen is displayed.

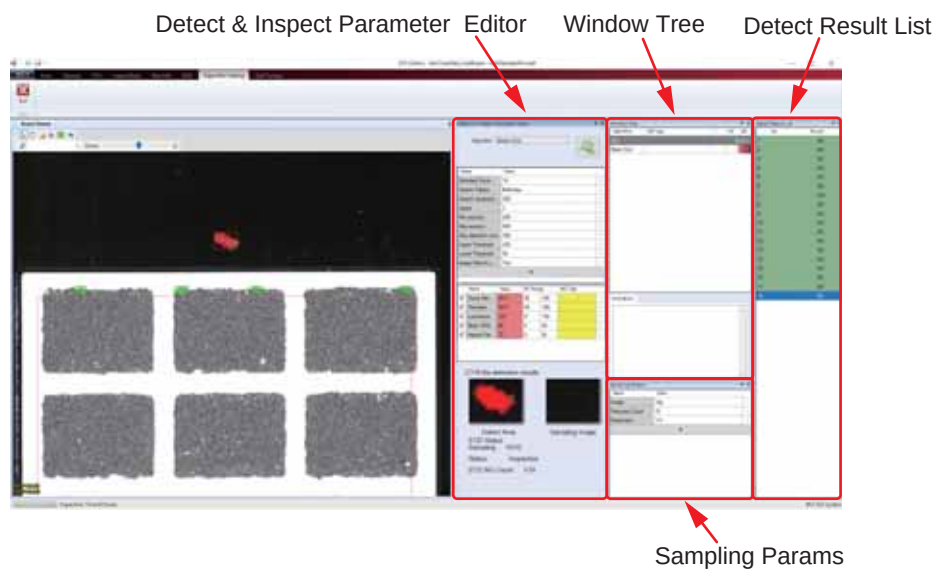


Figure 11-15 ECD Setting 11

Item	Description
Detect & Inspect Parameter Editor	Select algorithm for ECD inspection. Once a algorithm is selected, the detect parameter is displayed on upper list and the inspect parameter is displayed on lower list.
Sampling Params	The item is for Saki engineers only. Do not change the setting.
Detect Result List	Shows ECD inspection result. OK : The area which is detected as foreign matter candidate but judged as not foreign matter. NG :The area judged as foreign matter by inspection.
Window Tree	Displays ECD inspection window

Table 11-8 Descriptions

Step13: Select **Algorithm** on **Detect & Inspect Parameter Editor**, and set parameters if needed.
After parameter setting is completed, Click **Exit** from the ribbon to close editing screen.

NOTE

For details, refer to **11.5 ECD Algorithm Setting (Basic ECD)**.

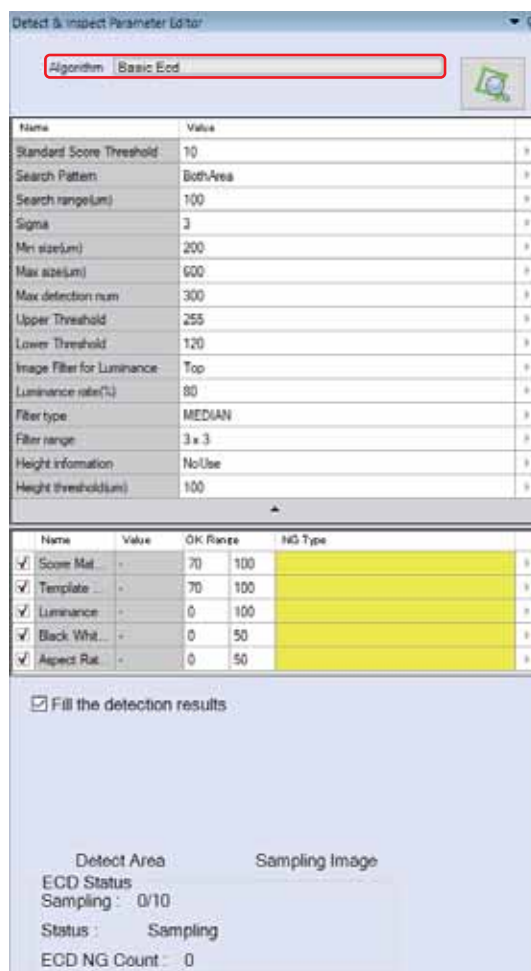


Figure 11-16 ECD Setting 12

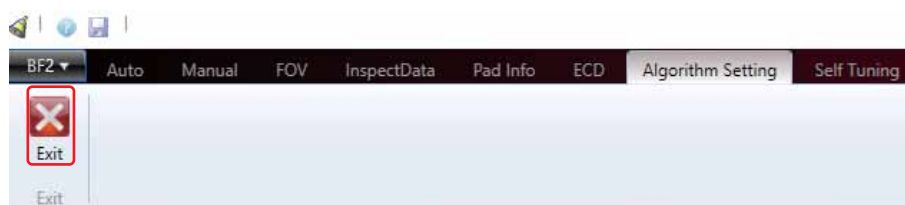


Figure 11-17 ECD Setting 13

11.4 Auto Mode

Description of ECD inspection on Auto mode screen.

On auto mode screen, **ECD Inspection Result** and **ECD Area list** to show status of each ECD area are displayed.

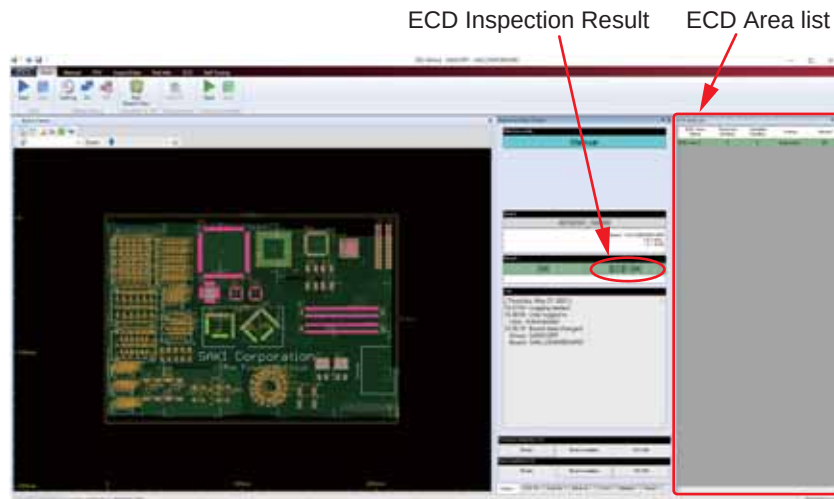


Figure 11-18 Auto Mode Screen

When ECD inspection result is NG, The number of detected foreign matters is shown as below.

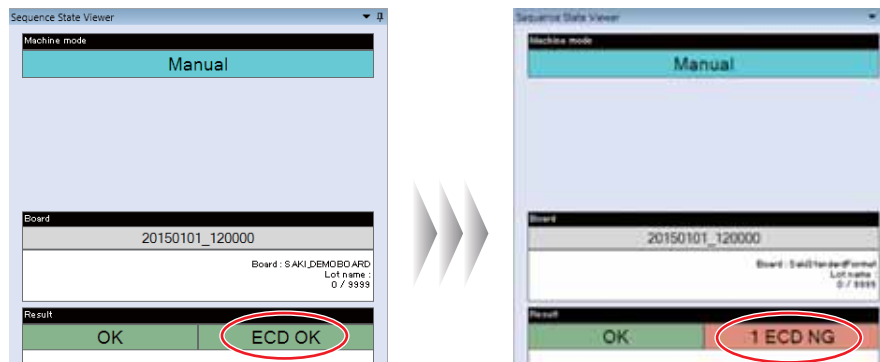


Figure 11-19 ECD Inspection Result

11.5 ECD Algorithm Setting (Basic ECD)

Basic ECD is an algorithm for ECD that samples several OK boards to generate a template image to be the judgement criteria, and detects unexpected foreign matters by differences from inspection board image.

11.5.1 Parameter Setting

Sampling Params

Sampling Params		
Name	Value	
Image	Top	▶
Required Count	10	▶
Resolution	1/4	▶
Output Image	false	▶

Figure 11-20 Sampling Params

Item	Description
Image	Select an image filter for sampling data generation. Select a lighting with different colors for foreign matter and boards.
Required Count	Set the number of sheets required for sampling. The higher the number of required counts, the better the foreign matter detection accuracy.
Resolution	Select a resolution for sampling data generation. The higher the value, the higher the resolution, and the smaller the value, the faster the inspection speed.
Output Image	The item is for Saki corporation engineers only. Do not change the setting. Default value is false .

Table 11-9 Descriptions

Detect & Inspect Parameter

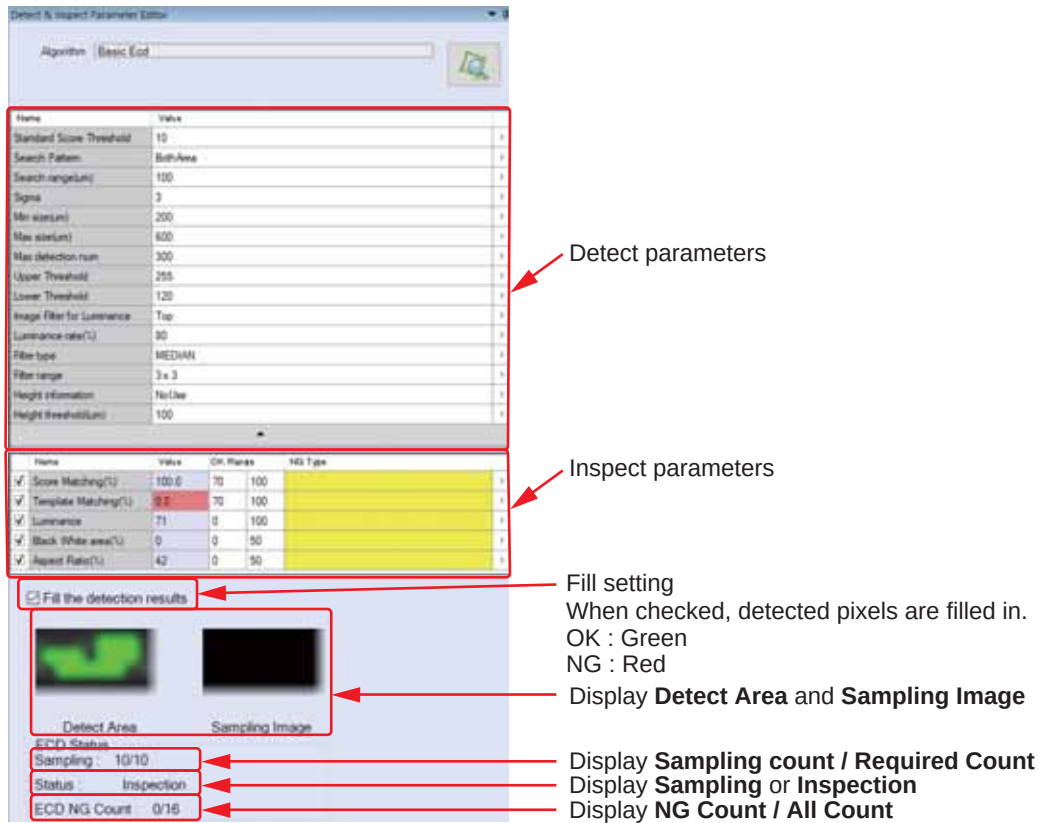


Figure 11-21 Detect and Inspect Parameter (Version 1.0.0.0)

Item	Description
Standard Score Threshold	Set a threshold for brightness variability of the inspection image to the sampling data.
Search Pattern	Specify the condition for the detection area. DarkArea detects only dark areas compared to sampling data. BrightArea detects only bright areas compared to sampling data. BothArea detects both areas.
Search range(um)	Expands the inspection area of Score Matching(%) and Template Matching(%) by the entered values, centering on the detection area. If there are many false detections due to board misalignment or the like, increasing the this value can reduce false detections.
Sigma	Set the variation tolerance after applying to the filter.
Min size(um)	Set the minimum size to detect. If the short side of the circumscribed rectangle of the detection area is shorter than this value, it will not be detected.
Max size(um)	Set the maximum size to detect. If the long side of the circumscribed rectangle of the detection area is longer than this value, it will not be detected.
Max detection num	Set the maximum number of detections. If the number of detections exceeds the set value, the detection process ends there.
Upper Threshold	Set the maximum brightness value of the pixel to be calculated for Black White area[%] . Pixels brighter than this value will not be calculated when calculating Black White area[%] .
Lower Threshold	Set the minimum darkness value of the pixel to be calculated for Black White area(%) . Pixels darker than this value will not be calculated when calculating Black White area(%) .
Image Filter for Luminance	Set an image filter used for Luminance .
Luminance rate(%)	Set a parameter used for Luminance . For details, refer to Luminance described later.

Item	Description
Filter type	Select the noise reduction process for the detected whole image.
Filter range	Specify a range of the noise reduction process. If the point-like difference range is large on a slit, increase this value.
Height information	Set whether height information is used for inspection. If UseHeight is selected, the area below the height set in Height threshold(um) will not be detected. Default value is NoUse . If height information is used, height information results may not be stable without the DOP compensation. For details on the DOP compensation, refer to 7 DOP Compensation .
Height threshold(um)	If Height information is set to UseHeight , set the height to be inspected. Areas where the height is less than this value will not be detected.

Table 11-10 Descriptions for Detect Parameter

Item	Description
Score Matching(%)	Shifts the position by Search range(um) centering on the detection area, detects foreign matters again, and shows the matching ratio. The closer the value is to 100%, the closer it is to what it detects around it.
Template Matching(%)	Shifts the position by Search range(um) centering on the detection area, performs image matching processing on the sampling data, and shows the matching ratio. The closer the value is to 100%, the closer it is to what it detects around it.
Luminance	Sorts pixel brightness that is detected in Image Filter for Luminance in ascending order, and outputs the pixel luminance of component set in Luminance rate(%) . This is effective to exclude areas that glow in the same way as foreign matters with the sampling parameter Image , but glow in the different way from foreign matters if they are separate lighting.
Black White area(%)	When the detection area is assumed to be 100%, outputs the ratio of pixels having luminance within Upper Threshold and Lower Threshold . This is effective to exclude areas that do not contain a certain level of brightness or more from the detection area.
Aspect Ratio(%)	The aspect ratio of the circumscribed rectangle of the detection area. Outputs the ratio when the longer width is taken as 100%. This is effective to prevent long and narrow area detection caused by subtle shift of solder paste. However, defects such as scratches may be missed.

Table 11-11 Descriptions for Inspect Parameter

11.5.2 Setting Procedure

- Step1: Select the **ECD** tab and set the ECD area.
For details, refer to **11.3 ECD Setting Procedure**.
- Step2: After the ECD area setting is complete, enter **ECD area edit mode**.
Select **Basic ECD** from the **Algorithm** drop-down list.
- Step3: Select the **Sampling Params** tab and set **Image**, **Required Count**, and **Resolution**.

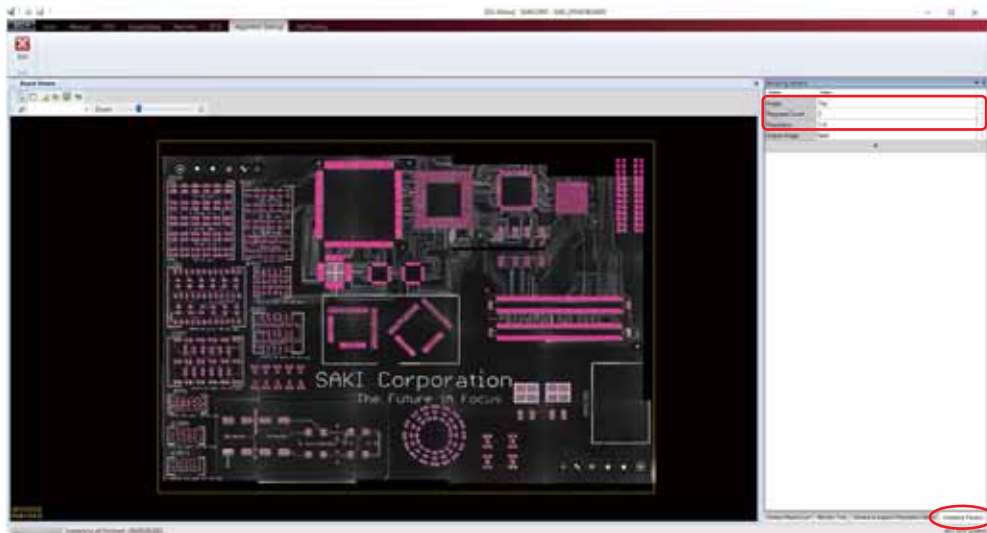


Figure 11-22 Sampling Params 1

- Step4: Prepare sampling boards according to the number of **Required Count** and sample them.
- Step5: After sampling is complete, select the **Detect and Inspect Parameter Editor** tab and set **Detect Parameters** and **Inspect Parameters** to detect foreign matters.

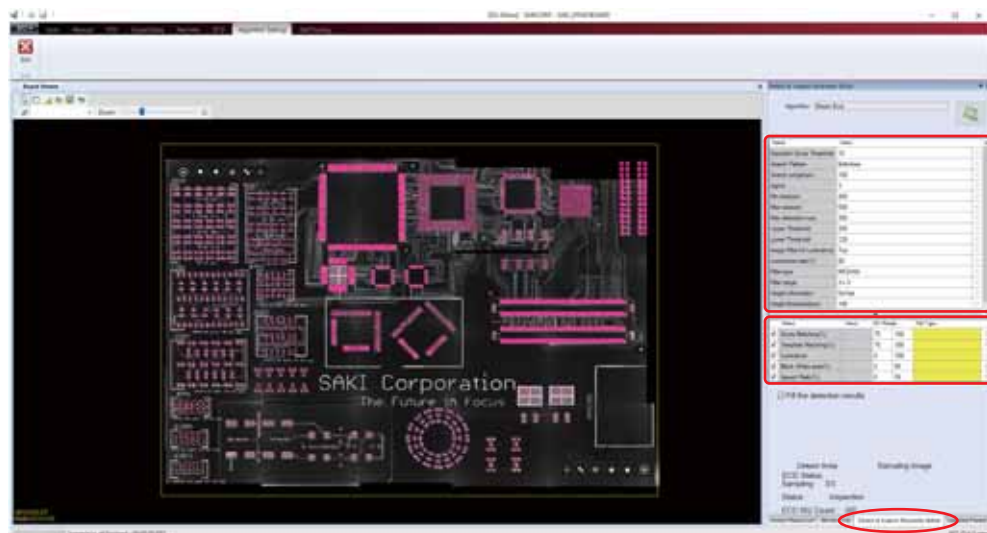


Figure 11-23 Sampling Params 2

- Step6: Click , and check the foreign matter is detected.
 The area judged as misplaced component is highlighted in red.
 And the area which is detected but judged as not foreign matters is highlighted in green.

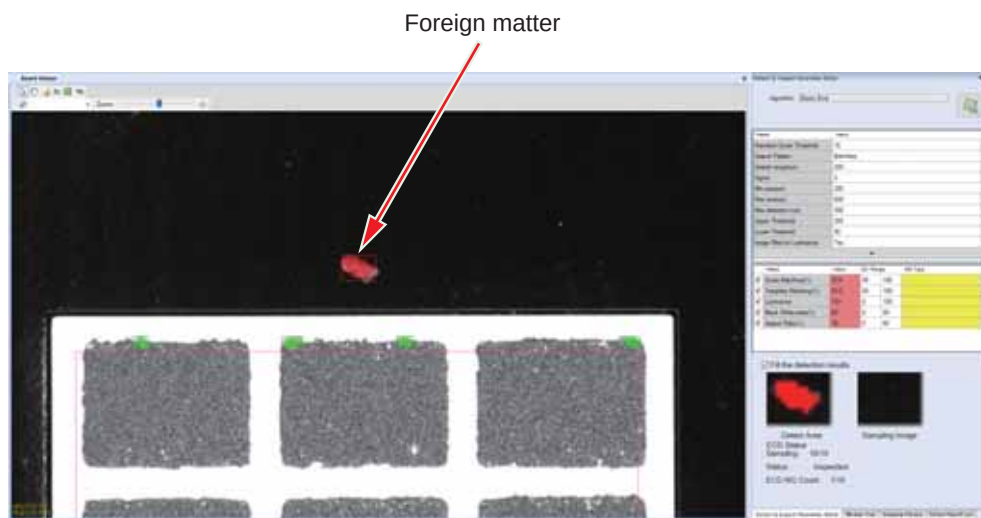


Figure 11-24 Detection of Foreign Matter

- Step7: If the foreign matter is not detected, adjust **Detect Parameters**.
 If the foreign matter is detected but judged as OK, adjust **Inspect Parameters**.

12 Self Tuning

Self Tuning is a function to automatically optimize the tolerances of inspection algorithms. Based on the selected criteria, this function sets tolerances of specific algorithm sample values from the inspection results collected from one or more boards.

12.1 Window Details

This section describes the Self Tuning screen.

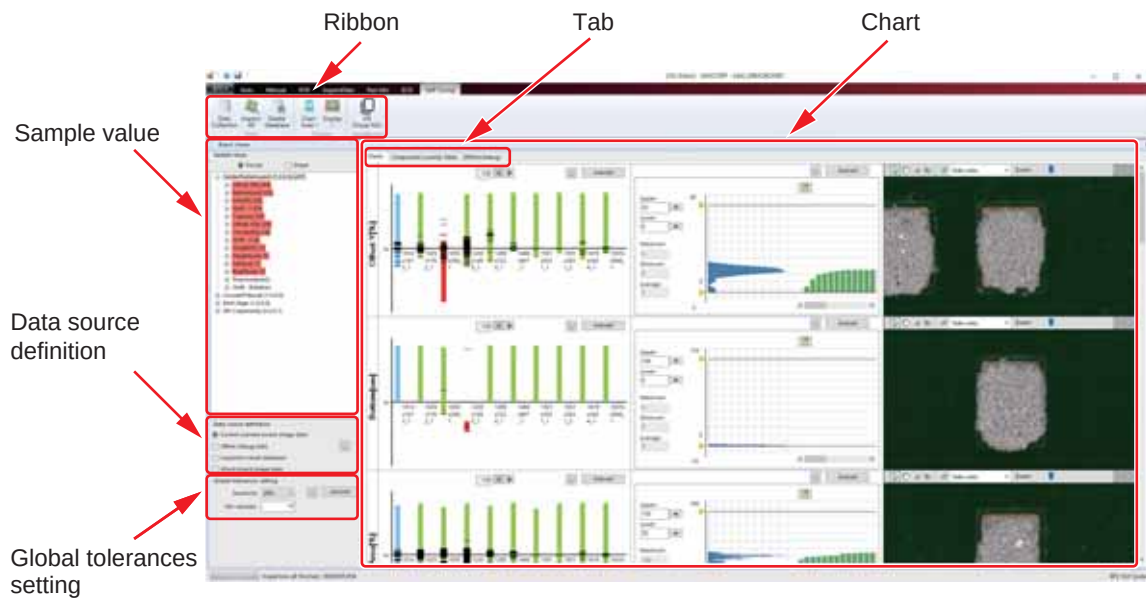
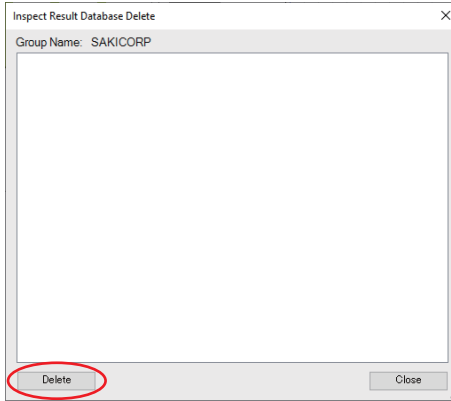
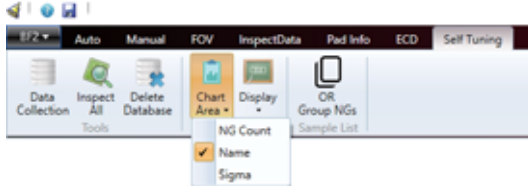


Figure 12-1 Window Details

Item		Description
Ribbon	Data Collection	Automatically collects sample values and inspection images of inspection results. For details, refer to 12.1.1 Data Collection .
	Inspect All	Performs inspection based on selection in Data source definition and updates sample values in the Charts tab.
	Delete Database	Deletes an existing inspect result database of the inspection group. Select a database and click Delete .  The dialog box is titled 'Inspect Result Database Delete'. It has a 'Group Name' field with 'SAKICORP' entered. Below the field is a large empty list box. At the bottom, there are two buttons: 'Delete' (highlighted with a red circle) and 'Close'.
	Chart Area	Sorts bar graphs of Recipes and Shapes in the Charts tab.  The image shows the 'Chart Area' drop-down menu in the software interface. The menu is open, showing options: 'NG Count' (selected with a checkmark), 'Name', and 'Sigma'. The background shows the software's ribbon with tabs like 'Auto', 'Manual', 'FOV', 'InspectData', 'Pad Info', 'ECD', and 'Self Tuning'. Figure 12-3 Chart Area Drop-down List NG Count : Shows Recipes and Shapes in order of the number of NGs. Name : Shows Recipes and Shapes alphabetically. Sigma : Shows Recipes and Shapes in order of standard deviation of sample values. (Select to determine Recipes and Shapes where input values of related algorithms should be checked due to a wide range of sample values.) NOTE The Charts tab show a list of Recipes and Shapes with the given sample value of algorithm. Sorting the contents of the Charts tab will make it easier to decide whether to focus the tolerance adjustment on the number of NG or the range of values examined (sigma). It also helps to decide what to optimize the algorithm tolerance for.
	Display	Select All Samples to show all sample values.
	OR Group NGs	Shows or hides NG counts of sample values if sample values exceeding the tolerance are in another algorithm in the OR group. For details, refer to 12.1.2 OR Group NGs .
Sample Value		Shows a list of algorithms used in the loaded inspection data. Shows sample values exceeding given tolerance in red if any. By selecting Recipe or Shape, graphs in the Charts tab are grouped by Recipe or Shape.
Data Source Definition		Set the data source to import inspection results and NG counts of sample values. For details, refer to 12.1.3 Data Source Definition .

Item		Description
Global Tolerances Settings		Set criteria for the lower tolerance level and upper tolerance level for given algorithms, sample values, and NG kinds. For details , refer to 12.1.4 Global Tolerances Setting .
Tab	Charts	Shows bar charts and histograms of sample values for selected algorithm on Sample Value .
	Component Lookup Table	Shows a list of sample values of each Recipe and the Self Tuning setting (Autotune).
	Offline Debug	Shows offline debug data. For details, refer to 9.5 Self Tuning Setting .

Table 12-1 Descriptions

12.1.1.1 Data Collection

Data collection is a function to automatically collect sample values and inspection images of inspection results.



Figure 12-4 Data Collection Dialog Box

Item	Description
Load/Unload	Click Load/Unload to load or unload boards into or from the inspection machine.
Scan only one PCB	Click Scan only one PCB to scan the loaded board.
Scan multiple PCBs	Click Scan multiple PCBs to scan the loaded board for the number specified on the Data Collection dialog box.
Output data path	Specify the destination to save whole board images. The destination is also used to load and inspect whole board images if Whole board image data is selected in Data source definition .
OK	Click OK to close the dialog box after saving settings.
Cancel	Click Cancel to close the dialog box without saving settings.

Table 12-2 Descriptions

12.1.2 OR Group NGs

This function hides or shows NG counts of sample values if sample values exceeding tolerances are in another algorithm of OR group.

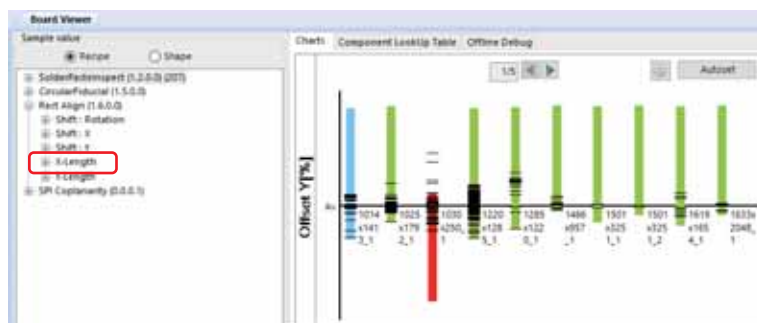


Figure 12-5 Example of Recipe with OR Group Results Hidden

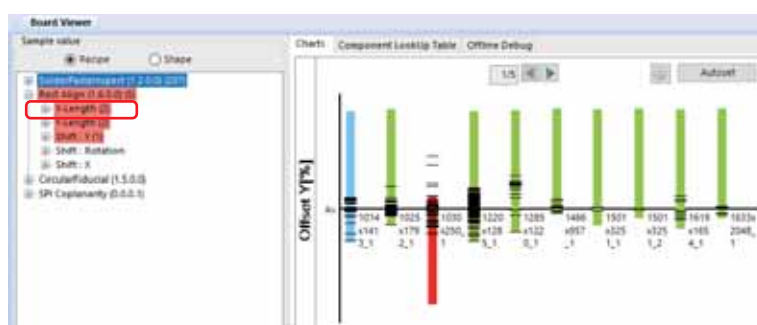


Figure 12-6 Example of Recipe with OR Group Results Shown

12.1.3 Data Source Definition

Data source definition sets the data source to import inspection results and NG counts of sample values. When **Inspect All** is executed or **Whole board image data** is selected, the system will search, load, and inspect images stored in the folder specified in the Data Collection dialog box.

All inspection results are also retained in the **Self Tuning** tab for further processing until **Inspect All** is executed next time.

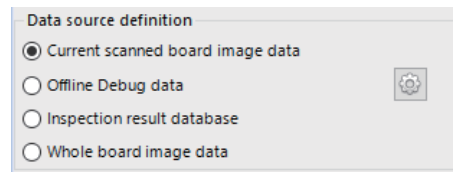


Figure 12-7 Data Source Definition

Item	Description
Current scanned board image data	Uses the currently loaded board image data as the data source.
Offline Debug Data	Uses offline debug data as the data source. For details, refer to 9.5 Self Tuning Setting .
Inspection result database	Uses the values of related Recipes and algorithms in the inspection result database as the data source. NOTE The data source is valid until the input parameters for the given Recipes change. If the input parameters are tuned, re-retrieve the data source. For details, refer to 12.1.1 Data Collection.
Whole board image data	Uses the whole board image data as the data source. Enables to load the sample values inspected on one or more whole board images for specific inspection data saved in the past.

Table 12-3 Definitions

12.1.4 Global Tolerances Setting

The Global tolerances setting screen enables to set a particular upper tolerance level and lower tolerance level for any algorithm, sample, and NG type relating to the loaded inspection data.

NOTE

The system does not differentiate between different algorithm versions. If the same sample values are in a different algorithm version, the upper tolerance level and lower tolerance level can be set.

Step1: Click  of **Global tolerances setting**.

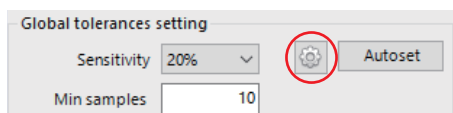


Figure 12-8 Global Tolerance Screen Setting 1

Step2: The Global Tolerances screen appears.

Set **NG Type**, **Lower**, **Upper** for **Algorithm** and **Sample** for tuning and click **OK**.

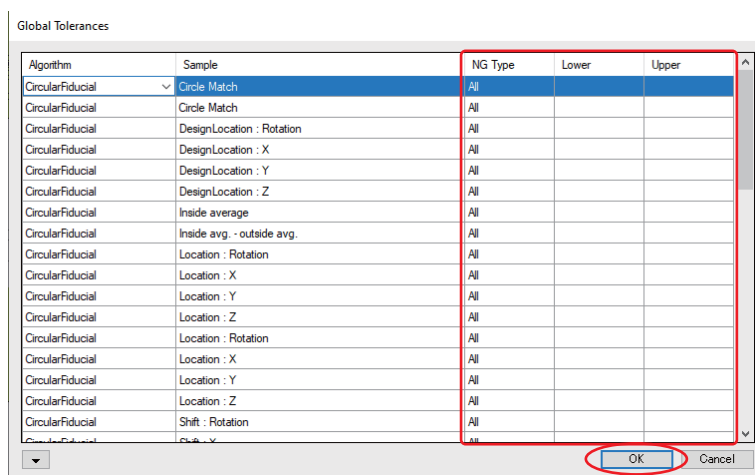


Figure 12-9 Global Tolerances Table Setting 2

Item		Description
Upper limit	Insert value	Insert the entered value.
	Av-3 σ	Calculates the upper limit of the sample value based on a sigma ratio and the average of the current sample values.
	Max-10%	Calculates 110% of the greatest value in the current sample values. The result is set as the upper limit of the sample value.
	Ok Av-3 σ	Calculates the upper limit of the sample value based on a sigma ratio and the average of the sample values of the inspection windows whose inspection results are OK.
	Ok Max-10%	Calculates 110% of the greatest value in the sample values of the inspection windows whose inspection results are OK. The result is set as the upper limit of the sample value.
Lower limit	Input value	Insert the entered value.
	Av-3 σ	Calculates the lower limit of the sample value based on a sigma ratio and the average of the current sample values.
	Min-10%	Calculates 90% of the smallest value in the current sample values. The result is set as the lower limit of the sample value.
	Ok Av-3 σ	Calculates the lower limit of the sample value based on a sigma ratio and the average of the sample values of the inspection windows whose inspection results are OK.
	Ok Min-10%	Calculates 90% of the smallest value in the sample values of the inspection windows whose inspection results are OK. The result is set as the lower limit of the sample value.

Table 12-4 Descriptions

12.1.5 Global Tolerances Autoset

Enables to automatically adjust values to the values set on the Global Tolerances screen.

Click **Autoset** to adjust all related Recipes/Shapes to the values set on the Global Tolerances screen.

If the number of samples are less than **Min samples**, **Autoset** does not adjust values to the values set on the Global Tolerances screen.

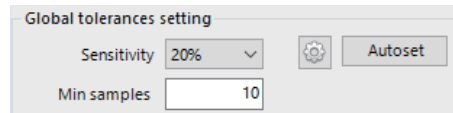


Figure 12-10 Global Tolerances Auto Set

Item	Description
Sensitivity	Set the adjustment ratio to apply to the current tolerance from the tolerance calculated by Autotune. For details, refer to Figure 12-11.
Min samples	Applies tolerances only for the number of samples greater than Min samples .
Autoset	Click to apply Upper Tolerance Level (UTL) and Lower Tolerance Level (LTL) for all sample values according to the criteria set on the Global Tolerances screen.

Table 12-5 Descriptions

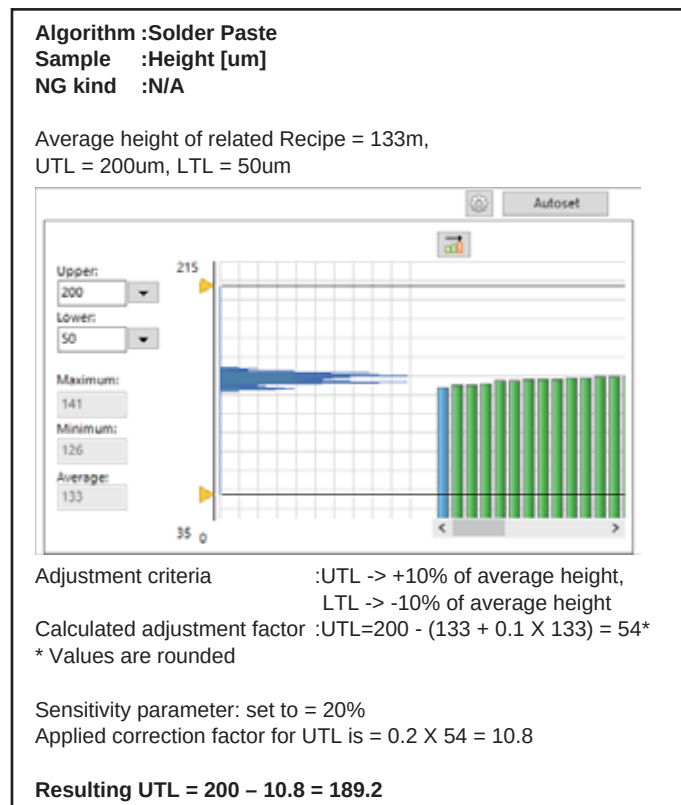


Figure 12-11 Example of **Sensitivity**

12.1.6 Charts Tab

The **Charts** tab shows bar charts and histograms of sample values for selected algorithm in **Sample value**.

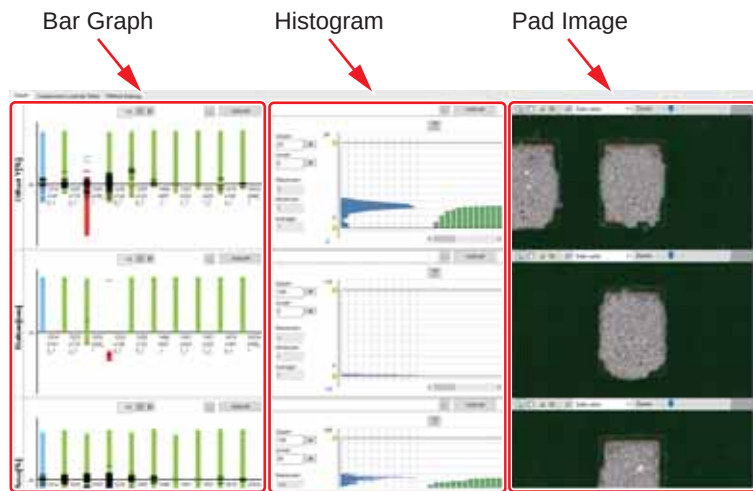


Figure 12-12 Charts Tab

Item	Description
Bar Graph	Shows sample values and tolerances for Recipes and Shapes with the algorithm selected in Sample Value .
Histogram	Shows histograms of sample values for first or selected bar graph in Bar Graph .
Pad Image	Shows pad images of first or the selected reference position in Histogram . Double-click to show the recipe edit screen and adjust parameters.

Table 12-6 Descriptions

12.1.7 Component Lookup Table Tab

The **Component Lookup Table** tab shows overview of currently used tolerances and giving information (if tolerances for sample values are not adjusted, adjusted manually, or adjusted by **Autoset**).

Uncheck **Autotune** so that the values are not changed by **Autoset**.

Charts		Component Lookup Table		Offline Debug					
Shape	Recipe	LinkID	Algorithm	Sample Name	Tolerance Low	Tolerance High	Modified	Autotune	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	CircularFiducial (1.5.0.0)	Inside average	180	255	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	CircularFiducial (1.5.0.0)	Inside avg. - outside avg.	100	255	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	Rect Align (1.6.0.0)	Shift: X	-300	300	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	Rect Align (1.6.0.0)	Shift: Y	-300	300	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	Rect Align (1.6.0.0)	Shift: Rotation	-5	5	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	Rect Align (1.6.0.0)	X-Length	1000	10000	None	☒	
	MARK_CIRCULAR_1100	MARK_CIRCULAR_1100	Rect Align (1.6.0.0)	Y-Length	1000	10000	None	☒	
2037x2307_1	2037x2307_1	2037x2307_1	SolderPasteInspect (1.2.0.0)	Height[um]	50	200	None	☒	
2037x2307_1	2037x2307_1	2037x2307_1	SolderPasteInspect (1.2.0.0)	Volume[%]	50	150	None	☒	
2037x2307_1	2037x2307_1	2037x2307_1	SolderPasteInspect (1.2.0.0)	Area[%]	50	150	None	☒	
2037x2307_1	2037x2307_1	2037x2307_1	SolderPasteInspect (1.2.0.0)	Shift: X	-100	100	None	☒	

Figure 12-13 Component Lookup Table Tab

12.2 Tolerance Adjustment

This function allows to adjust tolerances for related Recipes and Shapes.

- Adjustment by manual
- Correction by **Autoset** function

12.2.1 Adjustment by Manual

To adjust tolerances manually, drag with the mouse a bar graph on the **Charts** tab up or down.

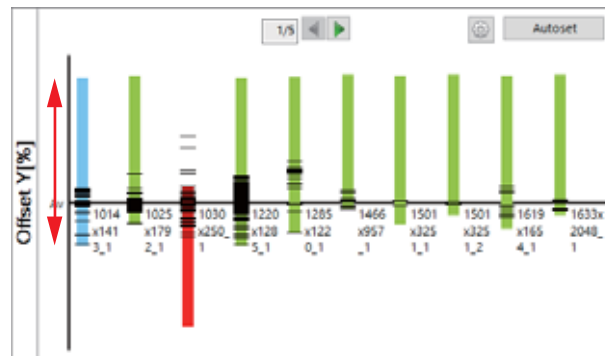


Figure 12-14 Bar Graph Adjustment

Tolerances for the selected Recipe and Shape can be also adjusted on **Histogram** of the **Charts** tab. Set the upper or lower tolerance by dragging ► or entering values in the text-box.

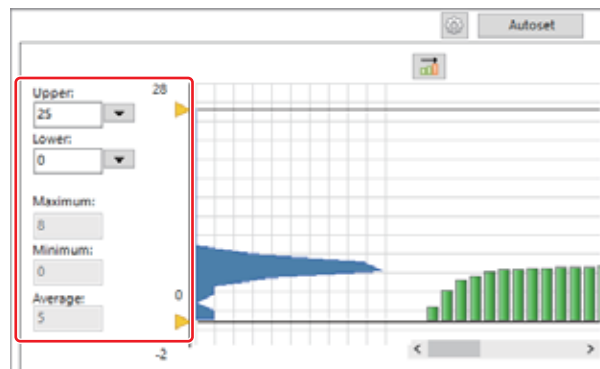


Figure 12-15 Histogram Setting

12.2.2 Adjustment by the Autoset Function

Clicking **Autoset** adjusts tolerances according to the criteria set in the Global Tolerances screen.

The tolerance can be adjusted by clicking each **Autoset** in the following range

- **Autoset** of **Global tolerance setting** : Sample values of all algorithms listed in **Sample value**.
- **Autoset** on bar graphs : Sample values of Recipes/Shapes with the algorithms selected in **Sample value**.
- **Autoset** on histograms : Sample values of Recipes/Shapes selected by a bar graph.

NOTE

If tolerances are adjusted by **Autoset**, **Global Tolerance Setting** is applied.
For details, refer to **12.1.5 Global Tolerance Autoset**.

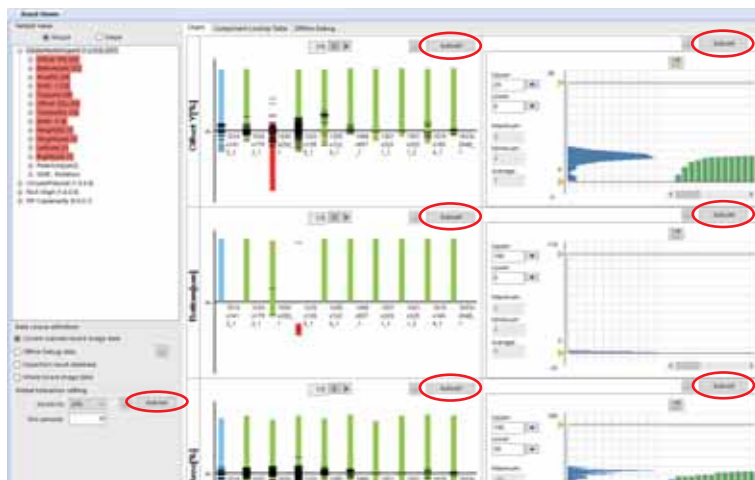


Figure 12-16 Adjustment by **Autoset**

12.2.3 Operation Example

Step1: Load inspection data and board images and select the **Self Tuning** tab.

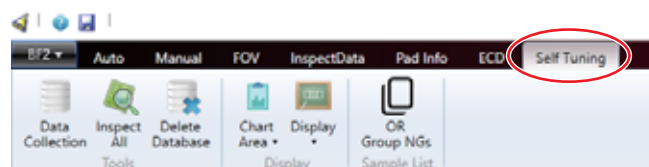


Figure 12-17 Operation Example 1

- Step2: Set necessary settings on the **Global Tolerances** screen.
 For Algorithm, Sample, and NG Type, refer to **12.1.4 Global Tolerances Setting**.
 e.g., Algorithm: Circular Fiducial, Sample: Circle Match,
 NG Type: All, Tolerance level: $\pm 3\sigma$ from average

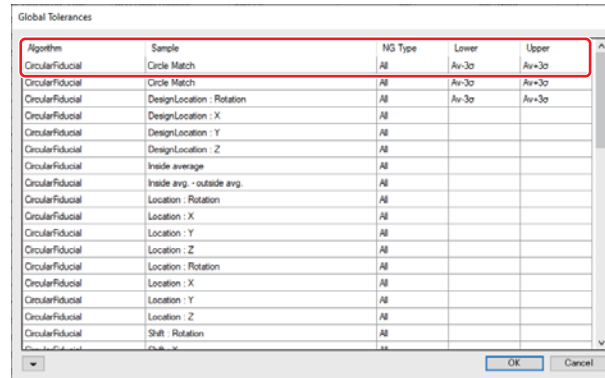


Figure 12-18 Operation Example 2

- Step3: Select the **Tools** group > **Inspect All**.

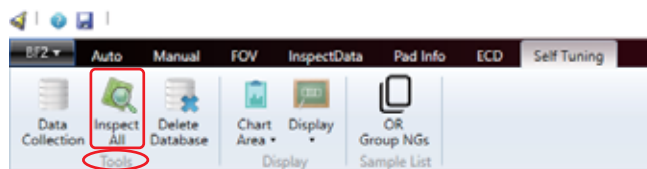


Figure 12-19 Operation Example 3

- Step4: Set **Sensitivity** to **20%**.

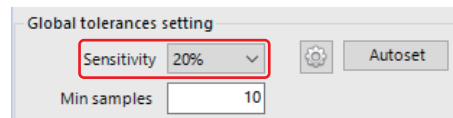


Figure 12-20 Operation Example 4

- Step5: Click **Autoset**.
Circle Match of Recipes using **CircularFiducial** is adjusted to Av. $\pm 3\sigma$ for the adjustment value 20% of the original tolerance level for each Recipe.

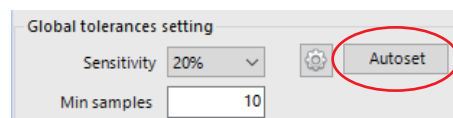


Figure 12-21 Operation Example 5

Inspection Algorithm

1 Solder Paste Inspection

1.1 Inspection Categories

This section describes inspection items of solder paste.

NOTE

Use the **Solder Paste Inspection** algorithm for solder paste inspections.

For details of algorithm, refer to **1.2 Solder Paste Inspection - Algorithm** and **1.3 Solder Paste Inspection - Setting Method**.

1.1.1 Height

Printed solder paste shape is generally recognized as truncated square pyramid shown in Figure 1-1. Therefore, a height measurement value is calculated using the top area of solder paste excluding the edge of solder paste.

In algorithm, specify the area (*1) to use height calculation shown in Figure 1-2. A sample value is calculated in the specified area by averaging heights of pixels detected as solder paste. Solder paste is detected based on brightness values of image filter set in the algorithm and height threshold (*2).

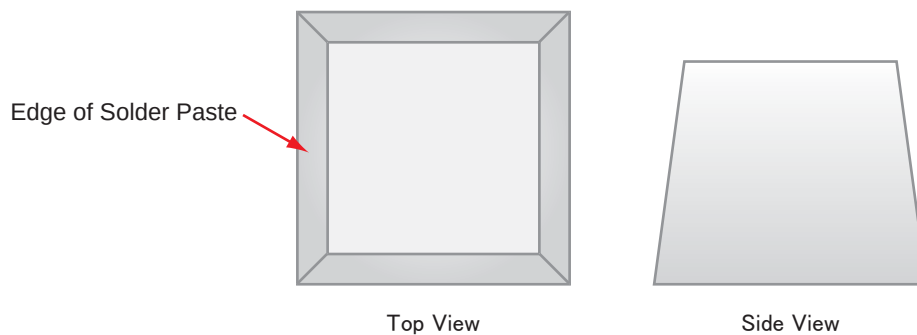


Figure 1-1 Conceptual Diagram of Printed Solder Paste Shape

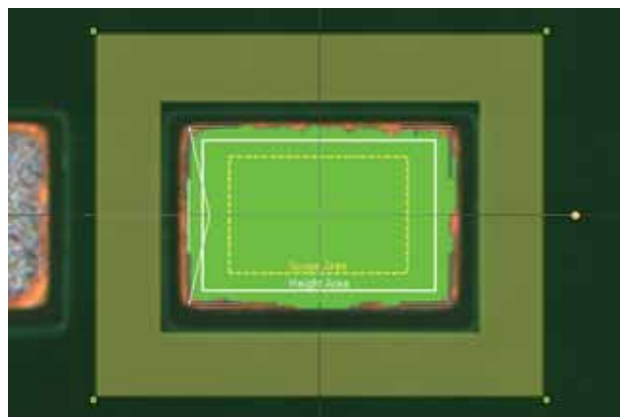


Figure 1-2 Area Using Height Sample Value Calculation

NOTE

(*1) It indicates **HeightArea** in the **Solder Paste Inspection** algorithm.

For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

(*2) In the **Solder Paste Inspection** algorithm, specifies the threshold of brightness by **Upper Limit** and **Lower Limit** and the threshold of height by **Minimum Paste Height**. For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

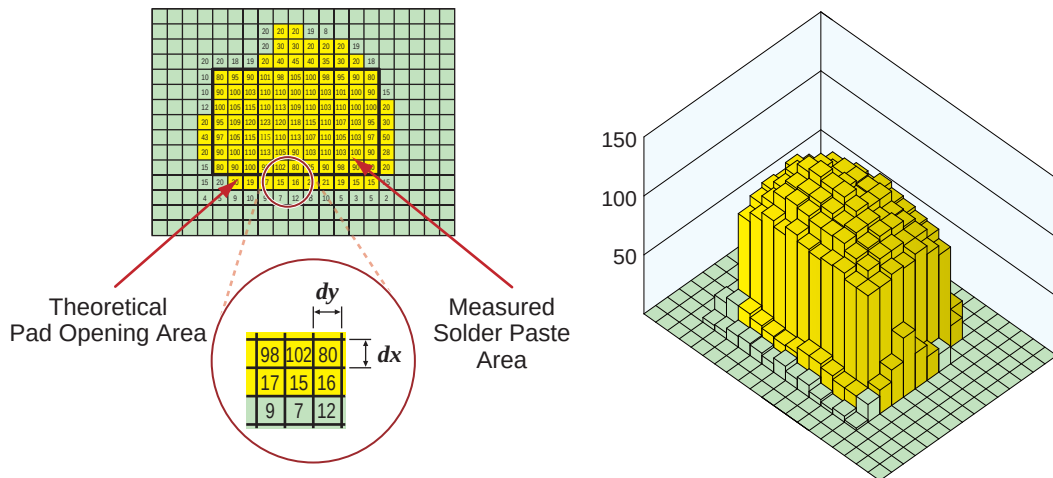
1.1.2 Volume

The volume of solder paste is calculated by integrating height values of all pixels detected as solder paste.

Solder paste is detected based on brightness values of image filter set in the algorithm and height threshold (*).

On the other hand, the volume of solder paste to be printed depends on the aperture area and the thickness of a stencil mask. By multiplying these, the theoretical volume of solder paste can be calculated at a rough estimate.

The volume of solder paste actually measured shall be a measured value, and the volume calculated by multiplying the area of Shape (opening area of a stencil mask) and thickness of a stencil mask shall be a theoretical value. The ratio of the measured value to the theoretical value is displayed with % in the algorithm.



If the H_{ij} assume as height of pixel(i, j),

Measurements volume V_M is calculated by

$$V_M = \left| \sum_{ij} H_{ij} dx dy \right|_{\text{solder_area}}$$

Theoretical value of the volume V_T is calculated by

$$V_T = \text{Area of Shape} \times \text{Stencil Mask Thickness}$$

Sample value of volume V_S is calculated by

$$V_S = V_M / V_T \times 100 [\%]$$

Figure 1-3 Volume Calculation of Solder Paste

NOTE

(*) In the **Solder Paste Inspection** algorithm, specifies the threshold of brightness by **Upper Limit** and **Lower Limit** and the threshold of height by **Minimum Paste Height**. For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

1.1.3 Area

Solder paste area is calculated as the number of all pixels detected as solder paste. Solder paste is detected based on brightness values of image filter set in the algorithm and height threshold (*).

The ratio of solder paste to the Shape area that is the aperture area of a stencil mask is displayed with % in the algorithm.

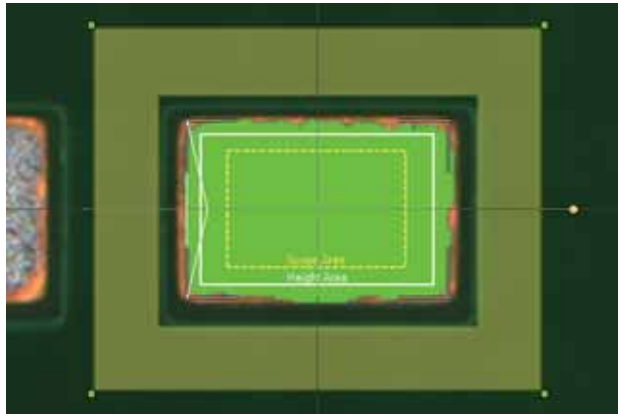


Figure 1-4 Area Detected as Solder Paste

NOTE

(*) In the **Solder Paste Inspection** algorithm, specifies the threshold of brightness by **Upper Limit** and **Lower Limit** and the threshold of height by **Minimum Paste Height**. For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

1.1.4 Bridge

The bridge of a printed solder paste is detected.

In algorithm, solder paste is judged as bridge when solder paste is on the side of an inspection window, the total amount of the area where bridge is narrowest is displayed as sample value.

Bridge can be judged at the top, bottom, left, and right of the inspection window as shown in Figure 1-6. Solder paste is detected based on brightness values of image filter set in the algorithm and height threshold (*).

NOTE

(*) In the **Solder Paste Inspection** algorithm, specifies the threshold of brightness by **Upper Limit** and **Lower Limit** and the threshold of height by **MinimumPaste Height**. For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

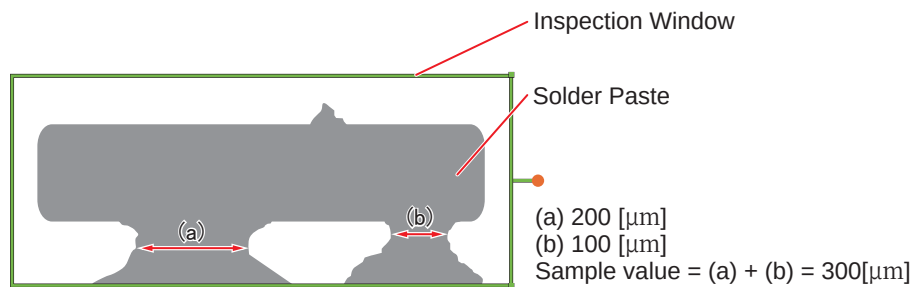


Figure 1-5 Conceptual Diagram of Bridge Inspection

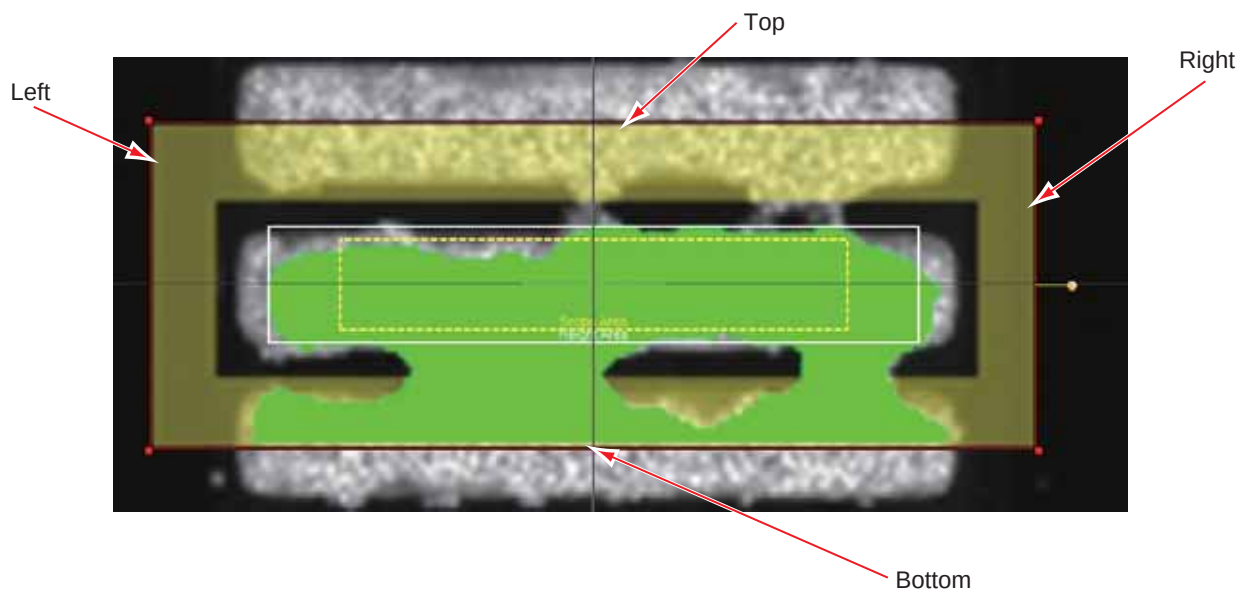


Figure 1-6 Area Detected as Solder Paste

1.1.5 Shift

Calculate the shift amount of printed solder paste from the pad.

In algorithm, shift amount is the distance between the center of gravity of all pixels detected as solder paste and the center of gravity of pad read from gerber data.

The detection of solder paste can be conducted by specifying brightness values and height threshold values of image filter set in the algorithm (*).

NOTE

- (*) In the **Solder Paste Inspection** algorithm, specifies the threshold of brightness by **Upper Limit** and **Lower Limit** and the threshold of height by **Minimum Paste Height**. For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

1.1.6 Form

A Form (shape) of printed solder paste is detected.

In algorithm, solder paste higher than the specified threshold (*1) is detected as a horn.

The cross-section of solder paste (*2) in the threshold is displayed as a sample value.

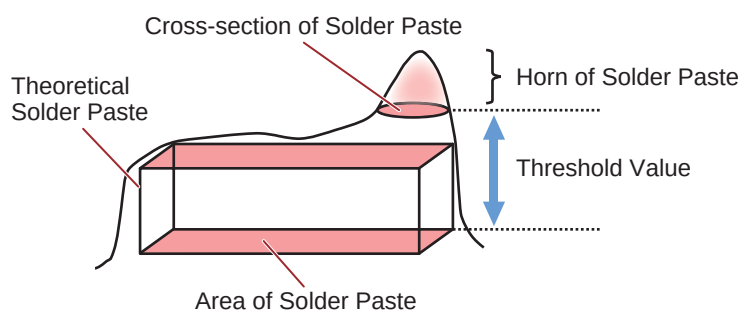


Figure 1-7 Conceptual Diagram of Solder Paste Form

NOTE

- (*1) It indicates **Peak Threshold** in the **Solder Paste Inspection** algorithm.
For details, refer to **1.2 Solder Paste Inspection - Algorithm**.
- (*2) It indicates **Peak Area** in the **Solder Paste Inspection** algorithm.
For details, refer to **1.2 Solder Paste Inspection - Algorithm**.

1.2 Solder Paste Inspection - Algorithm

This section describes the overview of **Solder Paste Inspection** algorithm which inspects solder paste.

1.2.1 Overview of Inspection

Solder Paste Inspection is algorithm to inspect solder paste inspection items (Height, Volume, Area, Bridge, Shift, and Horn) (*).

In algorithm, zero height compensation is conducted and each inspection item on inspected solder paste is calculated.

The figure below is the flow of inspections.

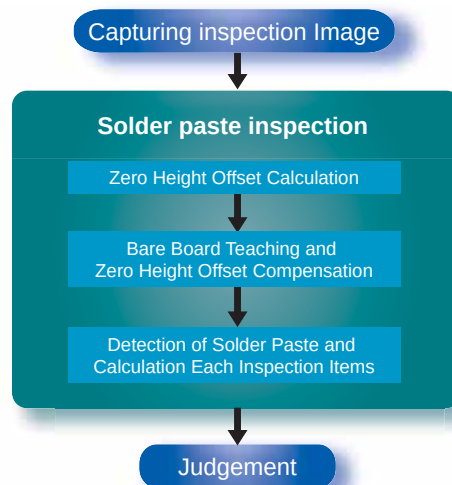


Figure 1-8 Flow of Inspection in Algorithm

NOTE

(*) For details of inspection items, refer to **1.1 Inspection Categories**.

1.2.2 Zero Height Offset Calculation

Solder Paste Inspection compensates the height of solder paste with offset value calculated by treat the height of the board around pad as zero point. (*1) (*2)

Refer to Figure 1-9.

Zero height calculation area is specified by a parameter for setting the size of an inspection window. (*3)

However, excluding adjacent solder paste and holes etc. (*4)

Height offset is calculated from the height of zero height calculation area. (*5)(*6)

NOTE

- (*1) Supported version is v1.0.0.0 or later.
- (*2) The height compensation works inside algorithm. So the inspection is possible even though board compensation is OFF.
- (*3) It uses **Expand Width** and **Expand Height** parameters in the **Solder Paste Inspection** algorithm. The area in the inspection window excepting the area where one-third of the specified value from a shape size is expanded is the zero height calculation area. For details, refer to **1.3 Solder Paste Inspection - Setting Method**.
- (*4) Among the areas specified by (*3), it is yellow area in Figure below, excluding solder paste and holes etc.
- (*5) The calculated offset value is displayed as Pad neighborhood zero offset on the **Board Viewer**. For details, refer to **1.3 Solder Paste Inspection - Setting Method**.
- (*6) If the inspection window overlaps other pad areas in version v1.2.0.0 or later, the overlapping areas are excluded from the zero height calculation area.

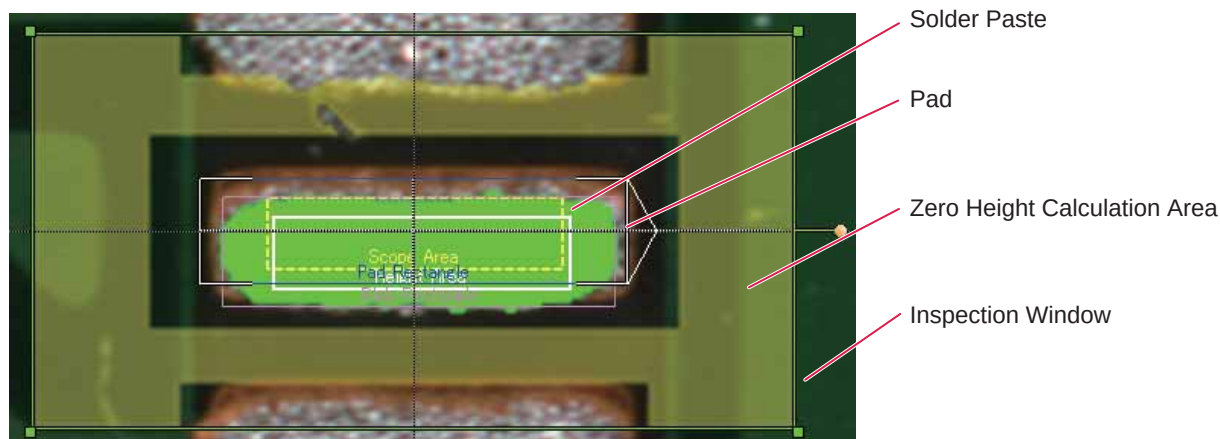


Figure 1-9 Zero Height Calculation Area

1.2.3 Bare Board Teaching

Solder Paste Inspection can compensate the height of solder paste by using the height offset value of pad and board which is obtained from bare board.

The height offset value of the board and pad is obtained from the **Bare Board Teaching** function. (*1)

The height offset is calculated from the height of the board and pad in the inspection window. (*2)

NOTE

- (*1) For details, refer to **Part II 6 Bare Board Teaching**.
- (*2) If it has the value which was registered in the previous version, it needs to register again after updating the latest algorithm version.

1.2.4 Detection of Solder Paste

In Solder Paste Inspection, solder paste is detected by specifying brightness values of set image filter and threshold values of height.

Brightness Value of Image Filter

In algorithm, it is set from **Upper level** and **Lower level** parameters. The recommended image filter is 2D image filter since the threshold value of height can be set separately.

Threshold Value of Height

In algorithm, it is set by **Minimum paste height** parameter. All the pixels with the height which is the same as the set value or lower are not used for inspections.

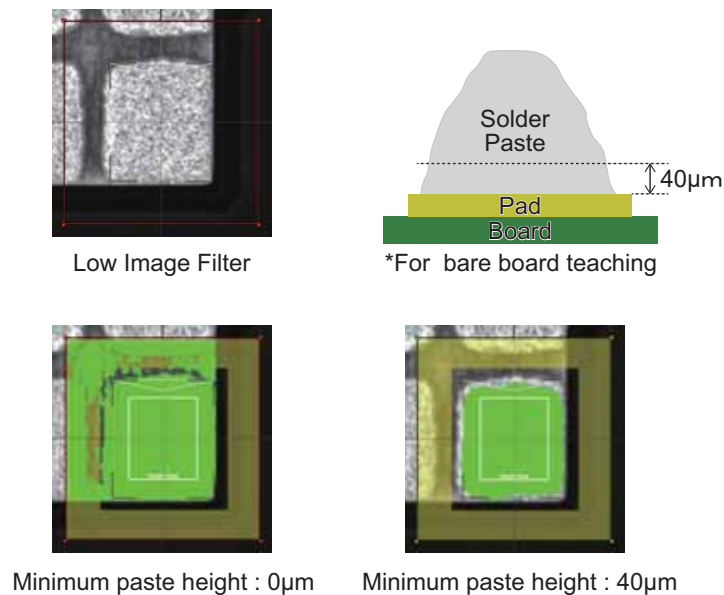


Figure 1-10 Detection of Solder Paste

The height image of the detected solder paste is displayed on the right side of the inspection window on **Board Viewer** of the recipe edit screen.



Figure 1-11 Height Image of Solder Paste

1.3 Solder Paste Inspection - Setting Method

This section describes how to set the **Solder Paste Inspection** algorithm to inspect solder paste.

1.3.1 Setting of Parameter

Here describes each parameter on a parameter edit window of a recipe edit screen.

For details of inspection items, refer to **1.1 Inspection Categories**. For the overview of algorithm, refer to **1.2 Solder Paste Inspection - Algorithm**.

Name	Value
Lower Level	40
Upper Level	255
Image Filter	Low
Height Area[%]	70
Scope Area[%]	70
Peak Threshold[um]	500
Expand Width[um]	200
Expand Height[um]	200
Minimum Paste Height[um]	40
SubPixelScale	3
Ratio[%]	10

	Name	Value	Warn Range	OK Range	NG Type
☒	Area[%]	-	0	0 50 150	Area
☒	Volume[%]	-	0	0 50 150	Volume
☒	Height[um]	-	0	0 50 200	Height
☒	Height[%]	-	0	0 50 150	Height
☒	PeakArea[um2]	-	0	0 0 32400	Form
☒	Top[um]	-	0	0 0 100	Bridge
☒	Right[um]	-	0	0 0 100	Bridge
☒	Bottom[um]	-	0	0 0 100	Bridge
☒	Left[um]	-	0	0 0 100	Bridge
☒	Shift	X:-um Y:-um R:-deg	0 0 0	0 0 -360 360	Shift
☒	Offset X[%]	-	0	0 0 25	Shift
☒	Offset Y[%]	-	0	0 0 25	Shift

Figure 1-12 Parameter (version 1.2.0.0)

Parameter	Description
Minimum Paste Height[um]	Sets minimum height of solder paste to be detected. Set the height so that it is above the height of a pad. The detected solder paste is determined depending on this parameter and Upper Level and Lower Level . For details, refer to 1.2.4 Detection of Solder Paste .
Image Filter	Select 2D Image Filter. (*1)
Lower Level	Specify a lower level value of brightness detected as solder paste. The detected solder paste is determined depending on this parameter, Upper Level and Minimum Paste Height . For details, refer to 1.2.4 Detection of Solder Paste .

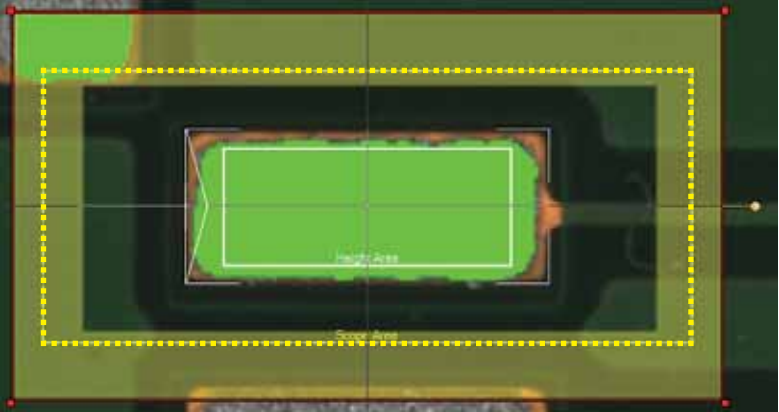
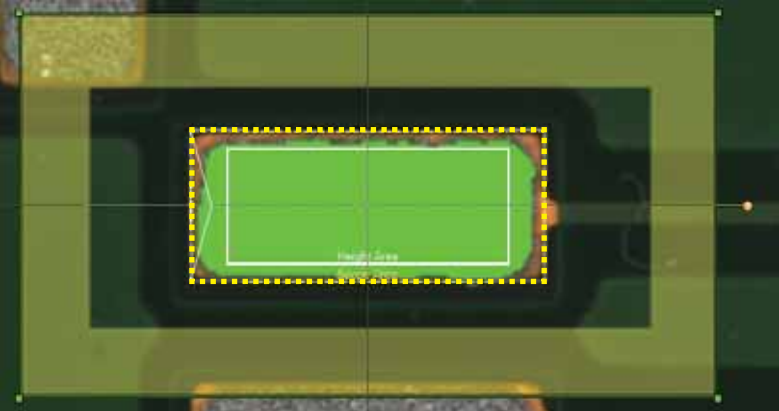
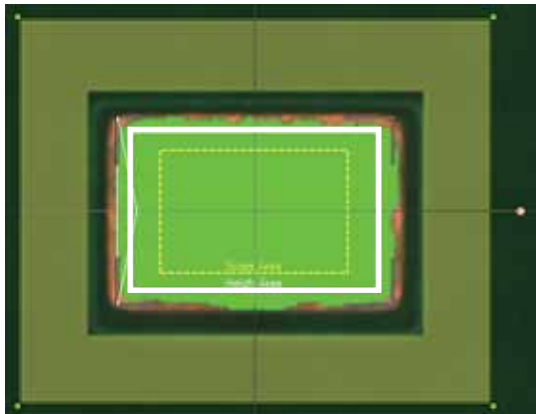
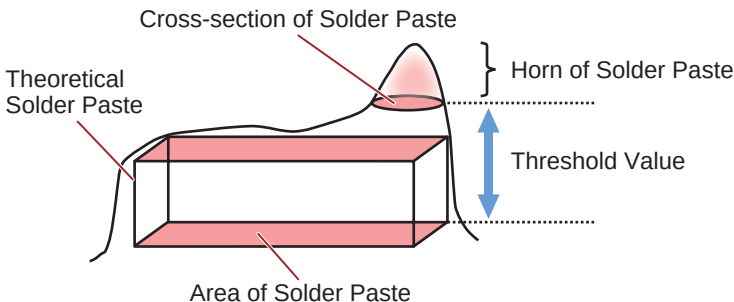
Parameter	Description
Upper Level	Specify an upper level value of brightness detected as solder paste. The detected solder paste is determined depending on this parameter, lower level and minimum paste height. For details, refer to 1.2.4 Detection of Solder Paste.
Scope Area[%]	Adjust this if adjacent solder paste is detected in the inspection window. Detected solder paste included in scope area or adjacent to scope area is used for inspections. This value is the specified value at the time when shape size is 100%. The set scope area is displayed with the yellow dotted line as the figure below. 180%  100% 

Figure 1-13 Scope Area

Parameter	Description
Height Area[%]	Set the area used for calculating a height sample value. This value is a specified value at the time when shape size is 100%. The set height area is displayed with white line as the figure below. The center position of the height area is the gravity position of the detected solder paste area.  Figure 1-14 Height Area
Peak Threshold[μm]	Specify the threshold of height for horn inspection. Solder paste higher than the specified value is detected as a horn.  Figure 1-15 Peak Threshold
Expand Width[μm]	Adjust the inspection window size of the right and left directions (*2). Resize the width based on shape size in micrometer. In addition, this parameter is used for setting zero height compensation. The area in the inspection window except the area which expanded one-third of the specified value from a shape size is the zero height calculation area. For details of zero height compensation, refer to 1.2.2 Zero Height Offset Calculation.
Expand Height[μm]	Adjust the inspection window size of the top and bottom directions (*3). Resize the height based on shape size in micrometer. In addition, this parameter is used for setting zero height compensation. The area in the inspection window except the area which expanded one-third of the specified value from a shape size is the zero height calculation area. For details of zero height compensation, refer to 1.2.2 Zero Height Offset Calculation.
SubPixelScale	Specifies the scale of sub pixel. The larger the setting value, the longer the inspection time but inspection with detailed pixel is enabled.

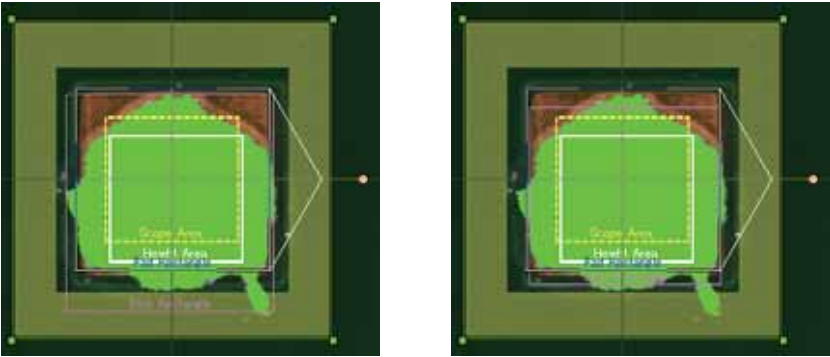
Parameter	Description
Ratio[%]	Specify the contact ratio between the detected solder paste and each side of the circumscribed rectangle that the solder paste contacts. If the solder paste has an uneven shape, increasing the value ignores the protrusion and the inner area is used.  Ratio [0%] Ratio [30%] Figure 1-16 Ratio

Table 1-1 Parameter

- NOTE

(*1) Any image filter can be used to inspect solder paste as long as solder paste is visible.

However, since height information has been used as filter inside, image filter without using height information is recommended.

(*2) As Figure 1-17, it is right and left directions (with blue arrow) based on the case where the yellow point beside the inspection window is on the right side.

(*3) As Figure 1-17, it is top and bottom directions (with red arrow) based on the case where the yellow point beside the inspection window is on the right side.

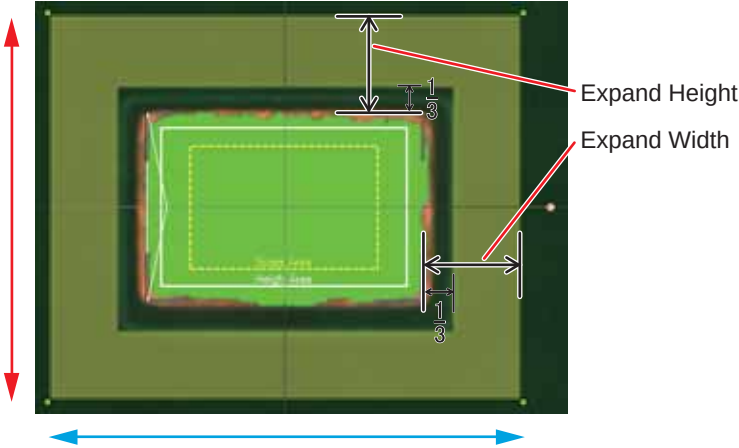
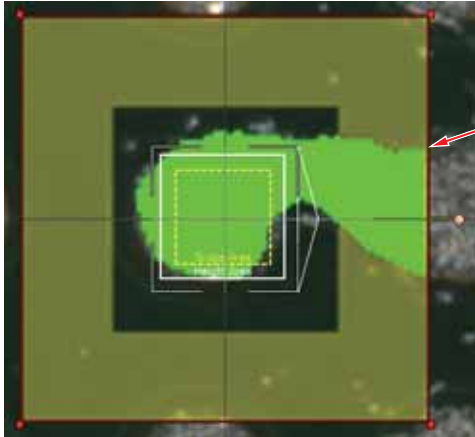
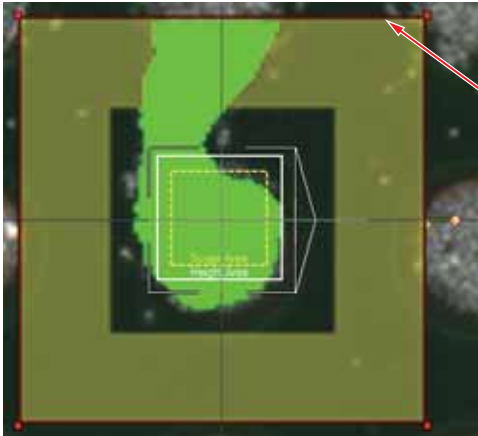
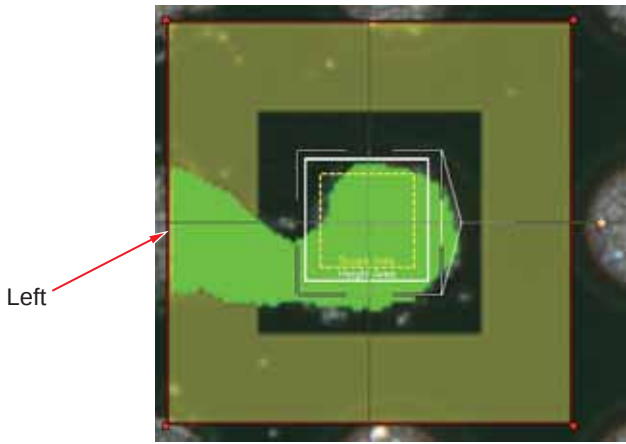
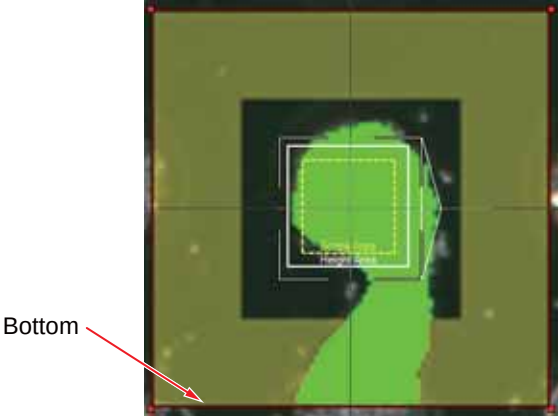
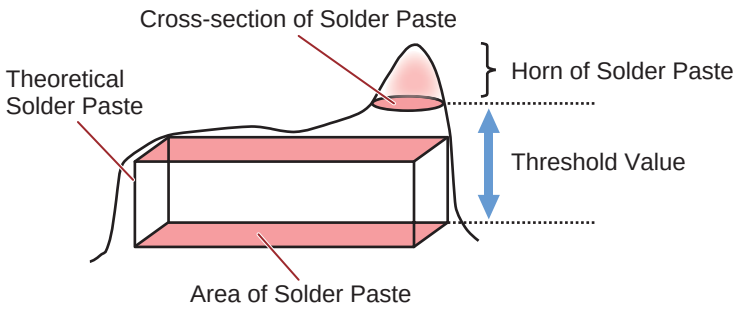
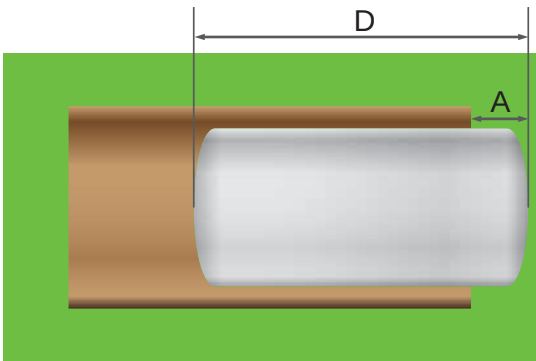


Figure 1-17 Direction of Inspection Window

The following parameters are values shown as sample value of algorithm.

Parameter		Description
Height[um]		Displays the average of height in the range specified in Height Area .
Height[%]		Displays the height in the range specified in Height Area as a percentage of the stencil thickness.
Volume[%]		An integral value of height in the range where solder paste is detected shall be a measured value, and a volume calculated from a gerber pad area value and thickness of a stencil shall be a theoretical value. The ratio (= measured value / theoretical value x 100[%]) is displayed as a sample value.
Area[%]		The percentage of where solder paste was detected in inspection window shall be a measured value, and the gerber pad area value shall be a theoretical value. The ratio (= measured value / theoretical value x 100[%]) is displayed as a sample value.
Shift	X	Displays the distance between the gravity center of detected solder paste and gravity position of gerber pad on X-axis.
	Y	Displays the distance between the gravity center of detected solder paste and gravity position of gerber pad on Y-axis.
	R	Not available. This value is always 0 because this algorithm does not detect the amount of rotation. Use OK Range and Warning Range as a default value.
Right		Displays the total of the narrow areas as a sample value in the case that solder paste is on the side (right) of an inspection window. If the yellow point beside the inspection window is on the right side, the right frame is a targeted area.  Figure 1-18 Right(Bridge)
Top		Displays the total of the narrow areas as a sample value in the case that solder paste is on the side (top) of an inspection window. If the yellow point beside the inspection window is on the right side, the top frame is a targeted area.  Figure 1-19 Top(Bridge)

Parameter	Description
Left	Displays the total of the narrow areas as a sample value in the case that solder paste is on the side (left) of an inspection window. If the yellow point beside the inspection window is on the right side, the left frame is a targeted area.  Figure 1-20 Left(Bridge)
Bottom	Displays the total of the narrow areas as a sample value in the case that solder paste is on the side (bottom) of an inspection window. If the yellow point beside the inspection window is on the right side, the bottom frame is a targeted area.  Figure 1-21 Bottom(Bridge)

Parameter	Description
PeakArea	Displays the solder paste cross-section area value in the threshold value specified in Peak Threshold. The detected PeakArea is displayed beside the inspection window of Board Viewer as shown in the figure below.  Figure 1-22 PeakArea 1  Figure 1-23 PeakArea 2
Offset X[%]	Calculates the overhang of solder paste in X direction by $A/D \times 100$ [%].  Figure 1-24 Offset X

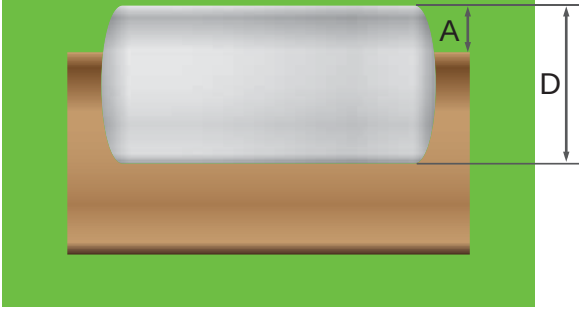
Parameter	Description
Offset Y[%]	Calculates the overhang of solder paste in Y direction by $A/D \times 100$ [%].  Figure 1-25 Offset Y

Table 1-2 OK Range Setting

The following three information items are displayed at the bottom left of **Board Viewer**.

Item	Description
Pad designed area	Area value calculated from a gerber pad.
Pad depth offset	Height offset calculated from Bare Board Teaching. When not using the Bare Board Teaching, height offset value is 0. Refer to 1.2.2 Zero Height Offset Calculation .
Pad neighborhood zero offset	Height offset calculated from pad neighborhood. Refer to 1.2.2 Zero Height Offset Calculation .
Stencil thickness	Stencil thickness value.

Table 1-3 Pad Information

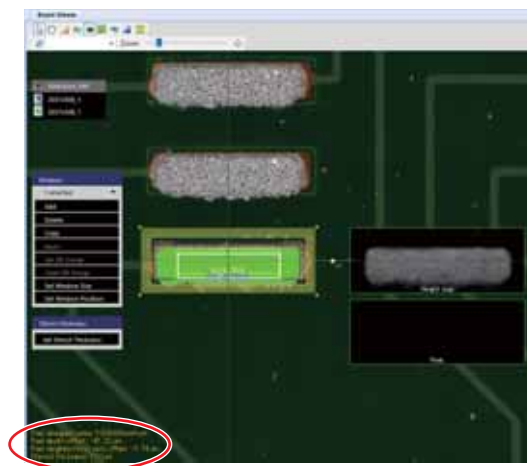


Figure 1-26 The Board Viewer of Recipe Edit Screen

1.3.2 Setting Procedure

Here describes how to set **Solder Paste Inspection** in Recipe edit screen.

NOTE

Parameter set of algorithm can be set and applied at once without opening Recipe edit screen. For details, refer to **Part II 4 Parameter Editor**.

Step1: Select an inspection window to which **Solder Paste Inspection** is assigned.

NOTE

Algorithm is automatically assigned when the data is created in the procedure of **Part II 1 Making an Inspection Data**.

NOTE

If the inspection window was re-created, select **Solder Paste Inspection** from the **Algorithm** drop-down list.

Step2: Set the parameter by reference to **1.3.1 Setting of Parameter**.

Step3: Click  and check if the inspection can be performed.

Step4: Specify OK Range of a sample value and set **NG Type** for the case where **OK Range**, **Warn Range**, and the sample value is judged as NG.

NOTE

If not performing the inspection, uncheck the item.

1.3.3 Aperture Reduction Setting

This setting is used to calculate **Offset X[%]** and **Offset Y[%]** of the solder paste inspection algorithm. In general, SPI inspection data use stencil aperture information to generate pad information, instead of board aperture information. Therefore, the pad sizes of board and inspection data may differ. If the pad sizes of board and inspection data are different, they can be fine-tuned by this setting.

Step1: From the **InspectData** tab, select the **Layout** group > **Board size**.

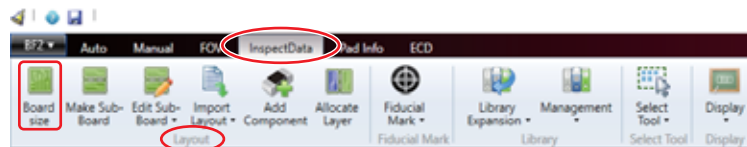


Figure 1-27 Aperture Reduction Setting 1

Step2: The dialog box shown below appears. Set **Aperture Reduction** and click **OK**.

NOTE

When **Aperture Reduction** is set to **10%**, **Offset X[%]** and **Offset Y [%]** are calculated based on the area where the stencil aperture information is enlarged by 10% in X and Y directions.

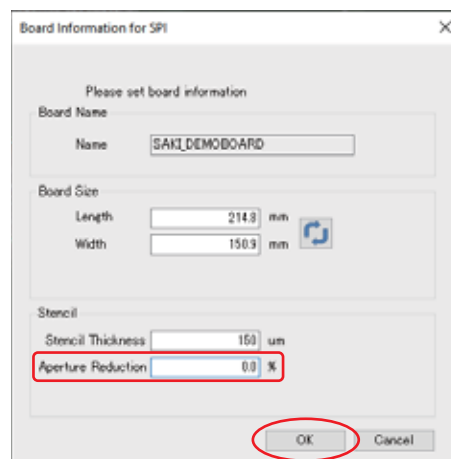


Figure 1-28 Aperture Reduction Setting 2

2 Position Recognition

2.1 Square Fiducial

2.1.1 Overview of Inspection

Square Fiducial is the algorithm to detect a square fiducial mark from an inspection window and calculate the distance between the detected mark and the center of the inspection window.

Square Fiducial is suitable for detecting a square fiducial mark.

2.1.2 Setting of Parameter



Figure 2-1 Dialog (Version 1.2.0.0)

Parameter	Description
Image filter	Select a lighting that displays the fiducial mark clearly.
Size	Measure the size of the fiducial mark accurately by reference to Part I 2.1.3 Measuring Dimensions of the Selected Area and specify the measured value.
Edge range	Specifies the distance between the inner and outer square window used for detecting a fiducial mark. Adjust the value so that every side of the square fiducial mark is included between the inner and outer window. If fiducial mark sides are uneven, increase the value.
Pattern	Select Bright mark in dark area if the brightness of the fiducial mark is brighter than its surroundings or Dark mark in bright area if it is darker than its surroundings.
Scale	With increasing this value, the precision increases because the position is acquired by finely partitioning the pixels. However, this increases the inspection time. Default value is 5 .
Search Method	Specifies the calculation method of the center position of the fiducial mark. Gravity calculates the center position of the fiducial mark from the gravity center of the detected area. FindRect calculates the center position of the fiducial mark from the circumscription of the detected area.

Table 2-1 Parameter

Parameter		Description
Shift	X : - μm	Specifies the acceptable range of the X-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	Y : - μm	Specifies the acceptable range of the Y-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	R : -deg	Not available. This algorithm is always set to 0 because it does not detect the amount of rotation. Use OK Rage as a default value.
Inside average		Specifies the average brightness value in the detected fiducial mark. Judges whether fiducial mark is correctly detected.
Inside avg. - outside avg.		Specifies the average brightness value in the detected fiducial mark and the acceptable range of the average brightness value. Judges whether fiducial mark is correctly detected.
Location		Displays the coordinate of the center position.

Table 2-2 Sample OK Range Setting

2.1.3 Setting Procedure

- Step1: Select **SquareFiducial** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the fiducial mark clearly from the **Image filter** drop-down list.
- Step3: Adjust the inspection window to surround the fiducial mark.

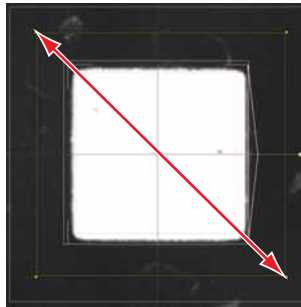


Figure 2-2 Adjusting the Inspection Window

Step4: To detect the fiducial mark, every side of the mark needs to be included between the inner and outer square windows. Adjust the value of **Size** and click  to check if the fiducial mark is detected.

NOTE Measure **Size** value accurately by reference to **Part I 2.1.3 Measuring Dimensions of the Selected Area**.

NOTE If fiducial mark sides are uneven, increase the **Edge range** value.

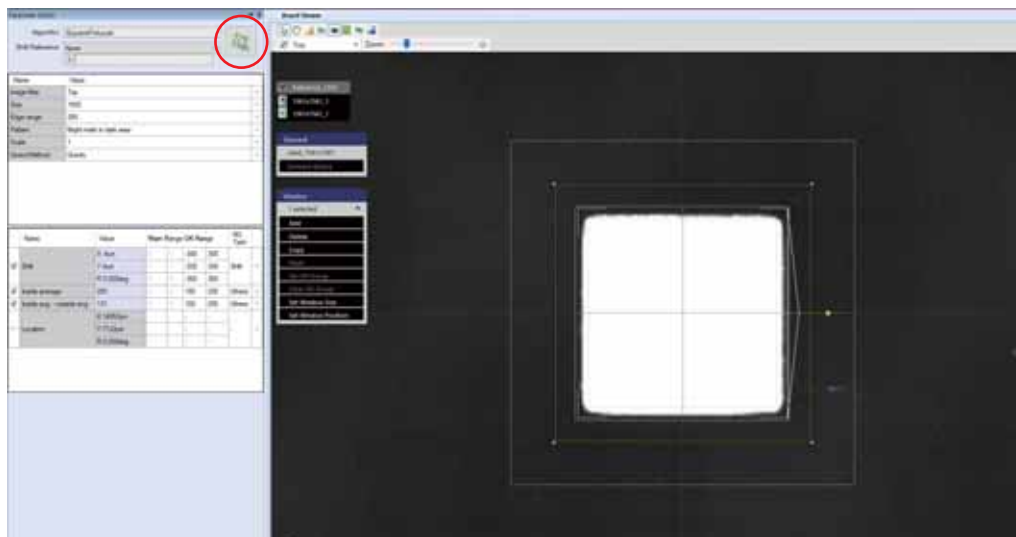


Figure 2-3 Detection of the Fiducial Mark

Step5: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.

Step6: Click  and check if the inspection is completed properly.

2.2 Rect Align

2.2.1 Overview of Inspection

Rect Align is the algorithm to detect a rectangular fiducial mark from an inspection window. The algorithm also calculates the mounting angle, size, and the distance between the detected fiducial mark and the center of the inspection window. Rect Align is suitable for detecting a rectangle shape fiducial mark.

2.2.2 Setting of Parameter

Parameter Editor

Algorithm

Rect Align

Shift Reference

None

Name	Value
Image filter	Height in 5um step
Lower limit	100
Upper limit	255
Option	Free
TopRatio[%]	50
BottomRatio[%]	50
LeftRatio[%]	50
RightRatio[%]	50
RotationOffset	0
Search Direction	Out -> In
SubPixelScale	1
RotationResolution	10
Terminal	by Object

	Name	Value	OK Range	NG Type
☒	Shift	X:-um	-300 300	-
		Y:-um	-300 300	
		R:-deg	-5 5	
☒	Result type	-	Found	-
☒	X-Length	-u	1000 10000	-
☒	Y-Length	-u	1000 10000	-
☐	End Overhang[um]	-um	-	-
☐	Side Overhang[%]	-%	-	-
☐	End Overlap[%]	-%	-	-

Body

Electrode

☐ Use Scope Area

Use Element Size

Width :

1 000

um

Length :

1 000

um

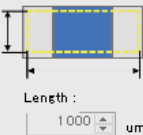


Figure 2-4 Dialog (Version 1.6.0.0)

Parameter		Description
Lower limit		Upper limit is the upper limit of the brightness levels. Lower limit is the lower limit of the brightness levels. Set 255 to Upper limit . Set the value which displays the fiducial mark clearly to Lower limit .
Upper limit		
Image filter		
Top Ratio		
Bottom Ratio		Each parameter indicates the contact area percentage between the detected fiducial mark and each side of the pink rectangle window. When the fiducial mark sides are uneven, increase the value. Protruding parts are ignored and inner area is recognized as the fiducial mark. To recognize a wider area as the body, decrease the value.
Left Ratio		
Right Ratio		
Option	Free	Recognizes the angle of the fiducial mark.
	No Rotate	Does not recognize the angle of the fiducial mark. Select this item when the fiducial mark cannot be detected by Free.
SubPixelScale		Specifies the scale of sub pixel. The larger the setting value, the longer the inspection time. But inspection with detailed pixel is enabled. Please set larger value for small fiducial mark.
Rotation Offset		If there are the specific angle for the fiducial mark, set this parameter.
RotationResolution		Specifies the resolution for calculation of rotation. More detailed calculation of angle is possible by setting larger value.
Search Direction		Out->In searches for the edge of the fiducial mark in the central direction from the outside of the inspection window. In->Out searches for the edge of the fiducial mark in the outward direction from the center of the inspection window. Usually, Out->In is specified. If an adjacent pad is mistakenly searched, specify In->Out . When specifying In->Out , a fiducial mark must be included in the center of the inspection window.
Terminal		The item is for AOI machines only. Do not change the setting.

Table 2-3 Parameter

Parameter		Description
Result type	Found	The inspection result becomes OK when the fiducial mark is detected.
	Not Found	The inspection result becomes OK when the fiducial mark is not detected.
	Too Near To Edge	The inspection result becomes OK when the fiducial mark is close to the inspection window corners.
Shift	X : -μm	Specifies the acceptable range of the X-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	Y : -μm	Specifies the acceptable range of the Y-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	R : -deg	Specifies the acceptable range of the angle of the detected fiducial mark. Angular value increases clockwise and decreases counter-clockwise.
X-Length		Specifies the acceptable range of the X-axis length of the detected fiducial mark.
Y-Length		Specifies the acceptable range of the Y-axis length of the detected fiducial mark.
Side Overhang[%]		The item is for AOI machines only. Do not change the setting.
End Overhang[μm]		
End Overlap[%]		

Table 2-4 OK Range Setting

NOTE

If multiple items are selected on **OK Range** of **Result type**, the inspection result becomes OK when one of the selected result types is detected.

Parameter	Description
Use Scope Area	When the noise except adjacent pad or target fiducial mark affects inspection result, check this. For details of scope area, refer to 2.2.4 Scope Area .
Width	Specifies scope area size. By clicking Use Element Size , the element size of body is set for Width and Length .
Length	

Table 2-5 Description of Body Tab

Parameter	Description
Use Electrode Size inside component	The item is for AO machines only. Do not change the setting.
Electrode Width	
Electrode Length	

Table 2-6 Description of Electrode Tab

2.2.3 Setting Procedure

- Step1: Select **Rect Align** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the fiducial mark clearly from the **Image filter** drop-down list.
- Step3: Adjust the inspection window to surround the fiducial mark.

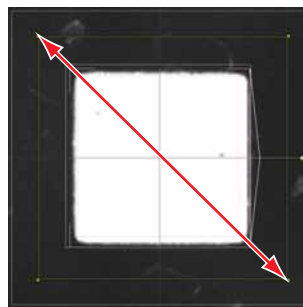


Figure 2-5 Adjusting the Inspection Window

- Step4: Check the values of **Lower limit** and **Upper limit**.
Set **255** to **Upper limit**. Set the value which displays the fiducial mark clearly to **Lower limit**.
- Step5: Check the values on **Top Ratio**, **Bottom Ratio**, **Left Ratio**, and **Right Ratio**.

Step6: Click  and check if the fiducial mark is detected.

The detected fiducial mark is displayed with the pink rectangle.

NOTE If an adjacent pad is mistakenly searched, set the **Search Direction** to **In->Out**. In this case, there must be a fiducial mark in the center of the inspection window.

NOTE If an adjacent pad is mistakenly searched, check **Use Scope Area**, and click **Use Element Size** to set scope area. In this case, fiducial mark needs to be included in scope area.

NOTE If small fiducial mark cannot be searched properly, set **SubPixelScale** to larger value. The larger the setting value, the longer the inspection time. But inspection with detailed pixel is enabled.

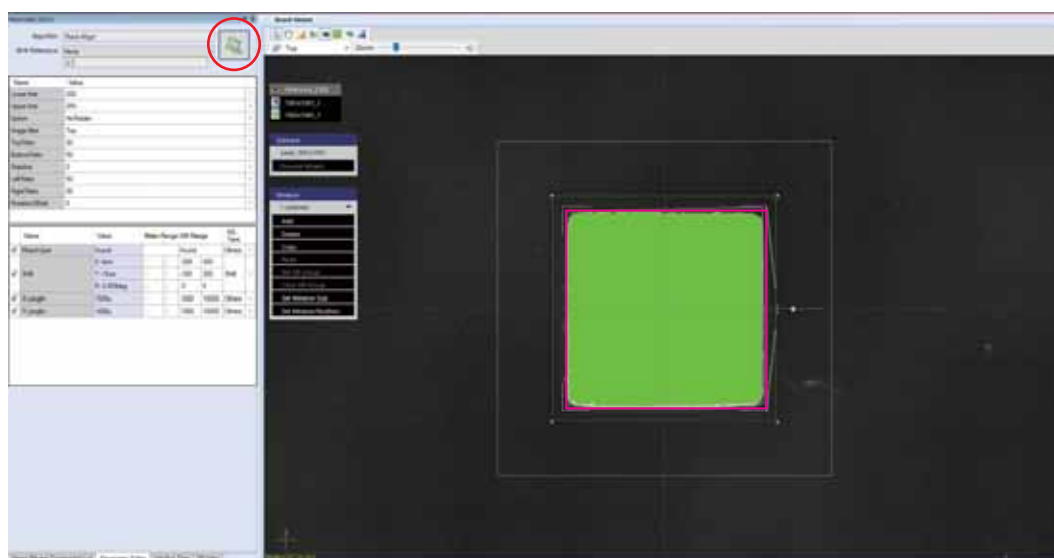


Figure 2-6 Detection of fiducial mark

Step7: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.

NOTE If not performing the inspection, uncheck the item.

Step8: Click  and check if the inspection is completed properly.

2.2.4 Scope Area

When scope area is used, the pixels that are not in contact with scope area are masked and treated as out of detection target.

By this function, fiducial mark can be searched with reducing the effect of noise from adjacent pad or board.

Scope area is displayed in dotted yellow line.

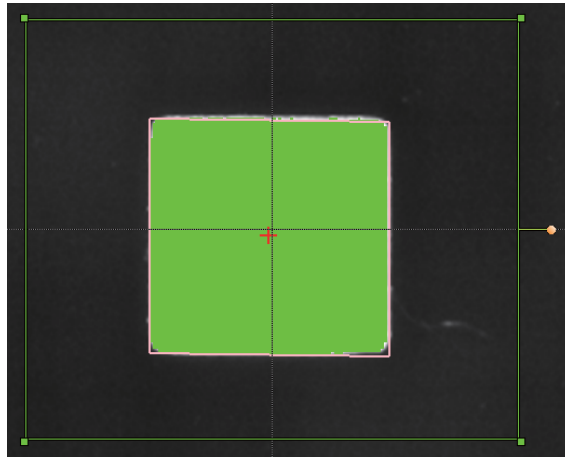


Figure 2-7 Without Scope Area Setting

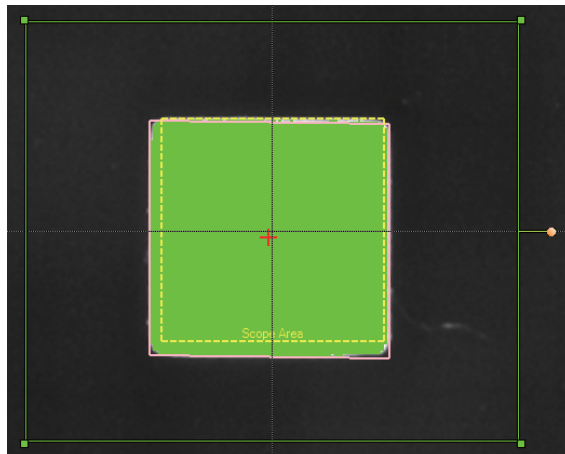


Figure 2-8 With Scope Area Setting

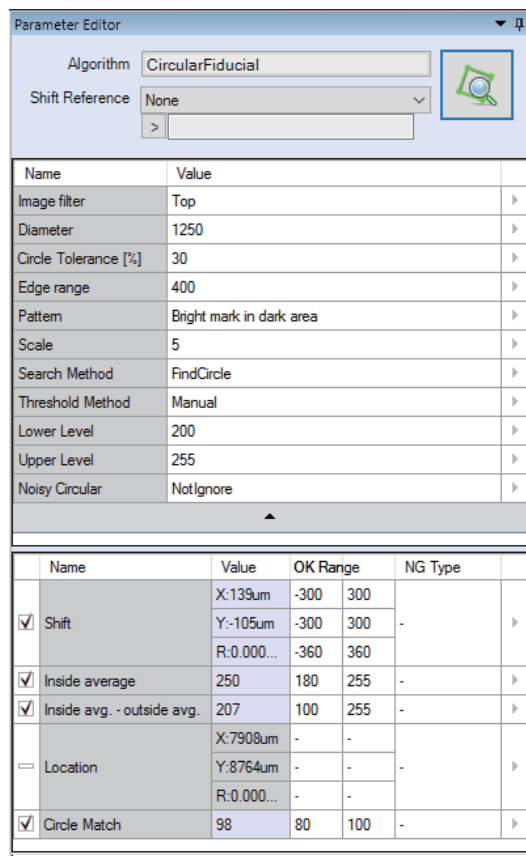
2.3 Circular Fiducial

2.3.1 Overview of Inspection

Circular Fiducial is the algorithm to detect a round fiducial mark from an image on an inspection window and calculate the distance between the detected mark and the center of the inspection window.

Circular Fiducial is suitable for detecting a round fiducial mark.

2.3.2 Setting of Parameter



Parameter Editor

Algorithm: CircularFiducial

Shift Reference: None

>

Name	Value
Image filter	Top
Diameter	1250
Circle Tolerance [%]	30
Edge range	400
Pattern	Bright mark in dark area
Scale	5
Search Method	FindCircle
Threshold Method	Manual
Lower Level	200
Upper Level	255
Noisy Circular	NotIgnore

	Name	Value	OK Range	NG Type
☒	Shift	X:139um	-300 300	
		Y:-105um	-300 300	
		R:0.000...	-360 360	
☒	Inside average	250	180 255	-
☒	Inside avg. - outside avg.	207	100 255	-
☐	Location	X:7908um	- -	
		Y:8764um	- -	
		R:0.000...	- -	
☒	Circle Match	98	80 100	-

Figure 2-9 Dialog (Version 1.5.0.0)

Parameter	Description
Image filter	Select a lighting that displays the fiducial mark clearly.
Diameter	Referring to Part I 2.1.3 Measuring Dimensions of the Selected Area , accurately measure the diameter of the circle of the fiducial mark and set the measured values.
Circle Tolerance[%]	This parameter is valid only if FindCircle has been set for Search Method . This parameter specifies the tolerance of circularity as percent. When the noise is contiguous with the fiducial mark which brightness is as same as the brightness of noise, please set this value higher to increase detection rate. However false call may be increased when this value is too much higher.
Edge range	This parameter is valid only if Gravity has been set for Search Method . When a fiducial mark is detected, two pink dashed circles are displayed around the fiducial mark. The inner circle shows the size of circle set in Diameter . The outer circle diameter is adding Diameter to Edge range . If there is unevenness around the fiducial mark, set higher value in this parameter to adjust so that the fiducial mark is included in the outer circle.
Pattern	Select Bright mark in dark area if the brightness of the fiducial mark is brighter than its surroundings or Dark mark in bright area if it is darker than its surroundings.
Scale	With increasing this value, the precision increases because the position is acquired by finely partitioning the pixels. However, this increases the inspection time. Default value is 5 .
Search Method	Specifies the calculation method of the center position of the fiducial mark. Gravity calculates the center position of the fiducial mark from the gravity center of the detected area. FindRect calculates the center position of the fiducial mark from the circumscription of the detected area.
Threshold Method	This parameter is valid only if FindCircle has been specified with Search method . Set Adaptive to automatically set the threshold values of the brightness level. Set Manual to manually set the threshold values. By setting the appropriate setting in Manual , the detection rate will be improved more than Adaptive .
Lower Level	This parameter is valid only if FindCircle has been set for Search Method and Manual has been set for Threshold Method . Set Upper Level to 255 . And set Lower Level to the value which shows the fiducial mark clearly visible.
Upper Level	
Noisy Circular	This parameter is valid only if FindCircle has been set for Search Method . When the circle that is similar size to the fiducial mark is existing near fiducial mark, and noise is existing inside/outside of the circle, the circle is set as out of detection target by setting Ignore for this parameter. Recommended value is NotIgnore .

Table 2-7 Parameter

Parameter	Description
Shift	X : - μ m
	Y : - μ m
	R : -deg
Inside average	Specifies the acceptable range of the average brightness level in the detected fiducial mark. Judges whether fiducial mark is correctly detected.
Inside avg. - outside avg.	Specifies the acceptable range of the average brightness level in the detected fiducial mark. Judged whether fiducial mark is correctly detected.
Location	Displays the coordinates of the center position.
Circle Match	Displays the matching rate when FindCircle has been specified with Search Method .

Table 2-8 OK Range Settings

2.3.3 Setting Procedure

- Step1: Select the **CircularFiducial** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the fiducial mark clearly from the **Image filter** drop-down list.
- Step3: Adjust the inspection window to surround the fiducial mark.

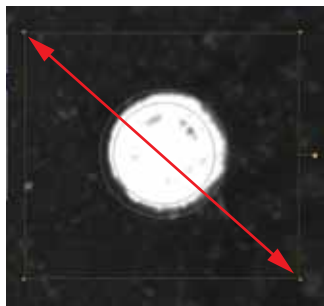


Figure 2-10 Adjusting the Inspection Window

- Step4: Select **Bright mark in dark area** if the brightness of the fiducial mark is brighter than its surroundings or **Dark mark in bright area** if it is darker in comparison from **Pattern** drop-down list.
- Step5: By reference to **Part I 2.1.3 Measuring Dimensions of the Selected Area**, accurately measure the diameter of the circle of the fiducial mark and set the measured values to **Diameter**.
- Step6: When **Gravity** is set for **Search Method**, adjust parameters by reference to **2.3.4 Search Method: Gravity**.
When **FindCircle** is set for **Search Method**, adjust parameters by reference to **2.3.5 Search Method: FindCircle**.
- Step7: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.
- Step8: Click  and check if the inspection is completed properly.

2.3.4 Search Method: Gravity

- Step1: Select **Gravity** from the **Search Method** drop-down list.
- Step2: If there is unevenness around the fiducial mark, set **Edge range** to higher value to adjust so that the fiducial mark is included in the outer circle.
- Step3: Click  to check if the fiducial mark is detected.
Two pink dashed circles are displayed around the fiducial mark.
The inner circle shows the size of circle set in **Diameter**. The outer circle **diameter** is adding Diameter to **Edge range**.



Figure 2-11 Search Method: Gravity

2.3.5 Search Method: FindCircle

- Step1: Select **FindCircle** from the **Search Method** drop-down list.
- Step2: When the noise is contiguous with the fiducial mark which brightness is as same as the brightness of noise, please set **Circle Tolerance[%]** higher. However false call may be increased when this value is too much higher.

NOTE

Circle Tolerance[%] judges if the target is circle or not based on the aspect ratio of circumscribed quadrangle.

When the following condition is met, the target is judged as NOT circle.

- $\text{Height} < \text{Width} \times (1 - (\text{Circle Tolerance} / 100))$
- $\text{Height} > \text{Width} \times (1 + (\text{Circle Tolerance} / 100))$

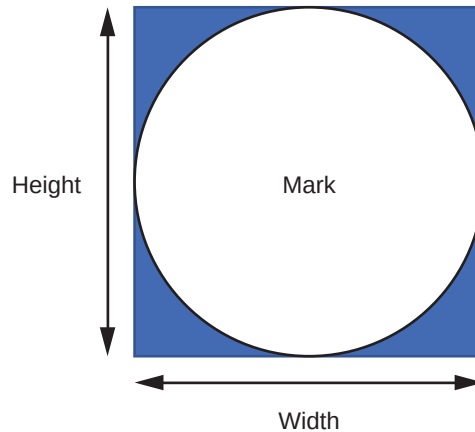


Figure 2-12 Circle Tolerance

When **Circle Tolerance[%]** value is higher, the target is judged as circle even if the aspect ratio of circumscribed quadrangle is higher. But when the value is too much higher, other circle than fiducial mark may be misrecognized.

When **Circle Tolerance[%]** value is smaller, only the circle closer to a complete round is recognized. But when the value is too much smaller, even slight noise add effect and target may not be judged as circle.

Please adjust the value of **Circle Tolerance[%]** depending on the image condition of fiducial mark and around of it.

Step3: Select **Manual** in **Threshold Method**.

Set **Upper Level** to **255**. And set **Lower Level** to the value which shows the fiducial mark clearly visible.

Step4: When the circle that is similar size to the fiducial mark is existing near fiducial mark, and noise is existing inside/outside of the circle, set **Noisy Circular** to **Ignore**.

NOTE

Noisy Circular judges if the target is circle with noise or not by using binarization or labeling. Treats the circle with noise as out of detection target. To increase the accuracy of binarization, parameter adjustment in Step3 is necessary.

Step5: Click  to check if the fiducial mark is detected.

When **Bright mark in dark area** is selected in **Pattern** drop-down list, white area in binarization is colored green.

When **Dark mark in bright area** is selected in **Pattern** drop-down list, black area in binarization is colored green.

And blue dashed circle which size is set in **Diameter** is displayed around the detected fiducial mark.



Figure 2-13 Detected Circle by FindCircle

NOTE

When the noise which is as same brightness as fiducial mark is contiguous with the fiducial mark, separate of fiducial mark and noise may be failed. In that case, set **Diameter** to the maximum length includes noise, or set **Circle Tolerance[%]** to higher.

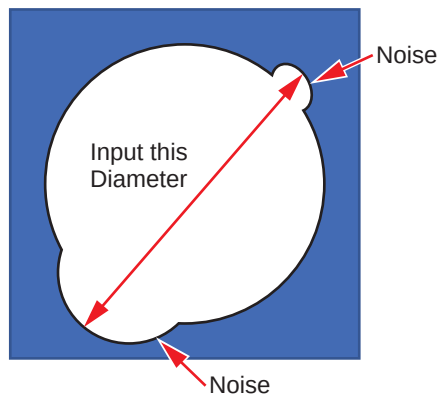


Figure 2-14 When The Noise Is Adjacent to The Fiducial Mark

NOTE

Please try the following steps when the fiducial mark is not detected correctly even if it is in the center of inspection window.

1. Check green area in Step5 and confirm the fiducial mark is colored green in a circle. And check the maximum length of fiducial mark is not changed by the noise.
2. When the fiducial mark is not colored green in a circle, adjust **Lower Limit** value. And re-measure the diameter of the circle and enter **Diameter** to it.
3. When the maximum length is changed in 1, adjust **Diameter** or **Circle Tolerance[%]** with considering maximum length.
4. When a circle with the same size as the fiducial mark exists in the inspection window, misdetection may occur. Select the lighting that clearly displays the difference between fiducial mark and misdetection circle in **Image filter**, and retry Step2 to Step5 in **2.3.5 Search Method: FindCircle**.

2.4 Circle Align

2.4.1 Overview of Inspection

Circle Align is the algorithm to detect a round fiducial mark from an inspection window.

The algorithm also calculates the diameter and the distance between the detected fiducial mark and the center of the inspection window. Circle Align is suitable for detecting a round fiducial mark.

2.4.2 Setting of Parameter

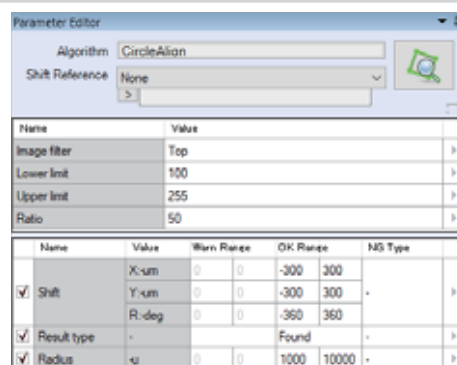


Figure 2-15 Dialog (Version 1.0.0.0)

Parameter	Description
Image filter	Select a lighting that displays the fiducial mark clearly.
Lower limit	Upper limit is the upper limit of the brightness level. Lower limit is the lower limit of the brightness level. Set 255 to Upper limit . Set the value which displays the fiducial mark clearly to Lower limit .
Upper limit	
Ratio	Indicates the contact area percentage between the detected fiducial mark and the pink circle. When the fiducial mark outer perimeter is uneven, increase the value. Protruding parts are ignored and inner area is recognized as the fiducial mark body. To recognize a wider area as the body, decrease the value.

Table 2-9 Parameter

Parameter	Description
Result type	Found The inspection result becomes OK when the fiducial mark is detected.
	Not Found The inspection result becomes OK when the fiducial mark is not detected.
	Too Near To Edge The inspection result becomes OK when the fiducial mark is close to the inspection window corners.
Shift	X : -μm Specifies the acceptable range of the X-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	Y : -μm Specifies the acceptable range of the Y-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	R : -deg Not available. This algorithm is always set to 0 because it does not detect the amount of rotation. Use OK Range as a default value.
Radius	Specifies the acceptable range of the radius of the detected fiducial mark.

Table 2-10 OK Range Setting

NOTE If multiple items are selected on **OK Range** of **Result type**, the inspection result becomes OK when one of the selected result types is detected.

2.4.3 Setting Procedure

- Step1: Select **CircleAlign** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the fiducial mark clearly from the **Image filter** drop-down list.
- Step3: Adjust the inspection window to surround the fiducial mark.

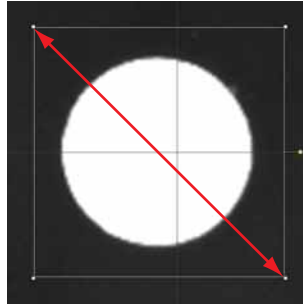


Figure 2-16 Adjusting the Inspection Window

- Step4: Adjust the values of **Lower limit** and **Upper limit**. Set **255** to **Upper limit**. Set the value which displays the fiducial mark clearly to **Lower limit**.
- Step5: Click  and check if the fiducial mark is detected. The detected fiducial mark is displayed with the pink circle.

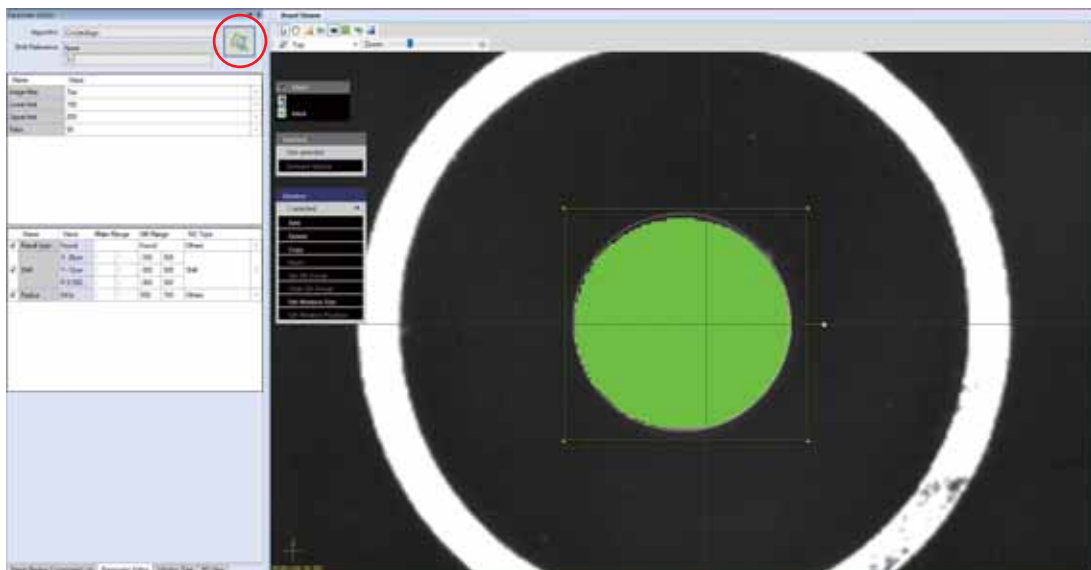


Figure 2-17 Detecting the Fiducial Mark

- Step6: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.

NOTE

If not performing the inspection, uncheck the item.

- Step7: Click  and check if the inspection is completed properly.

2.5 Gravity

2.5.1 Overview of Inspection

Gravity is the algorithm to detect a round fiducial mark from an inspection window and calculate the distance between the detected mark and the center of the inspection window.

Gravity is suitable for detecting a fiducial mark.

2.5.2 Setting of Parameter

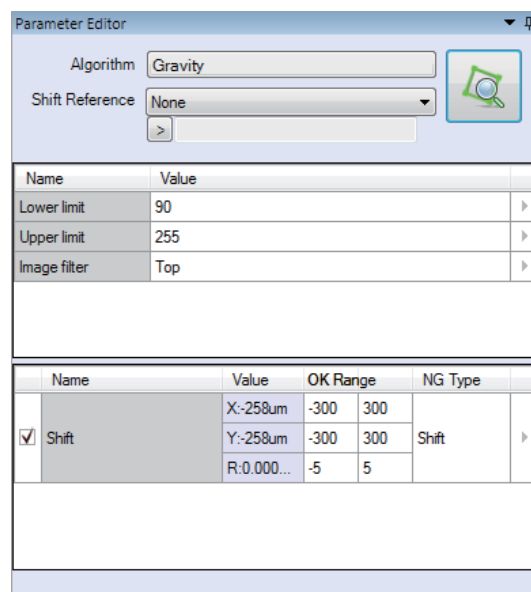


Figure 2-18 Dialog (Version 0.0.0.1)

Parameter	Description
Lower limit	Upper limit is the upper limit of the brightness level. Lower limit is the lower limit of the brightness level. Set the value which displays the round fiducial mark clearly.
Upper limit	
Image filter	Select a lighting that displays the fiducial mark clearly.

Table 2-11 Parameter

Parameter	Description
Shift	X : -μm Specifies the acceptable range of the X-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	Y : -μm Specifies the acceptable range of the Y-axis distance between the center of the detected fiducial mark and the center of the inspection window.
	R : -deg Not available. This algorithm is always set to 0 because it does not detect the amount of rotation. Use OK Range as a default value.

Table 2-12 OK Range Setting

2.5.3 Setting Procedure

- Step1: Select **Gravity** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the fiducial mark clearly from the **Image filter** drop-down list.
- Step3: Adjust the inspection window to surround the fiducial mark.

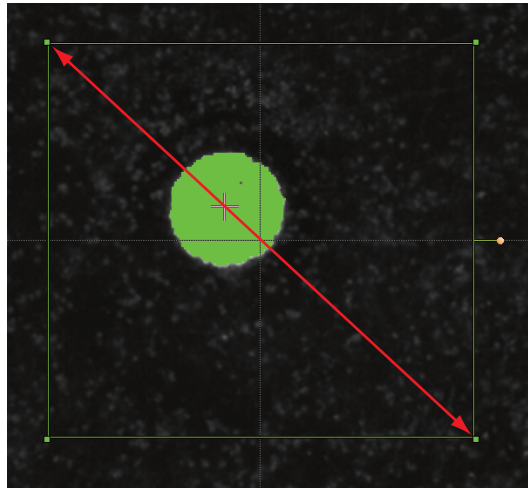


Figure 2-19 Adjusting the Inspection Window

- Step4: To detect the fiducial mark, outer perimeter of the mark needs to be included between the inner and outer round windows. Adjust the value of **Diameter** and click  to check if the fiducial mark is detected.

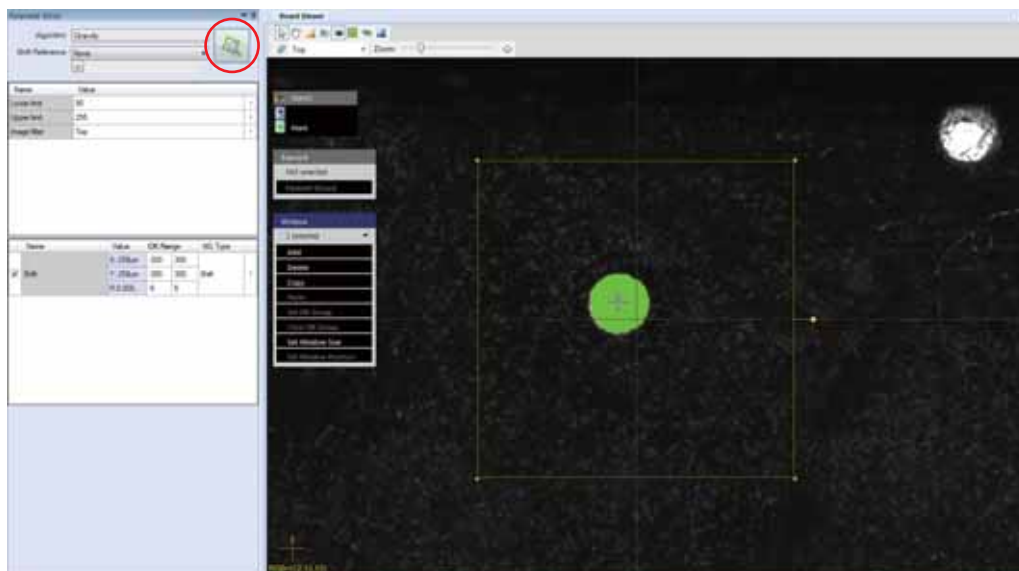


Figure 2-20 Detecting the Fiducial mark

- Step5: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.
- Step6: Click  and check if the inspection is completed properly.

2.6 Image Match

2.6.1 Overview of Inspection

Image Match is the algorithm to perform inspections with the same process as Image matching and used for inspecting a fiducial mark by Image Match.

2.6.2 Setting of Parameter



Figure 2-21 Dialog (Version 1.4.0.0)

Parameter	Description
Image filter	Select a lighting that displays the target clearly.
Matching exclusion area setting	Select Ignore Transparent to use the filter function or Fast to not use the filter.
Option	Select Normal .
Train	Saves the image of the specified area as the template image.
Edit	Changes the Filter settings.
Template	Displays the template image.

Table 2-13 Parameter

Parameter		Description
Shift	X : - μm	Specifies the acceptable range of the X-axis distance between the center of the detected image and the center of the inspection window.
	Y : - μm	Specifies the acceptable range of the Y-axis distance between the center of the detected image and the center of the inspection window.
	R : -deg	Specifies the acceptable range of the detected image angle. Angular value increases clockwise and decreases counter-clockwise.
Match Rate		Specifies the acceptable range of the matching rate.

Table 2-14 OK Range Setting

2.6.3 Setting Procedure

- Step1: Select **Image Match** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the inspection target clearly from the **Image Filter** drop-down list.
- Step3: Adjust the inspection window to match with the fiducial mark.

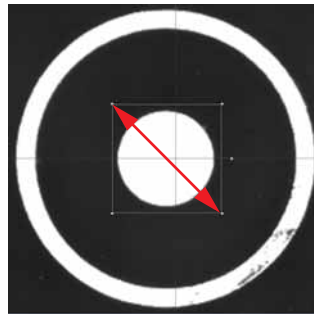


Figure 2-22 Adjusting the Inspection Window

- Step4: Click **Train** on the lower side of **Parameter Editor**.
- Step5: Specify the area to take the image and click **OK**.
The image taken in the specified area is displayed on **Template**.

NOTE

Window angles can be changed by dragging the circular mark on the right side of the window.

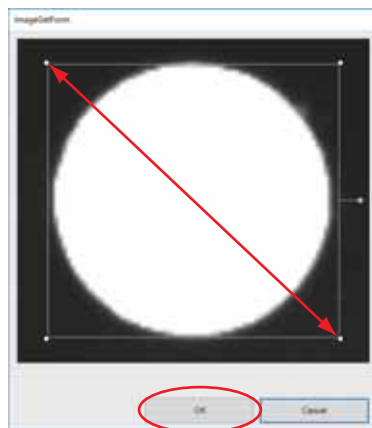


Figure 2-23 Loading the Template Image

Step6: Set the **Matching exclusion area setting** to **Ignore Transparent** to use the filter function and click **Edit** on the right of the **Train**.

With **Matching exclusion area setting** selected in the upper right, drag the section where the filter is to be applied.

The blue and black checkered area is excluded from the inspection.

After all the settings are completed, click **OK**.

NOTE This setting is effective for reducing noise effects.

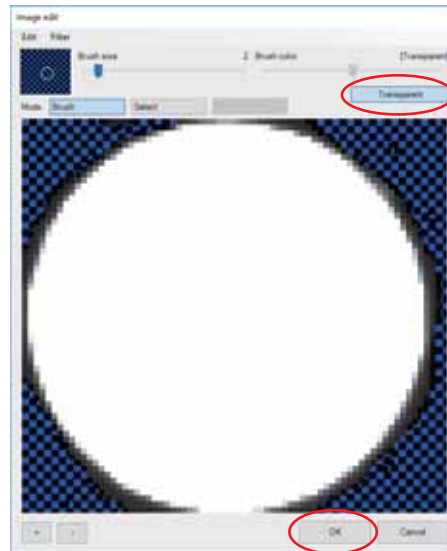


Figure 2-24 Image Filter Function

Step7: Click  and check if the inspection target is detected.

When the target is detected, the inspection window turns to green and the detected area is surrounded by the yellow dotted lines.

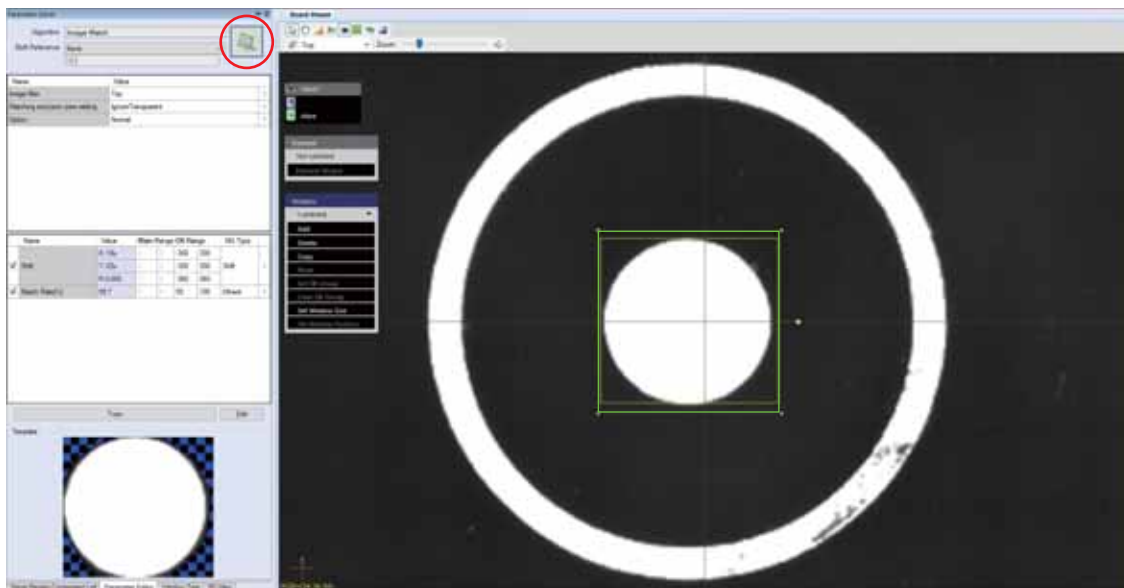


Figure 2-25 Detecting the Inspection Target

Step8: Specify **OK Range** of the sample value and set **NG Type** for the time when each **OK Range** and each sample value is judged as NG.

NOTE

If not performing the inspection, uncheck the item.

Step9: Click  and check if the inspection is completed properly.

3 Barcode Inspection

3.1 1D Code

3.1.1 Overview of Inspection

1D Code is the algorithm to recognize 1D barcodes.

The algorithm registers the barcode of the good PCB to compare with the barcode of the scanned PCB.

1D Code is suitable for traceability of PCBs with barcodes.

3.1.2 Setting of Parameter

Parameter Editor

Algorithm: 1D Code

Shift Reference: None

Name	Value
Image filter	Side color
Type	Code39
element_size_min	1.5
element_size_max	32
meas_thresh	0.05
meas_thresh_abs	5
max_diff_orient	10
Advanced Mode	Disable
Mask Height	3
Mask Width	4
Threshold Min	0
Threshold Max	160
Deconvolution filter	Disable
Blur amount[um]	200
Smooth[%]	60
Horizon filter mar...	10
Vertical filter marg...	10

	Name	Value	OK Range	NG Type
☒	ResultText	-		-

Figure 3-1 Dialog (Version 2.0.0.0)

Parameter	Description
Image filter	Select a lighting that displays the barcode clearly.
Type	Select a barcode type to be recognized.
element_size_min	The item is for Saki engineers only. Do not change the setting.
element_size_max	
meas_thresh	
meas_thresh_abs	
max_diff_orient	
Advanced Mode	
Mask Height	
Mask Width	
Threshold Min	
Threshold Max	

Parameter	Description
Deconvolution filter	Inspects with using deconvolution filter. Inspection is performed with the image inside of pink dotted line.
Blur amount[um]	Specifies the blur amount of deconvolution filter.
Smooth[%]	Specifies the parameter to remove noise occurred by deconvolution filter.
Horizon filter margin[%]	Specifies the horizontal margin of inspection area when using deconvolution filter. The margin is displayed as pink dotted line.
Vertical filter margin[%]	Specifies the vertical margin of inspection area when using deconvolution filter. The margin is displayed as pink dotted line.

Table 3-1 Parameter

Parameter	Description
Result Text	Displays the result of character recognition and registration.

Table 3-2 OK Range Setting

3.1.3 Setting Procedure

- Step1: Select **1D Code** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the barcode clearly from the **Image filter** drop-down list.
- Step3: Select the barcode type to be recognized from the **Type** drop-down list.
- Step4: Adjust the inspection window size to surround the barcode and move the yellow point beside the inspection window to the end of the barcode by mouse-dragging.



Figure 3-2 Adjusting the Inspection Window Angle

Step5: Click the **OK Range** cell.

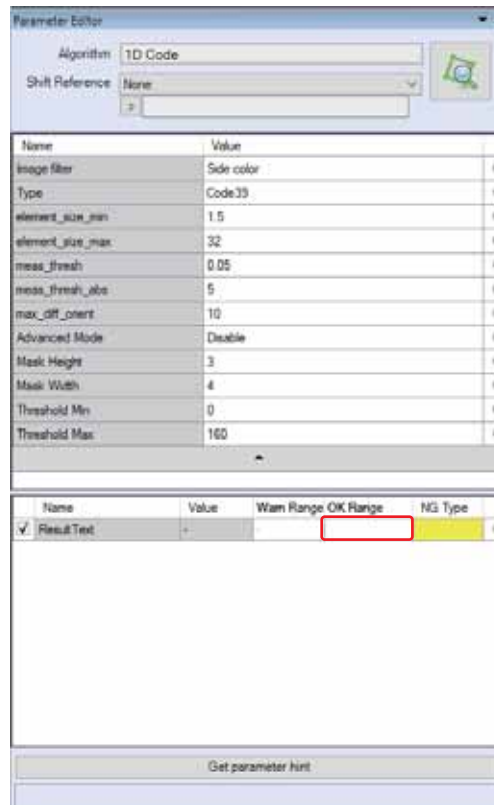


Figure 3-3 OK Range

Step6: Enter the characters to be registered into **Match what** and click  on the upper right side of the dialog.

NOTE

In **Match what**, wild cards can be used when **Use regular expressions** is not checked. Available wild cards are as follows.

“*”: Matches any characters which length is 0 or more.

e.g., “a*c” matches to “ac”, “aaac”, “abac” etc.

“?”: Matches any single character.

e.g., “a?c” matches to “aac”, “a1c”, “a-c” etc.

“!”: Matches any single number.

e.g., “a!c” matches to “a1c”, “a2c”, “a3c” etc.

“\$”: Matches any single alphabet.

e.g., “a\$c” matches to “aac”, “abc”, “acc” etc.

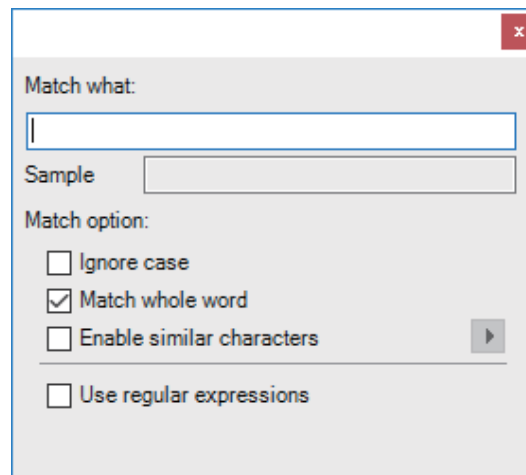


Figure 3-4 Match What

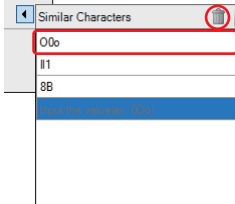
Item	Description
Ignore case	When checked, ignores upper and lower cases.
Match whole word	When checked, the inspection result is OK only when the characters completely matched with the registered characters. e.g., When 23 is entered in Match what and 1234 is detected, the inspection result becomes NG if Match whole word is checked. If Match whole word is unchecked, the inspection result becomes OK.
Enable similar characters	Use this item when the algorithm misrecognizes the characters which have the similar forms such as 1 (one) and l or 0 (zero) and O. When checked, the inspection result becomes OK whichever character is recognized. Click  to set similar characters in the dialog shown in Figure below.  Figure 3-5 Similar Characters Setting Enter similar characters with parenthesis into the text-box and click Enter . e.g., To register 0 and O, enter (OO). To delete the registered similar characters, click the recycle bin button  .
Use regular expressions	When checked, regular expressions can be used for recognizing. e.g., When 123[0-9] is entered in Match what , the inspection result becomes OK whichever numbers from 1230 to 1239 is detected.

Table 3-3 Descriptions

Step7: Set **NG Type** for the time when Sample value becomes NG.

Step8: Click  and check if the inspection is completed properly.



Figure 3-6 Confirming the Result

3.1.4 Deconvolution Filter

By using deconvolution filter, the blur of image can be removed.

This image filter is effective when the barcode is on shield and which is out of focus.

CAUTION

Deconvolution filter needs much calculation, so inspection time will be longer. It is recommended to enable this only the inspection which needs deconvolution filter.

Step1: Select **Enable** from **Deconvolution filter** drop-down list.

Step2: Set **Blur amount[μm]** and **Smooth[%]** as the image is displayed clearly.

NOTE

Set **Horizon filter margin[%]** and **Vertical filter margin[%]** and adjust inspection are if the noise is generated in the corner of image by deconvolution filter, and that effects inspection. Inspection is performed with the image inside of pink dotted line.

Step3: Complete the rest of settings and perform the inspection by reference to Step5 to Step8 of **3.1.3 Setting Procedure**.

3.2 2D Code

3.2.1 Overview of Inspection

2D Code is the algorithm to recognize 2D barcodes.

The algorithm registers the barcode of the good PCB to accord with the barcode of the scanned PCB.

2D Code is suitable for traceability of PCBs with barcodes.

3.2.2 Setting of Parameter

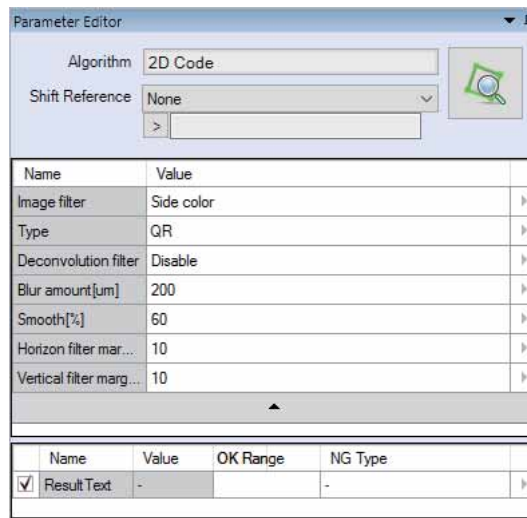


Figure 3-7 Dialog (Version 2.0.0.0)

Parameter	Description
Image filter	Select a lighting that displays the barcode clearly.
Type	Select a barcode type to be recognized.
Deconvolution filter	Inspects with using deconvolution filter. Inspection is performed with the image inside of pink dotted line.
Blur amount[um]	Specifies the blur amount of deconvolution filter.
Smooth[%]	Specifies the parameter to remove noise occurred by deconvolution filter.
Horizon filter margin[%]	Specifies the horizontal margin of inspection area when using deconvolution filter. The margin is displayed as pink dotted line.
Vertical filter margin[%]	Specifies the vertical margin of inspection area when using deconvolution filter. The margin is displayed as pink dotted line.

Table 3-4 Parameter

Parameter	Description
Result Text	Displays the result of character recognition and registration.

Table 3-5 OK Range Setting

3.2.3 Setting Procedure

- Step1: Select **2D Code** from the **Algorithm** drop-down list.
- Step2: Select a lighting that displays the barcode clearly from the **Image filter** drop-down list.
- Step3: Select the barcode type to be recognized from the **Type** drop-down list.
- Step4: Adjust the inspection window size to surround the barcode and adjust the position of the yellow point beside the side of the inspection window by mouse-dragging in accordance with the barcode direction.

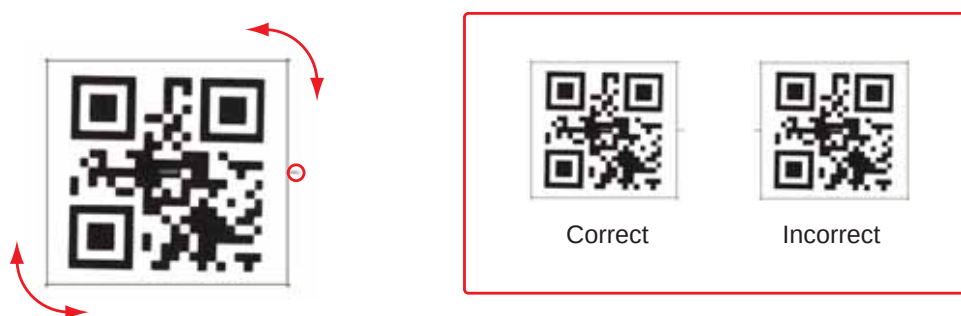


Figure 3-8 Adjusting the Inspection Window Angle

- Step5: Click the **OK Range** cell.

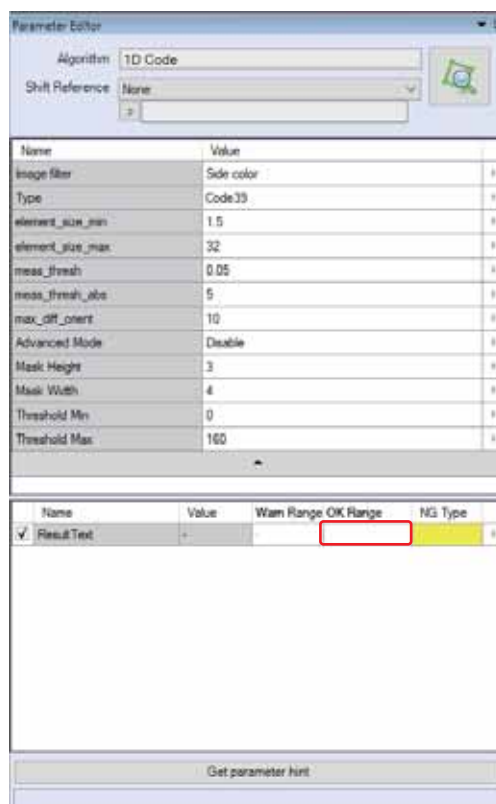


Figure 3-9 OK Range

Step6: Enter the characters to be registered into **Match what** text-box and click  on the upper right side of the dialog.

NOTE

In **Match what**, wild cards can be used when **Use regular expressions** is not checked. Available wild cards are as follows.

“*”: Matches any characters which length is 0 or more.

e.g., “a*c” matches to “ac”, “aac”, “abac” etc.

“?”: Matches any single character.

e.g., “a?c” matches to “aac”, “a1c”, “a-c” etc.

“!”: Matches any single number.

e.g., “a!c” matches to “a1c”, “a2c”, “a3c” etc.

“\$”: Matches any single alphabet.

e.g., “a\$c” matches to “aac”, “abc”, “acc” etc.

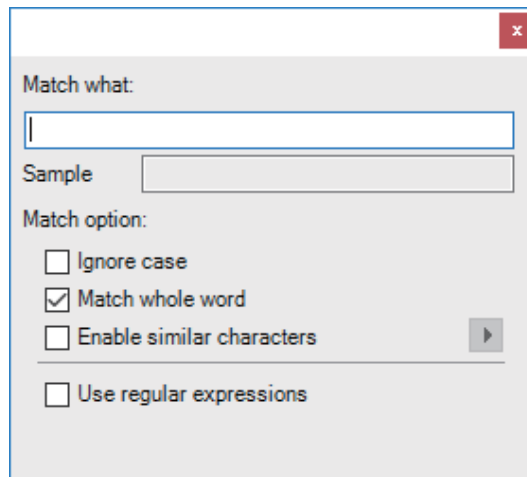


Figure 3-10 Match What

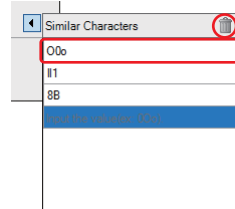
Parameter	Description
Ignore case	When checked, ignores upper and lower cases.
Match whole word	When checked, the inspection result is OK only when the characters completely matched with the registered characters. e.g., When 23 is entered in Match what and 1234 is detected, the inspection result becomes NG if Match whole word is checked. If Match whole word is unchecked, the inspection result becomes OK.
Enable similar characters	Use this item when the algorithm misrecognizes the characters which have the similar forms such as 1 (one) and l or 0 (zero) and O. When checked, the inspection result becomes OK whichever character is recognized. To set similar characters, click  to display the dialog shown in Figure below.  Figure 3-11 Similar Characters Setting Enter similar characters with parenthesis into the text-box and click Enter . e.g., To register 0 and O, enter (0O). To delete the registered similar characters, click the recycle bin button  .
Use regular expressions	When checked, regular expressions can be used for recognizing. e.g., When 123[0-9] is entered in Match what , the inspection result becomes OK whichever number from 1230 to 1239 is recognized.

Table 3-6 Descriptions

Step7: Set **NG Type** for the case where Sample value becomes NG.

Step8: Click  and check if the inspection is completed properly.

3.2.4 Deconvolution Filter

By using deconvolution filter, the blur of image can be removed.

This image filter is effective when the barcode is on shield and which is out of focus.

CAUTION

Deconvolution filter needs much calculation, so inspection time will be longer.
It is recommended to enable this only the inspection which needs deconvolution filter.

Step1: Select **Enable** from **Deconvolution filter** drop-down list.

Step2: Set **Blur amount[μm]** and **Smooth[%]** as the image is displayed clearly.

NOTE

Set **Horizon filter margin[%]** and **Vertical filter margin[%]** and adjust inspection are if the noise is generated in the corner of image by deconvolution filter, and that effects inspection. Inspection is performed with the image inside of pink dotted line.

Step3: Complete the rest of settings and perform the inspection by reference to Step5 to Step8 of **3.1.3 Setting Procedure**.

4 Other Inspections

4.1 Black White

4.1.1 Overview of Inspection

Black White is the algorithm to specify the acceptable range of the brightness level and to do an inspection by calculating the percentage of the area meeting the specified brightness level in the inspection window.

The algorithm judges the inspection result is OK if the percentage is higher than the specified value.

Black White is very simple algorithm and used for objects which cannot be detected in **Solder Paste Inspection** or algorithms for fiducial marks.

4.1.2 Setting of Parameter

Parameter Editor

Algorithm: Black White

Shift Reference: None

Name	Value
Lower limit	80
Upper limit	150
Image filter	Top

	Name	Value	Warn Range	OK Range	NG Type
☒	Sample value[%]	58	0	0	50
			100		Shift

Figure 4-1 Dialog (Version 0.0.0.1)

Parameter	Description
Lower limit	When the NG Brightness Level is Higher than the OK Brightness Level Enter the intermediate value of the OK and NG values to Upper limit . Enter 0 to Lower limit .
Upper limit	
Image filter	When the OK Brightness Level is Higher than the NG Brightness Level Enter 255 to Upper limit . Enter the intermediate value of the OK and NG values to Lower limit . Select a lighting that clearly displays the difference between the OK and NG brightness level. (When using this algorithm to the solder paste inspection, select Top .)

Table 4-1 Parameter

Parameter	Description
Sample value	Specifies the acceptable range of the percentage of the specified brightness level to the inspection window.

Table 4-2 OK Range Setting

4.1.3 Setting Procedure

- Step1: Select **Black White** from the **Algorithm** drop-down list.
- Step2: Select a lighting that clearly displays the difference between the OK and NG brightness level from the **Image filter** drop-down list.
- Step3: Adjust the inspection window size.

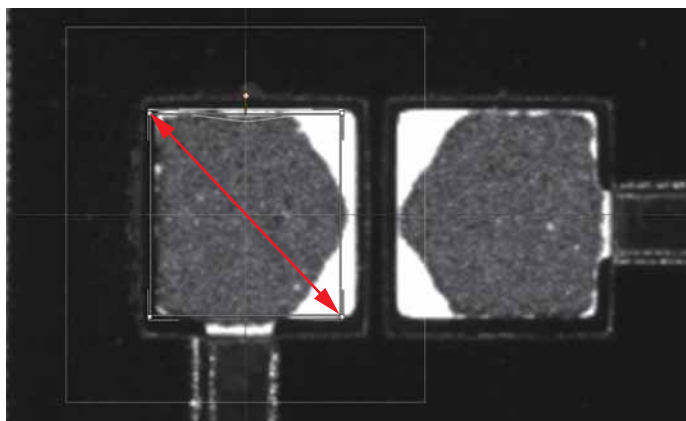


Figure 4-2 Adjusting the Inspection Window

- Step4: Check the values of Lower limit and Upper limit.

If the NG Brightness Level is Higher than the OK Brightness Level

Enter the intermediate value of the OK and NG values to **Upper limit**. Enter **0** to **Lower limit**.

If the OK Brightness Level is Higher than the NG Brightness Level

Enter **255** to **Upper limit**. Enter the intermediate value of the OK and NG values to **Lower limit**.

- Step5: Click  and check if the targeted objects can be detected.

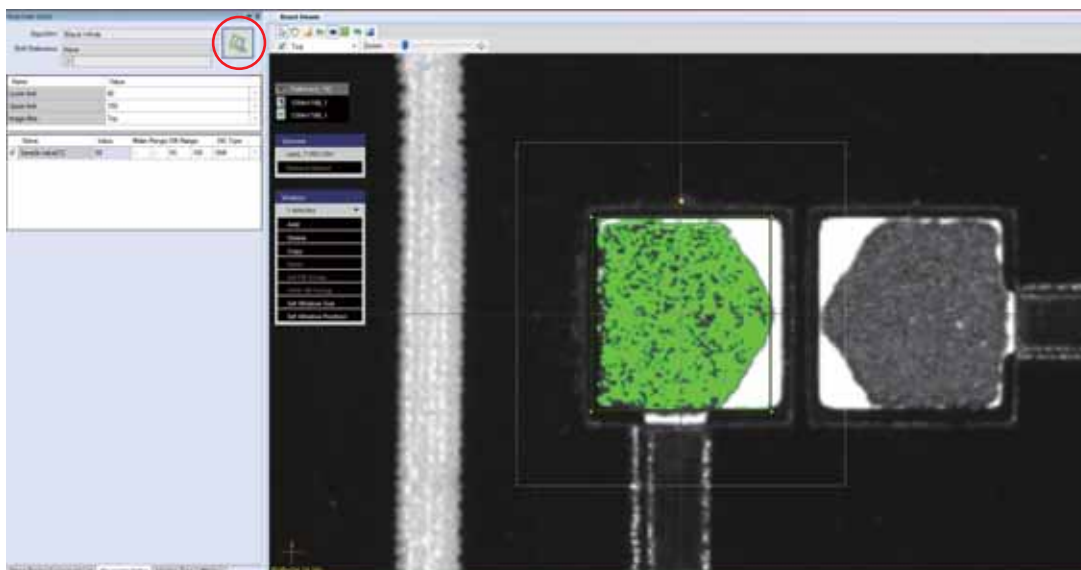


Figure 4-3 Detecting the Component

Step6: Set the acceptable range of the sample value.

NOTE

If not performing the inspection, uncheck the item.

Step7: Click  and check if the inspection is completed properly.

4.2 Jig Height

4.2.1 Overview of Inspection

Jig Height is algorithm to perform the same calculation as Repeatability and Reproducibility test on height jig during the calibration.

Jig Height is used for the recheck of a calibration result and the evaluation and not used for normal inspections.

NOTE To conduct proper measurements, use jig height instead of a PCB.

4.2.2 Setting of Parameter

Name	Value
FovSizeWidth(pix)	2048
HeightCoefCalibPointNumX	4
HeightCoefCalibPointNumY	4
FovSizeHeight(pix)	2048
TopPointStart(%)	20
TopPointEnd(%)	80
BottomPointStart(%)	60
BottomPointEnd(%)	80

Name	Value	Warn Range	OK Range	NG Type
Top Height [um]	246.0	-	-	-
Bottom Height [um]	134.1	-	-	-
Height Difference [um]	111.8	0	0	1000

Figure 4-4 Dialog (Version 0.0.0.1)

Item	Description
FovSizeWidth(pix)	The item is for Saki engineers only. Do not change the setting.
HeightCoefCalib PointNumX	
HeightCoefCalib PointNumY	
FovSizeHeight(pix)	
TopPointStart(%)	
TopPointEnd(%)	
BottomPointStart(%)	
BottomPointEnd(%)	

Table 4-3 Parameter

Parameter	Description
Top Height [um]	Displays the height of the top position detected on jig.
Bottom Height [um]	Displays the height of the bottom position detected on jig.
Height Difference [um]	Displays the difference between Top Height and Bottom Height .

Table 4-4 OK Range Setting

4.2.3 Setting Procedure

Step1: Select **Jig Height** from the **Algorithm** drop-down list.

Step2: When  is clicked and the inspection is performed, the four crosses appear for the targeted object.

Adjust every corner of the inspection window to the mark position.

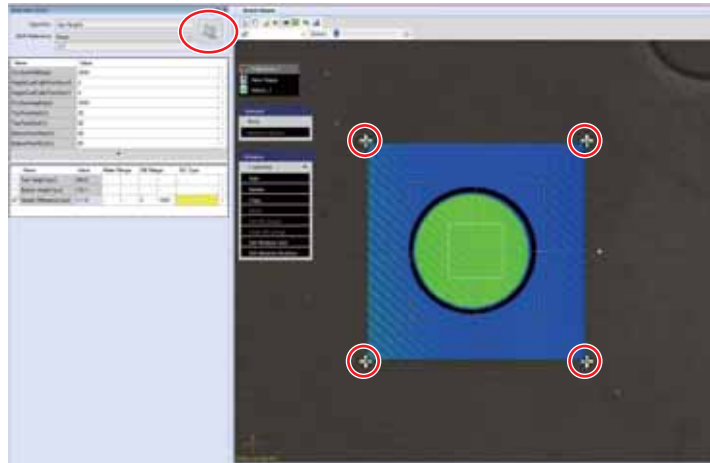


Figure 4-5 Adjusting the Inspection Window Angle

Step3: Click  to check if the measured value is displayed.

4.3 Color Channel

4.3.1 Inspection Overview

Color Channel calculates the ratio of the pre-registered color to the inspection window.

4.3.2 Parameter Setting

Parameter Editor

Algorithm

Color Channel

Shift Reference

None

>

Name	Value
Image Filter	SideColor
Channel Width	30
Channel Range	Inside

	Name	Value	OK Range	NG Type
☒	Area (%)	-	50 100	-

128, 128, 128

128, 128, 128

Figure 4-6 Dialog (Version 1.0.0.0)

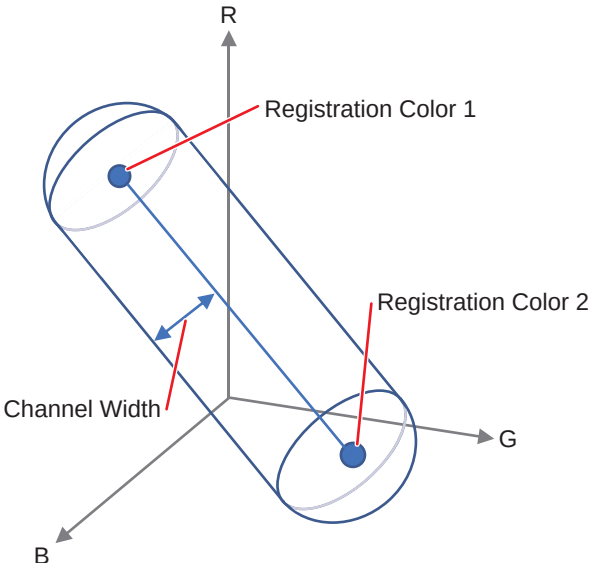
Parameter	Description
Image Filter	Select SideColor .
Channel Width	Acceptable range of color. Increasing the value expands the tolerance for colors that are close to the registration colors.  Figure 4-7 Channel Width
Channel Range	Inside outputs the area percentage within the registration color range. Outside outputs the area percentage out of the registration color range.
Color Registration Button 1	Registers two colors with the Color Registration buttons. A color between two colors is detected.  Figure 4-8 Color Registration Button
Color Registration Button 2	

Table 4-5 Parameter

Parameter	Description
Area (%)	The area percentage that the registered colors occupy for the color in the inspection window. If Channel Range is Outside , this is the area percentage that is not a registered color.

Table 4-6 OK Range Setting

4.3.3 Setting Procedure

Step1: Select **Color Channel** from the **Algorithm** drop-down list.

NOTE

To use same shift information by another algorithm, at the **Shift Reference** select **Body Shift** from the drop down list.

Step2: Select **SideColor** from the **Image Filter** drop-down list.

Step3: Adjust the inspection window size and position.

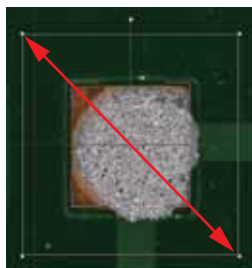


Figure 4-9 Adjusting the Inspection Window

Step4: Click the Registration Color button 1.

Step5: The dialog box shown below appears.

Drag the area to set for the registration color 1.

The average color of the dragged area is registered.



Figure 4-10 Color Registration 1

Step6: Click the button enclosed in the red frame.

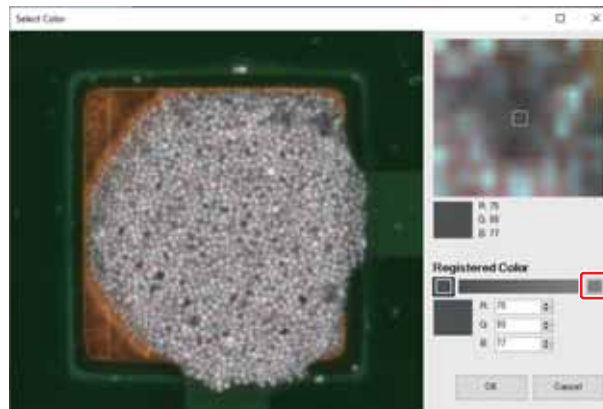


Figure 4-11 Color Registration 2

Step7: Drag the area to set for the registration color 2.
The average color of the dragged area is registered.
After the registration colors 1 and 2 are registered, click **OK**.

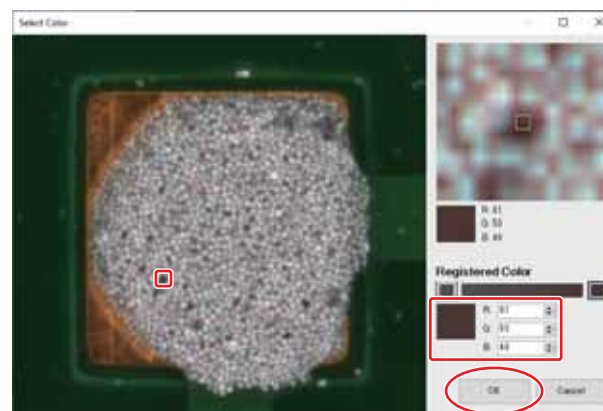


Figure 4-12 Color Registration 3

Step8: **For outputting the area percentage that the color in the inspection window is within the specified registration color range.**
Select **Inside** at **Channel Range**.

For outputting the area percentage that the color in the inspection window is out of the specified registration color range.
Select **Outside** at **Channel Range**.

Step9: Click  and check if the registration colors are detected.

Step10: Specify the tolerances of **Area** and set **NG Type** for when **OK Range** and **Area** are judged as NG.

NOTE If not performing the inspection, uncheck the item.

Step11: Click  and check if the inspection is completed properly.

Auto Inspection Mode and Judgement

1 Auto Inspection

1.1 Auto Inspection

A number of boards are inspected continuously. Even though the judgement result is OK, NG or Warning, the inspection continues. For details, refer to **Instruction Manual**.

1.2 Real-time Judgement Function

1.2.1 Overview

If the inspection result is NG or Warning, the judgement dialog box is displayed. Real-time Judgement Function requires operator's judgement.

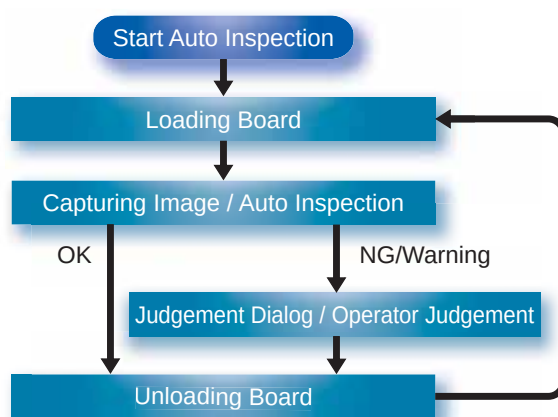


Figure 1-1 Flow Chart for Real-time Judgement

1.2.2 Settings of BF2 Software

Here describes how to set the Real-time Judgement Function.

Step1: Click the **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

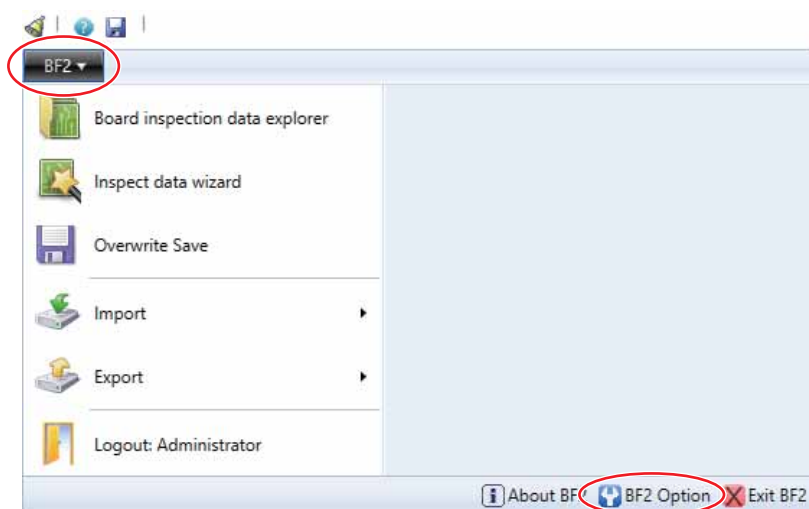


Figure 1-2 Real-time Judgement Function Setting 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

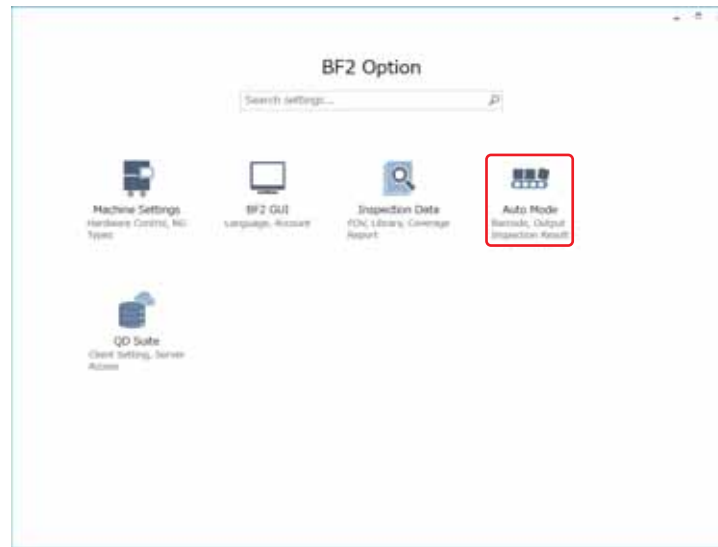


Figure 1-3 Real-time Judgement Function Setting 2

Step3: Select **Output Inspection Result** and set toggle switch of **BF2-Monitor: Real-time Judgment** to **ON**.

NOTE

When using BF2-Monitor used for the operator's repair terminal, or BF2-Monitor for RMS, refer to BF2-Monitor Manual.

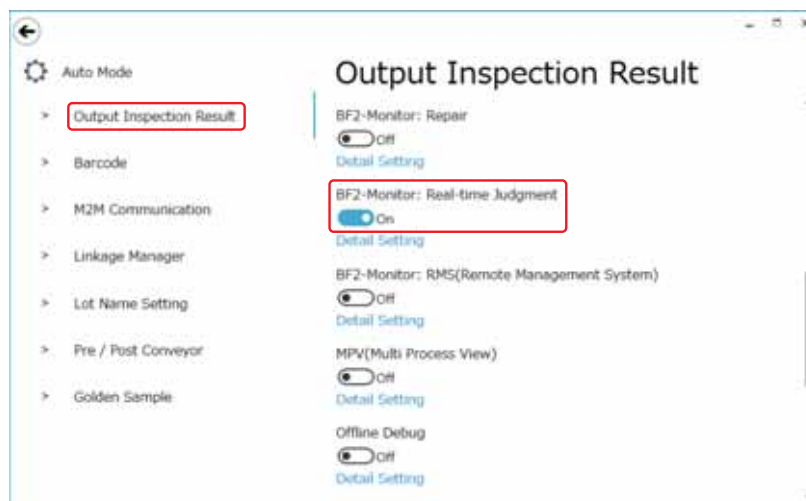


Figure 1-4 Real-time Judgement Function Setting 3

Step4: Click **Detail Setting** under the toggle switch of **BF2-Monitor: Real-time Judgment** to set detailed items.

Proceed the setup procedures, if necessary. Refer to the table below for detail settings.

After all the settings are completed, click on the upper right side of the dialog box and click **Yes** to close dialog box.

Item	Description
Local BF2-Monitor path	Specify the path to execute file of BF2-Monitor installed in the inspection machine. Default path is C:\BF2-Monitor\BF2-Monitor.exe .
Result data save location (for Real-time Judgment)	Specify the inspection result file location of real-time judgement. It is recommended to output the files located just below the C drive. e.g. C:\BF2MonitorData_RTJ
Reference data save location (for Real-time Judgment)	Specify the reference images and OK images location of real-time judgement. For detail of reference images, refer to 1.2.4 Reference Image Register Settings . It is recommended to output the files located just below the C drive. e.g. C:\BF2ReferenceData_RTJ
Maximum number of image file per board	If the number of NG windows on the same board exceeds a specified value, images at the portion of exceeded value will not be output. Default value is 50 .
Maximum number of image file per component	If the number of NG windows on the same pad exceeds a specified value, NG window image at the portion of exceeded value will not be output. Default value is 50 .
Output inspection light Image	Outputs inspection images to BF2-Monitor.
Output NG as unjudged	This setting is valid only when using the combination of inspection result output to BF2-Monitor for Repair Terminal and Real-Time Judgement. As for a board judged as NG in Real-time Judgement, when the operator requires the judgement on BF2-Monitor for Repair Terminal again, check this. In order to use this setting, select BF2 Option > Output Inspection Result and click Detail Setting under the toggle switch of BF2-Monitor: Repair . And check Output OK data as operator judgment done .

Table 1-1 Descriptions

1.2.3 BF2-Monitor Settings

Here describes how to set the BF2-Monitor.

The real-time judgement uses some of the BF2-Monitor's functions.

NOTE To start BF2-Monitor on the machine, BF2 must be started.

Step1: Complete BF2 settings according to **1.2.2 Settings of BF2 Software**.

Step2: In the **Auto** tab, select **BF2-Monitor**.

NOTE **BF2-Monitor** does not appear in the ribbon of BF2-Editor.

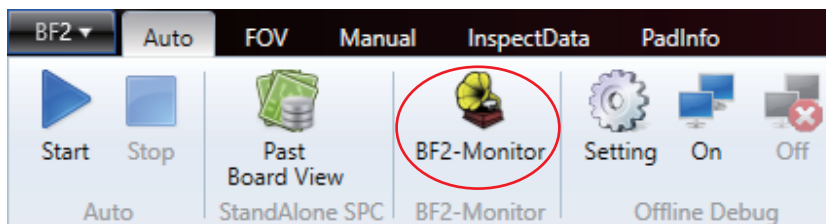


Figure 1-5 BF2-Monitor Start 1

Step3: If BF2-Monitor is not already started, the dialog box shown below appears. Click **OK**.

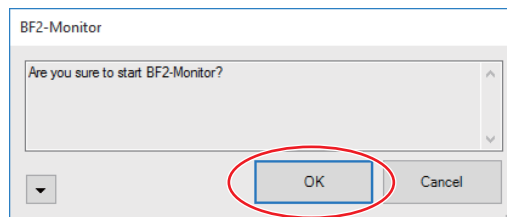


Figure 1-6 BF2-Monitor Start 2

Step4: Right-click the BF2-Monitor icon on the task bar and Select **Options**.

NOTE When the real-time judgement function is set to ON and the Auto operation is started, **BF2-Monitor** is started automatically. However, the right-click menu of BF2-Monitor is restricted.

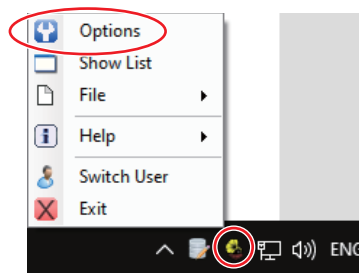


Figure 1-7 BF2-Monitor Start 3

Step5: The dialog box shown below appears.
Click **Judgement Option**.

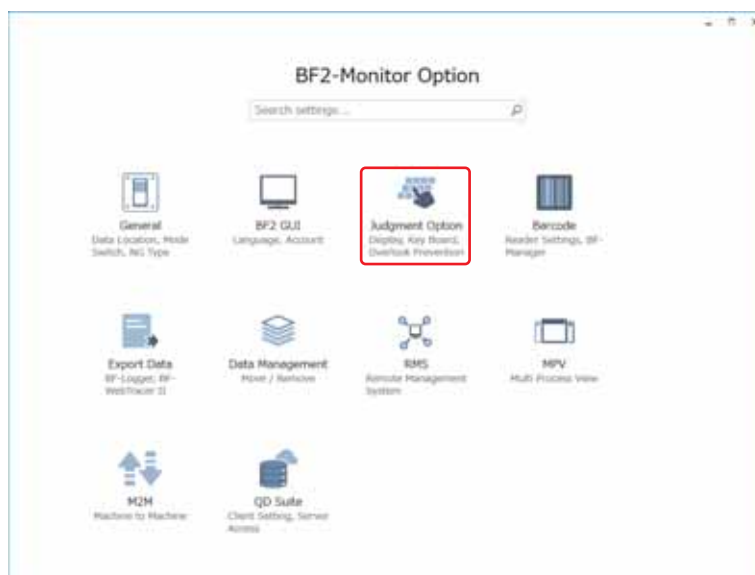


Figure 1-8 BF2-Monitor Start 4

Step6: Conduct the display setting for 3D image displayed on the judgement dialog box as follows.
Check **Use Expected Cube** of **Display > Judgment Window**.

NOTE Also, 3D images can be colored by the height and displayed. To set them, check **Use Height Color on 3D**.



Figure 1-9 BF2-Monitor Start 5

Step7: Conduct the inspected result display setting for the judgement dialog box as follows.
Select the drop down list to set the inspected result display in **Hard/Soft Limit mode** of **Display (*1) > Judgment Window**. For detail settings refer to table below.
In addition, set the **Override Password (*2)**.

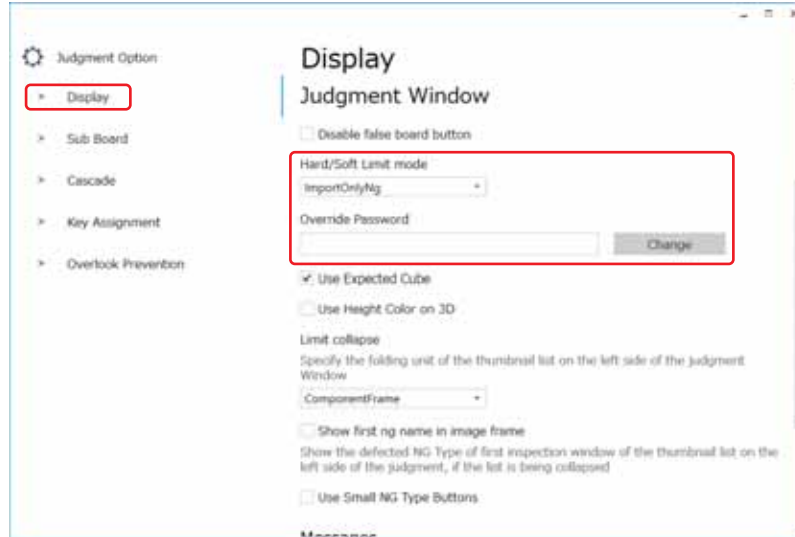


Figure 1-10 BF2-Monitor Start 6

Item	Description
ImportOnlyNg	Displays only the NG pad judged by an inspection machine. Normally, this mode is default.
JudgeAll	Displays the pad which was judged as NG and Warning by and inspection machine, for the operator.
JudgeOnlyWarnings	Displays the pad which was judged as NG and Warning by and inspection machine, for the operator. The False button is disabled. The False button is enabled by the password set to Override Password .
JudgeOnlyWarnings_UserAccountMode	This is the same as the above judgement, however, the False button is always effective in the administrator privilege settings.

Table 1-2 Hard/Soft Limit Setting

NOTE

- (*1) **Hard/Soft Limit mode** specifies whether the operator (soft) judges the pads again after the machine (hard) judged them as NG.
- (*2) **Override Password** is used for unlocking the **False** button.
In the NG pads, the False button is locked as the figure below during the use of **JudgeOnlyWarnings** or **JudgeOnlyWarnings_UserAccountMode**.
Click  button and enter the password set in **Override Password** to unlock.

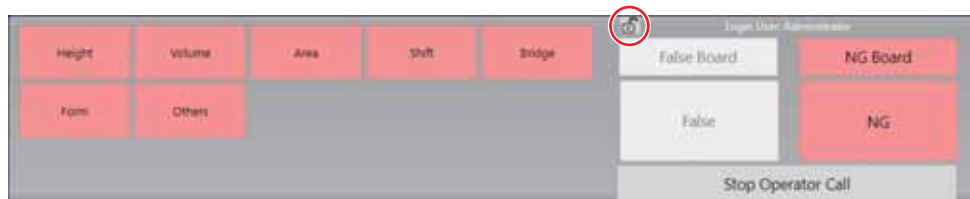


Figure 1-11 Hard/Soft Limit - Lock of False Button

1.2.4 Reference Image Register Settings

Here describes how to register a reference image displayed in the BF2-Monitor.

A registered reference image is transferred to BF2-Monitor and is displayed in the lower left side of the window.

Comparing with pad images on the upper side of a judgement window or 3D images makes it easier for an operator to make a visual judgement.

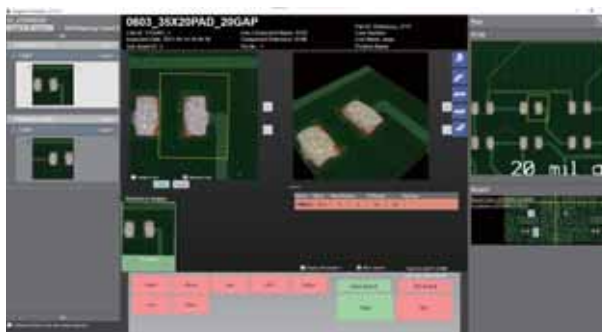


Figure 1-12 Reference Image Register 1

Step1: Open the Recipe editing screen of the pad to register a reference image in the inspection machine.

Step2: Click  for an inspection item to register the reference image as shown below and select **Recipe notes**.

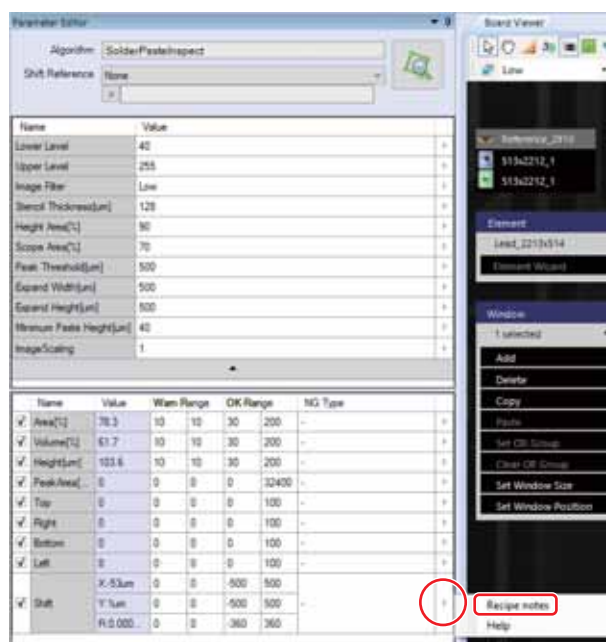


Figure 1-13 Reference Image Register 2

Step3: The dialog box shown below appears.
 Make the setting according to Table 1-3.
 After all settings are completed, click .

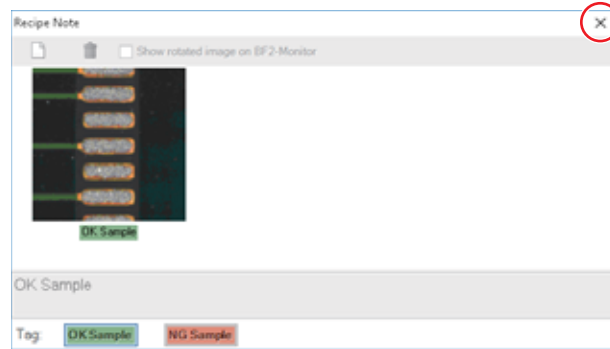


Figure 1-14 Reference Image Register 3

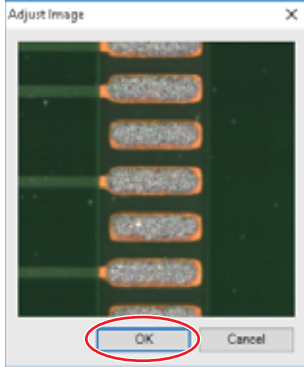
Item	Description
	Registers a new reference image. The dialog box shown in Figure 1-15 appears. Adjust the position of an image to register and click OK. NOTE The position can be adjusted by dragging the mouse and the magnification/reduction rate can be adjusted by scrolling the mouse wheel. 
	Deletes reference images which have been registered.
Show rotated image on BF2-Monitor	A reference image is displayed on the BF2-Monitor according to a rotation angle.
Text box	Entered content is displayed in the lower side of the reference image of BF2-Monitor.
Tag	Changes the frame color of the reference image in the BF2-Monitor. When registering OK images, select OK Sample. When registering NG images, select NG Sample. As for OK sample, frame color turns green. As for NG sample, frame color turns red. Default tag is OK Sample.

Table 1-3 Reference Image Register

1.2.5 Operation Procedure

Here describes how to operate the Real-time Judgement Function.

Step1: Make the setting according to **1.2.2 Settings of BF2 Software** and **1.2.3 BF2-Monitor Settings**.

Step2: Starts Auto inspection. For details, refer to **Instruction Manual**.

Step3: Auto inspection result is NG or Warning, buzzer is sounded and the **JudgementWindow** shown below appears. Click **Stop Operator Call** to stop buzzer sound. Judge the solder inspection result based on displaying information. For details of each items, refer to Table 1-4.

NOTE When the judgement result is OK, Auto inspection is continuing.

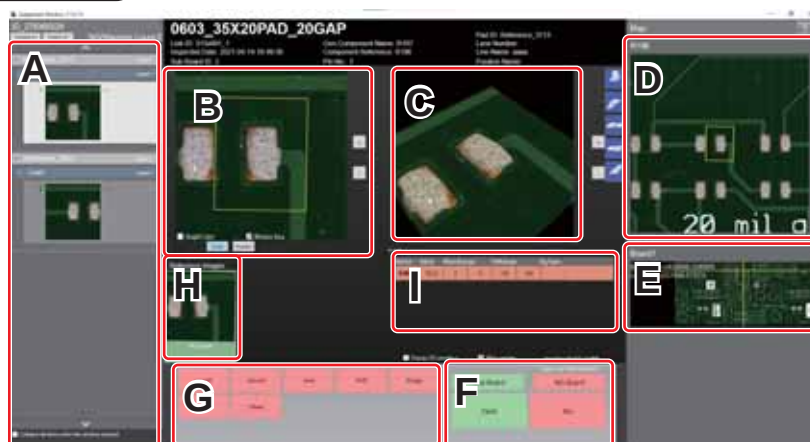


Figure 1-16 Judgement Window

Item	Description
A	Displays the NG pads.
B	Displays the 2D image of inspection window judged as NG.
C	Displays the 3D image of inspection window judged as NG. Changes the displaying angle of solder paste by mouseclicking or buttons on the right side.
D	Displays the position of displaying inspection window by yellow frame. Click to enlarge the image.
E	Displays the position of the board that displaying component. Click to enlarge the image.
F	Judgement button
G	Judgement button for NG Type
H	Displays the reference images.
I	Displays the sample value and judgement range of an inspection window.

Table 1-4 Descriptions

Step4: After judgement is completed, back to Auto inspection. For capturing by the Auto inspection, refer to **Instruction Manual**.

1.3 Remove Board Setting

Remove Board Setting is the function when the judgment result is NG or Warning, the board will stop at the PCB exit position of the machine and operator check is required.

1.3.1 Setting Procedure

Here describes how to set the **Remove Board Setting**.

Step1: Click the **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

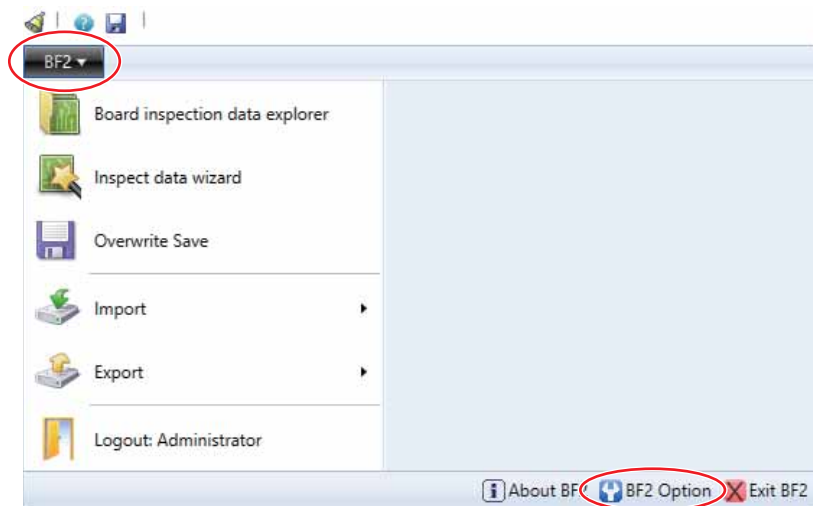


Figure 1-17 Remove Board Setting Procedure 1

Step2: The dialog box shown below appears.
Click **Machine Settings**.

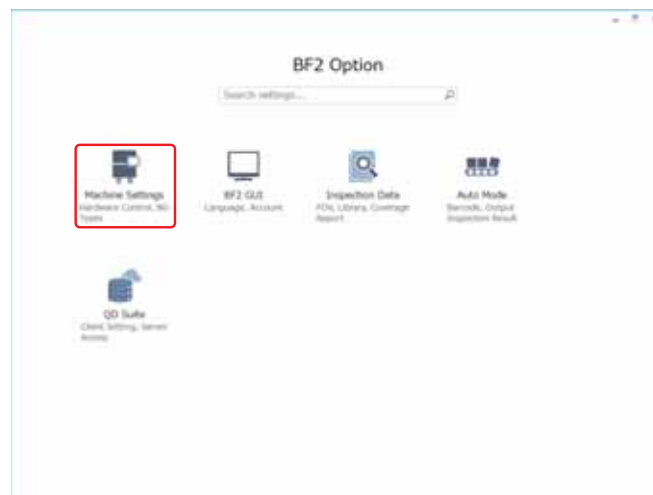


Figure 1-18 Remove Board Setting Procedure 2

Step3: Select **Machine > Hardware Control**. Select the **Sequence_ManualEjectOnNg.xml** in **Sequence configuration file**.

Select **Exit** or **Entrance** at **Board removal position**.

After all the settings are completed, click ☐ on the upper right side of the dialog box and click **Yes** to close dialog box.

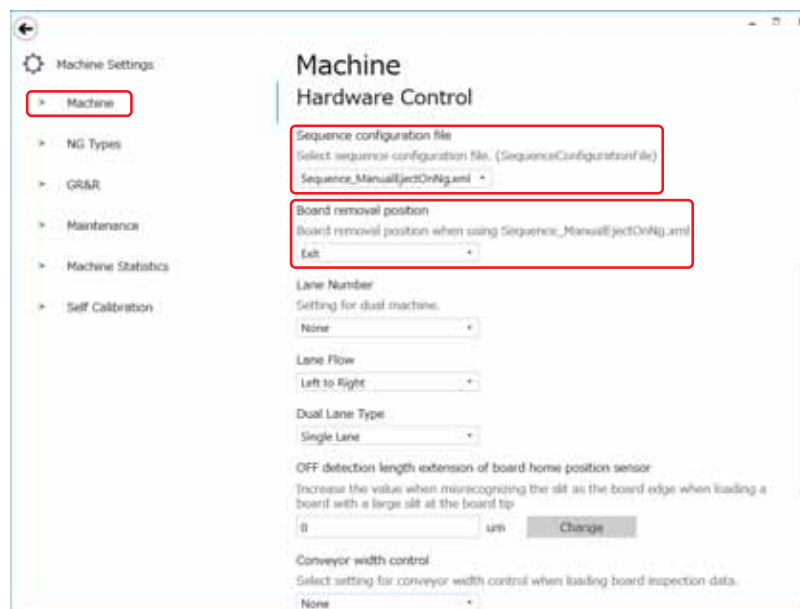


Figure 1-19 Remove Board Setting Procedure 3

1.3.2 Operating Procedure

Here describes how to operate **Remove Board Setting**.

Step1: Make the setting according to **1.3.1 Setting Procedure**.

Step2: Start Auto inspection. To start Auto inspection, refer to **Instruction Manual**.

Step3: Auto inspection result is NG or Warning, the dialog box shown in Figure 1-20 appears. The board is moved to PCB exit or entrance position of the machine. Remove the board manually and click **OK**.

NOTE

Click **OK** without removing the Board, and the dialog box shown in Figure 1-21 will appear.

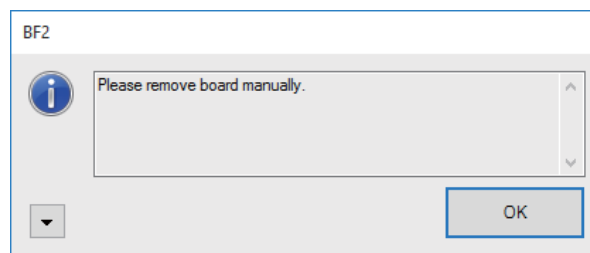


Figure 1-20 Remove Board Setting 1

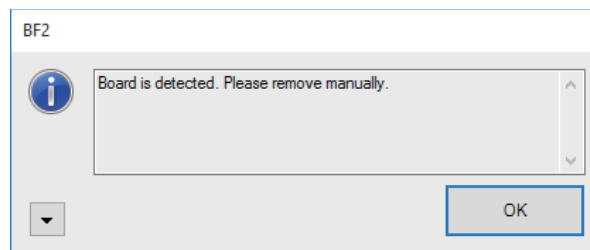


Figure 1-21 Remove Board Setting 2

Step4: Restart Auto inspection.

2 Warning Alarm Function

2.1 Warning Alarm Function

Warning Alarm function notifies you of that the situation during Auto Mode is satisfied with specified condition.

2.1.1 Overview of Function

Warning Alarm Function uses Warning Control Window (*) displayed on the **Auto** tab.

Warning Control Window shows the Warning Alarm Condition and notifies the detail information of warning when the alarm goes off.

NOTE

(*) To display Warning Control Window, the setting on BF2 Option is necessary.
For details, refer to **2.1.3 Setting Method**.

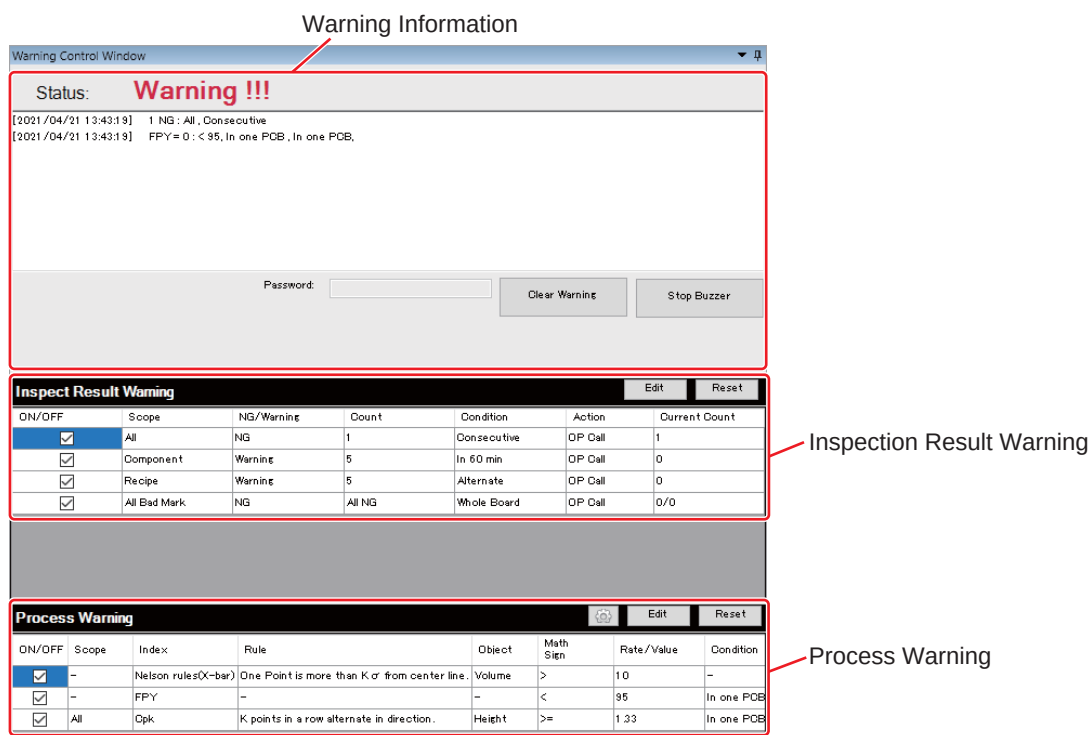


Figure 2-1 Warning Control Window

Item	Description
Warning Information	When warning alarm goes off, the warning condition and the judged time are displayed. The warning log is reset by clicking Clear Warning . Please enter password when it is set in BF2 Option. For details, refer to 2.1.3 Setting Method .
Inspection Result Warning	Displays the warning condition which is set for inspection result. The condition becomes invalid when ON/OFF is unchecked. Set the warning condition by clicking Edit . For details, refer to 2.1.2 Warning Alarm Conditions . The current Count is reset by clicking Reset .
Process Warning	Displays the warning conditions which is set for statistics processing. The condition becomes invalid when ON/OFF is unchecked. Define OK Range of Cp, Cpk, GR, and R by clicking  , and set the warning condition by clicking Edit . For details, refer to 2.1.2 Warning Alarm Conditions . To reset current Rate/Value , click the Reset .

Table 2-1 Descriptions

2.1.2 Warning Alarm Conditions

Here describe the warning alarm conditions.

The warning alarm conditions can be set on the dialog box shown below.

To display the dialog box, click the **Edit** button on the Warning Control Window shown in Figure 2-1.

Inspect Result Warning

The dialog box titled "Inspect Result Warning" contains a table with the following data:

	ON/OFF	Scope	NG/Warning	Count	Condition	Action
1	☒	All	NG	1	Consecutive	OP Call
2	☒	Component	Warning	5	In 60 min	OP Call
3	☒	Recipe	Warning	5	Alternate	OP Call
4	☒	All	NG	20	In one PCB	OP Call
* 5	☐					

Below the table is a large empty text area, a "Delete" button, and "OK" and "Cancel" buttons at the bottom right.

Figure 2-2 Warning Alarm Condition

Item	Description
ON/OFF	Check to enable the warning condition.
Scope	Select the scope to be judged from All , Component , Recipe , and All Bad Mark . The judgment for warning alarm is performed in in this scope.
NG/Warning	Select the judgment result to be specified from NG or Warning .
Count	Input the count number until the warning alarm occurring.
Condition	Specify the condition to go off the warning alarm. The warning alarm goes off when NG is occurred by the number set in Count during set condition. Item: Consecutive / Alternate / In one PCB / In 15 min / In 30 min / In 60 min / In 3 hr / In 6 hr / In 12 hr / In 18 hr / In 24 hr
Action	Specify the action when the warning alarm goes off. Select OP Call or OP Warning .

Table 2-2 Warning Alarm Condition

NOTE

Click the **Delete** button to delete the conditions of the selected row.

Process Warning

Process Warning

Please specify Warning conditions.

	ON/OFF	Scope	Index	Rule	Object	Math Sign	Rate/Value	Condition	Action
1	☒	-	Nelson rules(X-bar)	K points in a row are all within 1 σ .	Volume	>=	10	-	OP Call
2	☒	All	GRR	-	Height	<=	1	In one PCB	OP Call
3	☒	All	Cpk	-	Height	>=	1.33	In one PCB	OP Call
4	☐								

Delete OK Cancel

Figure 2-3 Process Warning

Item	Description
ON/OFF	Check to enable the warning condition.
Scope	Select the scope to be judged from All , Component , or Recipe . The judgement of the warning alarm is conducted within the scope specified here.
Index	Select a targeted index. Item: FPY / Cp / Cpk / GRR / Nelson rules(X-bar) / Nelson rules(S) / Nelson rules(R) / Nelson rules(RS) NOTE X-bar is average. S is standard deviation. R is range (Xmax - Xmin). RS is "average sample value of current board" - "average sample value of previous board".
Rule	Select a rule for triggering warning alarms. This option is only available if Nelson rules is specified in Index . Items: One Point is more than K σ from center line. / K points in a row are on the same side of center line. / K points in a row are continually increasing/decreasing. / K points in a row alternate in direction. / K points out of K + 1 Points in a row are more than 2 σ from center line in the same direction. / K points out of K + 1 Points in a row are more than 1 σ from center line in the same direction. / K points in a row are all within 1 σ . / K points in a row exist with none within 1 σ of center line.
Object	Select a targeted inspection item. Item: Height / Volume / Area / Shift X / Shift Y
Math Sign	Select a mathematical symbol. Item: < / > / <= / >=
Rate/Value	Enter a value to be warned.
Condition	Select conditions of the warning alarm activation. Item: In one PCB
Action	Specified an action at the time when the warning alarm is activated. Select OPCall .

Table 2-3 Process Warning

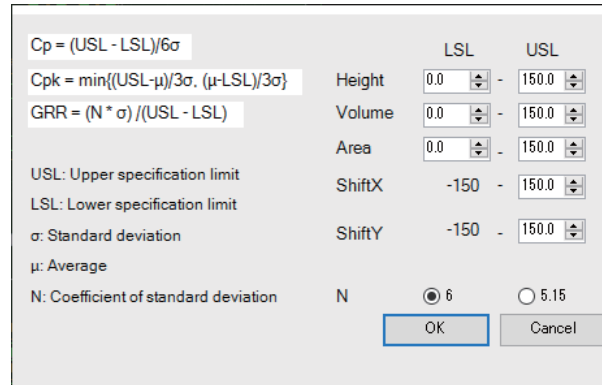
NOTE

Click the **Delete** button to delete the row of the selected conditions.

NOTE

On the statistic parameters setting dialog box, standard values and parameter coefficients can be set. Click the  button on the upper side of Figure 2-1 and click **Statistic params**.

For the details of the dialog box, refer to Figure 2-4.



$Cp = (USL - LSL) / 6\sigma$
 $Cpk = \min\{(USL - \mu) / 3\sigma, (\mu - LSL) / 3\sigma\}$
 $GRR = (N * \sigma) / (USL - LSL)$

USL: Upper specification limit
 LSL: Lower specification limit
 σ : Standard deviation
 μ : Average
 N: Coefficient of standard deviation

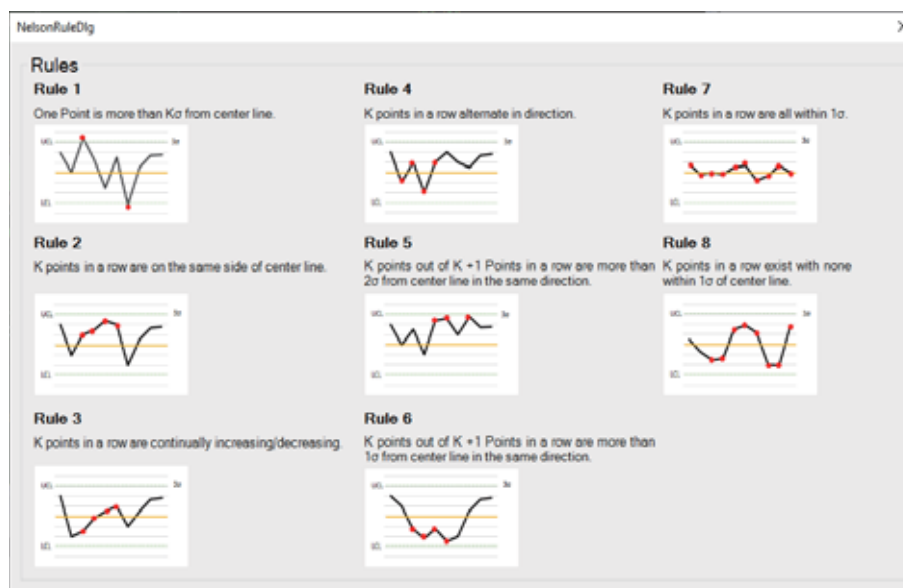
LSL: 0.0 - 150.0
 USL: 150.0
 Height: 0.0 - 150.0
 Volume: 0.0 - 150.0
 Area: 0.0 - 150.0
 ShiftX: -150 - 150.0
 ShiftY: -150 - 150.0
 N: ☒ 6 ☐ 5.15

OK Cancel

Figure 2-4 Statistic Parameters Setting Dialog Box

NOTE

On the Nelson Rule dialog box, **Rules** can be checked. Click  on the upper side of Figure 2-1 and click **Nelson rules**. For details, refer to Figure 2-5.



Rules

Rule 1
One Point is more than $K\sigma$ from center line.

Rule 2
 K points in a row are on the same side of center line.

Rule 3
 K points in a row are continually increasing/decreasing.

Rule 4
 K points in a row alternate in direction.

Rule 5
 K points out of $K + 1$ Points in a row are more than 2σ from center line in the same direction.

Rule 6
 K points out of $K + 1$ Points in a row are more than 1σ from center line in the same direction.

Rule 7
 K points in a row are all within 1σ .

Rule 8
 K points in a row exist with none within 1σ of center line.

Figure 2-5 Nelson Rule Dialog Box

2.1.3 Setting Method

Here describes how to set **Warning Alarm Function**.

Step1: Click the **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

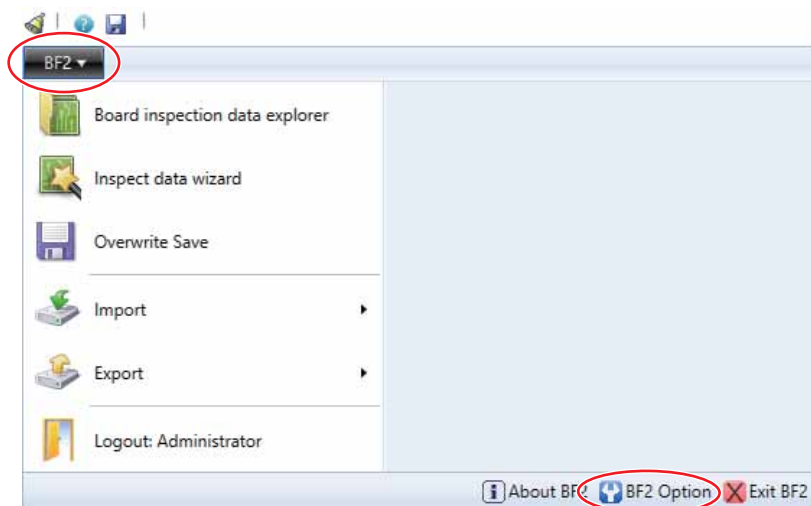


Figure 2-6 Warning Alarm Setting 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

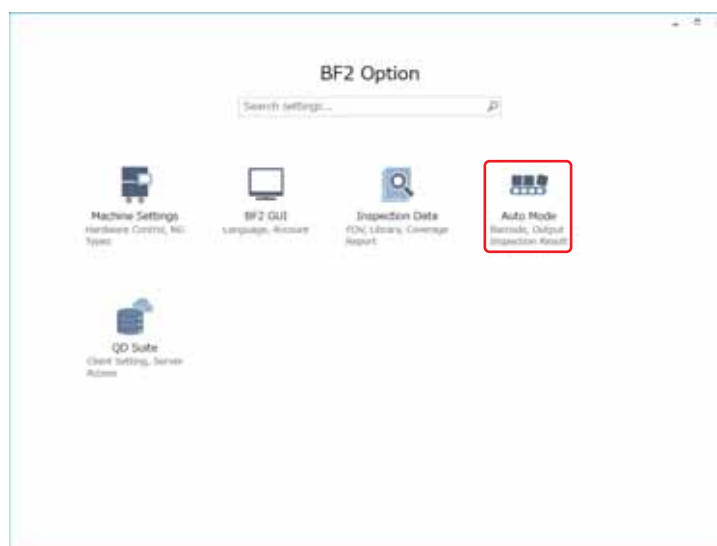


Figure 2-7 Warning Alarm Setting 2

Step3: Select **Output Inspection Result** and set toggle switch of **Warning Alarm** to **ON**.

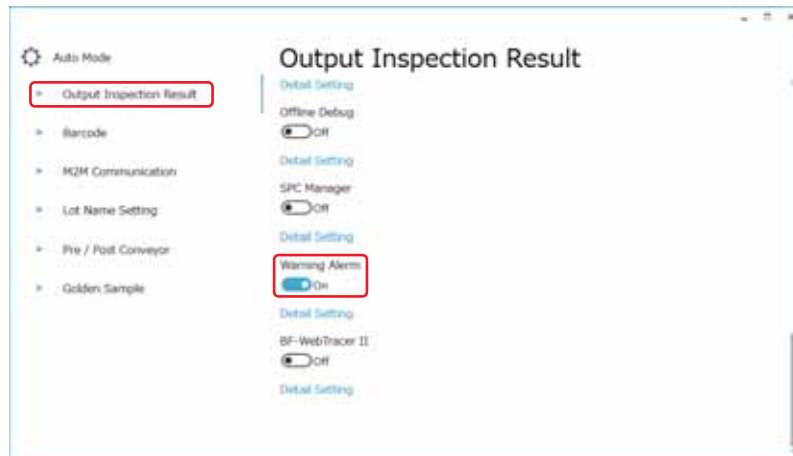


Figure 2-8 Warning Alarm Setting 3

Step4: Click **Detail Setting** under the toggle switch of **Warning Alarm** to set detailed items. Proceed the setup procedures, if necessary. Refer to the table below for detail settings. After all the settings are completed, click  on the upper right side of the dialog box and click **Yes** to close dialog box.

Item	Description
Override Password	Sets the password to call off warning alarm.
Reset count when OK Board	The count is reset when a board is judged as OK.
Stop Auto Mode	Specifies the operation when the warning alarm occurs. Stop Forcibly: Stops Auto Mode when the warning alarm goes off. User Select: The dialog box appears when the warning alarm goes off and the action can be selected. OFF: Auto Mode is not stopped.
Warning interval [sec]	Set the warning interval in seconds. This option is valid only when OP Warning is selected at Acton of Inspect Result Warning .

Table 2-4 Descriptions

Step5: **Warning Control Window** appears on **Auto** tab.

- Step6: Set the Warning Alarm Condition on **Warning Control Window**.
 Click **Edit** on **Inspect Result Warning** and **Process Warning** to set condition.
 For details, refer to **2.1.2 Warning Alarm Conditions**.

Warning Control Window

Status: **Warning !!!**

[2021/04/21 13:43:19] 1 NG: All, Consecutive
 [2021/04/21 13:43:19] FPY=0: < 95, In one PCB, In one PCB,

Password:

Inspect Result Warning

ON/OFF	Scope	NG/Warning	Count	Condition	Action	Current Count
☒	All	NG	1	Consecutive	OP Call	1
☒	Component	Warning	5	In 60 min	OP Call	0
☒	Recipe	Warning	5	Alternate	OP Call	0
☒	All Bad Mark	NG	All NG	Whole Board	OP Call	0/0

Process Warning

ON/OFF	Scope	Index	Rule	Object	Math Sign	Rate/Value	Condition
☒	-	Nelson rules(X-bar)	One Point is more than K σ from center line.	Volume	>	10	-
☒	-	FPY	-	-	<	95	In one PCB
☒	All	Opk	K points in a row alternate in direction.	Height	>=	1.33	In one PCB

Figure 2-9 Warning Control Window

2.2 Warning Alarm Operation

This section describes the operation of **Warning Alarm Function**.

Step1: Start Auto Mode.

Step2: When warning alarm occurs, detailed information is displayed on **Warning Information**.

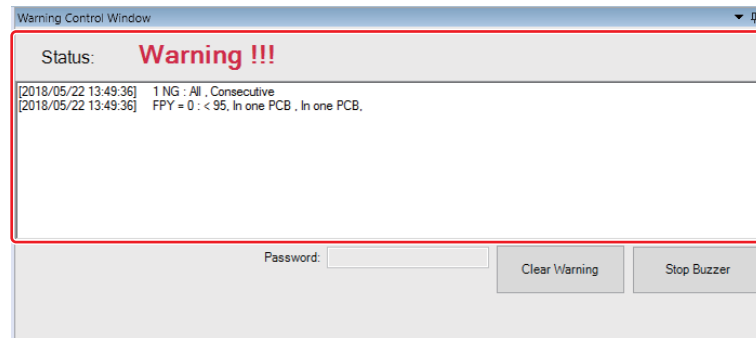


Figure 2-10 Warning Alarm Operation 1

Step3: The next step depends on the **Stop Auto Mode** setting of BF2 Option.

When **Stop Forcibly** is selected, please refer to **2.2.1 Forced Stoppage of Auto Mode**.

When **OFF** is selected, please refer to **2.2.2 Continuation of Auto Mode**.

When **User Select** is selected, please refer to **2.2.3 User Select**.

NOTE For details of **Stop Auto Mode** setting, refer to Step4 of **2.1.3 Setting Method**.

2.2.1 Forced Stoppage of Auto Mode

Step1: If **Stop Forcibly** is selected in **Stop Auto Mode** setting, Auto Mode is interrupted when warning alarm goes off.

Step2: Click **Stop Buzzer** to stop operator call.

Check the details of warning alarm and resolve the problem on production.

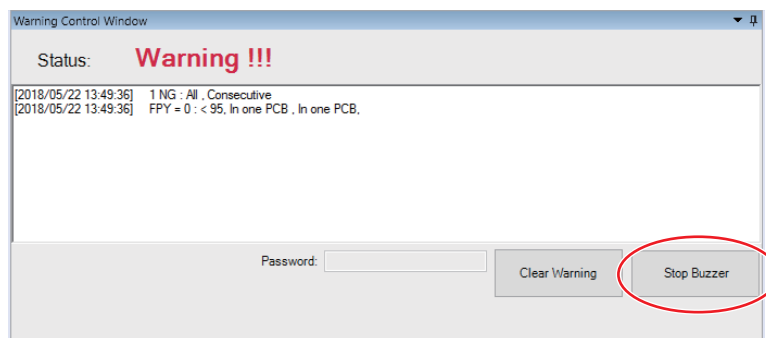


Figure 2-11 Forced Stoppage of Auto Mode 1

Step3: Enter the password which set in **Override Password** setting of BF2 Option, and click **Clear Warning** to clear Warning Alarm log.

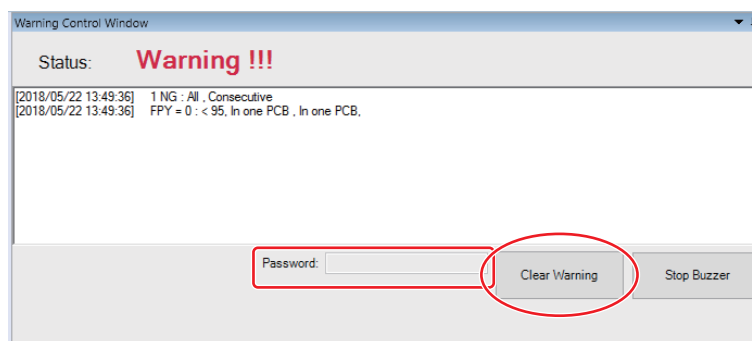


Figure 2-12 Forced Stoppage of Auto Mode 2

2.2.2 Continuation of Auto Mode

If **OFF** is selected in **Stop Auto Mode** setting, Auto Mode is continued.
Warning Log is displayed increasingly.

2.2.3 User Select

If **User Select** is selected in **Stop Auto Mode** setting, the dialog box shown below appears when warning alarm goes off.

Check the details of warning alarm and select the operation.

NOTE To select **Continue**, entering of the password which set in **Override Password** setting of BF2 Option is necessary.
Auto Mode is continued if **Continue** is selected.

NOTE For operation of the case when **Stop Auto Mode** is selected, please refer to **2.2.1 Forced Stoppage of Auto Mode**.

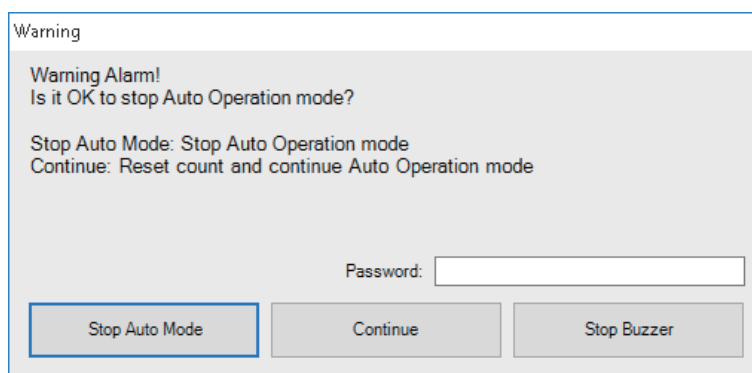


Figure 2-13 User Select

3 Golden Sample

Golden Sample Function can register the scan image and the inspection result based on the scan image together. By registering the sample image and inspection result of perfect board or the board which includes already-known issue in advance, reinspection by using registered sample image is performed at the Auto Mode is started after parameter tuning.

Overlooking because of parameter tuning or false call can be avoided by comparing the inspection result between reinspection and at the registration.

When the inspection results are not matched, it is notified to operator.

NOTE The board without NG is called **Golden Sample**. And the board with already-known NG is called **Silver Sample**.

3.1 Setting of Golden Sample Function

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

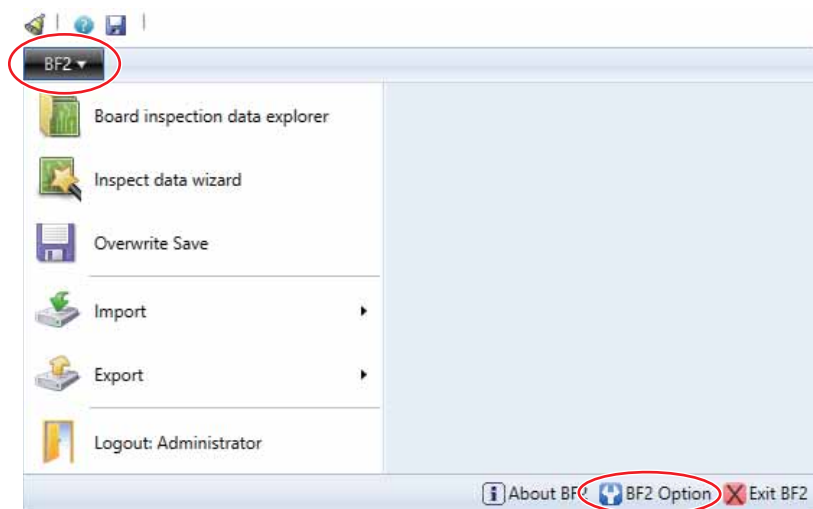


Figure 3-1 Setting of Golden Sample Function 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

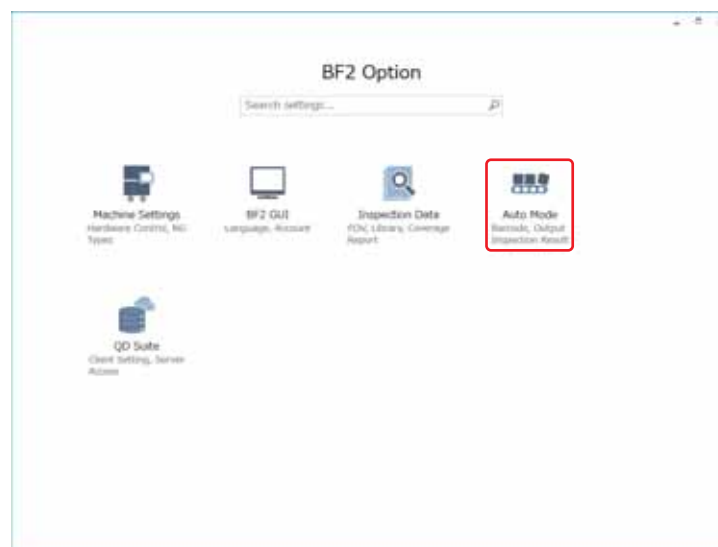


Figure 3-2 Setting of Golden Sample Function 2

Step3: The dialog box shown below appears.

Select **Golden Sample** and check **Enabled**.

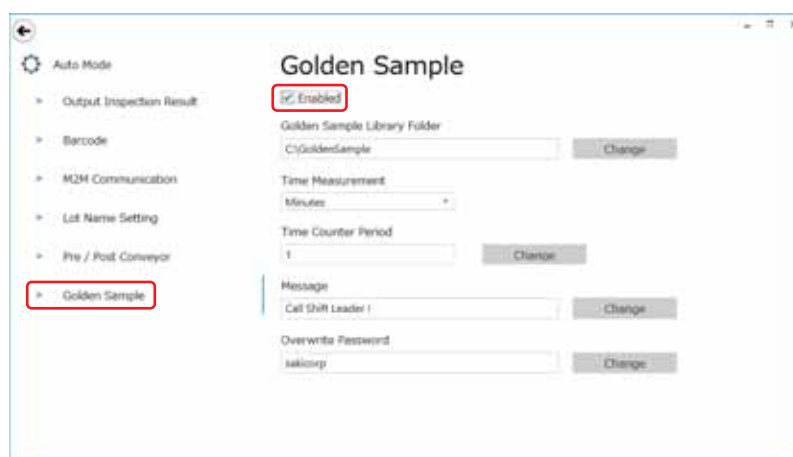


Figure 3-3 Setting of Golden Sample Function 3

Step4: Set detailed setting if necessary. For details refer to Table 3-1.

Item	Description
Golden Sample Library Folder	Sets the path to save the sample image and inspection result of Golden Sample and Silver Sample. Sample images are saved as bdmimg format, Inspection results are saved as gsr format. When the inspection data is shared with some machines, network path can be specified as saving location.
Time Measurement	Golden Sample function reinspect by using the sample image which is registered at the start of Auto Mode. Interval of reinspection can be specified here. Reinspection will not be performed until setting time of Time Measurement and Time Counter Period is past.
Time Counter Period	
Message	When the inspection results of reinspection and at the registration are not matched, it is notified to operator with displaying evaluation result. This setting can edit the message to notice operator.
Overwrite Password	When the inspection results of reinspection and at the registration are not matched, it is notified to operator with displaying evaluation result. If the account has Administrator right, Auto Mode is started by clicking Continue . If the account does not have Administrator right, password enter dialog box appears by clicking Continue. Auto Mode is started by entering the password set in Overwrite Password .

Table 3-1 Setting of Golden Sample Function

3.2 Registration of Sample Image and Inspection Result

Step1: Scan Golden Sample or Silver Sample.

Step2: From **InspectData** tab, select **Image File List > Add Golden sample**.

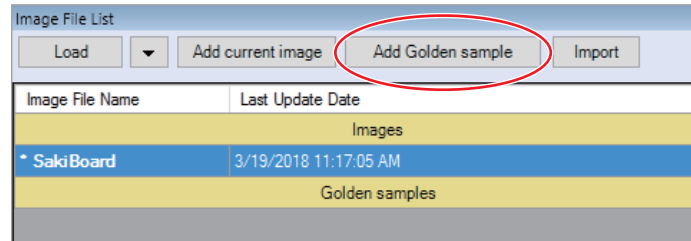


Figure 3-4 Registration of Sample Image and Inspection Result 1

Step3: The dialog box shown below appears.

- Yes** Registers images and NG inspection results.
No Registers images and NG and OK inspection result.
Cancel Cancels to register.

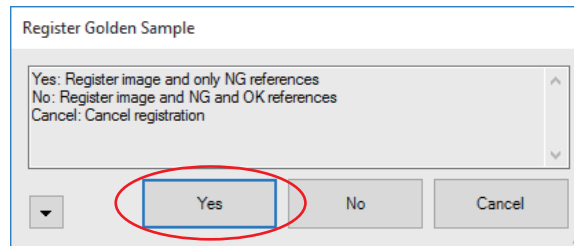


Figure 3-5 Registration of Sample Image and Inspection Result 2

Step4: The dialog box shown below appears by clicking **Yes** or **No** in Step3.
 Enter the file name of sample and click **OK**.

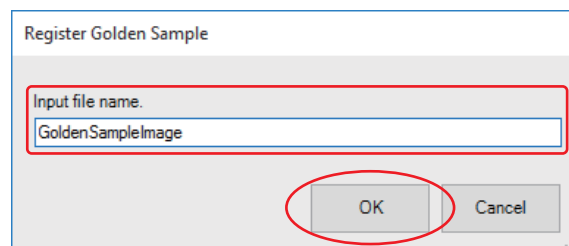


Figure 3-6 Registration of Sample Image and Inspection Result 3

Step5: The inspection is started automatically.

After inspection is completed, registered file name is displayed under **Golden samples** list.

NOTE Multiple sample can be registered.

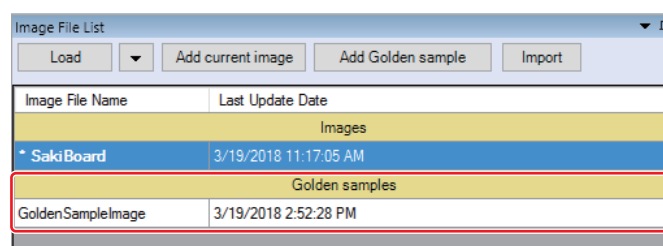


Figure 3-7 Registration of Sample Image and Inspection Result 4

3.3 Comparison of Inspection Result

At the start of Auto Mode, sample images are loaded automatically and perform inspection with current inspection data. When some samples are registered, the inspection is performed in succession. After the inspection is completed, the image which compares registered inspection result and newly inspected one is displayed.

3.3.1 When the Validation Result Is OK

The dialog box shown below appears.

When the inspection results of reinspection and at the registration are matched, **VALIDATION RESULT : PASS** is displayed.

Click **Continue** to start Auto Mode.

Click **Cancel** to cancel Auto Mode.

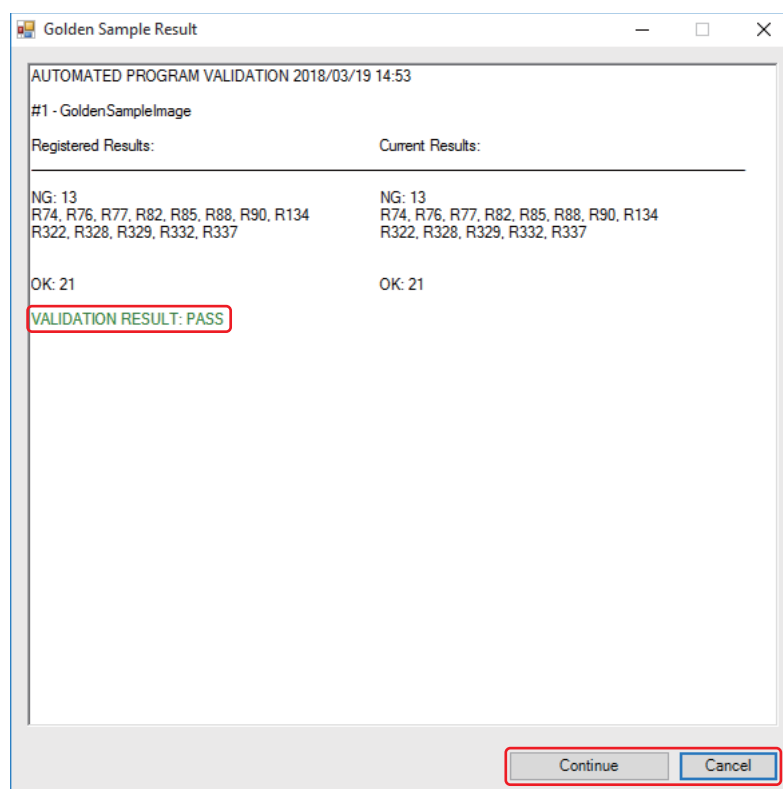


Figure 3-8 Comparison of Inspection Result 1

3.3.2 When the Validation Result Is NG

When Auto Mode is started by the account which has **Administrator** right, the dialog box shown below appears.

If the inspection results of reinspection and at the registration are not matched, **VALIDATION RESULT : FALL** is displayed.

The message set in Step4 of **3.1 Setting of Golden Sample Function** can be displayed at the last of validation result.

Click **Continue** to start Auto Mode.

Click **Cancel** to cancel Auto Mode.

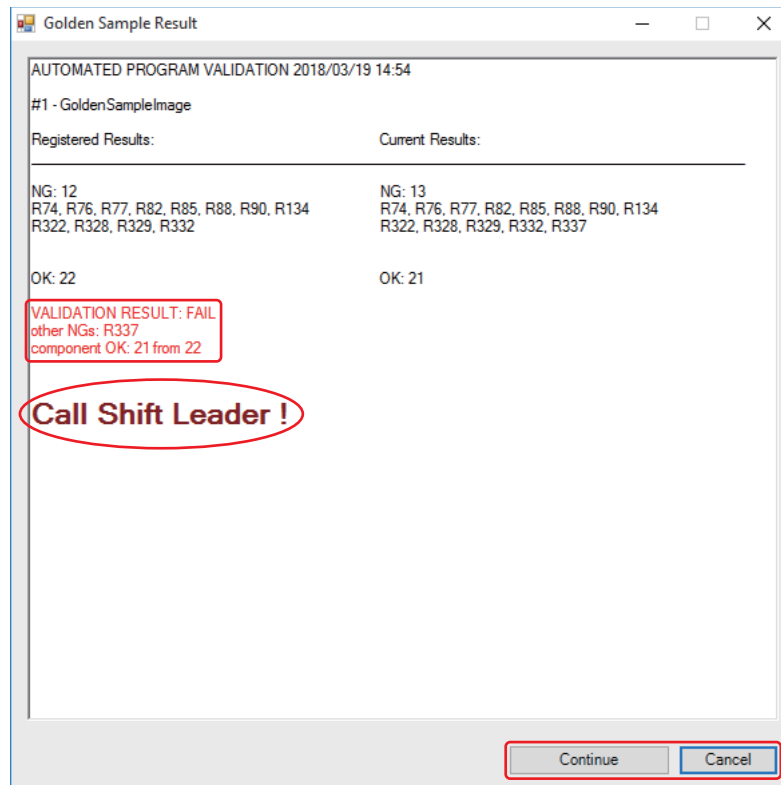


Figure 3-9 Comparison of Inspection Result 2

When Auto Mode is started by the account which does not have **Administrator** right, the dialog box shown below appears.

Continue is invalid but password enter dialog box appears by double-clicking **Continue**. Enter the password set in Step4 of **3.1 Setting of Golden Sample Function** and click **OK** to enable **Continue**.

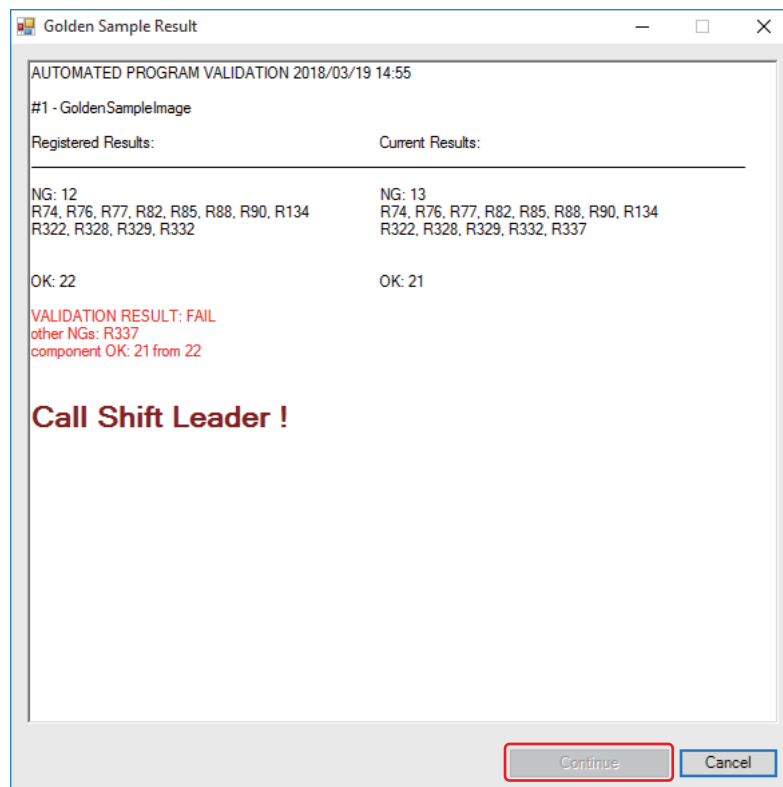


Figure 3-10 Comparison of Inspection Result 3

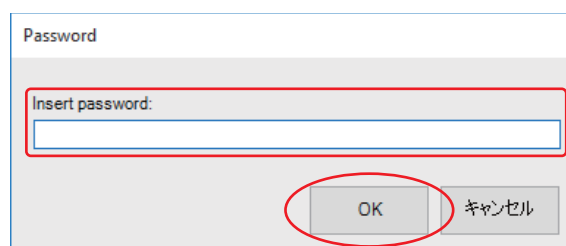


Figure 3-11 Comparison of Inspection Result 4

3.4 Golden Sample Log

When Golden Sample function is enabled, **Golden Sample Log** doc window appears on **Auto** tab.

The validation result of **3.3 Comparison of Inspection Result** is displayed. For details, refer to Table 3-2.

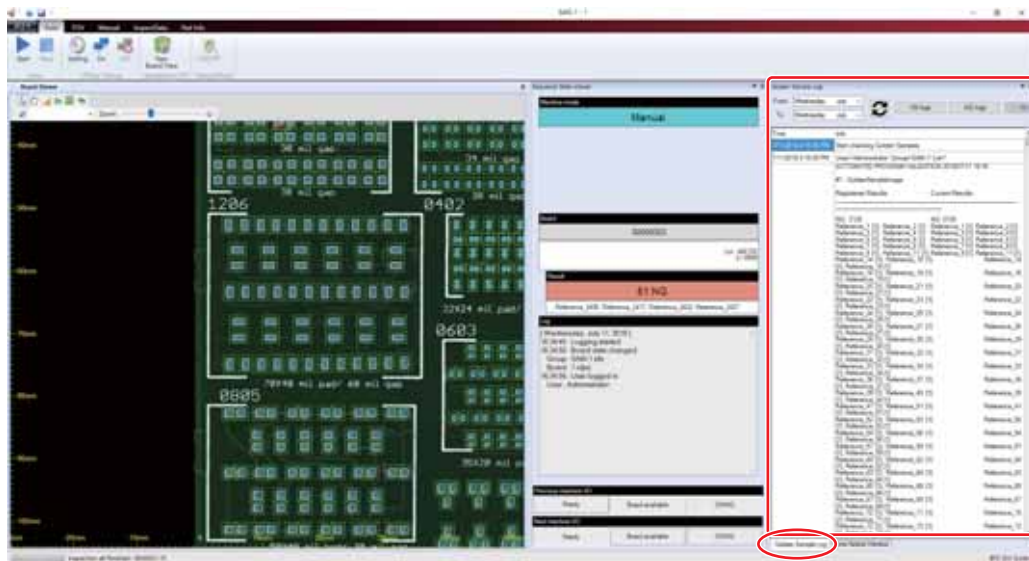


Figure 3-12 Golden Sample Log

Item	Description
From / To	Sets the period to show log. Log is updated, by clicking  .
OK logs	Displays the log for 3.3.1 When the Validation Result Is OK.
NG logs	Displays the log for 3.3.2 When the Validation Result Is NG.
All logs	Displays all logs.

Table 3-2 Golden Sample Log Setting

Stand Alone SPC

1 Stand Alone SPC Function

1.1 SPC

SPC (Statistical process control) is the method that controls the manufacturing process statistically. By using SPC, the quality of cleaning a printer can be maintained, and the manufacturing process can be controlled and improved.

1.2 Stand Alone SPC Function

1.2.1 Overview

The purpose of this function is to provide necessary information for manufacturing.

Stand Alone SPC shall be used for manufacturing lines. This function can be used without stopping auto inspection.

NOTE

This is called Stand Alone SPC due to the function to confirm only information on SPI machine. When managing the over all production line or the traceability function is required, please contact Saki Corporation.

1.2.2 Stand Alone SPC Functions

Stand Alone SPC function is comprised of two functions below.

Trend View

This is the function to manage and maintain the quality of printed solder paste.

This function monitors the inspection results and aggregates the sample values in real-time.

It also measures timing to clean a PCB screen printer, and provides information useful for tuning of the printer.

NOTE

For details, refer to **2 Trend View**.

Past Board View

This is the function to support the OK Range settings and to enable simplified traceability.

This function stores inspection results and sample values, and the information of boards inspected before and the results of the statistical analysis can be checked.

NOTE

For details, refer to **3 Past Board View**.

2 Trend View

2.1 Overview

Trend View is the function to maintain and manage the quality of printed solder paste.

It monitors the inspection results during Auto inspection and aggregates the sample values in real-time.

To maintain and manage the quality of printed solder paste, the information for the latest several hours is displayed on charts and tables. (*1)

Also, pad selection enables you to confirm only the pad of the specified size.

Use this function to grasp the timing of cleaning a printer or to get useful information for printer tuning.

NOTE

(*1) Data of 150 boards can be checked within 24 hours.

2.2 Trend View Function

This section describes each function of **Trend View**.

2.2.1 Function Structure and Name

This function is comprised of **Trend View** window as the main part for chart or table display, **Trend Color Map** for color map display, and **Pad Selection Panel** of a controller to select a pad.

Trend View

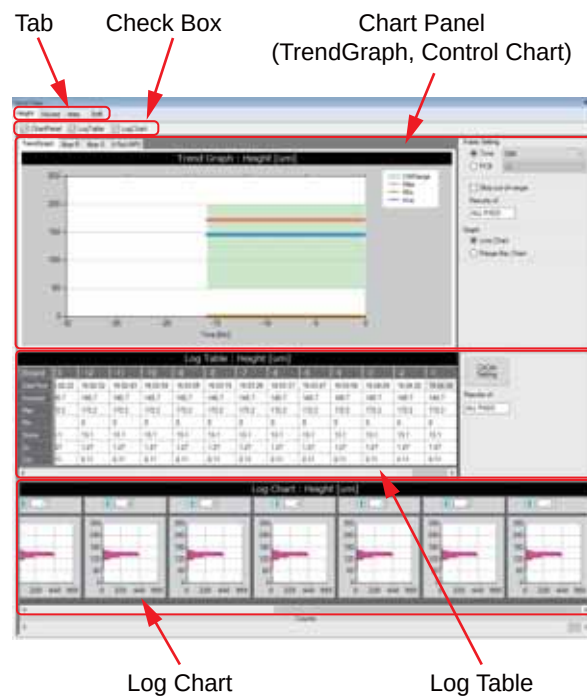


Figure 2-1 Trend View Function

Item	Description
Tab	Switches the displayed inspection items (Height, Volume, Area, and Shift).
Check Box	Selects to show or hide each chart.
Chart Panel	Displays Trend Graph , Xbar , Xbar-R , Xbar-S , and X-Rs (I-MR) . For details, refer to 2.2.2 Trend Graph and 2.2.3 Control Chart .
Log Table	Displays the Log Table . For details, refer to 2.2.4 Log Table .
Log Chart	Displays the Log Chart . For details, refer to 2.2.5 Log Chart .

Table 2-1 Descriptions

Trend Color Map

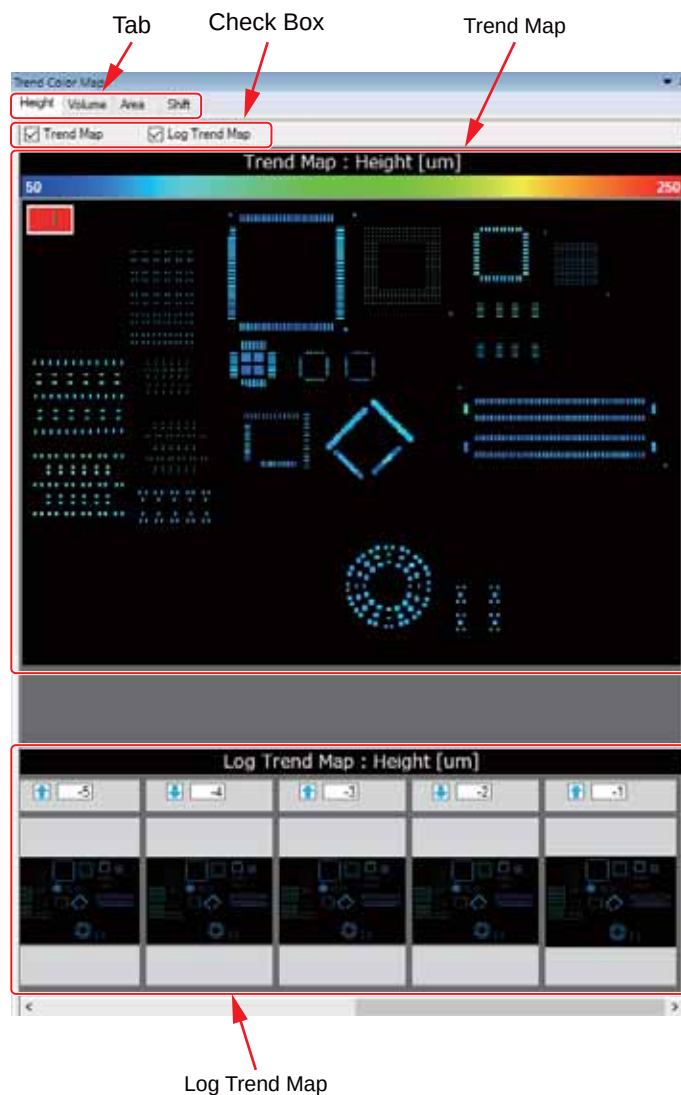


Figure 2-2 Trend Color Map

Item	Description
Tab	Switches the displayed inspection items (Height, Volume, Area, and Shift). It links with the tab of Trend View .
Check Box	Selects to show or hide each chart.
Trend Map	Displays Trend Map . For details, refer to 2.2.6 Trend Map .
Log Trend Map	Displays Log Trend Map . For details, refer to 2.2.7 Log Trend Map .

Table 2-2 Descriptions

Pad Selection Panel

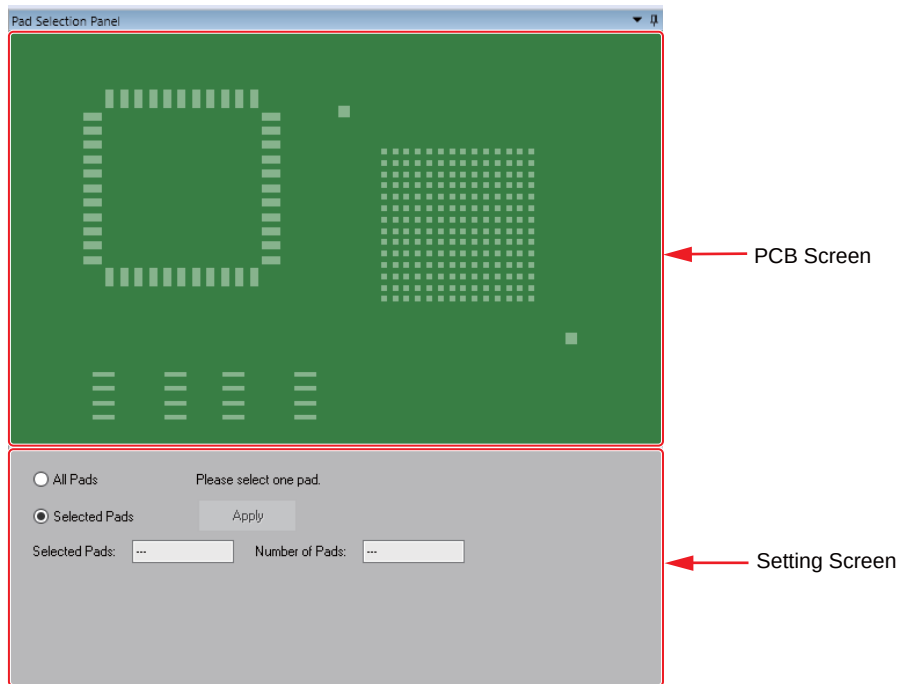


Figure 2-3 Pad Selection Panel

Item	Description
PCB Screen	Displays the pad inspected. Click a pad to select.
Setting Screen	Displays the pad information selected on the PCB screen. And switches display of Trend View window.

Table 2-3 Descriptions

2.2.2 Trend Graph

Trend Graph is the function to check the changing of sample values and tendency in charts.

The maximum, minimum and average values and OK Range (*1) of inspection items (Height, Volume, Area and Shift) are displayed on charts for each board.

The time scale on charts can be changed for every specified hours (0.5/1/3/6/9/12/18/24 hours) and for every specified number of boards (30/50/100/150 boards).

NOTE

(*1) The OK Range shall be displayed with the maximum and minimum of the range for the board being inspected now.

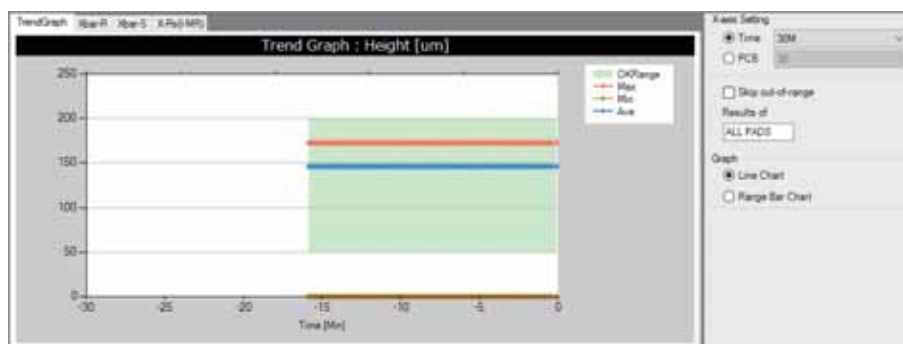


Figure 2-4 Trend Graph, Line Chart - All Data

If a pad is selected on Pad Selection Panel, the maximum, minimum, average values and OK range of inspection items (Height, Volume, Area and Shift) are displayed for all pads of the same size as the pad being selected, on charts. The size of the pad being selected is displayed on the right side.

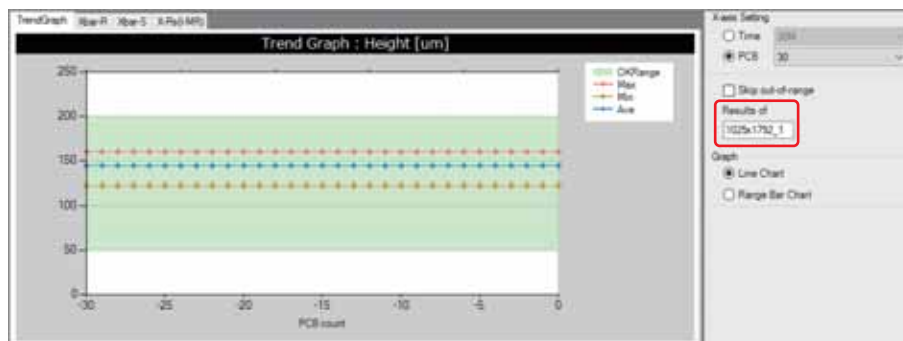


Figure 2-5 Trend Graph, Line Chart - Selected Pad

With this function, two types of charts (Line Chart and Range Bar Chart) can be drawn optionally.

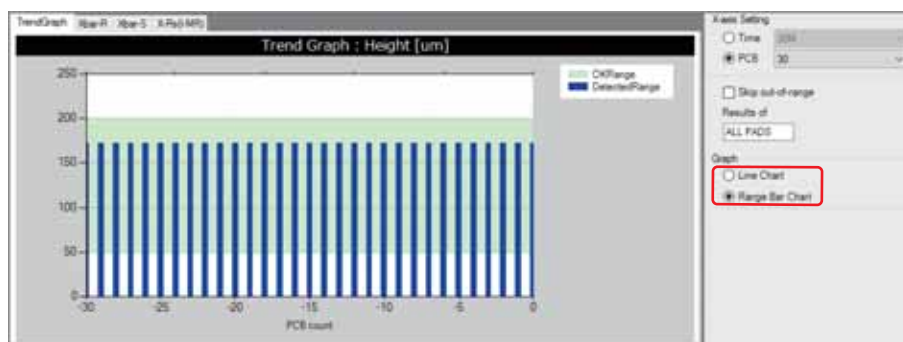


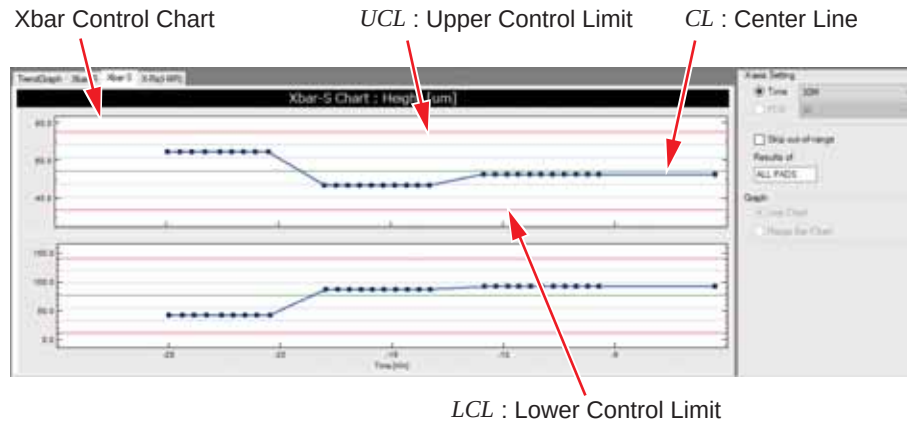
Figure 2-6 Trend Graph, Range Bar Chart - Selected Pad

2.2.3 Control Chart

Control Chart is the function to confirm the manufacturing quality.

The control chart of inspection items (Height, Volume, Area and Shift) is displayed for each board. The time scale of the chart can be changed for every specified hours (0.5/1/3/6/9/12/18/24hours).

Xbar Control Chart



$$UCL = \bar{\bar{X}} + 3\sigma$$

$$CL = \bar{\bar{X}}$$

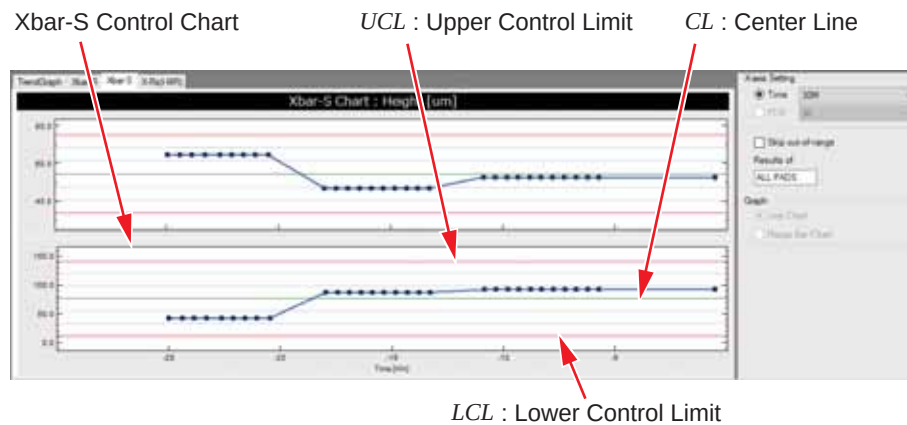
$$LCL = \bar{\bar{X}} - 3\sigma$$

$\bar{\bar{X}}$: Average value of all average value of samples x
(Calculate by each board)

3σ : Standard deviation of average value of samples x
(Calculate by each board) $\times 3$

Figure 2-7 Xbar Control Chart

Xbar-S Control Chart



$$UCL = \bar{\bar{s}} + 3\sigma$$

$$CL = \bar{\bar{s}}$$

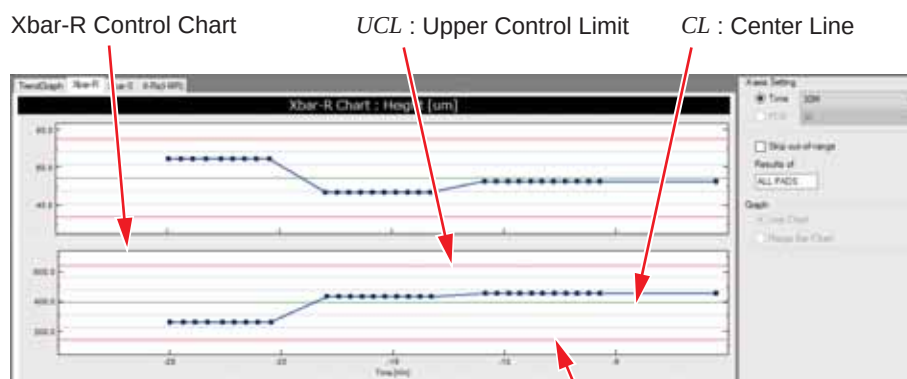
$$LCL = \bar{\bar{s}} - 3\sigma$$

$\bar{\bar{s}}$: Average value of standard deviation s of samples
(Calculate by each board)

3σ : Standard deviation of standard deviation s of samples
(Calculate by each board) $\times 3$

Figure 2-8 Xbar-S Control Chart

Xbar-R Control Chart



LCL : Lower Control Limit

$$UCL = \bar{R} + 3\sigma$$

$$CL = \bar{R}$$

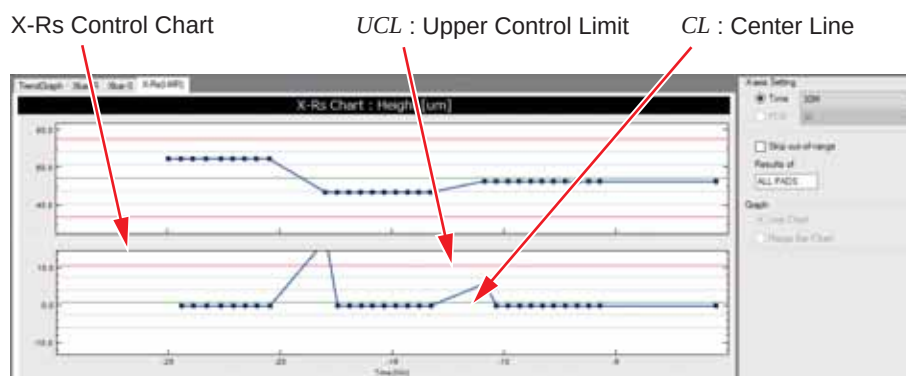
$$LCL = \bar{R} - 3\sigma$$

$\bar{R}$: Average value of range R of samples (Max - Min with samples x)
(Calculate by each board)

3σ : Standard deviation of range R of samples
(Calculate by each board) $\times 3$

Figure 2-9 Xbar-R Control Chart

X-Rs Control Chart



$$UCL = \bar{Rs} + 3\sigma$$

$$CL = \bar{Rs}$$

$\bar{Rs}$: Average value of absolute value of differences
from previous board

3σ : Standard deviation of absolute value of differences
from previous board $\times 3$

Figure 2-10 X-Rs Control Chart

NOTE

Differences from previous board: Average value of samples of current board - average value of samples of previous board.

2.2.4 Log Table

Log Table is the function to display inspection time, maximum value, minimum value, average value, Cp (process capability index), and Cpk (process capability index considering bias) of inspection items (Height, Volume, Area, and Shift) for each board, on table. The data of the latest 150 boards can be checked.

Log Table : Height [um]													
Board	3	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Date Time	16:02:22	16:02:32	16:02:43	16:02:54	16:03:05	16:03:15	16:03:26	16:03:37	16:03:47	16:03:58	16:04:09	16:04:20	16:04:38
Average	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7	145.7
Max	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2	172.2
Min	0	0	0	0	0	0	0	0	0	0	0	0	0
Sigma	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1
Cp	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27
Cpk	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11

Figure 2-11 Log Table - All Data Display

The Cp and Cpk limit values can be specified on the dialog box shown below by clicking **CpCpk Setting** button.

Cp, Cpk SettingForm

Set the LSL and USL for Cp, Cpk.

$Cp = (USL - LSL) / 6\sigma$

$Cpk = \min\{(USL - \mu) / 3\sigma, (\mu - LSL) / 3\sigma\}$

where,

USL : upper specification limit

LSL : lower specification limit

σ : standard deviation of the sample values

μ : average of sample values

Height LSL 50.0 - USL 150.0

Volume LSL 50.0 - USL 150.0

Area LSL 50.0 - USL 150.0

ShiftX LSL -100 - USL 100.0

ShiftY LSL -100 - USL 100.0

OK Cancel

Figure 2-12 Cp, Cpk Dialog Box

If a pad is selected on the pad selection panel, regarding all the pads of the same size as the pads being selected, Log Table is the function to display maximum values, minimum values, average values, Cp (process capability index), and Cpk (process capability index considering bias) on table. For Cp and Cpk limit values, OK range of the selected pad is used. (*1) The size of the selected pad is displayed on the right side.

NOTE

(*1) Limit values in this case can be checked on the dialog box displayed by clicking the **CpCpk Setting** button.

Log Table : Height [um]													
Board	3	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1
Date Time	16:03:26	16:03:37	16:03:47	16:03:58	16:04:09	16:04:20	16:04:38	16:04:49	16:04:59	16:05:10	16:05:21	16:05:33	16:05:43
Average	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3	144.3
Max	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8	159.8
Min	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4	122.4
Sigma	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6
Cp	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36	4.36
Cpk	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43	2.43

Figure 2-13 Log Table - Display for each pad

2.2.5 Log Chart

Log Chart is the function to check the changes and the tendency of sample values on charts. The sample values of inspection items (Height, Volume, Area and Shift) are displayed on histograms or scatter charts for each board. In addition, the squeegee direction of the PCB screen printer is displayed for each board. The data of the latest 150 boards can be checked.

When Inspection items are Height, Volume, and Area

Sample values are displayed on histogram shown below.

NOTE The vertical axis shows the sample values, and the horizontal axis shows the number of samples.

NOTE The green dashed lines indicate the upper limit and lower limit of OK range.

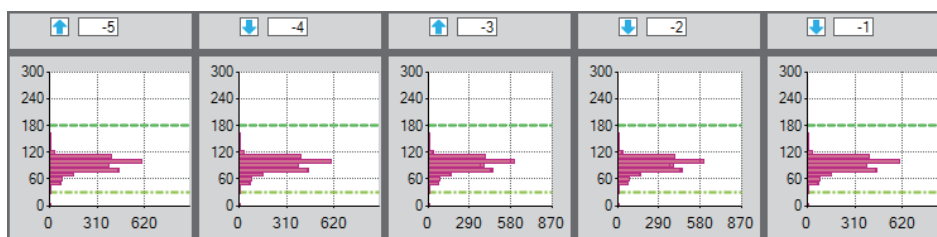


Figure 2-14 Log Chart - Histogram

When Inspection items is Shift

Shift amounts of X-axis and Y-axis are displayed on scatter charts shown below.

NOTE The average shift amounts for X and Y are shown in orange.

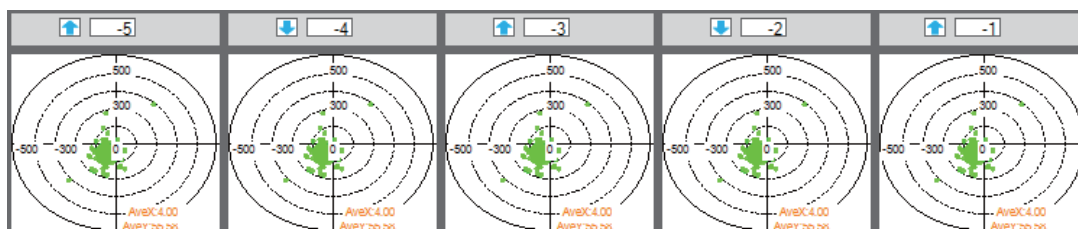


Figure 2-15 Log Chart - Scatter Chart

2.2.6 Trend Map

Trend Map is the function to check the tendency of printed solder paste on a board.

Distribution of printed solder paste on a board is displayed visually in colors based on sample values for each inspection item (Height, Volume, Area and Shift).

The limit value corresponding to the colors can be set on the dialog box of Figure 2-17 displayed by clicking scale color bar area at the top of the map.

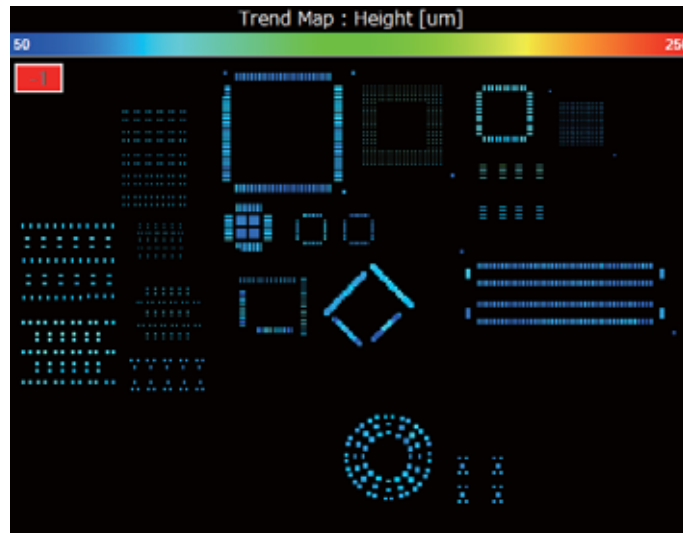


Figure 2-16 Overview of Trend Map

	Lower limit	Upper limit
Height	50	150
Volume	50	150
Area	50	150
Shift X	-100	100
Shift Y	-100	100

Figure 2-17 Map Scale Settings

2.2.7 Log Trend Map

Log Trend Map is the function to check the trend map for the last 10 sheets.

Click the image on Log Trend Map to display Trend Map.

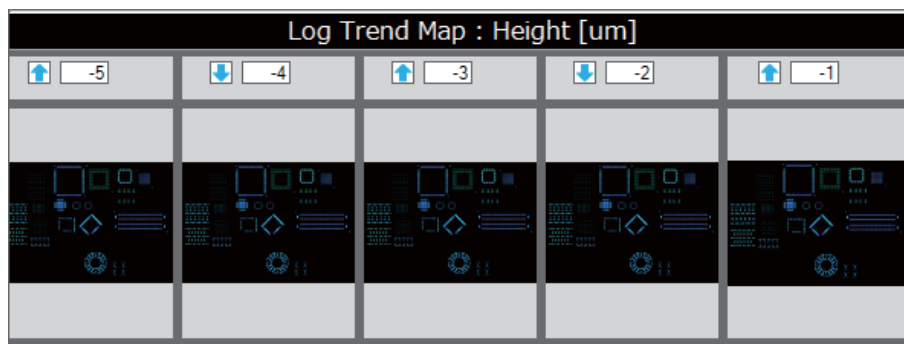


Figure 2-18 Log Trend Map

2.3 Setting

This section describes the operation procedure.

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

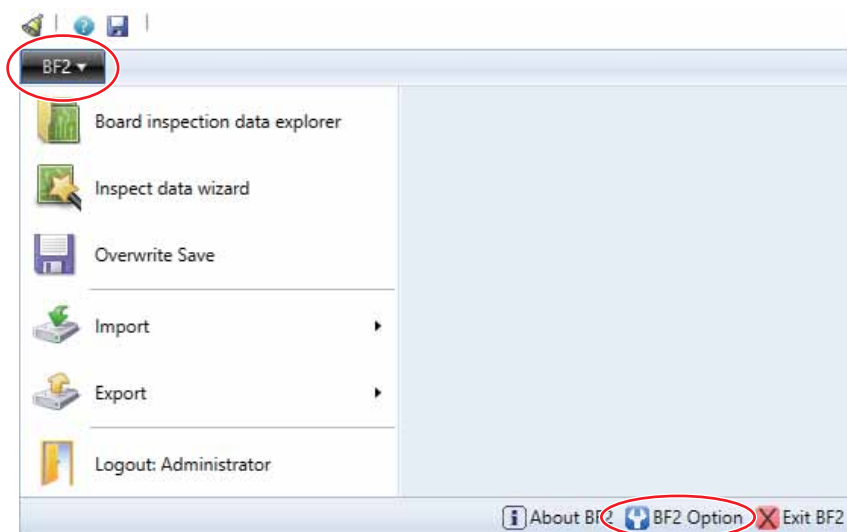


Figure 2-19 Trend View Setting 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

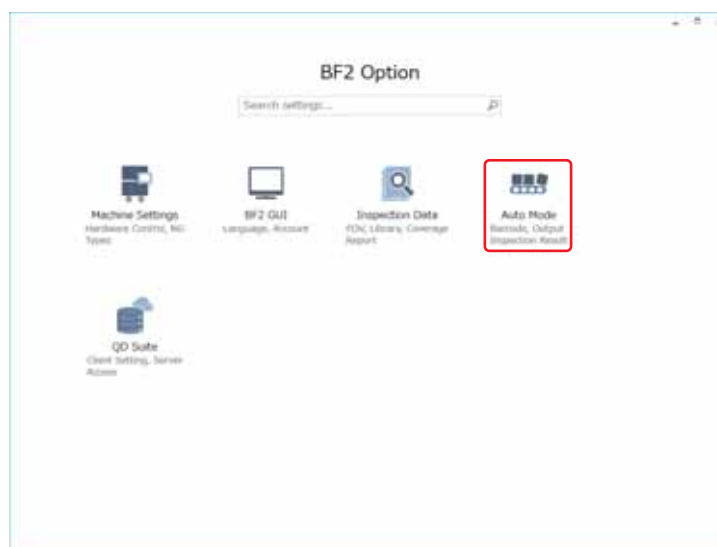


Figure 2-20 Trend View Setting 2

Step3: Select **Output Inspection Result** and set toggle switch of **Stand Alone SPC** to **ON**.

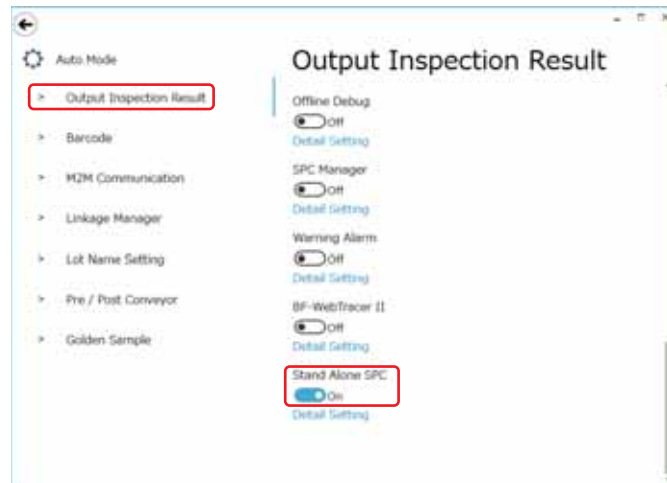


Figure 2-21 Trend View Setting 3

Step4: Click **Detail Setting** under the toggle switch of **Stand Alone SPC** to set detailed items. Proceed the setup procedures, if necessary. For details refer to Table 2-4. After all the settings are completed, click  on the upper right side of the dialog box and click **Yes** to close dialog box.

Item	Description
Enable Output Database	When checked, enables the Past Board View function.
Enable Trend View	When checked, enables the Trend View function.
Database Path	Set the saving data folder path. For the destination folder of the data, specify the folder on disk which has enough space.
Disk Space Limit[GB]	The disk space is checked at the time of starting the Auto inspection. If the specified space cannot be checked, the warning dialog box appears. Unit is [GB].
Output Images	When checked, enables the Output NG Images.
Delete expired date	Deletes data that has passed the specified number of days.
Interval between checks [days]	Checks for expired data at the specified interval (day).
EU Coordinate System	If enabled, data are recalculated to correspond with the EU coordinate system. + Y axis is in the upper direction.

Table 2-4 Descriptions

2.4 Operation Procedure

This section describes the main function of **Trend View** and the operation procedure of pad selection panel.

Step1: Start the Auto inspection. For the procedure to start the Auto inspection, refer to **Operation Manual**.

Display chart and table in the **Trend View** window. Display the distribution of sample values on the board in the **Trend Color Map** window. In addition, display the layout and setting screen of pads on the **Pad Selection Panel** window.

For charts, tables and settings, refer to **2.2 Trend View Function**.



Figure 2-22 Trend View, Trend Color Map and Pad Selection Panel window

Step2: To check only the pad of the specific size, use the pad selection panel.
Select **Selected Pads** in the **Pad Selection Panel** window.

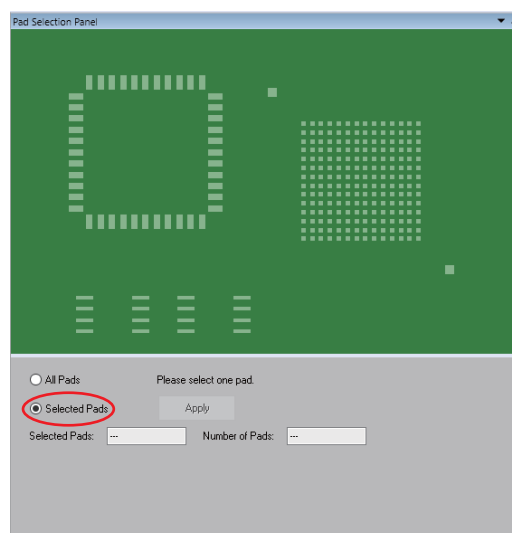


Figure 2-23 Pad Selection Panel 1

Step3: Click a pad to check on the PCB screen at the top of the **Pad Selection Panel** window. The size of the selected pad is displayed on **Selected Pads** and the number of the pads of the same size is displayed on **Number of Pads**. In addition, the same size pads are highlighted on the PCB screen.

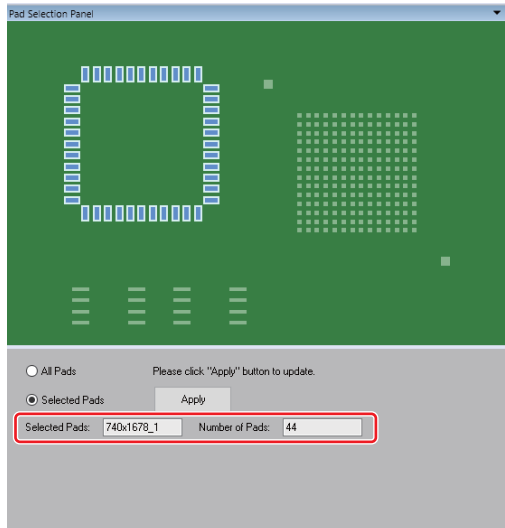


Figure 2-24 Pad Selection Panel 2

Step4: Click the **Apply** button. The information of the selected pad appears on the **Trend View** window.

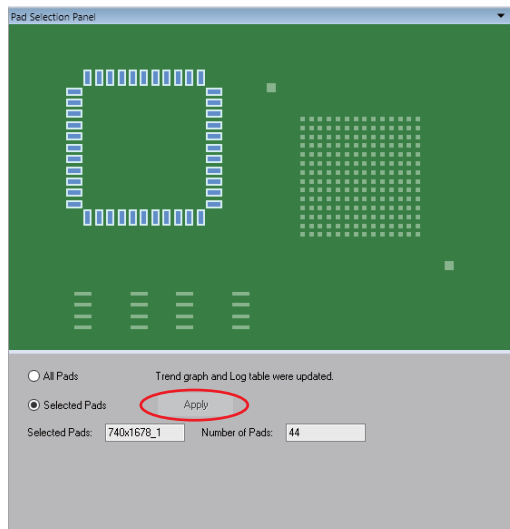


Figure 2-25 Pad Selection Panel 3

3 Past Board View

3.1 Overview

Past Board View is the function to support the OK Range settings and to enable the simplified traceability. This function can search the boards inspected in the past, and can display inspection results, sample values, NG types, NG images, and results of statistics and analysis.

3.2 Past Board View Function

This section describes each function of Past Board View.

3.2.1 Function Structure and Name

The Past Board View function is comprised of eight functions: Board List, Board View, Image View, Chart, Total Chart, Defect Analysis, Total Board Analysis, and Yield Chart.

The following figure and table explain each function name of Past Board View.

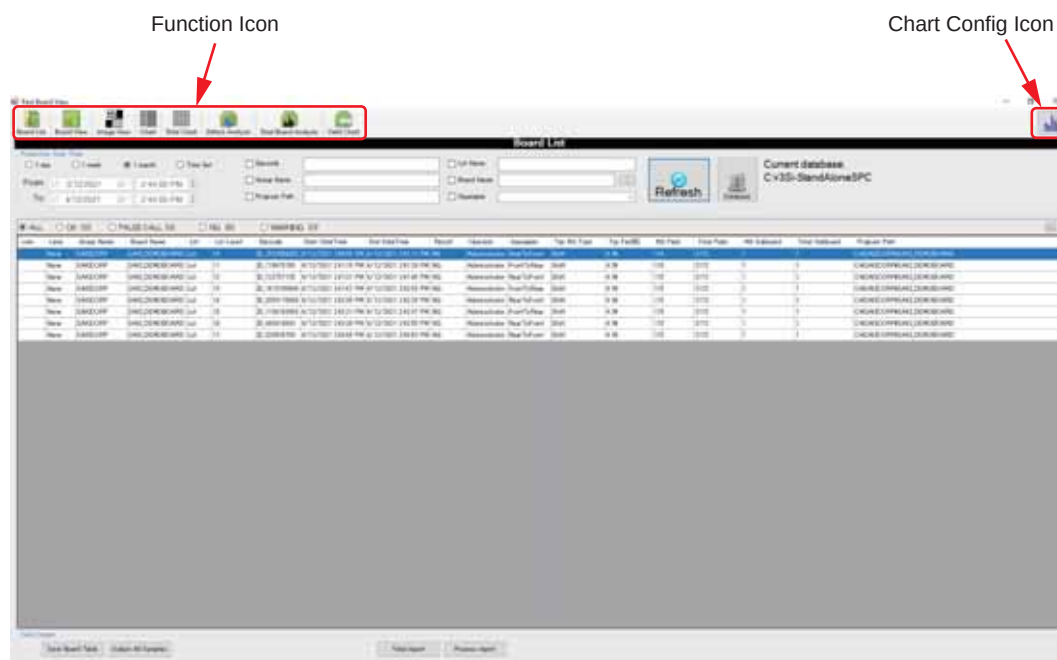


Figure 3-1 Past Board View Functions

Item	Icon	Descriptions
Board List		Searches board inspection results and lists them.
Board View		Details the board inspection results being selected in Board List .
Image View		Enlarges NG images being selected in Board View . Click NG image or icon to display it.
Chart		Displays a chart of the board inspection results being selected in Board List . Multiple boards can be selected. Select a board and click an icon to display it.
Total Chart		Displays total histogram of board inspection results being selected in Board List . Multiple boards can be selected. Select a board and click an icon to display it.
Defect Analysis		Displays defect analysis data of board inspection results selected in Board List . Multiple boards can be selected. Select an NG board and click the icon to display it.
Total Board Analysis		Displays analyzed data of board inspection results selected in Board List . Multiple boards can be selected. Select a board and click the icon to display it.
Yield Chart		Displays the yield chart of board inspection results specified by the filter in Board List . Select the Refresh button after specifying the filter and click the icon to display it.
Chart Config		Sets a chart displayed in Chart and Total Chart .

Table 3-1 Descriptions

3.2.2 Board List

With **Board List** window, past board inspection results can be searched and listed.

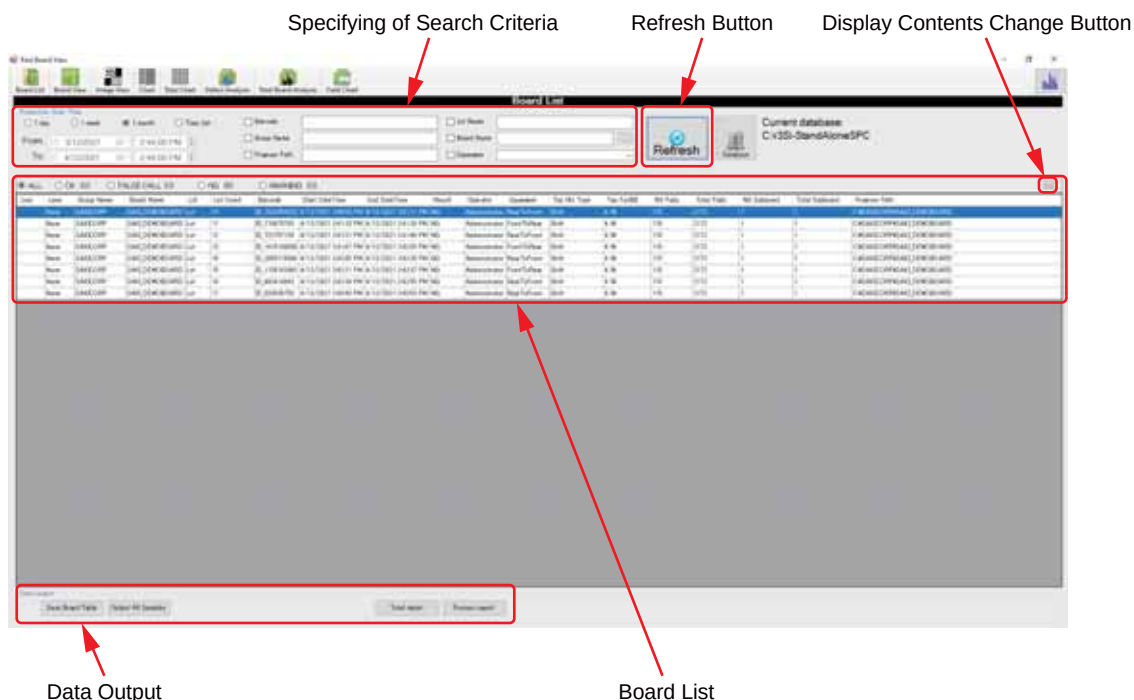


Figure 3-2 Board List Function

Item	Description	
Specifying of Search Criteria	Specifies the criteria to search the inspection results registered on database.	
Board List	Board List	Displays the searched board list. Double-click this on the list to switch to Board View . Right-clicking can delete data. The Display Contents Change button can select the column to be shown.
	Display Contents Change Button	Changes the columns shown in Board List .
Refresh Button	Searches inspection results and updates the board list.	
Data Output	Save Board Table	Outputs all the lists displayed on Board List to a text file.
	Output All Samples	Outputs all the sample values of the board selected on the list to a text file.
	Total report	Outputs the total report of the inspection results and NG type on all of the boards displayed in the Board List .
	Process report	Outputs the process report of all the sample values and the inspection results on all of the boards displayed in the Board List .

Table 3-2 Descriptions

3.2.3 Board View

The **Board View** window displays the details of board inspection results.

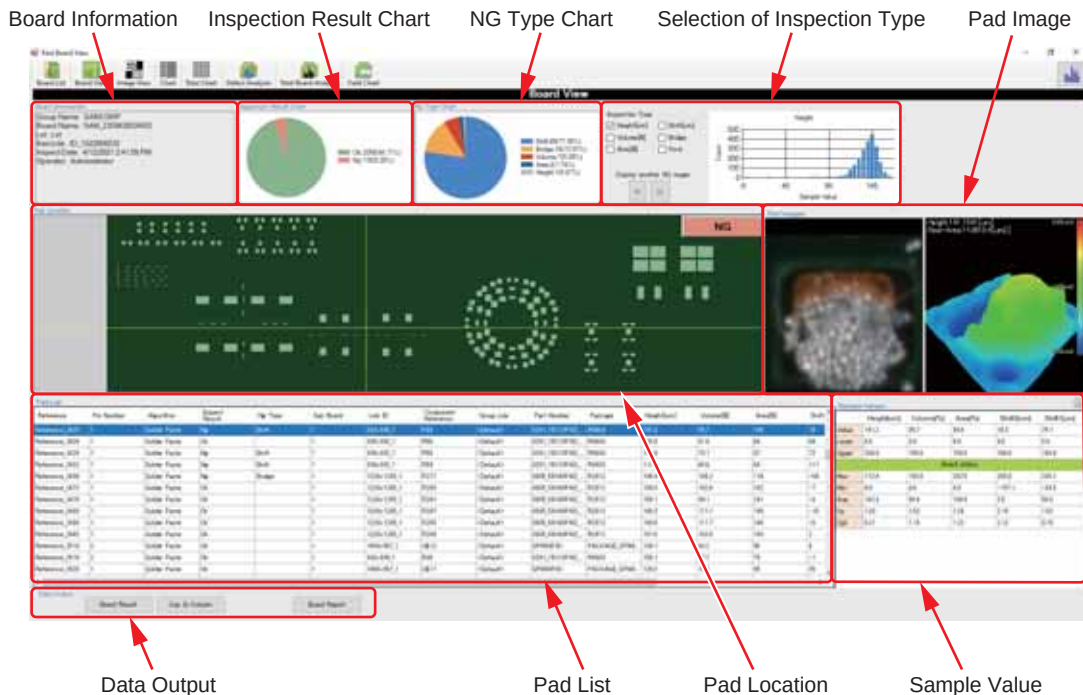


Figure 3-3 Board View Functions

Item	Description	
Board Information	Displays the basic information of the board being displayed.	
Inspection Result Chart	Displays the ratio of inspection results.	
NG Type Chart	Displays the ratio of NG types.	
Selection of Inspection Type	Switches NG images with   buttons regarding the item selected on a checkbox. In addition, displays sample values of inspection results for height, volume, area, and shift.	
Pad Location	Displays pad location on a board. By clicking a pad, NG image is displayed on Pad Image , and pad information and sample values are displayed on Pad List and Sample Value .	
Pad Image	Displays NG images of the pad selected in Pad Location or Pad List . Double-click the 2D image to display the enlarged image to Image View window.	
Pad List	Lists all the pads. By selecting a list, pad information and sample values are displayed on Sample Value , and NG image is displayed on Pad Image . In addition, the pad selected on Pad Location is displayed with the yellow cross bar.	
Sample Value	Displays sample values and limit values of the pad selected in Pad Location or Pad List and maximum value, minimum value, average, Cp, and Cpk in the whole board. Limit value of Cp and Cpk can be set from the setting icon ().	
Data Output	Result View	Outputs the data displayed on Pad List to a text file.
	Coplanarity Data	Outputs the data displayed on Pad List and the results of coplanarity inspection to a text file.
	Board Report	Outputs the board report of all the sample values and the inspection results on the boards displayed in the Board List .

Table 3-3 Descriptions

3.2.4 Image View

The **Image View** window displays an enlarged NG image of a pad selected on the **Board View** window.

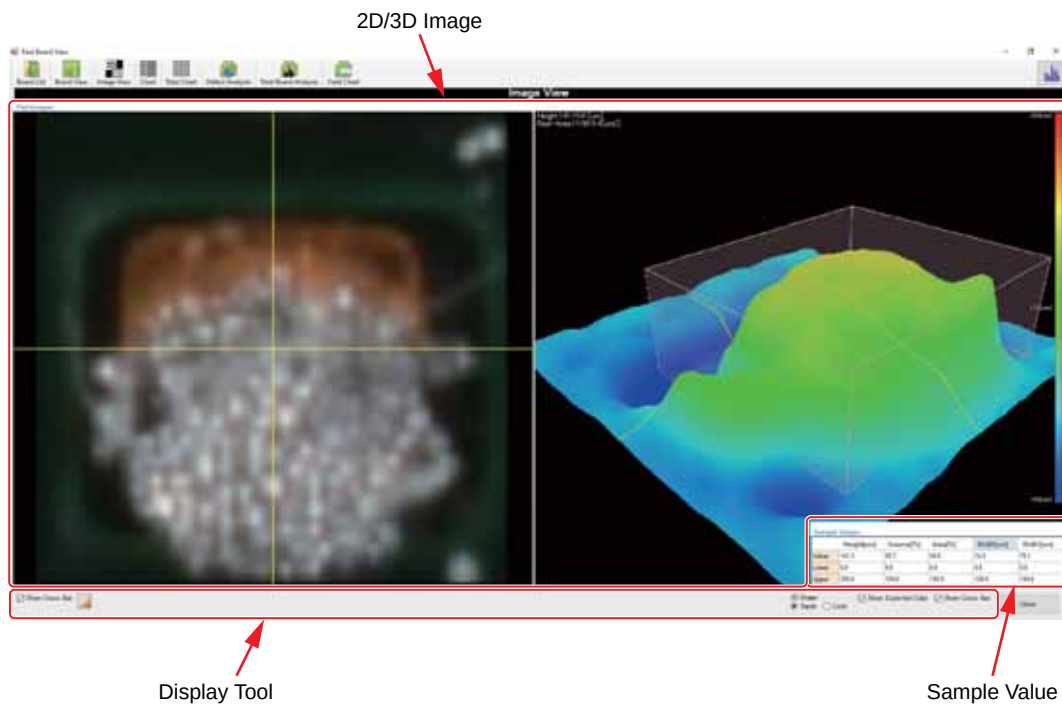


Figure 3-4 Image View Function

Item	Description
2D/3D Image	Displays the enlarged images of 2D and 3D images.
Sample Value	Display the sample value of the image being displayed.
Display Tool	Switches the display. For details, refer to the description of Table 3-5.

Table 3-4 Descriptions

Item	Description
Show Cross Bar (2D)	Displays the cross bar on 2D images.
	Displays the scale on the 2D image.
3D Image	Switches Depth (Color map for height) and Color (2D image).
Show Expected Cube	Display theoretical shape of solder on a 3D image.
Show Cross Bar (3D)	Show/Hide the cross bar on a 3D image.

Table 3-5 Descriptions

3.2.5 Chart

The **Chart** window displays board inspection result selected in **Board List** window for each item.

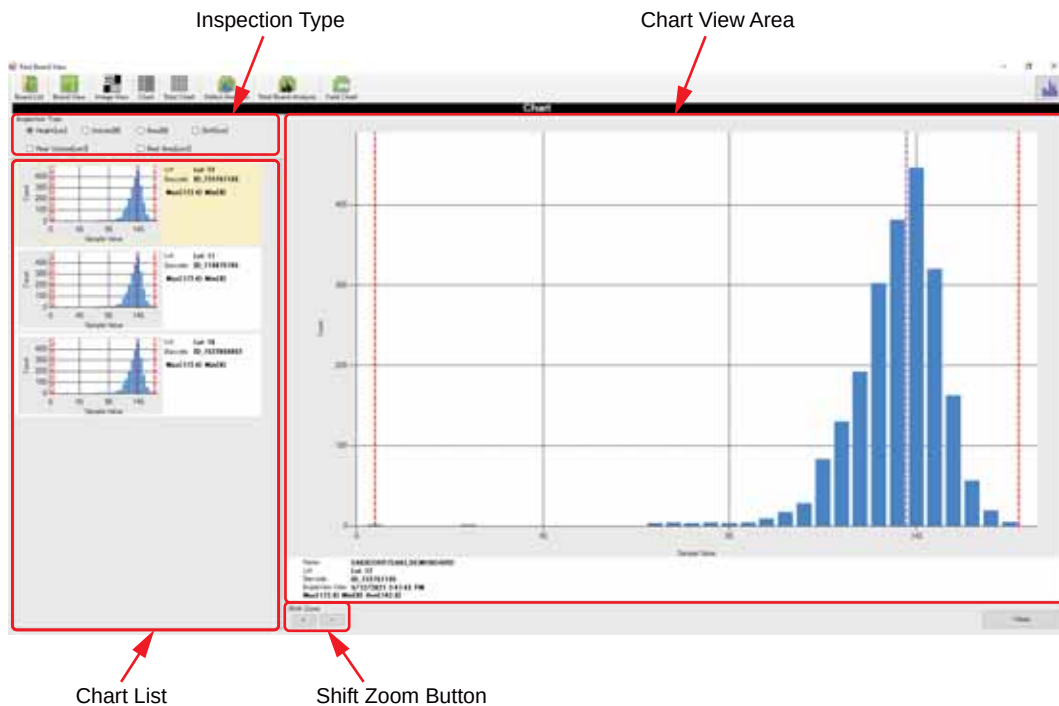


Figure 3-5 Chart Function

Item	Description
Inspection Type	Switches inspection items of charts to display.
Chart List	Lists charts for each board.
Chart View Area	Enlarges and displays the chart being selected on Chart List .
Shift Zoom Button	Zooms a chart view of shift. Valid only when selecting shift.

Table 3-6 Descriptions

3.2.6 Total Chart

In the **Total Chart** window, board inspection results selected in the **Board List** window are displayed on the list of histograms and scatter charts.

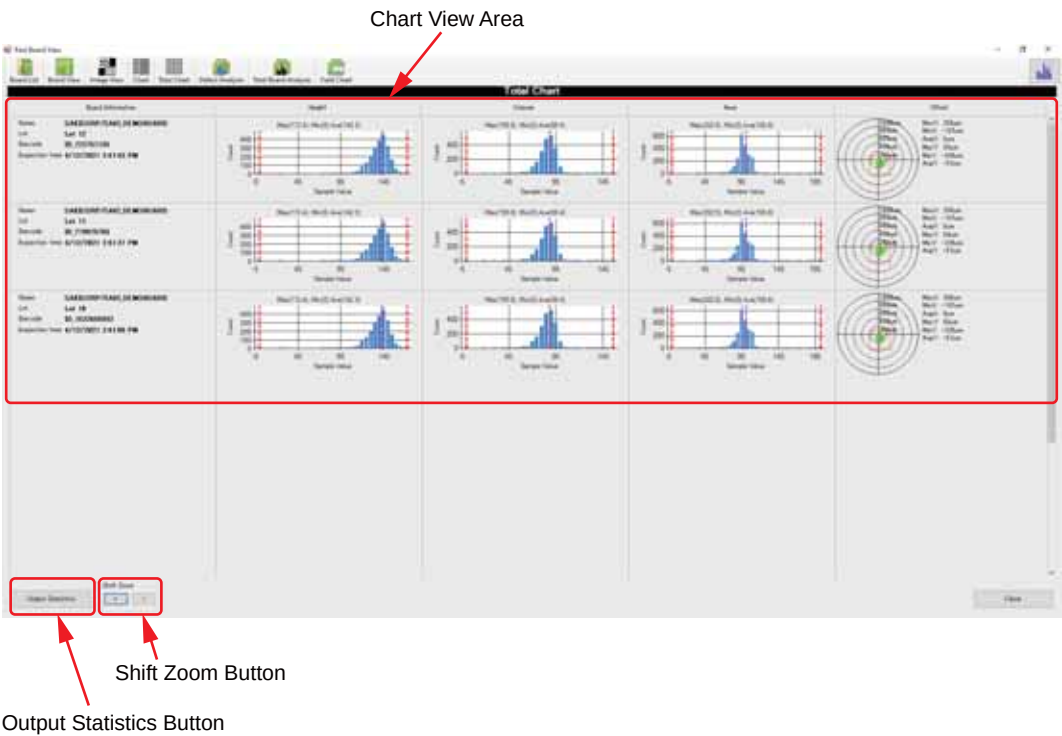


Figure 3-6 Total Chart Function

Item	Description
Chart View Area	Lists the charts of height, volume, area, and shift amount for each board.
Output Statistics Button	Outputs maximum value, minimum value, and average sample value for each board to a text file.
Shift Zoom Button	Zooms chart view of shift.

Table 3-7 Descriptions

3.2.7 Defect Analysis

The **Defect Analysis** window can analyze defects with boards selected on the **Board List** window.

Defect Information

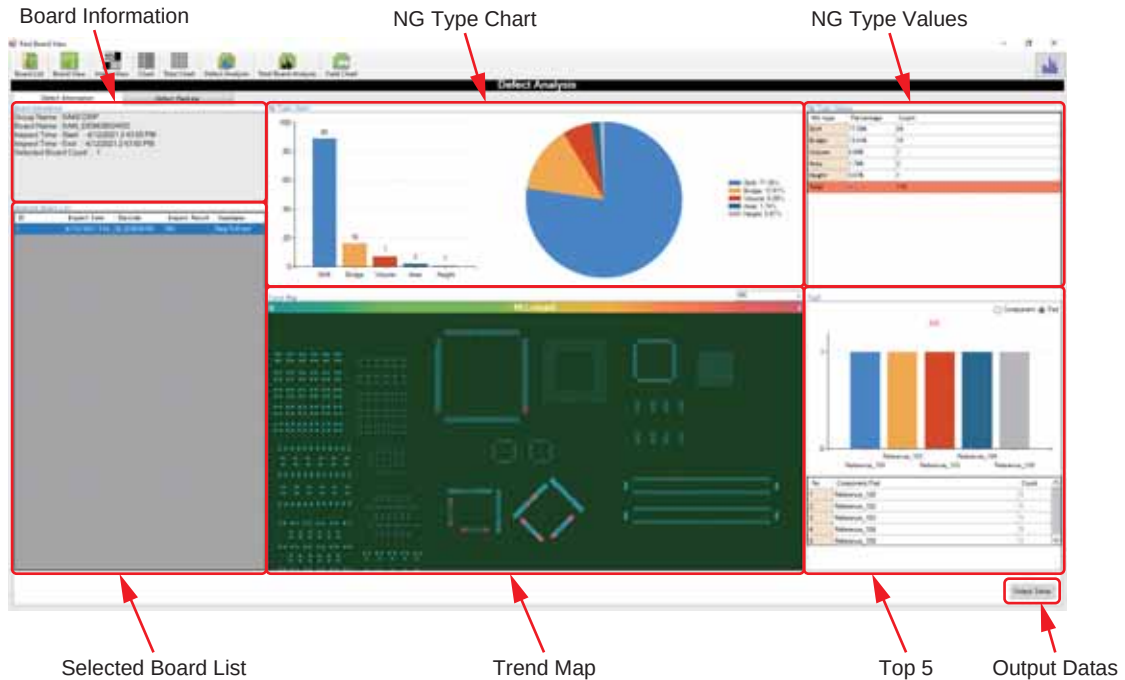


Figure 3-7 Defect Information

Item	Description
Board Information	Shows the basic information of the model being viewed.
Selected Board List	Shows the boards selected in Board List .
Ng Type Chart	Shows the ratio of NG types.
Ng Type Values	Shows values of the NG type chart.
Trend Map	Shows the number of NGs by color map. Blue indicates less NGs and red indicates more NGs.
Top5	Shows Top 5 components/pad with the most NGs.
Output Datas	Outputs the image of each screen.

Table 3-8 Descriptions

Defect Ranking

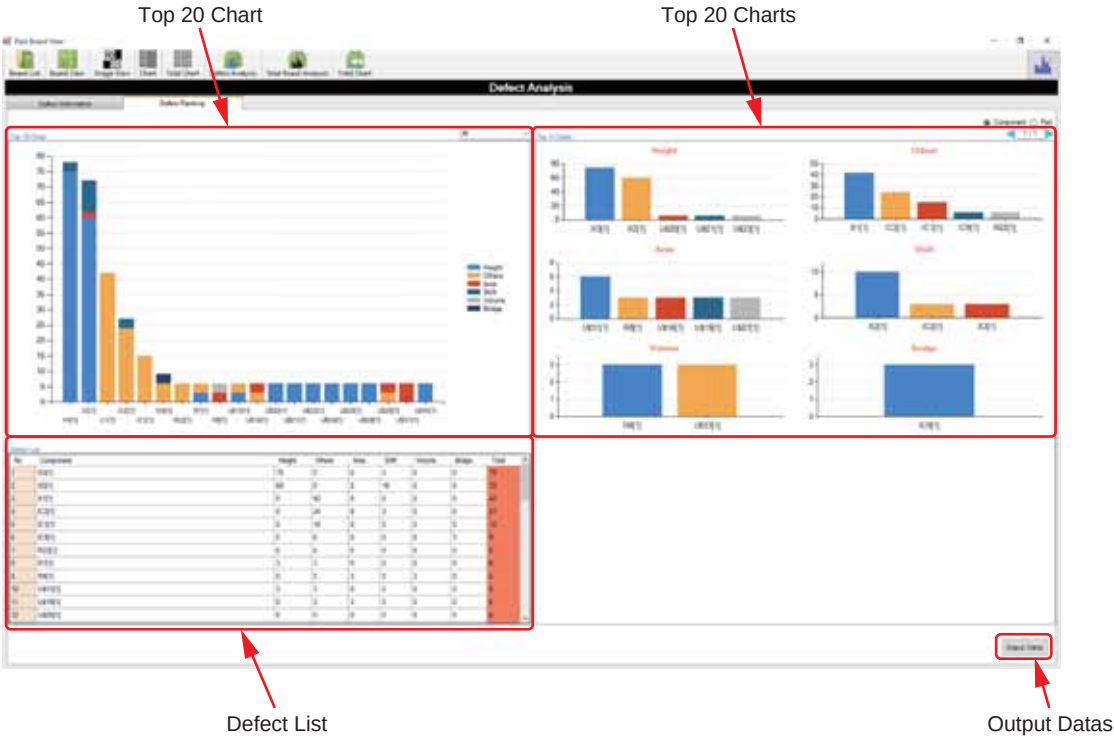


Figure 3-8 Defect Ranking

Item	Description
Top 20 Charts	Shows Top 20 components/pad with the most defects.
Top 5 Charts	Shows Top 5 components/pad with the most NGs for each NG type.
Defect List	Shows the number of NGs for all components/pads.
Output Datas	Outputs the image of each screen.

Table 3-9 Descriptions

3.2.8 Total Board Analysis

Total Board Analysis

The **Total Board Analysis** window can analyze grouped pads with multiple boards selected on **Board List**.

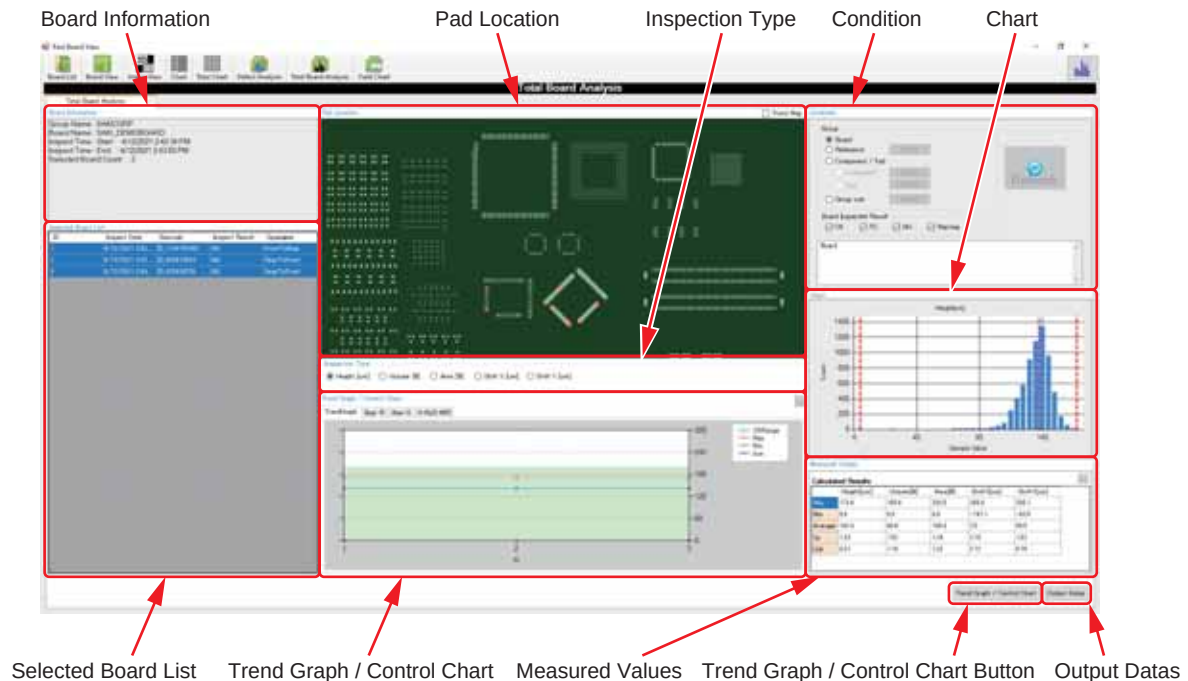


Figure 3-9 Total Board Analysis

Item	Description
Board Information	Shows the basic information of the model being viewed.
Selected Board List	Shows the boards selected in Board List .
Pad Location	Shows the pad location of the selected model. A panel to select pads and show them on a chart. Pads can be selected by dragging only when Reference in Condition is selected.
Inspection Type	Select the inspection type to show on the chart.
Trend Graph / Control Chart	Shows the sample value's trend for the selected pad on the chart, same as Trend View.
Condition	Set the selection criteria for the pads to show on the chart. For details, refer to Table 3-11.
Chart	With the selected pads, shows the histogram for Height/Volume/Area and the scatter chart for Shift.
Measured Values	Shows measured values from the selected pads.
Trend Graph / Control Chart Button	Shows trend graph/control chart of the selected pad on one screen. Clicking this button with trend graph selected shows a list of trend graphs. and clicking this button with control chart selected shows a list of control charts.
Output Datas	Outputs the image of each screen.

Table 3-10 Descriptions

Item		Description
Group	Board	Selects all pads on a board to use for measurements.
	Reference	Selects the reference to use for measurements. Click Select to select a reference from the list. Alternatively, the reference can be selected by clicking it on the Pad Location panel. Multiple references can be selected by clicking them on the Pad Location panel while pressing the Ctrl button. Multiple references can be also selected by dragging references on the Pad Location panel.
	Component / Pad	Selects the component/pad to use for measurements. Click Select to select a component/pad from the list. Alternatively the pad can be selected by clicking it on the Pad Location panel. Multiple components/pads can be selected by clicking them on the Pad Location panel while pressing the Ctrl button. NOTE The Component check box is enabled only when component information is available. NOTE Multiple components/pads cannot be selected with both Component and Pad check boxes checked.
	Group rule	Selects the group rule to use for measurements. Click Select to select a group rule from the list.
Board Inspected Result		Select the board inspection result to use for measurements.

Table 3-11 Condition

Trend Graph/Control Chart

The **Trend Graph/Control Chart** window shows the trend graphs/control charts of each inspection type on one screen and can analyze them.

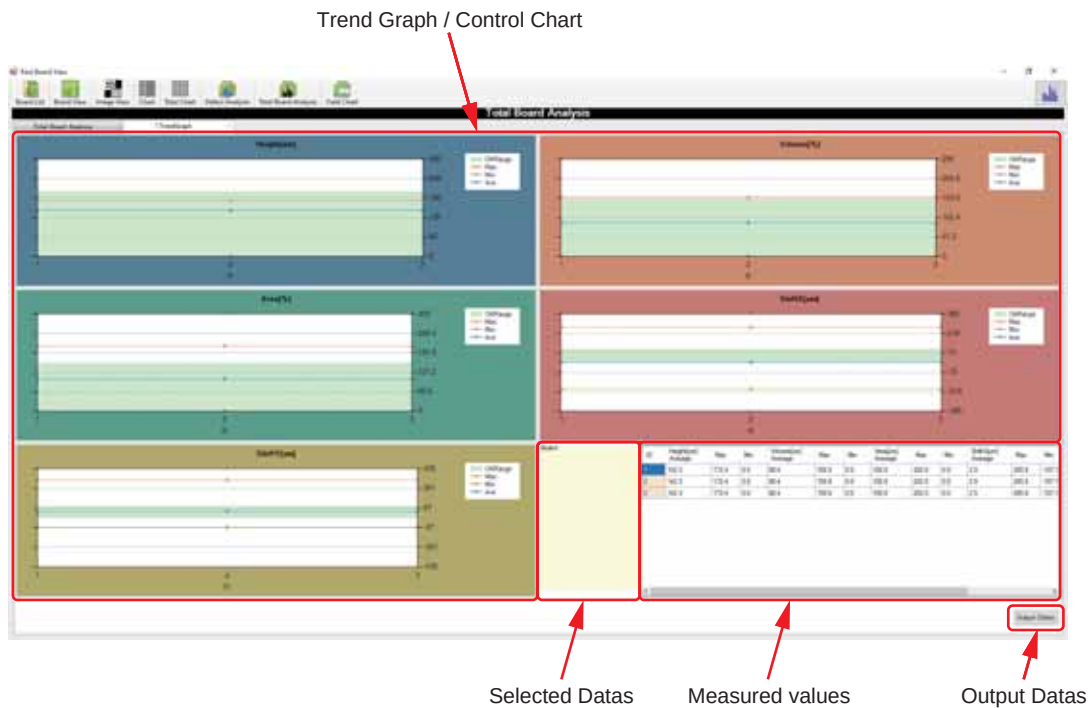


Figure 3-10 Trend Graph / Control Chart

Item	Description
Trend Graph / Control Chart	Shows each trend graph/control chart.
Selected Datas	Shows data set in Condition on the Total Board Analysis screen.
Measurement Values	Shows measured values.
Output Datas	Outputs the image of each screen.

Table 3-12 Descriptions

3.2.9 Yield Chart

The **Yield Chart** window shows the yield rate specified in the search criteria on **Board List**.



Figure 3-11 Yield Chart

Item	Description
Active Filter Information	Shows the filter criteria for data being viewed. Specify the filter criteria on Board List .
Yield Rate Chart	Shows yield rates (chart).
Yield Rate Value	Shows yield rates (values).
Yield Analysis	Shows yield rates in hours/days/weeks.
Output Datas	Outputs the image of each screen.

Table 3-13 Descriptions

3.2.10 Chart Setting

Here describes how to set charts displayed in the **Chart** window and the **Total Chart** window.

Click the **Chart Config** icon(), and the dialog box shown below appears. Set the bar step for histograms and the scale for scatter charts.

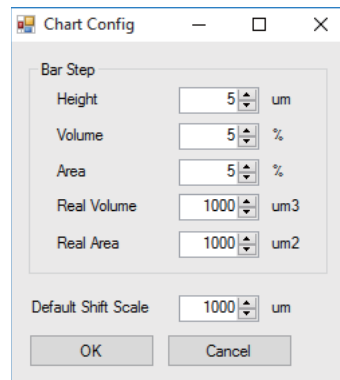


Figure 3-12 Chart Setting

3.3 Settings

This section describes how to set the settings and procedure.

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

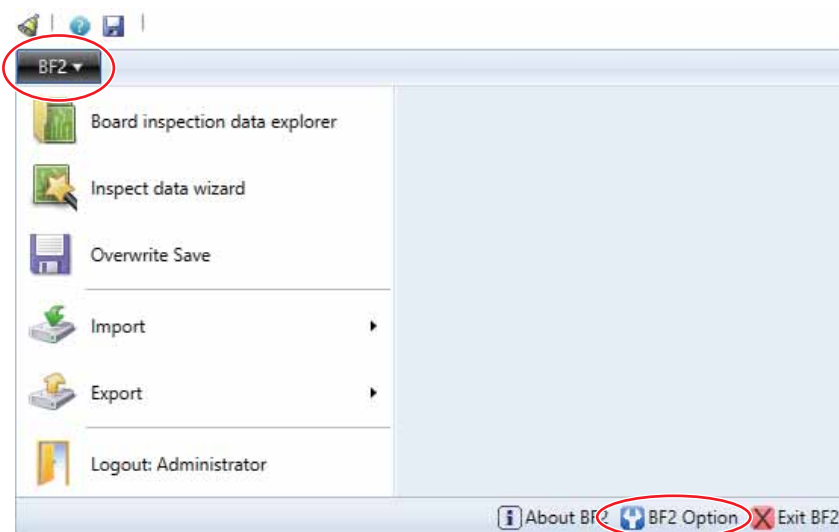


Figure 3-13 Past Board View Settings 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

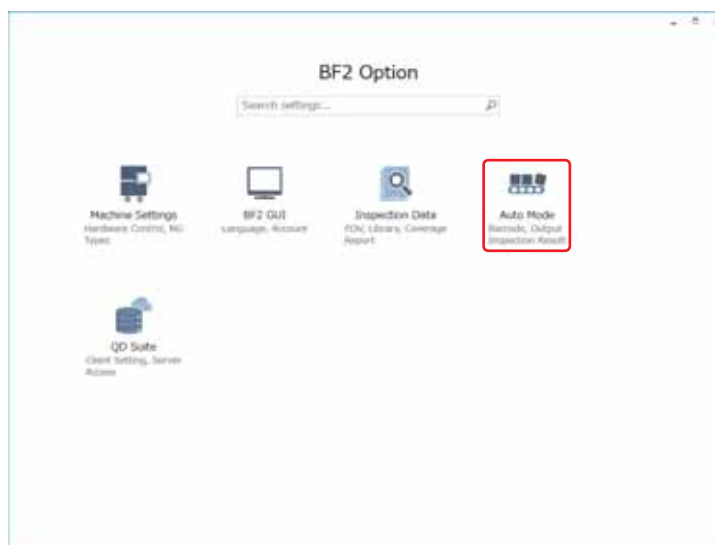


Figure 3-14 Past Board View Settings 2

Step3: Select **Output Inspection Result** and set toggle switch of **Stand Alone SPC** to ON.

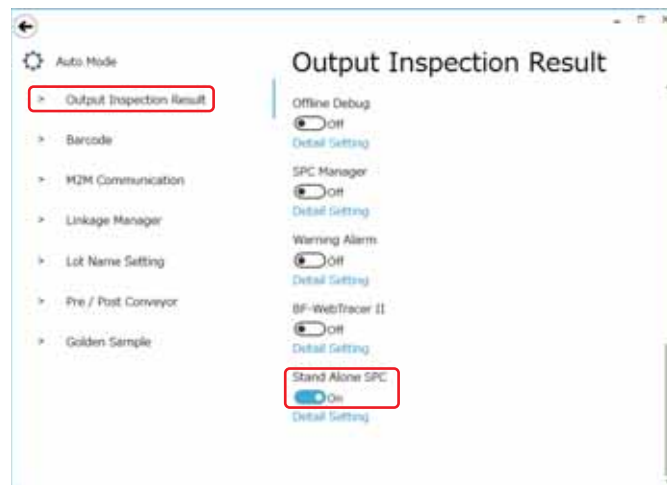


Figure 3-15 Past Board View Settings 3

Step4: Click **Detail Setting** under the toggle switch of **Stand Alone SPC** to set detailed items. Proceed the setup procedures, if necessary. For details refer to Table 3-14. After all the settings are completed, click  on the upper right side of the dialog box and click Yes to close dialog box.

Item	Description
Enable Output Database	When checked, enables the Past Board View function.
Enable Trend View	When checked, enables the Trend View function.
Database Path	Set the saving data folder path. For the destination folder of the data, specify the folder on disk which has enough space.
Disk Space Limit[GB]	The disk space is checked at the time of starting the Auto inspection. If the specified space cannot be checked, the warning dialog box appears. Unit is [GB].
Output Images	When checked, enables the Output NG Images.
Delete expired date	Deletes data that has passed a specified number of days.
Interval between checks [days]	Checks for expired data at a specified interval (day).
EU Coordinate System	If enabled, data are recalculated to correspond with the EU coordinate system. + Y axis is in the upper direction.

Table 3-14 Descriptions

3.4 Operation Procedure

This section describes the operation procedure of Past Board View.

Step1: Start the Auto inspection. For the procedure to start the Auto inspection, refer to **Operation Manual**.

NOTE

No need to start the Auto inspection to check the data that has already been saved. Proceed to Step2.

Step2: From the **Auto** tab, select the **StandAloneSPC** group > **Past Board View**.

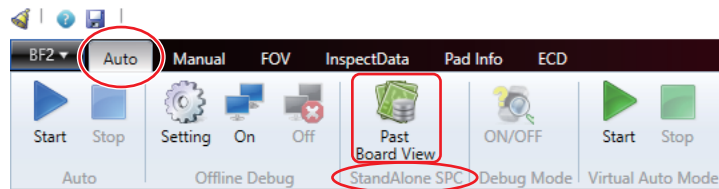


Figure 3-16 Past Board View Setting

Step3: The dialog box shown below appears.

Specify search criteria in the **Board List** window and click the **Refresh** button. The searched board data is listed.



Figure 3-17 Past Board View Dialog Box

Step4: To check the aggregate results in **Yield Chart**, click the **Yield Chart** icon (📊).

The screen switches to the **Yield Chart** window and the information appears.

Yield Chart Icon



Figure 3-18 Display Procedure of Yield Chart

Step5: To check the board data in **Board View**, select one item of board data from **Board List** and then click the **Board View** icon () or double-click it on **Board List**.

The **Board View** window is selected and the selected board information appears.

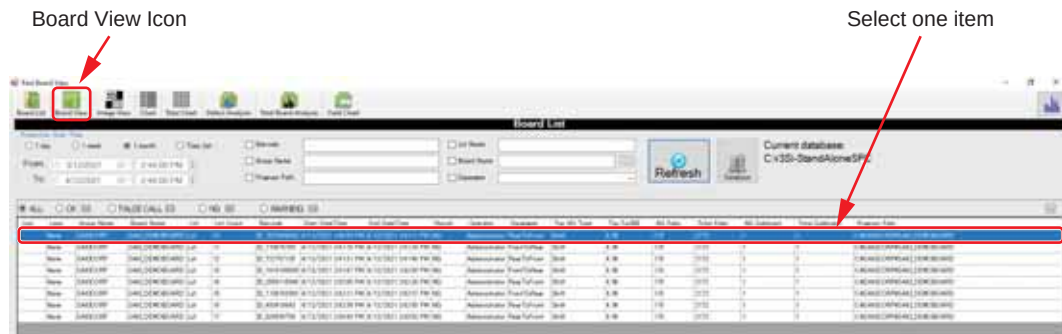


Figure 3-19 Display Procedure of Board View

Step6: To check the aggregate results in **Chart**, **Total Chart**, **Defect Analysis**, or **Total Board Analysis**, select multiple board data from **Board List** and then click the **Chart** icon (), the **Total Chart** icon (), the **Defect Analysis** icon (), or the **Total Board Analysis** icon ().

The window is switched to the **Chart**, **Total Chart**, **Defect Analysis**, or **Total Board Analysis** window and the selected board information appears.

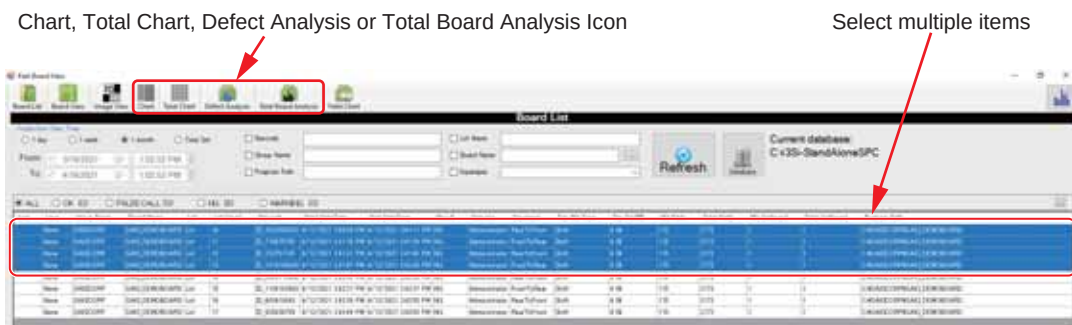


Figure 3-20 Display Procedure of Chart, Total Chart, Defect Analysis, or Total Board Analysis

3.5 Data Backup Procedure

This section describes data backup procedure saved by Stand Alone SPC function.

CAUTION

Continuing to save data reduces the disk space. It is recommended to make the backup as necessary.

Step1: Open the folder specified in database saving settings by explorer. Stand Alone SPC data is saved in folders for each month.

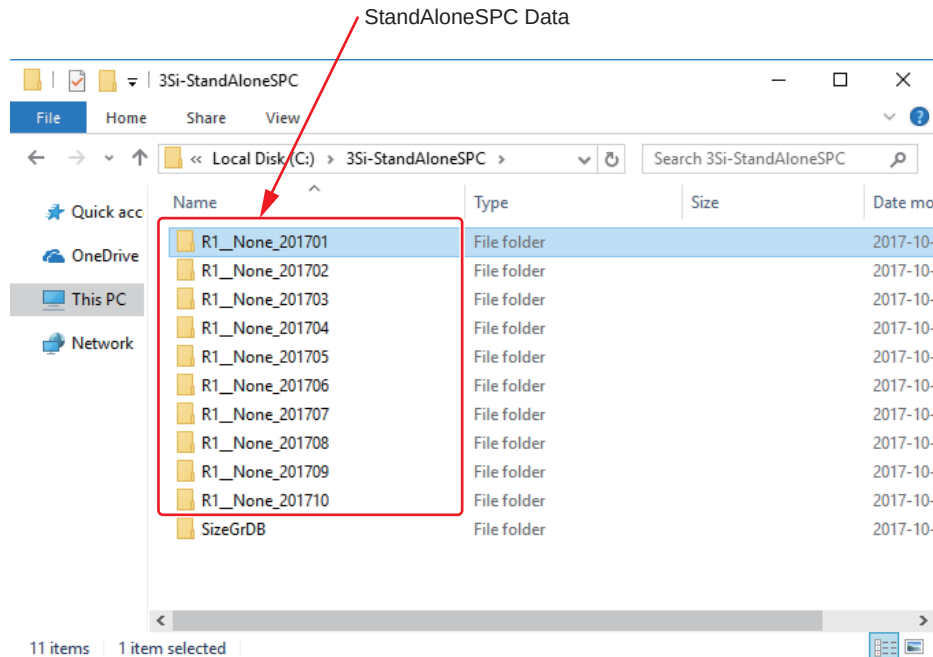


Figure 3-21 Stand Alone SPC Data Folder

Step2: Select folders whose dates are old and make a copy to backup disk. After the copy is completed, delete the copied folder from the saving folder.

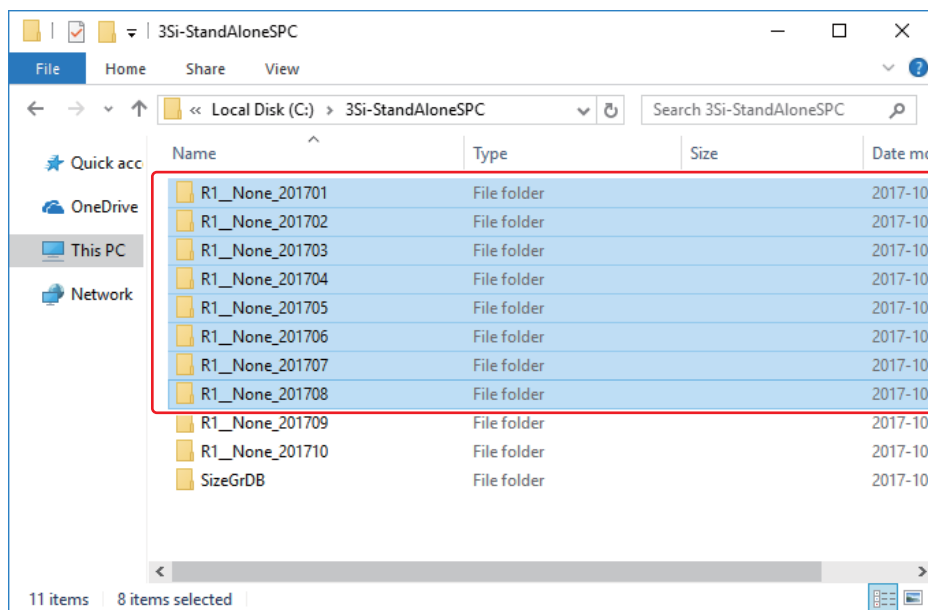


Figure 3-22 Backup

Other Functions

1 Making an Inspection Data on BF2-Editor

The image taken by inspection machines can be transferred to BF2-Editor to create inspection datas.

1.1 Transferring a Data to BF2-Editor

Step1: Scan a good PCB on an inspection machine according to **Part II 5 Acquiring a PCB image**.

Step2: From the dock window, select **Image File List > Add current image**.

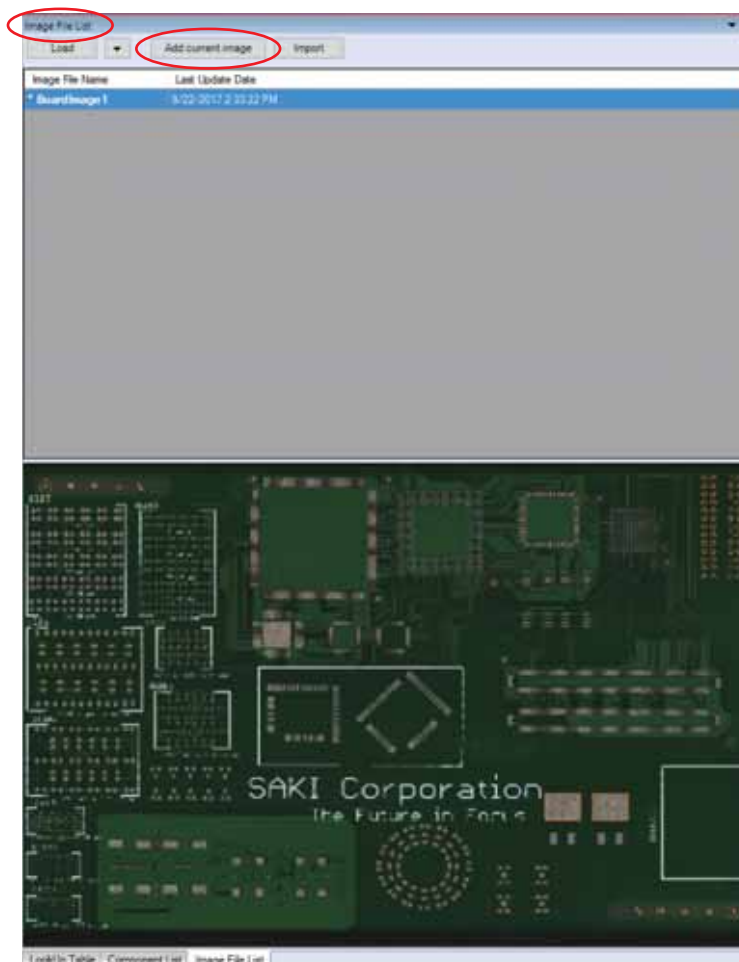


Figure 1-1 Add Current Image

Step3: Enter a desired file name and click **OK**.

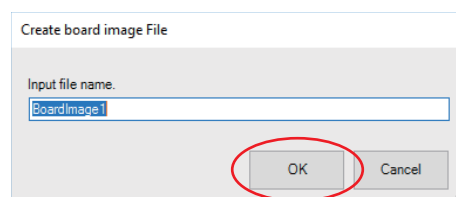


Figure 1-2 Entering the File Name

Step4: The PCB image is displayed on the lower side of the dock window.

When Transferring Only the Image File

Click  on the right side of **Load** and select **Export** from the drop-down list.

The file saving dialog box is displayed.

Save the data to a place where BF2-Editor can access or to an external device.

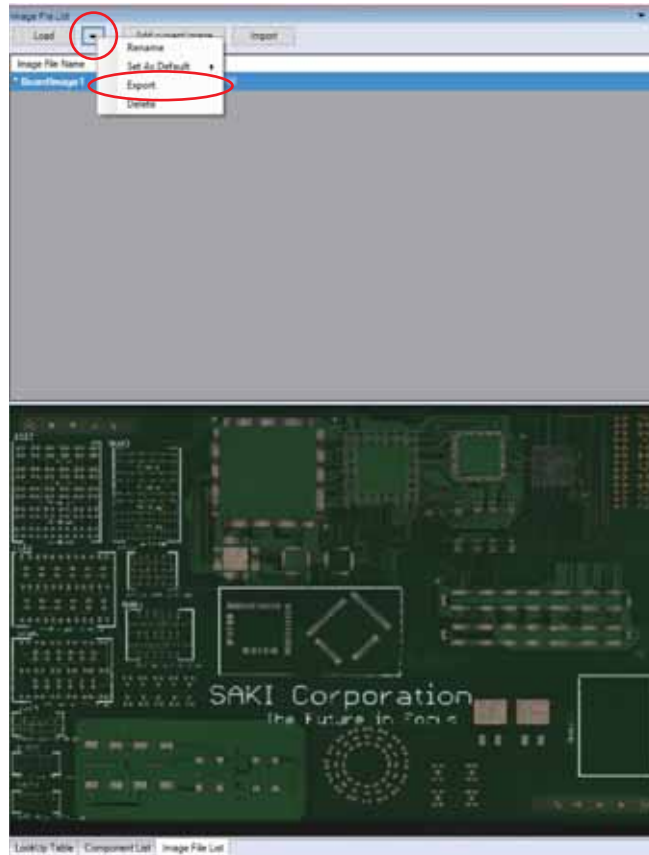


Figure 1-3 Export

When Transferring the Inspection Data

Click the **BF2** button on the upper left side of the window and select **Export > Copy Group**.

The dialog box shown in Figure 1-5 appears.

In this dialog box, all board data which is included in the selected group are displayed.

Select the board data to be transferred and click **Next**.

Save the data to a place where BF2-Editor can access or to an external device.

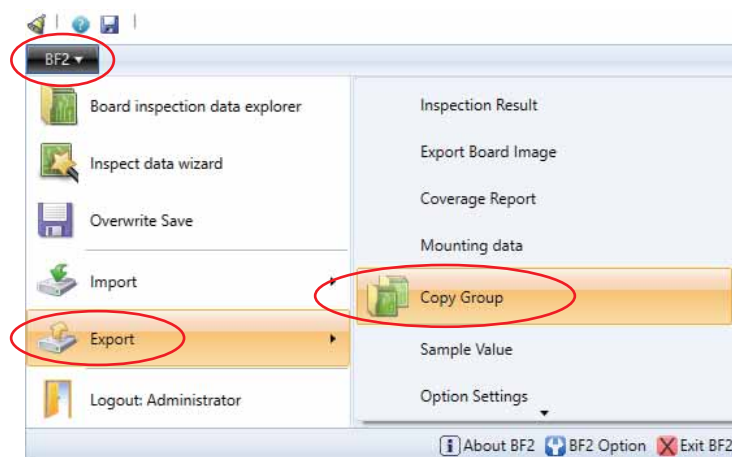


Figure 1-4 Copy Group 1

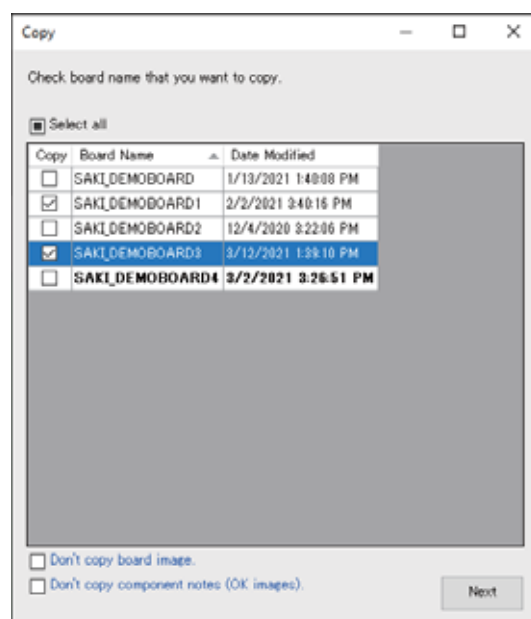


Figure 1-5 Copy Group 2

1.2 Making/Editing an Inspection Data on BF2-Editor

Procedures for making/editing an inspection data on BF2-Editor is the same as the procedure on inspection machine.

Step1: Save the inspection data or image file to the place where BF2-Editor can access or the external device according to **1.1 Transferring a Data to BF2-Editor**.

Step2: **When Only Editing the Inspection Data in BF2-Editor**

Select an inspection data according to **Part I 1.3 Selecting an Inspection Data File**.

Edit the inspection data according to **Part II 2 Edit Data**.

When Making the Inspection Data in BF2-Editor

Make the inspection data according to **Part II 1 Making an Inspection Data**.

NOTE

Scan is not displayed when loading a PCB image, because BF2-Editor is not capable of scanning the PCB. Select **Load Board Image** and load the file saved in Step1.

Step3: To edit the inspection data which is made/edited in BF2-Editor in an Inspection machine, save the inspection data to a place where the inspection machine can access or an external device according to step4 of **1.1 Transferring a Data to BF2-Editor**.

2 SPC Function

SPC (Statistical process control) is the method that controls the manufacturing process statistically. By using SPC, the quality of products can be maintained, and the manufacturing process can be controlled and improved.

2.1 Setting of SPC Function

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

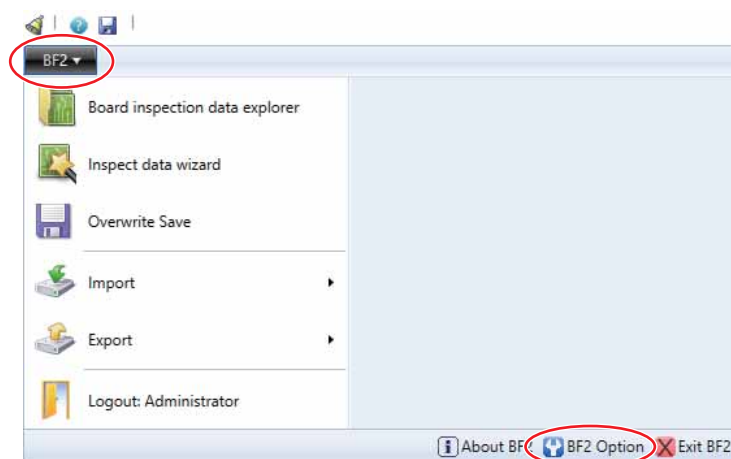


Figure 2-1 Setting of SPC Function 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

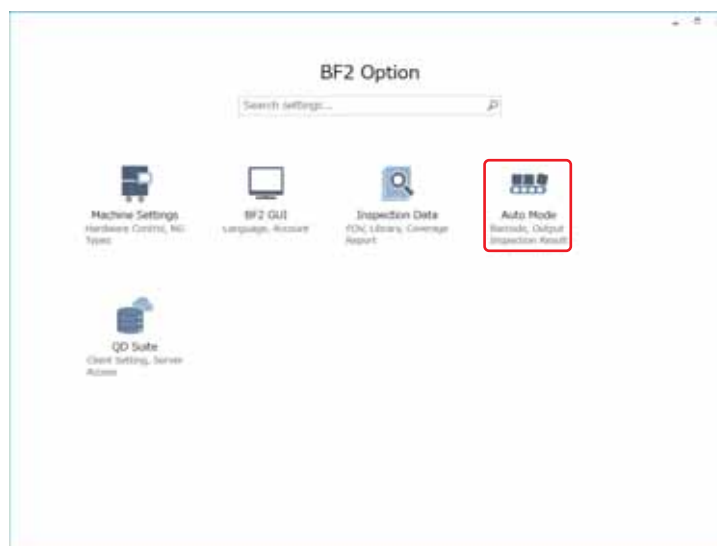


Figure 2-2 Setting of SPC Function 2

Step3: Select **Output Inspection Result** and set toggle switch of **SPC Manager** to **ON**.

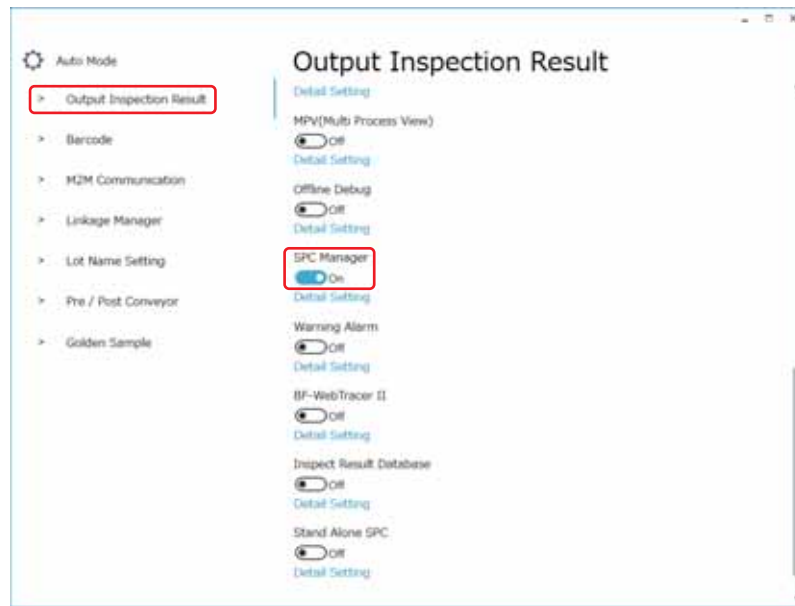


Figure 2-3 Setting of SPC Function 3

Step4: Click **Detail Setting** under the toggle switch of **SPC Manager** to set detailed items.

Enable a desired function. Refer to the table below for detail settings.

After all the settings are completed, click  on the upper right side of the dialog box and click Yes to close dialog box.

Item	Description
Inspection Sample Result Enable	The sample value is output as the csv file format to the Inspection Sample Result Folder .
Inspection Sample Result Folder	Sets a file location for the inspection sample result to be stored.
Simple Inspection Result Enable	The inspection result file is output as the xml file format to the Simple Inspection Result Folder .
Simple Inspection Result Folder	Sets a file location for the inspection sample result to be stored.
Enable SPC Window	The SPC window is updated during Auto Mode. The SPC window can be verified with the Auto tab. Displays the statistical information calculated from the inspection result of Auto Mode. For details, refer to 2.2 SPC Window .
Enable SPC Detail Window	The SPC Detail Window is updated during Auto Mode. The SPC Detail Window can be verified with the Auto tab. Displays the number of OK, NG, False Call and Total by unit of board, sub board unit, and pad unit. For details, refer to 2.2 SPC Window .
SPC Window Directory	Sets a file location for the SPC item result file to be stored.
SPC Window List	Sets the maximum display size of components, pads, and recipes displayed in the worst ranking of the SPC window. Default value is 15 .

Table 2-1 Setting of SPC Function

2.2 SPC Window

SPC Window and **SPC Detail Window** displays the statistical information calculated from the inspection results of **Auto Mode**.

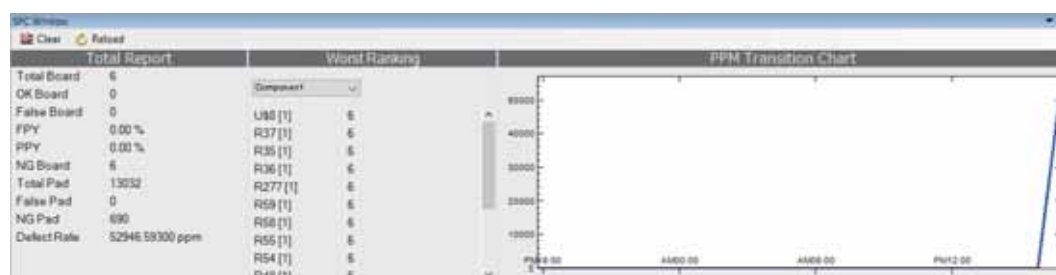
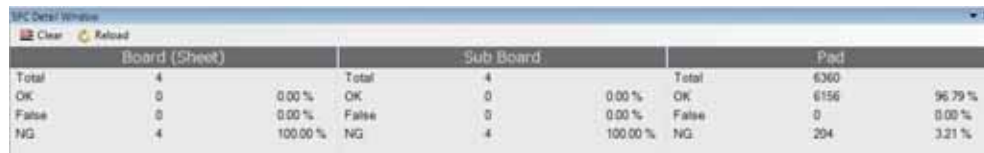


Figure 2-4 SPC Window

Item		Description
Clear		Clears the SPC Window Informations.
Reload		Updates the SPC Window Informations.
Total Report	Total Board	Shows the total number of the inspected PCBs.
	OK Board	Shows the total number of the OK PCBs.
	False Board	Shows the total number of boards judged to be excessive in Debug Mode and in Real-time Judgement.
	FPY	Shows the percentage of the number of OK boards out of all the boards. This is calculated with “(number of OK boards / total number of boards) x 100”.
	PPY	Shows the percentage of the number of OK boards and of the number of boards judged to be excessive in the total amount of boards. This is calculated with “((number of OK boards + number of boards judged to be excessive) / total number of boards) x 100”.
	NG Board	Shows the total number of the NG PCBs.
	Total Pad	Shows the total number of pads.
	False Pad	Shows the total number of pads judged to be falescall in Debug Mode and in Real-time Judgement.
	NG Pad	Shows the total number of NG pads.
Worst Ranking	Defect Rate	Shows the percentage of NG pads from the total number of pads. Calculated with “(number of NG pads / total number of pads) x 1000000”.
	Component	Shows the components, pads, and recipes at which many NGs were judged.
	Pad	
	Recipe	
PPM Transition Chart		Shows the trend of the percentage of defective boards through the course of time.

Table 2-2 SPC Window



Board (Sheet)			Sub Board			Pad		
Total	4		Total	4		Total	6360	
OK	0	0.00 %	OK	0	0.00 %	OK	6156	96.79 %
False	0	0.00 %	False	0	0.00 %	False	0	0.00 %
NG	4	100.00 %	NG	4	100.00 %	NG	204	3.21 %

Figure 2-5 SPC Detail Window

Item		Description
Clear		Clears the SPC Detail Window Informations.
Reload		Updates the SPC Detail Window Informations.
Board(Sheet)	Total	Shows the total number of the inspected PCBs.
	OK	Shows the total number of the OK PCBs.
	False	Shows the total number of PCBs judged as false call in Debug Mode and in Real-time Judgement.
	NG	Shows the total number of the NG PCBs.
Sub Board	Total	Shows the total number of the inspected sub boards.
	OK	Shows the total number of the OK sub boards.
	False	Shows the total number of sub boards judged as false call in Debug Mode and in Real-time Judgement.
	NG	Shows the total number of the NG sub boards.
Pad	Total	Shows the total number of the inspected pads.
	OK	Shows the total number of the OK pads.
	False	Shows the total number of pads judged as false call in Debug Mode and in Real-time Judgement.
	NG	Shows the total number of the NG pads.

Table 2-3 SPC Detail Window

3 Save Whole Board Image Function

The Save whole board image function is a function to outputs whole images of boards in Auto Mode.

3.1 Save Whole Board Image Setting

Step1: Click **BF2** button on the upper left side of the window to open the menu and click **BF2 Option**.

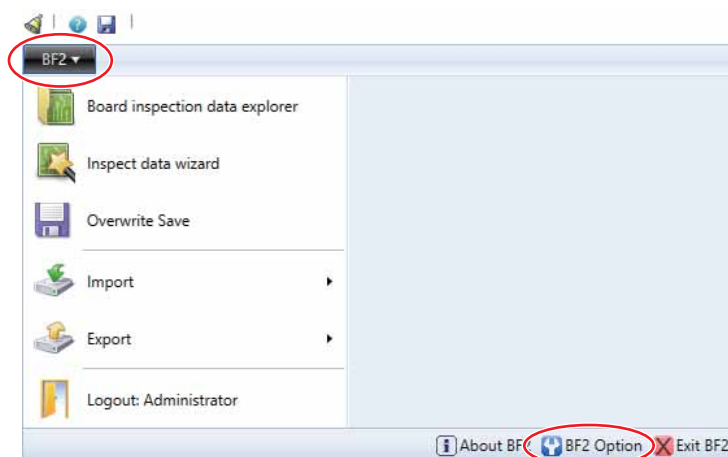


Figure 3-1 Whole Board Image Save Setting 1

Step2: The dialog box shown below appears.
Click **Auto Mode**.

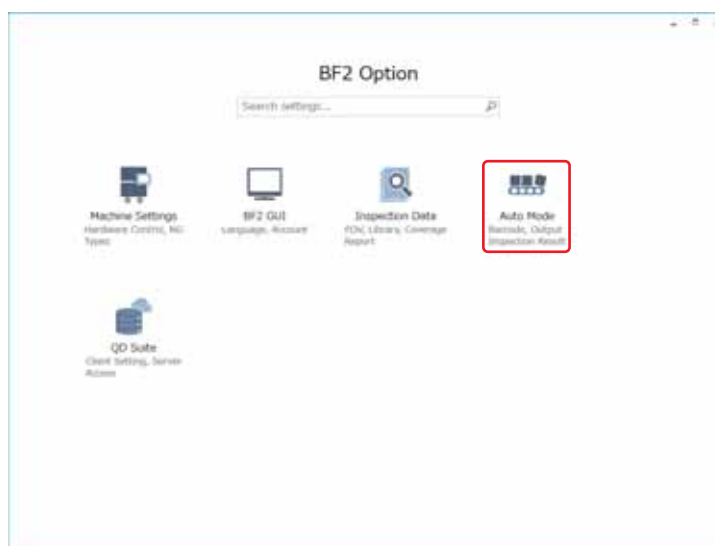


Figure 3-2 Whole Board Image Save Setting 2

Step3: Select **Output Inspection Result** and set toggle switch of **Whole Board Image Output** to **ON**.

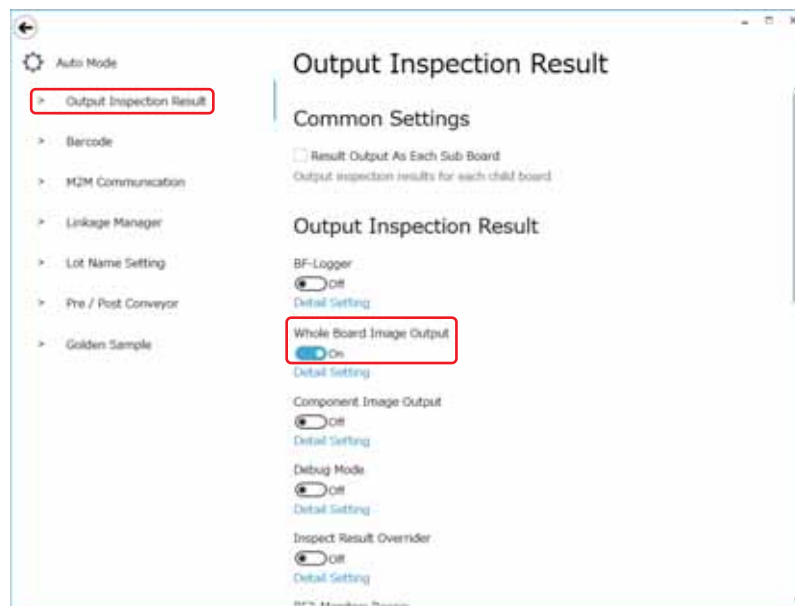


Figure 3-3 Whole Board Image Save Setting 3

Step4: Click **Detail Setting** under the toggle switch of **Whole Board Image Output** to set detailed items. Proceed the setup procedures, if necessary. Refer to the table below for detail settings.

After all the settings are completed, click  on the upper right side of the dialog box and click **Yes** to close dialog box.

Item	Description
Output Path	Set output destination of the image file.
File name	Set the image file output format. Select Board manage code to outputs the board manage code as the file name. Select Outputted time to outputs the outputted time as the file name. Select Group name_Board name_Board manage code_Outputted time to outputs Group name_Board name_Board manage code_Outputted time as the file name.
File type	Set the image file output format. Select Jpeg or Bmp . bding(BF2 Image) is an image used in inspection. bding(BF2 Image) image size is large so it takes time for output.
Lot name sub-folder	Create sub-folder of lot name.
Enable symbol output	Check to output after replacing only symbols that cannot be used in the file name due to OS restrictions. e.g., Original character string "B ¥ 12_3" Output file name "B_{BS} 12_3" Uncheck to output after replacing symbols that cannot be used in BF2. e.g., Original character string "B ¥ 12_3" Output file name "_{B_BS12_UB3}"
Compression rate [%]	Set the image file compression rate. If increase this value, compression rate also increase so that the output time will be reduced. Default value is 20[%] .
Save timing	Sets the output setting of the Save whole board image function. Select All boards to outputs fullscreen images of each board.
Output once in	Select Interval time to outputs fullscreen images at the Output once in by minutes. Select Interval boards to outputs fullscreen images at the Output once in by images.

Table 3-1 Descriptions

4 PAD Operation

4.1 Adding PAD Manually

This section describes how to add pads manually.

Step1: From the **Pad Info** tab, select the **Edit pads** group > **Add pads**.

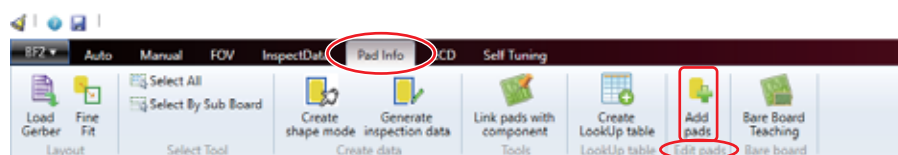


Figure 4-1 Adding Pads Manually 1

Step2: Set **Component Addition Mode** and **Shape** according to Table 4-1.
After all settings are complete, click where to add pads.

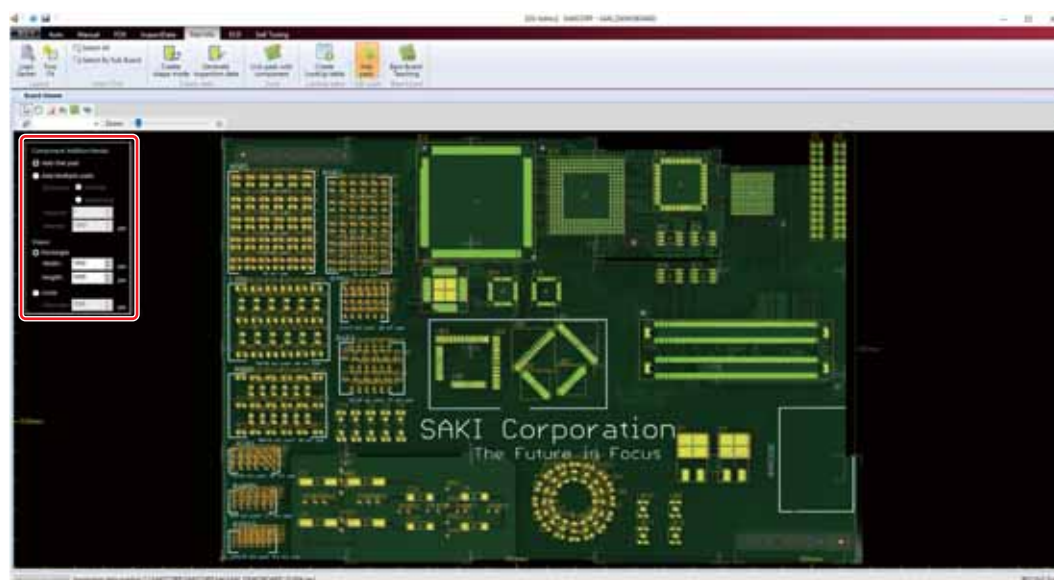


Figure 4-2 Adding Pads Manually 2

Item		Description
Component Addition Mode	Add One pad	Adds a single pad.
	Add Multiple pads	Adds more than one pad. Select to set the following items. Direction: Select the direction to add pads from Vertical or Horizontal . Number : Set the number of pads to add. Interval : Set the distance between pads to add.
Shape	Rectangle	Set Width and Height of the pad to add.
	Circle	Set Diameter of the pad to add.

Table 4-1 Descriptions

5 Exporting and Importing Option Settings

By exporting or importing option settings on other machines, it makes easier to share and compare option settings between machines.

NOTE Even if option settings are imported, the machine-specific settings (e.g., calibration information) are not reflected on other machines.

5.1 Exporting Option Settings

This section describes the procedure for exporting option settings.

Step1: Click **BF2** button on the upper left side of the window to open the menu and select **Export > Option Settings**.

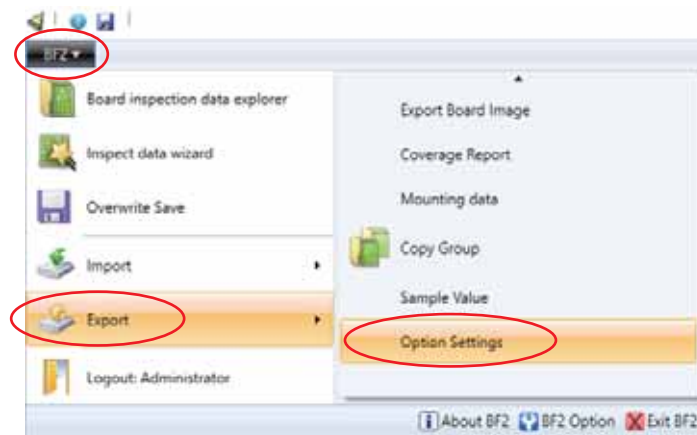


Figure 5-1 Exporting Option Settings 1

Step2: The dialog box shown below appears.
Check the option setting items to export and click **OK**.

NOTE Click the ☐ button on the upper left side of the window to show the settings for checking.
For details on the settings for checking, refer to Table 5-1.

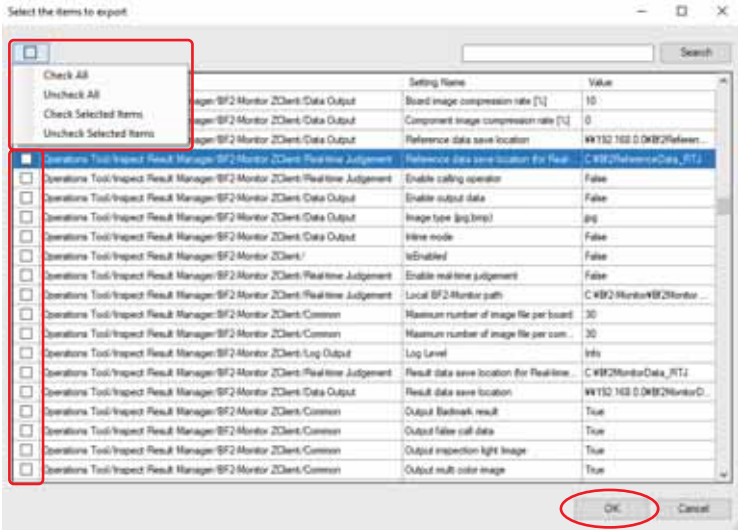


Figure 5-2 Exporting Option Settings 2

Item	Description
Check All	Checks all option settings.
Uncheck All	Unchecks all option settings.
Check Selected Items	Checks the selected option settings only.
Uncheck Selected Items	Unchecks the selected option settings only.

Table 5-1 Settings for Checking

Step3: The dialog box shown below appears.
Select the exporting destination and click **Save** to export files.

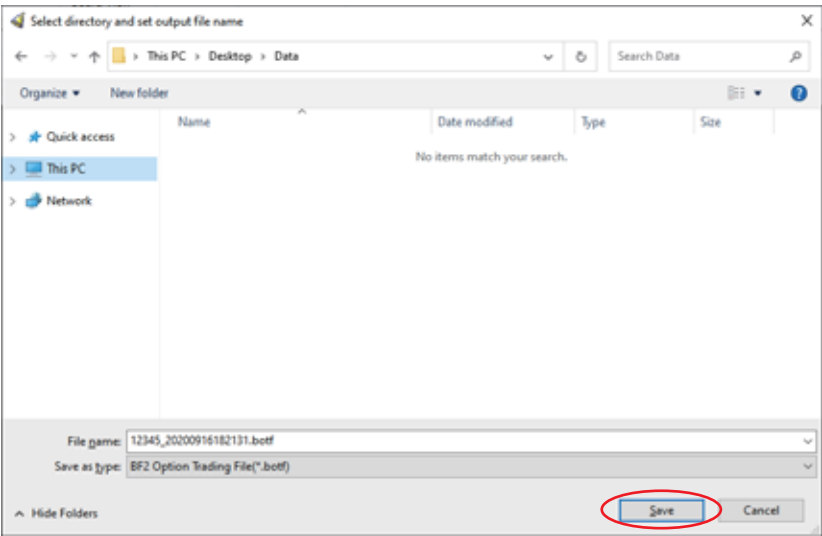


Figure 5-3 Exporting Option Settings 3

5.2 Importing Option Settings

This section describes the procedure for importing option settings.

Step1: Click **BF2** button on the upper left side of the window to open the menu and select **Import > Option Settings**.

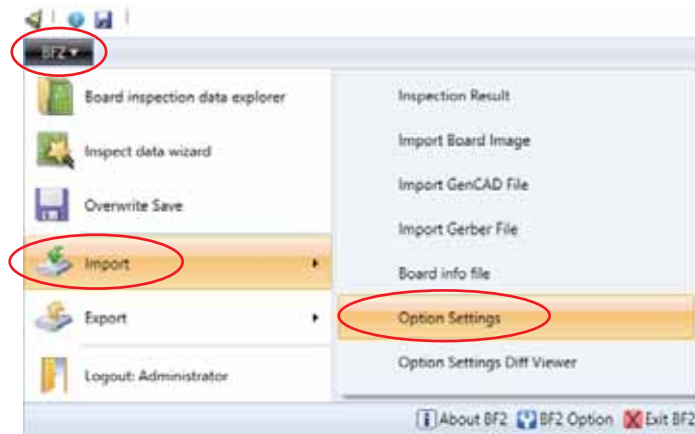


Figure 5-4 Importing Option Settings 1

Step2: The dialog box shown below appears.
Select the file exported in **5.1 Exporting Option Settings** and click **OK**.

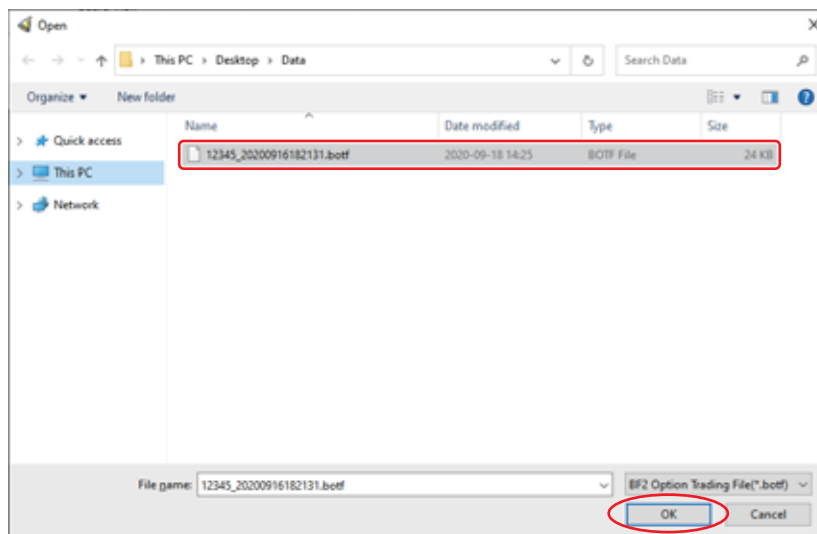


Figure 5-5 Importing Option Settings 2

Step3: The dialog box shown below appears.

Own option settings are shown in **Current Value**, and the option settings recorded in the file selected in Step2 are shown in **New Value**.

Basic Information shows information that identifies machines, such as lane numbers and line names.

In the center of the screen, the difference information for the option settings is shown in orange.

Check the option setting items to import and click **OK**.

The option setting values of the checked items are replaced with **New Value**.

- CAUTION**
- Option settings cannot be imported between machines which models are different, such as AOI and SPI.
- NOTE**
- Machine-specific information, such as **Basic Information** and calibration parameters, is not imported.
- NOTE**
- Click the ☐ button at the top left of the window to show the settings for checking.
For details on the settings for checking, refer to Table 5-1.
- NOTE**
- Check **Identical** at the bottom left of the screen to show items with no difference in option settings in white.
Check **Read Only** at the bottom left of the screen to show items that cannot import option settings in gray.

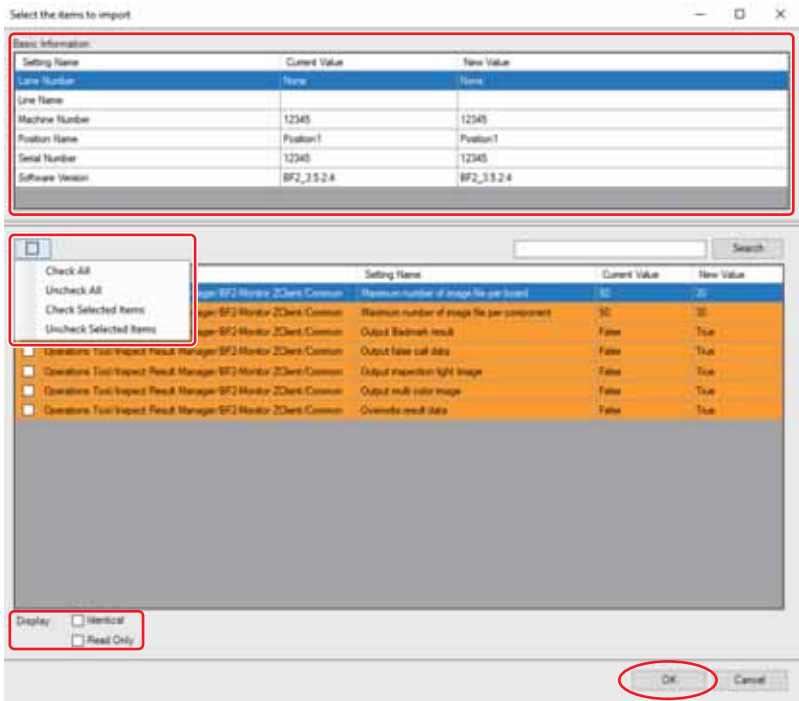


Figure 5-6 Importing Option Settings 3

5.3 Option Settings Difference View

This section describes the procedure to view differences in option settings.

Step1: Click **BF2** button on the upper left side of the window to open the menu and select **Import > Option Settings Diff Viewer**.

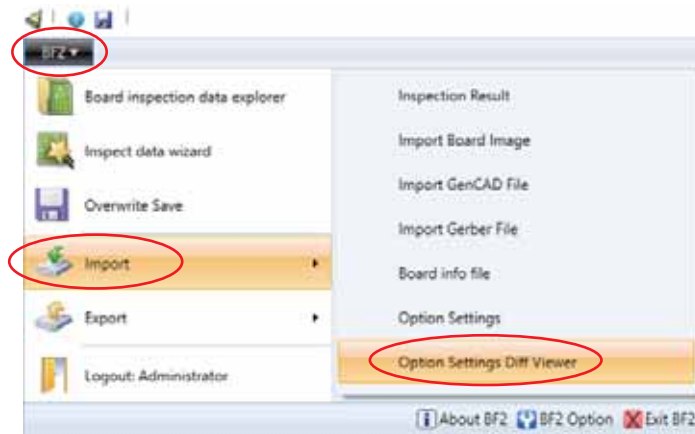


Figure 5-7 Option Settings Difference View 1

Step2: The dialog box shown below appears.
Click the  button on the right of **Right**.

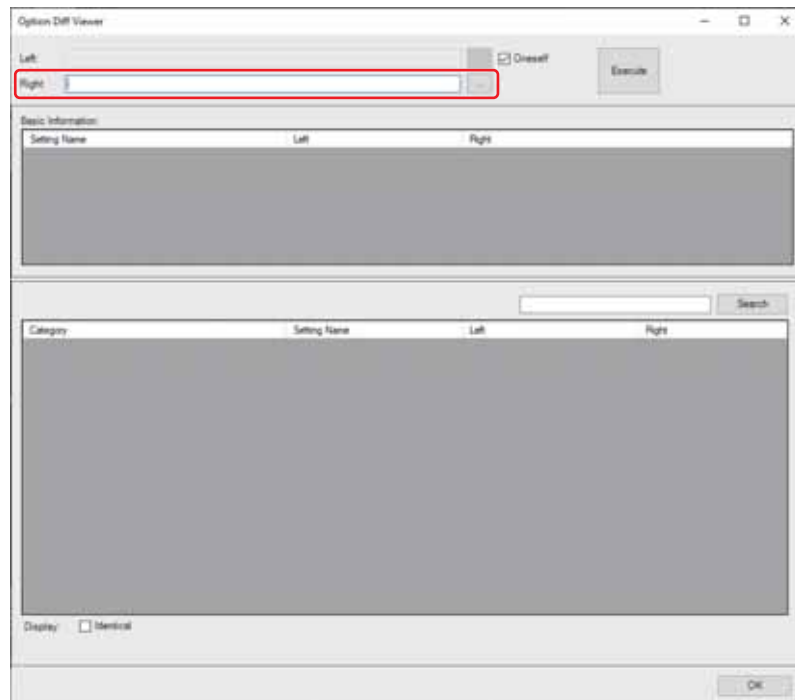


Figure 5-8 Option Settings Difference View 2

Step3: The dialog box shown below appears.

Select the file exported in **5.1 Exporting Option Settings** and click **OK**.

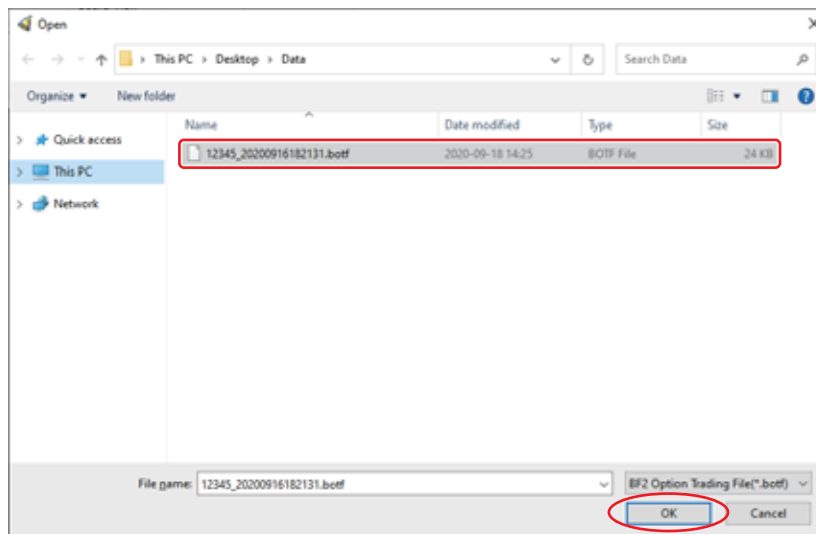


Figure 5-9 Option Settings Difference View 3

Step4: Click **Execute** to compare the option settings specified in **Left:** and the option setting specified in **Right:**. The difference information is shown in orange.

CAUTION The option setting difference information cannot be between machines which models are different, such as AOI and SPI.

NOTE **Oneself** is checked for **Left:** by default, but if **Oneself** is unchecked, any option setting file can be selected.

NOTE Check **Identical** at the bottom left of the screen to show items with no difference in option settings in white.

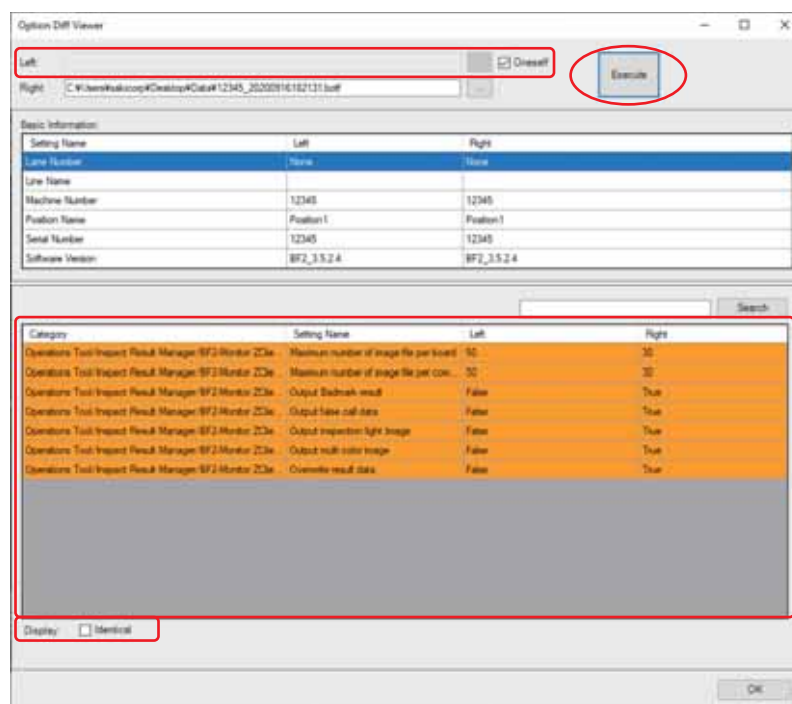


Figure 5-10 Option Settings Difference View 4

6 GR&R

6.1 Overview

GR&R is used to check whether measurement results meet the standards when several measurers inspect. The investigation items are as follows.

- Repeatability: Variation level in the measurement system caused by measurement machines
- Reproducibility: Variation level in the measurement system caused by differences between measurers
- Variation volume (small or not) in the measurement system compared to process variation
- Ability of the measurement system to identify differences between different components

GR&R analysis indicates whether inspection results of the same component by each measurer are consistent (repeatability) and whether the variation between measurers is consistent (reproducibility).

This function allows the GR&R analysis with the ANOVA and Type 1 methods.

6.1.1 ANOVA

Two-way ANOVA tests the difference in means between two factors.

For example, whether there is a difference between groups of each factor can be tested, such as two groups of **measurer A** and **measurer B**, and three groups of **first**, **second**, and **third** groups of boards.

This function evaluates the value of two-way ANOVA by applying it to GR&R.

The formulas are as follows.

Repeatability	$EV = \sqrt{\text{Error variance}}$
---------------	-------------------------------------

Reproducibility	$AV = \sqrt{\frac{\text{Measurer variance} - \text{Error variance}}{\text{Component count} \times \text{Repetition count}}}$
-----------------	--

Component Variability	$PV = \sqrt{\frac{\text{Component variance} - \text{Error variance}}{\text{Repetition count} \times \text{Measurer count}}}$
-----------------------	--

Repeatability/Reproducibility	$GRR = \sqrt{EV^2 + AV^2}$
-------------------------------	----------------------------

Total Variation	$TV = \sqrt{GRR^2 + PV^2}$
-----------------	----------------------------

The acceptance criteria for GR&R are as follows.

GR&R < 10%	: OK
10% ≤ GR&R < 30%	: WARNING
30% ≤ GR&R	: NG

6.1.2 Type 1

For Type 1, one measurer measures a single reference component several times so that it can evaluate the effect of bias (*1) and repeatability (*2) on the reference component measurements.

NOTE

(*1) Bias analyzes with null hypothesis t-test, where there is no bias.

Type 1 also evaluates gauge bias, which is the difference between the average measurement of the gauge and the official reference value (the component specification value).

Basically, this test is 1-sample t-test that judges whether mean measurement is significantly different from reference value. If there is a statistically significant bias, measurements from that gauge are consistently larger or smaller than correct values.

This function also calculates the measurement criterion Cgk of the process capability in order to evaluate repeatability and bias together. Cgk compares reference variation to tolerance, but it also considers whether the measurement is **achieving the target**. As the difference between the average measurement and the reference value of the gauge increases, Cgk decreases. The Cgk value of 1.33 is a common benchmark value indicating gauges with measuring capability that combine precision (excellent repeatability) and accuracy (small bias).

(*2) Repeatability is the ability of a gauge to consistently measure the same component. Even the gauge with measuring capability has some variation in measurements. However if the variation is too large for the tolerance of the component, the measurement system can be too variable for use.

To evaluate the repeatability of the gauge, the Cg criterion is calculated and the reference variation (the spread of the gauge measurement value) is compared with the percentage of tolerance.

The Cg value greater than 1.33 indicates that the spread of the gauge measurement value is narrow enough for the tolerance range.

6.2 Collecting Data for GR&R Calculation

To statistically identify variation, two or more batches on the same board are created and each batch should be inspected more than once.

NOTE A batch is a **collection of a certain amount of data for a certain period of time**.

NOTE When calculating, each batch must have the same number of inspection attempts (the number of scans).

Step1: From the **Manual** tab, select the **Calibration** group > **Capability test**.

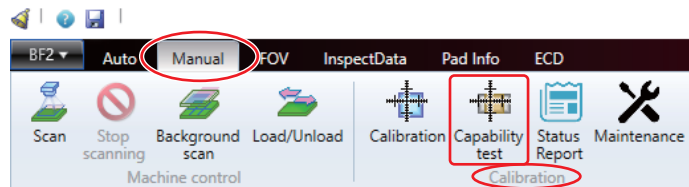


Figure 6-1 Capability Test

Step2: The dialog box shown below appears.
Click **Collect data for GR&R Calculation**.

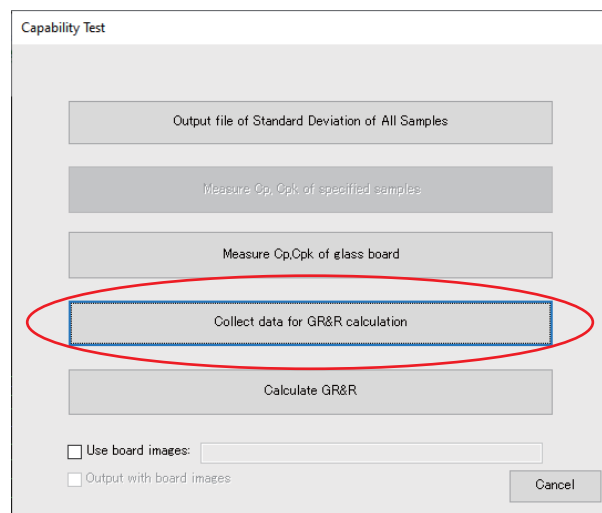


Figure 6-2 Collect Data for GR&R Calculation 1

Step3: The dialog box shown below appears.

For details, refer to Table 6-1.

After all settings are complete, click **START**.

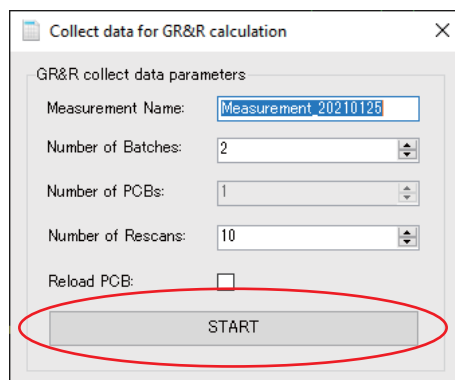


Figure 6-3 Collect Data for GR&R Calculation 2

Item	Description
Measurement Name	Enter a name to identify measurement data.
Number of Batches	Enter the number of batches.
Number of PCBs	Enter the number of boards used for evaluation. Default value is 1. This value cannot be changed. To set 2 or more, change Device Settings > GR&R > Allow enter number of PCBs to Enable in BF2 Option.
Number of Rescans	Enter the number of times to scan one board.
Reload PCB	Check to measure using the same board repeatedly.

Table 6-1 Descriptions

Step4: The dialog box shown below appears.

Enter a batch name and click **Continue**.

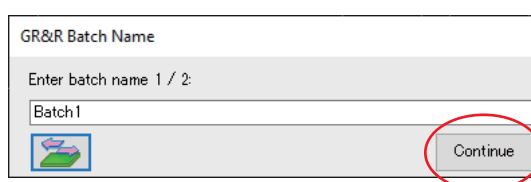


Figure 6-4 GR&R Batch Name

Step5: After all scans are complete, the **Method selection** dialog box appears.

To continue the GR&R calculation, refer to Step3 in **6.3 GR&R Calculation**.

6.3 GR&R Calculation

Step1: From the **Manual** tab, select the **Calibration** group > **Capability test**.

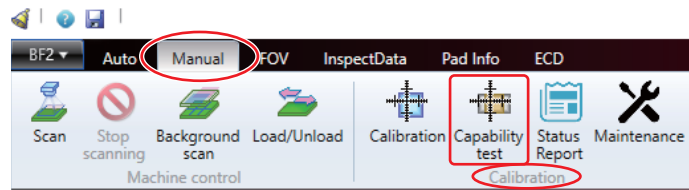


Figure 6-5 Capability Test

Step2: The dialog box shown below appears.

Click **Calculate GR&R**.

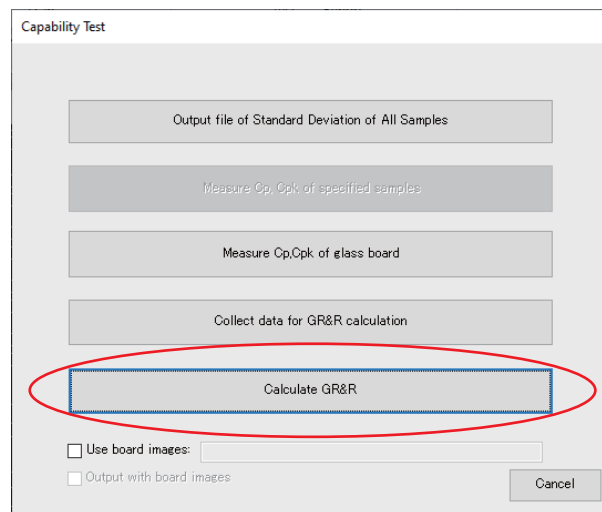


Figure 6-6 Calculate GR&R

Step3: The dialog box shown below appears.

To analyze with ANOVA, click **ANOVA**.

To analyze with Type 1, click **Type 1**.

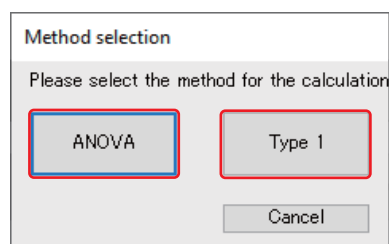


Figure 6-7 Method Selection

Step4: The dialog box shown below appears.

For details, refer to Table 6-2.

After setting is complete, click **CALCULATE**. Calculation results are shown on the right side of the screen.

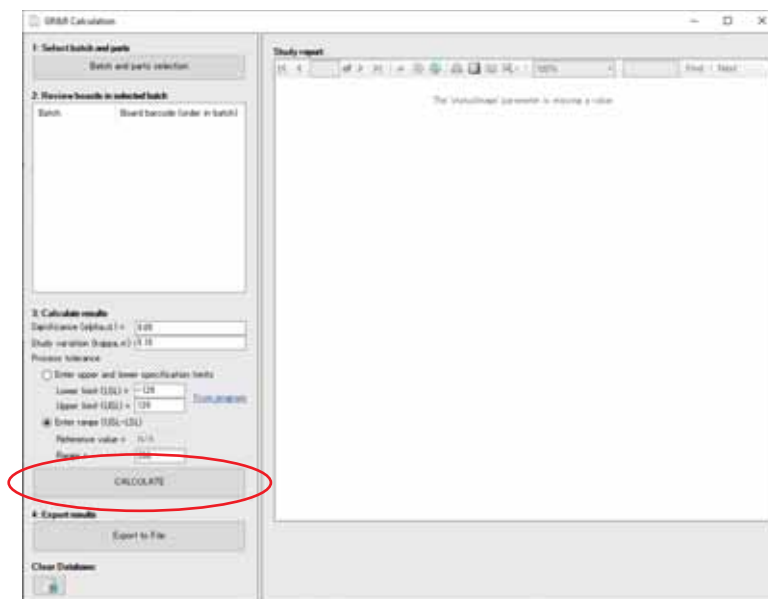


Figure 6-8 GR&R Calculation

Item		Description
1: Select batch and parts	Batch and parts selection	Click to show the GR&R Parts Selection screen. For details, refer to Figure 6-9.
2: Review boards in selected batch		Shows a list of boards in the batch selected on the GR&R Parts Selection screen. For details, refer to Figure 6-9.
3: Calculate results	Significance (alpha, α)	A significance level of 0.05 indicates that 5% is possible to conclude that the variation between batches is significant, even though the variation between batches is not significant. Default value is 0.005 .
	Study variation (kappa, K)	Multiplying the standard deviation can cover 99% of the standard deviation from the mean of the data. Default value is 5.15 .
	Lower limit (LSL)	Enter the lower acceptable limit.
	Upper limit (USL)	Enter the upper acceptable limit.
	From program	Click to use the OK range of an algorithm.
	Reference value	For the Type 1 analysis, enter the official reference value (the value that is the component specification).
	Range	Enter the difference between the standard limits if not calculating the discrimination error probability when there are both upper and lower acceptable limits.
	CALCULATE	Click to calculate.
4: Export results	Export to File	Exports calculation results to a file. Output formats:CSV (Comma Separated Values), DSV (Delimiter Separated Values), SSV (Semicolon Separated Values), TSV (Tab Separated Values).
Clear Database		Click to delete batch and component information in the database.
Study report		Shows CR&R calculation results.

Table 6-2 Descriptions

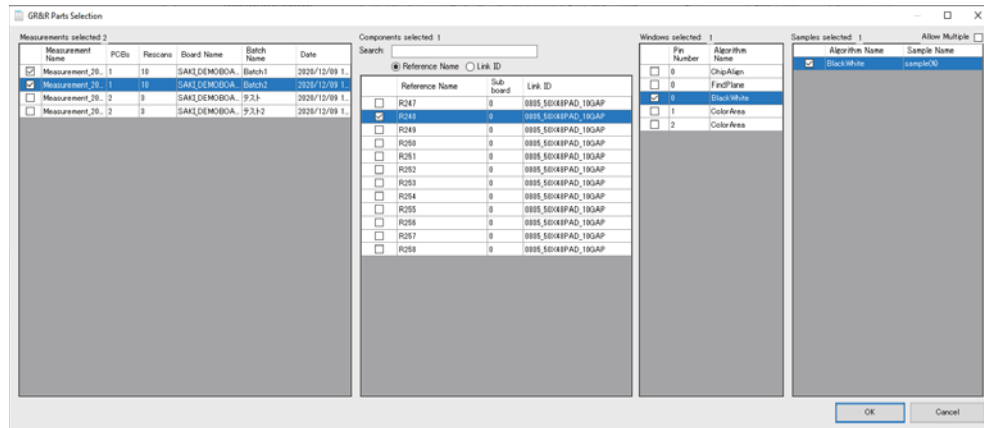


Figure 6-9 GR&R Parts Selection

Item	Description
Measurements selected	Select measurement data to use for the calculation.
Components selected	Select component data to use for the calculation.
Search	Searches for a component by Reference Name or Link ID.
Windows selected	Select an inspection window to use for the calculation.
Samples selected	Select samples to use for the calculation.
Allow Multiple	Check to select more than one sample.

Table 6-3 GR&R Parts Selection

Step5: Depending on the calculation method selected in Step3, the following dialog box will appear. When ANOVA is selected, the dialog box shown below appears. OK, WARNING, or NG is judged according to the number in the red frame. For details on the judgement criteria, refer to Table 6-4.

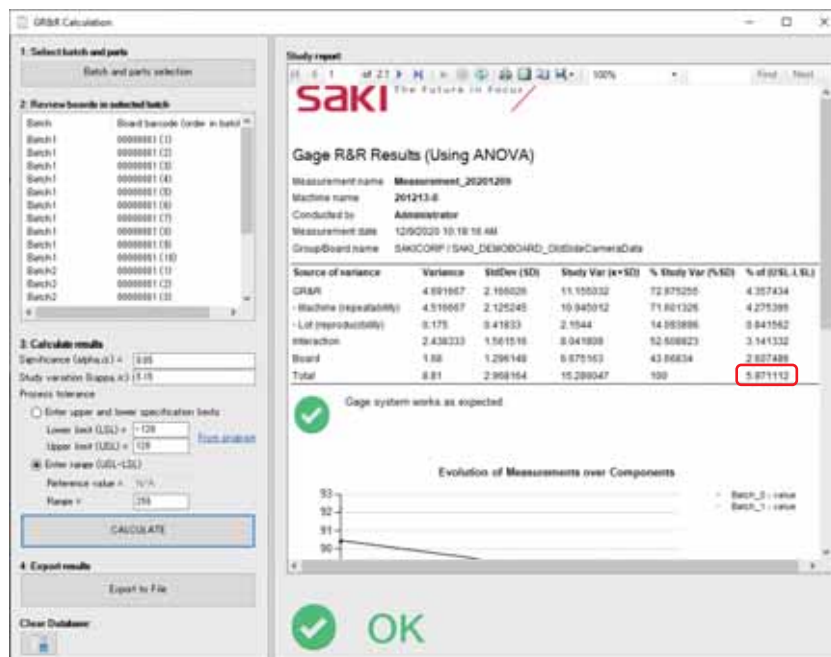


Figure 6-10 GR&R Calculation Results

Item	Description
OK	Judged as OK if the following condition is met. $\% \text{ of } (USL - LSL) < 10$
WARNING	Judged as WARNING if the following condition is met. $10 \leq \% \text{ of } (USL - LSL) < 30$
NG	Judged as NG if the following condition is met. $30 \leq \% \text{ of } (USL - LSL)$

Table 6-4 GR&R Judgement Criteria

When Type 1 is selected, the dialog box shown below appears.
OK or NG is judged according to the numbers in the red frames.
For details on the judgement criteria, refer to Table 6-5.

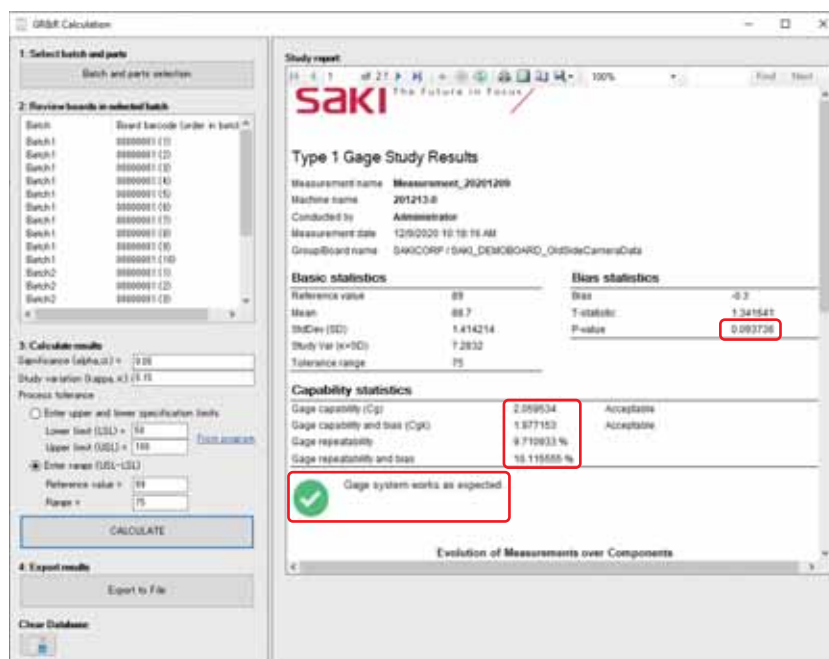


Figure 6-11 Type 1 Calculation Results

Item	Description
OK	Judged as OK if all of the following conditions are met. P-value $\geq$ Significance level (α) $Cg \geq 1.33$ or $Cgk \geq 1.33$ Gage repeatability ≤ 15 or Gage repeatability and bias ≤ 15
NG	Judged as NG if any of the following conditions is met. P-value $<$ Significance level (α) $Cg < 1.33$ or $Cgk < 1.33$ Gage repeatability > 15 or Gage repeatability and bias > 15

Table 6-5 Type 1 Judgement Criteria

Appendices and References

1 Appendix

1.1 Folder Structure of the PC

This section describes the folder structure of BF2.

CAUTION Do not move, delete, or rename files of the inspection software from the OS. It may result in damaging the inspection data.

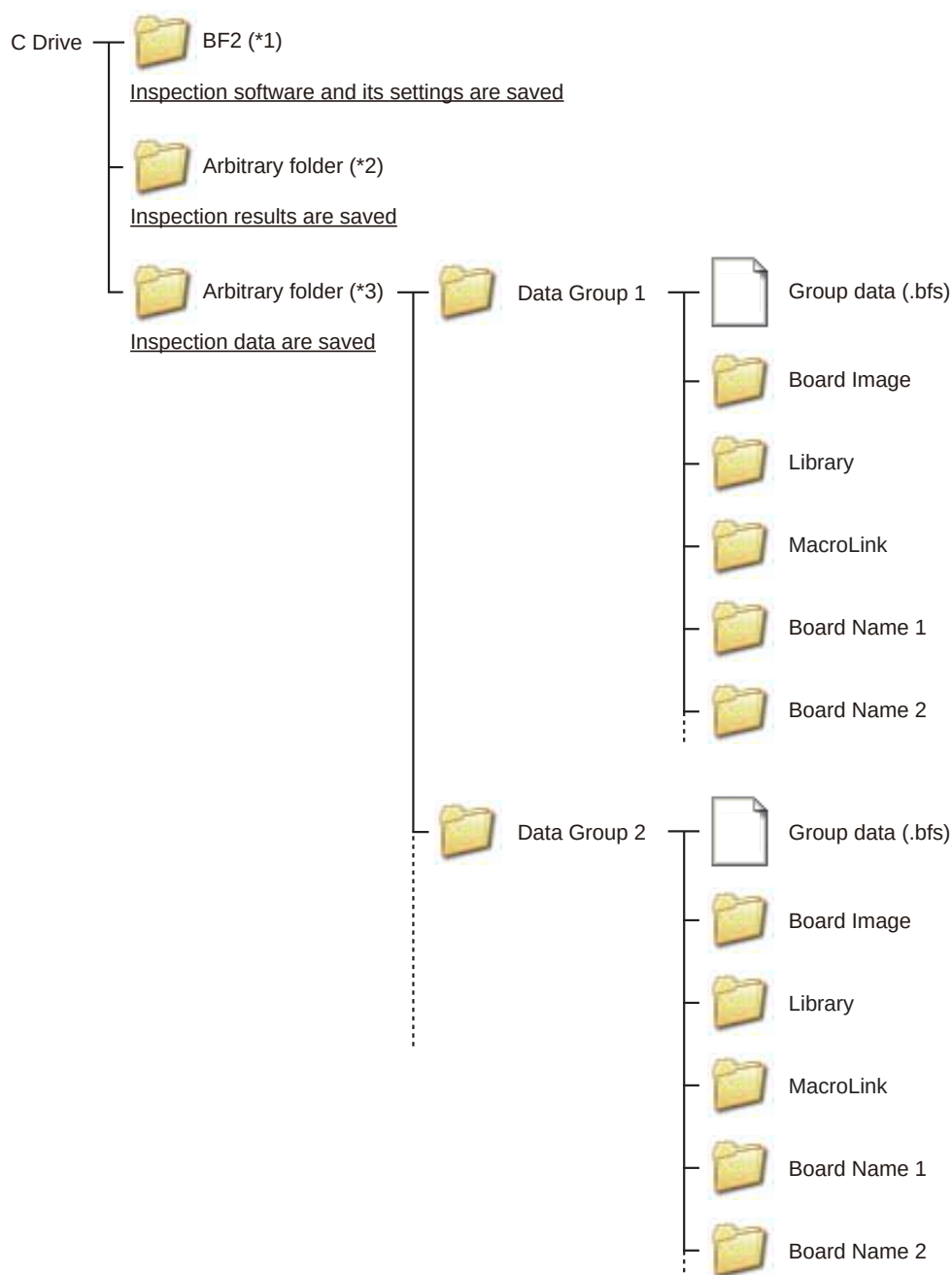


Figure 1-1 Folder Structure of the PC

NOTE

(*1) When using BF2-Editor, the folder name is **BF2-Editor**.

(*2) The data are saved in the folders which are set in **BF2 Options > Operations Tool > Inspect Result Manager > BF2-Monitor ZClient > Reference data save location** and **Result data save location**.

(*3) The data location is specified in an inspection data creation process.

1.2 Precautions When Using the Machine

CAUTION

If connecting an external memory device to the USB port on the machine, be sure to perform a virus-scan before connecting.

CAUTION

Do not use a two-byte character, space, tab, and symbolic character (\ < > : " / | ? * . , ; & % =). They may cause unexpected errors.

1.3 Warning Judgement

Can be specify **Warning Range** not only **OK Range**.

When sample value is in the Warning Range, judgement result is Warning.

Warning Range is specify in Recipe edit screen. Warning Range is determine as below.

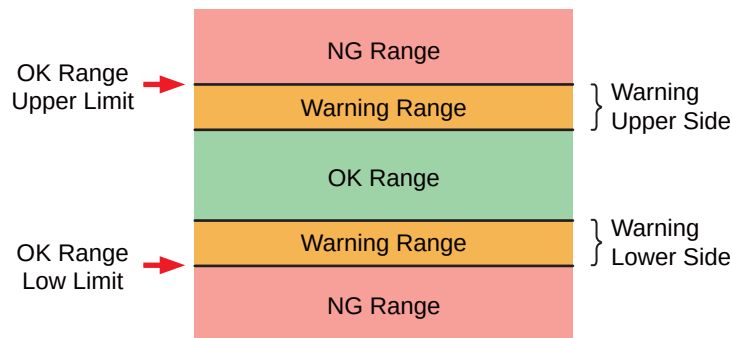


Figure 1-2 Conceptual Diagram of Judgement Range

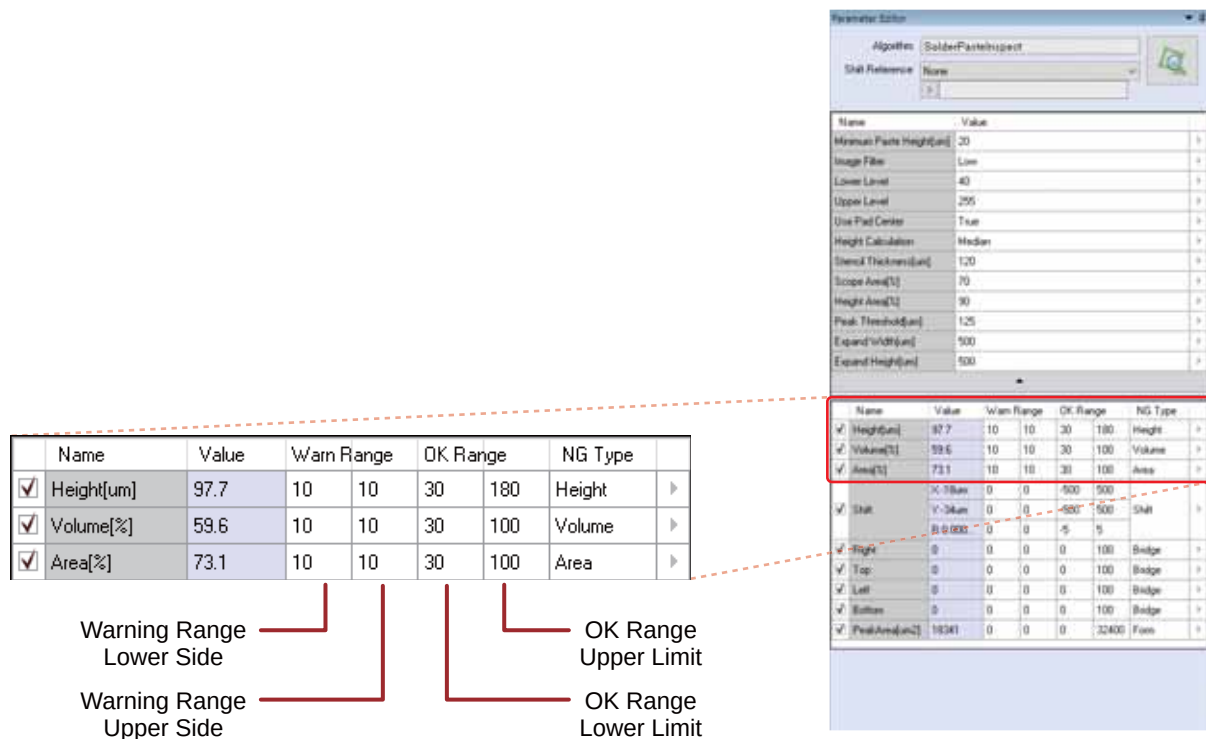


Figure 1-3 Warning Range Settings in Recipe Edit Screen

For example, OK Range: 50 to 180, Warning Range: Lower:10, Upper:20, and Sample value:167.1

	Name	Value	Warn Range		OK Range		NG Type
☒	Height[um]	167.1	10	20	50	180	
☒	Volume[%]	96.6	0	0	0	100	

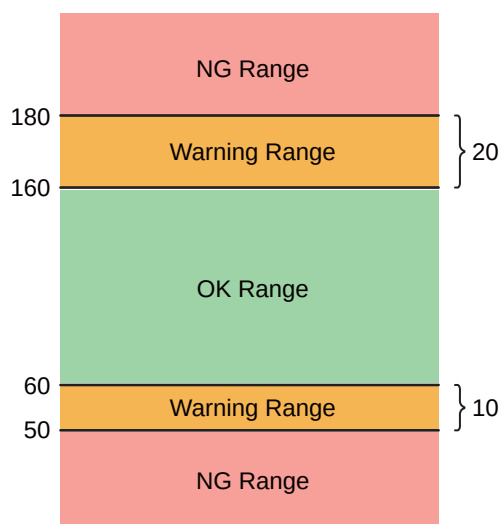


Figure 1-4 Example of Warning Judgement

1.4 Customer Support Contacts

Company Name	Saki Corporation	
Address	DMG MORI Tokyo Digital Innovation Center 3-1-4, Edagawa, Koto-ku, Tokyo, 135-0051, Japan	
URL	https://www.sakicorp.com/	
Phone	Inquiry about the products	+81-(0)3-6632-7901
	Inquiry about the technical service	+81-(0)3-6632-7906